

6500 Packet-Optical Platform

Fault Management - Alarm Clearing

Release 15.5

What's inside...

New in this release and documentation roadmap

Alarm and trouble clearing strategy

Alarm surveillance

Alarm hierarchies and alarm severities

Alarm clearing procedures

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New in this release and documentation roadmap

This Technical Publication supports 6500 Packet-Optical Platform (6500) Release 15.5 software and subsequent maintenance releases for Release 15.5.

You are reading *Fault Management - Alarm Clearing*, 323-1851-543. The following section details what's new in *Fault Management - Alarm Clearing*, 323-1851-543, Standard Issue 2 for Release 15.5.

Issue 2

This document is up-issued to include the “Secure Erase In Progress” alarm clearing procedure.

Issue 1

The following are the new/enhanced features covered in this document.

- New Hardware
 - Shelf Processor SP-3 (NTK555JA)
 - MDRLA 2x6 C-Band circuit pack (NTK553SA)
- Platform enhancements
 - Secure erase enhancements

Table 0-1
New and changed alarm clearing procedures in Release 15.5

Alarm procedure	New/Changed	Related feature
“Automatic Power Reduction Active”	Changed	MDRLA 2x6
“Automatic Shutoff”	Changed	MDRLA 2x6
“Automatic Shutoff Compromised”	Changed	MDRLA 2x6
“Automatic Shutoff Disabled”	Changed	MDRLA 2x6

Table 0-1
New and changed alarm clearing procedures in Release 15.5 (continued)

Alarm procedure	New/Changed	Related feature
"Channel Controller: Failure Detected"	Changed	MDRLA 2x6
"Channel Controller: Unexpected Loss Detected"	Changed	MDRLA 2x6
"Channel Degrade"	Changed	MDRLA 2x6
"Channel Opacity Error"	Changed	MDRLA 2x6
"Circuit Pack Failed"	Changed	SP-3
"Control Mode Provisioning Mismatch"	Changed	SP-3
"Cross-connection Mismatch"	Changed	SP-3
"Database Restore Failed"	Changed	SP-3
"High Fiber Loss"	Changed	MDRLA 2x6
"High Optical Power"	Changed	MDRLA 2x6
"High Received Span Loss"	Changed	MDRLA 2x6
"Input Loss Of Signal"	Changed	MDRLA 2x6
"Line Adjacency Manual Provisioning Required"	Changed	MDRLA 2x6
"Loss Of Signal (OPTMON, VOA)"	Changed	MDRLA 2x6
"Low Optical Return Loss at Output"	Changed	MDRLA 2x6
"Low Received Span Loss"	Changed	MDRLA 2x6
"Minimum Gain"	Changed	MDRLA 2x6
"Output Loss Of Signal"	Changed	MDRLA 2x6
"Resources Above Threshold"	New	SP-3
"Resources At Limit"	New	SP-3
"Secure Erase In Progress"	Changed	SP-3
"Server Validation Certificate About to Expire"	New	SP-3
"Server Validation Certificate Expired"	New	SP-3
"Unassigned Channel Detected"	Changed	MDRLA 2x6

Table 0-2
New/Changed alarms history

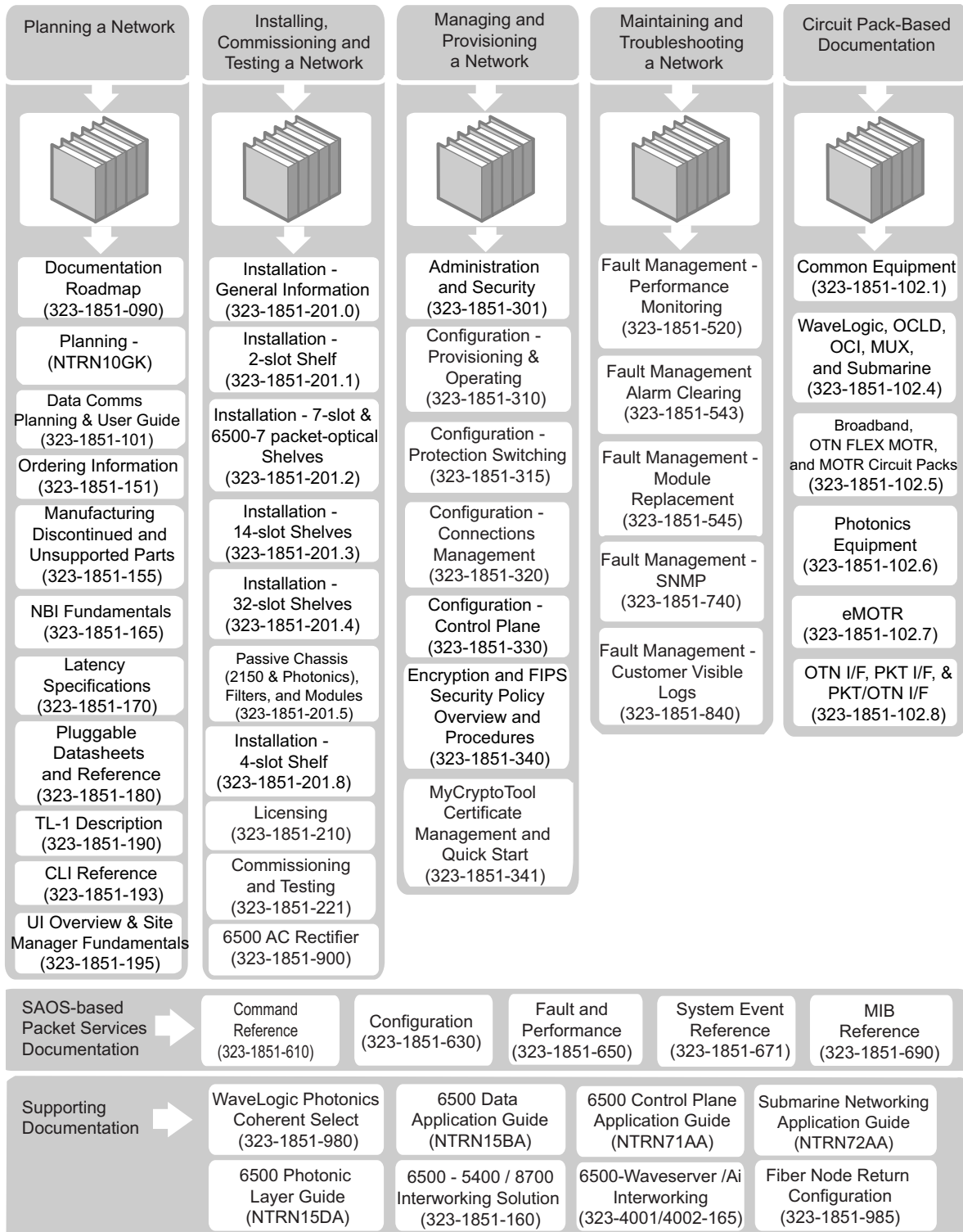
Alarm ID	Alarm Class	Alarm Text	Changes	Release
2125	EQPT	Resources Above Threshold	New	15.5
2128	EQPT	Resources At Limit	New	15.5
2254	SECU	Server Validation Certificate About to Expire	New	15.5
2253	SECU	Server Validation Certificate Expired	New	15.5
2136	CBRTTP	Frequency Out of Range	Deprecated	15.5
2137	CBRTTP	Tx Frequency Out of Range	Deprecated	15.5
2265, 2266	SNC, SNCG	Link Not Protected	New	12.85
2263, 2264	SNC, SNCG	No Protection Route Available	New	12.85
2248	COM	Decommissioning In Progress	New	12.8
2228	EQPT	Mate to Mate Interconnection Provisioning Required	New	12.8
2229	ETTP	Facility Provisioning Failure	New	12.8
2237	COM	Secure Erase In Progress	New	12.8
2240	DOC	Dark Section Conditioning Disabled	New	12.8
2241	PTP	Connection Validation Testing In Progress	New	12.8
2242	NMCC	Target Unachievable	New	12.8
2246	COM	Software Install In Progress	New	12.8
2247	COM	Software Install Failed	New	12.8
2250	SECU	Client Certificate About to Expire	New	12.8
2251	SECU	Client Certificate Expired	New	12.8
2255,2259	PTP,OTM	Optical Signal Failed	New	12.8
124	EQPT	Software Upgrade Failed	Changed Alarm Text	12.8
2196	CHC	ASE Channel Source Active	New	12.72
2221	COM	TIDc Shelf Provisioning Near Limit	New	12.72
2222	COM	TIDc Shelf Provisioning Limit Reached	New	12.72
2223	COM	Streaming Telemetry Limit Reached	New	12.72

Table 0-2
New/Changed alarms history (continued)

Alarm ID	Alarm Class	Alarm Text	Changes	Release
2224	PTP	Facility Provisioning Out Of Range	New	12.72
2225	SECU	TIDc Primary Authentication Server Unavailable	New	12.72
2226	SNC	SNC Home Path Failed - Auto Reversion Inhibited	New	12.72
2227	SNCG	SNCG Home Path Failed - Auto Reversion Inhibited	New	12.72
941	RAMAN	Automatic Power Reduction Active	Deprecated	12.72
1241	COM	Database Corruption	Changed Alarm Text, condtype	12.72

6500 technical publications

The following roadmap identifies the technical publications that support the 6500 platform for Release 15.5.

6500 roadmap

Alarm and trouble clearing strategy

This release of 6500 Packet-Optical Platform (6500) supports the Broadband, PKT/OTN transport and switches, and Photonic services for different circuit pack. The combination of two or all services is also supported.

For more information on the services (and the circuit packs related to each service), refer to the *6500 Planning*, NTRN10GK.

Note: Unless otherwise specified, eMOTR in this document refers to eMOTR (NTK536AA, NTK536AB, NTK536FA, NTK536FB) and eMOTR Edge (NTK536BE) variants.

Site level alarm correlation

The site level alarm correlation feature reduces the number of alarms reported at each site to a minimum. It does not reduce the number of sites in a network reporting alarms. This is performed by the Network level alarm correlation feature. Once Alarm Correlation is enabled, it automatically enables both Site level and Network level alarm correlation. One cannot exist without the other.

The feature requires the system to have full knowledge of topology and connectivity within the network, including channel routing at OADM sites. The physical topology of the network is represented by adjacency objects. The shelf level correlation uses these adjacency objects to notify downstream facilities that an upstream failure has occurred and suppresses alarms on circuit packs within the same site. The service photonic layer interoperability module is responsible for messaging and auditing the fault information which spans shelves within the site.

Alarm correlation requires that physical adjacency information be provisioned between Photonic equipment at a site. For systems controlled by the Domain Optical Controller (DOC), no extra information is required from a user perspective as this adjacency information already exists in order to build the channel topology information (to automatically add/delete photonic channels). For passive photonic systems, which do not use DOC, the adjacency information is necessary to enable features such as Service and Photonic Layer Interoperability (SPLI) and Site Level Alarm correlation. For alarm

correlation to work all the way down to the service layer (that is, service circuit packs connected to the photonic equipment), physical adjacency information must be provisioned. The Physical Adjacency information between the service layer equipment and the CMD is provided by the SPLI feature. The adjacency information between service layer equipment (for example, between a 40G MUX OCI and a 10G OTR) must be manually provisioned. Also, Alarm Correlation requires that the Transponder connections between an OCLD and an OCI be provisioned. Transponder connections are also required to carry traffic.

Alarm correlation has the ability to correlate the service circuit pack faults “backwards” to the CMD44, OMD4, OMX, OMD4, OMD4, BS1, BS2, BS3, or BS5 when all “in-use” channels are faulted. Alarm Correlation software considers a channel “in-use” when the channel’s service circuit pack facility connected to the CMD44, OMD4, OMX, OMD4, or OMD4 Channel Output port has a non-nil discovered far-end address. A non-nil discovered far-end address implies a SPLI match exists.

Note: In passive photonic configurations, the DSCM, which is provisioned as excess loss on a given filter port, is not involved in alarm correlation.

If the discovered far-end address is nil then that channel is not considered when calculating the “all in-use channels failed” condition.

At least two channels must be in-use before the “backwards” alarm correlation is initiated.

A new virtual OPTMON facility is created for the CMD44, OMD4, OMX, OMD4, OMD4, BS1, BS2, BS3, or BS5 Common Input (or NTKW Rx) port and the Common In (or NTKW Rx) fault is reported as an OPTMON Loss of Signal, or OPTMON Group Loss of Signal alarm, depending upon the scenario.

Unlike CMD44 Channel Input ports, the Common In port defaults to “IS” rather than “IS-ANR,MON” when no channel is provisioned and changes to “OOS-AU, FAF” when the LOS condition is asserted.

For the Optical Bridge and Broadcast (OBB) and Optical Broadband Mux/Demux (OBMD 1x8), “Circuit pack Missing” or “Circuit pack Mismatch” alarms mask pre-existing signal alarms on module alarms. For example, LOS on Far-end OCLD RX alarms that were masked by either NLC or SLC prior to a Comms failure remain masked.

For the Optical Protection Switch (OPS) circuit packs, Loss of Signal (LOS) at SW1 IN/SW2 IN does not result in an alarm against OPS SW1/SW2 ports if SLC (Site Level Correlation) is enabled and a correlated upstream alarm within the same site co-exists with the OPS LOS fault.

The Alarm Correlation parameter within the Site Manager Node Information application and the System tab must be set to On for the “backwards” alarm correlation to function.

Backwards alarm correlation only functions if all service circuit packs connected to the CMD44, OMD4, OMX, OMD4F, OMD4F8, BS1, BS2, BS3, or BS5 support alarm correlation.

Site level alarm correlation is supported on the OMX, CMD44, OMD4F, OMD4F8, BS1, BS2, BS3, BS5, and OMD4 configurations provided that all of the following conditions apply:

- Alarm correlation is set to On in the Site Manager Node Information application and the System tab.
- All adjacencies are manually provisioned. Refer to the “Editing facility parameters” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.
- All cascading equipment (, CMD44, OMD4, OMX, OMD4F, OMD4F8, BS1, BS2, BS3, BS5) are provisioned in the same shelf.
- Service circuit packs used in the configuration must support alarm correlation.

See 6500 Packet-Optical Platform *Photonic Layer Guide*, NTRN15DA, for detailed information on OMD4 based single-span applications, and passive photonic layer applications.

Site level alarm correlation is supported on all L0 Photonic circuit packs (Line Interface Module (LIM), WSS, CMD, OSC).

Site level alarm correlation is supported on the RAMAN amplifiers in the following configurations.

- SRA - XLA – SRA
- SRA - XLA – ESAM/SAM
- ESAM/SAM - XLA – SRA
- ESAM/SAM – XLA – ESAM/SAM (any combination)

All traffic related alarms are masked when the “OTDR Trace in Progress” alarm is raised.

SPLI based Alarm Correlation is supported on the PTP, ODUTTP and the OTUTTP facilities for 40G, 100G, WLAi MOTR, and eMOTR circuit packs in the following configurations:

- 40G Broadband circuit packs (OCLD, OCI, and MUX) in POTS provisioning mode (mated with 40G OTN XCIF)
- 100G Broadband circuit packs (OCLD and OCI) in POTS provisioning mode (mated with 100G PKT/OTN XCIF).
- 40G OCLD in POTS provisioning mode (mated with eMOTR)
- PTP on eMOTR circuit packs (40G and 100G)

When alarm correlation is enabled, the OTMn, PTP, ODUTTP, and OTUTTP alarms are masked. The masked alarms can be viewed in Site Manager 'Active Disabled Alarms' window.

The following alarms are masked downstream when Alarm Correlation is on and when an upstream channel, port level or service layer fault occurs:

- Loss Of Signal
- Loss of Clock
- Loss Of Frame
- Loss Of Multiframe
- ODU BDI (masked by a special algorithm when alarm correlation is on, and not by upstream faults)
- OTU BDI (masked by a special algorithm when alarm correlation is on, and not by upstream faults)
- OTU Trace Identifier Mismatch
- ODU Signal Fail
- ODU Trace Identifier Mismatch
- OPU Payload Type Mismatch
- Pre-FEC Signal Fail
- Mapping Mismatch
- OPU AIS
- ODU AIS
- ODU LCK
- ODU OCI
- OTU Signal Fail
- MSI Mismatch
- Far End Client Signal Fail

- Loss Of Channel
- OTU Skew Out Of Range
- OTL Skew Out Of Range
- Multiplexed Rate Mismatch

The following faults trigger downstream masking when Alarm Correlation is on. These are port level issues, where all channels are affected, which will continue to propagate the downstream masking in the Demux direction:

- Loss Of Signal
- Loss of Clock
- Loss Of Frame
- Loss of Channel
- Loss Of Multiframe
- ODU AIS
- ODU LCK
- ODU OCI
- Pre-FEC Signal Fail
- OTU Signal Fail
- OTU Skew Out Of Range
- ODU Signal Fail
- Circuit Pack Failed
- Circuit Pack Missing
- Circuit Pack Mismatch
- Circuit Pack Mismatch – Pluggable
- Circuit Pack Missing – Pluggable
- Circuit Pack Failed – Pluggable

The following client interface circuit packs must be enabled to support SPLI on 10x10G MUX and 40G MUX OCI circuit packs on client interfaces:

- 40G MUX OCI and 10x10G MUX Broadband circuit packs
- 10x10G PKT/OTN I/F circuit packs (POTS)
- 40G MUX OCI interface with 100G PKT/OTN XCIF (POTS)

[Figure 1-1 on page 1-6](#) and [Figure 1-2 on page 1-7](#) show an example of alarms to be raised in a line fiber cut scenario with and without alarm correlation on a DOC controlled system.

Figure 1-1
Line fiber cut without alarm correlation on a DOC-controlled system

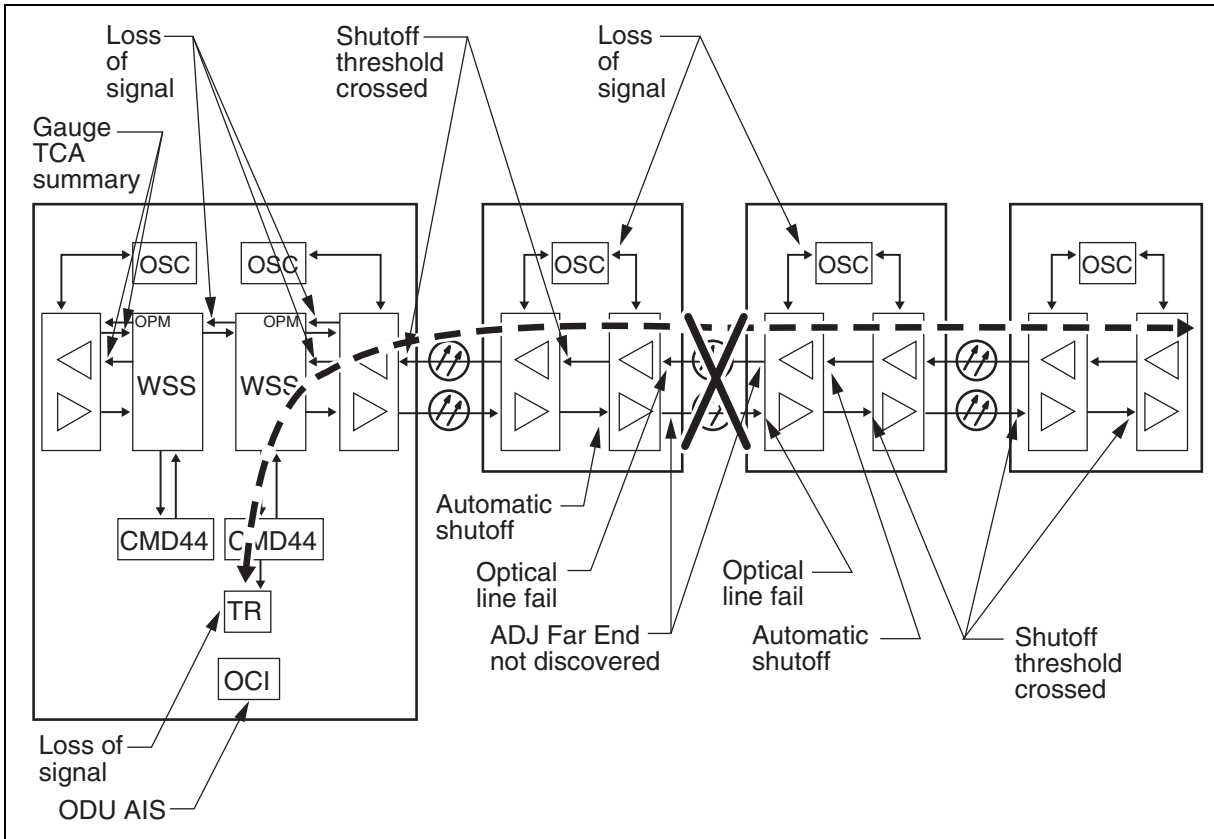
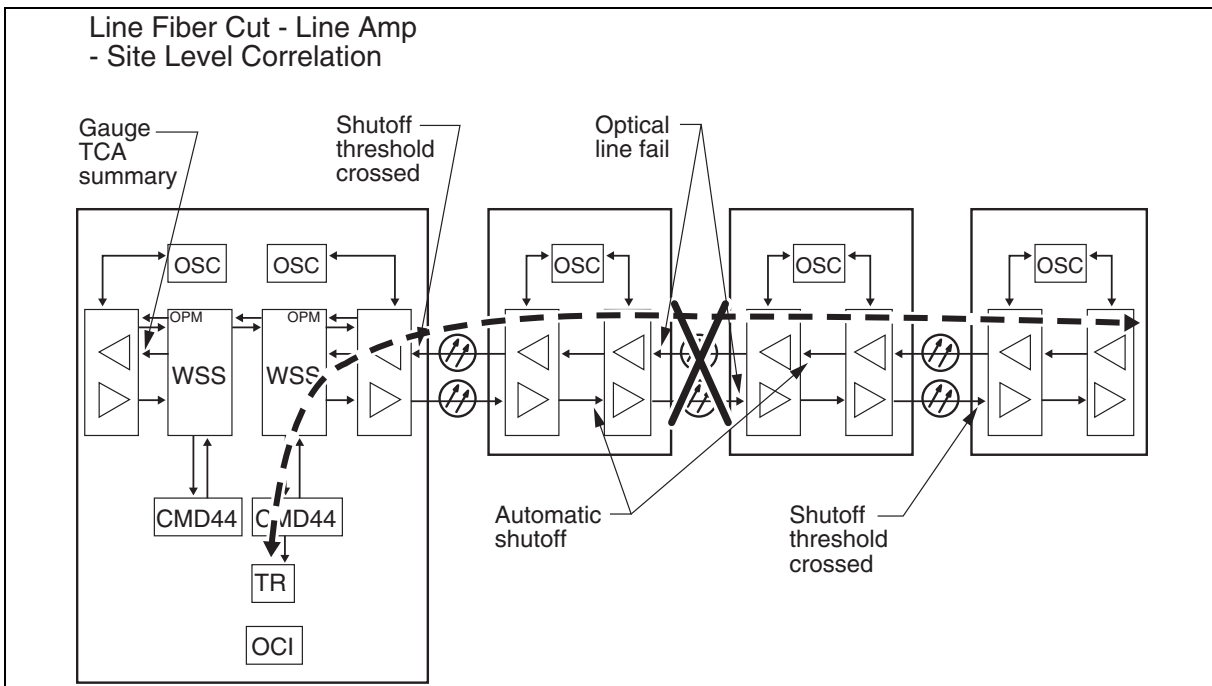


Figure 1-2
Line fiber cut with alarm correlation on a DOC-controlled system



For the list of supported circuit pack combinations refer to the “CMD Tx/Rx type discovery, Tx power & Tx wavelength autoprovisioning, network & site alarm correlation per circuit pack type support” table in *6500 Planning*, NTRN10GK.

Alarm Correlation is supported for DIA OADM configurations.

The alarm correlation functionality can be enabled/disabled on a per shelf basis by editing the Alarm correlation parameter using the Site Manager Node Information application and the System tab. Refer to the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301 for more information.

Network level alarm correlation

Network level alarm correlation (NLC) builds on the Site Level Alarm Correlation feature. Site Level Alarm Correlation masks alarms within a site based on a detected fault. However, Site Level Alarm Correlation does not share fault information with neighboring nodes in the network. It also does not track per-channel failures when channels are muxed and mixed with non-failed channels.

Essentially, NLC will perform site level alarm correlation but it will also perform network level alarm correlation. Network Level Alarm Correlation addresses the gaps of the SLC feature based on wavelength topology. Network Level Alarm Correlation expands adjacency discovery (AD) messaging to support OTS to OTS (intra-node and inter-node) Network Level Alarm Correlation messaging. NLC shares and collects channel status information with neighbors. With knowledge of the channel statuses, NLC performs alarm correlation and masks alarms.

The Network Level Alarm Correlation:

- Suppresses symptomatic (downstream and lower level) alarms
- Suppresses “Hard” faults if all channels on that port are failed (Hard faults are the faults that affect all channels on a port; for example, LOS, AIS, Shutoff Crossed)
- Suppresses “Soft” faults if any channel on that port is failed (Soft faults are the faults that affect only some channels or partial power loss; for example, PM TCAs, Unexpected Loss)
- Suppresses alarms in the network within one minute of the failure

When the photonic network is single-channel, the NLC may be unable to distinguish between a hard fault and a soft fault since there is only one channel within the line port. In these situations, manual isolation of the fault may be required to determine where the root cause of the failure is within the photonic network.

Network level alarm correlation is supported on all L0 Photonic circuit packs (Line Interface Module (LIM), WSS, CMD, OSC).

For the OPS circuit packs, Loss Of Signal (LOS) at SW1 IN/SW2 IN does not result in an alarm against OPS SW1/SW2 ports if NLC is enabled and a correlated upstream alarm co-exists with the OPS LOS fault.

[Figure 1-3 on page 1-9](#) shows an example of Network Level Alarm Correlation.

[Figure 1-4 on page 1-9](#) and [Figure 1-5 on page 1-10](#) show examples of data flow and alarm suppression using channel status indicators.

Figure 1-3
Network level alarm correlation

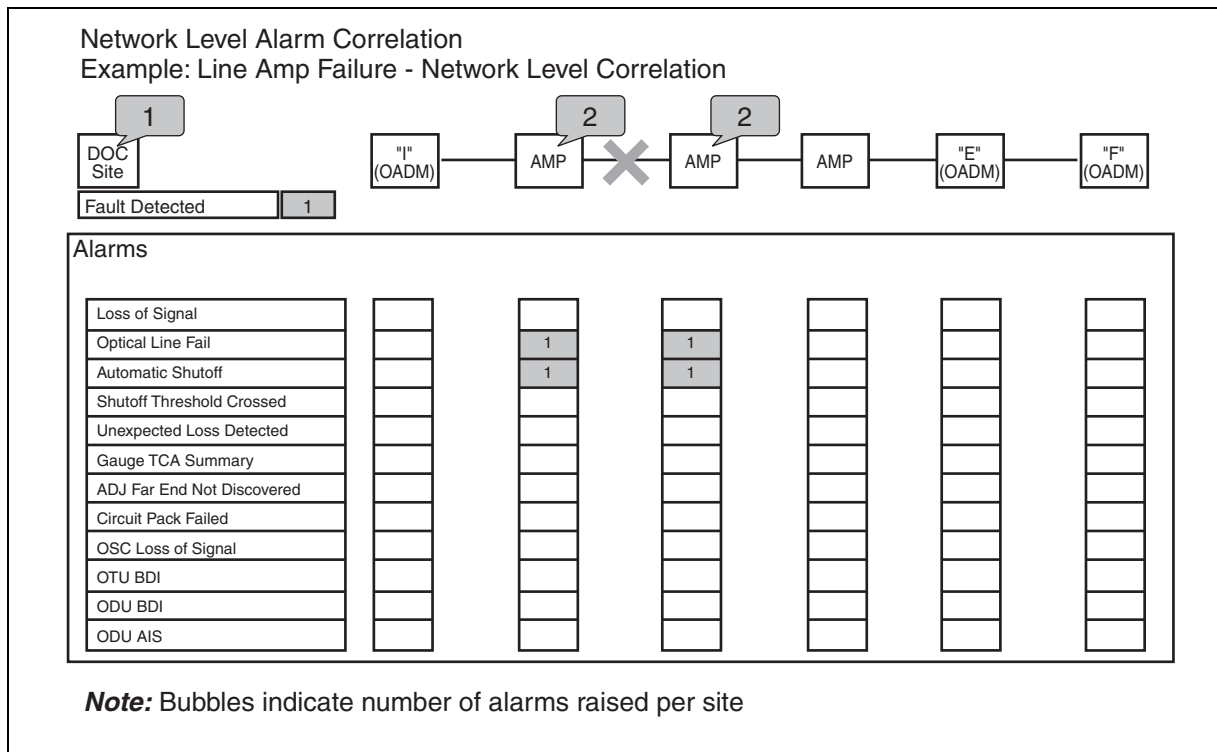


Figure 1-4
Data flow of channel status indicators

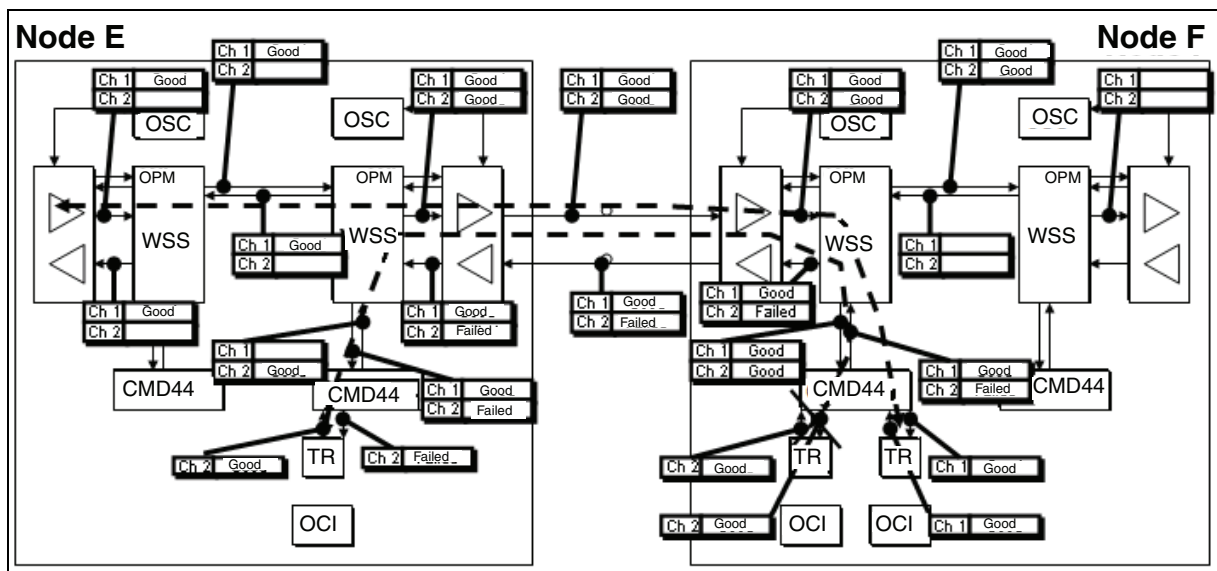
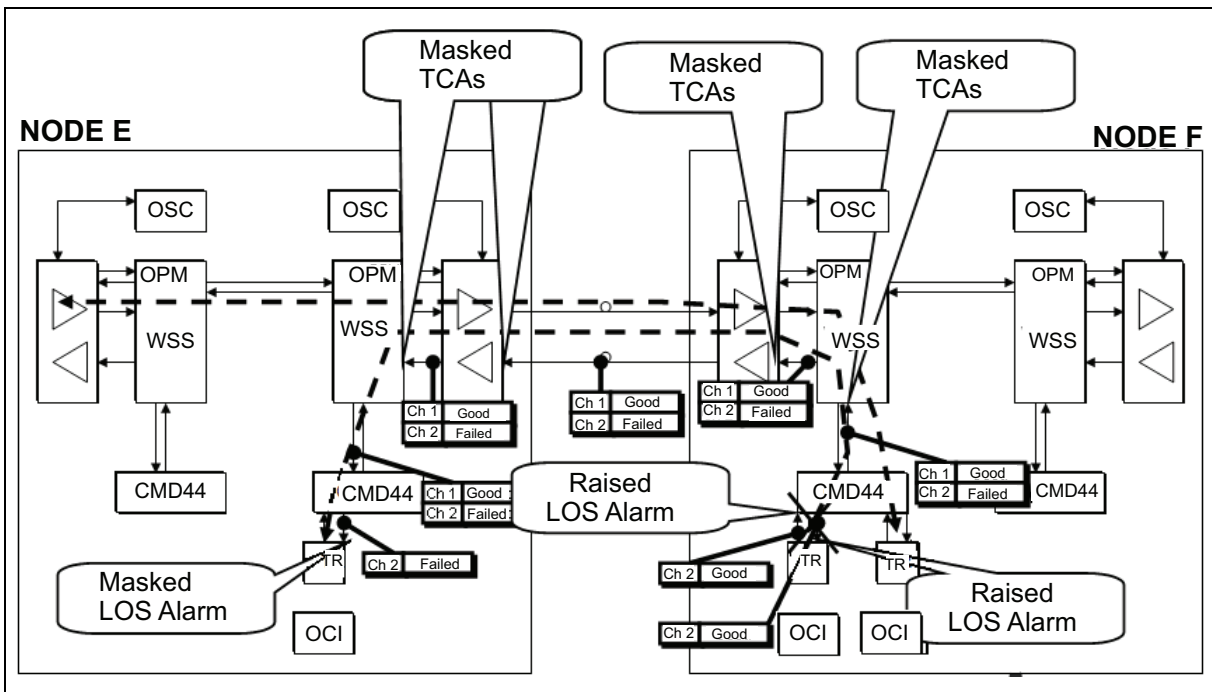


Figure 1-5
Alarm suppression using channel status indicators



Backwards Defect Indicators (BDI)

OTMn facilities raise the “BDI” alarm to indicate a problem exists at the other end of the link. By definition, “BDI” is always a “symptomatic” alarm. When Alarm Correlation is On (default), Conditioning Override can be Standing Condition, Profile (default) or Off. When Alarm Correlation is Off, Conditioning Override can be Standing Condition, or Profile. Refer to the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301.

Before the Conditioning Override parameter is edited, the shelf must be alarm free. Refer to the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301. If the shelf is not alarm free, you must perform a warm restart on the shelf processor after the change is completed to force a re-evaluation of the raised alarms. For details on the shelf processor restart, refer to [Procedure 2-11, “Restarting a circuit pack or shelf processor”](#).

Provisionable severity for alarms and events

Two types of events are supported:

- Standing Conditions (SC), for which a raised and a clear Autonomous Output (AO) is generated
- Transient Conditions (TC), for which a raised AO is generated and cleared

Site Manager supports Critical, Major, Minor, and Warning severity types for alarms. If a Warning severity is selected for a given alarm, the alarm will be raised by the network element as an SC. The Warning severity can be applied to all alarms independently of their default severity (Critical, Major, or Minor). By default, the Warning severity of Standing Conditions is also provisionable in the Alarms Profile application to any of Critical, Major or Minor.

By default each alarm has two severities assigned to it, a Service Affecting (SA) and a Non-Service Affecting (NSA) severity, both of which are generally provisionable in the Alarms Profile application. Environmental alarm severities can be provisioned by changing the attributes of the specific environmental alarm. Some alarms on the 6500 support only one severity, which could be either SA or NSA. In the Alarms Profile application, Site Manager displays “-” for a non-provisionable severity and disables the capability to edit the field. If for any of the SA or NSA severities a Warning is selected, the alarm would be raised as a Standing Condition (SC) for the selected severity.

The severity of an alarm is indicated by the following designations: Critical (C), Major (M) or minor (m). A warning (w) is used for Standing Conditions. It is important to note that the impact of a SC, SA or NSA, is not reported by the network element, hence the Active Alarms application in Site Manager would report the SC as a “w” and displays a “-” in the Service column.

The capability to provision the severity does not apply to Transient Conditions, non provisionable alarms, or output external alarms.

Alarm severity toggling

The WLAi MOTR/FOTR w/ OPS can have PTP alarms which toggle in severity. When a WLAi MOTR w/ OPS (NTK538DS) or WLAi FOTR w/ OPS (NTK538FS) has OPS protection provisioned, and a PTP facility alarm in the following list which exists within both line ports 1P1 and 1P2, a protection switch will occur every minute, generating an event for the protection switch and causing the alarm to toggle between high and low severities.

- Loss Of Channel
- Loss Of Frame
- Loss Of Multiframe
- Loss Of Clock

- Pre-FEC Signal Fail

Workaround: Place one PTP facility Out-Of-Service (OOS) or operate a user protection switch to force the traffic onto one line.

Alarm clearing strategy

The 6500 hardware and software perform automatic fault detection and identification. When a network element detects a fault, it issues autonomous alarms, activates office alarms, and displays alarms through LEDs.

In addition to raising alarms on a shelf, 6500 also supports the generation of northbound simple network management protocol (SNMP) traps for network element alarms and events. Refer to the “Editing and deleting SNMP trap destinations” procedure in *Fault Management - SNMP*, 323-1851-740.

The alarm clearing strategy is based on several assumptions:

- No external problem causes the alarm, such as power fluctuation.
- Primary fault generates primary and secondary alarms that you can clear with a single fault clearing procedure.
- The network element is provisioned correctly and works until the time of the alarm.
- If protection circuitry exists, traffic is switched before performing the alarm clearing.

The network elements report alarms in the following ways:

- alarm LEDs on the network element and circuit packs
- lamps and audible alarms on the
 - MIC and BIP in a 14-slot 6500 shelf. The MIC indicates alarms for the shelf and the BIP indicates alarms for the 6500 shelves in a bay.
 - access panel in a 32-slot 6500 shelf
 - cooling unit in a 4-slot shelf, 7-slot 6500 shelf or 6500-7 packet-optical shelf
- northbound SNMP trap reporting for the NE-level alarms
- network element alarm messages retrieved locally through Site Manager or TL1
- office alarms (optional)

The following steps make up the strategy for fault and alarm clearing:

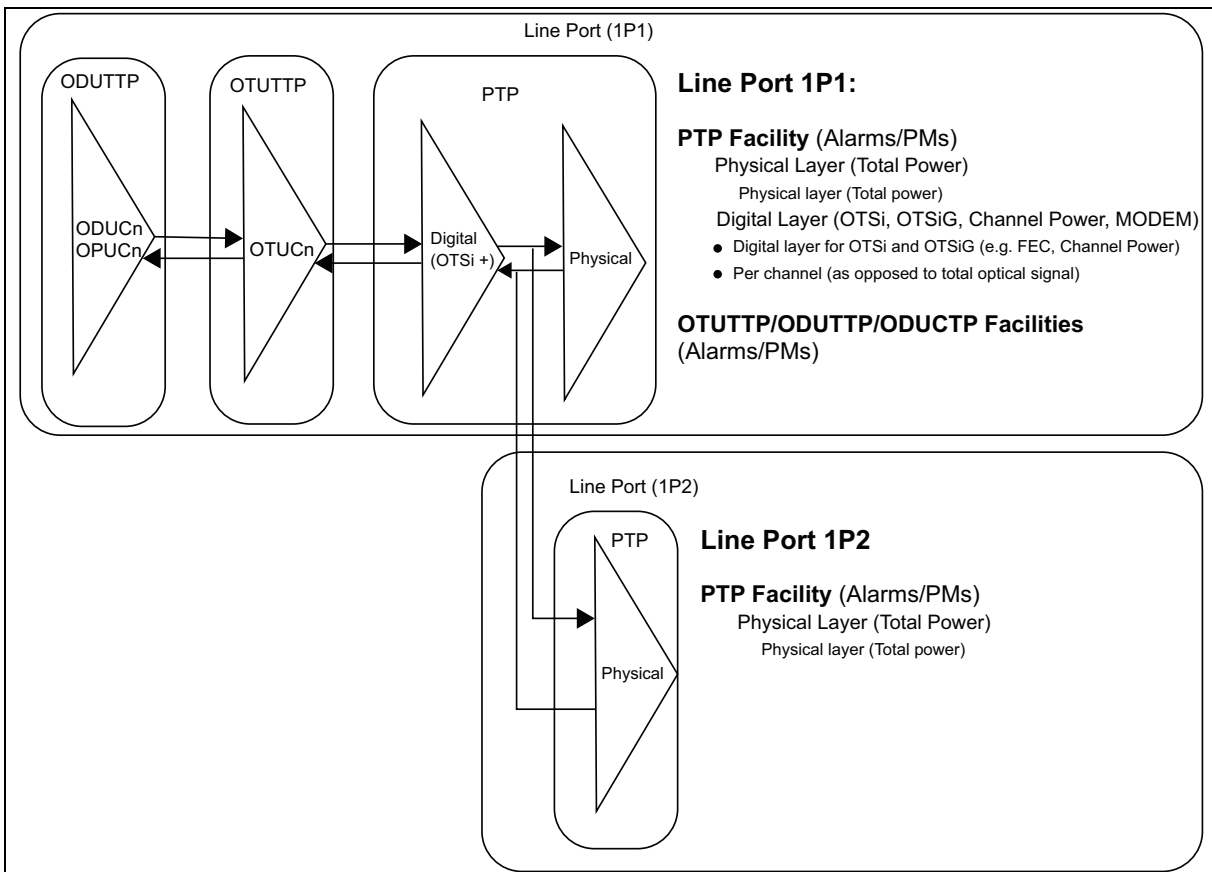
- Detect there is a fault.
- Identify the network element that raised the alarm.
- Check for illuminated Fail LEDs on the circuit packs or modules.

- If a Fail LED is illuminated, perform the procedure to replace a failed circuit pack, pluggable module, or pluggable transceiver.
- If the Fail LED is not illuminated, retrieve alarm messages through Site Manager.
- Identify the local and remote alarms during the procedure.
- Identify the alarm severity.
- Identify which network element to clear.
- Perform trouble-clearing procedures.
- Determine if there are additional alarms.
- If alarms continue to be active, begin the process again.
- If the alarms are cleared, end the process.

For more information on the steps, see [Figure 1-7 on page 1-16](#).

Note: The OTSi/OTUCn/ODUCn/OPUCn layers are only visible on port 1P1 and not on 1P2. Even after a protection switch is performed to 1P2, Site Manager and MCP will show the OTSi/OTUCn/ODUCn/OPUCn on 1P1. Total optical power is visible on both 1P1 and 1P2 regardless of which port is active. The OTSi layer includes the channel power. This power reading should not be used to determine if a protection switch is active or not. Use the protection status application to determine the active path. See [Figure 1-6](#) for alarm/PM reporting of the iOPS circuit packs.

Figure 1-6
Line iOPS monitoring



Alarm priority

Critical alarms have the highest priority and are reported before Major, minor or Warning alarms. Major alarms are reported before minor alarms and minor alarms are reported before warnings. Clear alarms in order of severity:

- Critical, service-affecting (C, SA) alarm
- Major, service-affecting (M, SA) alarm
- Major, non-service-affecting (M, NSA) alarm
- failed circuit pack non-service-affecting (NSA) alarm
- minor, service-affecting (m, SA) alarm
- minor, non-service-affecting (m, NSA) alarm

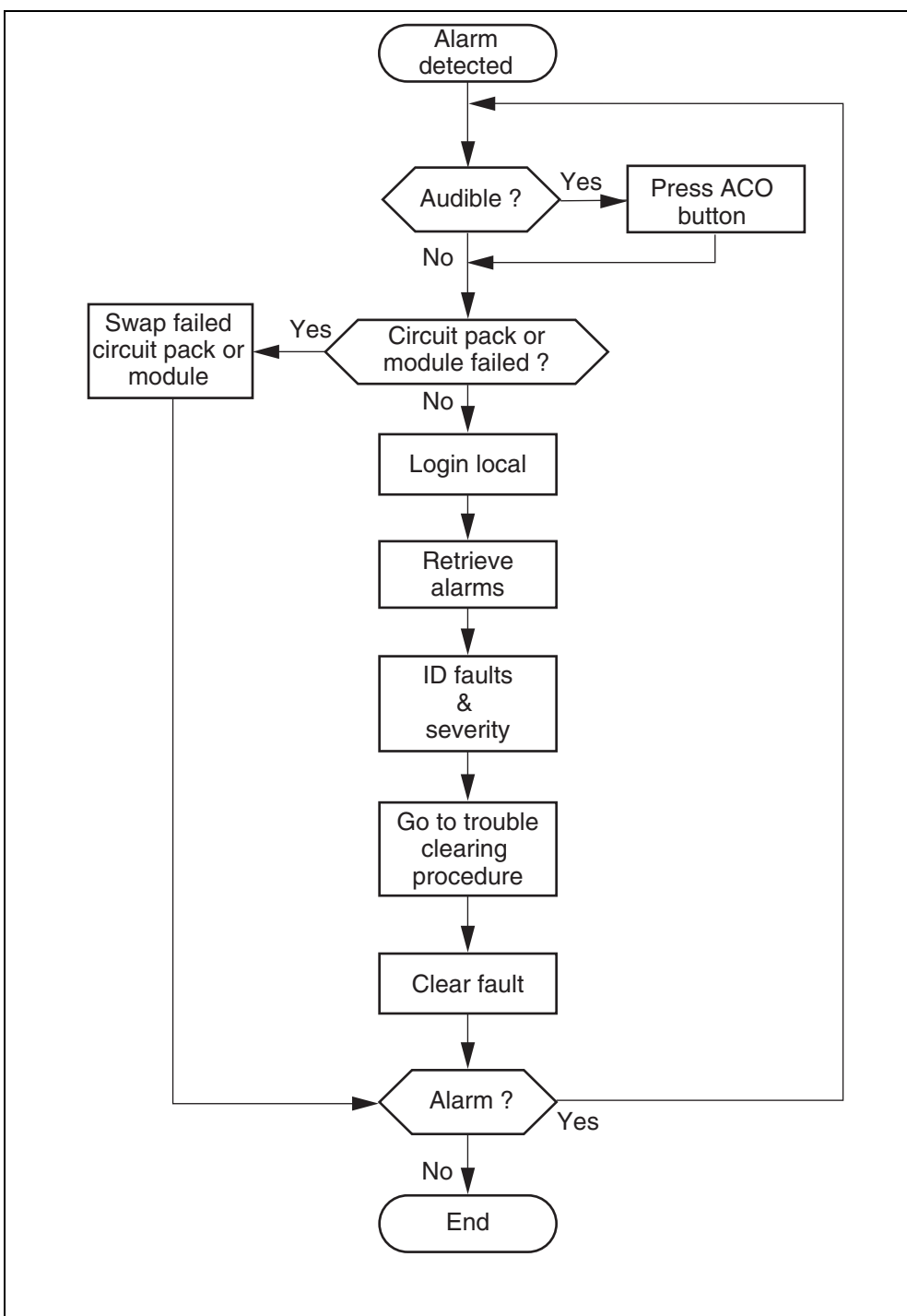
One fault can cause more than one alarm. Clearing the alarm with the highest severity can clear other alarms.

You can change the alarm severity of both high and low entities using the **Alarms Profile** application in Site Manager. Refer to [Procedure 2-8, “Editing an alarm profile”](#) in [“Alarm surveillance”](#).

The alarm notification codes (service affecting, SA, or non-service affecting, NSA) are unaffected by change of alarm severity. The alarm severities in the ALL ENABLED, ALL DISABLED, and FACTORY DEFAULT alarm profiles cannot be changed.

Refer to the [“Service-affecting and non-service-affecting severities”](#) on page [3-2](#) for further details on alarm severities.

Figure 1-7
Fault clearing strategy



LED indications

The 6500 uses an LED indication scheme, where:

- red indicates failure (requires replacement) (temporarily on when circuit pack is powering up)
- green indicates active (powered and operational).
- green flashing indicates circuit pack software is booting or restarting (warm/cold)
- yellow indicates warning (something missing or activity in progress)
- blue indicates circuit pack is in use. Do not unseat (removing the circuit pack will impact service)

Alarm unit

The alarm unit (AU) contains LEDs to provide visual indication of alarms active for the 6500 installed on the rack. The alarm unit illuminates the appropriate LEDs (Critical, Major and minor) on the front panel of the BIP based on alarm signals received from the 6500 shelves through the alarm input interface and close the appropriate alarm relay contacts on the alarm output interface.

The alarm unit contains the following LEDs:

- red and yellow alarm LEDs fed from the input alarm signals illuminate to indicate the following conditions:
 - Critical alarm (red)
 - Major alarm (red)
 - minor alarm (yellow)
- red power LED illuminates to indicate either or both of the following conditions:
 - tripped circuit breaker on either side (A or B)
 - open alarm unit fuse

The alarm unit contains a lamp test button on the front panel of the BIP. When pressed, all LEDs illuminate. The alarm unit also contains a push button switch (ACO) on the BIP front panel to reset all BIP alarm relays. The alarm unit is field replaceable on the NTK599BA variants.

Note that the 1U BIP/FIP does not have an alarm unit.

Circuit pack/Module LEDs

After a circuit pack is inserted, reseated, or a cold restart is performed, all the LEDs turn on for approximately five seconds.

Each circuit pack contains the following status LEDs:

- triangular red Fail LED
- rectangular green Ready LED
- diamond blue In Use LED

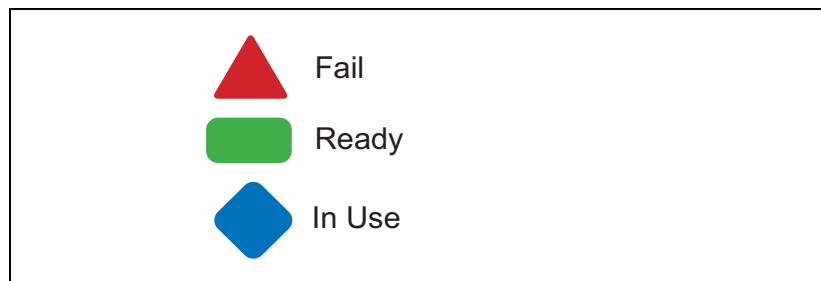
Exceptions include the Power Input Card, fan module, MIC, filler cards, and any equipment connected to external slots (with an RJ45 inventory cable) such as a PPC6, CMD44, OMD4, OMX, BMD2, UBMD2, MBMD2, or DSCM module.

The Shelf Level View application in Site Manager displays the current color-coded LED status for modules, sub-modules, and ports on 6500 shelves. A checkbox allows you to enable or disable the LED status display.

Some Broadband circuit packs also contain a:

- circular yellow LOS LED
- circular yellow Synchronization status LED
- circular red/yellow bi-color Optical Fail/LOS LED

Figure 1-8
Circuit pack LEDs



Red Fail LED

The triangular red Fail LED indicates a hardware or software failure state of the circuit pack.

The red LED turns on when an equipment failure has been isolated to that circuit pack and must be replaced (for example, a Circuit Pack Failed alarm is active on the circuit pack).

The red LED turns on for approximately 30 seconds when a lamp test is initiated.

For the X-Conn 1600G PKT/OTN (NTK616AA), X-Conn 3200G PKT/OTN Type 2 (NTK616BA), X-Conn 600G PKT/OTN (NTK615AA), and X-Conn 1200G PKT/OTN Type 2 (NTK615BA) circuit packs, the red LED turns on for approximately five seconds before a restart is performed on the circuit pack.

The red LED flashes when a user intervention flash test is being performed.

Green Ready LED

The rectangular green Ready LED indicates the circuit pack is powered-up and does not have a circuit pack fail condition.

The green LED turns on when the circuit pack initialization is completed and a circuit pack fail condition is not present.

The green LED turns off when a circuit pack fail condition is detected on the circuit pack.

The green LED turns on for approximately 30 seconds when a lamp test is initiated.

The green LED flashes when:

- a circuit pack is initializing
- a software auto-upgrade is being performed
- a user intervention flash test is being performed

Blue In Use LED

The diamond blue In Use LED indicates the circuit pack is in-service.

The blue LED turns on when the circuit pack is in-service.

ATTENTION

When the shelf processor or the OTN cross-connect circuit packs power up in a shelf (after the shelf is decommissioned), the blue/green LED on both cross-connect circuit packs will turn on.

The blue LED turns on for approximately 30 seconds when a lamp test is initiated.

The blue LED flashes when a user intervention flash test is being performed.

The blue LED turns off when the:

- circuit pack is out-of-service
- circuit pack has been inserted or cold restarted and cannot be provisioned by the shelf processor. This occurs for failed or mismatched circuit packs, or circuit packs that have their database rebuilt.



CAUTION

Risk of traffic loss

Do not remove a circuit pack while the blue In Use LED is illuminated.

Yellow LOS LED

The circular LOS LED is used to indicate Rx Loss of Signal condition.

The yellow LED turns on for approximately 30 seconds when a lamp test is initiated.

The yellow LED flashes when a user intervention flash test is being performed.

Red/yellow bi-color circle (Fail/LOS) LED

The circular LOS LED located to the left of each port is used to communicate optical module (SFP/XFP/SFP+) fail and receive loss of signal (for the CFP module this LED is located below the module):

- The red LED turns on when a pluggable is present and the pluggable has failed.
- The yellow LED turns on when a receive loss of signal condition is detected on the port (and the corresponding facility is provisioned) and the pluggable is present and has not failed.

Some circuit packs contain additional LEDs as detailed in the following sections.

Power Input Card

All 6500 shelves use field-replaceable Power Input Cards to interface between the shelf power connection points and the backplane. Depending on the shelf or equipped Power Input Card variant, the power may be supplied by -48/60 V dc feeds, 24 V dc feeds or 100-240 V ac (50 Hz or 60 Hz) feeds.

The Power Input Cards have one or more Power OK LEDs (circular shape except the NTK605CAE5, NTK605EAE5 variants of the 3x60A Power Input Cards where a rectangle is used) and a circular yellow Low current LED. The green LED turns on to indicate the power feed is active and a minimum voltage is detected and flowing through to the backplane (through any breaker, fuse or power converter if applicable to the equipped Power Input Card). For DC Power Input Cards, the feed must also be in the correct polarity for its associated Power OK LED to illuminate. In any case where the Power OK LED is not illuminated and the Power Input Card is working correctly, an operational shelf processor should assert any applicable alarms such as Power Failure - A/B, Power Failure - Low Voltage or Power Failure - Fuse Blown. However, in

some cases a Power OK LED may be illuminated indicating that a minimum voltage is detected on a feed but the shelf processor may still assert one or more of the following alarms based on the shelf's operating specification: Power Failure - Low Voltage or Power Failure - Fuse Blown (if applicable). See alarm clearing procedures in this document for details.

With the exception of the NTK605CAE5 and NTK605EAE5 variants of the 3x60A Power Input Cards, Power Input Cards that terminate more than one feed have a Power OK LED per feed (for example, Power OK for each of feed L1, L2, ...). For the NTK605CAE5 and NTK605EAE5 variants of the 3x60A Power Input Card (breakered and fused), the green LED turns on to indicate that all three feeds (L1, L2 and L3) are at least approximately -40 Vdc and all three breakers are on (or all 3 fuse cartridges with fuses are in place and working). Power OK LED ON means that there are no "Power Failure" alarms against any feeds on that module.

For the 3x60A Power Input Cards Type 2 (NTK605FA, NTK605FE) and the 4x60A Power Input Cards Type 2 (NTK605GA and NTK605GE), the Lx Power OK LED (Lx=L1, L2, L3 for 3x60A and Lx=L1, L2, L3, L4 for 4x60A) indicates that the feed is live (has a minimum voltage applied). The SP raises an alarm if the voltage is not at least -40V on a feed). The LED may remain illuminated until there is insufficient voltage on the feed to illuminate it.

For the NTK605CAE5 and NTK605EAE5 variants of the 3x60A Power Input Card, the rectangular amber LED labeled "Low voltage" indicates that at least one of the feed voltages is below approximately -40 Vdc. The LED turns off when all feed voltages fall below approximately -24 Vdc. The "Low Voltage" LED could also mean that one or two breakers are off/tripped (or one or two fuses missing/blown) and the remaining feed is at least approximately -24 Vdc. If all three breakers are off/tripped (or fuses missing/blown) both LEDs will be off (because there is no voltage to activate the Low Voltage LED). Low Voltage LED ON means that there is at least one "Power Failure" alarm against a feed on that module.

ATTENTION

Use of the alarm indicator fuse provided with Fused Power Input Card kits (NTK505EAE5, NTK605EAE5, or NTK505EEE5) is optional and requires careful consideration. The fuse cartridge connects the main fuse and alarm indicator fuse in parallel. If the main fuse is missing or already blown on installation, the electrical path through the parallel alarm indicator fuse may provide sufficient current to activate the Power OK LED on the card but insufficient current to blow the fuse. Neither a Power Failure nor a Power Failure - Fuse Blown alarm would be asserted giving the impression of power redundancy. Alarm indicator fuses are intended for use where required by the installation practice. Otherwise, the indicator fuse position must be covered with the label provided with the Power Input Card.

Maintenance interface card

The maintenance interface card (MIC) contains five circular LEDs that indicate the alarm status of the shelf:

- The green Power LED turns on if power is present on the shelf.
- The red Critical LED turns on if one or more Critical alarm exists on the shelf.
- The red Major LED turns on if one or more Major alarm exists on the shelf.
- The yellow Minor LED turns on if one or more minor alarm exists on the shelf.
- The white ACO LED turns on when the ACO button is pressed and at least one Critical, Major, or Minor alarm exists on the shelf. The ACO LED turns off if:
 - no alarms exist on the shelf
 - a new Major or Critical alarm is raised

Fan modules

The fan modules have a triangular red LED and a rectangular green LED.

- The red LED turns on to indicate a fan module failure.
- The green LED turns on to indicate an operational fan module.

The fan modules in the 4-Slot Shelf, 7-slot shelf or 6500-7 packet-optical shelf include alarm LEDs and Alarm Cut-off (ACO)/Lamp test button.

Ethernet ports

Each Ethernet port on the shelf processor and access panel has two LEDs (one green and one yellow) integrated in the RJ45 connector.

- The green LED turns on when a link pulse is received.
- The yellow LED turns on when data is being received on the receive port.

The Ethernet port LEDs are not turned on during a power-up or a cold restart sequence of the shelf processor.

Interface circuit packs without pluggable optics

Circuit pack without pluggable optics (such as, 40G OCL, 40G OCLD or Wavelength-selective 40G OCLD) have a circular yellow LED. The yellow LED turns on when a receive loss of signal condition is detected on the port and the corresponding facility is provisioned.

If a terminal loopback is applied to the optical facility while a loss of signal condition exists, the yellow LED turns off as the loss of signal detection circuit is bypassed.

Interface circuit packs with pluggable optics

Circuit packs with pluggable optics (for example, ports 2-5 on a WLAi MOTR) have a circular bi-color red/yellow LED located to the left of each port.

The red LED turns on when a pluggable is present and the pluggable has failed.

The yellow LEDs turns on when a receive loss of signal condition is detected on the port (and the corresponding facility is provisioned) and the pluggable is present and has not failed.

If a terminal loopback is applied to the optical facility while a loss of signal condition exists, the yellow LED turns off as the loss of signal detection circuit is bypassed.

Cross-connect circuit pack

The cross-connect circuit packs, have four circular yellow LEDs (LOS Ref 1, Ref 2, Ref 3, and Ref 4), one for each of the four possible reference sources in the timing generation hierarchy. The yellow LED turns on when a loss of synchronization reference is detected for the associated provisioned timing reference source.

LED sequences

During a power up (circuit pack insertion or reseal) or a restart, the red Fail, the green Ready, and the blue In Use status LEDs on the front of the circuit pack go through a sequence as detailed in the following sections.

Circuit pack insertion/reseat and cold restart LED sequence

For a Photonic circuit pack insertion/reseat or a cold restart, the red Fail, the green Ready, and the blue In Use status LEDs go through the following sequence:

- The red LED turns on momentarily.
- After a few seconds, the red, green and blue LEDs turn on.

- After a few seconds, the red and blue LEDs turn off and the green LED flashes to indicate the software is initializing.
- When software initialization is complete, the green status LED turns on (does not flash).
- If the circuit pack is in-service, the blue LED turns on after the green LED while the circuit pack is in use. If the circuit pack has failed, the green LED turns off and the red LED turns on.

During a circuit pack insertion/reseat or a cold restart, the port and Ethernet port status LEDs remain off.

When the green LED is steadily lit (after the software initialization), an auto-upgrade of the circuit pack may occur. During an auto-upgrade, the green LED flashes and the blue LED remains off (a Software Auto-Upgrade in progress alarm is raised and clears after the auto-upgrade is complete).

Warm restart sequence

For a warm restart, the red Fail, the green Ready and the blue In Use status LEDs go through the following sequence:

- The red and blue LEDs remain in the state they were before the restart.
- After a few seconds, the green LED flashes to indicate the software is initializing.
- When software initialization is complete, the green status LED turns on (does not flash).
- If the circuit pack has failed, the green LED turns off and the red LED turns on.

Lamp test

A lamp test can be initiated using the ACO/Lamp Test (labeled ACO/Lamp Test, ACO/LT or ACO/LPT depending on the shelf type) button found near the shelf's Critical, Major and Minor LEDs (on all 6500 shelves except for 2-slot variants) or using Site Manager.

When initiated, the lamp test is performed on all the applicable LEDs on the network element (does not apply to any power input LEDs, LEDs on shelf peripherals provisioned in virtual slots or connected to RJ45 external slot inventory interfaces, LEDs on RJ-45 ports used for Telemetry In/Out or any equipped circuit pack that is not in a ready state). A lamp test times out after approximately 30 seconds and the LEDs revert to the previous status. For bi-color port LEDs, one color is lit for first 15 seconds and the other color is lit for the remaining 15 seconds.

For the 32-slot shelf, the lamp test button (labeled ACO/LT) is on the Access Panel (AP). For the 14-slot shelf, the lamp test button (labeled ACO/Lamp Test) is on the maintenance interface card (MIC). For the 4-slot shelf, 6500-7

packet-optical shelf, the lamp test button (labeled ACO/Lamp Test) is on the fan module. The lamp test function will activate all applicable LEDs in the system if at least one SP is present and in a ready state. Depressing the button will have no effect if both SPs are missing or are not ready.

For the 7-slot shelf, the lamp test button (labeled ACO/Lamp Test) is on the fan module. The lamp test function will activate all applicable LEDs in the system if the SP (or SPAP-2 in the case of a 7-slot Type 2 shelf) is present and in a ready state. Depressing the button will have no effect if the SP (or SPAP-2) is missing or is not ready.

For the 2-slot shelf, there is no lamp test button on the shelf. The software initiated lamp test is performed using Site Manager.

When initiated from Site Manager, the test can be performed on all the applicable LEDs on the network element or on a single circuit pack.

ATTENTION

You cannot perform a lamp test on the LEDs on the Power Input Cards. On some access panels, the RJ45 external slot inventory interfaces appear to have two LEDs but there is only one that is used during normal operation and that will illuminate during a lamp test.

In addition, the user has the option to perform a user intervention flash test on a slot or port basis that causes the LEDs to flash, allowing a user at the site to identify a circuit pack, pluggable module, or pluggable transceiver. When initiated for a slot, the status LEDs (red Fail, green Ready, and blue In Use) on the specified circuit pack flash for 15 minutes and the status of all port LEDs remain the same. When initiated for a port, the status LEDs (red Fail, green Ready, and blue In Use) on the circuit pack, the pluggable module, and the specified port LED flash for 15 minutes (if the port LED is bi-color, only the yellow color flashes).

When user flash test is initiated on a port of an eMOTR circuit pack, Port yellow LEDs are switched on without any flashing.

Consider the following:

- Only ports on circuit packs with pluggables physically present can be selected for a port user intervention flash test.
- A user intervention flash test cannot be performed on the LEDs on the MIC, the Power Input Card, the access panel, the fan modules, or the LAN port on the shelf processor.
- A user intervention flash test can be stopped by performing a non-flash lamp test or using the on/off radio buttons on Site Manager.

For information on initiating a lamp test:

- using the ACO button, refer to [Procedure 2-13, “Clearing audible alarms and performing lamp tests”](#).
- from Site Manager, refer to the “Performing a lamp test and clearing audible alarms using the Visualization tool” procedures in *Administration and Security*, 323-1851-301.

Viewing active alarms and events

Active alarms are indicated on the 6500 equipment and are visible from the Site Manager user interface. Alarm history, events, and logs are stored on the network element. Login sessions using Site Manager craft user interface, Optical Manager Element Adapter, and Optical Application Platform provide details of network element alarms.

The types of 6500 network element alarms are:

- equipment
- common equipment
- facility

Color-coded alarm severity

Color is used to highlight the severity value in the application summary tables for the following Site Manager applications:

- Consolidated Alarms
- Active Alarms
- Historical Fault Browser
- Active Disabled Alarms
- Visualization tool display of filtered active alarms

Alarm severity color codes can be configured by modifying the values of color codes defined in the “gui.properties” file, which can be found at the base directory located on the “Site Manager” installation directory on your computer.

The format of color codes in “gui.properties” should be in Hex format. The standard alarm severity color codes are shown below.

```
# Standard alarm severity colors.  
COLOR_ALARM_CRITICAL=0xFF0000  
COLOR_ALARM_MAJOR=0xFF0000  
COLOR_ALARM_MINOR=0xFFB200  
COLOR_ALARM_WARNING=0xFFCC00  
COLOR_ALARM_INFO=0xE0E0E0
```

```
COLOR_ALARM_INDETERMINATE=0xFFFFFFFF
COLOR_ALARM_DEFAULT=0xE0E0E0
COLOR_ALARM_ALERT=0xFFCC00
```

If the above color codes are commented in “gui.properties”, the default alarm color codes are displayed.

The color coding is defined as follows in [Table 1-1](#).

Table 1-1
Alarm severity color coding

Alarm severity		Site Manager default alarm color coding
Critical, service-affecting	(C, SA)	red
Critical, non-service-affecting	(C, NSA)	red
Major, service-affecting	(M, SA)	red
Major, non-service-affecting	(M, NSA)	red
Minor, service-affecting	(m, SA)	orange
Minor, non-service-affecting	(m, NSA)	orange
Warning	(w, -)	yellow
Note: Site Manager Alarm Colors are user configurable. Refer to “Modifying Site Manager Alarm Color” procedure in <i>User Interface Overview and Site Manager Fundamentals</i> , 323-1851-195. When Site Manager is launched from MCP, the MCP Color definitions are used as defined in the Alarm Indicators and severities section of <i>MCP Unified Management System Standard Operations Guide</i> , 450-3201-301.		

Viewing active alarms

Site Manager provides the user with a visual summary of all active alarms for all 6500 network elements logged in to, through the alarm banner. The user views a list of active alarms on a 6500 shelf by selecting the **Active Alarms** application on the **Fault** menu of Site Manager. A maximum of 4600 active alarms are supported on the 6500 network element.

The active alarm application provides the user with the ability to filter and sort alarms and perform manual or automatic refresh of the active alarm list. Alarms details are available for each active alarm in the list.

Alarm reports can be affected by the primary state of the circuit pack and facilities. For example, an alarm is not reported until the primary state of a circuit pack or facility changes from out-of-service to in-service.

Auto In Service (AINS)

AINS is the ability to set up traffic on a network without alarms prior to the end user signal being applied and error-free. This allows entities to be put in-service but remain alarm free until a circuit pack is inserted or cables (traffic) are plugged into the circuit pack.

When an equipment/facility was created with equipment/facility AINS enabled or edited to enable AINS, and the equipment or clean facility signal is missing, the equipment/facility enters the AINS state. When in the AINS state, the equipment/facility primary and secondary states indicate the equipment/facility is in an alarmed condition but alarms are suppressed for that equipment/facility.

When a valid circuit pack is inserted or a clean signal is detected for an equipment or facility in AINS, a timer starts. If the equipment or facility remains fault free for the specified AINS timeout value, the equipment or facility is automatically transitioned from the AINS state. Subsequent conditions that result in alarms will result in alarms being generated on the NE.

If a circuit pack supports equipment AINS, then any pluggable modules or pluggable transceivers supported on that circuit pack also support equipment AINS.

Two independent system-wide AINS timeout parameters are supported - one for Facility AINS, and the other for Equipment AINS, applicable to all entities supporting the AINS on the shelf. The supported AINS timeout value range is from five minutes (default) to 96 hours. The AINS timeout value can be set in increments of five minutes. Refer to “Editing the nodal system parameters” in *Administration and Security*, 323-1851-301.

When an equipment or facility is in an AINS state, suppressed alarms can be retrieved in the Active Disabled Alarms application within Site Manager.

Traffic alarms and events are suppressed for the facility with the AINS state, explicitly including the following:

- Traffic faults, near end and far end
- PM TCA reports
- WAN alarms/PMs on a LAN facility
- Alarms which require manual action to clear, such as “Loopback Active”

Protection switch alarms/events are not suppressed.

Facility AINS does not permanently mask an ODU OCI OTM3 alarm on Broadband circuit packs. The facility AINS timer counts down when the fault is active, and the alarm is unmasked after the timer expires.

The ODUk OCI, ODUk LCK, and AIS alarms are considered valid signals from an AINS perspective. Therefore, if an OTMn facility transitions from a hard failure (such as LOS, LOF, and LOC) to an ODUk OCI, ODUk LCK, or AIS alarm, then the timer starts counting down just as if traffic had been restored. This applies to all OTMn facilities and on all circuit packs that support these facilities.

Facility alarms for received signal-affecting faults such as the following alarms, as well as others, prevent the AINS timer from counting down:

- Loss of Signal
- Loss of frame
- Loss of clock
- Loss of Data Sync
- Frequency Out Of Range
- Signal Fail
- Signal Degrade
- Loss Of MultiFrame
- Excessive Error Ratio
- Link Down
- Rx Power Out Of Range

The following equipment alarms are not AINS impacting and do not reset AINS timer:

- Circuit Pack Failed
- Cold Restart Required
- Intercard Suspected
- Internal Mgmt Comms Suspected
- High Received Span Loss
- Low Received Span Loss
- Autoprovisioning Mismatch - Pluggable
- Circuit Pack Mismatch - Pluggable
- Circuit Pack Failed - Pluggable
- Circuit Pack Unknown - Pluggable
- Intercard Suspected - Pluggable
- Provisioning Incompatible - Pluggable

PM collection is also inhibited when a facility is in an AINS state. PM counts will not be recorded/calculated for facilities and paths. Analog PMs, such as power levels which are measured, will continue to be recorded. Bins will be set to IDF when the AINS state is active.

Viewing events

The **Historical Fault Browser** application supports the following functionality:

- viewing of historical (current and cleared) alarms for the 6500 network element
- viewing of logs
- filtering of alarms based on severity
- details for specific events (alarms or logs)

The 6500 also supports the generation of northbound SNMP traps for network element alarms and events. For more details on supported SNMP functionality, refer to the:

- “SNMP support” section in *Fault Management - SNMP*, 323-1851-740
- external alarm provisioning, external controls in [“Alarm surveillance”](#)

The 6500 network element stores up to 5500 events. 5000 of those are from alarms/events that were enabled, and the other 500 are for alarms/events that were disabled at the time they were generated. The Historical Fault Browser application provides the user with the ability to filter and sort events and perform manual refresh of the event list.

Alarm profiles

The alarm profiles application gives the user the ability to view, edit, and delete alarm profiles. You can display alarm profiles from the Configuration menu or the Shelf Level View application.

A profile contains all the alarm points applicable for the alarm class and a status, enabled or disabled, for each alarm point. A profile can be applied to an individual facility or circuit pack of that alarm class to quickly disable multiple alarm points. A default profile can be set for an alarm class so that when a new facility or circuit pack of that class is first provisioned, the default alarm profile is applied to it automatically.

Note: In some Alarm Classes, a few alarm points appear twice in the Profile Content table. Selecting one alarm point automatically selects the other alarm point with the same name. Editing performs changes on both alarm points.

The alarm profile also contains the severity (C, M, m, w) for each alarm, which can be edited by the user.

The 6500 network element provides two non-editable predefined profiles (All Enabled which is the default, and All Disabled) and allows for three more predefined profiles to be user editable on the network element.

The Common alarm class supports an additional non-editable profile, Factory Default. For the Common alarm class, the Factory Default profile enables all common alarms except for the Disk 75 percent Full, Disk 90 percent Full, LAN-15 Port Failure, and LAN-16 Port Failure alarms, which are disabled.

Note: On L0 control plane, active alarm profile is not applicable for OSRPNODE, OSRPLINK, OSRPLINE/ OSRPLINE0, SNC, and SNCG objects. Alarms against these objects are raised according to the default alarm profile.

The following is a list of alarms that do not appear in any alarm profile (which means these alarm points cannot be disabled):

- Alarm and Event Throttling Active
- Automatic shutoff disabled
- Autoprovisioning Mismatch
- Autoprovisioning Mismatch - pluggable
- Circuit pack unknown
- Circuit pack unknown - pluggable
- Equipment Configuration Mismatch
- Database Not Recovered for Slot
- Fan Failed
- Slot Empty
- Manual Area Address Dropped
- Number of Level 1 NEs Exceeded
- Fan Missing
- Laser Failed
- Provisioning Incompatible
- Provisioning Incompatible - pluggable
- Redundant Release Synch Failed
- Shelf Data Missing
- High Temperature
- High Temperature Warning

Alarm hold-off

Alarm hold-off period is the time delay between the time that the alarm condition occurs and the time that the alarm is raised. The user can manually change the alarm hold-off from 2.5 seconds (default) to 0 seconds for alarms associated with Broadband, OTN I/F, PKT I/F, and Photonic circuit pack. The alarm hold-off feature applies to AMP, OPTMON, NMCC and OSC facilities in Photonic circuit packs, and all facilities on Broadband, PKT I/F and OTN I/F circuit packs including OTN Flex MOTR circuit packs.

This feature allows you to see alarms for fast transient conditions that would normally be filtered by the 2.5 second hold-off. The alarm hold-off period is changed using the **System** tab in the **Node Information** application in Site Manager. Refer to the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301.

ATTENTION

When Alarm hold-off is set to 2.5, alarms which are raised and cleared within 2.5 seconds time-frame are not acknowledged.

Viewing disabled alarms

Alarms that have been disabled from the alarm profiles application, AINS, or SP alarm correlation are not displayed in the **Active Alarms** application. Active alarms that have been disabled can be viewed in the **Active Disabled Alarms** application in the **Faults** menu of Site Manager. A manual refresh is required to see the latest Active Disabled Alarms.

ATTENTION

Disabled alarms will not be displayed in the **Active Disabled Alarms** application against a WAN, OTM0 or OTM1 mapping layer facility when the mapping layer of the port is connected in a client hairpin.

External alarm provisioning and controls

The 6500 network element has 16 parallel telemetry input points and four external control relays.

The input points allow remote monitoring of other equipment in the office in which the network element is located. For example, the input points can monitor room temperature alarms or office door open alarms. Specific external alarms must be set up during provisioning. The alarm types are assigned to a specific contact pin.

The **External Alarm Provisioning** application in the **Configuration** menu of Site Manager supports the following functionalities:

- displaying external alarm attributes (telemetry input points)

- editing the environmental alarm attributes on the network element
- deleting defined environmental alarm attributes on the network element

The external controls application allows the user to retrieve and display the labels and status of the four external controls relays for the 6500 network element. The 6500 network elements allow the user to operate or release these relays to turn external equipment on and off (for example, air conditioning, fan, sprinkler) and edit the labels of the relays.

ODU layer alarm monitoring

For the (1+8)xOTN Flex MOTR circuit pack, the ODU facility monitors the ODU layer alarms for protection switching for A-SNCP connections. The ODU layer alarms monitored on the ODU facility are the same as those monitored on the OTM Mapping Layer facility that is on the same ODU connection.

By default, ODU alarm monitoring is only enabled in one location depending on the ODU connection type. For 1+1 TPT protected connections, the ODU alarm monitoring is enabled on the OTN Mapping Layer facility. For A-SNCP connections, the ODU alarm monitor is enabled on the ODU facility.

To provide full flexibility, the 1+8 OTN Flex MOTR circuit pack allows ODU monitoring to be enabled or disabled on the other endpoint facility (the OTM Mapping Layer facility or ODU facility) on the same ODU connection.

Enabling the ODU monitoring on both endpoint facilities can result in duplicate ODU alarms being raised on the ODU connection if any ODU fault condition is present.

The ODU monitoring flag on the OTN mapping layer facility is enabled when it is first created and there is no ODU connection provisioned on it.

Power cycling of the network element

If the network element time is not provisioned to retrieve its time of day from an NTP server and a power cycle occurs on the network element, you must reprovision the network element date and time as the date and time are reset to the default values. If the time of day (TOD) synchronization feature is enabled, you do not have to reprovision the date and time. Refer to the “Editing the nodal general parameters” procedure in *Administration and Security*, 323-1851-301.

Signal conditioning for 40G circuit packs

40G OCLD/Wavelength-selective 40G OCLD unprotected configuration

The following table presents the signal conditioning that is applied to the 40G OCLD and Wavelength-selective 40G OCLD circuit packs in an unprotected configuration.

ATTENTION

When using a 1+1 protection configuration, the OCLD conditioning towards its mated circuit pack may not trigger conditioning on the mated circuit pack if that circuit pack can successfully switch to the protection path. However, if both the working and the protection OCLD circuit packs are conditioning towards the mated circuit pack, then conditioning will be triggered on the mated circuit pack.

Table 1-2
40G OCLD/Wavelength-selective 40G OCLD signal conditioning (unprotected)

Fault Condition	Signal conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Line Port
No ODU Connection	None (no connection)	ODU3 OCI
p1 SF (OTM3 on-ramp)	No conditioning	ODU3 AIS
p1 Facility OOS	ODU3 LCK	ODU3 LCK
p1 LOS, LOC, LOCH (Note), LOF, LOMF, Pre-FEC SF	ODU3 AIS	OTU3 SM-BDI
Note: "Loss of Channel" alarm is only applicable when colorless capable circuit packs deployed on colorless line system.		

40G MUX OCI signal conditioning

The following table presents the signal conditioning that is applied to the 40G MUX OCI circuit pack.

Table 1-3
40G MUX OCI signal conditioning

Fault Condition	40G MUX OCI p1-4 Facility	Signal Conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Client Port (Laser off if LOFEF enabled)
No ODU3 Connection	OC192/STM64 ETH10G OTM2 FC800 FC1200	None (no connection) None (no connection) None (no connection) None (no connection) None (no connection)	AIS-L LF ODU2 AIS NOS LF

Table 1-3
40G MUX OCI signal conditioning (continued)

Fault Condition	40G MUX OCI p1-4 Facility	Signal Conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Client Port (Laser off if LOFEF enabled)
p100 ODU3 AIS, ODU3 LCK, ODU3 OCI, SF (OTM3 off-ramp)	OC192/STM64 ETH10G OTM2 FC800 FC1200	ODU3 PM-BDI ODU3 PM-BDI ODU3 PM-BDI ODU3 PM-BDI ODU3 PM-BDI	AIS-L LF ODU2 AIS NOS LF
p100 OPU3 PT Mismatch, OPU3 AIS (OTM3 off-ramp)	OC192/STM64 ETH10G OTM2 FC800 FC1200	No conditioning No conditioning No conditioning No conditioning No conditioning	AIS-L LF ODU2 AIS NOS LF
p1-4 ODU2 AIS, ODU2 LCK, ODU2 OCI (OTM2 off-ramp)	OC192/STM64 ETH10G OTM2 FC800 FC1200	ODU2 PM-BDI ODU2 PM-BDI No conditioning ODU2 PM-BDI ODU2 PM-BDI	AIS-L LF No conditioning (Note 1) NOS LF
p1-4 OPU2 PT Mismatch, OPU2 AIS (OTM2 off-ramp)	OC192/STM64 ETH10G OTM2 FC800 FC1200	No conditioning No conditioning No conditioning No conditioning No conditioning	AIS-L LF No conditioning NOS LF
p1-4 WAN Link Down, WAN Loss of Frame Delineation (WAN off- ramp, GFP only)	ETH10G	No conditioning	LF
p1-4 Facility OOS	OC192/STM64 ETH10G OTM2 FC800 FC1200	ODU2 LCK ODU2 LCK ODU2 LCK ODU2 LCK ODU2 LCK	AIS-L LF ODU2 LCK NOS LF
p1-4 LOS, LOC, LOCH (Note 3), LOF	OC192/STM64 FC800	OPU2 AIS OPU2 AIS	No conditioning No conditioning

Table 1-3
40G MUX OCI signal conditioning (continued)

Fault Condition	40G MUX OCI p1-4 Facility	Signal Conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Client Port (Laser off if LOFEF enabled)
p1-4 LOS, LOC, LOCH (Note 3), LOF LOMF, Pre-FEC SF	OTM2	OPU2 AIS	OTU2 PM-BDI
p1-4 LOS, LOC, LOCH (Note 3), LOF, HIBER	ETH10G FC1200	ODU2 AIS ODU2 AIS	No conditioning (Note 2) No conditioning
Service-affecting Circuit Pack Fail	OC192/STM64, ETH10G OTM2 FC800 FC1200	ODU3 AIS ODU3 AIS ODU3 AIS ODU3 AIS ODU3 AIS	AIS-L LF ODU2 AIS NOS LF
Note 1: There is no conditioning unless LOFEF is enabled, in which case, the laser is turned off. Note 2: For the NTK525CA, there is no conditioning when operating in 11.05 and 11.09 proportional wrap. However, in GFP mode RF is conditioned. For the NTK525CF, there is no conditioning. Note 3: “Loss of Channel” alarm is only applicable when colorless capable circuit packs deployed on colorless line system.			

40G OCI signal conditioning

The following table presents the signal conditioning that is applied for the 40G OCI circuit packs.

Table 1-4
40G OCI signal conditioning

Fault Condition	40G OCI p1 Facility	Signal Conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Client Port (Laser off if LOFEF enabled)
No ODU Connection	OC768/STM256 OTM3 ETH40G	None (no connection) None (no connection) None (no connection)	AIS-L ODU3 OCI LF
p1 SF (OTM3 off-ramp)	OC768/STM256 OTM3 ETH40G	ODU3 PM-BDI No conditioning ODU PM-BDI	AIS-L ODU3 OCI LF

Table 1-4
40G OCI signal conditioning (continued)

Fault Condition	40G OCI p1 Facility	Signal Conditioning Towards Mated Circuit Pack	Signal Conditioning Towards Client Port (Laser off if LOFEF enabled)
p1 ODU3 AIS, ODU3 LCK, ODU3 OCI (OTM3 off-ramp)	OC768/STM256 OTM3 ETH40G	ODU3 PM-BDI No conditioning ODU3 PM-BDI	AIS-L No conditioning (Note 1) LF
p1 OPU PT Mismatch, OPU3 AIS (OTM3 off-ramp)	OC768/STM256 OTM3	None (no connection) None (no connection)	AIS-L No conditioning
p1 OPU PT Mismatch, OPU3 AIS, OPU3 CSF (OTM3 off-ramp) (Note 2)	ETH40G	None (no connection)	LF
p1 Facility OOS	OC768/STM256 OTM3 ETH40G	ODU3 LCK ODU3 LCK ODU3 LCK	AIS-L ODU3 LCK LF
p1 LOS, LOF	OC768/STM256	OPU3 AIS	No conditioning
p1 LOS, LOC, LOCH (Note 3), LOF, LOM, Pre-FEC SF	OTM3	ODU3 AIS	OTU3 SM-BDI
p1 LOS, LOF, HiBER	ETH40G	OPU3 CSF	No conditioning
Service-affecting Circuit Pack Fail	OC768/STM256 OTM3 ETH40G	ODU3 AIS ODU3 AIS ODU3 AIS	AIS-L ODU3 AIS LF
Note 1: There is no conditioning unless LOFEF is enabled, in which case, the laser is turned off. Note 2: OPU CSF is alarmed as Far End Client Signal Fail. Note 3: "Loss of Channel" alarm is only applicable when colorless capable circuit packs deployed on colorless line system.			

Signal conditioning for 100G circuit packs

Optical muxponder configuration (10x10GE MUX, 10x10G MUX, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD)

The following tables present the signal conditioning that is applied for the 10x10GE MUX, 10x10G MUX, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD circuit packs in the optical muxponder configuration.

Table 1-5
10x10GE MUX, 10x10G MUX, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD signal conditioning—optical muxponder configuration

Trigger		Tx conditioning	
Condition	10x10GE MUX/10x10G MUX, p1-10 Facility	100G OCLD/100G WL3/WL3e OCLD/ Flex2 WL3/WL3e OCLD/ Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes) p1 (OTM4), Signal output	10x10GE MUX/ 10x10G MUX, p1-10 (client) Signal output, (Laser off if LOFEF enabled)
No Transponder Connection	OC192/STM64 ETH10G OTM2	ODU4 OCI ODU4 OCI ODU4 OCI	AIS-L LF ODU2 AIS
p1 (100G OCLD) Facility OOS	OC192/STM64 ETH10G OTM2	ODU4 LCK ODU4 LCK ODU4 LCK	AIS-L LF ODU2 AIS
p1 (100G OCLD) LOS, LOC, LOCH (Note 2), LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	OC192/STM64 ETH10G OTM2	OTU4 BDI, ODU4 BDI OTU4 BDI, ODU4 BDI OTU4 BDI, ODU4 BDI	AIS-L LF ODU2 AIS
p1 (100G OCLD) ODU AIS, ODU LCK, ODU OCI	OC192/STM64 ETH10G OTM2	ODU4 BDI ODU4 BDI ODU4 BDI	AIS-L LF ODU2 AIS
p1 (100G OCLD) OPU AIS	OC192/STM64 ETH10G OTM2	No conditioning No conditioning No conditioning	AIS-L LF ODU2 AIS

Table 1-5
10x10GE MUX, 10x10G MUX, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD signal conditioning—optical muxponder configuration (continued)

Trigger		Tx conditioning	
Condition	10x10GE MUX/10x10G MUX, p1-10 Facility	100G OCLD/100G WL3/WL3e OCLD/ Flex2 WL3/WL3e OCLD/ Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes) p1 (OTM4), Signal output	10x10GE MUX/ 10x10G MUX, p1-10 (client) Signal output, (Laser off if LOFEF enabled)
p1-10 (10x10GE MUX/10x10G MUX) OTM2 OPU2 PT Mismatch	OC192/STM64 ETH10G OTM2	No conditioning No conditioning No conditioning	AIS-L LF No conditioning
p1-10 (10x10G MUX) WAN Link Down	WAN	No conditioning	LF
p1-10 (10x10G MUX) WAN Loss of Frame Delineation	WAN	No conditioning	LF
p1-10 (10x10GE MUX/10x10G MUX) HiBER	ETH10G	OPU2 AIS	No conditioning
p1-10 (10x10GE MUX/10x10G MUX) Facility OOS	OC192/STM64 ETH10G OTM2	ODU2 LCK ODU2 LCK ODU2 LCK	AIS-L LF ODU2 LCK
p1-10 (10x10G MUX) ODU AIS, ODU LCK, ODU OCI (onramp, at OTM2 client facility Rx)	OTM2	ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning
p1-10 (10x10GE MUX/10x10G MUX) ODU AIS, ODU LCK, ODU OCI (offramp, at OTM2 layer facility)	OC192/STM64, ETH10G OTM2	OC192/STM64: p1-10 = ODU2 BDI, ETH10G: p1-10 = ODU2 BDI, OTM2: p1-10 = No conditioning	OC192/STM64: p1-10 = AIS-L ETH10G: p1-10 = LF OTM2: p1-10 = No conditioning (laser off) (Note 1)
p1-10 (10x10GE MUX/10x10G MUX) LOS, LOC, LOCH (Note 2), LOF, LOMF (OTM2 client)	OC192/STM64 ETH10G OTM2	OPU2 AIS OPU2 AIS ODU2 AIS	No conditioning No conditioning, OTU2 BDI
p1-10 (10x10G MUX) Pre-FEC SF	OTM2	ODU2 AIS	OTU2 BDI

Table 1-5
10x10GE MUX, 10x10G MUX, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD signal conditioning—optical muxponder configuration (continued)

Trigger		Tx conditioning	
Condition	10x10GE MUX/10x10G MUX, p1-10 Facility	100G OCLD/100G WL3/WL3e OCLD/ Flex2 WL3/WL3e OCLD/ Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes) p1 (OTM4), Signal output	10x10GE MUX/ 10x10G MUX, p1-10 (client) Signal output, (Laser off if LOFEF enabled)
Client facility LOC, LOS, LOF, HiBER	OTM2 ETH10G	OPU2_CSF, OPU2_AIS, GSF CMF_CSF	OPU2_CSF
Service-affecting Circuit Pack Fail	OC192/STM64 ETH10G OTM2	ODU4 AIS ODU4 AIS ODU4 AIS	AIS-L LF ODU2 AIS
Note 1: There is no conditioning unless LOFEF is enabled, in which case, the laser is turned off. Note 2: “Loss of Channel” alarm is only applicable when colorless capable circuit packs deployed on colorless line system.			

Table 1-6
100G WL3n MOTR signal conditioning—OTM2 client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU2 OCI	N/A
Line Facility OOS	ODU4 LCK	ODU2 AIS
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	ODU2 AIS
Line ODU4 AIS ODU4 LCK ODU4 OCI	ODU4 BDI	ODU2 AIS
Line OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	ODU2 AIS
Client Facility Deletion	MSI MM	N/A

Table 1-6
100G WL3n MOTR signal conditioning—OTM2 client (continued)

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
Client Facility OOS	ODU2 LCK	ODU2 LCK
Client LOS, LOC, LOF, OTU PreFEC-SF	ODU2 AIS	OTU2 BDI
Client PTM, MSI MM, OPU AIS, OPU2 CSF	No conditioning	No conditioning
Client MR MM	No conditioning	No conditioning
Client ODU2 AIS, ODU2 LCK, ODU2 OCI	ODU-2 AIS, ODU2 LCK, ODU2 OCI	No conditioning
Line Port: P1 Client Port: P2-11		

Table 1-7
100G WL3n MOTR signal conditioning—OC192 client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU2 OCI	N/A
Line Facility OOS	ODU4 LCK	AIS_L
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	AIS_L
Line ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 BDI	AIS_L
Line OPU4 AIS, OPU4 CSF, PTMM	No conditioning	AIS_L
Client Facility Deletion	MSI MM	N/A
Client Facility OOS	ODU2 LCK	AIS_L
Client LOS, LOC, LOF	OPU2 CSF	No conditioning

Table 1-7
100G WL3n MOTR signal conditioning—OC192 client (continued)

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
Client ODU2 AIS, ODU2 LCK, ODU2 OCI	ODU-2 BDI	AIS_L
Client MSI MM, PTM, OPU2-AIS, OPU2-CSF	No conditioning	AIS_L
Line Port: P1 Client Port: P2-11		

Table 1-8
100G WL3n MOTR signal conditioning—ETH10G client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU2 OCI	N/A
Line Facility OOS	ODU4 LCK	LF
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	LF
Line ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF
Line OPU4-AIS, OPU4-CSF, PTMM	No conditioning	LF
Client Facility Deletion	MSI MM	N/A
Client Facility OOS	ODU2 LCK	LF
Client LOS, LOC, LOF HiBER LODS	OPU2 CSF	No conditioning
Client ODU2-AIS, ODU2-LCK, ODU2-OCI	ODU-2 BDI	LF
Client MSI MM, PTM, OPU2-AIS, OPU2-CSF	No conditioning	LF

Table 1-8
100G WL3n MOTR signal conditioning—ETH10G client (continued)

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
Client WAN LINKDOWN	No conditioning	LF
Client WAN Loss of frame delineation	No conditioning	LF
Line Port: P1 Client Port: P2-11		

Table 1-9
100G WL3n MOTR signal conditioning—FC-1200 client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU2 OCI	N/A
Line Facility OOS	ODU4 LCK	LF
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	LF
Line ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF
Line OPU4-AIS, OPU4-CSF, OPU4-PTM	No conditioning	LF
Client Facility Deletion	MSI MMI	N/A
Client Facility OOS	ODU2 LCK	LF
Client LOS, LOC, LODS, HiBER	OPU2 AIS	No conditioning
Client ODU2-AIS, ODU2-LCK, ODU2-OCI	ODU 2 BDI	LF
Client PTM, MSI MM, OPU2-AIS, OPU2-CSF	No conditioning	LF
Line Port: P1 Client Port: P2-11		

Table 1-10
100G WL3n MOTR signal conditioning—FC-800 client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU2 OCI	N/A
Line Facility OOS	ODU4 LCK	NOS
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	NOS
Line ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	NOS
Line OPU4-AIS, OPU4-CSF, MSI MM	No conditioning	NOS
Client Facility Deletion	MSI MM	N/A
Client Facility OOS	ODU2 LCK	NOS
Client LOS, LOC, LODS, HiBER	OPU2 CSF	RF
Client ODU2-AIS, ODU2-LCK, ODU2-OCI	ODU-2 BDI	NOS
Client PTM, MSI MM, OPU2-AIS, OPU2-CSF	No conditioning	NOS
Line Port: P1 Client Port: P2-11		

Table 1-11
100G WL3n MOTR signal conditioning—ETH40G client

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
No Client Port connection	ODU3 OCI	N/A
Line Facility OOS	ODU4 LCK	LF
Line LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI, ODU4 BDI	LF
Line ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF

Table 1-11
100G WL3n MOTR signal conditioning—ETH40G client (continued)

Trigger	Tx conditioning	
Condition	Line Port Signal output	Client Port Signal output, (Laser off if LOFEF enabled)
Line OPU4-AIS, OPU4-CSF, PTMM	No conditioning	LF
Client Facility Deletion	ODU3 OCI	N/A
Client Facility OOS	ODU3 LCK	LF
Client LOS, LOC, LOF, HiBER LODS	OPU3 CSF	No conditioning
Client ODU3-AIS, ODU3-LCK, ODU3-OCI	ODU3 BDI	LF
Client MSI MM, PTM, OPU3-AIS, OPU3-CSF	No conditioning	LF
Client WAN LINKDOWN	No conditioning	LF
Client WAN Loss of frame delineation	No conditioning	LF
Line Port: P1 Client Port: P4-11, 104, 108		

Table 1-12
2C-BPSK 100G MOTR signal conditioning—FC1200 client

Trigger	Tx conditioning		
Condition	OCLD P1Signal output		10x10G MUX P1-10, Signal output (Laser off if LOFEF enabled)
	Prime	Member	
No ODU connection	ODU4 OCI	ODU4 OCI	LF
Prime and/or member OCLD P1 Facility OOS	ODU4 LCK	ODU4 LCK	LF
Prime OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI	No OTM4 conditioning	LF
Prime and/or member OCLD circuit pack Missing	No OTM4 conditioning	No OTM4 conditioning	LF
Member OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF	No OTM4 conditioning	OTU4 BDI	LF

Table 1-12**2C-BPSK 100G MOTR signal conditioning—FC1200 client (continued)**

Trigger	Tx conditioning		
Condition	OCLD P1 Signal output		10x10G MUX P1-10, Signal output (Laser off if LOFEF enabled)
	Prime	Member	
Mux P100 OPU4-AIS, OPU4-CSF	No OTM4 conditioning	No OTM4 conditioning	LF
Mux P100 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	ODU4 BDI	LF
Mux P2-10 No Connection	No OTM4 conditioning ODU2-OCI	No OTM4 conditioning ODU2-OCI	LF
Mux P2-10 Facility OOS	No OTM4 conditioning ODU2-LCK	No OTM4 conditioning ODU2-LCK	LF
Mux P2-10 LOS, LOC, LODS, HiBER	No OTM4 conditioning OPU2-AIS	No OTM4 conditioning OPU2-AIS	RF
Mux circuit pack Missing	ODU4-AIS	ODU4-AIS	No conditioning
Mux P2-10 PTM	No OTM4 conditioning	No OTM4 conditioning	LF
Mux P2-10 ODU-2 AIS, ODU2-LCK, ODU2-OCI	No OTM4 conditioning	No OTM4 conditioning	LF

Table 1-13**2C-BPSK 100G MOTR signal conditioning—FC800 client**

Trigger	Tx conditioning		
Condition	2C- BPSK OCLD P1 Signal output		10x10G MUX P1-10, Signal output (Laser off if LOFEF enabled)
	Prime	Member	
No ODU Connection	ODU4 OCI	ODU4 OCI	NOS
Prime and/or member OCLD P1 Facility OOS	ODU4 LCK	ODU4 LCK	NOS
Prime OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF	OTU4 BDI	No OTM4 conditioning	NOS
Prime and/or member OCLD circuit pack Missing	No OTM4 conditioning	No OTM4 conditioning	NOS

Table 1-13
2C-BPSK 100G MOTR signal conditioning—FC800 client (continued)

Trigger	Tx conditioning		
Condition	2C- BPSK OCLD P1 Signal output		10x10G MUX P1-10, Signal output (Laser off if LOFEF enabled)
	Prime	Member	
Member OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF	No OTM4 conditioning	OTU4-BDI	NOS
MUX P100 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4-BDI	ODU4-BDI	NOS
MUX P100 OPU4-AIS, OPU4-CSF	No OTM4 conditioning	No OTM4 conditioning	NOS
MUX P1-10 No Connection	No OTM4 conditioning ODU2-OCI	No OTM4 conditioning ODU2-OCI	NOS
MUX P1-10 Facility OOS	No OTM4 conditioning ODU2-LCK	No OTM4 conditioning ODU2-LCK	NOS
MUX P1-10 LOS, LOC, LODS	No OTM4 conditioning OPU2-AIS	No OTM4 conditioning OPU2-AIS	No conditioning
MUC Circuit Pack Missing	ODU4 AIS	ODU4 AIS	No conditioning
MUX P2-10 PTM	No OTM4 conditioning	No OTM4 conditioning	NOS
MUX P1-10 ODU2-AIS, ODU2-LCK, ODU2-OCI	No OTM4 conditioning	No OTM4 conditioning	NOS

Table 1-14
100G WL3n OTR signal conditioning—100GE client

Trigger	Tx conditioning	
Condition	Line Port P1 Signal output	Client Port P2 Signal output (Laser off if LOFEF enabled)
P1 Facility OOS	ODU4 LCK	LF
P1 LOS, LOC, LOCH, LOF, PreFEC-SF, LOMF	OTU4 BDI, ODU4 BDI	LF
P1 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF

Table 1-14
100G WL3n OTR signal conditioning—100GE client (continued)

Trigger	Tx conditioning	
Condition	Line Port P1 Signal output	Client Port P2 Signal output (Laser off if LOFEF enabled)
P1 PreFEC SD	No conditioning	No conditioning
P2 No facility	ODU4 OCI	N/A
P2 Facility OOS	ODU4 LCK	LF
P2 LOS, LOC, LODS, LOF, HiBER	OPU4 CSF	No conditioning
P2 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF
P2 OPU CSF, PTM	No conditioning	LF
ODU4 SF, ODU4 SD	No conditioning	No conditioning

Table 1-15
100G WL3n OTR signal conditioning—100G OTU4 client

Trigger	Tx conditioning	
Condition	Line Port P1 Signal output	Client Port P2 Signal output, (Laser off if LOFEF enabled)
P1 Facility OOS	ODU4 LCK	ODU4 LCK
P1 LOS, LOC, LOF, PreFEC-SF, LOMF	OTU4 BDI, ODU4 BDI	ODU4 AIS
P1 ODU4-AIS, ODU4-LCK, ODU4-OCI	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI
P1 PreFEC SD	No conditioning	No conditioning
P2 No facility	ODU4 OCI	N/A
P2 Facility OOS	ODU4 LCK	ODU4 LCK
P2 LOS, LOC, LOF, PreFEC-SF, LOMF	OPU4 CSF	OTU4 BDI, ODU4 BDI
P2 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning

Signal conditioning for WLAi MOTR/FOTR and WL5e MOTR circuit packs

The following tables present the signal conditioning that is applied for the WLAi MOTR/FOTR and WL5e MOTR circuit packs.

Table 1-16
Off-Ramp Triggers

Off-ramp Trigger (Line Facing)		Conditioning					
Layer	Trigger	on-ramp	Off-ramp Trigger (Client Tx)				OTUTTP (OTUK)
			All Client Types (LOFE Enabled)	ETTP (Ethernet)	STTP (OCn)	CBRTTP (FC16/32)	
PTP (OCH/OTSi)	Rx CR (Critical) Faults	PTP-RDI OTUCn-BDI ODUCn-BDI	Laser Off	LF	AIL_L	NOS	AIS
	Facility OOS (OOS-MA)	ODUCn-LCK					LCK
OTUTTP (OTUCn)	Rx CR (Critical) Faults	OTUCn-BDI ODUCn-BDI					AIS
	Facility OOS (OOS-MA)	ODUCn-LCK					LCK
	OTUTTP Facility deleted	OTUCn-AIS					AIS
	Facility Loopback	None					LCK
	Terminal Loopback	ODUCn-LCK	None	None	None	None	None
ODUTTP (ODUCn)	Rx CR (Critical) Faults	ODUCn-BDI	Laser Off	LF	AIL_L	NOS	AIS
	ODUTTP Facility deleted	ODUCn-AIS					AIS
	Facility OOS (OOS-MA)	ODUCn-LCK					LCK

Table 1-16
Off-Ramp Triggers (continued)

Off-ramp Trigger (Line Facing)		Conditioning					
Layer	Trigger	on-ramp	Off-ramp Trigger (Client Tx)				OTUTTP (OTUK)
			All Client Types (LOFE Enabled)	ETTP (Ethernet)	STTP (OCn)	CBRTTP (FC16/ 32)	
ODUCTP (ODUk)	OPUk-CSF (FECSF)	None	Laser Off	LF	AIS_L	NOS	n/a
	OPUk-PT Byte Mismatch						
	MSI Mismatch	None					AIS
	ODUCTP Connection deleted	ODUk-OCI					OCI
	ODUk-AIS/LCK/ OCI (none OTUk Client)	ODUk-BDI					n/a
	ODUk-AIS/LCK/ OCI (OTUk Client)	None					None
LOFE Enabled: PTP CONDTYPE = LASEROFF PTP Rx Critical Faults: Loss of Signal (LOS) OTUK OTUTTP Rx Critical Faults: AIS Deskew Fail, Loss Of Frame (LOF), Loss Of Multiframe (LOMF), Loss Of Clock (LOC), Pre-FEC SF							

Table 1-17
On-Ramp Triggers

On-ramp Trigger (Client Facing)		Conditioning								
Layer	Trigger	on-ramp			OTUTTP (OTUK)	off-ramp				
		ETTP (Ethernet)	STTP (OCn)	Flex (FC16/ 32)		All Client Types (LOFE Enabled)	ETTP (Ether- net)	STTP (OCn)	CBRTTP (fc16/32)	OTUTTP (OTUK)
PTP (OCH)	Rx CR (Critical) Faults (1) ODUCTP CONDTYPE= OPUk_CSF (2) ODUCTP CONDTYPE= ODUk_AIS	(1) OPUk- CSF+LF (2) ODUk AIS	(1) OPUk- CSF+AIS (2) ODUk AIS	(1) OPUk- CSF+ NOS (2) ODUk AIS	ODUk AIS	None	None	None	None	OTU BDI
	Facility OOS (OOS-MA) (1) ODUCTP CONDTYPE= OPUk_CSF (2) ODUCTP CONDTYPE= ODUk_AIS	(1) OPUk- CSF+LF (2) ODUk AIS	(1) OPUk- CSF+AIS (2) ODUk AIS	(1) OPUk- CSF+ NOS (2) ODUk AIS	ODUk LCK	Laser Off	LF	AIL-L	NOS	ODUk LCK
	No xTTP facility	ODUk OCI	ODUk OCI	ODUk OCI	ODUk OCI	Laser Off	Laser Off	Laser Off	Laser Off	Laser Off
	No wavelength provisioned (DWDM only)	None	None	None	None	Laser Off	Laser Off	Laser Off	Laser Off	Laser Off

Table 1-17
On-Ramp Triggers (continued)

On-ramp Trigger (Client Facing)		Conditioning								
Layer	Trigger	on-ramp				off-ramp				
		ETTP (Ethernet)	STTP (OCn)	Flex (FC16/ 32)		All Client Types (LOFE Enabled)	ETTP (Ether- net)	STTP (OCn)	CBRTTP (fc16/32)	OTUTTP (OTUK)
xTTP (service)	Rx CR (Critical) Faults (1) ODUCTP CONDTYPE= OPUk_CSF (2) ODUCTP CONDTYPE= ODUk_AIS	(1) OPUk- CSF+LF (2) ODUk AIS	(1) OPUk- CSF+AIS (2) ODUk AIS	(1) OPUk- CSF+ NOS (2) ODUk AIS	ODUk AIS	None	None	None	None	OTU BDI
	Facility OOS (OOS-MA) (1) ODUCTP CONDTYPE= OPUk_CSF (2) ODUCTP CONDTYPE= ODUk_AIS	(1) OPUk- CSF+LF (2) ODUk AIS	(1) OPUk- CSF+AIS (2) ODUk AIS	(1) OPUk- CSF+ NOS (2) ODUk AIS	ODUk LCK	Laser Off	LF	AIL-L	NOS	ODUk- LCK
	No ODUCTP facility	ODUk-OCI	ODUk- OCI	ODUk- OCI	ODUk- OCI	Laser Off	LF	AIL_L	NOS	ODUk- AIS
	Facility Loopback	(1) OPUk- CSF+LF (2) ODUk AIS	(1) OPUk- CSF+AIS (2) ODUk AIS	(1) OPUk- CSF+ NOS (2) ODUk AIS	ODUk LCK	None	None	None	None	None
	Terminal Loopback	None	None	None	None	Laser Off	Laser Off	Laser Off	Laser Off	Laser Off

Table 1-17
On-Ramp Triggers (continued)

On-ramp Trigger (Client Facing)		Conditioning								
Layer	Trigger	on-ramp				off-ramp				
		ETTP (Ethernet)	STTP (OCn)	Flex (FC16/ 32)	OTUTTP (OTUK)	All Client Types (LOFE Enabled)	ETTP (Ether- net)	STTP (OCn)	CBRTTP (fc16/32)	OTUTTP (OTUK)
ODUCTP (ODUK)	Facility OOS (OOS-MA)	ODUK-LCK	ODUK-LCK	ODUK-LCK	ODUK-LCK	Laser Off	LF	AIL_L	NOS	ODUK-AIS
	Terminal Loopback	None	None	None	None	Laser Off	LF	AIL_L	NOS	n/a
	ODUCTP Connection deleted	ODUK-OCI	ODUK-OCI	ODUK-OCI	ODUK-OCI	Laser Off	LF	AIL_L	NOS	ODUK-AIS
LOFE Enabled: PTP CONDTYPE = LASEROFF PTP Rx Critical Faults: Loss of Signal (LOS) OTUK OTUTTP Rx Critical Faults: AIS Deskew Fail, Loss Of Frame (LOF), Loss Of Multiframe (LOMF), Loss Of Clock (LOC), Pre-FEC SF										

Table 1-18
ODUCn Regen Triggers

Trigger		Conditioning	
Layer	Trigger	on-ramp	Off-ramp
P1 PTP (OCH/OTSi)	Rx CR (Critical) Faults	PTP-RDI OTUCn-BDI	ODUCn-AIS
	Facility OOS (OOS-MA)	ODUCn-LCK	ODUCn-LCK
P1 OTUTTP (OTUCn)	Rx CR (Critical) Faults	OTUCn-BDI	ODUCn-AIS
	Facility OOS (OOS-MA)	ODUCn-LCK	ODUCn-LCK
	Facility Loopback	None	ODUCn-LCK
	Terminal Loopback	ODUCn-LCK	None
P1 ODUTTP (ODUCn)	LCK/OCI/LCK	None	None
	ODUTTP Connection Deleted	ODUCn-OCI	ODUCn-OCI
	Facility OOS (OOS-MA)	ODUCn-LCK	ODUCn-LCK
P2-5 PTP (OCH/OTSi)	Rx CR (Critical) Faults	OTUCn-BDI	OTUCn-AIS
		ODUCn-LCK	ODUCn-LCK
LOFE Enabled: PTP CONDDTYPE = LASEROFF PTP Rx Critical Faults: Loss Of Signal (LOS), Loss Of Frame (LOF), Loss Of Multiframe (LOMF), Loss Of Clock (LOC), Loss Of Channel (LOCH), Pre-FEC SF, GID Mismatch, PHY Map Mismatch, Reach Violation PTP Tx Critical Faults: Channel Contention OTUCn Rx Critical Faults: Loss Of lane Alignment (LOLA), Loss Of Frame and Multiframe (LOFLOM), AIS, Service Mismatch			

Optical transponder configuration (100GE OCI, 100G WL3/WL3e OCLD, and 100G OCLD)

The following table presents the signal conditioning that is applied for the 100GE OCI, 100G WL3/WL3e OCLD, and 100G OCLD circuit packs in the optical transponder configuration.

Table 1-19
100GE OCI, 100G WL3/WL3e OCLD, and 100G OCLD signal conditioning—optical transponder configuration

Trigger		Tx conditioning	
Condition	100GE OCI, p1 (OTM4) Facility	100G OCLD/100G WL3/WL3e OCLD, p1 (OTM4) Signal output	100GE OCI (client) Signal output, (Laser off if LOFEF enabled)
No Transponder Connection	ETH100G	ODU OCI	LF
p1 (100GE OCI and 100G OCLD) Facility OOS	ETH100G	ODU LCK	LF
p1 (100G OCLD) OCLD LOS, LOC, LOCH (Note 1), LOF, LOMF, OTU PreFEC-SF	ETH100G	OTU BDI, ODU BDI	LF
p1 (100G OCLD) ODU AIS, ODU LCK, ODU OCI	ETH100G	ODU BDI	LF
p1 (100GE OCI) OPU4 CSF	ETH100G	No conditioning	LF
p1 (100GE OCI) LOS, LOC, LODS (Note 2), LOF, HiBER	ETH100G	OPU CSF	No conditioning
Service-affecting Circuit Pack Fail	ETH100G	ODU AIS	LF
Note 1: “Loss of Channel” alarm is only applicable when colorless capable circuit packs deployed on colorless line system. Note 2: “Loss Of Data Sync” is raised when IEEE Std 802.3ba skew limit is crossed. However, the signal conditioning is triggered only if the hardware is not capable to deskew.			

Optical transponder configuration (100G OCI, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), 100G WL3/WL3e OTR, and 100G OCLD)

The following tables present the signal conditioning that is applied for the 100G OCI, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G WL3/WL3e OCLD OTR circuit packs.

Table 1-20

100G OCI, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 /WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD signal conditioning—optical transponder configuration

Trigger		Tx conditioning	
Condition	100G OCI P1	100G OCLD/100G WL3/ WL3e OCLD/ Flex2 WL3/ WL3e OCLD/ Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes) p1 Signal output	100G OCI p1 Signal output (Laser off if LOFEF enabled), (Note 1)
No Transponder Connection	OTM4	ODU OCI	ODU OCI
p1 (100G OCLD) Facility OOS	OTM4	ODU LCK	ODU LCK
p1 (100G OCLD) LOS, LOC, LOCH (Note 2), LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	OTM4	OTU BDI	ODU AIS
OCLD CP MISSING	OTM4	No conditioning	ODU AIS
100G OCLD ODU AIS, ODU LCK, ODU OCI	OTM4	No conditioning	ODU AIS, ODU LCK, ODU OCI
P1 (OCI) Facility OOS	OTM4	ODU LCK	ODU LCK
P1 (OCI) LOS, LOF, LOMF, OTU PreFEC-SF, OTL Skew OOR	OTM4	ODU AIS	OTU BDI
OCI CP MISSING	OTM4	ODU AIS	No conditioning
P1 (OCI) ODU AIS, ODU LCK, ODU OCI	OTM4	ODU AIS, ODU LCK, ODU OCI	No conditioning
No Transponder Connection	ETH100G	ODU OCI	LF
p1 (100G OCLD) Facility OOS	ETH100G	ODU LCK	LF
p1 (100G OCLD) LOS, LOC, LOCH (Note 2), LOMF, OTU PreFEC-SF, OTU Skew OOR	ETH100G	OTU BDI, ODU BDI	LF
100G OCLD ODU AIS, ODU LCK, ODU OCI	ETH100G	ODU BDI	LF
p1 (100G OCLD) OPU CSF, PTM	ETH100G	No conditioning	LF

Table 1-20

100G OCI, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes), and 100G OCLD signal conditioning—optical transponder configuration (continued)

Trigger		Tx conditioning	
Condition	100G OCI P1	100G OCLD/100G WL3/WL3e OCLD/ Flex2 WL3/WL3e OCLD/ Flex3 WL3e OCLD, Flex4 WL3e OCLD (QPSK and BPSK modes) p1 Signal output	100G OCI p1 Signal output (Laser off if LOFEF enabled), (Note 1)
P1 (OCI) Facility OOS	ETH100G	ODU LCK	LF
P1 (OCI) LOS, LOC, LODS (Note 3), LOF, HiBER	ETH100G	OPU CSF	No conditioning
Note 1: No RF conditioning on 100G OCI Tx when LOS is received. Note 2: “Loss of Channel” alarm is only applicable when colorless capable circuit packs deployed on colorless line system. Note 3: “Loss Of Data Sync” is raised when IEEE Std 802.3ba skew limit is crossed. However, the signal conditioning is triggered only if the hardware is not capable to deskew.			

Table 1-21

Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G MOTR configuration (OTM2 client)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	ODU2 AIS	ODU2 AIS
P1 OCLD Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	ODU2 AIS	ODU2 AIS
P1 (OCLD) LOS, LOC, LOF, LOMF, OTU PreFEC-SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	ODU2 AIS	ODU2 AIS
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	No conditioning	No conditioning	No conditioning	ODU2 AIS	ODU2 AIS

Table 1-21**Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G MOTR configuration (OTM2 client) (continued)**

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
OCLD P1, OTUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	ODU2 AIS	ODU2 AIS
Left MUX P100 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU2 AIS	No conditioning
Left MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	ODU2 AIS	No conditioning
Left MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	ODU2 OCI	No conditioning
Left MUX P1-10, facility OOS	No conditioning	No conditioning	No conditioning	ODU2 LCK	No conditioning
Left MUX P1-10 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	No conditioning	No conditioning	No conditioning	ODU2 BDI	No conditioning
Left MUX CP Missing	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning
Left MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Right MUX P100 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	No conditioning	ODU2 AIS
Right MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	No conditioning	ODU2 AIS
Right MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	No conditioning	ODU2 OCI

Table 1-21

Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G MOTR configuration (OTM2 client) (continued)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
Right MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	No conditioning	ODU2 LCK
Right MUX P1-10 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	No conditioning	No conditioning	No conditioning	No conditioning	ODU2 BDI
Right MUX CP Missing	No conditioning	No conditioning	ODU4 AIS	No conditioning	No conditioning
Right MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Right MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning

Table 1-22

Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G OTR configuration (OTM4 client)

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
No OCLD Connection	ODU4 OCI	ODU4 OCI	No conditioning	ODU4 OCI	ODU4 OCI
P1 OCLD Facility OOS	ODU4 LCK	ODU4 LCK	ODUG LCK	ODU4 LCK	ODU4 LCK
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	ODU4 AIS	ODU4 AIS	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS

Table 1-22**Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G OTR configuration (OTM4 client) (continued)**

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
OCLD P1, OTUG SD, ODU4 SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	ODU4 AIS	ODU4 AIS	No conditioning	No conditioning	No conditioning
OCLD P1, OTUG AIS, ODU4 LCK	ODU4 AIS	ODU4 AIS	ODUG BDI	ODU4 AIS	ODU4 AIS
Left OCI P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCI No connection	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCI P1 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	ODU4 BDI	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left OCI CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4 AIS ODU4 LCK ODU4 OCI	No conditioning
Right OCI P1 Facility OOS	No conditioning	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK
Left OCI No Connection	No conditioning	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI
Right OCI P1 LOS, LOF, LOMF, OTU PreFEC SF, OTU PostFEC SF	No conditioning	ODU4 BDI	No conditioning	No conditioning	ODU4 AIS

Table 1-22**Flex3 WL3e OCLD/Flex4 WL3e OCLD Signal Conditioning—200G OTR configuration (OTM4 client) (continued)**

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
Right OCI P1 PostFEC ODU SF PostFEC OTU SD PostFEC ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Right OCI CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right OCI P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS ODU4 LCK ODU4 OCI

Table 1-23**Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (OTU4) and 100G MOTR (ETH10G) Signal Conditioning**

Trigger	Tx conditioning				
Condition	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
No OCLD Connection	LF	LF	No conditioning	ODU4 OCI	ODU4 OCI
P1 OCLD Facility OOS	LF	LF	ODUG LCK	ODU4 LCK	ODU4 LCK
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	LF	LF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD P1,OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	LF	LF	No conditioning	No conditioning	No conditioning

Table 1-23

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (OTU4) and 100G MOTR (ETH10G) Signal Conditioning (continued)

Trigger	Tx conditioning				
	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
OCLD P1, ODU4 AIS, ODU4 LCK	LF	LF	ODUG BDI	ODU4 AIS	ODU4 AIS
Left OCI P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCI, No connection	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCI P1 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	ODU4 BDI	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left OCI CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4 AIS ODU4 LCK ODU4 OCI	No conditioning
Right MUX P100 OPU4 AIS, OPU4 CSF	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P100 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning	No conditioning	ODU4 BDI
Right MUX P1-10 No Connection	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P1-10 Facility OOS	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P1-10 LOS, LOC, LOF, HiBER	No conditioning	RF	No conditioning	No conditioning	No conditioning

Table 1-23

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (OTU4) and 100G MOTR (ETH10G) Signal Conditioning (continued)

Trigger	Tx conditioning				
Condition	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
Right MUX CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right MUX P1-10 PTM	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P1-10 WAN LINKDOWN	No conditioning	LF	No conditioning	No conditioning	No conditioning
Right MUX P1-10 WAN Loss of Frame Delineation	No conditioning	LF	No conditioning	No conditioning	No conditioning

Table 1-24

200G OTR Signal Conditioning (ETH100G Client)

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output (Laser off if LOFEF enabled)		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
No OCLD Connection	LF	LF	No conditioning	ODU4 OCI	ODU4 OCI
P1 OCLD Facility OOS	LF	LF	ODUG LCK	ODU4 LCK	ODU4 LCK
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	LF	LF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS

Table 1-24
200G OTR Signal Conditioning (ETH100G Client) (continued)

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output (Laser off if LOFEF enabled)		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
OCLD P1, OTUG SD, ODU4 SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	LF	LF	No conditioning	No conditioning	No conditioning
OCLD P1, ODU4 AIS, ODU4 LCK	LF	LF	ODUG BDI	ODU4 AIS	ODU4 AIS
Left OCI P1 Facility OOS	LF	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCI, No connection	LF	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCI P1 LOS, LOF, LOC, HiBER	LF	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, ODU4 AIS, ODU4 LCK, ODU4 OCI	LF	No conditioning	No conditioning	ODU4 BDI	No conditioning
Left OCI P1, OPU4 CSF	LF	No conditioning	No conditioning	No conditioning	No conditioning
Right OCI P1 Facility OOS	No conditioning	LF	No conditioning	No conditioning	ODU4 LCK
Right OCI No Connection	No conditioning	LF	No conditioning	No conditioning	ODU4 OCI
Right OCI P1 LOS, LOC, LOF, HiBER	No conditioning	LF	No conditioning	No conditioning	ODU4 AIS

Table 1-24
200G OTR Signal Conditioning (ETH100G Client) (continued)

Trigger	Tx conditioning				
Condition	100G OCI P1 Signal output (Laser off if LOFEF enabled)		Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
	Left	Right	P1	P1-1	P1-2
Right OCI CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right OCI P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning	No conditioning	ODU4 BDI
Right OCI P1 OPU4 CSF	No conditioning	LF	No conditioning	No conditioning	No conditioning

Table 1-25
Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (ETH100G) + 100G MOTR (OC192) Signal Conditioning

Trigger	Tx conditioning				
Condition	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
No OCLD Connection	LF	LF	No conditioning	ODU4 OCI	ODU4 OCI
P1 OCLD Facility OOS	LF	LF	ODUG LCK	ODU4 LCK	ODU4 LCK
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	LF	LF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	LF	LF	No conditioning	No conditioning	No conditioning
OCLD P1, ODUG AIS, ODUG LCK	LF	LF	ODUG BDI	ODU4 AIS	ODU4 AIS

Table 1-25

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (ETH100G) + 100G MOTR (OC192) Signal Conditioning (continued)

Trigger	Tx conditioning				
Condition	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
Left OCI P1 Facility OOS	LF	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCI, No connection	LF	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCI P1 LOS, LOF, LOC, HiBER	LF	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, ODU4 AIS, ODU4 LCK, ODU4 OCI	LF	No conditioning	No conditioning	ODU4 BDI	No conditioning
Left OCI P1, OPU4 CSF	LF	No conditioning	No conditioning	No conditioning	No conditioning
Right MUX P100 OPU4 AIS, OPU4 CSF	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning
Right MUX P100 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning
Right MUX P1-10 No Connection	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning
Right MUX P1-10 Facility OOS	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning
Right MUX P1-10 LOS, LOC, LOF	No conditioning	RDI-L	No conditioning	No conditioning	No conditioning

Table 1-25

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (ETH100G) + 100G MOTR (OC192) Signal Conditioning (continued)

Trigger	Tx conditioning				
Condition	Left 100G OCI P1 Signal output (Laser off if LOFEF enabled)	Right 10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	Flex3 WL3e OCLD/Flex4 WL3e OCLD P1 Signal output		
			P1	P1-1	P1-2
Right MUX CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right MUX P1-10 PTM	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning
Right MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	AIS-L	No conditioning	No conditioning	No conditioning

Table 1-26

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (OTU4) + 100GE OTR (100GE) Signal Conditioning

Trigger	Tx conditioning				
Condition	100G OCI Signal output (Laser off if LOFEF enabled)		Flex3 WL3e OCLD P1 Signal output		
	Left P1	Right P1	P1	P1-1	P1-2
No OCLD Connection	LF	LF	No conditioning	ODU4 OCI	ODU4 OCI
P1 OCLD Facility OOS	LF	LF	ODUG LCK	ODU4 LCK	ODU4 LCK
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	LF	LF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	LF	LF	No conditioning	No conditioning	No conditioning
OCLD P1, ODUG AIS, ODUG LCK	LF	LF	ODUG BDI	ODU4 AIS	ODU4 AIS

Table 1-26

Flex3 WL3e OCLD/Flex4 WL3e OCLD 100G OTR (OTU4) + 100GE OTR (100GE) Signal Conditioning (continued)

Trigger	Tx conditioning				
Condition	100G OCI Signal output (Laser off if LOFEF enabled)		Flex3 WL3e OCLD P1 Signal output		
	Left P1	Right P1	P1	P1-1	P1-2
Left OCI P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCI, No connection	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCI P1 LOS, LOF, LOMF, OTU PreFEC SF, ODU SF	ODU4 BDI	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1 OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left OCI CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCI P1, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning
Right OCI P1 Facility OOS	No conditioning	LF	No conditioning	No conditioning	ODU4 LCK
Right OCI No Connection	No conditioning	LF	No conditioning	No conditioning	ODU4 OCI
Right OCI P1 LOS, LOC, LOF, HiBER	No conditioning	LF	No conditioning	No conditioning	ODU4 AIS
Right OCI CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right OCI P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning	No conditioning	ODU4 BDI
Right OCI P1 OPU4 CSF	No conditioning	LF	No conditioning	No conditioning	No conditioning

Table 1-27
100G WL3e OTR (100GE OTR) signal conditioning

Trigger	Tx conditioning	
Condition	Line Port p1 Signal output	Client Port p2 Signal output (Laser off if LOFEF enabled)
P1Facility OOS	ODU4 LCK	LF
P1LOS, LOC, LOF, PreFEC-SF, LOMF	OTU4 BDI, ODU4 BDI	LF
P1 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF
P1 PreFEC-SD	No conditioning	No conditioning
P2 No facility	ODU4 OCI	NA
P2 Facility OOS	ODU4 LCK	LF
P2 LOS, LOC, LODS, LOF, HIBER	OPU4-CSF	No conditioning
P2 ODU4-AIS, ODU4-LCK, ODU4-OCI	ODU4 BDI	LF
P2 OPU-CSF, PTM	No conditioning	LF

Table 1-28
100G WL3e OTR (100G OTU4) signal conditioning

Trigger	Tx conditioning	
Condition	Line Port P1 Signal output	Client Port P2 Signal output (Laser off if LOFEF enabled)
P1Facility OOS	ODU4-LCK	ODU4-LCK
P1LOS, LOC, LOF, PreFEC-SF	OTU4 BDI, ODU4 BDI	ODU4 AIS
P1 ODU4-AIS, ODU4-LCK, ODU4-OCI	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI
P1 PreFEC-SD, LOMF	No conditioning	No conditioning
P2 No facility	ODU4 OCI	NA
P2 Facility OOS	ODU4 LCK	ODU4 LCK
P2 LOS, LOC, LOF, PreFEC-SF	ODU4 AIS	OTU4 BDI, ODU4 BDI
P2 ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning
P2 PreFEC-SD, LOMF	No conditioning	No conditioning

Table 1-29

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OTM2 client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD/Flex4 WL3e OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
No OCLD Connection	No conditioning	ODU2 AIS	ODU2 AIS
P1 OCLD Facility OOS	ODUC2 LCK	ODU2 AIS	ODU2 AIS
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	ODU2 AIS	ODU2 AIS
OCLD CP Missing	N/A	ODU2 AIS	ODU2 AIS
OCLD P1 OTUC2 AIS, ODUC2 LCK	ODUC2 BDI	ODU2 AIS	ODU2 AIS
OCLD CP Missing	No conditioning	N/A	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU2 AIS	No conditioning
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	ODU2 AIS	No conditioning
OCI P1-10 No connection	No conditioning	MSI PSI BYTES=0, ODU2 OCI	No conditioning
OCI P1-10 facility OOS	No conditioning	ODU2 LCK	No conditioning
OCI P1-10 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	No conditioning	ODU2 BDI	No conditioning
OCI P1-10 MSI MM, MR MM	No conditioning	ODU2 AIS	No conditioning
OCI P1-10 PTM OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	No conditioning
OCI P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU2 AIS

Table 1-29

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OTM2 client) (continued)

Trigger	Tx conditioning		
Condition	Flex3 OCLD/Flex4 WL3e OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
OCI P102 OPU4 AIS, OPU4 LCK, OPU4 PTM	No conditioning	No conditioning	ODU2 AIS
OCI P11-20 No connection	No conditioning	No conditioning	MSI PSI BYTES=0, ODU2 OCI
OCI P11-20 facility OOS	No conditioning	No conditioning	ODU2 LCK
OCI P11-20 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	No conditioning	No conditioning	ODU2 BDI
OCI P11-20 MSI MM, MR MM	No conditioning	No conditioning	ODU2 AIS
OCI P11-20 PTM OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	No conditioning
OCI P11-20 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning

Table 1-30

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OC192 client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
No OCLD Connection	No conditioning	AIS-L	AIS-L
P1 OCLD Facility OOS	ODUC2 LCK	AIS-L	AIS-L
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	AIS-L	AIS-L
OCLD CP Missing	N/A	AIS-L	AIS-L

Table 1-30

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OC192 client) (continued)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
OCLD P1 OTUC2 AIS, ODU2 LCK	ODUC2 BDI	AIS-L	AIS-L
OCI CP Missing	No conditioning	N/A	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	AIS-L	No conditioning
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	AIS-L	No conditioning
OCI P1-10 No connection	No conditioning	AIS-L	No conditioning
OCI P1-10 facility OOS	No conditioning	AIS-L	No conditioning
OCI P1-10 LOS, LOC, LOF	No conditioning	No conditioning	No conditioning
OCI P1-10 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	AIS-L	No conditioning
OCI P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	AIS-L	No conditioning
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	AIS-L
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	No conditioning	AIS-L
OCI P11-20 No connection	No conditioning	No conditioning	AIS-L
OCI P11-20 facility OOS	No conditioning	No conditioning	AIS-L
OCI P11-20 LOS, LOC, LOF	No conditioning	No conditioning	No conditioning
OCI P11-20 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	AIS-L
OCI P11-20 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	AIS-L

Table 1-31
200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR
configuration (OC768 client)

Trigger	Tx conditioning	
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)
	P1	P1-5
No OCLD Connection	No conditioning	AIS-L
P1 OCLD Facility OOS	ODUC2 LCK	AIS-L
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	AIS-L
OCLD CP Missing	N/A	AIS-L
OCLD P1 OTUC2 AIS, ODUC2 LCK	ODUC2 BDI	AIS-L
OCI CP Missing	No conditioning	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	AIS-L (P1-3)
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	AIS-L (P1-3)
OCI P1-5 No connection	No conditioning	AIS-L
OCI P1-5 facility OOS	No conditioning	AIS-L
OCI P1-5 LOS, LOC, LOF	No conditioning	No conditioning
OCI P1-5 PTM, MSI MM, OPU3 AIS, OPU3 CSF	No conditioning	AIS-L
OCI P1-5 ODU3 AIS, ODU3 LCK, ODU3 OCI	No conditioning	AIS-L
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	AIS-L (P3-5)
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	AIS-L (P3-5)

Table 1-32

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (ETH10G client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
No OCLD Connection	No conditioning	LF	LF
P1 OCLD Facility OOS	ODUC2 LCK	LF	LF
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	LF	LF
OCLD CP Missing	N/A	LF	LF
OCLD P1 OTUC2 AIS, ODU2 LCK	ODUC2 BDI	LF	LF
OCI CP Missing	No conditioning	N/A	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	LF	No conditioning
OCI P1-10 No connection	No conditioning	LF	No conditioning
OCI P1-10 facility OOS	No conditioning	LF	No conditioning
OCI P1-10 LOS, LOC, LOF, HiBER	No conditioning	No conditioning	No conditioning
OCI P1-10 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	LF	No conditioning
OCI P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	LF	No conditioning
OCI P1-10 WAN LINKDOWN	No conditioning	LF	No conditioning
OCI P1-10 WAN Loss of Frame Delineation	No conditioning	LF	No conditioning
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	LF

Table 1-32

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (ETH10G client) (continued)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	No conditioning	LF
OCI P11-20 No connection	No conditioning	No conditioning	LF
OCI P11-20 facility OOS	No conditioning	No conditioning	LF
OCI P11-20 LOS, LOC, LOF, HiBER	No conditioning	No conditioning	No conditioning
OCI P11-20 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	LF
OCI P11-20 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	LF
OCI P11-20 WAN LINKDOWN	No conditioning	No conditioning	LF
OCI P11-20 WAN Loss of Frame Delineation	No conditioning	No conditioning	LF

Table 1-33

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (ETH40G client)

Trigger	Tx conditioning	
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)
	P1	P1-5
No OCLD Connection	No conditioning	LF
P1 OCLD Facility OOS	ODUC2 LCK	LF
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	LF
OCLD CP Missing	N/A	LF

Table 1-33

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (ETH40G client) (continued)

Trigger	Tx conditioning	
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)
	P1	P1-5
OCLD P1 OTUC2 AIS, ODUC2 LCK	ODUC2 BDI	LF
OCI CP Missing	No conditioning	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF (P1-3)
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	LF (P1-3)
OCI P1-5 No connection	No conditioning	LF
OCI P1-5 facility OOS	No conditioning	LF
OCI P1-5 LOS, LOC, LOF, HiBER	No conditioning	No conditioning
OCI P1-5 PTM, MSI MM, OPU3 AIS, OPU3 CSF	No conditioning	LF
OCI P1-5 ODU3 AIS, ODU3 LCK, ODU3 OCI	No conditioning	LF
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF (P3-5)
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	LF (P3-5)

Table 1-34
200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR
configuration (ETH100G client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1	P2
No OCLD Connection	No conditioning	LF	LF
P1 OCLD Facility OOS	ODUG LCK	LF	LF
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUG PreFEC-SF	OTUG BDI ODUG BDI	LF	LF
P1 (OCLD) OTUC2 SD, ODUC2 SD	No conditioning	No conditioning	No conditioning
OCLD CP Missing	N/A	LF	LF
OCLD P1 ODUC2 AIS, ODUC2 LCK	ODUG BDI	LF	LF
OCI CP Missing	No conditioning	N/A	N/A
OCI P1 facility OOS	No conditioning	LF	No conditioning
OCI P1 No connection	No conditioning	LF	No conditioning
OCI P1 LOS, LOC, LOF, HiBER	No conditioning	LF	No conditioning
OCI P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning
OCI P1 OPU AIS, OPU4 CSF,	No conditioning	No conditioning	No conditioning
OCI P2 facility OOS	No conditioning	No conditioning	LF
OCI P2 No connection	No conditioning	No conditioning	LF
OCI P2 LOS, LOC, LOF, HiBER	No conditioning	No conditioning	LF
OCI P2 ODU4 AIS, ODU4 OCI	No conditioning	No conditioning	LF
OCI P2 OPU AIS, OPU4 CSF	No conditioning	No conditioning	LF

Table 1-35

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (FC1200 client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
No OCLD Connection	No conditioning	LF	LF
P1 OCLD Facility OOS	ODUC2 LCK	LF	LF
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	LF	LF
OCLD CP Missing	N/A	No conditioning	No conditioning
OCLD P1 OTUC2 AIS, ODU2 LCK	ODUC2 BDI	LF	LF
OCI CP Missing	No conditioning	N/A	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	LF	No conditioning
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	LF	No conditioning
OCI P1-10 No connection	No conditioning	LF	No conditioning
OCI P1-10 facility OOS	No conditioning	LF	No conditioning
OCI P1-10 LOS, LOC, LOF, HiBER	No conditioning	No conditioning	No conditioning
OCI P1-10 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	LF	No conditioning
OCI P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	LF	No conditioning
OCI P102 ODU4 AIS, ODU4 CSF, ODU4 OCI	No conditioning	No conditioning	LF
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	No conditioning	LF
OCI P11-20 No connection	No conditioning	No conditioning	LF
OCI P11-20 facility OOS	No conditioning	No conditioning	LF

Table 1-35

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (FC1200 client) (continued)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
OCI P11-20 LOS, LOC, LODS, HiBER	No conditioning	No conditioning	No conditioning
OCI P11-20 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	LF
OCI P11-20 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	LF

Table 1-36

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (FC800 client)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
No OCLD Connection	No conditioning	NOS	NOS
P1 OCLD Facility OOS	ODUC2 LCK	NOS	NOS
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	NOS	NOS
OCLD CP Missing	N/A	No conditioning	No conditioning
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	NOS	No conditioning
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	NOS	No conditioning
OCI P1-10 No connection	No conditioning	NOS	No conditioning
OCI P1-10 facility OOS	No conditioning	NOS	No conditioning
OCI P1-10 LOS, LOC, HiBER	No conditioning	No conditioning	No conditioning

Table 1-36

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (FC800 client) (continued)

Trigger	Tx conditioning		
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)	
	P1	P1-10	P11-20
OCI P1-10 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	NOS	No conditioning
OCI P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	NOS	No conditioning
OCI P102 ODU4 AIS, ODU4 CSF, ODU4 OCI	No conditioning	No conditioning	NOS
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	No conditioning	NOS
OCI P11-20 No connection	No conditioning	No conditioning	NOS
OCI P11-20 facility OOS	No conditioning	No conditioning	NOS
OCI P11-20 LOS, LOC, HiBER	No conditioning	No conditioning	No conditioning
OCI P11-20 PTM, MSI MM, OPU2 AIS, OPU2 CSF	No conditioning	No conditioning	NOS
OCI P11-20 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	NOS

Table 1-37

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OTM3 client)

Trigger	Tx conditioning	
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)
	P1	P1-5
No OCLD Connection	No conditioning	ODU3 AIS
P1 OCLD Facility OOS	ODUC2 LCK	ODU3 AIS
P1 (OCLD) LOS, LOC, LOF, LOMF, OTUC2 PreFEC-SF	OTUC2 BDI ODUC2 BDI	ODU3 AIS

Table 1-37

200G (2x100G/5x40G) MUX over OTUC2 line interface Signal Conditioning—200G MOTR configuration (OTM3 client) (continued)

Trigger	Tx conditioning	
Condition	Flex3 OCLD P1 Signal output	200G (2x100G/5x40G) MUX Signal Output (Laser off if LOFEF enabled)
	P1	P1-5
OCLD CP Missing	N/A	ODU3 AIS
OCLD P1 OTUC2 AIS, ODUC2 LCK	ODUC2 BDI	ODU3 AIS
OCI CP Missing	No conditioning	N/A
OCI P101 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU3 AIS (P1-3)
OCI P101 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	ODU3 AIS (P1-3)
OCI P1-5 No connection	No conditioning	MSI PSI BYTES=0, ODU3 OCI
OCI P1-5 facility OOS	No conditioning	ODU3 LCK
OCI P1-5 LOS, LOF, LOMF, OTU PreFEC-SF, OTU SF, ODU SF	No conditioning	OTU3 BDI, ODU3 BDI
OCI P1-5 PTM, MSI MM, OPU3 AIS, OPU3 CSF	No conditioning	No conditioning
OCI P1-5 MR MM	No conditioning	No conditioning
OCI P1-5 ODU3 AIS, ODU3 LCK, ODU3 OCI	No conditioning	No conditioning
OCI P102 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU3 AIS (P3-5)
OCI P102 OPU4 AIS, OPU4 CSF, OPU4 PTM	No conditioning	ODU3 AIS (P3-5)

Optical Regen configuration (100G OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD (QPSK and BPSK modes), and 100G WL3/WL3e OCLD)

The following tables present the signal conditioning that is applied for the 100G OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD (QPSK and BPSK modes), and 100G WL3/WL3e OCLD circuit packs in optical regen configuration.

Table 1-38

100G OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD (QPSK and BPSK modes), and 100G WL3/WL3e OCLD signal conditioning—optical Regen configuration

Trigger	Tx conditioning	
Condition	100G OCLD/Flex2 WL3/WL3e OCLD/Flex3 WL3e OCLD (QPSK and BPSK modes)/100G WL3/WL3e OCLD p1 (OTM4) Signal output	Facility
No Transponder Connection	ODU OCI ODU OCI	p1 (left mate) p1 (right mate)
Left slot service-affecting Circuit Pack Fail	ODU AIS ODU AIS	p1 (left mate) p1 (right mate)
Left slot p1 Facility OOS	ODU LCK ODU LCK	p1 (left mate) p1 (right mate)
Left slot p1 LOS, LOF, LOMF, PreFEC SF	OTU BDI ODU AIS	p1 (left mate) p1 (right mate)
Right slot service-affecting Circuit Pack Fail	ODU AIS ODU AIS	p1 (left mate) p1 (right mate)
Right slot p1 Facility OOS	ODU LCK ODU LCK	p1 (left mate) p1 (right mate)
Right slot p1 LOS, LOF, LOMF, PreFEC SF	ODU AIS OTU BDI	p1 (left mate) p1 (right mate)

Table 1-39
2C-BPSK 100G regen signal conditioning—optical transponder configuration

Trigger	Tx conditioning			
	Left 2C- BPSK OCLD Signal output		Right 2C- BPSK OCLD Signal output	
	Prime	Member	Prime	Member
No ODU connection	ODU4 OCI	ODU4 OCI	ODU4 OCI	ODU4 OCI
Left Prime and/or member OCLD P1 Facility OOS	ODU4 LCK	ODU4 LCK	ODU4 LCK	ODU4 LCK
Left Prime OCLD P1 Facility LOS, LOC, LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	OTU4 BDI	No conditioning	ODU4 AIS	OTU4 AIS
Left Prime and/or member OCLD circuit pack Missing	No conditioning	No conditioning	ODU4 AIS	OTU4 AIS
Left Prime and/or member OCLD P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI
Left Member OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	No conditioning	OTU4 BDI	ODU4 AIS	OTU4 AIS
Right Prime and/or member OCLD P1 Facility OOS	ODU4 LCK	ODU4 LCK	ODU4 LCK	ODU4 LCK
Right Prime OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	ODU4 AIS	ODU4 AIS	OTU4 BDI	No conditioning
Right Prime and/or member OCLD circuit pack Missing	OTU4 AIS	OTU4 AIS	No conditioning	No conditioning
Right Prime and/or member OCLD P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning
Right Member OCLD P1 LOS, LOC, LOF, LOMF, OTU PreFEC-SF, OTU Skew OOR	ODU4 AIS	ODU4 AIS	No conditioning	OTU4 BDI

Table 1-40
200G (16QAM) Regen Signal Conditioning

Trigger	Tx conditioning					
Condition	Left OCLD Signal output			Right OCLD Signal output		
	P1	P1-1	P1-2	P1	P1-1	P1-2
Left OCLD No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	No conditioning	ODU4 OCI	ODU4 OCI
Left OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	ODUG LCK	ODU4 LCK	ODU4 LCK
Left OCLD P1 LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	No conditioning	ODU4 AIS	ODU4 AIS
Left OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left OCLD CP Missing	No conditioning	No conditioning	No conditioning	ODUG AIS	ODU4 AIS	ODU4 AIS
Left OCLD P1, ODUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	No conditioning	ODU4 AIS	ODU4 AIS
Right OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	ODUG LCK	ODU4 LCK	ODU4 LCK
Right OCLD P1 LOS, LOF, LOC, LOMF, OTU PreFEC SF, OTU SF	No conditioning	ODU4 AIS	ODU4 AIS	OTUG BDI, ODUG BDI	ODU4 AIS	ODU4 AIS
Right OCLD P1 OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Right OCLD CP Missing	ODUG AIS	ODU4 AIS	ODU4 AIS	No conditioning	No conditioning	No conditioning
Right OCLD P1 ODUG AIS, ODUG LCK	No conditioning	ODU4 AIS	ODU4 AIS	ODU4 BDI	ODU4 AIS	ODU4 AIS

Table 1-41
200G (QPSK<=>16QAM) Regen Signal Conditioning

Trigger	Tx conditioning				
Condition	OCLD (in QPSK) Signal output		Flex3 WL3e OCLD (in 16QAM) Signal output		
	Left	Right	P1	P1-1	P1-2
OCLD (16QAM) No Connection	ODU4 OCI	ODU4 OCI	No conditioning	ODU4 OCI	ODU4 OCI
OCLD (16QAM) Facility OOS	ODU4 LCK	ODU4 LCK	ODUG LCK	ODU4 LCK	ODU4 LCK
OCLD (16QAM) P1 LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG S	OTU4 BDI	No conditioning	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD (16QAM) P1 OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD (16QAM) CP Missing	ODU4 AIS	ODU4 AIS	No conditioning	No conditioning	No conditioning
OCLD (16QAM) P1 ODUG AIS, ODUG LCK	ODU4 AIS	ODU4 AIS	ODUG BDI	ODU4 AIS	ODU4 AIS
Left OCLD (QPSK) P1 LOS, LOF, LOMF, OTU4 PreFEC SF, OTUSF, ODU SF	ODU4 BDI	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCLD (QPSK) P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCLD (QPSK) P1, No connection	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	No conditioning
Left OCLD (QPSK) CP Missing	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning

Table 1-41
200G (QPSK<=>16QAM) Regen Signal Conditioning (continued)

Trigger	Tx conditioning				
Condition	OCLD (in QPSK) Signal output		Flex3 WL3e OCLD (in 16QAM) Signal output		
	Left	Right	P1	P1-1	P1-2
Left OCLD (QPSK) P1, ODU AIS, ODU LCK, ODU OCI	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning
Right OCLD (QPSK) P1 LOS, LOC, LOF, LOMF, OTU4, PreFEC SF, OTU SF, ODU SF	No conditioning	OTU4 BDI	No conditioning	No conditioning	ODU4 AIS
Right OCLD (QPSK) P1 Facility OOS	No conditioning	ODU4 LCK	No conditioning	No conditioning	ODU4 LCK
Left OCLD (QPSK) No Connection	No conditioning	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI
Right OCLD (QPSK) CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right OCLD (QPSK) P1, ODU AIS, ODU LCK, ODU OCI	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI

Table 1-42
200G (BPSK<=>16QAM) Regen Signal Conditioning

Trigger	Tx conditioning						
Condition	Left OCLD (in BPSK) Signal output		Right OCLD (in BPSK) Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
	Prime P1	Member P1	Prime P1	Member P1	P1	P1-1	P1-2
OCLD (16QAM) No Connection	ODU4 OCI	No conditioning	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	ODU4 OCI
OCLD (16QAM) Facility OOS	ODU4 LCK	No conditioning	ODU4 LCK	No conditioning	ODUG LCK	ODU4 LCK	ODU4 LCK
OCLD (16QAM) P1 LOS, LOC, LOF, LOMF, OTUG PreFE C SF, OTUG SF, ODUG SF	ODU4 AIS	No conditioning	ODU4 AIS	No conditioning	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD (16QAM) P1 OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD (16QAM) CP Missing	ODU4 AIS	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning	No conditioning
OCLD (16QAM) P1 ODUG AIS, ODUG LCK	ODU4 AIS	No conditioning	ODU4 AIS	No conditioning	ODUG BDI	ODU4 AIS	ODU4 AIS
Left Prime OCLD (BPSK) P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left Prime OCLD (BPSK) P1 No Connection	ODU4 OCI	No conditioning	No conditioning	No conditioning	No conditioning	ODU4OCI	No conditioning

Table 1-42
200G (BPSK<=>16QAM) Regen Signal Conditioning (continued)

Trigger	Tx conditioning						
	Left OCLD (in BPSK) Signal output		Right OCLD (in BPSK) Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
	Prime P1	Member P1	Prime P1	Member P1	P1	P1-1	P1-2
Left Prime OCLD (BPSK) P1 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	ODU4 BDI	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left Prime OCLD (BPSK) P1 OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left Prime OCLD (BPSK) CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left Prime OCLD (BPSK) P1, ODU4 AIS, ODU4 LCK, ODU4 OC	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OC	No conditioning
Left Member OCLD (BPSK) P1 LOS, LOC, LOF, LOMF, OTU, PreFEC SF, OTU SF, ODU SF	No conditioning	OTU4 BDI	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left Member OCLD (BPSK) P1 OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left Member OCLD (BPSK) CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning

Table 1-42
200G (BPSK<=>16QAM) Regen Signal Conditioning (continued)

Trigger	Tx conditioning						
Condition	Left OCLD (in BPSK) Signal output		Right OCLD (in BPSK) Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
	Prime P1	Member P1	Prime P1	Member P1	P1	P1-1	P1-2
Right Prime OCLD (BPSK) P1 Facility OOS	No conditioning	No conditioning	ODU4 LCK	No conditioning	No conditioning	No conditioning	ODU4 LCK
Right Prime OCLD (BPSK) No Connection	No conditioning	No conditioning	ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4 OCI
Right Prime OCLD (BPSK) P1 LOS, LOC, LOF, LOMF, OTU, PreFEC SF, OTU SF, ODU SF	No conditioning	No conditioning	OTU4 BDI	No conditioning	No conditioning	No conditioning	ODU4 AIS
Right OCLD (BPSK) P1, OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning

Table 1-43
200G (BPSK+QPSK)<=>16QAM Regen Signal Conditioning (OTM4 Client)

Trigger	Tx conditioning					
	Left OCLD (in QPSK) P1 Signal output	Right OCLD (in BPSK) P1 Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
		Prime	Member	P1	P1-1	P1-2
OCLD (16QAM) No Connection	ODU4 OCI	ODU4 OCI	No conditioning	No conditioning	ODU4 OCI	ODU4 OCI
OCLD (16QAM) P1 Facility OOS	ODU4 LCK	ODU4 LCK	No conditioning	ODUG LCK	ODU4 LCK	ODU4 LCK
OCLD (16QAM) P1 LOS, LOC, LOF, LOMF, OTUG PreFEC SF, OTUG SF, ODUG SF	ODU4 AIS	ODU4 AIS	No conditioning	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS
OCLD (16QAM) P1 OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD (16QAM) CP Missing	ODU4 AIS	ODU4 AIS	No conditioning	No conditioning	No conditioning	No conditioning
OCLD (16QAM) P1 ODUG AIS, ODUG LCK	ODU4 AIS	ODU4 AIS	No conditioning	ODUG BDI	ODU4 AIS	ODU4 AIS
Left OCLD (QPSK) P1 Facility OOS	ODU4 LCK	No conditioning	No conditioning	No conditioning	ODU4 LCK	No conditioning
Left OCLD (QPSK) P1 No Connection	ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4OCI	No conditioning
Left OCLD (QPSK) P1 LOS, LOF, LOMF, OTU PreFEC SF, OTU SF, ODU SF	ODU4 BDI	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning

Table 1-43
200G (BPSK+QPSK)<=>16QAM Regen Signal Conditioning (OTM4 Client) (continued)

Trigger	Tx conditioning					
Condition	Left OCLD (in QPSK) P1 Signal output	Right OCLD (in BPSK) P1 Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
		Prime	Member	P1	P1-1	P1-2
Left OCI P1 OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Left OCLD (QPSK) CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS	No conditioning
Left OCLD (QPSK) P1, ODU4 AIS, ODU4 LCK, ODU4 OC	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning
Prime OCLD (BPSK) P1 Facility OOS	No conditioning	ODU4 LCK	No conditioning	No conditioning	No conditioning	ODU4 LCK
Prime OCLD (BPSK) No Connection	No conditioning	ODU4 OCI	No conditioning	No conditioning	No conditioning	ODU4 OCI
Prime OCLD (BPSK) P1 LOS, LOC, LOF, LOMF, OTU, PreFEC SF, OTU SF, ODU SF	No conditioning	OTU4 BDI	No conditioning	No conditioning	No conditioning	ODU4 AIS
Prime OCLD (BPSK) P1, OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Prime OCLD (BPSK) CP Missing	ODU4 AIS	ODU4 AIS	No conditioning	No conditioning	No conditioning	ODU4 AIS

Table 1-43**200G (BPSK+QPSK)<=>16QAM Regen Signal Conditioning (OTM4 Client) (continued)**

Trigger	Tx conditioning					
Condition	Left OCLD (in QPSK) P1 Signal output	Right OCLD (in BPSK) P1 Signal output		Flex3 WL3e OCLD (in 16QAM) P1 Signal output		
		Prime	Member	P1	P1-1	P1-2
Prime OCLD (BPSK) P1 ODU4 AIS, ODU4 LCK, ODU4 OCI	ODU4 AIS	ODU4 AIS	No conditioning	ODUG BDI	ODU4 AIS	ODU4 AIS
Member OCLD (BPSK) P1 LOS, LOC, LOF, LOMF, OTU, PreFEC SF, OTU SF, ODU SF	No conditioning	No conditioning	OTU4 BDI	No conditioning	No conditioning	ODU4 AIS
Member OCLD (BPSK) P1, OTU SD, ODU SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Member OCLD (BPSK) CP Missing	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning	ODU4 AIS

Table 1-44
Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (OC192 client)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	AIS-L	AIS-L
OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	AIS-L	AIS-L
OCLD P1 LOS, LOC, LOF, LOMF, OTUG PreFEC-SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	AIS-L	AIS-L
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	No conditioning	No conditioning	No conditioning	AIS-L	AIS-L
OCLD P1, OTUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	AIS-L	AIS-L
Left MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning
Left MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning
Left MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning
Left MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning
Left MUX P1-10 LOS, LOC, LOF	No conditioning	No conditioning	No conditioning	RDI-L	No conditioning
Left MUX CP Missing	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning
Left MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning

Table 1-44**Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (OC192 client) (continued)**

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
Left MUX P1-10, ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	AIS-L	No conditioning
Right MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L
Right MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L
Right MUX P1-10 No connection	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L
Right MUX P1-10 Facility OOS	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L
Right MUX P1-10 LOS, LOC, LOF	No conditioning	No conditioning	No conditioning	No conditioning	RDI-L
Right MUX CP Missing	No conditioning	No conditioning	ODU4 AIS	No conditioning	No conditioning
Right MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L
Right MUX P1-10, ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	AIS-L

Table 1-45
Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (ETH10G client)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	LF	LF
OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	LF	LF
OCLD P1 LOS, LOC, LOF, LOMF, OTUG PreFEC-SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	LF	LF
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	No conditioning	No conditioning	No conditioning	LF	LF
OCLD P1, OTUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	LF	LF
Left MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU4 BDI	No conditioning	LF	No conditioning
Left MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10 LOS, LOF, HiBER	No conditioning	No conditioning	No conditioning	RF	No conditioning
Left MUX CP Missing	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning
Left MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	LF	No conditioning

Table 1-45**Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (ETH10G client) (continued)**

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
Left MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10 WAN LINKDOWN	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10 Loss of Frame Delineation	No conditioning	No conditioning	No conditioning	LF	No conditioning
Right MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10 LOS, LOF, HiBER	No conditioning	No conditioning	No conditioning	No conditioning	RF
Right MUX CP Missing	No conditioning	No conditioning	ODU4 AIS	No conditioning	No conditioning
Right MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	LF

Table 1-46

Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (FC1200 client)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	LF	LF
OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	LF	LF
OCLD P1 LOS, LOC, LOF, LOMF, OTUG PreFEC-SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	LF	LF
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	No conditioning	No conditioning	No conditioning	LF	LF
OCLD P1, OTUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	LF	LF
Left MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU4 BDI	No conditioning	LF	No conditioning
Left MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	LF	No conditioning
Left MUX P1-10 LOS, LOF, HiBER	No conditioning	No conditioning	No conditioning	RF	No conditioning
Left MUX CP Missing	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning
Left MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	LF	No conditioning

Table 1-46**Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (FC1200 client) (continued)**

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
Left MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	LF	No conditioning
Right MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU4 BDI	No conditioning	LF
Right MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10 LOS, LODS, HiBER	No conditioning	No conditioning	No conditioning	No conditioning	RF
Right MUX CP Missing	No conditioning	No conditioning	ODU4 AIS	No conditioning	No conditioning
Right MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	LF
Right MUX P1-10 ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	LF

Table 1-47

Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (FC800 client)

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
No OCLD Connection	No conditioning	ODU4 OCI	ODU4 OCI	NOS	NOS
OCLD P1 Facility OOS	ODUG LCK	ODU4 LCK	ODU4 LCK	NOS	NOS
OCLD P1 LOS, LOC, LOF, LOMF, OTUG PreFEC-SF, OTUG SF, ODUG SF	OTUG BDI ODUG BDI	ODU4 AIS	ODU4 AIS	NOS	NOS
OCLD P1, OTUG SD, ODUG SD	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
OCLD CP Missing	No conditioning	No conditioning	No conditioning	NOS	NOS
OCLD P1, OTUG AIS, ODUG LCK	ODUG BDI	ODU4 AIS	ODU4 AIS	NOS	NOS
Left MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	NOS	No conditioning
Left MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	ODU4 BDI	No conditioning	NOS	No conditioning
Left MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	NOS	No conditioning
Left MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	NOS	No conditioning
Left MUX P1-10 LOS, LOC, LODS	No conditioning	No conditioning	No conditioning	NOS	No conditioning
Left MUX CP Missing	No conditioning	ODU4 AIS	No conditioning	No conditioning	No conditioning
Left MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	NOS	No conditioning

Table 1-47**Flex3 WL3e OCLD Signal Conditioning—200G MOTR configuration (FC800 client) (continued)**

Trigger	Tx conditioning				
Condition	Flex3 WL3e OCLD P1 Signal output			10x10G MUX P1-10 Signal output (Laser off if LOFEF enabled)	
	P1	P1-1	P1-2	Left	Right
Left MUX P1-10, ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	NOS	No conditioning
Right MUX P100, OPU4 AIS, OPU4 CSF	No conditioning	No conditioning	No conditioning	No conditioning	NOS
Right MUX P100, ODU4 AIS, ODU4 LCK, ODU4 OCI	No conditioning	No conditioning	ODU4 BDI	No conditioning	NOS
Right MUX P1-10, No connection	No conditioning	No conditioning	No conditioning	No conditioning	NOS
Right MUX P1-10, Facility OOS	No conditioning	No conditioning	No conditioning	No conditioning	NOS
Right MUX P1-10 LOS, LODS	No conditioning	No conditioning	No conditioning	No conditioning	NOS
Right MUX CP Missing	No conditioning	No conditioning	ODU4 AIS	No conditioning	NOS
Right MUX P1-10 PTM	No conditioning	No conditioning	No conditioning	No conditioning	No conditioning
Right MUX P1-10, ODU2 AIS, ODU2 LCK, ODU2 OCI	No conditioning	No conditioning	No conditioning	No conditioning	NOS

Signal conditioning for 2x10G OTR 4x10-11.3G XFP and 4x10G OTR circuit packs

Terminal configuration

[Table 1-48 on page 1-102](#) presents the signal conditioning that is applied for the 2x10G OTR and 4x10G OTR circuit packs in the terminal configuration.

Note that when FLEX facilities are set to OOS, the client laser is shutdown irrespective of the LOFEF or TXCON settings (not applicable to FC800 or FC1200).

ATTENTION

No conditioning is performed for Ethernet when the client receives a Local Fault or a Remote Fault condition.

For the 2x10G OTR circuit pack, the client ports are ports 2 and 4 and the line ports are ports 1 and 3.

The 4x10G OTR 8xXFP (NTK530QM, NTK530QP) circuit pack implements a different port association between ULL equipment mode and OTR equipment mode. The port associations are 1-5, 2-6, 3-7, 4-8 for OTR equipment mode and 1-2, 3-4, 5-6, 7-8 for ULL equipment mode. For the 4x10G OTR 4xXFP/4xSFP+ (NTK530QA) circuit pack, port associations are 1-5, 2-6, 3-7, 4-8 for both the OTR equipment mode and the ULL equipment mode.

Table 1-48
2x10G OTR and 4x10G OTR signal conditioning—terminal configuration

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
No client port facility	UAS	ODU OCI	Laser Off
Line port facility OOS	OC192	ODU LCK	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7) (Note 2)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		ODU LCK
Line Port 5/6/7/8 (for 4x10G OTR) ODUk Post-FEC SD	ODU2	ODUk Post-FEC SD + Switch	ODU2 SD
Client Port 1/2/3/4 (for 4x10G OTR) SD	ODU2	ODUk Post-FEC SD	ODU2 SD
Port 1/3 (for 2x10G OTR), Port 5/6/7/8 (for 4x10G OTR) LOS/LOF/LOMF/OTU SF	OC192	OTU BDI, ODU BDI, RFI-L OTU BDI (laser off does not apply)	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		ODU AIS

Table 1-48**2x10G OTR and 4x10G OTR signal conditioning—terminal configuration (continued)**

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
Port 1/3 (for 2x10G OTR), Port 5/6/7/8 (for 4x10G OTR) ODU AIS/ODU LCK/ODU OCI	OC192	ODU BDI, RFI-L (laser off does not apply) None	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		None
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), OPU AIS (Note 4)	OC192	None	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		None

Table 1-48**2x10G OTR and 4x10G OTR signal conditioning—terminal configuration (continued)**

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), OTM2 OPU2 PT Mismatch (Note 4)	OC192	None	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		None
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), (WAN) Link Down	WAN	None	LF
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), (WAN) Loss of Frame Delineation	WAN	None	LF
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), EBER	ETH10G (10.7)	P-AIS	RF (2x10G OTR only)
	ETH10G (11.05) (Note 1)		None
	ETH10G (11.09)		None
	ETH10G Hairpin		LF
	ETH10G ULL		LF

Table 1-48**2x10G OTR and 4x10G OTR signal conditioning—terminal configuration (continued)**

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), facility OOS	OC192	ODU LCK	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		ODU LCK
Port 2/4 (for 2x10G OTR) Frequency Out Of Range	OC192	None	None
	OC-192 Hairpin		
	ETH10G (10.7)		
	ETH10G (11.05)		
	ETH10G (11.09)		
	OTM2		
	FC800/FC1200		
Port 2/4 (for 2x10G OTR), Loss Of Lock	OC192	OPU2 AIS	None
	OC-192 Hairpin		
	ETH10G (10.7)		
	ETH10G (11.05)	ODU AIS	OTU BDI (laser off does not apply)
	ETH10G (11.09)		
	OTM2		
	FC800/FC1200		None

Table 1-48**2x10G OTR and 4x10G OTR signal conditioning—terminal configuration (continued)**

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), LOS/LOF/LOMF	ETH10G ULL	OPU AIS (Note 6) ODU AIS	None
	OTM2 (Note 3)		AIS-L
	ETH10G (10.7)		For NTK530PAE5: RF (GFP only) (laser off does not apply) For NTK530PGE5 and NTK530PME5: None
	ETH10G (11.05) (Note 1)		None
	ETH10G (11.09)		None
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		OTU BDI (laser off does not apply)
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), Pre-FEC SF	OTM2 (Note 3)	ODU AIS	OTU BDI (laser off does not apply)
Service affecting circuit pack fail	OC192	ODU AIS	AIS-L
	OC-192 Hairpin		AIS-L
	ETH10G (10.7)		LF
	ETH10G (11.05) (Note 1)		LF
	ETH10G (11.09)		LF
	ETH10G Hairpin		LF
	ETH10G ULL		LF
	OTM2 (Note 3)		ODU AIS

Table 1-48**2x10G OTR and 4x10G OTR signal conditioning—terminal configuration (continued)**

Trigger condition	Client Port	Line Port Conditioning (Laser off if LOFEF enabled)	Client Port Conditioning (Laser off if LOFEF enabled)
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR), DWDM XFP pluggable fail	OC192	None	Laser off
	OC-192 Hairpin		
	ETH10G (10.7)		
	ETH10G (11.05) (Note 1)		
	ETH10G (11.09)		
	ETH10G Hairpin		
	ETH10G ULL		
	OTM2 (Note 3)		
Port 1/3 (for 2x10G OTR), Port 5/6/7/8 (for 4x10G OTR), DWDM XFP pluggable fail	OC192	Laser off	None
	OC-192 Hairpin		
	ETH10G (10.7)		
	ETH10G (11.05) (Note 1)		
	ETH10G (11.09)		
	ETH10G Hairpin		
	ETH10G ULL		
	OTM2 (Note 3)		
Note 1: Not supported on the 4x10G OTR circuit packs.			
Note 2: In the special case of a 10GE to GFP to STS192c to OPU2 mapping a P-UEQ would be conditioned instead of an ODU LCK.			
Note 3: Not supported on the client ports (1,2,3,4) of the NTK530QA 4x10G OTR circuit packs.			
Note 4: By default, these faults received from the line-facing equipment will be alarmed against the OTM2 layer facility on P2/P4 as the ODU monitor flag is enabled by default for P2/P4. The ODU monitor flag can be changed to enabled on the OTM2 facility on P1/P3 and the alarm can be seen at these ports.			
Note 5: Any conditioning that is not explicitly identified in the table should not be assumed.			
Note 6: In the special case of a 10GE to GFP to STS192c to OPU2 mapping, a P-AIS would be conditioned instead of an OPU AIS.			

Regenerator configuration

Table 1-49 presents the signal conditioning that is applied for the 2x10G OTR and 4x10G OTR (NTK530QM, NTK530QP variants only) circuit packs in regenerator configuration.

Table 1-49

2x10G OTR and 4x10G OTR (NTK530QM, NTK530QP variants only) signal conditioning—regen configuration

Trigger condition	Conditioning	
	Port 1/3 (for 2x10G OTR) Port 1/2/3/4 (for 4x10G OTR)	Port 2/4 (for 2x10G OTR) Port 1/2/3/4 (for 4x10G OTR)
Service affecting circuit pack fail	ODU AIS	ODU AIS
Port 1/3 (for 2x10G OTR), Port 5/6/7/8 (for 4x10G OTR) facility OOS	ODU LCK	ODU LCK
Port 1/3 (for 2x10G OTR), Port 5/6/7/8 (for 4x10G OTR) LOS/LOF/LOMF/OTU SF	OTU BDI	ODU AIS
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR) facility OOS	ODU LCK	ODU LCK
Port 2/4 (for 2x10G OTR), Port 1/2/3/4 (for 4x10G OTR) LOS/LOF/LOMF/OTU SF	ODU AIS	OTU BDI

Signal conditioning for FC400/FC800/FC1200 (applicable to 2x10G OTR (NTK530PME5), 4x10G OTR, the 4x10G MUX (NTK525CFE5), and the 10x10G MUX (NTK529BAE5/NTK529BBE5))

Table 1-50 presents the signal conditioning that is applied for FC400/FC800 and FC1200. FC400 applies to the 4x10G OTR NTK530QA variants only.

Table 1-50
FC800 and FC1200 signal conditioning

Trigger condition	FC400/FC800 Signal Conditioning		FC1200 Signal Conditioning	
	TXCOND = NOS	TXCOND = LOFEF	TXCOND = LF	TXCOND = LOFEF
OTU LOS, OTU LOF, OTU LOMF or pre-FEC SF detected on ingress of corresponding OTM2/OTM3/OTM4 line facility	Egress from Client Port: NOS	Client Laser Off	Egress from Client Port: LF	Client Laser Off
Corresponding OTM2/OTM3/OTM4 line facility is put out-of-service	Egress from Client Port: NOS	Client Laser Off	Egress from Client Port: LF	Client Laser Off
ODU AIS, ODU LCK, ODU OCI, OPU AIS or OPU PT Mismatch detected from the connection facing direction of a corresponding OTM2 layer facility	Egress from Client Port: NOS	Client Laser Off	Egress from Client Port: LF	Client Laser Off
Client (FLEX) facility is put out-of-service	Egress from Client Port: NOS	Client Laser Off	Egress from Client Port: LF	Client Laser Off
	ODU LCK in the connection facing direction of the corresponding OTM2 layer facility			
LOS, LOC, or Loss of Data Synch detected on ingress of an FC400/FC800 client (FLEX) facility	OPU AIS in the connection facing direction of the corresponding OTM2 layer facility		N/A	
NOS or OLS detected on ingress of an FC400/FC800 client (FLEX) facility	Passed transparently through the corresponding OTM2 line/layer facility in the Tx direction		N/A	
NOS or OLS detected on ingress of corresponding OTM2 line/layer facility	Passed transparently through the corresponding FC400/FC800 client facility in the Tx direction		N/A	

Table 1-50
FC800 and FC1200 signal conditioning (continued)

Trigger condition	FC400/FC800 Signal Conditioning		FC1200 Signal Conditioning	
	TXCOND = NOS	TXCOND = LOFEF	TXCOND = LF	TXCOND = LOFEF
LOS, LOC, LOF, EBER detected on ingress of an FC1200 client (FLEX) facility	N/A		OPU AIS in the connection facing direction of the corresponding OTM2 layer facility	
LF or RF detected on ingress of an FC1200 client (FLEX) facility	N/A		Passed transparently through the corresponding OTM2 line/layer facility in the Tx direction	
LF or RF detected on ingress of corresponding OTM2 line/layer facility	N/A		Passed transparently through the corresponding FC1200 client facility in the Tx direction	
OTM2 mapping facility LOS, LOC, LOF/HiBER or Loss of Data Synch detected on ingress of an FC400/FC800 client facility	on-ramp: OPUk_AIS, OPU2-CSF is conditioned for the OTM2 mapping facility to the line transmit direction off-ramp: FECSF		LF	
OPU2-CSF is received by the OTM2 mapping facility	FC800 parent facility is provisioned with Conditioning Type = LOFEF, laser shut off is initiated		NOS	

10G/40G/100G client signal fail alarming and signal conditioning

Table 1-51 describes client signal fail on-ramp signal conditioning/off-ramp signal conditioning and alarming requirements for the 10G/40G/100G mappings.

Table 1-51
10G/40G/100G Client Signal Fail requirement (On-Ramp)

Client Ingress Fault (on-Ramp)				
Mapping	Ingress Fault or Signal	OTMx Layer CONDTYPE	On-Ramp Signal Conditioning	
			WAN (GFP) Layer	OTMx Layer
10GE GFP (GFPSTD, GFPMACOSTER, GFPMACTR)	LOS/LOC/HiBER/LOF	OPUk-AIS	None	OPU2-AIS+OPU2-CSF
		ODUk-AIS	None	ODU2-AIS
		OPUk-CSF (default) Note 4	GFP(CMF-CSF) Note 1, Note 2	OPU2-CSF
10GE GFP (GFPSTD, GFPMACTR)	LF	N/A	GFP(CMF-FDI) Note 5	None
	RF	N/A	GFP(CMF-FDI) Note 5	None
	All Clear	N/A	GFP(CMF-DCI) Note 5	None
10GE GFP (GFPMACOSTER)	LF	N/A	LF encapsulated into GFP CDF	None
	RF	N/A	RF encapsulated into GFP CDF	None
	All Clear	N/A	All Clear encapsulated into GFP CDF	None
10GE PROP237/ PROP238	LOS/LOC/HiBER/LOF	OPUk-AIS	None	OPU2-AIS+OPU2-CSF
		ODUk-AIS	None	ODU2-AIS
		OPUk-CSF (default) Note 4	N/A	OPU2-CSF+100GE LF Note 3
	LF/RF	N/A	N/A	None

Table 1-51
10G/40G/100G Client Signal Fail requirement (On-Ramp) (continued)

Client Ingress Fault (on-Ramp)				
Mapping	Ingress Fault or Signal	OTMx Layer CONDTYPE	On-Ramp Signal Conditioning	
			WAN (GFP) Layer	OTMx Layer
40GE	LOS/LOC/HiBER/LOF	OPUk-CSF (default)	N/A	OPU3-CSF+40GE LF
		ODUk-AIS	N/A	ODU3-AIS
OC768	LOS/LOC/LOF	OPUk-AIS (default)	N/A	OPU3-AIS+OPUk-CSF
		ODUk-AIS	N/A	ODU3-AIS
100GE	LOS/LOC/HiBER/LOF	OPUk-CSF (default)	N/A	OPU4-CSF+100GE LF
		ODUk-AIS	N/A	ODU4-AIS
OCn/FC1200/800/ PSIFB	LOS/LOC/LOF LossOfCodeWordSync	OPUk-AIS (default)	N/A	OPUk-AIS+OPUk-CSF
	LOS/LOC/LOF LossOfCodeWordSync	ODUk-AIS	N/A	ODUk-AIS
OTMX Client	LOS/LOC/LOF	ODUk-CSF (default)	N/A	ODUx-AIS
All Client	No Client facility	N/A	N/A	ODUx-OCI
All Client (MUX)	No Client facility	N/A	N/A	MSI PSI byte = 0+ODUx-OCI
All Client	FAC OOS	N/A	N/A	ODUx-LCK
All Client	Plug fail	N/A	None	None

Note 1: Send GFP CMF-CSF (OPI=0x1) for client LOS/HiBER/LOF, send GFP CMF-CSF (OPI=0x2) for client LOC. Send GFP CMF-CSF every 100ms.

Note 2: Send GFP CMF-DCI (OPI=0x3) immediately when client fault clears.

Note 3: 10GE LF (Replace the 10GBASE-R signal by a stream of 66B blocks when each block carrying two local fault sequence ordered sets (as specified in IEEE Std 802.3). This replacement signal is then mapped into the OPU2e.

Note 4:

- Default CONDTYPE (Pre-release 10.0 based circuit packs): OPUk-AIS for 10G clients.
- Default CONDTYPE (Release 10.1+ based circuit packs): OPUk-CSF for 10G clients.

Note 5: 83=LF, 84=RF, 85=DCI if WAN parameter OSTTRAN=LEGACYCMF, LEGACYCMF is supported only on 2x10G OTR, 10X10G MR Mux OCI, 4x10G Mux OCI (CF) and 4x10G OTR Circuit packs

Signal conditioning for 5G IBM PSIFB on 2x10GOTR (NTK530PME5 variants only) and 5G IBM PSIFB or ISC3 on 4x10G OTR (NTK530QA variants only)

Table 1-52 presents the signal conditioning that is applied to 5G IBM PSIFB on the 2x10G OTR, and 5G IBM PSIFB or ISC3 on the 4x10G OTR.

Table 1-52
5G IBM PSIFB on 2x10GOTR (NTK530PME5 variants only) and 5G IBM PSIFB or ISC3 on 4x10G OTR (NTK530QA variants only) signal conditioning

Trigger condition	Signal Conditioning
OOS, OTU LOS, OTU LOF, OTU LOMF or pre-FEC SF detected on ingress of corresponding OTM2 line facility	Egress from the client port: client facility laser is shutoff (Note 1)
ODU AIS, ODU LCK, ODU OCI, OPU AIS or OPU PT Mismatch detected from the connection facing direction of a corresponding OTM2 layer facility	Egress from the client port: client facility laser is shutoff (Note 1)
LOC detected on ingress of a client facility (Note 2)	OPU AIS in the connection facing direction of the corresponding OTM2 layer facility
Client facility is put out-of-service	ODU LCK in the connection facing direction of the corresponding OTM2 layer facility
Client facility is put out-of-service	Egress from the client port: client facility laser is shutoff (Note 1)
K30.7 signal conditioning detected on ingress of a client facility	Passed transparently through the corresponding OTM2 line facility in the Tx direction
K30.7 signal conditioning detected on ingress of corresponding OTM2 line facility	Passed transparently through the corresponding client facility in the Tx direction
Note 1: The client facility laser is shutoff when LOFEF is enabled.	
Note 2: LOC is not supported on the 4x10G OTR circuit packs.	

Signal conditioning for 8xOTN Flex MOTR (NTK532BA), (1+8)xOTN Flex MOTR (NTK532DA/NTK532DE), circuit packs

The following tables list the signal conditioning for the 8xOTN Flex MOTR and (1+8)xOTN Flex MOTR circuit packs.

Table 1-53
8xOTN Flex MOTR and (1+8)xOTN Flex MOTR Tx conditioning parameters

Protocol	Tx conditioning parameter	
	Options (Note 1 to Note 4)	Default
OC3 OC12 OC48	LOFEF AIS-L	AIS-L
STM1 STM4 STM16	LOEFEF MS-AIS	MS-AIS
SRM250G ESRM250G	LOFEF AIS-L	AIS-L
1000Base-X 1000Base-T	LOEFEF 8B10B K30_7 LNKFLTC1C2	LOFEF
100Base-X	LOEFEF	LOEFEF
Fiber Channel 100 FICON	LOEFEF 8B10B NOS	NOS
Fiber Channel 200 FICON Express	LOEFEF 8B10B NOS	NOS
Fiber Channel 400 FICON 4G	LOEEF 8B10B NOS	NOS
IBM ISC-3 Peer Mode 1 Gbps and 2Gbps	LOEEF 8B10B	LOEEF

Table 1-53**8xOTN Flex MOTR and (1+8)xOTN Flex MOTR Tx conditioning parameters (continued)**

Protocol	Tx conditioning parameter	
	Options (Note 1 to Note 4)	Default
ESCON	LOEEF 8B10B	8B10B
DVB-AIS	LOEEF 8B10B	8B10B
Note 1: LOFEF is “Laser Off Far End Failure” and means the pluggable transmitter will be disabled Note 2: 8B10B is an invalid 10B codeword as specified in G.7041 (sometimes referred to as 10B_ERR) Note 3: K30_7 is the 802.3 specified Error Propagation signal /K30.7/ Note 4: NOS is the Fiber Channel defined “Not Operational” ordered SET		

Table 1-54
8xOTN Flex MOTR and (1+8)xOTN Flex MOTR Hold-off parameters

Protocol	Hold-off Parameter (in ms)					
	Range	Default	Signal output During Hold-Off	Recommended Setting for Protected configurations		
				Unprotected	1+1 Line (Note 1)	1+1 Port TPT (Note 2) and (Note 5)
OC3 OC12 OC48	0 to 1000 in 100 intervals	0	AIS-L	0	100	100
STM1 STM4 STM16		0	AIS-L	0	100	100
SRM250G ESRM250G		0	AIS-L	0	100	100
1000Base-X 1000Base-T		0	802.3 Idle	0	100	100
100Base-X		0	PN11	0	100	100
Fiber Channel 100 FICON		100 (Note 3)	Fibre Channel IDLE	100 (Note 4)	100	100
Fiber Channel 200 FICON Express		100 (Note 3)	Fiber Channel IDLE	100 (Note 4)	100	100
Fiber Channel 400 FICON 4G		100 (Note 3)	Fiber Channel IDLE	100 (Note 4)	100	100

Table 1-54
8xOTN Flex MOTR and (1+8)xOTN Flex MOTR Hold-off parameters (continued)

Protocol	Hold-off Parameter (in ms)					
	Range	Default	Signal output During Hold-Off	Recommended Setting for Protected configurations		
				Unprotected	1+1 Line (Note 1)	1+1 Port TPT (Note 2) and (Note 5)
IBM ISC-3 Peer Mode (1 Gbps and 2 Gbps)	0 to 1000 in 100 intervals	0	8B10B	0	100	100
ESCON		0	8B10B	0	100	100
DVB-AIS		0	8B10B	0	100	100

Note 1: For a 1+1 Line protected client service, holdoff is set to 100ms to allow the protection switch to complete without applying a Tx conditioning treatment to the client.

Note 2: During a protection switch for a 1+1 Port TPT protected client, there is a brief period where the transmitter is off on both ports. This may cause an interruption to be detected at the subtending device.

Note 3: The hold-off value for a FC interface is 100ms to align with FC standard R_T_TOV value in order to provide a seamless transition during short failures, such as that of a protection switch.

Note 4: Unprotected Fibre Channel hold-off should be 0ms if the subtending FC switch is providing protection.

Note 5: For a 1+1 Port TPT protected client service, the holdoff timer does not delay the system's ability to disable/enable the transmitter during a switch. The holdoff timer will only have an impact if both paths are failed -- in which case, the configured Tx Conditioning treatment is not applied until the defect lasts longer than the holdoff timer.

Table 1-55
8xOTN Flex MOTR (NTK532BA) (Port with Condition)

Trigger			Port with Condition		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Line Facility	OTM1 p1-p4	Facility Unconnected	OTM1 p1-p4	ODU1 OCI	N/A
	OTM1 p1-p4	Facility Partially Unconnected (Note 1)	OTM1 p1-p4	Unconnected TS: ODU0 OCI Connected TS: Traffic	N/A
	OTM1 p1-p4	Facility OOS	OTM1 p1-p4	ODU1 LCK	Switch to standby OTM1
	OTM1 p1-p4	LOS, LOF, LOMF, Pre-FEC SF	OTM1 p1-p4	Traffic + OTU1 BDI + ODU1 BDI	Switch to standby OTM1
	OTM1 p1-p4	Pre-FEC SD	OTM1 p1-p4	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OTU1 BDI	OTM1 p1-p4	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OTU1 TTI Mismatch	OTM1 p1-p4	Traffic + No Conditioning	No switch
	OTM1 p1-p4	ODU1 AIS, OCI, LCK	OTM1 p1-p4	Traffic + ODU1 BDI	Switch to standby OTM1
	OTM1 p1-p4	ODU1 BDI	OTM1 p1-p4	Traffic + No Conditioning	No switch
	OTM1 p1-p4	ODU1 TTI Mismatch	OTM1 p1-p4	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OPU1 PT Mismatch (Note 2)	OTM1 p1-p4	Traffic + No Conditioning	Switch to standby OTM1
	OTM1 p1-p4	OPU1 Far End Client Signal Failure	OTM1 p1-p4	Traffic + No Conditioning	Switch to standby OTM1
	OTM1 p1-p4	OPU1 AIS	OTM1 p1-p4	Traffic + No Conditioning	Switch to standby OTM1

Table 1-55
8xOTN Flex MOTR (NTK532BA) (Port with Condition) (continued)

Trigger			Port with Condition		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility	OTM1 p1-p8	Facility Unconnected	OTM1 p1-p8	No Action	N/A
	OTM1 p1-p8	Facility OOS	OTM1 p1-p8	ODU1 LCK	N/A
	OTM1 p1-p8	ODU1 AIS, OCI, LCK	OTM1 p1-p8	Traffic + ODU1 BDI	N/A
	OTM1 p1-p8	ODU1 BDI	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OTM1 p1-p8	ODU1 TTI Mismatch	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OTM1 p1-p8	OPU1 PT Mismatch	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OTM1 p1-p8	OPU1 Far End Client Signal Failure	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OTM1 p1-p8	OPU1 AIS	OTM1 p1-p8	Traffic + No Conditioning	N/A
	FLEX p1-p8	LOS, LOL, LODS	FLEX p1-p8	OPU1 AIS + CSF	N/A
	FLEX p1-p8	Rx Power OOR	FLEX p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	LOF, SF	OTM1 p1-p8	OPU1 AIS + CSF	N/A
	OC-n p1-p8	SF, SD	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	AIS-L	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	RDI-L	OTM1 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	Section Trace Mismatch	OTM1 p1-p8	Traffic + No Conditioning	N/A

Table 1-55
8xOTN Flex MOTR (NTK532BA) (Port with Condition) (continued)

Trigger			Port with Condition		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility (continued)	OTM0 p1-p8	Facility Unconnected	OTM0 p1-p8	No Action	N/A
	OTM0 p1-p8	Facility OOS	OTM0 p1-p8	ODU0 LCK	N/A
	OTM0 p1-p8	ODU0 AIS, OCI, LCK	OTM0 p1-p8	Traffic + ODU0 BDI	N/A
	OTM0 p1-p8	ODU0 BDI	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OTM0 p1-p8	ODU0 TTI Mismatch	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OTM0 p1-p8	OPU0 PT Mismatch	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OTM0 p1-p8	OPU0 Far End Client Signal Failure	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OTM0 p1-p8	OPU0 AIS	OTM0 p1-p8	Traffic + No Conditioning	N/A
	FLEX p1-p8	LOS, LOL, LODS	OTM0 p1-p8	GE: OPU0 CSF non-GE: OPU0 AIS + CSF	N/A
	FLEX p1-p8	Rx Power OOR	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	LOF, SF	OTM0 p1-p8	OPU0 AIS + CSF	N/A
	OC-n p1-p8	SF, SD	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	AIS-L	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	RDI-L	OTM0 p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	Section Trace Mismatch	OTM0 p1-p8	Traffic + No Conditioning	N/A

Table 1-55
8xOTN Flex MOTR (NTK532BA) (Port with Condition) (continued)

Trigger			Port with Condition		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility (continued)	WAN p1-p8	Loss of Frame Delineation	WAN p1-p8	Traffic + No Conditioning	N/A
	WAN p1-p8	Client Service Mismatch	WAN p1-p8	Traffic + No Conditioning	N/A
	WAN p1-p8	Far End Client Signal Failure	WAN p1-p8	N/A	N/A
	FLEX p1-p8	Facility OOS	WAN p1-p8	GFP CMF LOS (or user configured CMF)	N/A
	FLEX p1-p8	LOS, LOL, LODS	WAN p1-p8	Traffic + No Conditioning	N/A
	FLEX p1-p8	Rx Power OOR	WAN p1-p8	Traffic + No Conditioning	N/A
Client Facility	FLEX p1-p8	Facility Unconnected	FLEX p1-p8	FLEX TXCON setting	N/A
	FLEX p1-p8	Facility OOS	FLEX p1-p8	FLEX TXCON setting	N/A
	FLEX p1-p8	LOS, LOL	FLEX p1-p8	Traffic + No Conditioning	N/A
	FLEX p1-p8	Loss of Data Synch	FLEX p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	LOF	OC-n p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	SF, SD	OC-n p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	AIS-L	OC-n p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	RDI-L	OC-n p1-p8	Traffic + No Conditioning	N/A
	OC-n p1-p8	Section Trace Mismatch	OC-n p1-p8	Traffic + No Conditioning	N/A
Note 1: This means 1xODU0 connection is in place. Note 2: Not applicable if OTM1 p1-p4 is an ODU1 connected to another OTM1 p1-p4 line interface.					

Table 1-56
8xOTN Flex MOTR (NTK532BA) Connected/Associated Client Port

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection
Line Facility	OTM1 p1-p4	Facility Unconnected	N/A	N/A	N/A
	OTM1 p1-p4	Facility Partially Unconnected (Note 1)	N/A	N/A	N/A
	OTM1 p1-p4	Facility OOS	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p4	LOS, LOF, LOMF, Pre-FEC SF	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p4	Pre-FEC SD	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OTU1 BDI	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OTU1 TTI Mismatch	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p4	ODU1 AIS, OCI, LCK	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p4	ODU1 BDI	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p4	ODU1 TTI Mismatch	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p4	OPU1 PT Mismatch (Note 2)	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p4	OPU1 Far End Client Signal Failure	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p4	OPU1 AIS	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX

Table 1-56
8xOTN Flex MOTR (NTK532BA) Connected/Associated Client Port (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility	OTM1 p1-p8	Facility Unconnected	FLEX p1-p8	FLEX TXCON setting	N/A
	OTM1 p1-p8	Facility OOS	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p8	ODU1 AIS, OCI, LCK	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p8	ODU1 BDI	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	ODU1 TTI Mismatch	FLEX p1-p8	Traffic + No Conditioning	Switch to standby FLEX
	OTM1 p1-p8	OPU1 PT Mismatch	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p8	OPU1 Far End Client Signal Failure	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p1-p8	OPU1 AIS	FLEX p1-p8	Traffic + No Conditioning	No switch
	FLEX p1-p8	LOS, LOL, LODS	FLEX p1-p8	Traffic + No Conditioning	No switch
	FLEX p1-p8	Rx Power OOR	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	LOF, SF	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	SF, SD	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	AIS-L	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	RDI-L	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM1 p1-p8	Section Trace Mismatch	FLEX p1-p8	Traffic + No Conditioning	No switch

Table 1-56
8xOTN Flex MOTR (NTK532BA) Connected/Associated Client Port (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility (continued)	OTM0 p1-p8	Facility Unconnected	FLEX p1-p8	FLEX TXCON setting	N/A
	OTM0 p1-p8	Facility OOS	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p1-p8	ODU0 AIS, OCI, LCK	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p1-p8	ODU0 BDI	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	ODU0 TTI Mismatch	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	OPU0 PT Mismatch	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p1-p8	OPU0 Far End Client Signal Failure	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p1-p8	OPU0 AIS	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	FLEX p1-p8	LOS, LOL, LODS	FLEX p1-p8	Traffic + No Conditioning	No switch
	FLEX p1-p8	Rx Power OOR	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	LOF, SF	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	SF, SD	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	AIS-L	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	RDI-L	FLEX p1-p8	Traffic + No Conditioning	No switch
	OTM0 p1-p8	Section Trace Mismatch	FLEX p1-p8	Traffic + No Conditioning	No switch

Table 1-56
8xOTN Flex MOTR (NTK532BA) Connected/Associated Client Port (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility (continued)	WAN p1-p8	Loss of Frame Delineation	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	WAN p1-p8	Client Service Mismatch	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	WAN p1-p8	Far End Client Signal Failure	FLEX p1-p8	FLEX TXCON setting	Switch to standby FLEX
	FLEX p1-p8	Facility OOS	FLEX p1-p8	No action	No switch
	FLEX p1-p8	LOS, LOL, LODS	FLEX p1-p8	No action	No switch
	FLEX p1-p8	Rx Power OOR	FLEX p1-p8	No action	No switch
Client-to-Client Connection (hairpin)					
Client Facility	FLEX p1-p8	Facility Unconnected	FLEX p1-p8	FLEX TXCON setting	Not Supported
	FLEX p1-p8	Facility OOS	FLEX p1-p8	FLEX TXCON setting	Not Supported
	FLEX p1-p8	LOS, LOL	FLEX p1-p8	FLEX TXCON setting	Not Supported
	FLEX p1-p8	Loss of Data Synch	FLEX p1-p8	FLEX TXCON setting	Not Supported
	OC-n p1-p8	LOF	FLEX p1-p8	FLEX TXCON setting	Not Supported
	OC-n p1-p8	SF, SD	FLEX p1-p8	Traffic + No Conditioning	Not Supported
	OC-n p1-p8	AIS-L	FLEX p1-p8	Traffic + No Conditioning	Not Supported
	OC-n p1-p8	RDI-L	FLEX p1-p8	Traffic + No Conditioning	Not Supported
	OC-n p1-p8	Section Trace Mismatch	FLEX p1-p8	Traffic + No Conditioning	Not Supported
Note 1: This means 1xODU0 connection is in place. Note 2: Not applicable if OTM1 p1-p4 is an ODU1 connected to another OTM1 p1-p4 line interface.					

Table 1-57
8xOTN Flex MOTR (Connected Line Port)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Line Facility	OTM1 p1-p4	Facility Unconnected	N/A	N/A	N/A	N/A
	OTM1 p1-p4	Facility Partially Unconnected (Note)	N/A	N/A	N/A	N/A
	OTM1 p1-p4	Facility OOS	OTM1 p1-p4	ODU0 ODU1	ODU0 AIS ODU1 LCK	
	OTM1 p1-p4	LOS, LOF, LOMF, Pre-FEC SF	OTM1 p1-p4	ODU0 ODU1	ODU0 AIS ODU1 AIS	
	OTM1 p1-p4	Pre-FEC SD	OTM1 p1-p4	ODU0 ODU1	No Conditioning	
	OTM1 p1-p4	OTU1 BDI	OTM1 p1-p4	ODU0 ODU1	No Conditioning	
	OTM1 p1-p4	OTU1 TTI Mismatch	OTM1 p1-p4	ODU0 ODU1	No Conditioning	
	OTM1 p1-p4	ODU1 AIS, OCI, LCK	OTM1 p1-p4	ODU0 ODU1	ODU0 AIS Received signal passes through	
	OTM1 p1-p4	ODU1 BDI	OTM1 p1-p4	ODU0 ODU1	No Conditioning	
	OTM1 p1-p4	ODU1 TTI Mismatch	OTM1 p1-p4	ODU0 ODU1	N/A	
	OTM1 p1-p4	OPU1 PT Mismatch	OTM1 p1-p4	ODU0 ODU1	ODU0 AIS No Conditioning	
	OTM1 p1-p4	OPU1 Far End Client Signal Failure	OTM1 p1-p4	ODU0 ODU1	N/A	
	OTM1 p1-p4	OPU1 AIS	OTM1 p1-p4	ODU0 ODU1	N/A	

Table 1-57
8xOTN Flex MOTR (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
ODU1 Encapsulated Client connected to OTM1 line						
Client Mapping Layer Facility	OTM1 p1-p8	Facility Unconnected	N/A	N/A	N/A	N/A
	OTM1 p1-p8	Facility OOS	OTM1 p1-p4	ODU1	Signal from mapping layer	No switch
	OTM1 p1-p8	ODU1 AIS, OCI, LCK	OTM1 p1-p4	ODU1	Signal from mapping layer	No switch
	OTM1 p1-p8	ODU1 BDI	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	ODU1 TTI Mismatch	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	OPU1 PT Mismatch	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	OPU1 Far End Client Signal Failure	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	OPU1 AIS	OTM1 p1-p4	ODU1	No action	No switch
	FLEX p1-p8	LOS, LOL, LODS	OTM1 p1-p4	ODU1	Signal from mapping layer	No switch
	FLEX p1-p8	Rx Power OOR	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	LOF, SF	OTM1 p1-p4	ODU1	Signal from mapping layer	No switch
	OTM1 p1-p8	SF, SD	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	AIS-L	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	RDI-L	OTM1 p1-p4	ODU1	No action	No switch
	OTM1 p1-p8	Section Trace Mismatch	FLEX p1-p8	ODU1	No action	No switch

Table 1-57
8xOTN Flex MOTR (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
ODU0 Encapsulated Client connected to OTM1 line						
Client Mapping Layer Facility	OTM0 p1-p8	Facility Unconnected	N/A	N/A	N/A	N/A
	OTM0 p1-p8	Facility OOS	OTM1 p1-p4	ODU0	Signal from mapping layer	No switch
	OTM0 p1-p8	ODU0 AIS, OCI, LCK	OTM1 p1-p4	ODU0	Signal from mapping layer	No switch
	OTM0 p1-p8	ODU0 BDI	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	ODU0 TTI Mismatch	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	OPU0 PT Mismatch	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	OPU0 Far End Client Signal Failure	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	OPU0 AIS	OTM1 p1-p4	ODU0	No action	No switch
	FLEX p1-p8	LOS, LOL, LODS	OTM1 p1-p4	ODU0	Signal from mapping layer	No switch
	FLEX p1-p8	Rx Power OOR	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	LOF, SF	OTM1 p1-p4	ODU0	Signal from mapping layer	No switch
	OTM0 p1-p8	SF, SD	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	AIS-L	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	RDI-L	OTM1 p1-p4	ODU0	No action	No switch
	OTM0 p1-p8	Section Trace Mismatch	OTM1 p1-p4	ODU0	No action	No switch
	WAN p1-p8	Loss of Frame Delineation	OTM1 p1-p4	ODU0	No action	No switch
	WAN p1-p8	Client Service Mismatch	OTM1 p1-p4	ODU0	No action	No switch

Table 1-57
8xOTN Flex MOTR (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Client Mapping Layer Facility (continued)	WAN p1-p8	Far End Client Signal Failure	OTM1 p1-p4	ODU0	No action	No switch
	FLEX p1-p8	Facility OOS	OTM1 p1-p4	ODU0	No action	No switch
	FLEX p1-p8	LOS, LOL, LODS	OTM1 p1-p4	ODU0	No action	No switch
	FLEX p1-p8	Rx Power OOR	OTM1 p1-p4	ODU0	No action	No switch
Note: 1+1 Port TPT Protection is not supported.						

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Line Facility	OTM2 p1	Facility Unconnected	OTM2 p1	ODU2 OCI	Not Supported
	OTM2 p1	Facility Partially Unconnected (PT= 0x20)	OTM2 p1	Unconnected TS: ODU1 OCI Connected TS: Traffic	Not Supported
	OTM2 p1	Facility Partially Unconnected (PT= 0x21)	OTM2 p1	Unconnected TS: MSI=Unused Connected TS: Traffic	Not Supported
	OTM2 p1	Facility OOS	OTM2 p1	ODU2 LCK	Not Supported
	OTM2 p1	LOS, LOF, LOMF, Pre-FEC SF	OTM2 p1	Traffic+ OTU2 BDI + ODU2 BDI	Not Supported

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Line Facility (continued)	OTM2 p1	Pre-FEC SD	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	ODUk Post-FEC SD	OTM2 p1	ODU2 SD	Not Supported
	OTM2 p1	OTU2 BDI	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	OTU2 TTI Mismatch	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	ODU2 AIS, OCI, LCK	OTM2 p1	Traffic + ODU2 BDI	Not Supported
	OTM2 p1	OTU2 BDI	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	OTU2 TTI Mismatch	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	OPU2 PT Mismatch	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	OPU2 Far End Client Signal Failure	OTM2 p1	Traffic + No Conditioning	Not Supported
	OTM2 p1	OPU2 AIS	OTM2 p1	Traffic + No Conditioning	Not Supported

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Line Facility (continued)	OTM1 p2-p9	Facility Unconnected	OTM1 p2-p9	ODU1 OCI	N/A
	OTM1 p2-p9	Facility Partially Unconnected	OTM1 p2-p9	Unconnected TS: ODU0 OCI Connected TS: Traffic	N/A
	OTM1 p2-p9	Facility OOS	OTM1 p2-p9	ODU1 LCK	Switch to standby OTM1
	OTM1 p2-p9	LOS, LOF, LOMF, Pre-FEC SF	OTM1 p2-p9	Traffic + OTU1 BDI + ODU1 BDI	Switch to standby OTM1
	OTM1 p2-p9	Pre-FEC SD	OTM1 p2-p9	Traffic + No Conditioning	No switch
	OTM1 p2-p9	OTU1 BDI	OTM1 p2-p9	Traffic + No Conditioning	No switch
	OTM1 p2-p9	OTU1 TTI Mismatch	OTM1 p2-p9	Traffic + No Conditioning	No switch
	OTM1 p2-p9	ODU1 AIS, OCI, LCK	OTM1 p2-p9	Traffic + ODU1 BDI	Switch to standby OTM1
	OTM1 p2-p9	ODU1 BDI	OTM1 p2-p9	Traffic + No Conditioning	No switch
	OTM1 p2-p9	ODU1 TTI Mismatch	OTM1 p2-p9	Traffic + No Conditioning	No switch
	OTM1 p2-p9	OPU1 PT Mismatch	OTM1 p2-p9	Traffic + No Conditioning	Switch to standby OTM1
	OTM1 p2-p9	OPU1 Far End Client Signal Failure	OTM1 p2-p9	Traffic + No Conditioning	Switch to standby OTM1
	OTM1 p2-p9	OPU1 AIS	OTM1 p2-p9	Traffic + No Conditioning	Switch to standby OTM1

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility	OTMFLEX p2-p9	Facility Unconnected	OTMFLEX p2-p9	No Action	N/A
	OTMFLEX p2-p9	Facility OOS	OTMFLEX p2-p9	ODUflex LCK	N/A
	OTMFLEX p2-p9	ODUflex AIS, OCI, LCK, Loss of Frame and Multiframe	OTMFLEX p2-p9	Traffic+ ODUflex BDI	N/A
	OTMFLEX p2-p9	ODUflex BDI	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTMFLEX p2-p9	ODUflex TTI Mismatch	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTMFLEX p2-p9	ODUflex PT Mismatch	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTMFLEX p2-p9	OPUflex AIS	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTMFLEX p2-p9	MSI Mismatch	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTMFLEX p2-p9	MSI Mismatch	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	FLEX p2-p9	LOS, LOL, LODS	OTMFLEX p2-p9	OPUflex AIS+CSF	N/A
	FLEX p2-p9	Rx Power OOR	OTMFLEX p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	Facility Unconnected	OTM1 p2-p9	No action	N/A
	OTM1 p2-p9	Facility OOS	OTM1 p2-p9	ODU1 LCK	N/A
	OTM1 p2-p9	ODU1 AIS, OCI, LCK, Loss of Frame and Multiframe	OTM1 p2-p9	Traffic + ODU1 BDI	N/A

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility (continued)	OTM1 p2-p9	ODU1 BDI	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	ODU1 TTI Mismatch	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	OPU1 PT Mismatch	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	OPU1 Far End Client Signal Failure	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	OPU1 AIS	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OTM1 p2-p9	MSI Mismatch	OTM1 p2-p9	Traffic + No Conditioning	N/A
	FLEX p2-p9	LOS, LOL, LODS	OTM1 p2-p9	OPU1 AIS + CSF	N/A
	FLEX p2-p9	Rx Power OOR	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	LOF, SF	OTM1 p2-p9	OPU1 AIS + CSF	N/A
	OC-n p2-p9	SF, SD	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	AIS-L	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	RDI-L	OTM1 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	Section Trace Mismatch	OTM1 p2-p9	Traffic + No Conditioning	N/A

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility (continued)	OTM0 p2-p9	Facility Unconnected	OTM0 p2-p9	No action	N/A
	OTM0 p2-p9	Facility OOS	OTM0 p2-p9	ODU0 LCK	N/A
	OTM0 p2-p9	ODU0 AIS, OCI, LCK, Loss of Frame and Multiframe	OTM0 p2-p9	Traffic + ODU0 BDI	N/A
	OTM0 p2-p9	ODU0 BDI	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OTM0 p2-p9	ODU0 TTI Mismatch	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OTM0 p2-p9	OPU0 PT Mismatch	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OTM0 p2-p9	OPU0 Far End Client Signal Failure	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OTM0 p2-p9	OPU0 AIS	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OTM0 p2-p9	MSI Mismatch	OTM0 p2-p9	Traffic + No Conditioning	N/A
	FLEX p2-p9	LOS, LOL, LODS	OTM0 p2-p9	OPU0 AIS + CSF	N/A
	FLEX p2-p9	Rx Power OOR	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	LOF, SF	OTM0 p2-p9	OPU0 AIS + CSF	N/A
	OC-n p2-p9	SF, SD	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	AIS-L	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	RDI-L	OTM0 p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	Section Trace Mismatch	OTM0 p2-p9	Traffic + No Conditioning	N/A

Table 1-58
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) Signal Conditioning (continued)

Trigger			Signal Conditioning		
Facility		Condition	Facility	Signal output (toward line)	1+1 Line Protection
Client Mapping Layer Facility (continued)	WAN p2-p9	Loss of Frame Delineation	WAN p2-p9	Traffic + No Conditioning	N/A
	WAN p2-p9	Client Service Mismatch	WAN p2-p9	Traffic + No Conditioning	N/A
	WAN p2-p9	Far End Client Signal Failure	WAN p2-p9	Traffic + No Conditioning	N/A
	FLEX p2-p9	Facility OOS	WAN p2-p9	N/A	N/A
	FLEX p2-p9	LOS, LOL, LODS	WAN p2-p9	GFP CMF LOS (or user configured CMF)	N/A
	FLEX p2-p9	Rx Power OOR	WAN p2-p9	Traffic + No Conditioning	N/A
Client Facility	FLEX p2-p9	Facility Unconnected	FLEX p2-p9	FLEX TXCON setting	N/A
	FLEX p2-p9	Facility OOS	FLEX p2-p9	FLEX TXCON setting	N/A
	FLEX p2-p9	LOS, LOL, LODS	FLEX p2-p9	Traffic + No Conditioning	N/A
	FLEX p2-p9	Loss of Data Synch	FLEX p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	LOF	OC-n p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	SF, SD	OC-n p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	AIS-L	OC-n p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	RDI-L	OC-n p2-p9	Traffic + No Conditioning	N/A
	OC-n p2-p9	Section Trace Mismatch	OC-n p2-p9	Traffic + No Conditioning	N/A

Table 1-59**(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port)**

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Line Facility	OTM2 p1	Facility Unconnected	N/A	N/A	N/A
	OTM2 p1	Facility Partially Unconnected (PT= 0x20)	N/A	N/A	N/A
	OTM2 p1	Facility Partially Unconnected (PT= 0x21)	N/A	N/A	N/A
	OTM2 p1	Facility OOS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM2 p1	LOS, LOF, LOMF, Pre-FEC SF	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM2 p1	Pre-FEC SD	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM2 p1	OTU2 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM2 p1	OTU2 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM2 p1	ODU2 AIS, OCI, LCK	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM2 p1	OTU2 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM2 p1	OTU2 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM2 p1	OPU2 PT Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM2 p1	OPU2 Far End Client Signal Failure	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX

Table 1-59
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port) (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Line Facility (continued)	OTM2 p1	OPU2 AIS	FLEX p2-p9	Traffic + No Conditioning	Switch to standby FLEX
	OTM1 p2-p9	Facility Unconnected	N/A	N/A	N/A
	OTM1 p2-p9	Facility Partially Unconnected	N/A	N/A	N/A
	OTM1 p2-p9	Facility OOS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	LOS, LOF, LOMF, Pre-FEC SF	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	Pre-FEC SD	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	OTU1 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	OTU1 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	ODU1 AIS, OCI, LCK	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	ODU1 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	ODU1 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	OPU1 PT Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	OPU1 Far End Client Signal Failure	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	OPU1 AIS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX

Table 1-59**(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port) (continued)**

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility	OTMFLEX p2-p9	Facility Unconnected	FLEX p2-p9	FLEX TXCON setting	N/A
	OTMFLEX p2-p9	Facility OOS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTMFLEX p2-p9	ODUflex AIS, OCI, LCK	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTMFLEX p2-p9	ODUflex BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTMFLEX p2-p9	ODUflex TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTMFLEX p2-p9	ODUflex PT Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTMFLEX p2-p9	OPUflex AIS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTMFLEX p2-p9	MSI Mismatch	FLEX p2-p9	FLEX TXCON settingS	Switch to standby FLEX
	OTMFLEX p2-p9	MSI Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	FLEX p2-p9	LOS, LOL, LODS	FLEX p2-p9	Traffic + No Conditioning	No Switch
	FLEX p2-p9	Rx Power OOR	FLEX p2-p9	Traffic + No Conditioning	No Switch

Table 1-59
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port) (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility (continued)	OTM1 p2-p9	Facility Unconnected	FLEX p2-p9	FLEX TXCON setting	N/A
	OTM1 p2-p9	Facility OOS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	ODU1 AIS, OCI, LCK	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	ODU1 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	ODU1 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM1 p2-p9	OPU1 PT Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	OPU1 Far End Client Signal Failure	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	OPU1 AIS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM1 p2-p9	MSI Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	FLEX p2-p9	LOS, LOL, LODS	FLEX p2-p9	Traffic + No Conditioning	No Switch
	FLEX p2-p9	Rx Power OOR	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	LOF, SF	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	SF, SD	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	AIS-L	FLEX p2-p9	Traffic + No Conditioning	No Switch

Table 1-59**(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port) (continued)**

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Client Mapping Layer Facility (continued)	OC-n p2-p9	RDI-L	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	Section Trace Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility	OTM0 p2-p9	Facility Unconnected	FLEX p2-p9	FLEX TXCON setting	N/A
	OTM0 p2-p9	Facility OOS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p2-p9	ODU0 AIS, OCI, LCK	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p2-p9	ODU0 BDI	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM0 p2-p9	ODU0 TTI Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OTM0 p2-p9	OPU0 PT Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p2-p9	OPU0 Far End Client Signal Failure	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p2-p9	OPU0 AIS	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	OTM0 p2-p9	MSI Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	FLEX p2-p9	LOS, LOL, LODS	FLEX p2-p9	OPU0 AIS + CSF	No Switch
	FLEX p2-p9	Rx Power OOR	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	LOF, SF	FLEX p2-p9	Traffic + No Conditioning	No Switch

Table 1-59
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected/Associated Client Port) (continued)

Trigger			Connected/Associated Client Port		
Facility		Condition	Facility	Treatment	1+1 Port TPT Protection (w/ 1+1 Line)
Client Mapping Layer Facility (continued)	OC-n p2-p9	SF, SD	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	AIS-L	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	RDI-L	FLEX p2-p9	Traffic + No Conditioning	No Switch
	OC-n p2-p9	Section Trace Mismatch	FLEX p2-p9	Traffic + No Conditioning	No Switch
Client Facility associated with this mapping layer facility					
Client Mapping Layer Facility	WAN p2-p9	Loss of Frame Delineation	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	WAN p2-p9	Client Service Mismatch	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	WAN p2-p9	Far End Client Signal Failure	FLEX p2-p9	FLEX TXCON setting	Switch to standby FLEX
	FLEX p2-p9	Facility OOS	FLEX p2-p9	No Action	No Switch
	FLEX p2-p9	LOS, LOL, LODS	FLEX p2-p9	No Action	No Switch
	FLEX p2-p9	Rx Power OOR	FLEX p2-p9	No Action	No Switch

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Line Facility	OTM2 p1	Facility Unconnected	N/A	N/A	N/A	Not Supported
	OTM2 p1	Facility Partially Unconnected (PT= 0x20)	N/A	N/A	N/A	Not Supported
	OTM2 p1	Facility Partially Unconnected (PT= 0x21)	N/A	N/A	N/A	Not Supported
	OTM2 p1	Facility OOS	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	ODU0 AIS ODU1 AIS ODUflex AIS Not Supported	Not Supported
	OTM2 p1	LOS, LOF, LOMF, Pre-FEC SF	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	ODU0 AIS ODU1 AIS ODUflex AIS Not Supported	Not Supported
	OTM2 p1	Pre-FEC SD	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	No Conditioning No Conditioning No Conditioning Not Supported	Not Supported
	OTM2 p1	OTU2 BDI	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	No Conditioning No Conditioning No Conditioning Not Supported	Not Supported
	OTM2 p1	OTU2 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	No Conditioning No Conditioning No Conditioning Not Supported	Not Supported

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Line Facility (continued)	OTM2 p1	ODU2 AIS, OCI, LCK	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	ODU0 AIS ODU1 AIS ODUflex AIS Not Supported	Not Supported
	OTM2 p1	OTU2 BDI	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	No Conditioning No Conditioning No Conditioning Not Supported	Not Supported
	OTM2 p1	OTU2 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	No Conditioning No Conditioning No Conditioning Not Supported	Not Supported
	OTM2 p1	OPU2 PT Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	ODU0 AIS ODU1 AIS ODUflex AIS Not Supported	Not Supported
	OTM2 p1	OPU2 Far End Client Signal Failure	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	N/A N/A N/A Not Supported	Not Supported
	OTM2 p1	OPU2 AIS	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	N/A N/A N/A Not Supported	Not Supported
	OTM2 p1	Service Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1 ODUflex ODU2	ODU0 AIS ODU1 AIS ODUflex AIS Not Supported	Not Supported
	OTM1 p2-p9	Facility Unconnected	N/A	N/A	N/A	N/A

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Line Facility (continued)	OTM1 p2-p9	Facility Partially Unconnected	N/A	N/A	N/A	N/A
	OTM1 p2-p9	Facility OOS	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	ODU0 AIS ODU1 LCK	Not Supported
	OTM1 p2-p9	LOS, LOF, LOMF, Pre-FEC SF	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	ODU0 AIS ODU1 AIS	Not Supported
	OTM1 p2-p9	Pre-FEC SD	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	No Conditioning	Not Supported
	OTM1 p2-p9	OTU1 BDI	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	No Conditioning	Not Supported
	OTM1 p2-p9	OTU1 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	No Conditioning	Not Supported
	OTM1 p2-p9	ODU1 AIS, OCI, LCK	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	ODU0 AIS Rx signal passes through	Not Supported
	OTM1 p2-p9	ODU1 BDI	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	No Conditioning	Not Supported
	OTM1 p2-p9	ODU1 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	N/A	N/A
	OTM1 p2-p9	OPU1 PT Mismatch	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	ODU0 AIS No Conditioning	Not Supported
	OTM1 p2-p9	OPU1 Far End Client Signal Failure	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	N/A	N/Ad
	OTM1 p2-p9	OPU1 AIS	OTM1 p2-p9 OTM2 p1	ODU0 ODU1	N/A	N/A

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
ODUflex Encapsulated Client connected to OTM1/OTM2 line						
Client Mapping Layer Facility (continued)	OTMFLEX p2-p9	Facility Unconnected	N/A	N/A	N/A	N/A
	OTMFLEX p2-p9	Facility OOS	OTM1 p2-p9 OTM2 p1	ODUflex	Signal from mapping layer	Not Supported
	OTMFLEX p2-p9	ODUflex AIS, OCI, LCK	OTM1 p2-p9 OTM2 p1	ODUflex	Signal from mapping layer	Not Supported
	OTMFLEX p2-p9	ODUflex BDI	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	OTMFLEX p2-p9	ODUflex TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	OTMFLEX p2-p9	ODUflex PT Mismatch	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	OTMFLEX p2-p9	OPUflex AIS	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	OTMFLEX p2-p9	MSI Mismatch	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	OTMFLEX p2-p9	MSI Mismatch	OTM1 p2-p9 OTM2 p1	ODUflex	No Action	Not Supported
	FLEX p2-p9	LOS, LOL, LODS	OTM1 p2-p9 OTM2 p1	ODUflex	Signal from mapping layer	Not Supported
	FLEX p2-p9	Rx Power OOR	OTM1 p2-p9 OTM2 p1	ODUflex	Signal from mapping layer	Not Supported

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
ODU1 Encapsulated Client connected to OTM1/OTM2 line						
Client Mapping Layer Facility	OTM1 p2-p9	Facility Unconnected	N/A	N/A	N/A	N/A
	OTM1 p2-p9	Facility OOS	OTM1 p2-p9 OTM2 p1	ODU1	Signal from mapping layer	Not Supported
	OTM1 p2-p9	ODU1 AIS, OCI, LCK	OTM1 p2-p9 OTM2 p1	ODU1	Signal from mapping layer	Not Supported
	OTM1 p2-p9	ODU1 BDI	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OTM1 p2-p9	ODU1 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OTM1 p2-p9	OPU1 PT Mismatch	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Client Mapping Layer Facility (continued)	OTM1 p2-p9	OPU1 Far End Client Signal Failure	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OTM1 p2-p9	OPU1 AIS	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OTM1 p2-p9	MSI Mismatch	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	FLEX p2-p9	LOS, LOL, LODS	OTM1 p2-p9 OTM2 p1	ODU1	Signal from mapping layer	Not Supported
	FLEX p2-p9	Rx Power OOR	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OC-n p2-p9	LOF, SF	OTM1 p2-p9 OTM2 p1	ODU1	Signal from mapping layer	Not Supported
	OC-n p2-p9	SF, SD	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OC-n p2-p9	AIS-L	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OC-n p2-p9	RDI-L	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
	OC-n p2-p9	Section Trace Mismatch	OTM1 p2-p9 OTM2 p1	ODU1	No Action	Not Supported
ODU0 Encapsulated Client connected to OTM1/OTM2 line						
Client Mapping Layer Facility	OTM0 p2-p9	Facility Unconnected	N/A	N/A	N/A	N/A
	OTM0 p2-p9	Facility OOS	OTM1 p2-p9 OTM2 p1	ODU0	Signal from mapping layer	Not Supported
	OTM0 p2-p9	ODU0 AIS, OCI, LCK	OTM1 p2-p9 OTM2 p1	ODU0	Signal from mapping layer	Not Supported

Table 1-60**(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)**

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Client Mapping Layer Facility (continued)	OTM0 p2-p9	ODU0 BDI	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OTM0 p2-p9	ODU0 TTI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OTM0 p2-p9	OPU0 PT Mismatch	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OTM0 p2-p9	OPU0 Far End Client Signal Failure	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OTM0 p2-p9	OPU0 AIS	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OTM0 p2-p9	MSI Mismatch	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	FLEX p2-p9	LOS, LOL, LODS	OTM1 p2-p9 OTM2 p1	ODU0	Signal from mapping layer	Not Supported
	FLEX p2-p9	Rx Power OOR	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OC-n p2-p9	LOF, SF	OTM1 p2-p9 OTM2 p1	ODU0	Signal from mapping layer	Not Supported
	OC-n p2-p9	SF, SD	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OC-n p2-p9	AIS-L	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OC-n p2-p9	RDI-L	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	OC-n p2-p9	Section Trace Mismatch	OTM1 p2-p9 OTM2 p1	ODU0	No Action	Not Supported
	WAN p2-p9	Loss of Frame Delineation	OTM1 p2-p9	ODU0	No Action	Not Supported

Table 1-60
(1+8)xOTN Flex MOTR (NTK532DA/NTK532DE) (Connected Line Port) (continued)

Trigger			Connected Line Port			
Facility		Condition	Facility	Connection Type	Treatment	1+1 Line Protection
Client Mapping Layer Facility (continued)	WAN p2-p9	Client Service Mismatch	OTM1 p2-p9	ODU0	No Action	Not Supported
	WAN p2-p9	Far End Client Signal Failure	OTM1 p2-p9	ODU0	No Action	Not Supported
	FLEX p2-p9	Facility OOS	OTM1 p2-p9	ODU0	No Action	Not Supported
	FLEX p2-p9	LOS, LOL, LODS	OTM1 p2-p9	ODU0	No Action	Not Supported
	FLEX p2-p9	Rx Power OOR	OTM1 p2-p9	ODU0	No Action	Not Supported

Signal conditioning for 2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) circuit packs

The following tables list the signal conditioning for the 2xCFP2 OTN Flex MOTR (NTK532GA) and 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) circuit packs

Table 1-61

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) Tx conditioning parameters

Conditioning Type	Values	Conditioning Signal	Notes
PTP_CONDDTYPE	None	Does not turn the laser off	
	LASEROFF	Turns the laser off	
ETTP_CONDDTYPE	8B10B	10B_ERR	Applies to ETH1G
	K30_7	K30_7	Applies to ETH1G
	LF_RF	Local Fault	Applies to ETH10G
WAN_CONDDTYPE	GFPCMF	ITU-T G.7041 GFP Client Management Frames	Applies to ETH1G, ETH10G
	GFPCMFUSER	GFP CMF UPI value within GFP Client Management Frames	Applies to ETH1G
	GFP10BERR	10B_ERR in GFP frames	Applies to ETH1G
CTP_CONDDTYPE	OPUK_AIS	OPU AIS (Generic AIS) and assert OPU CSF	Applies to OCn, STMn
	OPUK_CSF	Client specific replacement signal and assert OPU CSF	Applies to ETH1G, ETH10G
	OPUK_AIS	ODU AIS	Applies to ETH1G, ETH10G, OCn, STMn

Table 1-62

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—PTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	PTP_CONDDTYPE	Not applicable	
Terminal Loopback	Laser off	Not applicable	Occurs when OTU TTP, ETTP or STTP is in terminal loopback
Facility Loopback	None	Not applicable	Occurs when OTU TTP, ETTP or STTP is in facility loopback
OOS-MA	PTP_CONDDTYPE	Not applicable	
Child ODU term CTP OOS-MA	PTP_CONDDTYPE	Not applicable	
No Cross Connections on child ODU term-CTP	Laser off	Not applicable	Occurs when child is ETTP or STTP
No child ODU CTP	PTP_CONDDTYPE	Not applicable	No child ODU CTPs present
Loss of Signal	None	Send Faceplate Defect indication to children	
Receive at least one Fabric Defect indication from children	PTP_CONDDTYPE	Not applicable	Could be WAN defect, ODU defect or ODU term-CTP defect

Table 1-63

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—OTUTTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	None	Not applicable	
Loopback	None	Not applicable	Clears any active conditioning when either a facility or terminal loopback is active
OOS-MA	None	Not applicable	

Table 1-63

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—OTUTTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
Faceplate Signal Fail	OTU BDI	Send Faceplate Defect indication to children	Faceplate Signal Fail defects are: OTU LOF, LOC, LOM, SF, TIM, AIS
OTU IAE	OTU BIAE	Not applicable	
OTU Signal Degrade	OTU BEI	Not applicable	No child ODU CTP provisioned

Table 1-64

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—ETTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ETTP_CONDTYPE	Not applicable	
Loopback	None	Not applicable	Clears any active hold timer or conditioning when either a facility or terminal loopback is active
OOS-MA	ETTP_CONDTYPE	Not applicable	
Child ODU term CTP OOS-MA	ETTP_CONDTYPE	Not applicable	
Faceplate Signal Fail	None	Send Faceplate Defect indication to children	Faceplate Signal Fail defects are: LOS, LOC, LODS, Link Down, EER
Receive at least one Fabric Defect from children (WAN or ODU term-CTP)	Holdoff/ETTP_CONDTYPE	Not applicable	ETH IDLE conditioning during holdoff period

Table 1-65

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—STTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	AIS-L	Not applicable	
Loopback	None	Not applicable	Clears any active conditioning when either a facility or terminal loopback is active
OOS-MA	AIS-L	Not applicable	
Child ODU term CTP OOS-MA	AIS-L	Not applicable	
Faceplate Signal Fail	None	Send Faceplate Defect indication to children	Faceplate Signal Fail defects are: LOS, LOF, LOC, SF/B2 BER
Receive at least one Fabric Defect from children (ODU term-CTP)	AIS-L	Not applicable	

Table 1-66

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—ODUTTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ODU AIS	Not applicable	
OOS-MA	ODU LCK	Not applicable	
Receive Fabric Defect Indication from parents	ODU BDI	Not applicable	
Faceplate ODU Signal Fail	ODU BDI	Send Faceplate Defect indication to children	Faceplate ODU Signal Fail Defects are: ODU LCK, OCI, AIS, LOF, TIM
Faceplate OPU MSIM	ODU BDI	Send Faceplate Defect indication to children	Detected at Higher-Order ODU TTP
Faceplate OPU Signal Fail	None	Send Faceplate Defect indication to children	Faceplate ODU Signal Fail Defects are: OPU PTM, MSIM, CSF, AIS

Table 1-67

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—WAN facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	None	
OOS-MA	Not applicable	None	
Receive Faceplate Defect indication from parents	Not applicable	WAN_CONDTYPE	
Fabric FESF	Send Fabric Defect indication to parents (ETTP & PTP)	None	
Fabric CSM	Send Fabric Defect indication to parents (ETTP & PTP)	None	
Fabric LOFD	Send Fabric Defect indication to parents (ETTP & PTP)	None	

Table 1-68

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—ODU monitor-CTP, ODU transparent-CTP facilities

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ODU AIS	ODU AIS	
OOS-MA	ODU LCK	ODU LCK	
OPU MSIM	None	ODU AIS	Detected at Higher-Order ODU TTP
Receive Faceplate Defect indication from parents	ODU BDI	ODU AIS	

Table 1-68

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—ODU monitor-CTP, ODU transparent-CTP facilities

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
No cross Connect	ODU OCI	None	
Fabric ODU defect	None	None	ODU defects are passed from internal fabric to faceplate
Fabric OPU defect	None	None	ODU defects are passed from internal fabric to faceplate

Table 1-69

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—ODU terminated-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	ODU AIS	
OOS-MA	Not applicable	ODU LCK	
Receive Faceplate Defect indication from parents	Not applicable	CTP_CONDTYPE	
No cross Connect	None	ODU OCI	
Fabric ODU defect	Send Fabric Defect indication to parents	None	ODU defects are: ODU LCK, OCI, AIS, LOF, TIM
Fabric OPU defect	Send Fabric Defect indication to parents	None	ODU defects are: OPU PTM, MSIM, CSF, AIS

Table 1-70

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—TCM TTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	TCM AIS	Not applicable	
OOS-MA	TCM LCK	Not applicable	
Receive Faceplate Defect indication from parents	TCM BDI	Not applicable	
No Cross connect on sibling ODU	TCM OCI	Not applicable	
Faceplate TCM defect	TCM BDI	Not applicable	TCM defects are: TCM AIS, LCK, OCI, LTC, LOF, TIM
Faceplate TCM IAE	TCM BIAE	Not applicable	
Faceplate TCM Signal Degrade	TCM BEI	Not applicable	

Table 1-71

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—TCM monitor-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	TCM AIS	TCM AIS	
OOS	TCM LCK	TCM LCK	
Receive Faceplate Defect indication from parents	None	None	
No Cross connect on sibling ODU	TCM OCI	TCM OCI	
Faceplate TCM defect	None	None	TCM defects are: TCM AIS, LCK, OCI, LTC, LOF, TIM - all defects are passed through from faceplate to internal fabric
Faceplate TCM IAE	None	None	Passed through from faceplate to internal fabric

Table 1-72

2xCFP2 OTN Flex MOTR (NTK532GA), 2xQSFP28 OTN Flex MOTR (NTK532GN), 2x100G MOTR (NTK537BA), and 200G MOTR (NTK537CA) signal conditioning—TCM terminated-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	TCM AIS	
OOS	Not applicable	TCM LCK	
Receive Faceplate Defect indication from parents	Not applicable	TCM AIS	
No Cross Connect on sibling ODU	Not applicable	TCM OCI	
Fabric TCM Defect	Not applicable	TCM BDI	TCM defects are: TCM AIS, LCK, OCI, LTC, LOF, TIM
Fabric TCM IAE	Not applicable	TCM BIAE, TCM BEI	
Fabric TCM Signal Degrade	Not applicable	TCM BIAE, TCM BEI	

Signal conditioning for 100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F, and 40G OTN XCIF circuit packs

ATTENTION

No defects and no conditioning are applicable to the FTTP facility object.

The following signal conditioning is applied for 16xFLEX OTN I/F, and 40G OTN XCIF triggered events.

The 10x10G PKT/OTN I/F circuit pack does not raise an “OPU AIS” alarm when interworking with 40G MUX OCI, 10x10G Mux, or 2x10G OTR circuit packs through an OC192/STM64 client. You must change the conditioning from OPU AIS to ODU AIS on these circuit packs.

Also, the 10x10G PKT/OTN I/F circuit pack does not raise an OPU AIS alarm when interworking with 40G MUX OCI, 10x10G Mux, or 2x10G OTR circuit packs through a 10GE client and the mapping mode is:

10.7G - GFP/OPU2+7 (Preamble/ Ordered Set/ MAC transparent)

You must set the conditioning type to ODU AIS (versus the default OPU AIS) on these circuit packs.

The OPU AIS alarm is raised when the 10x10G PKT/OTN I/F circuit pack is interworking with 40G MUX OCI, 10x10G Mux, or 2x10G OTR circuit packs through a 10GE client and the mapping mode is:

- 11.09G - OPU2e (PCS transparent)
- or
- 10.7G - GFP/OPU2 (Standard MAC transparent)

No conditioning changes are required in this case.

Table 1-73

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F signal conditioning—PTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	PTP_CONDDTYPE	Not applicable	
Terminal Loopback	Laser off	Not applicable	Occurs when OTU TTP, ETTP or STTP is in terminal loopback
Facility Loopback	None (Laser on)	Not applicable	Occurs when OTU TTP, ETTP or STTP is in facility loopback
OOS-MA	PTP_CONDDTYPE	Not applicable	
Child ODU term CTP OOS-MA	PTP_CONDDTYPE	Not applicable	
No Cross Connections on child ODU term-CTP	Laser off	Not applicable	PTP service type of ETTP, STTP, etc. (non-OTU)
	None (no conditioning as to not interfere with GCC0 (OTU) and GCC 1 & 2 (ODU))	Not applicable	PTP service type of OTU
No CTP	PTP_CONDDTYPE	Not applicable	No child ODU CTPs present
Loss of Signal	None	Send Faceplate Defect indication to children	
XC Intercard Fail	PTP_CONDDTYPE	Not applicable	
Receive at least one Fabric Defect indication from children	PTP_CONDDTYPE	Not applicable	WAN defect, Rx ODU defect on ODU term-CTP only

Table 1-74

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—OTUTTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	None	Not applicable	
Loopback	None	Not applicable	Clears active conditioning when a facility or terminal loopback is active
OOS-MA	None	Not applicable	
Faceplate Signal Fail	OTU BDI	Send Faceplate Defect indication to children	OTU LOS, LOF, LOC, LOM, SF, TIM, AIS
OTU IAE	OTU BIAE	Not applicable	
OTU Signal Degrade	OTU BEI	Not applicable	

Table 1-75

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—ETTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ETTP_CONDTYPE	Not applicable	
Loopback	None	Not applicable	Clears any active hold timer or conditioning when a facility or terminal loopback is active
OOS-MA	ETTP_CONDTYPE	Not applicable	
Child ODU term CTP OOS-MA	ETTP_CONDTYPE	Not applicable	
XC Intercard Fail	Holdoff/ETTP_CONDTYPE	Not applicable	ETH IDLE conditioning during holdoff period
Faceplate Signal Fail	None	Send Faceplate Defect indication to children	LOS, LOC, LODS, Link Down, EER
Receive at least one Fabric Defect from children (WAN or ODU term-CTP)	Holdoff/ETTP_CONDTYPE	Not applicable	ETH IDLE conditioning during holdoff period

Table 1-76
100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—STTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	AIS-L	Not applicable	
Loopback	None	Not applicable	Clears active conditioning when a facility or terminal loopback is active
OOS-MA	AIS-L	Not applicable	
Child ODU term CTP OOS-MA		Not applicable	
XC Intercard Fail	AIS-L	Not applicable	
Faceplate Signal Fail	AIS-L	Send Faceplate Defect indication to children	LOS, LOF, LOC, SF/B2 BER
Receive at least one Fabric Defect from children (WAN or ODU term-CTP)	AIS-L	Not applicable	

Table 1-77
100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—ODUTTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ODU AIS	Not applicable	
Loopback	None	Not applicable	Clears active conditioning when a facility or terminal loopback is active
OOS-MA	ODU LCK	Not applicable	
XC Intercard Fail	ODU AIS	Not applicable	
PST = USPC	None	Not applicable	Placeholder ODU TTP until ODU TTP/CTP is provisioned by end user
Receive Faceplate Defect from parents	ODU BDI	Not applicable	

Table 1-77

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—ODUTTP facility (continued)

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
Faceplate ODU Signal Fail	ODU BDI	Send Faceplate Defect indication to children	Rx ODU LCK, OCI, AIS, LOF, TIM
Faceplate OPU MSIM	ODU BDI	Send Faceplate Defect indication to children	Detected at Higher-Order ODU TTP
Faceplate OPU Signal Fail	None	Send Faceplate Defect indication to children	Rx OPU PTM, MSIM, CSF, AIS

Table 1-78

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—WAN facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	None	
OOS-MA	Not applicable	None	
Terminal Loopback	Not applicable	None	Clears active conditioning when a facility or terminal loopback is active
Facility Loopback	Not applicable	WAN_CONDTYPE	
Receive Faceplate Defect from parent	Not applicable	WAN_CONDTYPE	
Fabric FESF	Send Fabric Defect indication to parents (ETTP & PTP)	None	
Fabric CSM	Send Fabric Defect indication to parents (ETTP & PTP)	None	
Fabric LOFD	Send Fabric Defect indication to parents (ETTP & PTP)	None	

Table 1-79

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—ODU monitor-CTP, ODU transparent-CTP facilities

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	ODU AIS	ODU AIS	
Facility Loopback	None	ODU AIS	Faceplate traffic is looped back
Terminal Loopback	None	None	All active conditioning is cleared when a facility or terminal loopback is active
OOS-MA	ODU LCK	ODU LCK	
OPU MSIM	None	ODU AIS	Detected at Higher-Order ODU TTP
XC Intercard Fail	ODU AIS	None	
Mate Intercard Fail	None	ODU AIS	XCIF only
Receive Faceplate Defect from parent	ODU BDI	ODU AIS	
No rx_conn	None	ODU OCI	No CRS from XCIF/trib faceplate to XC
No tx_conn	ODU OCI	None	No CRS from XC to XCIF/trib faceplate
No cross-connect	ODU OCI	ODU OCI	No CRS in either direction
Fabric ODU defect	None	None	
Fabric OPU defect	None	None	ODU/OPU Defects are passed from fabric to faceplate

Table 1-80

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—ODU terminated-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	ODU AIS	
Facility Loopback	Not applicable	None	Faceplate traffic is looped back
Terminal Loopback	Not applicable	ODU AIS	All active conditioning is cleared when a facility or terminal loopback is active
OOS-MA	Not applicable	ODU LCK	
Mate Inter-card Fail	Not applicable	ODU AIS	XCIF only
Receive at least one Faceplate Defect from parent	Not applicable	CTP_CONDTYPE (If CTP_CONDTYPE is 'OPUK_NONE', conditioning is ODU AIS)	
Fabric ODU Signal Fail	Send Fabric Defect indication to parents	ODU BDI	ODU LCK, OCI, AIS, LOF, TIM
Fabric OPU Signal Fail	Send Fabric Defect indication to parents	None	OPU PTM, MSIM, CSF, AIS
No rx_conn & existing tx_conn	Not applicable	ODU OCI	No CRS from XC to XCIF/trib faceplate
Existing rx_conn & no tx_conn	Not applicable	None	
No cross-connect	Tell parent PTP and xTTP there are no CRSs	ODU OCI	

Table 1-81
100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—TCM TTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	TCM AIS	Not applicable	
Loopback	None	Not applicable	All active conditioning is cleared when a facility or terminal loopback is active
OOS-MA	TCM LCK	Not applicable	
No Connection on sibling ODU	TCM OCI	Not applicable	
XC Intercard Fail	TCM AIS	Not applicable	
Receive Faceplate Defect from parents	TCM BDI	Not applicable	
Faceplate TCM defect	TCM BDI	Not applicable	TCM AIS, LCK, OCI, LTC, LOF, TIM
Faceplate TCM IAE	TCM BIAE	Not applicable	
Faceplate TCM Signal Degrade	TCM BEI	Not applicable	

Table 1-82
100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—TCM monitor-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	TCM AIS	TCM AIS	
Terminal Loopback	None	None	All active conditioning is cleared when a facility or terminal loopback is active
Facility Loopback	None	TCM AIS	
OOS	TCM LCK	TCM LCK	
No Connection on sibling ODU	TCM OCI	TCM OCI	
XC Intercard Fail	TCM AIS	TCM AIS	
Mate intercard Fail	None	None	

Table 1-82

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—TCM monitor-CTP facility (continued)

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
Receive Faceplate Defect from parents	None	None	
Faceplate TCM defect	None	None	TCM AIS, LCK, OCI, LTC, LOF, TIM - all defects are passed through from faceplate to fabric
Faceplate TCM IAE	None	None	Passed through from faceplate to fabric
Fabric TCM Defect	None	None	TCM AIS, LCK, OCI, LTC, LOF, TIM - all defects are passed through from fabric to faceplate
Fabric TCM IAE	None	None	Passed through from fabric to faceplate

Table 1-83

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—TCM terminated-CTP facility

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
SA CP Fail	Not applicable	TCM AIS	
Terminal Loopback	Not applicable	None	All active conditioning is cleared when a facility or terminal loopback is active
Facility Loopback	Not applicable	TCM AIS	
OOS	Not applicable	TCM LCK	
No Connection on sibling ODU	Not applicable	TCM OCI	
Mate Inter-card Fail	Not applicable	TCM AIS	
Receive Faceplate Defect from parents	Not applicable	TCM AIS	

Table 1-83

100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 16xFLEX OTN I/F and 40G OTN XCIF signal conditioning—TCM terminated-CTP facility (continued)

Defect	Faceplate Conditioning	Fabric Conditioning	Notes
Fabric TCM Defect	Not applicable	TCM BDI	TCM AIS, LCK, OCI, LTC, LOF, TIM
Fabric TCM IAE	Not applicable	TCM BIAE TCM BEI	
Fabric TCM Signal Degrade	Not applicable	TCM BIAE TCM BEI	

Signal conditioning for L2 MOTR 2xXFP/10xSFP circuit packs

L2 MOTR OTM2 port 1 and 2 trigger

Signal conditioning behavior on L2 MOTR is affected by the following:

- ETH versus ETH10G facility
- OTM2 mapped with ETH10G in a particular mode (11.05/11.09 versus 10.709)
- ETH10G in ByPass-OTN mode (EVPL versus L2 mode)
- TXCON attribute on ETH or ETH10G facilities in EVPL mode
- OTM2 1+1 Line Protection provisioned or not
- The ETH and ETH100 facilities will raise two additional alarms under the same conditioning scenarios compared to the ETH10G facilities.
Conditioning example: Remote Port Unreachable.
 - ETH10G: Alarms raised, Remote Port Unreachable
 - ETH, ETH100, Alarms raised, Remote Port Unreachable, Link Down, Loss Of Datasync

Table 1-84 presents the signal conditioning that is applied for OTM2 port 1 and 2 triggered events.

Table 1-84
L2 MOTR signal conditioning—OTM2 port 1 and 2 trigger

Trigger		Tx conditioning				
Condition	Facility	ETH10G	OTM2 1+1 Provisioned	TxCon/ EVPL	Signal output	Facility
OTM2 p1-p2 Facility OOS	OTM2 p1-p2	No conditioning (Note 1)	Switched to Standby OTM2	OTM2/ ETH10G conditioning (Note 2)	ODU2 LCK	OTM2 p1-p2
OTM2 p1-p2 Rx ODU2 AIS	OTM2 p1-p2	Remote Fault conditioning (Note 3)	Switched to Standby OTM2	OTM2/ ETH10G conditioning (Note 2)	ODU2 BDI ETH10G conditioning	OTM2 p1-p2
OTM2 p1-p2 Rx OTU2 BDI	OTM2 p1-p2	BDI No conditioning	Traffic remain on the Active OTM2	No conditioning (Note 4)	ODU2 BDI alarm No conditioning	OTM2 p1-p2
OTM2 p1-p2 Rx OTU2 LOS, LOF, LOMF, or PreFEC SF	OTM2 p1-p2	Remote Fault conditioning (Note 3)	Switched to Standby OTM2	OTM2/ ETH10G conditioning (Note 2)	OTU2 BDI and ODU2 BDI, ETH10G conditioning	OTM2 p1-p2
OTM2 p1-p2 Rx OPU2 PN11	OTM2 p1-p2	Remote Fault conditioning (Note 3)	Traffic remain on the Active OTM2	OTM2/ ETH10G conditioning (Note 2)	No conditioning	OTM2 p1-p2

Table 1-84
L2 MOTR signal conditioning—OTM2 port 1 and 2 trigger (continued)

Trigger		Tx conditioning				
Condition	Facility	ETH10G	OTM2 1+1 Provisioned	TxCon/ EVPL	Signal output	Facility
OTM2 p1-p2 Rx OTU Signal Degrade	OTM2 p1-p2	No conditioning (SD conditioning)	Switched to Standby OTM2	OTM2/ ETH10G conditioning (Note 2)	OTU2 SD alarm No conditioning when FEC is disabled	OTM2 p1-p2
OTM2 p1-p2 Rx ODU2 OCI	OTM2 p1-p2	Remote Fault conditioning (Note 3)	Switched to Standby OTM2	OTM2/ ETH10G conditioning (Note 2)	ODU2 BDI	OTM2 p1-p2
Note 1: If the ETH10G is mapped into this OTM2 being placed OOS-MA, the Ethernet datapath is disabled such that no L2 traffic can be carried in either direction on the endpoint. Note 2: If the OTM2/ETH10G is the NNI Line for any ETH EVPL UNI clients (by way of EVPL connections with ENT-CRS-VCE), those client facilities can be conditioned based on their TXCON attribute value (either laser conditioning or 10B Error injection). Note 3: ETH10G is in a Rx Local Fault state which disables datapath in both directions and sends Remote Fault to far-end. Note 4: No ETH EVPL UNI client conditioning occurs if the OTM2/ETH10G NNI Line for this ETH client receives BDI.						

Signal conditioning for 20G OC-n/STM-n circuit packs

Signal conditioning behavior on the following 20G OC-n/STM-n is affected by the ODU LCK Signal conditioning when port 1 is OOS:

- (2+8)xOC-n/STM-n STS-1/VC-3 20G
- (2+8)xOC-n/STM-n STS-1/VC-3 OTN 20G
- (2+8)xOC-n/STM-n LO 20G
- (2+8)xOC-n/STM-n LO OTN 20G

Automatic laser shutdown (Broadband services)

Automatic laser shutdown (ALS) on the optical interface circuit pack shuts down a laser source if a Loss of Signal (LOS) alarm condition is detected on the associated receive interface. ALS can be enabled/disabled on an optical port basis (default is disabled for SONET mode and enabled for SDH/SDH-J mode enabled).

If enabled, the ALS circuit shuts down the laser in the event of an LOS alarm on the associated receive interface. The laser remains shutdown until the LOS alarm clears. If the alarm does not clear, an automatic restart is attempted:

- For non-DWDM circuit packs, the laser is turned on for two seconds every 100 seconds.
- For DWDM circuit packs, the laser is turned on for five seconds ($100 + 6 \times \text{Lambda index}$) seconds) after the LOS signal is detected and then every 379 seconds. The Lambda index depends on the wavelength of the optical interface and ensures that all the DWDM optical interfaces feeding an optical amplifier are not all turned on/off at the same time.

Automatic Power Reduction (APR) (Photonic services)

The APR feature is a controlled ramp-down and recovery mechanism used to limit potential exposure to instances of high optical power with a view of protecting personnel and preventing equipment damage on detection of high reflections, breaks, or disconnects in the optical line. It is non-provisionable, and is activated when an amplifier circuit pack is placed in service.

When the Optical Return Loss (ORL) of an amplifier falls below the threshold, the system reduces the amplifier output power level to minimize the danger of personal eye injury. A regulatory-deemed safe level of optical power is transmitted in the period of optical discontinuity on the line to facilitate automatic detection of line restoration and recovery to normal state. It is used by lowering the optical output to a residual level suitable for making OR measurements and facilitating auto recovery when normal system connectivity resumes.

The 6500 Amplifier Module EDFA facility APR system is triggered by low return loss detected at the line out. This detection affects the EDFA immediately preceding the reflection point invoking APR on this EDFA. The low return loss condition may be due to:

- poor connection at output connector or subsequent connections in the line
- fiber break downstream

When an APR condition is triggered, an Automatic Power Reduction Active alarm is raised against any amplifier with reduced power.

ATTENTION

As of Release 10.2, the MuxAmp configuration disables the Automatic Power Reduction (APR) function. Certain C-band amplifier circuit packs used in a MuxAmp configuration with one or more wavelengths in the IEC 60825-1 Class 1M range (such as those from a WL3n source) may have a Class 1 Hazard level label but should have a Class 1M Hazard Level label because APR is disabled.

Circuit packs that were originally manufactured with a Hazard Level 1 warning label can be re-labeled with the Level 1M label kit (part number 415-2818-001).

The following C-band amplifier circuit packs may have a Hazard Level 1 warning label.

- NTK552AAE5, Single Line Amplifier (SLA C-Band), Revision 19 and lower
- NTK552BAE5, Midstage Line Amplifier (MLA C-Band), Revision 19 and lower
- NTK552FAE5, Midstage Line Amplifier 2 (MLA2 C-Band), Revision 17 and lower
- NTK552FB, Midstage Line Amplifier 2 (MLA2 C-Band) with variable optical attenuator (VOA), Revision 05 and lower
- NTK552GAE5, Midstage Line Amplifier 3 (MLA3 C-Band), Revision 06 and lower

For more information and the procedure to apply the Level 1M label on these circuit packs, see the section on observing product and personnel safety guidelines in *Installation - General Information*, 323-1851-201.0.

Automatic Line Shut Off (ALSO) (Photonic services)

The ALSO feature is a safety laser shutdown mechanism. It is non-provisionable, and is activated when an amplifier circuit pack is placed in service or during some SLAT procedures. Amplifier power levels are turned down or turned off when a fiber break or intermediate connector disconnect occurs between two neighboring sites, where optical radiation is being fed into both ends of the optical fiber and generating a hazard on both ends of a fiber break.

When an ALSO condition is triggered, an Automatic Shutoff alarm is raised against any transmitter that was shut off.

ATTENTION

For the ALSO feature to function correctly, you must properly configure the OTS first.

ALSO can be disabled on the SAM, ESAM or LIM circuit packs (C-Band and L-Band) by setting the ALSO_Disable flag to TRUE for the OPTMON facility. This action must only be performed when there is no OSC in the OTS for automatic recovery from optical line fail condition.

Alarm surveillance

This section provides information and procedures about alarm profiles, external alarm provisioning, network alarm monitoring and events surveillance.

Note: Unless otherwise specified, eMOTR in this document refers to eMOTR (NTK536AA, NTK536AB, NTK536FA, NTK536FB) and eMOTR Edge (NTK536BE) variants.

For a description of the 6500 alarm features, refer to [“Alarm and trouble clearing strategy”](#) in this document.

Abbreviations used in this document

ACO	Alarm Cut-Off
CFP2	C form-factor pluggable generation 2
CFP2-ACO	C form-factor pluggable generation 2-Digital Coherent Optical transceiver
CFP	100G transceiver form factor pluggable
DCC	Data Communications Channel
BDI	Backward Defect Identifier
GCC	General Communications Channel
IPv4/6	Internet Protocol version 4/6
LAN	Local Area Network
LED	Light-emitting Diode
MIC	Maintenance Interface Card
MS	Multiplex Section
QSFP+	Enhanced quad (4-channel) small form-factor pluggable

QSFP28	Enhanced quad (4-channel) small form-factor pluggable 28
SFP	Small form Factor Pluggable
SNMP	Simple Network Management Protocol
S/R	Save/Restore
SSM	Synchronization Status Messaging
TOD	Time Of Day
UPC	User Privilege Code
VC	Virtual Container
WAN	Wide Area Network
XFP	10G transceiver form factor pluggable

Alarm parameters

External control types

The external control relays support the external control types listed in the following table.

Table 2-1
External control labels

External control label	External control type
Air conditioning	Air conditioning
Engine	Engine
Fan	Fan
Generator	Generator
Heat	Heater
Light	Lighting
Miscellaneous	Miscellaneous
Sprinkler	Sprinkler
(Null)	No label is associated with the specific relay. However, some external equipment can be connected to this relay.

Environmental alarm labels

The following table lists the labels and associated condition types available for the environmental alarms.

Table 2-2
Environmental alarm labels and associated condition types

Alarm label (default description)	Condition Type
48-V power supply failure (Note)	PWR-48
Air compressor failure	AIRCOMPR
Air conditioning failure	AIRCOND
Air dryer failure	AIRDRYR
Battery discharging	BATDSCHRG
Battery failure	BATTERY
Commercial power failure	POWER
Cooling fan failure	CLFAN
Engine failure	ENGINE
Engine operating	ENGOPRG
Explosive gas	EXPLGS
Fire	FIRE
Fire detector failure	FIRDETR
Flood	FLOOD
Fuse failure	FUSE
Generator failure	GEN
High airflow	HIAIR
High humidity	HIHUM
High temperature	HITEMP
High water	HIWTR
Intrusion	INTRUDER
Low battery voltage	LWBATVG
Low cable pressure	LWPRES
Low fuel	LWFUEL
Low humidity	LWHUM

Table 2-2
Environmental alarm labels and associated condition types (continued)

Low temperature	LWTEMP
Low water	LWWTR
Miscellaneous	MISC
Open door	OPENDR
Pump failure	PUMP
Rectifier failure	RECT
Rectifier high voltage	RECTHI
Rectifier low voltage	RECTLO
Smoke	SMOKE
Toxic gas	TOXICGAS
Ventilation system failure	VENTN
Note: “-48 Vdc” is expressed as “48-V” in the label and default description.	

Autonomous events

Autonomous events are faults raised with a severity of “Log”. Events report the activity status on the network elements, and do not always require user action. To retrieve events, refer to [“Retrieving events for a network element” on page 2-15](#). The events listed in the **Historical Fault Browser** application include the alarms that have been raised, both cleared or not cleared, and the logged warnings and events. For more information about the logged events, refer to *Fault Management - Customer Visible Logs*, 323-1851-840.

The following table lists logged events. For a complete list of alarms that can be raised, refer to the [“List of alarms” on page 4-6](#).

Table 2-3
Autonomous events

Event category (Note)	Description
Files	Remote transfer of files for '<release>' - <#> of <#> MB, <#> of <#> Files Release '<release>' Successfully delivered Remote transfer of file '<path>/<filename>' failed Removing files for '<release>' started Release '<release>' has been deleted
Upgrades	Redundant Release Synch Complete
Load installation	Load Installation: Cancel Passed Load Installation: Check Failed Load Installation: Check Passed Load Installation: Commit Failed Load Installation: Commit Passed Load Installation: Committing New Release Load Installation: Invoke Failed Load Installation: Invoke Passed Load Installation: Load Failed Load Installation: Load Passed Load Installation: Programming Load to FLASH Load Installation: Running from incorrect FLASH bank Load Installation: Unable to Access Release Files Load Installation: Unable to Program Load to FLASH
Performance Monitoring	15-Min Threshold Crossing 1-Day Threshold Crossing Untimed Threshold Crossing
Photonics	DOC Channel Add Completed DOC Channel Delete Completed DOC Pre-Check Fail DOC Pre-Check Pass DOC Reset TCA Baselines Failed
Protection	Protection Exerciser Complete Protection Switch Initiated

Table 2-3
Autonomous events (continued)

Event category (Note)	Description
Save and restore	Cancel S/R Completed Cancel S/R Failed Cancel S/R Failed: Save/Restore not in progress Check S/R Failed: Blocked by another application Check S/R Failed: Blocked by presence of alarms Check S/R Failed: Could not connect to destination Cancel SR Failed: Software Subsystem Check S/R Failed: FTP access denied Check S/R Failed: SFTP access denied Check S/R Failed: Invalid destination Check S/R Completed Check S/R Failed Check S/R Failed: Blocked by remote application Check S/R: Restore Blocked by another application Check S/R: Save Blocked by another application Database Commit Failed Database Commit Failed: CPU2 Application Failed Database Commit Failed: Restored backup is corrupt Database Commit: Restart in progress Database Commit Failed: Software Subsystem Database Restore Completed Database Restore Failed Database Restore Failed: Backup not from this node Database Restore Failed: Backup not from this shelf type Database Restore Failed: Backup not from this SP family Database Restore Failed: Backup requires SP protection provisioning Database Restore Failed: Blocked by another application Database Restore Failed: CPU2 Application Unsupported Database Restore Failed: Incompatible Shelf Assembly Database Restore Failed: Incompatible SP Database Restore Failed: SP Protection Not Provisioned Database Restore Failed: Blocked by presence of alarms Database Restore Failed: Blocked by remote application

Table 2-3
Autonomous events (continued)

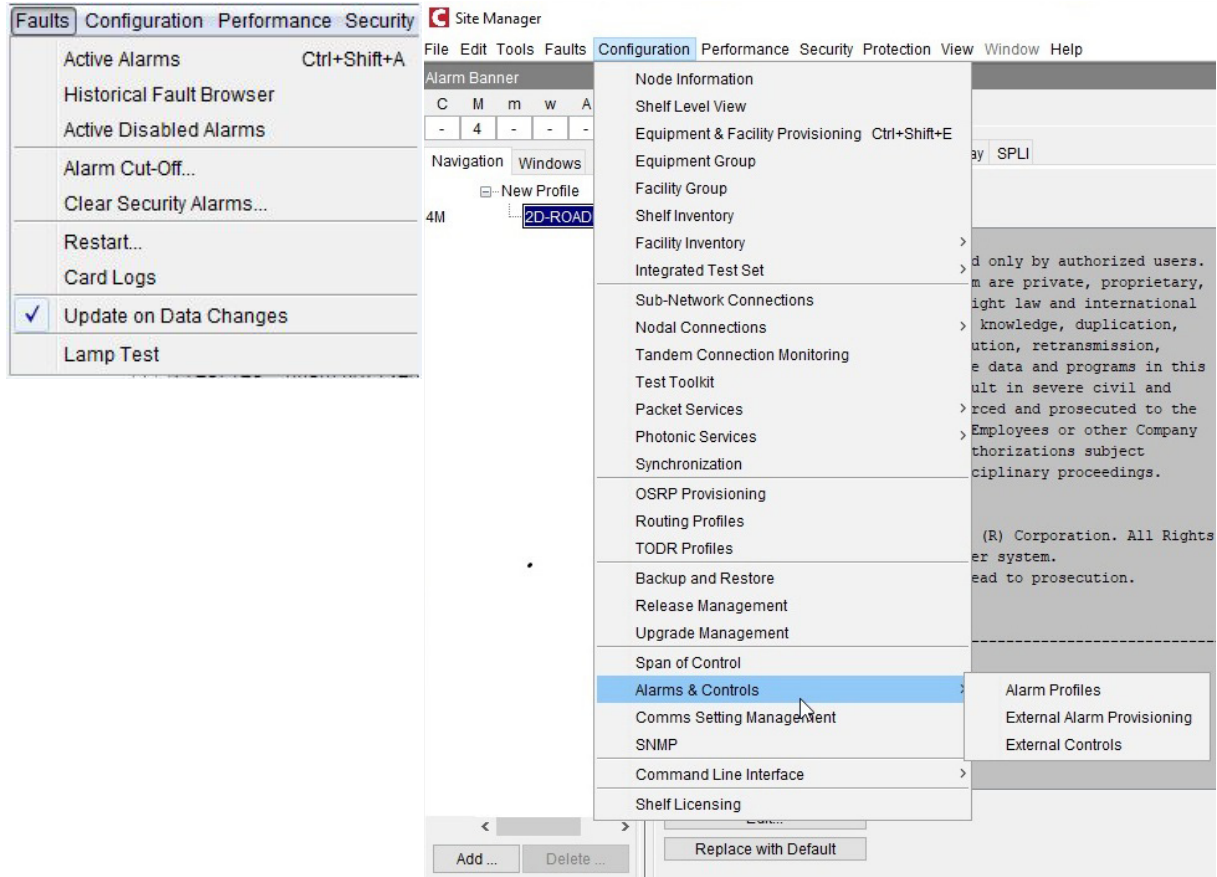
Event category (Note)	Description
Save and restore (continued)	Database Restore Failed: Could not connect to source Database Restore Failed: CPU2 Application Failed Database Restore Failed: Failure transferring file Database Restore Failed: FTP access denied Database Restore Failed: SFTP access denied Database Restore Failed: Incompatible options Database Restore Failed: Incompatible S/R options specified Database Restore Failed: Interrupted by card restart Database Restore Failed: Invalid source Database Restore Failed: Mismatched Software Releases Database Restore Failed: Restored backup is corrupt Database Restore in progress Database Save Failed Database Restore Failed Database Commit Failed Database Save Completed Database Save Failed: Blocked by another application Database Save Failed: Blocked by presence of alarms Database Save Failed: Blocked by remote application Database Save Failed: Could not connect to destination Database Save Failed: CPU2 Application Failed Database Save Failed: Failure transferring file Database Save Failed: FTP access denied Database Save Failed: SFTP access denied Database Save Failed: Interrupted by card restart Database Save Failed: Invalid destination Database Validate Failed Database Validate Failed: Restored backup is corrupt Redundant database synch complete Redundant database synch in progress

Table 2-3
Autonomous events (continued)

Event category (Note)	Description
Security	Password has expired. Contact your Admin Password has expired. Change it now Password will expire in <n> day(s) Value of n can be between 1 to 14 days. Password will expire today Intrusion Attempt: <n> time(s) by "intruder" n= number of intrusion attempt(s) Secure Erase Completed Secure Erase Failed
Timing and synchronization	Secondary SETS Locking to Primary Timing Distribution Reference Switch Complete Timing Generation Reference Switch Complete Timing Generation Entry to Freerun Timing Generation Entry to Holdover
Restart	Cold Restart Completed Warm Restart Completed
Note: Site Manager does not display the events as event categories. Event categories are used to organize this table only. In Site Manager, the severity of alerts and events is displayed as Log.	

Site Manager navigation

The following figures provide an overview of the Site Manager navigation associated with the **Faults** and **Configuration** menus for the 6500. The figures show the path from the Site Manager menu bar.



Procedures for alarms and events

Table 2-4
Alarms and events

Action
Procedure 2-1, "Setting the time zone for network element or Site Manager timestamps"
Procedure 2-2, "Retrieving active alarms for one or more network elements"
Procedure 2-3, "Retrieving events for a network element"
Procedure 2-4, "Retrieving active disabled alarms"
Procedure 2-5, "Allowing or inhibiting the display of log, inventory, and database change events"
Procedure 2-6, "Clearing security alarms"

Procedures for alarm provisioning and alarm profiles

Table 2-5
Alarm provisioning and alarm profiles

Action
Procedure 2-7, "Retrieving alarm profiles"
Procedure 2-8, "Editing an alarm profile"
Procedure 2-9, "Setting a default profile"
Procedure 2-10, "Setting a profile as active"

Procedures for alarm monitoring and management

Table 2-6
Alarm monitoring and management

Action
Procedure 2-11, "Restarting a circuit pack or shelf processor"
Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm"
Procedure 2-13, "Clearing audible alarms and performing lamp tests"

Procedures for external alarm provisioning and external controls

Table 2-7
External alarm provisioning and external controls

Action
Procedure 2-14, "Provisioning environmental alarm attributes"
Procedure 2-15, "Provisioning, operating, and releasing external controls"

Procedures for Photonic system maintenance

Table 2-8
Photonic system maintenance

Action
Procedure 2-16, "Locating a reflective event"
Procedure 2-17, "Preparing to perform fiber work on a Photonic system"
Procedure 2-18, "Measuring Photonic amplifier output power"

Associated procedures

Some procedures require the user to perform procedures relating to other topics. Before performing a procedure, if necessary ensure that the information about the associated procedures is available.

All procedures assume that you have logged in to the network element. Refer to the interface login and logout procedures in *Administration and Security*, 323-1851-301.

Procedure 2-1

Setting the time zone for network element or Site Manager timestamps

Use this procedure to set the time zone used for displaying timestamps. You can select either the network element time zone or the local operating system (OS) time zone.

Note: Diagnostics-related timestamps are based on the Network Element time zone, and not on the Site Manager Local OS time zone. If these time zones do not match, it results in a time difference compared to timestamps for other network element-generated events (for example, alarms, events, and refresh times), which are translated to the Site Manager Local OS time zone. The user must cross-reference the network element timestamp with the Site Manager timestamp. Ciena recommends that timestamps use the Network Element time zone.

Changes do not take effect until the next launch of Site Manager.

Step	Action
1	Select the network element in the navigation tree.
2	Select Preferences from the Edit drop-down menu. The Preferences dialog box opens.
3	Select 6500 from the Nodal Manager option in the navigation area on the left of the Preferences dialog box.
4	Select the required radio button (Network Element or Local OS) to set the time zone used for displaying timestamps. — Select the Set Defaults button to return the setting to the default (Network Element). — The OK and Apply buttons are disabled if the current active time zone setting is selected.
5	Click OK . The applied changes do not take effect until the next launch of the Site Manager. To restart Site Manager, refer to the interface login and logout procedures in chapter 1 of <i>Administration and Security</i> , 323-1851-301.

—end—

Procedure 2-2

Retrieving active alarms for one or more network elements

Use this procedure to:

- retrieve the active alarms and alarm details
- access the alarm clearing procedure
- sort active alarms
- filter active alarms
- update active alarms

Step	Action												
1	Select the desired network element in the navigation tree.												
2	Select Consolidated Alarms from the Tools drop-down menu or Active Alarms from the Fault drop-down menu. The Consolidated Alarms application opens in a separate window, and displays the active alarms according to your last filter settings and the alarm points that are not disabled. Alarms from all logged in network elements are displayed by default. You can choose to show alarms for one or a set of specific network elements. The Active Alarms application opens in a separate tab, and displays the active alarms according to your last filter settings and the alarm points that are not disabled. Alarms from all shelves are displayed by default. You can choose to show alarms from one specific shelf.												
3	<table> <tr> <th>If you</th><th>Then</th></tr> <tr> <td>want to sort the active alarms</td><td>go to step 4</td></tr> <tr> <td>want to filter the active alarms</td><td>go to step 6</td></tr> <tr> <td>want to update the active alarms</td><td>go to step 8</td></tr> <tr> <td>want to view details of an active alarm</td><td>go to step 9</td></tr> <tr> <td>have the required information displayed</td><td>the procedure is complete</td></tr> </table>	If you	Then	want to sort the active alarms	go to step 4	want to filter the active alarms	go to step 6	want to update the active alarms	go to step 8	want to view details of an active alarm	go to step 9	have the required information displayed	the procedure is complete
If you	Then												
want to sort the active alarms	go to step 4												
want to filter the active alarms	go to step 6												
want to update the active alarms	go to step 8												
want to view details of an active alarm	go to step 9												
have the required information displayed	the procedure is complete												
4	Click on a column header to sort the alarms by that column, in ascending order. When you first open the Consolidated Alarms application, the columns are sorted from highest to lowest severity and then from most recent to oldest. All columns are sorted in alphabetical order except the Time Raised column. The Time Raised column is sorted by date, then time.												

Procedure 2-2 (continued)

Retrieving active alarms for one or more network elements

Step	Action
5	Click again on the same column header to sort the alarms in descending order. Go to step 3.
6	To hide alarms of a specific severity from the Alarm List, clear the appropriate check box in the Show area. By default, the Consolidated Alarms or Active Alarms application displays active alarms of all severities. The Consolidated Alarms or Active Alarms application updates and no longer shows the alarms of that severity.
7	To display alarms filtered from the list, select the appropriate check box again in the Show area. The Consolidated Alarms or Active Alarms application updates. Go to step 3.
8	By default, the Auto refresh check box is checked and the alarm list is updated automatically. To manually update the active alarms, clear the Auto refresh check box to enable the Refresh button. Click Refresh. The Last refresh field displays the date (yyyy-mm-dd) and time (hh:mm:ss) of the most recent update of the Consolidated Alarms or Active Alarms application. Go to step 3.
9	To view the details of an alarm, click on the row for the alarm that you want to see in detail from the list of active alarms. The Alarm details area at the bottom of the Consolidated Alarms or Active Alarms application displays the details of the alarm. You can view the details of only one alarm at a time. In the Active alarms application, you can click on the How to Clear button to access the alarm clearing procedure for the selected alarm. Go to step 3.

—end—

Procedure 2-3

Retrieving events for a network element

Use this procedure to:

- retrieve all events or only disabled alarm events
- retrieve detailed information about an event. For a list of autonomous events, refer to [Table 2-3 on page 2-5](#)
- sort the event list
- filter the events to display (event severities are Critical, Major, minor, warning, cleared, and logged)
- update the events

The time stamp of alarm entries in the **Historical Fault Browser** application represents the date and time of the alarm being raised or cleared and not the actual system occurrences (alarm timestamps include the hold-on and hold-off periods). As a result, there may be a time discrepancy between an event and the corresponding alarms being raised or cleared. See the following:

- Time Raised column will be updated for all raise Events.
- Clear Time column will be updated for all Clear Events and for those Raise Events, which got Cleared.
- Date Time column indicated the date and time of both Raise and Clear events. For Raise events, this column will be same as Time Raised column and for Clear events, this column will be same as Clear Time column.

Note: When retrieving event counts, the **Historical Fault Browser** application includes Events + Events (disabled alarms only) and provides the status bar message as the total. For example, if the maximum count is 500 for retrieving events, the status bar message indicates 521. That is, it includes Events + Events (disabled alarms only).

Step	Action
1	Select the network element in the navigation tree.
2	Select Historical Fault Browser from the Faults drop-down menu. You can identify the events by looking for Log in the Severity column. Alerts also have the Log severity. If you return to the Historical Fault Browser application during a session, the application displays the events according to the previous filter settings.

Procedure 2-3 (continued)

Retrieving events for a network element

Step	Action												
3	In the Show area, select the: • Events radio button to display all events • Events (Disabled Alarms Only) radio button to display only events associated with disabled alarms Note: If the Events (Disabled Alarms Only) option is selected for a consolidated node, the Historical Fault Browser application retrieves and displays information for the primary shelf only. To retrieve events associated with disabled alarms on member shelves, you must execute the appropriate TL1 command on individual shelves.												
4	<table> <tr> <th>If you</th><th>Then</th></tr> <tr> <td>want to sort the event list</td><td>go to step 5</td></tr> <tr> <td>want to filter events</td><td>go to step 7</td></tr> <tr> <td>want to update the Historical Fault Browser application</td><td>go to step 9</td></tr> <tr> <td>want to retrieve events details</td><td>go to step 10</td></tr> <tr> <td>have the required information displayed</td><td>the procedure is complete</td></tr> </table>	If you	Then	want to sort the event list	go to step 5	want to filter events	go to step 7	want to update the Historical Fault Browser application	go to step 9	want to retrieve events details	go to step 10	have the required information displayed	the procedure is complete
If you	Then												
want to sort the event list	go to step 5												
want to filter events	go to step 7												
want to update the Historical Fault Browser application	go to step 9												
want to retrieve events details	go to step 10												
have the required information displayed	the procedure is complete												
5	Click on a column header to sort the events by that column, in ascending order. When you first open the Historical Fault Browser application, the columns are sorted from most recent to oldest. All columns are sorted in alphabetical order except the Time Raised column, Clear Time column, and Date, Time column. The Time Raised column and the Clear Time columns are sorted by date and then time.												
6	Click again on the same column header to sort the events in descending order. Go to step 4.												
7	To hide events of a specific severity from the events list, clear the appropriate check box in the Show area. The Historical Fault Browser application updates and no longer shows the events of that severity.												
8	To display events filtered from the list, select the appropriate check box again in the Show area. Go to step 4.												

Procedure 2-3 (continued)

Retrieving events for a network element

Step	Action
9	If the Events radio button is selected, by default, the Auto refresh check box is checked and the event list is updated automatically. To manually update the events, clear the Auto refresh check box to enable the Refresh button. Click Refresh. If the Events (Disabled Alarms Only) radio button is selected, only manual refresh is supported. Click Refresh. The Last refresh field displays the date (yyyy-mm-dd) and time (hh:mm:ss) of the most recent update of the Historical Fault Browser application. Go to step 4.
10	From the event list, click on the row for the event that you want to see in detail. The Event details area at the bottom of the Historical Fault Browser application displays the details of the event. You can view the details of only one event at a time. Go to step 4.

—end—

Procedure 2-4

Retrieving active disabled alarms

Use this procedure to:

- retrieve list of active alarms that are raised against disabled alarm points
- retrieve detailed information about a disabled alarm
- access the alarm clearing procedure
- sort the alarms list
- update the **Active Disabled Alarms** application manually

Step	Action										
1	Select the network element in the navigation tree.										
2	Select Active Disabled Alarms from the Faults drop-down menu. The Active Disabled Alarms application opens, and displays a list of active alarms that are raised against disabled alarm points.										
3	<table> <tr> <th>If you</th><th>Then</th></tr> <tr> <td>want to sort the active alarms</td><td>go to step 4</td></tr> <tr> <td>want to update the active alarms</td><td>go to step 6</td></tr> <tr> <td>want to view details of an active alarm</td><td>go to step 7</td></tr> <tr> <td>have the required information displayed</td><td>the procedure is complete</td></tr> </table>	If you	Then	want to sort the active alarms	go to step 4	want to update the active alarms	go to step 6	want to view details of an active alarm	go to step 7	have the required information displayed	the procedure is complete
If you	Then										
want to sort the active alarms	go to step 4										
want to update the active alarms	go to step 6										
want to view details of an active alarm	go to step 7										
have the required information displayed	the procedure is complete										
4	Click on a column header to sort the alarms by that column, in ascending order.										
5	Click again on the same column header to sort the alarms in descending order. Go to step 3 .										
6	To update the active alarms, click Refresh . The Last refresh field displays the date (yyyy-mm-dd) and time (hh:mm:ss) of the most recent update of the Active Disabled Alarms application. Go to step 3 .										
7	To view the details of an alarm, click on the row for the alarm that you want to see in detail from the list of inhibited alarms. The Alarm details area at the bottom of the Active Disabled Alarms application displays the details of the alarm. You can view the details of only one alarm at a time. Click on the How to Clear button to access the alarm clearing procedure for the selected alarm. Go to step 3 .										

—end—

Procedure 2-5

Allowing or inhibiting the display of log, inventory, and database change events

Use this procedure to allow or inhibit TL1 autonomous events used for application refreshes (except for alarms that operate independently of this option). The TL1 autonomous events are allowed by default.

Step	Action
1	Select the network element in the navigation tree.
2	Select the Faults drop-down menu. The Update on Data Changes menu option appears at the bottom of this menu. If a checkmark appears next to the Update on Data Changes option, the option is enabled and the TL1 autonomous events used for application refreshes are allowed. If a checkmark does not appear next to the Update on Data Changes option, the option is disabled and the TL1 autonomous events used for application refreshes are inhibited.
3	To change the status of the Update on Data Changes menu option (to either allowed or inhibited), select this option from the Faults drop-down menu.

—end—

Procedure 2-6

Clearing security alarms

Use this procedure to clear security alarms (except those raised against the Primary and Secondary RADIUS servers) on a network element (for example, Intrusion Attempt).

Step	Action
------	--------

- | | |
|---|---|
| 1 | Select the network element in the navigation tree. |
| 2 | Select Clear Security Alarms from the Faults drop-down menu. |
| 3 | Click Yes in the confirmation dialog box.

All security alarms (except those raised against the Primary and Secondary RADIUS servers) are cleared.

The procedure does not unlock channels associated with an Intrusion Attempt alarm. Refer to the “Unlocking source addresses/users” procedure in <i>Administration and Security</i> , 323-1851-301. |

—end—

Procedure 2-7

Retrieving alarm profiles

Use this procedure to retrieve information about alarm profiles and profile details.

Step	Action
1	If you want to retrieve the alarm profile from the Top-Level Menu Then go to step 2 Shelf Level View Application step 7
2	Select the network element in the navigation tree.
3	Select Alarms & Controls from the Configuration drop-down menu.
4	Select Alarm Profiles .
5	Select an alarm class from the Alarm Class drop-down list. Profile Names and default Profiles are retrieved from network element and will be visible in Profiles table. Profile Content tab is selected by default.
6	If you want to display details about an alarm profile, select the profile name from the Profiles table in the upper section of the Alarm Profiles application. The Profile Content table at the bottom of the Alarm Profiles application displays all the alarm points applicable to the selected profile and their status. Go to step 9
7	Select Shelf Level View from the Configuration drop-down menu.
8	Right Click on a slot or Port and select Show Alarm Profile option in the drop-down menu. Alarm class combo is displayed with the first facility type available at selected slot or port. Active Profile tab is selected by default. Active profiles for first facility type is retrieved and displayed.
9	To retrieve Alarm Points for an Alarm Profile, select a row in Alarm Profiles table. All Alarm points will be retrieved from network element and displayed under Profile Content tab.
10	To retrieve Active Alarm Profile , click on the Active Profiles tab.
11	Select desired values in Type, Slot, Sub-slot, Port and Sub-Port drop-down list. Active profiles will be retrieved for selected values and listed in table under the Active profile tab.

—end—

Procedure 2-8

Editing an alarm profile

**CAUTION****Risk of unidentified problem conditions**

Disabling an alarm point prevents alarm notification if a fault occurs.

Use this procedure to edit an existing alarm profile. This procedure allows you to change the:

- name of the alarm profile
- status (enabled/disabled) of the alarm points
- **Service affecting severity (SA)** or **Not service affecting (NSA)** severities (Critical, Major, minor, warning)

Alarm provisioning only affects alarm notification and has no effect on the alarm function.

Note: When Shelf Synch is enabled on a consolidated node, the alarm profile cannot be edited for member shelves. For information on Shelf Synch, refer to the “Shelf Synch” section in *Administration and Security*, 323-1851-301.

ATTENTION

You cannot edit the ALL ENABLED, ALL DISABLED, or FACTORY DEFAULT profiles that the system has defined.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Note: An account with a level 4 UPC or higher is required to edit the Alarm class of Security alarm points.

Step	Action
1	Retrieve the alarm profiles of the network element. Refer to “Retrieving alarm profiles” on page 2-21 .
2	Select the profile to edit. You cannot edit the ALL ENABLED, ALL DISABLED, and FACTORY DEFAULT profiles that the system has defined.

Procedure 2-8 (continued)
Editing an alarm profile

Step	Action														
3	<table> <tr> <th>If you</th><th>Then</th></tr> <tr> <td>want to change the alarm profile name</td><td>go to step 4</td></tr> <tr> <td>want to Set default Alarm Profile</td><td>go to step 10</td></tr> <tr> <td>want to change the status of the alarm points</td><td>go to step 12</td></tr> <tr> <td>Set Active Alarm Profile</td><td>go to step 18</td></tr> <tr> <td>want to change the SA and NSA severity</td><td>go to step 22</td></tr> <tr> <td>have completed the required changes</td><td>the procedure is complete</td></tr> </table>	If you	Then	want to change the alarm profile name	go to step 4	want to Set default Alarm Profile	go to step 10	want to change the status of the alarm points	go to step 12	Set Active Alarm Profile	go to step 18	want to change the SA and NSA severity	go to step 22	have completed the required changes	the procedure is complete
If you	Then														
want to change the alarm profile name	go to step 4														
want to Set default Alarm Profile	go to step 10														
want to change the status of the alarm points	go to step 12														
Set Active Alarm Profile	go to step 18														
want to change the SA and NSA severity	go to step 22														
have completed the required changes	the procedure is complete														
4	Right click the profile name to open the drop-down menu.														
5	Select the Edit name in the menu to open the Edit Profile Name dialog box. The Edit Profile Name dialog box contains the current name of the selected profile.														
6	Click on the Label field and highlight the profile name.														
7	Type in the new alarm profile name. The alarm profile name can be up to 20 characters.														
8	If shelf sync is not enabled, select the Apply to all available shelves within the TID check box if you want to apply the profile name change to all available shelves.														
9	Click OK . Go to step 3 .														
10	Right click on the selected alarm profile to open the drop-down menu.														
11	Click on Set as default menu button Go to step 3 .														
12	Select one or more editable row in table under Profile Content tab that you want to enable or disable. To select multiple alarm points, hold down the Shift or Ctrl key when you select the alarm points.														
13	Right click the selected row(s) and select Edit to open the Edit Alarm Point dialog box.														
14	Click Enabled or Disabled as applicable from the Alarm status drop-down list.														
15	If shelf sync is not enabled select the Apply to all available shelves within the TID check box if you want to apply the alarm point change to all available shelves.														
16	Click OK .														

Procedure 2-8 (continued)

Editing an alarm profile

Step	Action
17	Repeat step 12 to step 14 until you have finished editing all the alarm points. The edited alarm point status is displayed in the Profile details table at the bottom of the Alarm Profiles application. Go to step 3 .
18	Click on the Active Profiles tab.
19	Select row(s) in table under Active Profiles table.
20	Right click on selected row and click on Set as active... menu button.
21	Select a profile from the list in a dialog box and click on OK . Selected Alarm profile will set as active profile for selected unit. Go to step 3 .
22	Select one or more alarm points that you want to change the severity. To select multiple alarm points, hold down the Shift or Ctrl key when you select the alarm points.
23	Right click on selected row and click on Edit Value... menu button to open the Edit Alarm Point dialog box.
24	Select the desired severity for the alarm from the Service affecting severity or Not service affecting severity drop-down list. You can provision alarm severities (SA or NSA) as Critical, Major, minor or warning. Note: In case of unprotected configurations, the majority of the alarms will raise as service affecting (SA). In such cases, changing the NSA severity has no impact.
25	If shelf sync is not enabled, select the Apply to all available shelves within the TID check box if you want to apply the alarm point change to all available shelves.
26	Click OK . Go to step 3 .

—end—

Procedure 2-9

Setting a default profile

Use this procedure to set the alarm profile of an alarm class as the default profile.



CAUTION

Risk of unidentified problem conditions

Disabling an alarm point prevents alarm notification if a fault occurs.

Alarm provisioning only affects alarm notification and has no effect on the alarm function.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Note that an account with a level 4 UPC or higher is required to edit the Alarm class of Security alarm points.

Step	Action
1	Retrieve the alarm profiles of the network element by alarm class. Refer to “Retrieving alarm profiles” on page 2-21 .
2	Select a profile from the profiles table in the upper section of the Alarm Profiles application.
3	Click Set As Default .
4	If you want to apply the default profile to all available shelves, click Yes in the confirmation dialog.
5	If you do not want to apply the default profile to all available shelves, click No in the confirmation dialog.
	The word Default is displayed in the Alarm Class Default column on the row of the selected profile.

—end—

Procedure 2-10

Setting a profile as active



CAUTION

Risk of unidentified problem conditions

Disabling an alarm point prevents alarm notification if a fault occurs.

Use this procedure to set the alarm profile of an alarm class as the active profile.

Consider the following:

- Alarm provisioning only affects alarm notification and has no effect on the alarm function.
- Selecting the ALL DISABLED profile for an alarm class or a specific equipment/facility disables all alarms for the alarm class or the specific equipment or facility.
- Selecting the FACTORY DEFAULT profile for Common alarm class enables all alarms in this class (except LAN-15 and LAN-16, which are disabled).

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

An account with a level 4 UPC or higher is required to edit the Alarm class of Security alarm points.

Step	Action
1	Retrieve the alarm profiles of the network element. Refer to “Retrieving alarm profiles” on page 2-21 .
2	Select a profile from the profiles table.
3	Click Set as Active . The Active profiles table in the center section of the Alarm Profiles application displays the active profiles for the selected alarm class. The Set as Active button is disabled if there are no equipment/facilities provisioned for the selected alarm class.

—end—

Procedure 2-11

Restarting a circuit pack or shelf processor

**CAUTION****Risk of traffic loss**

A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic.

As cold restarts can be traffic affecting, you must only perform a cold restart to restore functionality when all other trouble clearing procedures have been performed. Before performing a cold restart, if possible, put the circuit pack out-of-service and unless it contains unprotected services, contact your next level of support or your Ciena support group for assistance.

**CAUTION****Risk of traffic loss**

A warm or cold restart of the 2xOSC circuit pack causes a Wayside Channel (WSC) traffic hit.

**CAUTION****Risk of traffic loss**

The 4x10G OTR (NTK530QE variant) circuit pack requires a valid Time of Day (TOD) in order to perform the certificate validation before bringing up traffic. The TOD information is received from the SP on every circuit pack restart.

If the 4x10G OTR (NTK530QE variant) circuit pack is reseated while an SP is not present, then traffic does not recover automatically on this circuit pack until the SP is available to provide the system TOD. Ensure that the SP is installed in the chassis before you reseat the 4x10G OTR circuit pack.

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor



CAUTION

Risk of traffic loss

For Broadband circuit packs that support 1+1 TPT protection, disconnect the client fibers from the inactive circuit pack prior to a circuit pack cold restart to avoid causing traffic hits on the active circuit pack in the protection group.

Use this procedure to initialize a circuit pack or the shelf processor (SP) in a warm restart or cold restart mode.

Consider the following:

- A loss of connectivity to the shelf processor occurs when you restart the shelf processor. You must wait up to 10 minutes before logging back in.
- In case of SP redundancy, a “Redundant Database Synch Failed” alarm is expected after a warm or cold restart. The alarm will clear automatically.
- If SP redundancy is provisioned, both SPs are healthy and available, and all provisioning data is synchronized between the SP pair, a user-initiated cold restart will result in a shelf processor switch of activity.
 - a user-initiated warm restart will not result in a shelf processor switch of activity.
 - a user-initiated cold restart will result in a shelf processor switch of activity.
- If SP redundancy is provisioned and the mate/inactive SP is unhealthy, unavailable or is unsynchronized:
 - a user-initiated warm restart will not result in a switch of activity.
 - a user-initiated cold restart will not result in a switch of activity. The exception is if the mate/inactive SP is an “IS-ANR,FLT” (In-service, Abnormal, Fault Detected) state, in which case, there will be a switch of activity.
- If SP redundancy is not provisioned, a user-initiated restart will not result in a switch of activity.

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor

- Warm restart is supported on the eMOTR circuit packs. In Release 10.1 and older, restarts on the eMOTR always escalated to cold restarts, impacting traffic for the duration of the restart. The warm restart on the eMOTR circuit pack can be performed through TL-1/Site Manager or automatically triggered (for example, during software upgrades). Note that upgrades to Release 10.2 still require a cold restart of the eMOTR circuit pack. However, upgrades to releases following Release 10.2 will only require a warm restart, avoiding impacts to traffic.
- It is an expected behavior that after a shelf processor restart is performed on a remote network element (RNE), the first login attempt to the RNE will fail. An “Operation Failed” error message appears and you need to log back in a second time to establish the connection to the RNE.

Impact of circuit pack restart**Impact on PM counts**

- For Broadband circuit packs, current and previous bin PM counts are stored on the circuit pack.
- For Photonic circuit packs, current and previous bin PM statistics are stored on the shelf processor. Therefore, a restart of any type on a MSPP and Broadband circuit packs clears untimed, current and previous bin PM counts, and displays “0” for each monitor type along with the invalid data flag (?). Similarly, a restart of the shelf processor clears untimed, current and previous bin PM counts for Photonic circuit packs provisioned on the shelf.

Impact on alarms and time of day

- After a cross-connect is restarted, the Duplicate Shelf alarm is masked for 20 minutes.
- If the time of day (TOD) synchronization feature is enabled, you do not have to reprovision the date and time.
- A warm restart of the SP causes the passive Photonic modules which are connected to the access panel on a SPAP2 w/2xOSC to temporarily raise a “Circuit Pack Missing” alarm.

Impact on cross-connects and timing

- A cold restart of the cross-connect (XC) circuit pack can affect traffic if the timing on the mate cross-connect circuit pack is not locked (Secondary SETS Locking to Primary warning is present).

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor

- If the shelf is equipped with only one working XC or equipped with one working XC and one out-of-service XC, a cold-restart of the working XC will result in a loss of all traffic on cross-connections provisioned through the XC and a loss of shelf timing. Additionally, any OTN I/F, PKT I/F or PKT/OTN I/F circuit pack may automatically restart when it detects that the shelf has no working PKT/OTN XC.

Prerequisites

For Broadband services, a warm or cold restart of an interface circuit pack or shelf processor with GCC0/GCC1 enabled causes a GCC0/GCC1 Link Fail alarm on the adjacent network elements. The GCC0/GCC1 Link Fail alarm will clear automatically.

For Photonic services, restarts are only supported on the LIM (SLA, MLA, MLA2, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, and LIM (C-Band and L-Band)), SAM, ESAM, XLA, SRA, WSS w/OPM, and 2xOSC circuit packs.

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action										
1	Select the network element in the navigation tree.										
2	<table> <tr> <th>If you want to perform a</th><th>Then go to</th></tr> <tr> <td>cold restart</td><td>step 3</td></tr> <tr> <td>warm restart</td><td>step 13</td></tr> </table>	If you want to perform a	Then go to	cold restart	step 3	warm restart	step 13				
If you want to perform a	Then go to										
cold restart	step 3										
warm restart	step 13										
3	<table> <tr> <th>If the circuit pack you are performing a cold restart on</th><th>Then go to</th></tr> <tr> <td>contains facilities in a 1+1 TPT, 1+1 port TPT, configuration</td><td>step 4</td></tr> <tr> <td>is an interface circuit pack</td><td>step 11</td></tr> <tr> <td>is a cross-connect circuit pack</td><td>step 12</td></tr> <tr> <td>is a shelf processor</td><td>step 13</td></tr> </table>	If the circuit pack you are performing a cold restart on	Then go to	contains facilities in a 1+1 TPT, 1+1 port TPT, configuration	step 4	is an interface circuit pack	step 11	is a cross-connect circuit pack	step 12	is a shelf processor	step 13
If the circuit pack you are performing a cold restart on	Then go to										
contains facilities in a 1+1 TPT, 1+1 port TPT, configuration	step 4										
is an interface circuit pack	step 11										
is a cross-connect circuit pack	step 12										
is a shelf processor	step 13										
4	For each facility/path on the circuit pack in a 1+1 TPT, 1+1 port TPT configuration, perform a manual switch to verify the inactive path. Refer to the "Operating a protection switch" procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. If traffic switches back autonomously to the active path, contact your next level of support or your Ciena support group.										

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor

Step	Action
5	For each facility on the circuit pack in a 1+1 port TPT, or 1+1 TPT configuration, perform a protection switch. Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. • If the circuit pack to be restarted is the working one (odd slot), operate a forced switch. • If the circuit pack to be restarted is the protection one (even slot), operate a lockout.
6	If necessary, perform a synchronization protection switch for any line facility on the circuit pack that is a synchronization reference. • Retrieve the synchronization details for timing generation and timing distribution of the circuit pack. Refer to the “Retrieving synchronization protection status details” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. • Perform a manual synchronization protection switch for any active synchronization facility. Refer to the “Operating a synchronization protection switch” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. • Perform a lockout synchronization protection switch for the facility. Refer to the “Operating a synchronization protection switch” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315.
7	If you are performing cold restart on a Broadband circuit pack that supports 1+1 TPT protection, disconnect the client fibers from the inactive circuit pack prior to the cold restart to avoid causing traffic hits on the active circuit pack in the protection group. Go to step 8.

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor

Step	Action
8	If the circuit pack you are performing a cold restart on contains facilities which are all in 1+1 TPT configurations is a protection circuit pack Then go to step 9 step 11
9	Change the facility states to out-of-service for all facilities on the circuit pack. Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. To display the Facility details, you must first select the equipment from the Equipment area of the Equipment & Facility Provisioning application.
10	Change the equipment state of the circuit pack to out-of-service. Refer to the “Changing the primary state of a circuit pack, module, or pluggable” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. When you place the circuit pack out-of-service, any pluggable modules and pluggable transceivers on that circuit pack are automatically placed out-of-service.
11	Select Restart from the Faults drop-down menu.
12	If applicable, select the required shelf from the Shelf drop-down list.
13	Select the circuit pack or shelf processor you want to restart from the Card drop-down list.
14	Select the restart type (warm or cold) from the Restart type drop-down list.
15	Click OK .
16	A Confirm Restart dialog box appears. If you have chosen the ACTIVE SP to be restarted in the previous menu, then the confirmation dialog box appears with “CPU1 (main CPU)”, “CPU2” and “Capture logs before restart” checkboxes. If you have chosen any other circuit pack to be restarted in the previous menu, then the confirmation dialog box appears with only “Capture logs before restart” checkbox. This box is unchecked by default if you have chosen a warm restart. For cold restart, the box is checked by default. For more information about the “Capture logs” feature, refer to <i>Fault Management - Customer Visible Logs</i>, 323-1851-840.

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor**Step Action****ATTENTION**

Some of the L0 circuit packs do not support “Capture logs before restart” option in Site Manager.

- 17 Click **Restart**. For an active SP, the restart takes 8 to 15 minutes. For other circuit packs, the restart will take 4 to 10 minutes to complete. If after the expected time the retrieve log is not completed, Site Manager automatically issues the restart command.

For a warm restart, the red Fail, the green Ready and the blue In Use status LEDs on the front of the circuit pack go through the following sequence:

- The red and blue LEDs remain in the state they were before the restart.
- After a few seconds, the green LED flashes to indicate the software is initializing.
- When software initialization is complete, the green status LED turns on (does not flash).
- If the circuit pack has failed, the green LED turns off and the red LED turns on.

For a cold restart, the red Fail, the green Ready and the blue In Use status LEDs on the front of the circuit pack go through the following sequence:

- The red LED turns on momentarily.
- After a few seconds, the red, green and blue LEDs turn on.
- After a few seconds, the red and blue LEDs turn off and the green LED flashes to indicate the software is initializing.
- When software initialization is complete, the green status LED turns on (does not flash).
- If the circuit pack is in-service, the blue LED turns on after the green LED once the circuit pack is in use. If the circuit pack has failed, the green LED turns off and the red LED turns on.

Procedure 2-11 (continued)

Restarting a circuit pack or shelf processor

Step	Action										
18	<table><tr><th>If you have performed a</th><th>Then</th></tr><tr><td>warm restart</td><td>the procedure is complete</td></tr><tr><td>cold restart on a shelf processor</td><td>the procedure is complete</td></tr><tr><td>cold restart on a cross-connect circuit pack</td><td>go to step 19</td></tr><tr><td>cold restart on an interface circuit pack (converged Broadband and Photonic services)</td><td>go to step 19</td></tr></table>	If you have performed a	Then	warm restart	the procedure is complete	cold restart on a shelf processor	the procedure is complete	cold restart on a cross-connect circuit pack	go to step 19	cold restart on an interface circuit pack (converged Broadband and Photonic services)	go to step 19
If you have performed a	Then										
warm restart	the procedure is complete										
cold restart on a shelf processor	the procedure is complete										
cold restart on a cross-connect circuit pack	go to step 19										
cold restart on an interface circuit pack (converged Broadband and Photonic services)	go to step 19										
19	Change the equipment state of the circuit pack to in-service. Refer to the “Changing the primary state of a circuit pack, module, or pluggable” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.										
20	Change the facility states to in-service for all facilities on the circuit pack. Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.										
21	If applicable, return traffic to the circuit pack by releasing the protection switch for all facilities on the circuit pack that you performed in step 5 . Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. The procedure is complete.										

—end—

Procedure 2-12

Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm

Use this procedure to identify which circuit pack, module/port, or facility has raised an alarm.

Step	Action
1	Retrieve the active alarms on the network element. Refer to Procedure 2-2, "Retrieving active alarms for one or more network elements" .
2	Identify the alarm in the Alarm List. The Unit column in the Consolidated Alarms or Active Alarms application specifies the circuit pack and circuit pack slot/port using the following format: <circuit pack>-<shelf-id>-slot#-subslot# (for subslot equipment alarms) or <facility type>-<shelf-id>-slot#-subslot# (for subslot facility alarms). For example, PTP-8-5-1 refers to an alarm raised by port 1 of the circuit pack in slot 5 on shelf 8. For circuit packs that do not have SFP/SFP+/XFP/CFP/DPO/QSFP/QSFP+/QSFP28, the port number is used instead of the SFP/SFP+/XFP/CFP/DPO/QSFP/QSFP+/QSFP28 number. For example, TRM-7-4-1 refers to an alarm raised by the WLAi MOTR circuit pack in shelf 7, slot 4, port 1. To view details of an alarm, select the alarm in the alarm list. The Alarm details area displays the details of the alarm.

—end—

Procedure 2-13

Clearing audible alarms and performing lamp tests

Use this procedure to clear audible alarms and perform lamp tests on network elements using the ACO button. When you clear an audible alarm, the alarmed LEDs and fault are not cleared.

This procedure does not apply to 2-slot shelf variants. The lamp test on the 2-slot shelf can only be performed using Site Manager.

ATTENTION

Audible alarms are not supported on the 4-slot shelf and 6500-7 Packet-Optical Shelf.

As of Release 9.2, the SPAP-2 w/2xOSC 2xSFP (NTK555NA, NTK555NB) works in the 7-slot optical Type 2 shelf (NTK503KA); however, the ACO/Lamp Test button and LED are located on the fan. Therefore, the Lamp Test button can be applied, but there will be no Audible alarms to cut-off since there will be no audible alarm contacts on a shelf with an SPAP-2 w/2xOSC 2xSFP. Note that for the SPAP-2 w/2xOSC 2xSFP (NTK555NB), the console port does not light up when the lamp test is performed.

The network element has relay contacts that you can connect to both visual and audible alarms. There are four contact pairs for a network element. Therefore, you can connect Critical, Major, minor, and remote alarms to separate audible alarms for a network element.

To clear audible alarms and perform lamp tests using the Site Manager **Visualization** tool, refer to the “Performing a lamp test and clearing audible alarms using the Visualization tool” procedure in *Administration and Security*, 323-1851-301.

Prerequisites

To perform the clearing audible alarms using Site Manager steps, you require an account with at least a level 2 UPC.

Procedure 2-13 (continued)

Clearing audible alarms and performing lamp tests

Step	Action	
1	If you want to	Then go to
	clear audible alarms manually	step 2
	clear audible alarms using Site Manager Faults menu	step 4
	perform a lamp test using the ACO button	step 9
	perform a lamp test using Site Manager	see Note
	Note: Refer to the “Performing a lamp test and clearing audible alarms using the Visualization tool” procedure in <i>Administration and Security</i> , 323-1851-301.	

Clearing audible alarms manually

- 2 Locate the network element with the audible alarm.
- 3 Press the ACO button once to reset the audible alarm relays for the network element.

The ACO LED is lit.

ATTENTION

The ACO button and LED is located with the shelf Critical, Major and Minor alarm LEDs. For example, they are found on the cooling fan module Type 2 in a 4-slot shelf, 7-slot shelf or 6500-7 packet-optical shelf, on the maintenance interface card (MIC) in a 14-slot shelf or on the access panel in a 32-slot shelf.

The procedure is complete.

Procedure 2-13 (continued)

Clearing audible alarms and performing lamp tests

Step	Action
------	--------

Clearing audible alarms using Site Manager Faults menu

- | | |
|---|--|
| 4 | Select the network element in the navigation tree. |
| 5 | Select Alarm Cut-Off from the Faults drop-down menu.
The Alarm Cut-Off dialog box is displayed. |
| 6 | If applicable, select the required shelf from the Shelf drop-down list. |
| 7 | Select All from the Source drop-down list. |
| 8 | Click OK .
The procedure is complete. |

Performing a lamp test using the ACO button

- | | |
|----|--|
| 9 | Locate the required network element. |
| 10 | Press the ACO button for two seconds, and then release (See the “Attention” below). If there are: <ul style="list-style-type: none">• no audible alarms, pressing the ACO button activates a lamp test.• audible alarms, pressing the ACO button clears the audible alarms (ACO LED is lit). To perform a lamp test, you must press the ACO button a second time. |

ATTENTION

The ACO button and LED is located with the shelf Critical, Major and Minor alarm LEDs. For example, they are found on the cooling fan module Type 2 in a 4-slot shelf, 7-slot shelf or 6500-7 packet-optical shelf, on the maintenance interface card (MIC) in a 14-slot shelf or on the access panel in a 32-slot shelf.

The ACO LED remains lit until a new Major alarm is raised. Upon detection of a Major alarm, the ACO releases and the LED turns off.

ATTENTION

On some access panels, the RJ45 external slot inventory interfaces appear to have two LEDs but only one is activated during a lamp test or when the external slot is in use (properly connected to provisioned external equipment such as a PPC6, CMD44, DSCM, OMD4, BMD2, UBMD2, MBMD2, or DSCM).

—end—

Procedure 2-14

Provisioning environmental alarm attributes

Use this procedure to:

- retrieve environmental alarm attributes
- set up or change environmental alarm attributes on the network element
- delete defined environmental alarm attributes on the network element

Environmental alarm attributes require resetting if you replace an existing environmental alarm with a different type of input. For example, when you replace a humidity alarm with a toxic gas detector, you must edit the environmental alarm attributes.

ATTENTION

When you remove a device for detecting an environmental alarm, delete the environmental alarm attributes.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	Select the network element in the navigation tree.						
2	Select Alarms & Controls from the Configuration menu.						
3	Select External Alarm Provisioning .						
4	Select a shelf from the Shelf drop-down list.						
5	Select a source from the Source drop-down list.						
6	<table border="0"> <tr> <td>If you want to</td><td>Then go to</td></tr> <tr> <td>edit the environmental alarm attributes</td><td>step 7</td></tr> <tr> <td>delete the environmental alarm attributes</td><td>step 16</td></tr> </table>	If you want to	Then go to	edit the environmental alarm attributes	step 7	delete the environmental alarm attributes	step 16
If you want to	Then go to						
edit the environmental alarm attributes	step 7						
delete the environmental alarm attributes	step 16						
7	Select any entry in the contact list to enable the Edit button.						
8	Click Edit to open the Edit External Alarm dialog box.						
9	Select the contact you want to set or edit from the Contact drop-down list.						

Procedure 2-14 (continued)

Provisioning environmental alarm attributes

Step	Action								
10	Select the label from the Label drop-down list. Refer to “Environmental alarm labels” on page 2-3 .								
11	Select the severity from the Severity drop-down list.								
12	Edit the description if you want to describe the alarm with specific text. The description can contain a maximum of 40 characters.								
ATTENTION Do not use apostrophes “ ‘ ” in the alarm text. Using apostrophes in the alarm text will cause display errors in the MCP active events list (AEL).									
13	Click Apply .								
14	Repeat step 9 through step 13 if you want to set or edit more contacts.								
15	Click OK . The procedure is complete.								
16	Select the provisioned contact from which you want to delete attributes in the contact list as follows:								
<table> <tr> <th>If you want to delete</th><th>Then</th></tr> <tr> <td>one entry</td><td>click the entry you want to delete</td></tr> <tr> <td>some, but not all entries</td><td>select the first entry in the list and hold down the Ctrl key while individually clicking on each required entry</td></tr> <tr> <td>all entries</td><td>select the first entry in the list and hold down the Shift key while clicking once on the last entry in the list or select any desired entry in the list and then Ctrl+A to select all entries</td></tr> </table>		If you want to delete	Then	one entry	click the entry you want to delete	some, but not all entries	select the first entry in the list and hold down the Ctrl key while individually clicking on each required entry	all entries	select the first entry in the list and hold down the Shift key while clicking once on the last entry in the list or select any desired entry in the list and then Ctrl+A to select all entries
If you want to delete	Then								
one entry	click the entry you want to delete								
some, but not all entries	select the first entry in the list and hold down the Ctrl key while individually clicking on each required entry								
all entries	select the first entry in the list and hold down the Shift key while clicking once on the last entry in the list or select any desired entry in the list and then Ctrl+A to select all entries								
17	Click Clear Entry .								
18	Click Yes in the confirmation dialog box.								

—end—

Procedure 2-15

Provisioning, operating, and releasing external controls

Use this procedure to:

- retrieve the labels and status of all external controls
- provision control labels and types to control relays on the network element. The network element allows four external control relays to turn external equipment on and off.
- operate external controls
- release external controls

Prerequisites

To edit external controls, you require an account with at least a level 3 UPC.

To operate or release external control equipment, you require an account with at least a level 2 UPC.

Step	Action								
1	Select the required network element in the navigation tree.								
2	Select Alarms & Controls from the Configuration drop-down menu.								
3	Select External Controls .								
4	Select a shelf from the Shelf drop-down list.								
5	Select a source from the Source drop-down list.								
6	<table> <tr> <th>If you want to</th><th>Then go to</th></tr> <tr> <td>edit external control attributes</td><td>step 7</td></tr> <tr> <td>operate (turn on) external control equipment</td><td>step 11</td></tr> <tr> <td>release external control equipment</td><td>step 14</td></tr> </table>	If you want to	Then go to	edit external control attributes	step 7	operate (turn on) external control equipment	step 11	release external control equipment	step 14
If you want to	Then go to								
edit external control attributes	step 7								
operate (turn on) external control equipment	step 11								
release external control equipment	step 14								
7	Click Edit .								
8	Select the relay label from the drop-down list at the relay for which you want to set or edit attributes. Refer to “External control types” on page 2-2 .								
9	Repeat step 8 if you want to set or edit more relays.								
10	Click OK .								
	The procedure is complete.								

Procedure 2-15 (continued)

Provisioning, operating, and releasing external controls

Step	Action
11	Select the required relay.
12	Click Operate .
13	Click Yes in the confirmation dialog box. The procedure is complete.
14	Select the required relay.
15	Click Release .
16	Click Yes in the confirmation dialog box.

—end—

Procedure 2-16

Locating a reflective event

Use this procedure to locate connector losses or a reflective event at a line amplifier site.

In this procedure you will attempt to isolate the connector losses by starting at the connector that is farthest from the alarmed amplifier and then working back towards the alarmed amplifier. Refer to [Figure 2-1 on page 2-46](#) for an example of an alarmed amplifier at a line amplifier site.

ATTENTION

This procedure involves wrapping optical fiber around a mandrel to create optical power attenuations. Winding the fiber too tightly will damage optical fibers and the optical power attenuations can generate additional alarms on the system.

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in *Installation - General Information*, 323-1851-201.0.
- have a network diagram that identifies all connection points at the site of the alarmed module that can be cleaned
- have the appropriate personal grounding device to dissipate electrostatic charges
- have a mandrel or screwdriver

Step	Action
1	CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage.
2	Review the network diagram, and locate the connector that is farthest from the alarmed amplifier within the site.



Connect the wrist strap to the ESD jack on the shelf or module.

Review the network diagram, and locate the connector that is farthest from the alarmed amplifier within the site.

Procedure 2-16 (continued)
Locating a reflective event

Step	Action								
3	Wind the optical fiber patchcord that is attached to the connector, identified in step 2, four turns around a 15-mm mandrel. You can use a screwdriver handle or other similar sized cylinder in place of a mandrel. This step checks for high reflection (low return loss) by causing attenuation of the optical power in the direction back towards the amplifier output.								
4	<table> <tr> <th>If the original alarm</th><th>Then</th></tr> <tr> <td>does not clear, and there are more optical fiber patchcords before the alarmed amplifier</td><td> the reflective event is closer to the alarmed amplifier. Go to step 5. </td></tr> <tr> <td>does not clear, and there are no more optical fiber patchcords before the alarmed amplifier at this site (you are now at the output of the alarmed amplifier)</td><td> If you were sent from another procedure, return to the step in the procedure that referred you to this procedure. Otherwise, contact your next level of support or your Ciena support group. </td></tr> <tr> <td>clears</td><td> Go to step 6. </td></tr> </table>	If the original alarm	Then	does not clear, and there are more optical fiber patchcords before the alarmed amplifier	the reflective event is closer to the alarmed amplifier. Go to step 5.	does not clear, and there are no more optical fiber patchcords before the alarmed amplifier at this site (you are now at the output of the alarmed amplifier)	If you were sent from another procedure, return to the step in the procedure that referred you to this procedure. Otherwise, contact your next level of support or your Ciena support group.	clears	Go to step 6.
If the original alarm	Then								
does not clear, and there are more optical fiber patchcords before the alarmed amplifier	the reflective event is closer to the alarmed amplifier. Go to step 5.								
does not clear, and there are no more optical fiber patchcords before the alarmed amplifier at this site (you are now at the output of the alarmed amplifier)	If you were sent from another procedure, return to the step in the procedure that referred you to this procedure. Otherwise, contact your next level of support or your Ciena support group.								
clears	Go to step 6.								
5	Locate the next farthest connector from the alarmed amplifier and go to step 3.								
6	Place the alarmed amplifier out-of-service (OOS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.								

Procedure 2-16 (continued)
Locating a reflective event

Step	Action
------	--------

7

**DANGER****Risk of laser radiation exposure**

Do not look directly into the optical beam. Invisible light can severely damage your eyes.

**CAUTION****Risk of damage to circuit packs**

Never disconnect an optical fiber that is connected to an active or powered up optical amplifier. To disconnect or reconnect an optical fiber, make sure the optical amplifier is out of service (OOS), then disconnect or reconnect the fiber.

**CAUTION****Risk of damage to circuit packs**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

Clean and then reconnect the output fibers and connectors at the amplifier. Refer to the “Inspecting and cleaning optical interface connectors” and “Cleaning optical connectors and adapters on patchcords” procedures in *Installation - General Information*, 323-1851-201.0 .

8

Place the amplifier back in-service (IS). Refer to the “Changing the primary state of a facility” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.

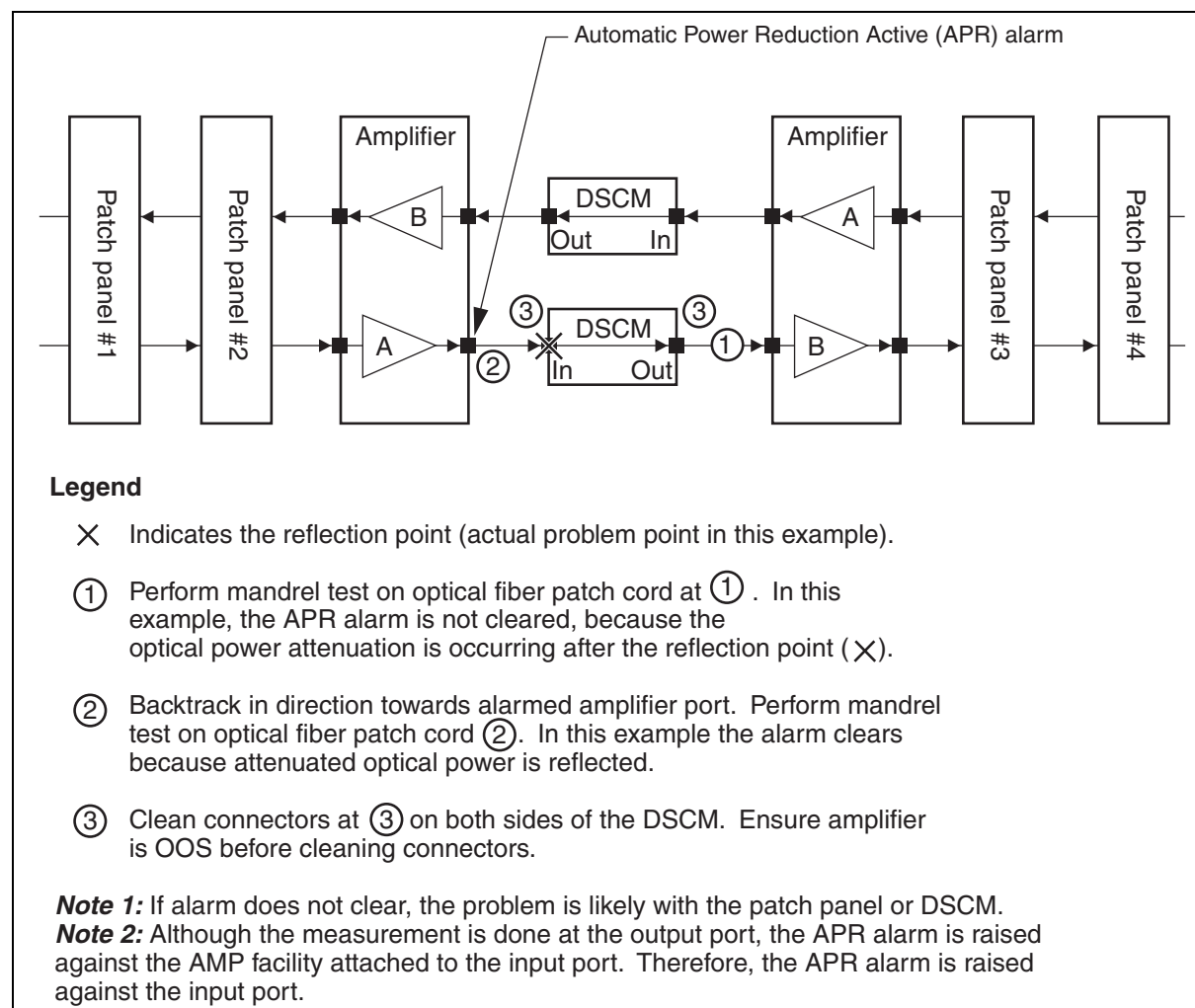
9

If the original alarm Then

clears	the procedure is complete
does not clear	return to the step in the procedure that referred you to this procedure or contact your next level of support or your Ciena support group

—end—

Figure 2-1
Locating connector losses at a Photonic line amplifier site (with DSCM)



Procedure 2-17

Preparing to perform fiber work on a Photonic system

Use this procedure to record existing power levels and amplifier gain settings before performing fiber maintenance/repair and to re-adjust these parameters if the fiber maintenance/repair causes power levels to change.

Note that this procedure cannot be used to repair a fiber cut that occurred. This procedure can only be used to prepare in advance for a fiber cut that you know will occur (for example, related to scheduled maintenance activities).



CAUTION

Risk of traffic loss

Placing an amplifier circuit pack OOS causes a traffic loss. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure.



CAUTION

Risk of damage to downstream amplifier

You must complete this procedure on an amplifier that is downstream of the location where the fiber cut or maintenance activity will be performed. Not performing this procedure can result in power levels at the amplifier that are high enough to damage the module.

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in *Installation - General Information*, 323-1851-201.0
- have a network diagram that identifies all connection points at the site of the alarmed module that can be cleaned

Step	Action
1	Log into the network element containing the amplifier that is downstream of the location where the fiber cut or maintenance activity will be performed.
2	From the Configuration menu, select Equipment & Facility Provisioning .

Procedure 2-17 (continued)

Preparing to perform fiber work on a Photonic system

Step	Action
3	Select the LIM circuit pack supporting the AMP facility (of the amplifier that is downstream of the location where the fiber cut or maintenance activity will be performed) in the Equipment area.
4	From the Facility Type drop-down menu, select AMP.
5	Record the Target Gain for the AMP facility of the amplifier that is downstream of the location where the fiber cut or maintenance activity will be performed.
6	From the Performance menu, select Performance Monitoring and then New .
7	From the Type drop-down list, select AMP.
8	From the Facility drop-down list, select the AMP facility of the amplifier that is downstream of the location where the fiber cut or maintenance activity will be performed.
9	Retrieve the PMs for the AMP facility.
10	Record the current (Untimed) Input Power (OPIN).
11	From the Configuration menu, select Photonics Services and then Domain Optical Controller (DOC) .
12	Change the DOC Primary State to out of service (OOS). Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
13	 CAUTION Risk of traffic loss Placing an amplifier circuit pack OOS causes traffic loss. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure. Place the amplifier downstream of the fiber cut/maintenance OOS. Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.
14	Perform the maintenance/repair fiber according to the safety requirements of your company and the safety requirements described in <i>Installation - General Information</i> , 323-1851-201.0 .
15	Repeat step 1 to step 10 .
16	Compare the current (Untimed) Input Power (OPIN) to the previously recorded value in step 10 .

Procedure 2-17 (continued)

Preparing to perform fiber work on a Photonic system

Step	Action
17	If the Input Power (OPIN) value has decreased by more than 3 dB Then check the fiber splice and fiber connections. It is likely that the splice is poor or there is a dirty connection. Correct the problem and go to step 18. increased by more than 3 dB provision the downstream OOS amplifier Target Gain downwards by the difference in the input power. For example, if the input power increased by 4 dB and the previous amplifier Target Gain was 18 dB, the new amplifier Target Gain will be 18 dB - 4 dB = 14 dB. otherwise Go to step 18. go to step 18
18	Place the amplifier downstream of the fiber cut/maintenance from step 13 back in service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
19	Place the DOC Primary state back in service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
20	Reset all threshold crossing alert (TCA) baselines for the DOC domain. Refer to the “Resetting TCA baselines” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
21	Record the current DOC Automation mode . Change the DOC Automation mode to “Enhanced”. Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
22	Wait for DOC to perform optimization. This may take up to five minutes to begin.
23	Once optimization is complete, return the DOC Primary state and DOC Automation mode to their original settings.

—end—

Procedure 2-18

Measuring Photonic amplifier output power

Use this procedure to measure the output power of an amplifier to facilitate Photonic line troubleshooting.

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements in *Installation - General Information*, 323-1851-201.0
- have a network diagram that identifies all connection points at the site of the alarmed module that can be cleaned
- have a high-powered optical power meter with the same optical connectors as the network element that can read power levels as high as +11 dB
- have a 10 dB fixed-pad optical attenuator (ensure the attenuator loss value has been calibrated)
- have a personal grounding device



CAUTION

Risk of traffic loss

Disconnecting fibers causes traffic loss on the associated facilities. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure.

Step	Action
------	--------

- | | |
|---|--|
| 1 | <p>In the Domain Optical Controller (DOC) application in Site Manager, record the current Automation mode for the two DOC network elements that manage each optical direction of the amplifier.</p> <p>Change the Automation mode to “Enhanced Auto Monitor Only” (if not already). Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.</p> |
|---|--|

Procedure 2-18 (continued)
Measuring Photonic amplifier output power

Step	Action				
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.				
	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage.				
3	Place a 10 dB fixed-pad optical attenuator at the input to the optical power meter (to prevent raising the Automatic Power Reduction Active alarm).				
4	If you want to measure the amplifier output power of Then go to <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 60%;">port 5 (Line B)</td><td style="text-align: right;">step 5</td></tr> <tr> <td>port 7 (Line A)</td><td style="text-align: right;">step 9</td></tr> </table>	port 5 (Line B)	step 5	port 7 (Line A)	step 9
port 5 (Line B)	step 5				
port 7 (Line A)	step 9				
5	In the Equipment & Facility Provisioning application in Site Manager, record the current AMP facility Shut Off Threshold (dBm) value for Line A (port 8) on both line-facing ends of the amplifier. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Provision the AMP facility Shutoff Threshold to -60 dBm for Line A (port 8) on both line-facing ends of the amplifier. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. ATTENTION Provisioning the SAM, ESAM or AMP facility Shutoff Threshold to -60 dBm disables the ALSO safety feature, and injury can occur. The Automatic Shutoff Disabled alarm is raised. ATTENTION ALSO can be disabled on the LIM cards (C-Band and L-Band) by setting the ALSO Disable flag to TRUE for the OPTMON facility. This action should only be performed when there is no OSC in the OTS for automatic recovery from optical line fail condition.				
6	Remove the OSC B In (port 3) fiber from the amplifier. This allows you to measure the amplifier output only. If the OSC B In (port 3) fiber is not removed, you are also including the OSC channel in the measurement.				

Procedure 2-18 (continued)

Measuring Photonic amplifier output power

Step	Action
7	DANGER Risk of laser radiation exposure Do not look directly into the optical beam. Invisible light can severely damage your eyes.
	CAUTION Risk of traffic loss Disconnecting fibers causes traffic loss on the associated facilities. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure.
	Measure the port 5 (Line B) amplifier output power using the high-powered optical power meter. Add the calibrated attenuator loss to the measured value. This total power is 10 dB lower due to the attenuator inserted into the power meter.
8	Clean and reinstall the OSC B In (port 3) fiber removed in step 6. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0 Go to step 10.
9	DANGER Risk of laser radiation exposure Do not look directly into the optical beam. Invisible light can severely damage your eyes.
	CAUTION Risk of traffic loss Disconnecting fibers causes traffic loss on the associated facilities. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure.



Procedure 2-18 (continued)

Measuring Photonic amplifier output power

Step	Action
	Measure the port 7 (Line A) amplifier output power using the optical power meter. Add the calibrated attenuator loss to the measured value. This total power is 10 dB lower due to the attenuator inserted into the power meter.
10	When the power measurement is complete, clean all connections of the optical fiber link following your company standards, and restore the fiber connection to the amplifier output port. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0 .
11	If port 5 (Line B) was measured, then restore the AMP facility Shut Off Threshold (dBm) value for Line A (port 8) to the original settings recorded in step 5 for both line-facing ends of the amplifier. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
12	In the Domain Optical Controller (DOC) application in Site Manager, restore the Automation mode to the original settings recorded in step 1 for both DOC network elements. Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

—end—

Alarm hierarchies and alarm severities

ATTENTION

Alarm severities described in this section are the default alarm severities provisioned on the system. Alarm severities can be modified by using different alarm profiles. Refer to [Chapter 2, “Alarm surveillance”](#) of this document for details on how to change alarm severities.

Alarm severities

The level of severity for alarms are Critical (C), Major (M), and minor (m). Alarm reports always contain a notification code that identifies the alarm severity, or the code CL to indicate that the fault has been cleared. The w code indicates a warning. The A code indicates an alert (only applicable to alarm banner). Event reports have a logged (Log) severity.

Note: Alarm severities can be modified (protected/unprotected). Refer to [Procedure 2-8, “Editing an alarm profile”](#).

Critical alarms (C)

Critical alarms are the most severe. Critical alarms always indicate a service-affecting fault. For example, unprotected facility losses and unprotected facility-carrying equipment failures raise critical alarms.

Major alarms (M)

Major alarms are less severe than critical alarms but can be service-affecting or non-service-affecting.

Major alarms are raised when something has an effect on a low-speed facility. For example, a Major alarm is raised when tributary signals fail or unprotected provisioned circuit packs are missing.

Minor alarms (m)

Minor alarms are less severe than Major alarms, but can be service-affecting or non-service-affecting.

For example, a non-service-affecting minor alarm is raised when a protected circuit fails. However, an AIS service-affecting minor alarm raises when a 1+1/MSP linear protected configuration does not have protection available.

Cleared alarm notification (CL)

The cleared notification code indicates that the fault no longer exists. The automatic output cache stores the cleared alarm reports.

Warning (w)

Warning events are raised by the network element as Standing Conditions (SC), and are less severe than minor alarms. A warning is an indication a problem exists on the network element that can eventually escalate into an alarm of higher severity.

Alert (A)

Threshold-crossing alerts are less severe than alarms. An alert can indicate that the threshold crossed does not affect service but requires further investigation.

Alerts are indicated in the alarm banner and appear with a Logged (Log) severity in the Events window.

Note: Performance high and low threshold crossings are not displayed in Site Manager as alarms, but they are displayed in MCP as alarms.

Logged (Log)

Event reports are generated from changes of state and other important transient conditions.

Service-affecting and non-service-affecting severities

By default, alarms generally have both a service-affecting (high impact) severity and non-service-affecting (low impact) severity. The triggered severity depends on the system state when the alarmed conditions occurred. For some alarms, there is only one severity. The user can perform alarm provisioning, which can be used to enable/disable alarm points, and change the service-affecting (high impact) severity and non-service-affecting (low impact) severities of alarms. Refer to [“Procedures for alarm provisioning and alarm profiles” on page 2-10](#).

For example, a Circuit Pack Fail alarm can be raised with a:

- high impact of Critical, service-affecting (C, SA) for a circuit pack in an unprotected configuration with cross-connects
- low impact of minor, non-service-affecting (m, NSA) for a circuit pack without cross-connects

Broadband services circuit pack alarms

The service-affecting (high impact) severity is applicable when the circuit pack is unprotected (that is, either it cannot be protected, or can be protected but currently is not) and has a connected traffic facility.

The non-service-affecting (low impact) severity is applicable when the circuit pack is:

- protected but inactive
- unprotected and has a traffic facility that is either disconnected or protected but inactive

Broadband services facility alarms

The service-affecting (high impact) severity is applicable when the facility is connected (Broadband facilities are connected but not through the cross-connect circuit pack) and unprotected.

The non-service-affecting (low impact) severity is applicable when the facility is:

- disconnected
- protected, but inactive

Photonic services alarms

Photonic alarms do not support multiple severity levels. That is, there is only one severity per alarm.

Alarm hierarchies

The following alarm hierarchy diagrams are made to be as generic and simple as possible. Therefore, not every alarm or circumstance shown applies to all circuit packs.

- [“Overall alarm hierarchy” on page 3-8](#)
- [“Equipment alarm hierarchy \(circuit packs and modules\)” on page 3-9](#)
- [“Shelf Processor alarm hierarchy” on page 3-10](#)
- [“Equipment alarm hierarchy \(provisioned pluggable\)” on page 3-10](#)
- [“Equipment alarm hierarchy \(unprovisioned pluggable\)” on page 3-11](#)
- [“Shelf equipment alarm hierarchy” on page 3-12](#)
- [“ETH10G and FC1200 facility alarm hierarchy” on page 3-13](#)
- [“FC1200 alarm hierarchy for 2x10G OTR, 4x10G OTR, 10x10G MUX, and 4x10G MUX circuit packs” on page 3-14](#)
- [“FC400/FC800 alarm hierarchy for 2x10G OTR, 4x10G OTR, 10x10G MUX, and 4x10G MUX circuit packs” on page 3-15](#)
- [“OTM2 Lines alarm hierarchy on 2x10G OTR circuit pack” on page 3-16](#)

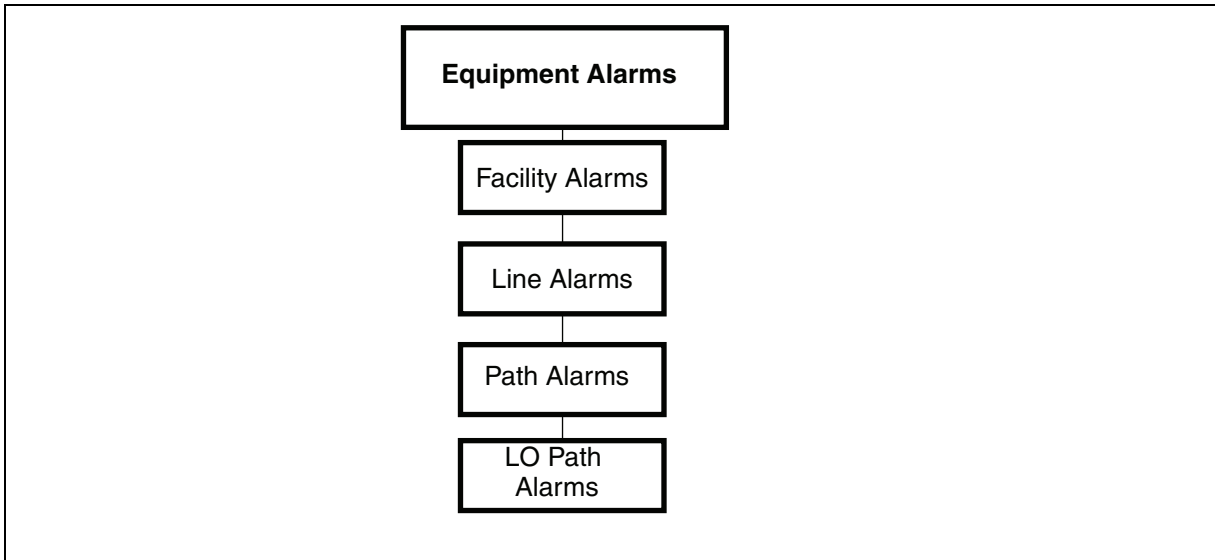
- “4x10G OTR (NTK530QA) circuit pack infiniband alarm hierarchy” on page 3-16
- “4x10G OTR w/ Encryption (NTK530QE) circuit pack alarm hierarchy” on page 3-17
- “100G OCI and 100G WL3n/WL3e OTR circuit pack ETH100G facility alarm hierarchy” on page 3-18
- “100G WL3n MOTR circuit pack ETH10G facility alarm hierarchy” on page 3-19
- “40G OCI+ CFP circuit pack ETH40G facility alarm hierarchy” on page 3-20
- “40G MUX OCI circuit pack 10GbE WAN facility alarm hierarchy” on page 3-20
- “LAN/WAN EPL facilities alarm hierarchy (ETH, ETH100, FC100/FC200/FC400, GE, and WAN facilities)” on page 3-21
- “FLEX MOTR and Broadband circuit packs Fiber Channel, 8B10B, and Transparent client protocols alarm hierarchy” on page 3-23
- “8xOTN Flex MOTR and (1+8)xOTN Flex MOTR circuit packs Fiber Channel, 8B10B, and Transparent client protocols alarm hierarchy” on page 3-24
- “8xOTN Flex MOTR and (1+8)xOTN Flex MOTR circuit packs WAN facility alarm hierarchy” on page 3-25
- “Broadband circuit packs client port facility OTM mapping layer alarm hierarchy” on page 3-26
- “(1+8)xOTN Flex MOTR client port facility (low-order ODU0, ODU1, and ODU FLEX facilities) alarm hierarchy” on page 3-27
- “OPU-CSF/GFP-CSF client port alarm hierarchy (OTM and Wan facilities)” on page 3-28
- “OTN PKT facility alarm hierarchy - faceplate to fabric direction (RX)” on page 3-29
- “Alarm hierarchy for eMOTR circuit pack” on page 3-30
- “ETTP facility alarm hierarchy for eMOTR circuit pack” on page 3-31
- “ODUk CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)” on page 3-32
- “ODUj CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)” on page 3-33
- “ODUi CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)” on page 3-34
- “OTN facility alarm hierarchy - fabric to faceplate direction (TX)” on page 3-35

- “OTN facility alarm hierarchy - fabric to faceplate direction (TX)” on page 3-35
- “ETTP facility alarm hierarchy - monitored” on page 3-36
- “ETTP facility alarm hierarchy - terminated” on page 3-37
- “ETTP facility alarm hierarchy mapped within OTN port” on page 3-38
- “ODUTTP ETH facility alarm hierarchy” on page 3-39
- “PKT facility alarm hierarchy - faceplate to fabric direction (RX)” on page 3-40
- “Mapped ETTP facility alarm hierarchy” on page 3-41
- “100G OCLD circuit packs line port OTM4 facility alarm hierarchy” on page 3-42
- “100G WL3n MOTR circuit packs client port OTM2 facility alarm hierarchy” on page 3-43
- “100G WL3n MOTR circuit packs line port OTM4 facility alarm hierarchy” on page 3-44
- “100G WL3n MOTR circuit pack ETH40G client facility alarm hierarchy” on page 3-45
- “100G WL3n MOTR circuit pack Flex FC800/1200 facility alarm hierarchy” on page 3-46
- “ODU Terminated CTP facility alarm hierarchy” on page 3-47
- “WLAi circuit pack transponder equipment alarm hierarchy” on page 3-48
- “WLAi circuit pack QSFP pluggable equipment alarm hierarchy” on page 3-49
- “WLAi MOTR circuit pack 100GE client port alarm hierarchy” on page 3-50
- “WLAi circuit pack STTP facility alarm hierarchy” on page 3-52
- “WLAi MOTR circuit pack OTU4 client port alarm hierarchy” on page 3-53
- “WLAi circuit pack OTUTTP and ODUCTP monitored client port alarm hierarchy” on page 3-54
- “WLAi circuit pack Terminated ODUCTP facility alarm hierarchy” on page 3-55
- “WLAi circuit pack CBRTTP facility alarm hierarchy” on page 3-56
- “WLAi and WL5e circuit packs TCM CTP (monitor) facility alarm hierarchy” on page 3-57
- “WLAi and WL5e circuit packs TCM CTP (terminated) facility alarm hierarchy” on page 3-58
- “WLAi and WL5e circuit pack TCM TTP facility alarm hierarchy” on page 3-59

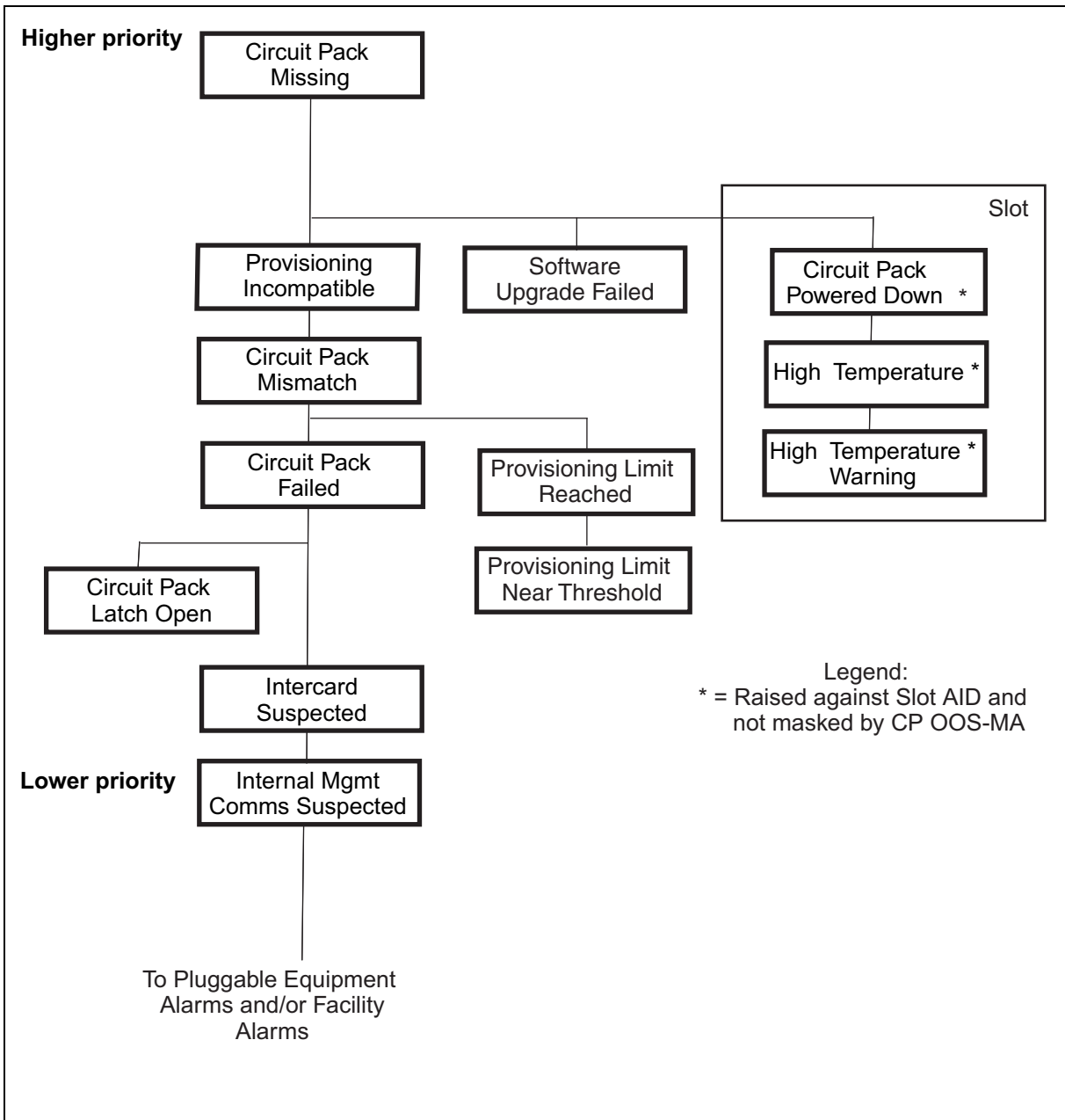
- “WLAi circuit pack PTP and ODUCTP facility for ODUk on line port alarm hierarchy” on page 3-60
- “Flex3 WL3e OCLD circuit packs line port OTMC2 facility alarm hierarchy” on page 3-67
- “Flex3 WL3e OCLD circuit packs line port ODU4 facility in 16QAM alarm hierarchy” on page 3-68
- “100G WL3n/WL3e OTR circuit packs line port OTM4 facility in QPSK alarm hierarchy” on page 3-69
- “100G WL3n/WL3e OTR circuit packs client port OTM4 facility alarm hierarchy” on page 3-70
- “100G WL3n/WL3e OTR circuit packs OTM4 mapping facility on client port alarm hierarchy” on page 3-71
- “10x10GE MUX OCI circuit pack backplane port 100 OTM4 facility alarm hierarchy” on page 3-72
- “100G OCI circuit pack client port OTM4 facility alarm hierarchy” on page 3-73
- “100G OCI circuit pack client port OTM4 mapping layer facility alarm hierarchy” on page 3-74
- “200G (2x100G/5x40G) MUX circuit pack client port ETH10G facility alarm hierarchy” on page 3-75
- “200G (2x100G/5x40G) MUX circuit pack client port ETH40G facility alarm hierarchy” on page 3-76
- “200G (2x100G/5x40G) MUX circuit pack client port ETH100G facility alarm hierarchy” on page 3-77
- “200G (2x100G/5x40G) MUX circuit pack client port OC192/OC768 facility alarm hierarchy” on page 3-78
- “200G (2x100G/5x40G) MUX circuit pack client port OTM2 mapping layer facility alarm hierarchy” on page 3-79
- “200G (2x100G/5x40G) MUX circuit pack client port OTM3 mapping layer facility alarm hierarchy” on page 3-80
- “200G (2x100G/5x40G) MUX circuit pack client port OTM4 facility alarm hierarchy” on page 3-81
- “200G (2x100G/5x40G) MUX circuit pack client port OTM4 mapping layer facility alarm hierarchy” on page 3-82
- “200G (2x100G/5x40G) MUX circuit pack client port OTM4 muxing layer facility alarm hierarchy” on page 3-83
- “200G (2x100G/5x40G) MUX circuit pack WAN facility alarm hierarchy” on page 3-84

- “OTM2 mapping facility (associated with FC1200, OC-192/STM-64, and ETH10G facility) alarm hierarchy” on page 3-85
- “TCMCTP (terminated) facility alarm hierarchy - backplane to faceplate direction” on page 3-86
- “Photonic optical signal facilities alarm hierarchy” on page 3-87
- “SRA circuit pack RAMAN facility alarm hierarchy” on page 3-88
- “SRA circuit pack OSC facility alarm hierarchy” on page 3-89
- “SRA circuit pack ADJ facility alarm hierarchy” on page 3-89
- “SAM and ESAM circuit packs OPTMON facility alarm hierarchy” on page 3-90
- “CCMD12, MLA3 L-Band, LIM L-Band, and WSS Flex L-Band w/OPM 8x1 circuit packs OPTMON facility alarm hierarchy” on page 3-90
- “CCMD12, GMD10 L-Band, MLA3 L-Band, LIM L-Band, UCS, and WSS Flex L-Band w/OPM 8x1 circuit packs adjacency facility alarm hierarchy” on page 3-91

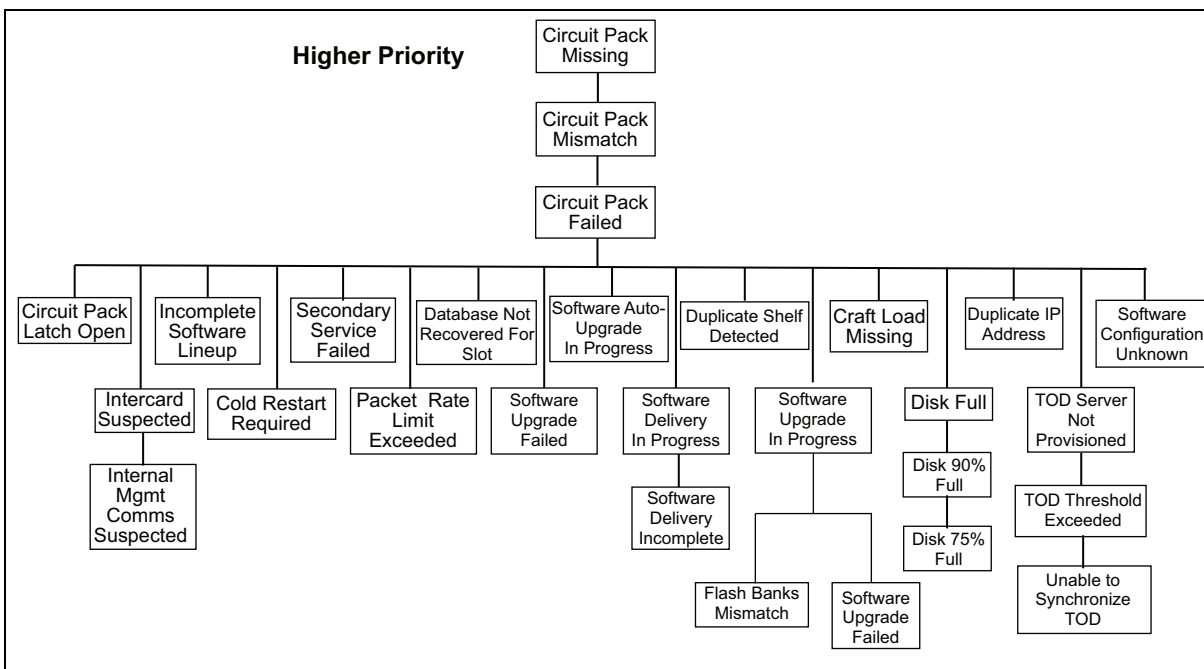
Overall alarm hierarchy



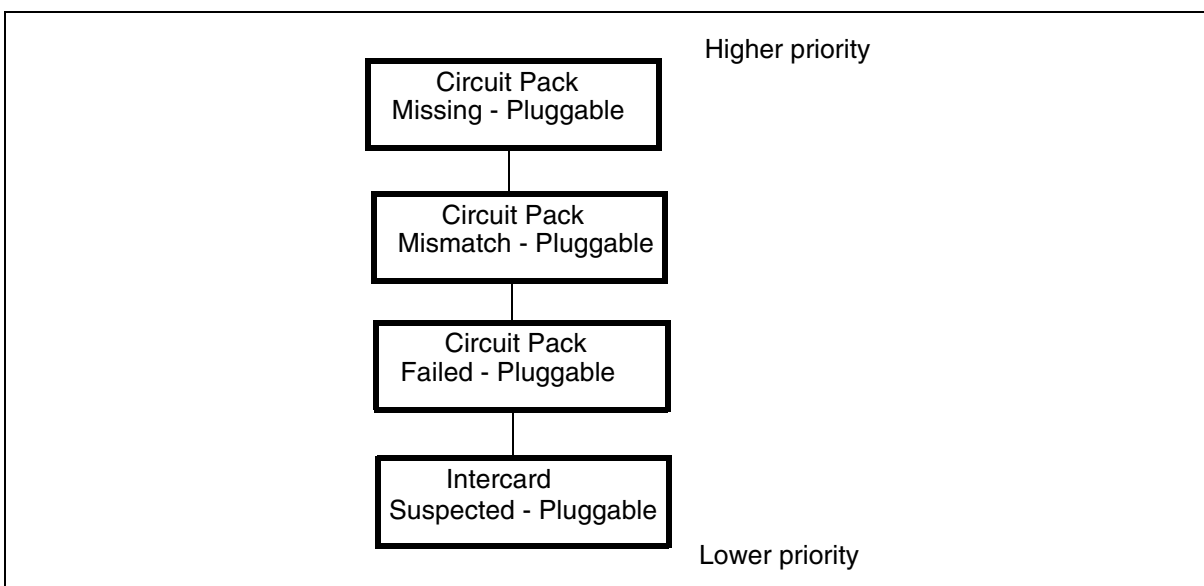
Equipment alarm hierarchy (circuit packs and modules)



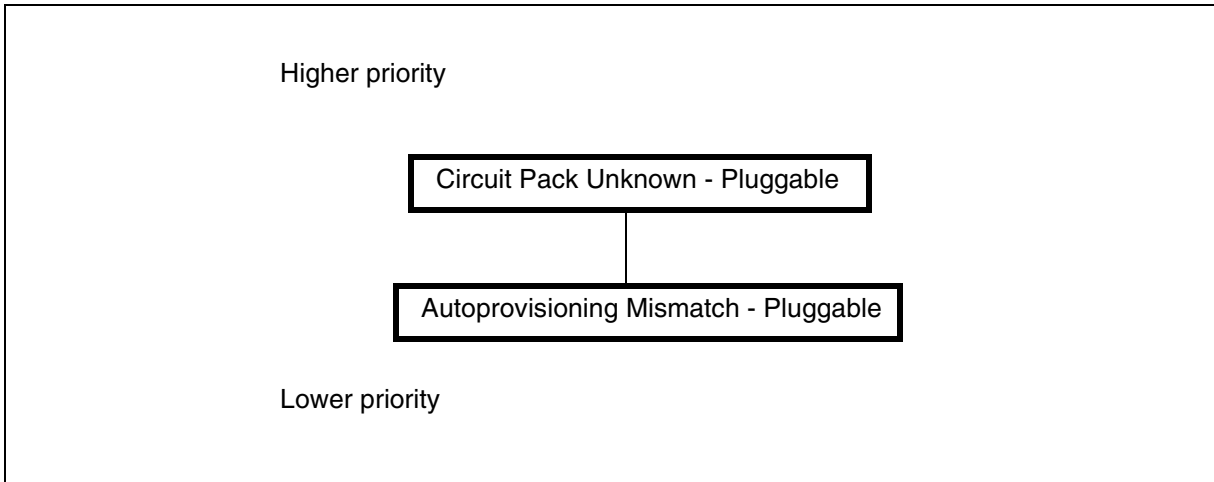
Shelf Processor alarm hierarchy



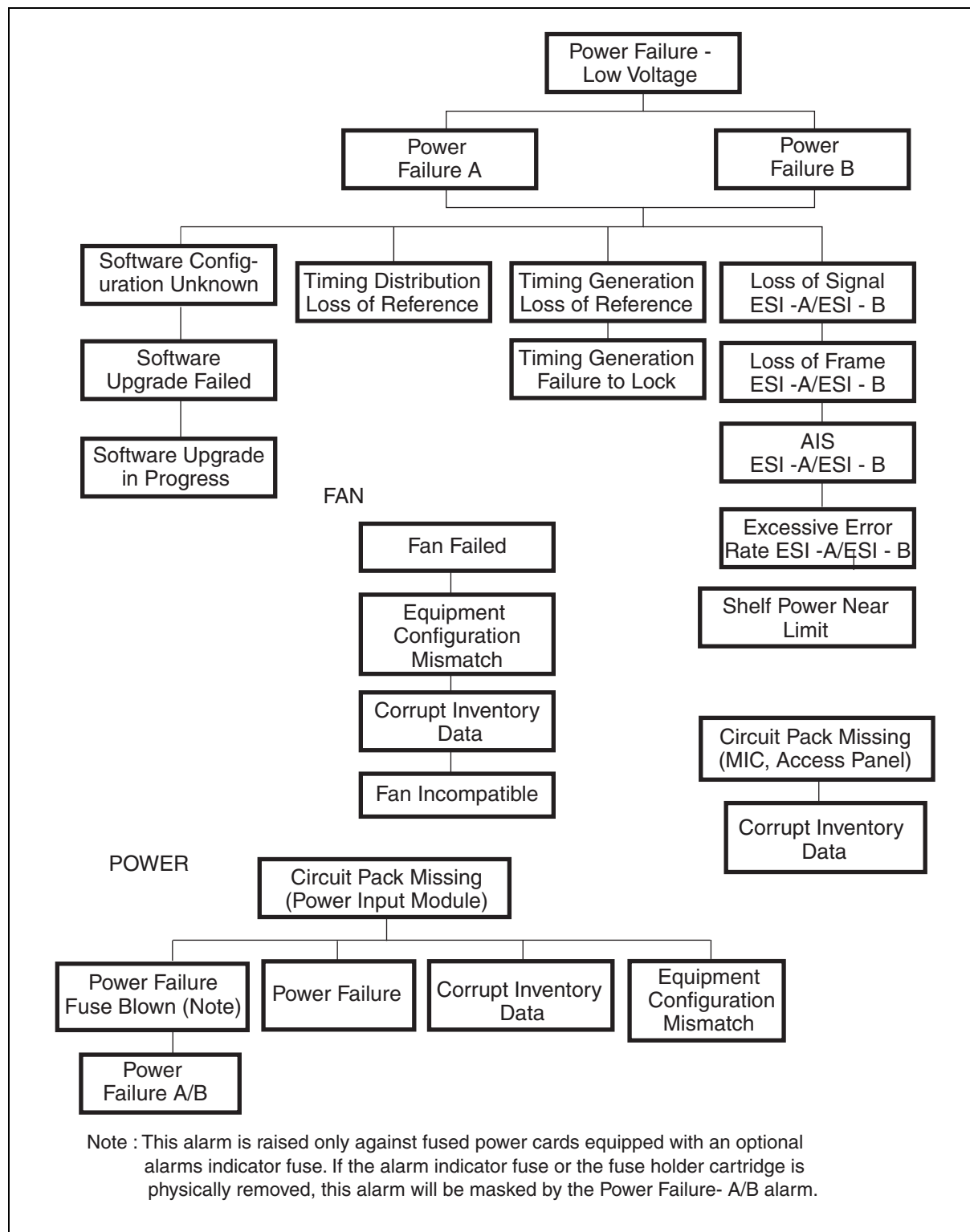
Equipment alarm hierarchy (provisioned pluggable)



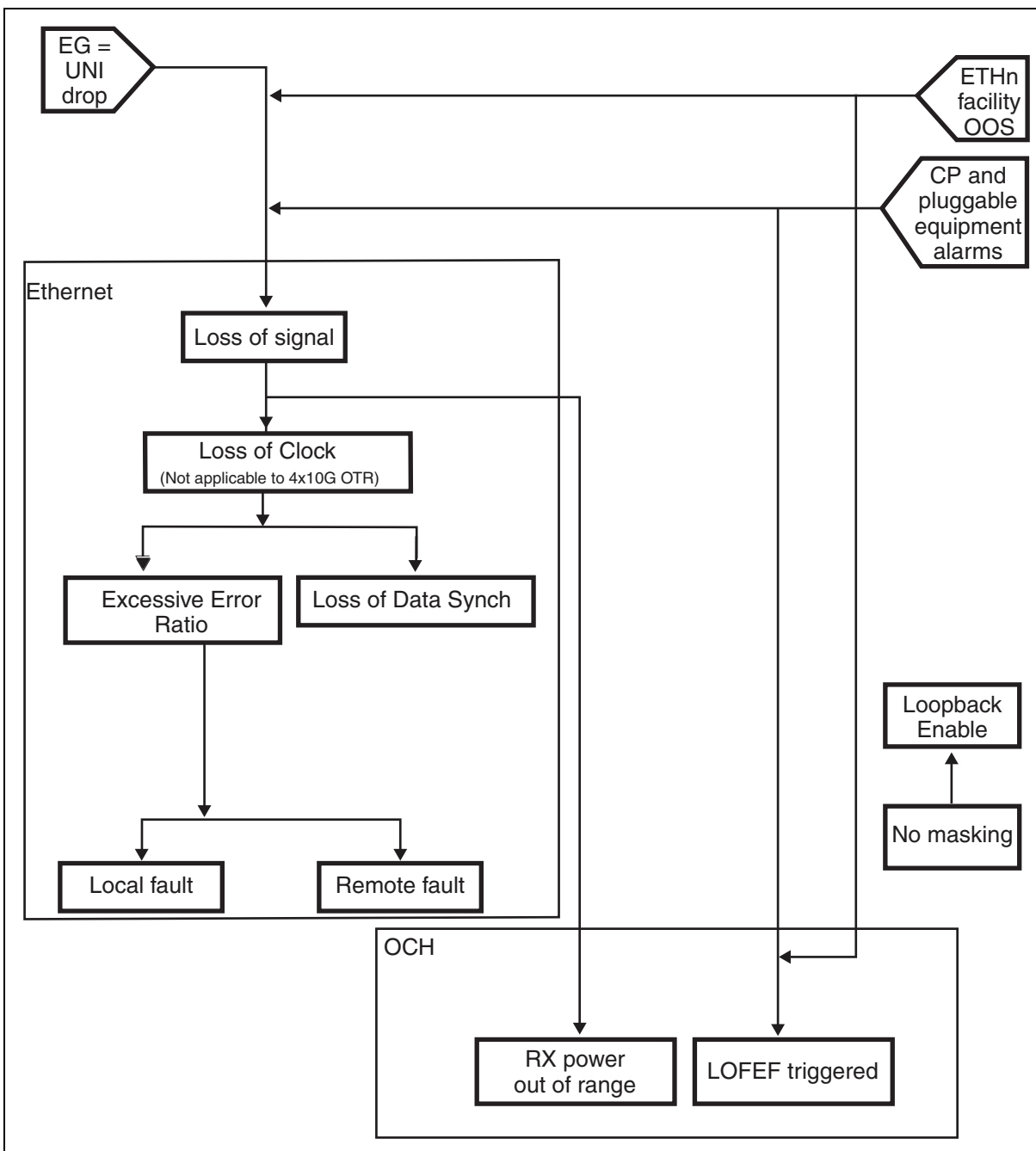
Equipment alarm hierarchy (unprovisioned pluggable)



Shelf equipment alarm hierarchy

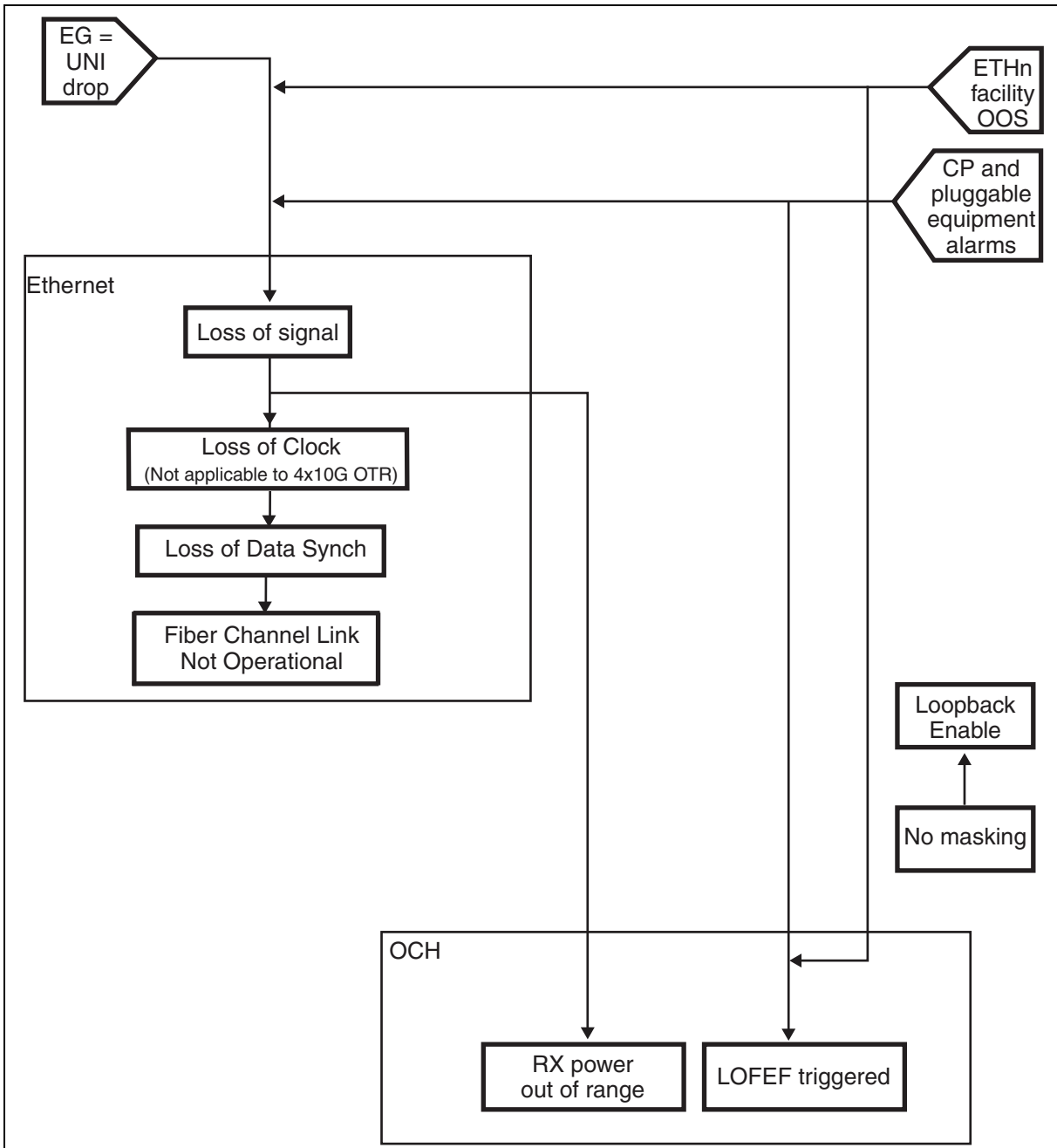


FC1200 alarm hierarchy for 2x10G OTR, 4x10G OTR, 10x10G MUX, and 4x10G MUX circuit packs



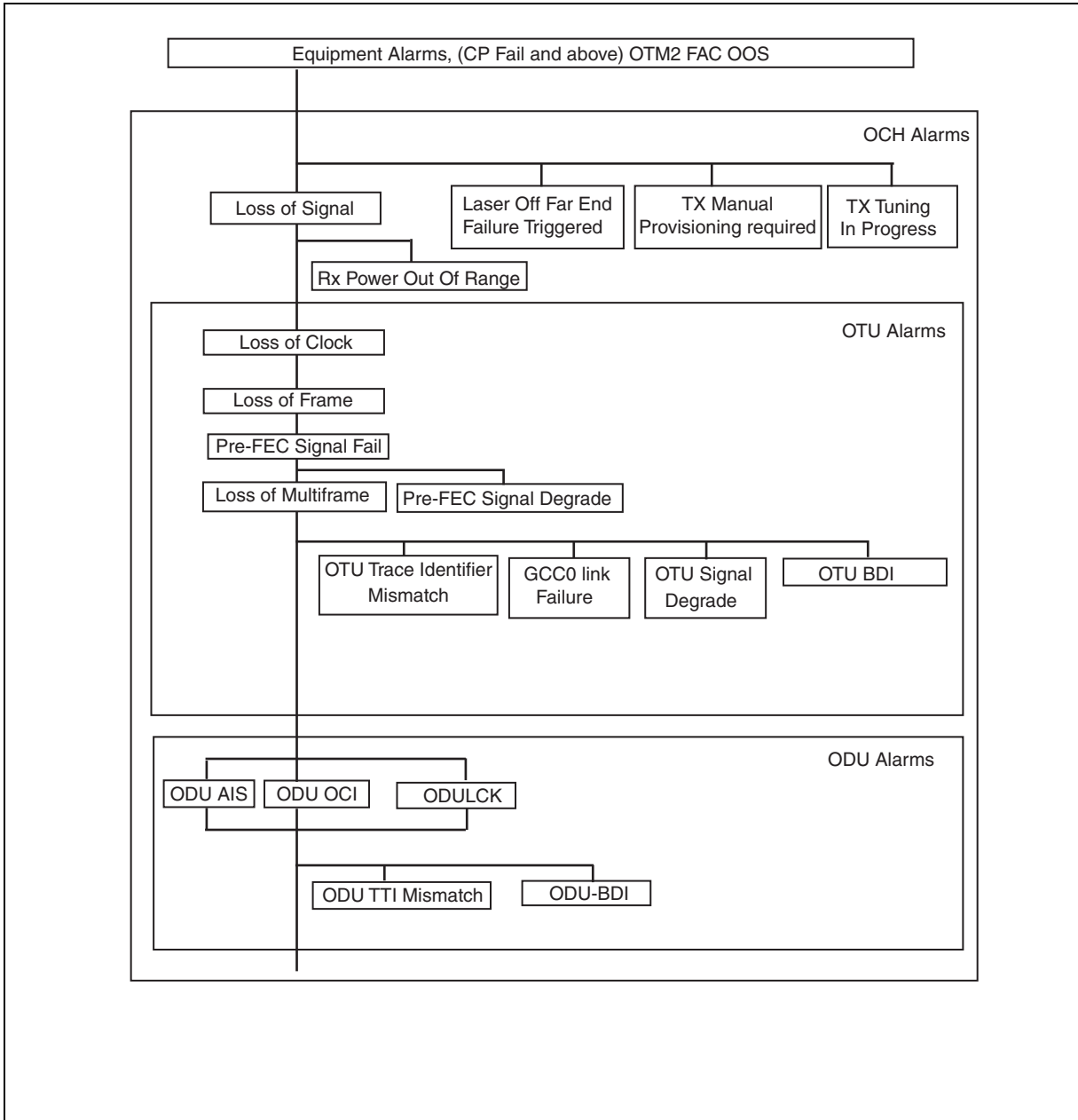
FC400/FC800 alarm hierarchy for 2x10G OTR, 4x10G OTR, 10x10G MUX, and 4x10G MUX circuit packs

Note that FC400 applies to the 4x10G OTR NTK530QA variant only.



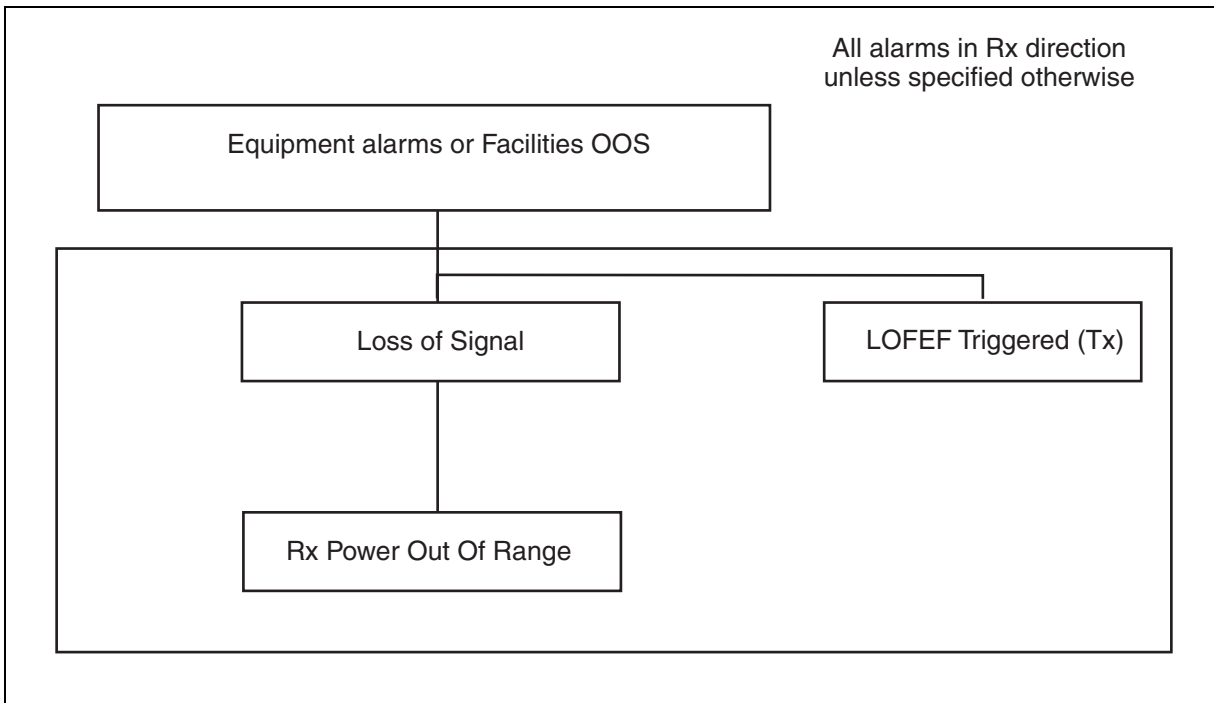
OTM2 Lines alarm hierarchy on 2x10G OTR circuit pack

The OTM2 lines alarm hierarchy for the 2x10G OTR circuit pack is as follows:.

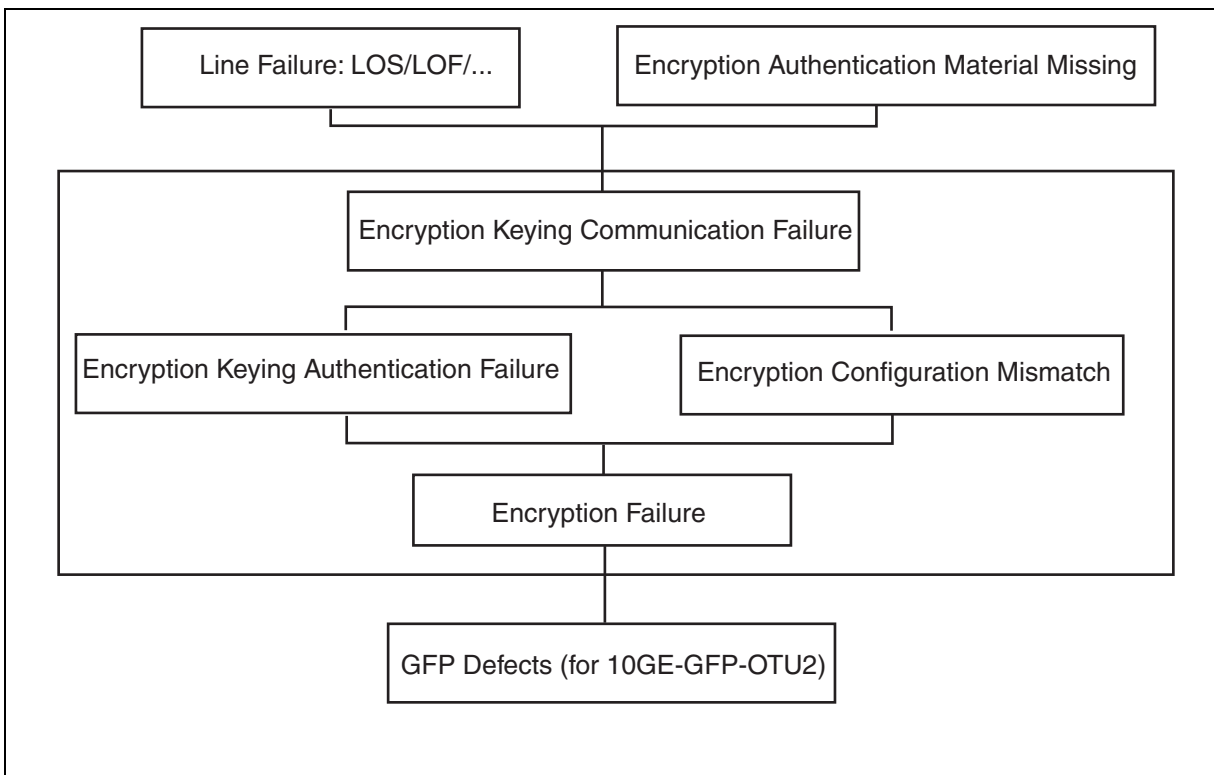


4x10G OTR (NTK530QA) circuit pack infiniband alarm hierarchy

The infiniband alarm hierarchy for the 4x10G OTR NTK530QA variant circuit pack applies to PSIFB PROTOCOL FLEX facilities.

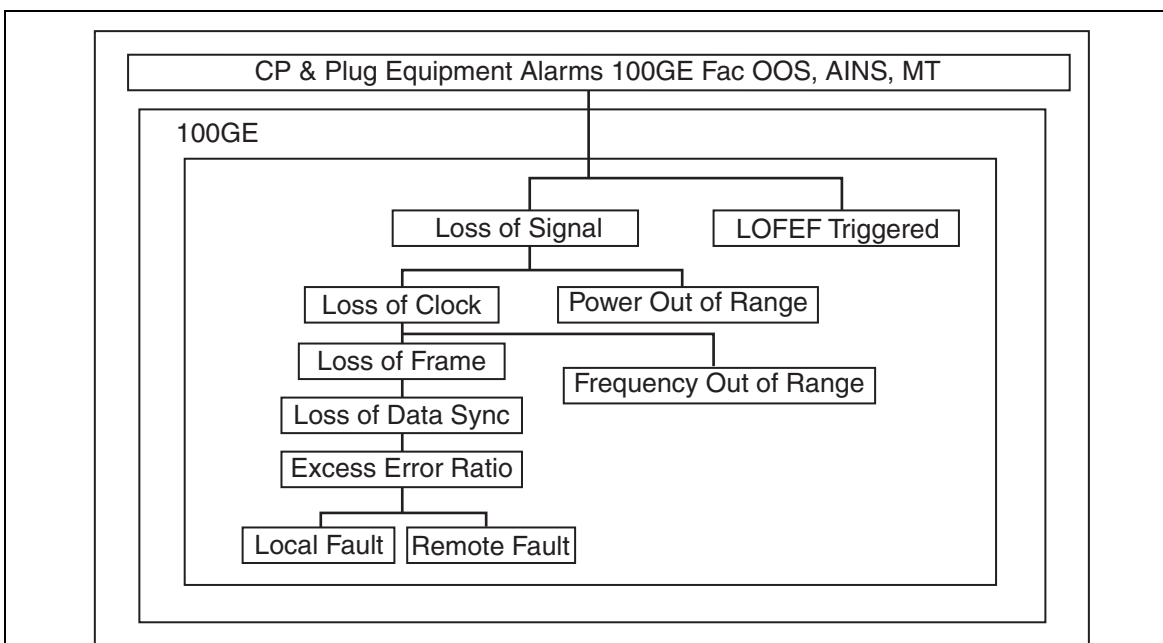


4x10G OTR w/ Encryption (NTK530QE) circuit pack alarm hierarchy



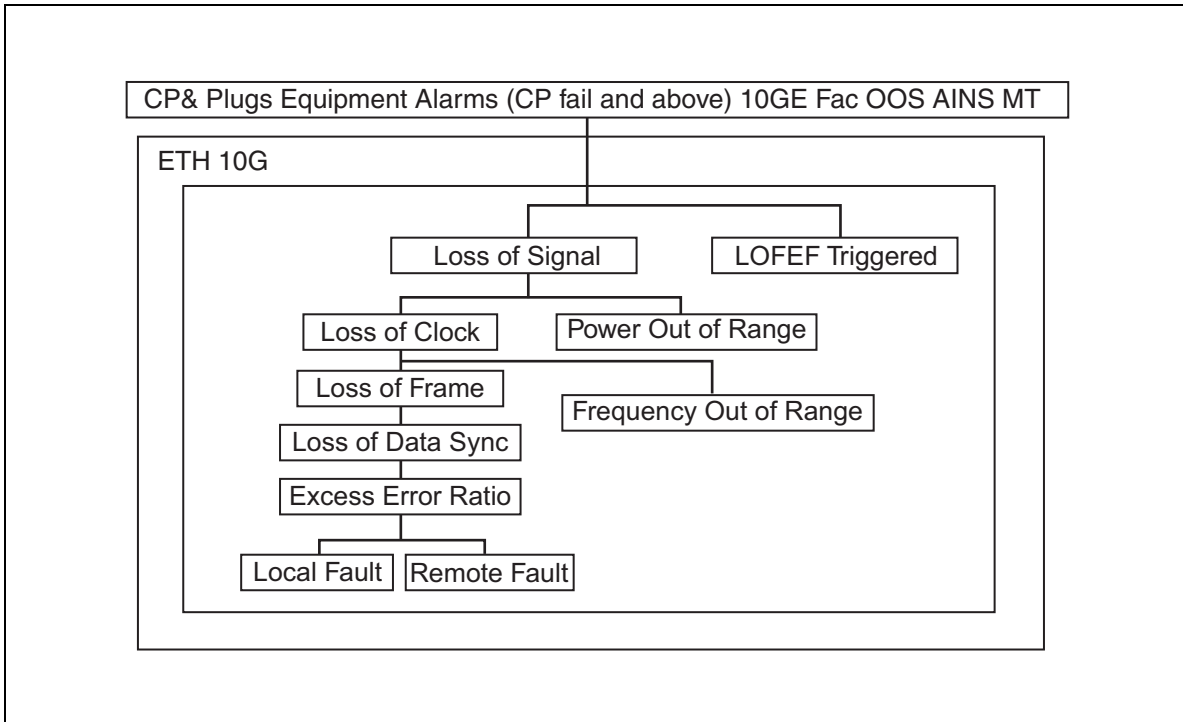
100G OCI and 100G WL3n/WL3e OTR circuit pack ETH100G facility alarm hierarchy

The ETH100G facility on the client port of 100G OCI and 100G WL3n/WL3e OTR circuit pack supports the following alarm hierarchy.



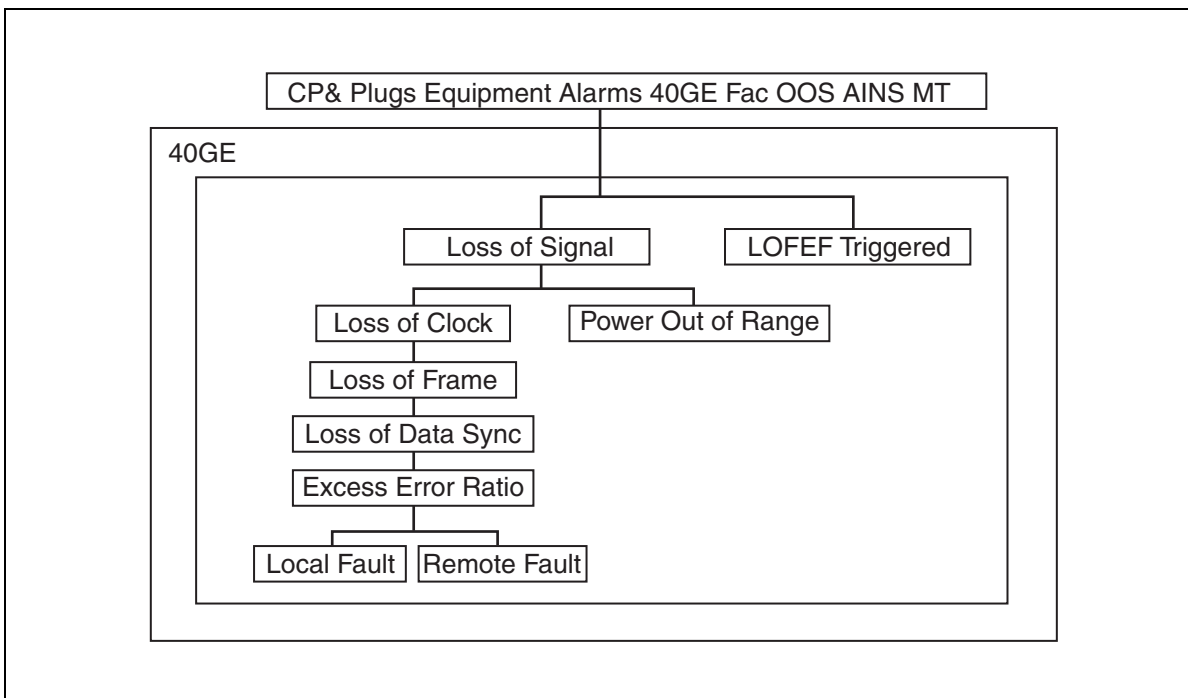
100G WL3n MOTR circuit pack ETH10G facility alarm hierarchy

The ETH10G facility on the client port of 100G WL3n MOTR circuit pack supports the following alarm hierarchy.

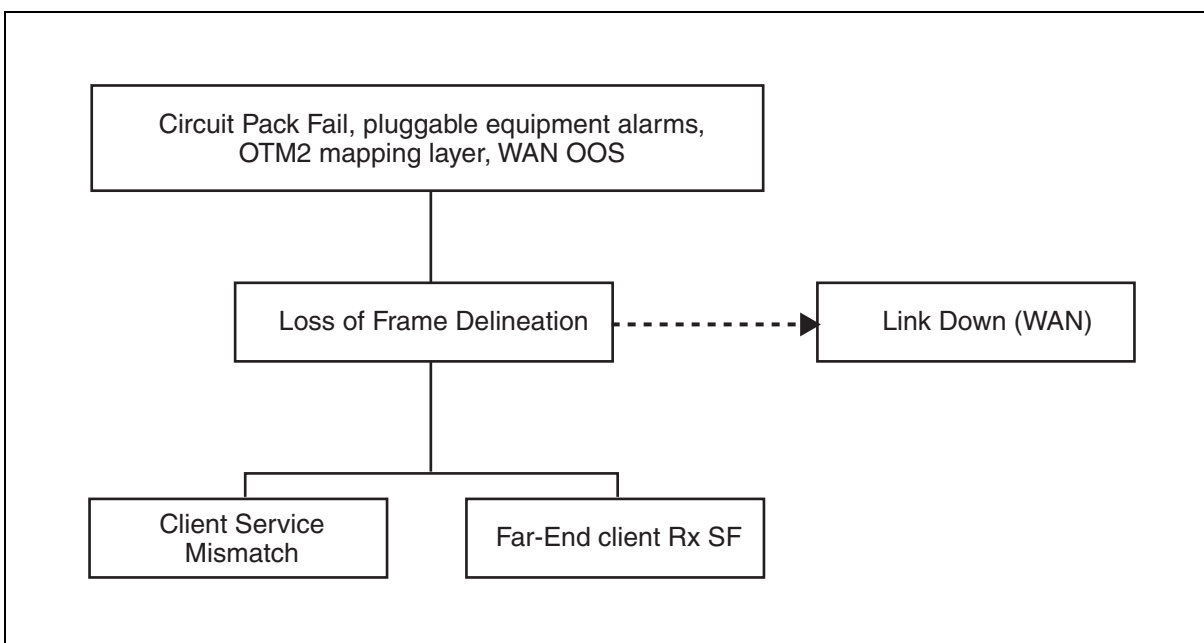


40G OCI+ CFP circuit pack ETH40G facility alarm hierarchy

The ETH40G facility on the client port of 40G OCI+ CFP circuit pack supports the following alarm hierarchy.



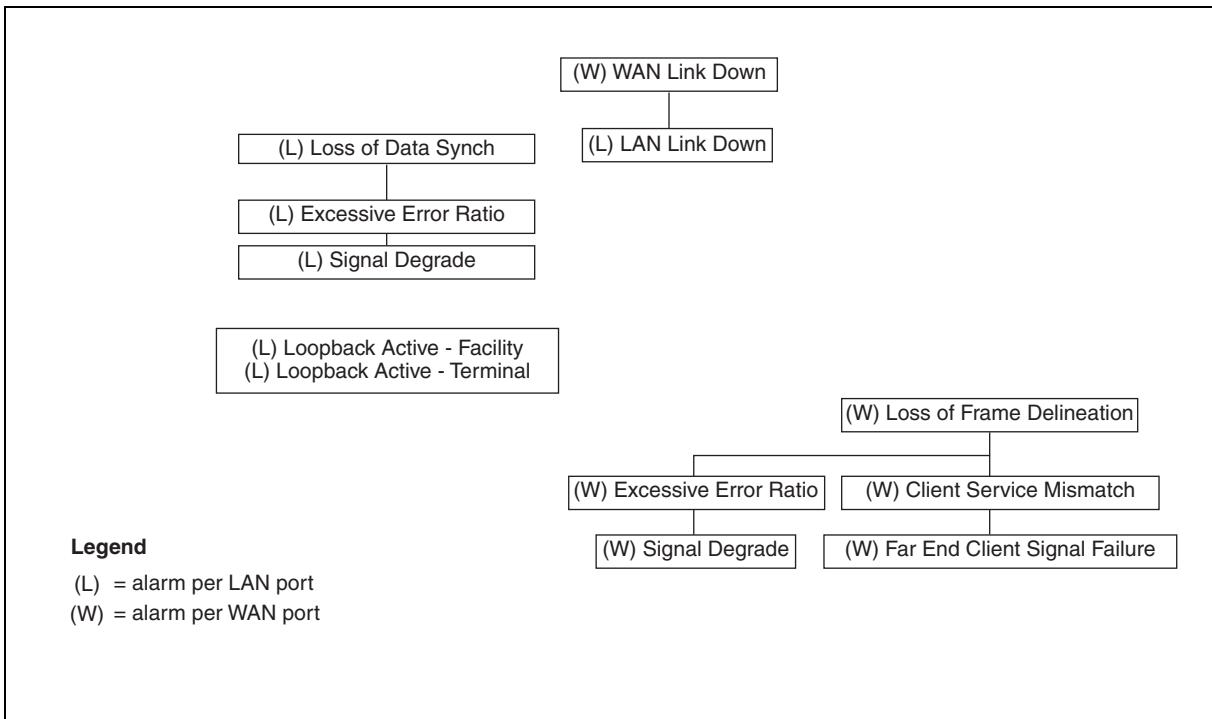
40G MUX OCI circuit pack 10GbE WAN facility alarm hierarchy



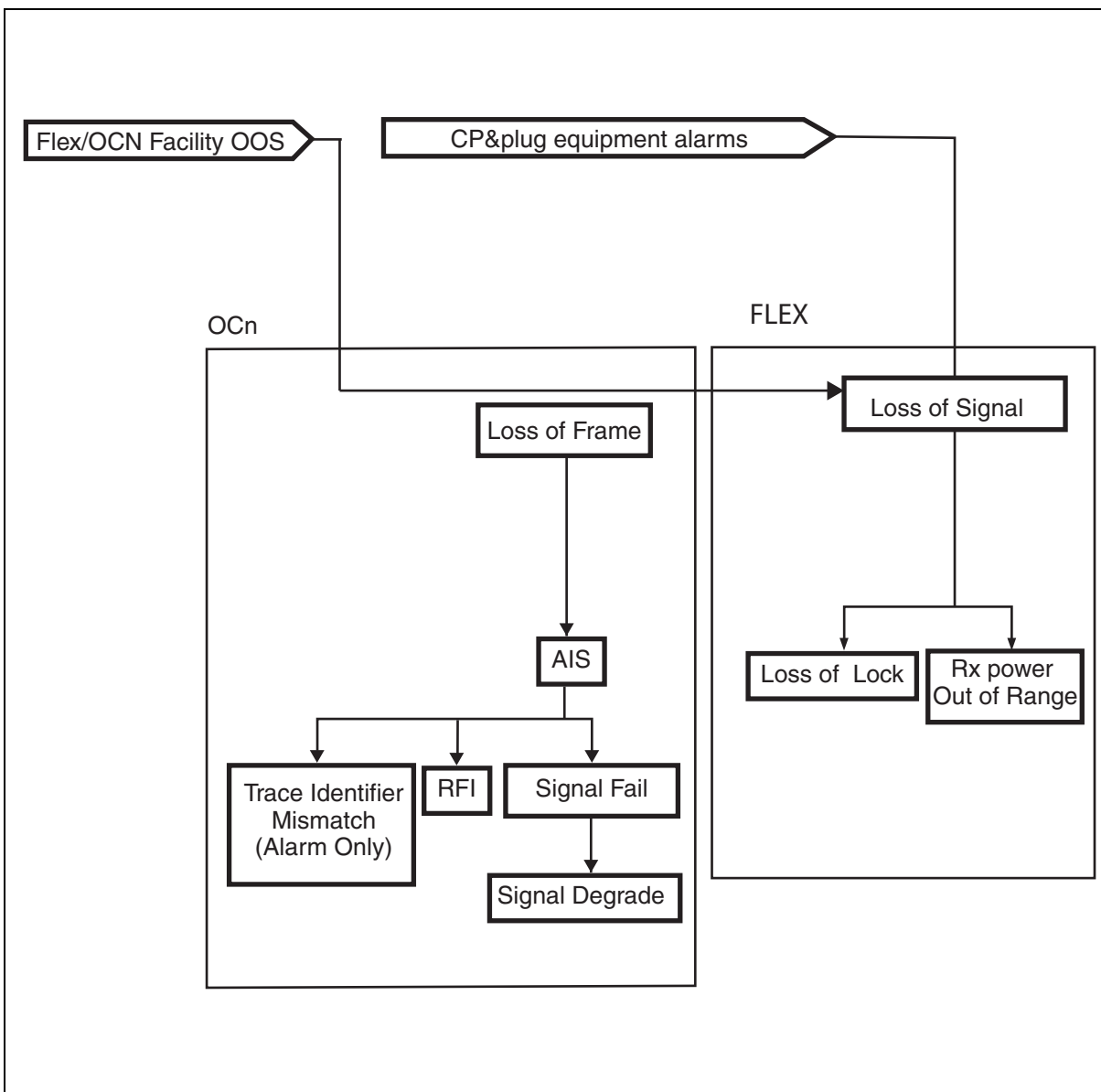
LAN/WAN EPL facilities alarm hierarchy (ETH, ETH100, FC100/FC200/FC400, GE, and WAN facilities)

For the ETH10G alarm hierarchy, refer to the “ETH10G and FC1200 facility alarm hierarchy” on page 3-13.

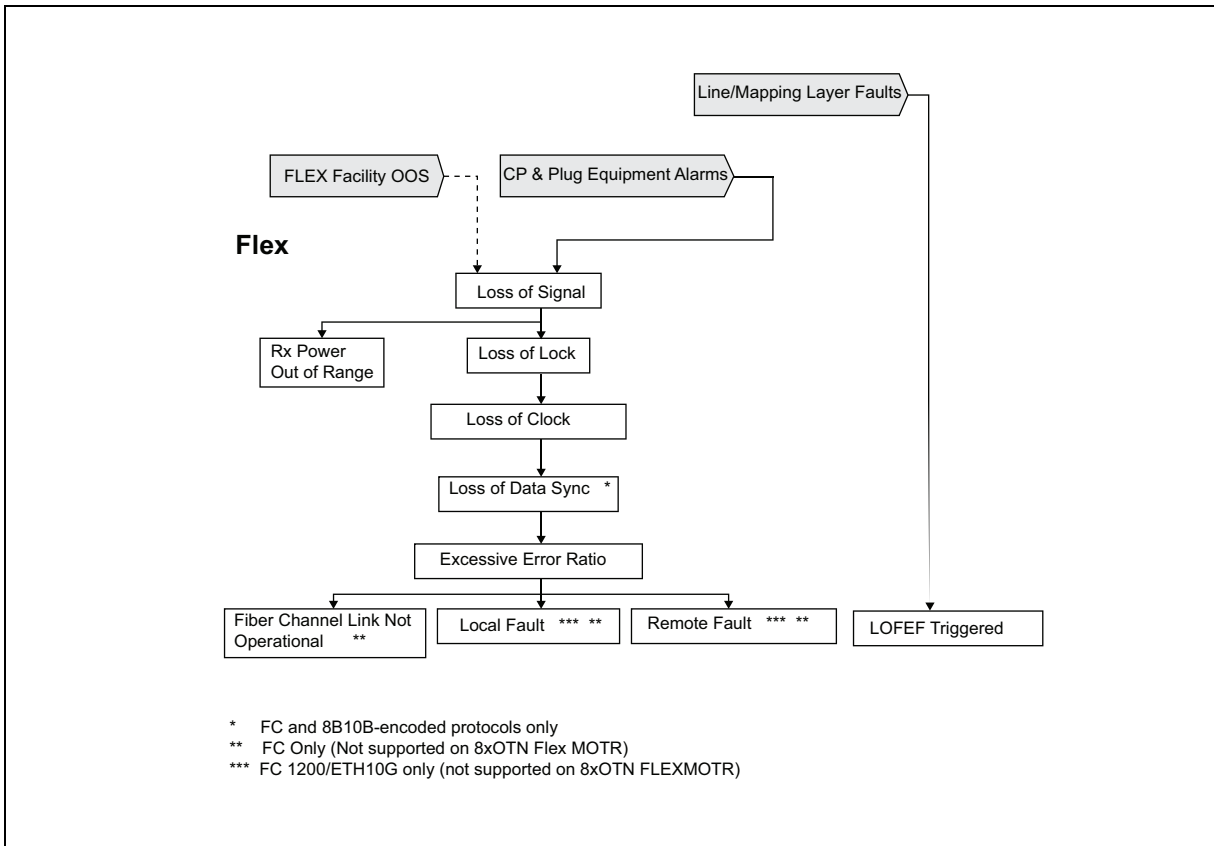
For the ETH100G alarm hierarchy, refer to the “FC1200 alarm hierarchy for 2x10G OTR, 4x10G OTR, 10x10G MUX, and 4x10G MUX circuit packs” on page 3-14.



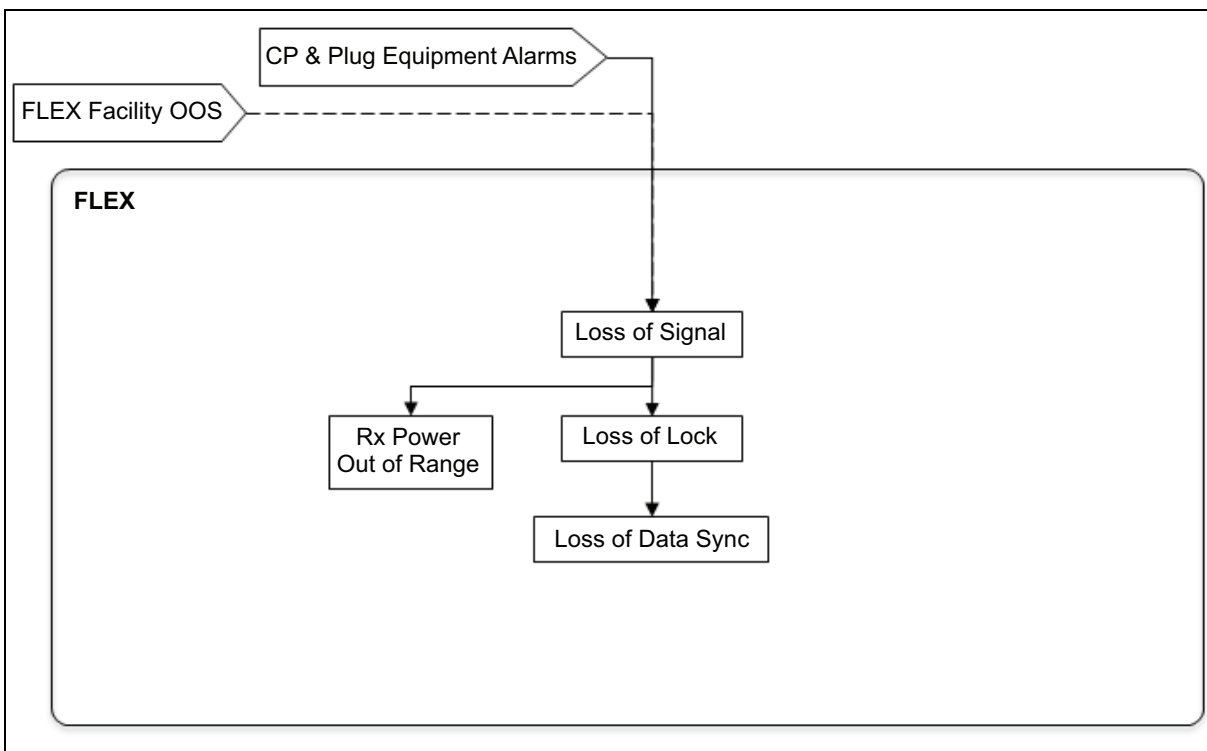
8xOTN Flex MOTR, and (1+8)xOTN Flex MOTR circuit packs OC-n/ STM-n client protocols alarm hierarchy



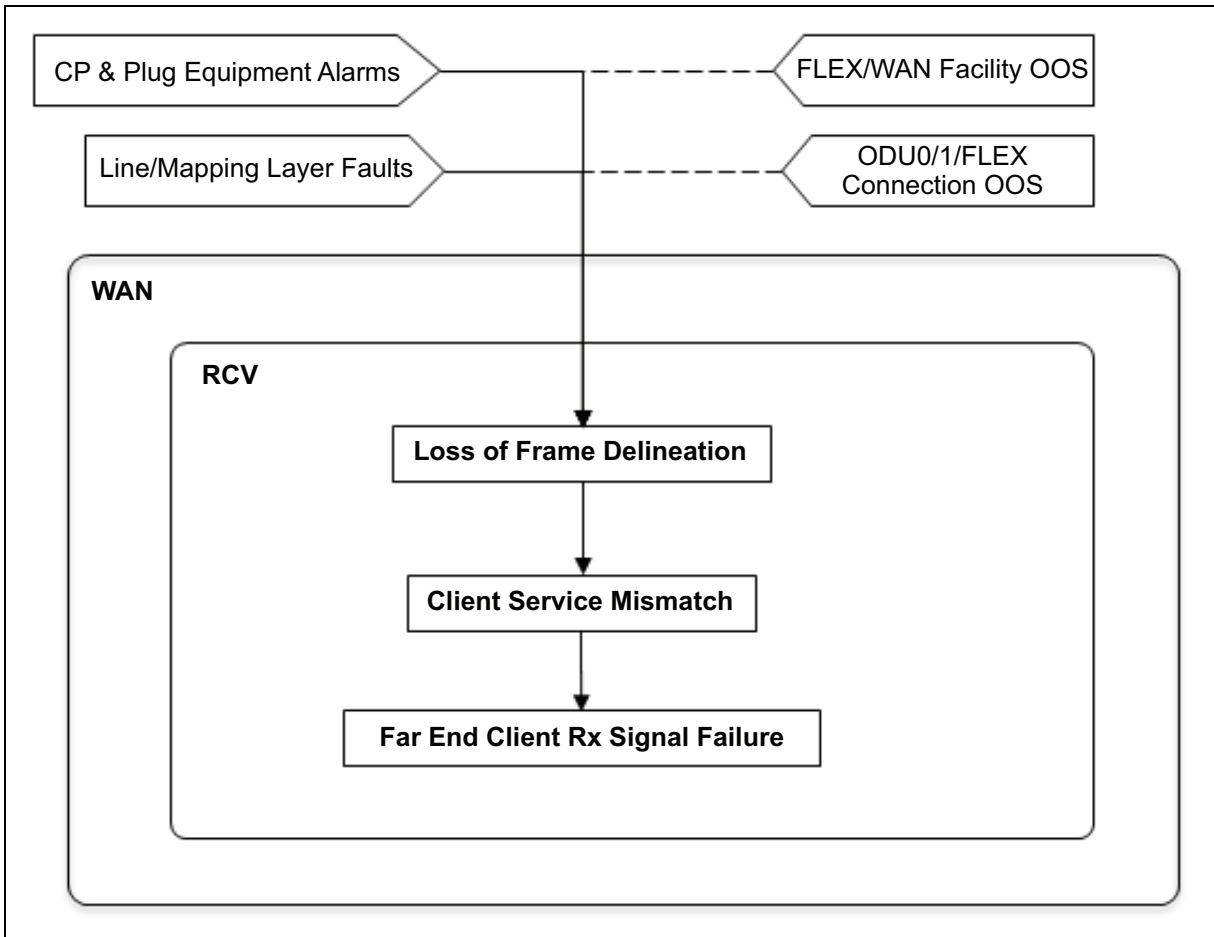
FLEX MOTR and Broadband circuit packs Fiber Channel, 8B10B, and Transparent client protocols alarm hierarchy



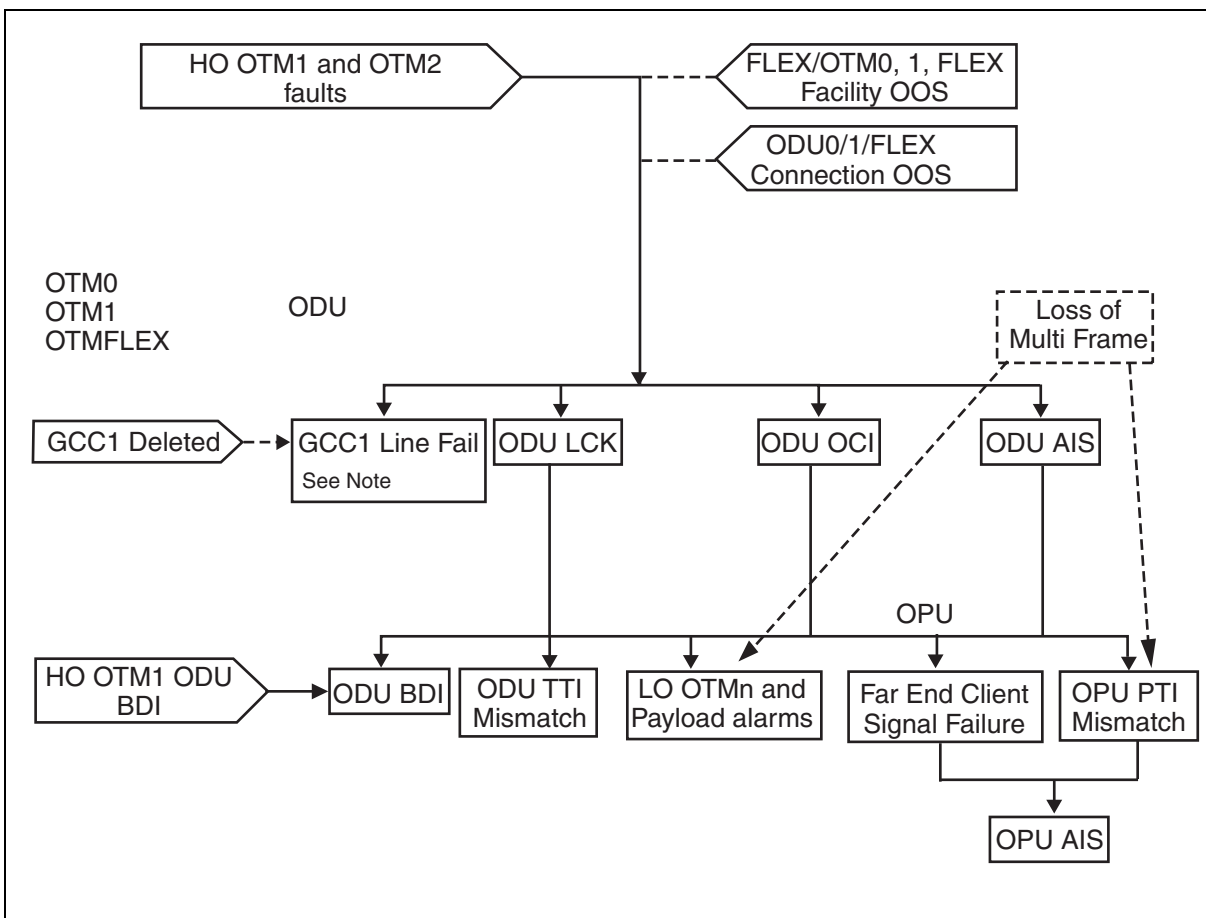
8xOTN Flex MOTR and (1+8)xOTN Flex MOTR circuit packs Fiber Channel, 8B10B, and Transparent client protocols alarm hierarchy



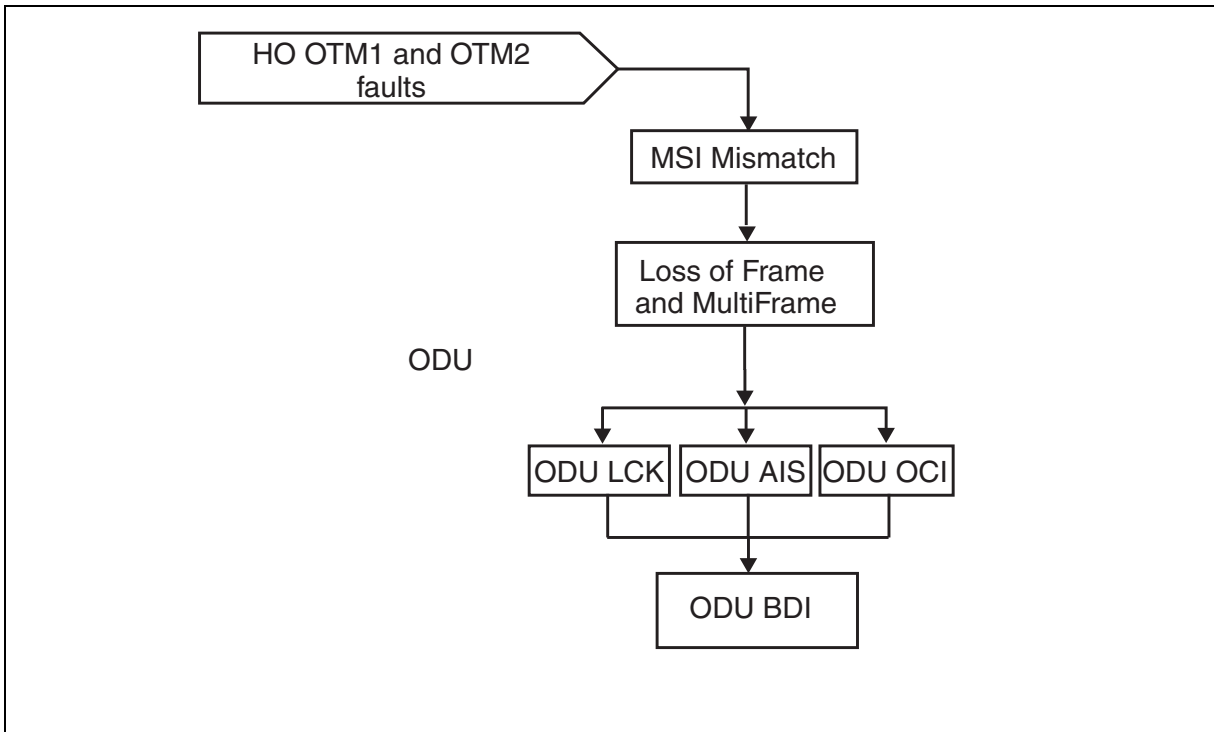
8xOTN Flex MOTR and (1+8)xOTN Flex MOTR circuit packs WAN facility alarm hierarchy



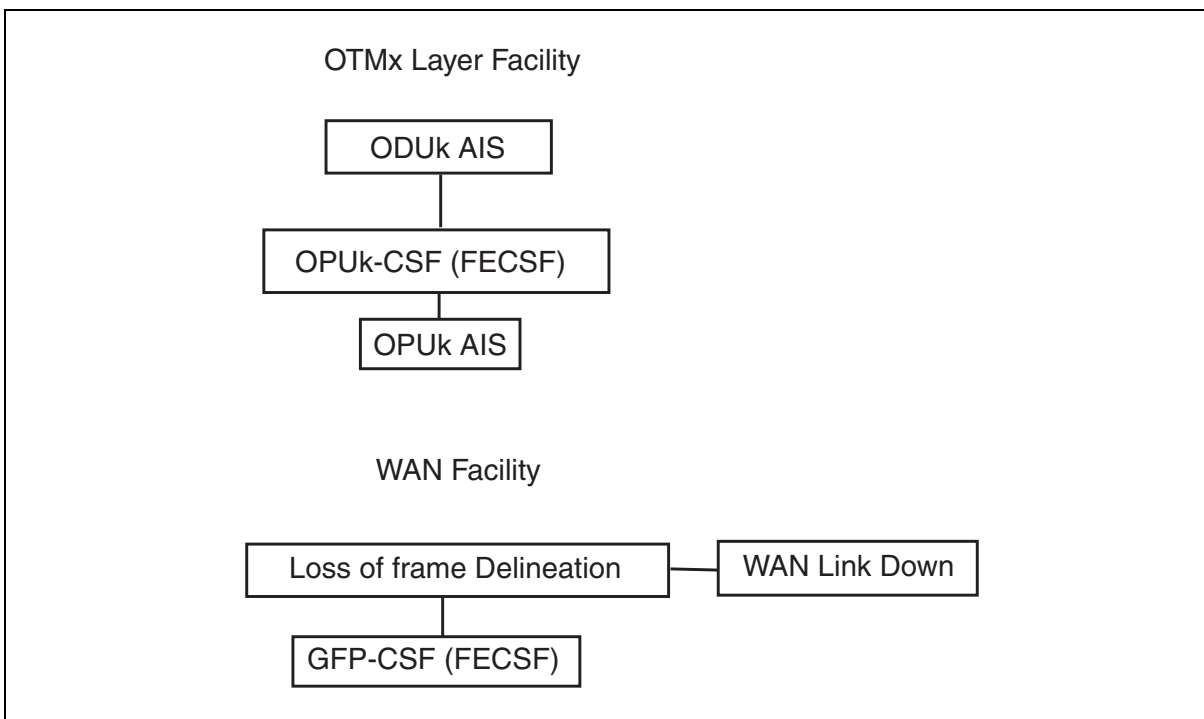
Broadband circuit packs client port facility OTM mapping layer alarm hierarchy



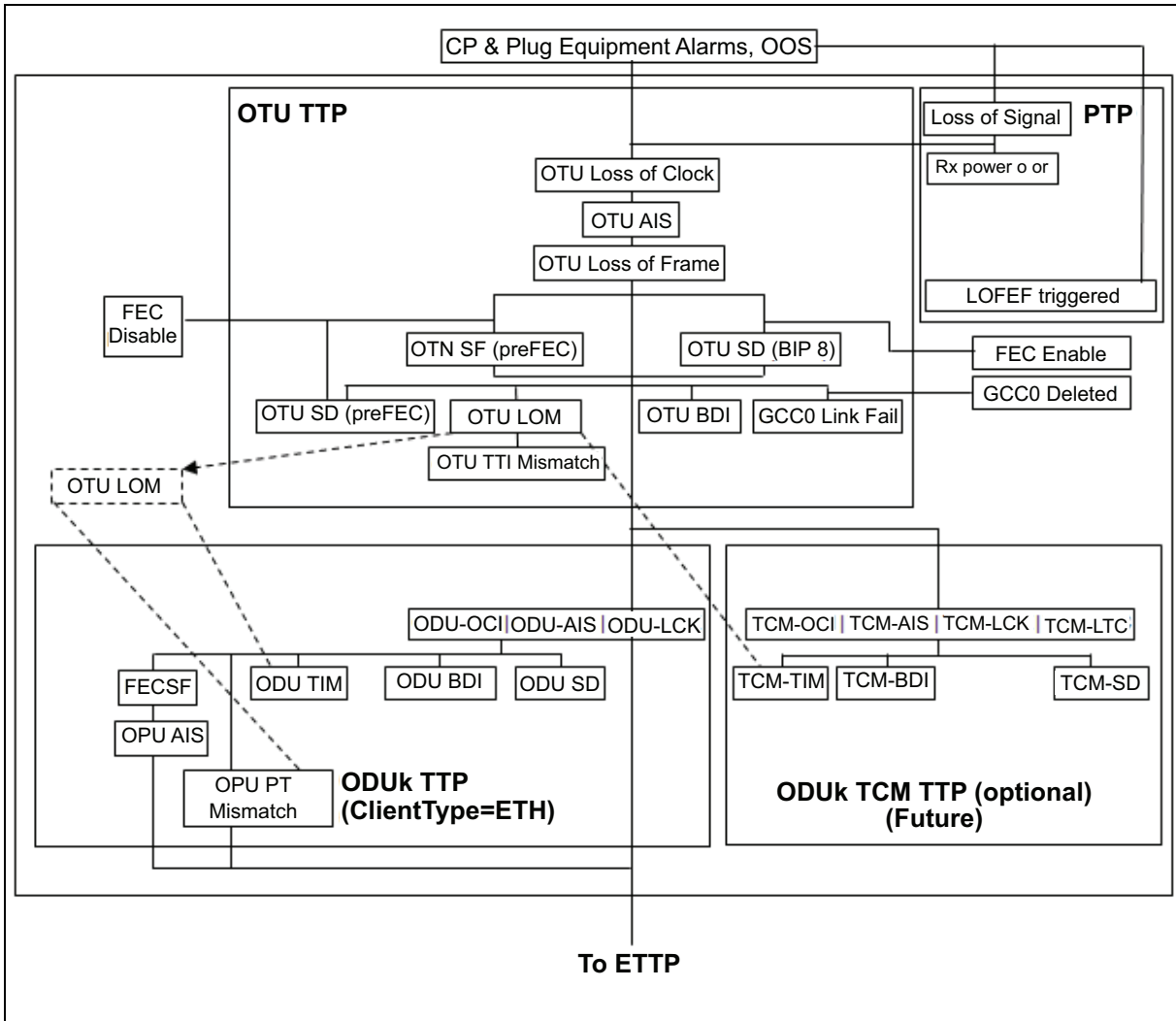
(1+8)xOTN Flex MOTR client port facility (low-order ODU0, ODU1, and ODUFLEX facilities) alarm hierarchy



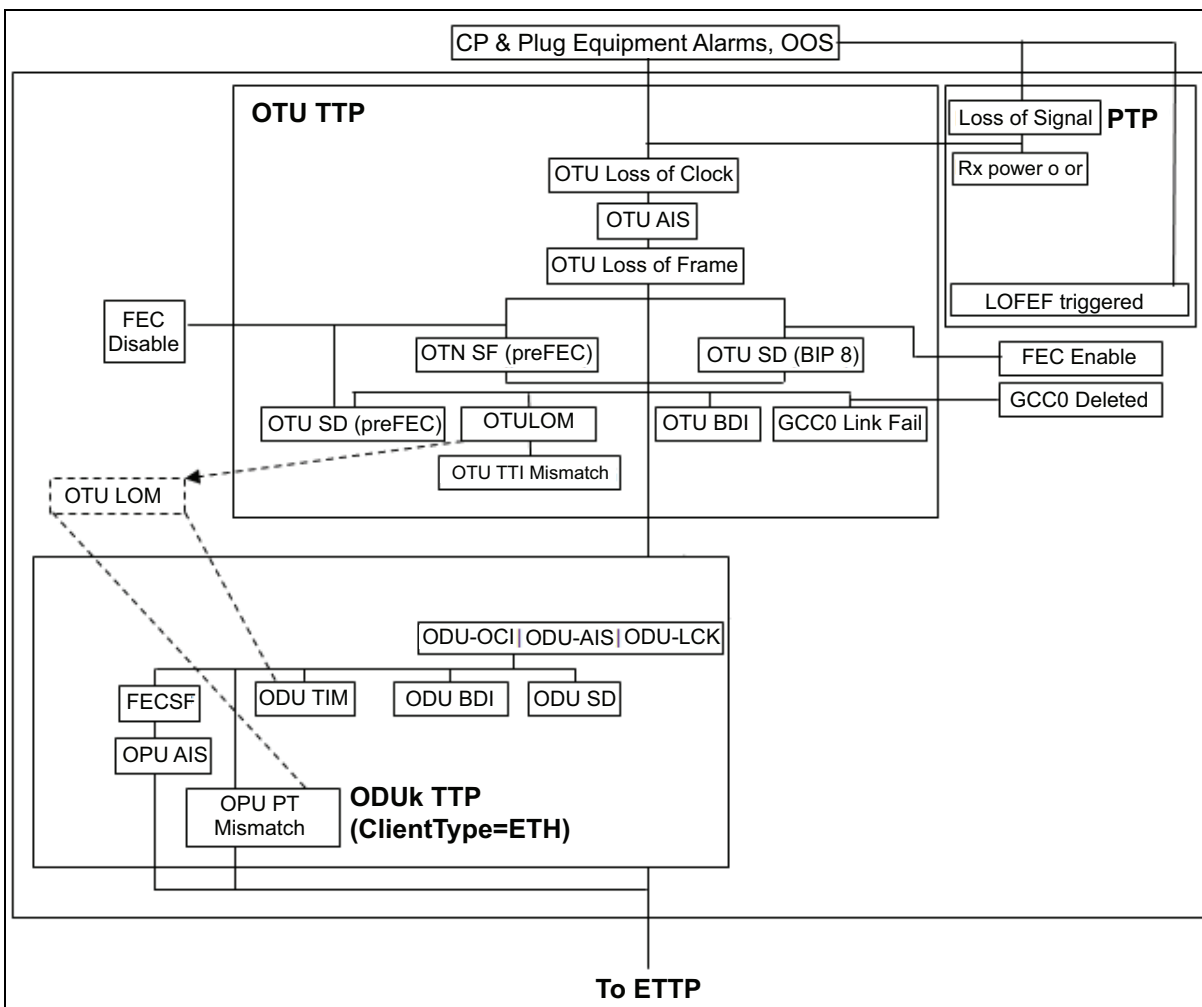
OPU-CSF/GFP-CSF client port alarm hierarchy (OTM and Wan facilities)



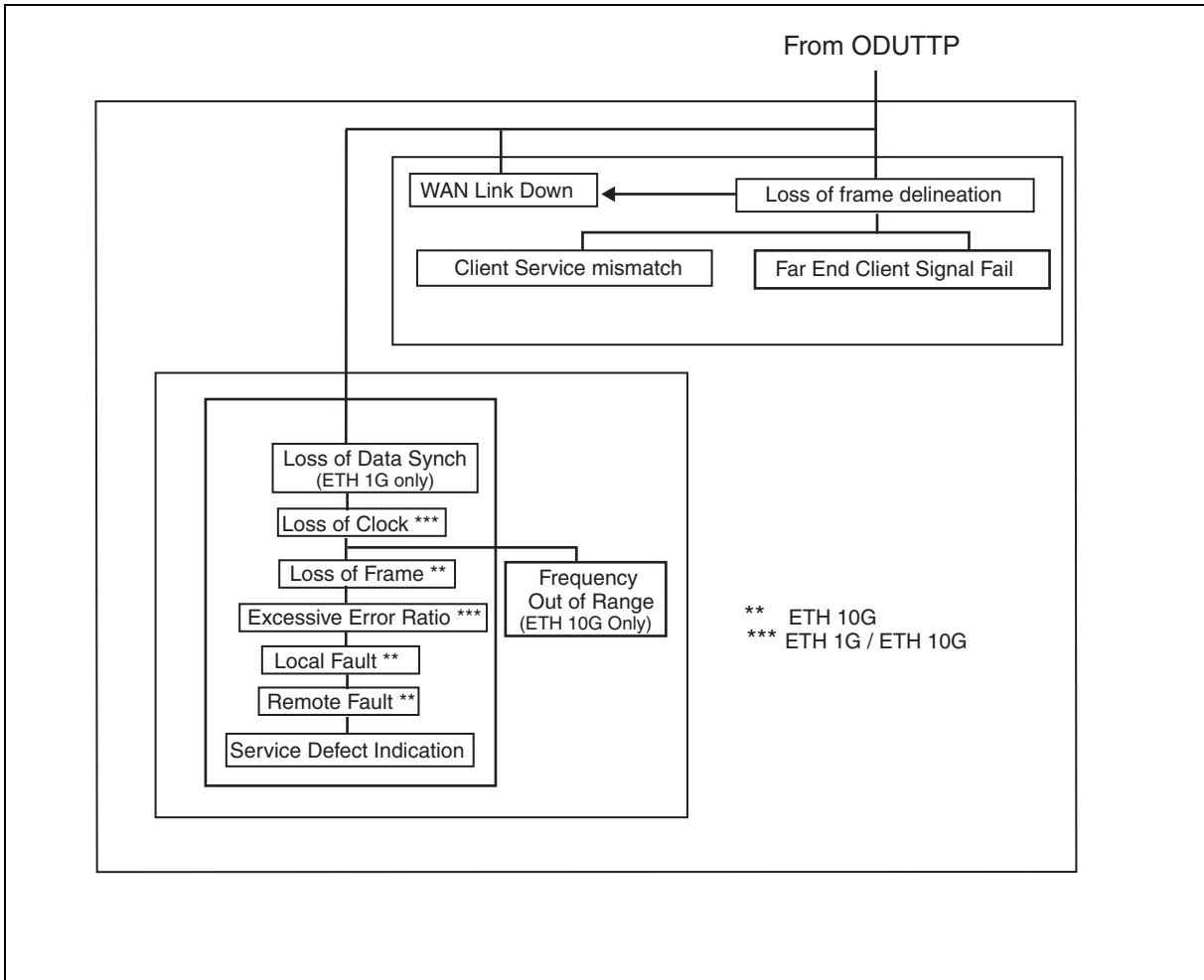
OTN PKT facility alarm hierarchy - faceplate to fabric direction (RX)



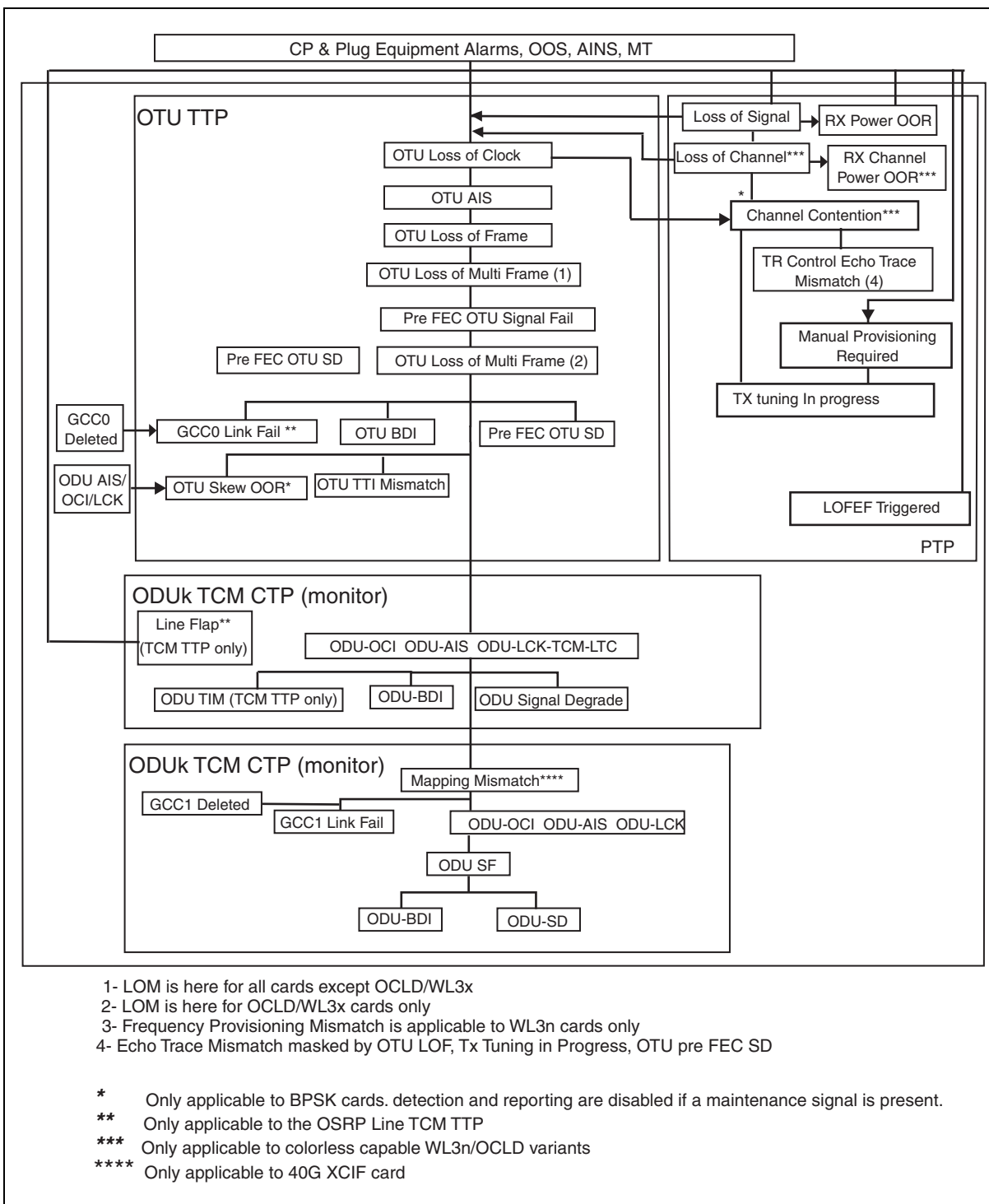
Alarm hierarchy for eMOTR circuit pack



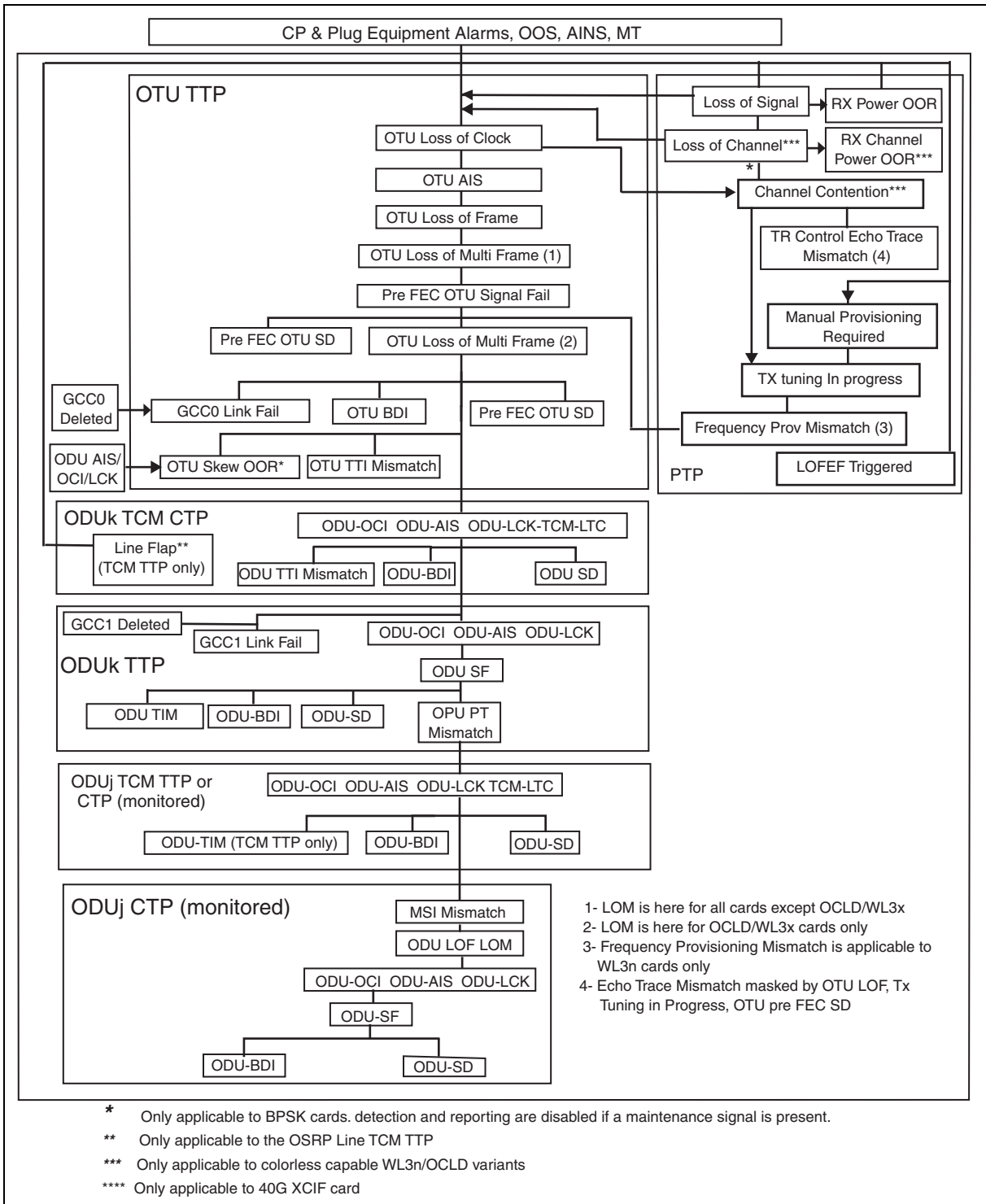
ETTP facility alarm hierarchy for eMOTR circuit pack



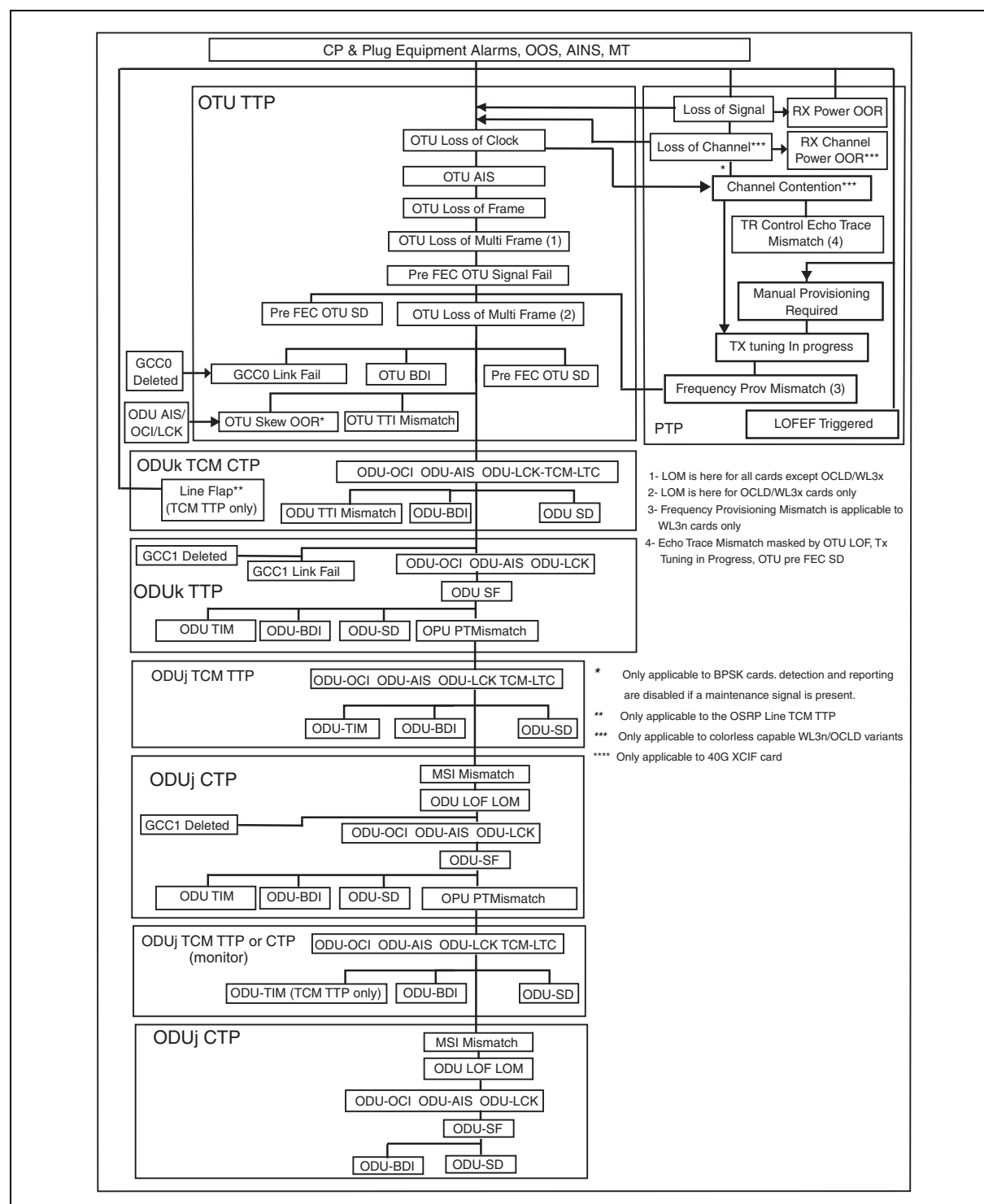
ODUk CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)



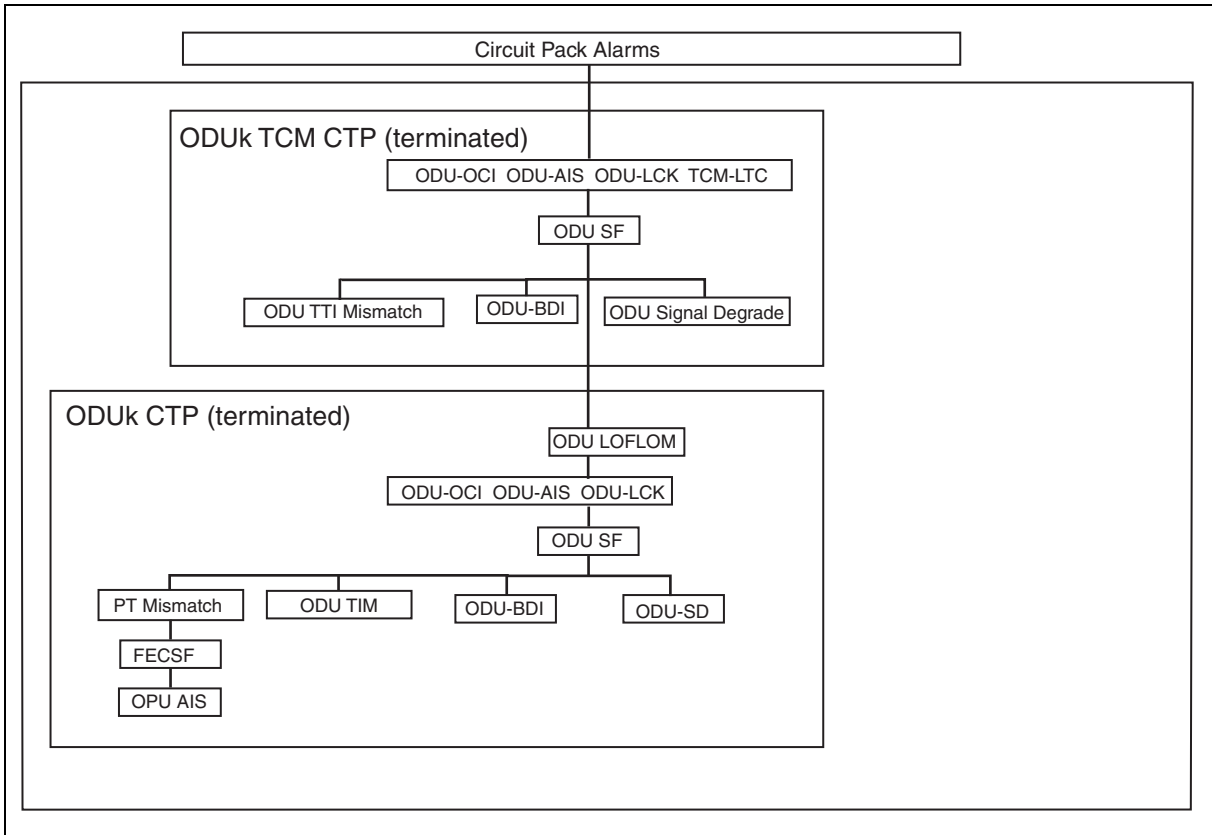
ODUj CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)



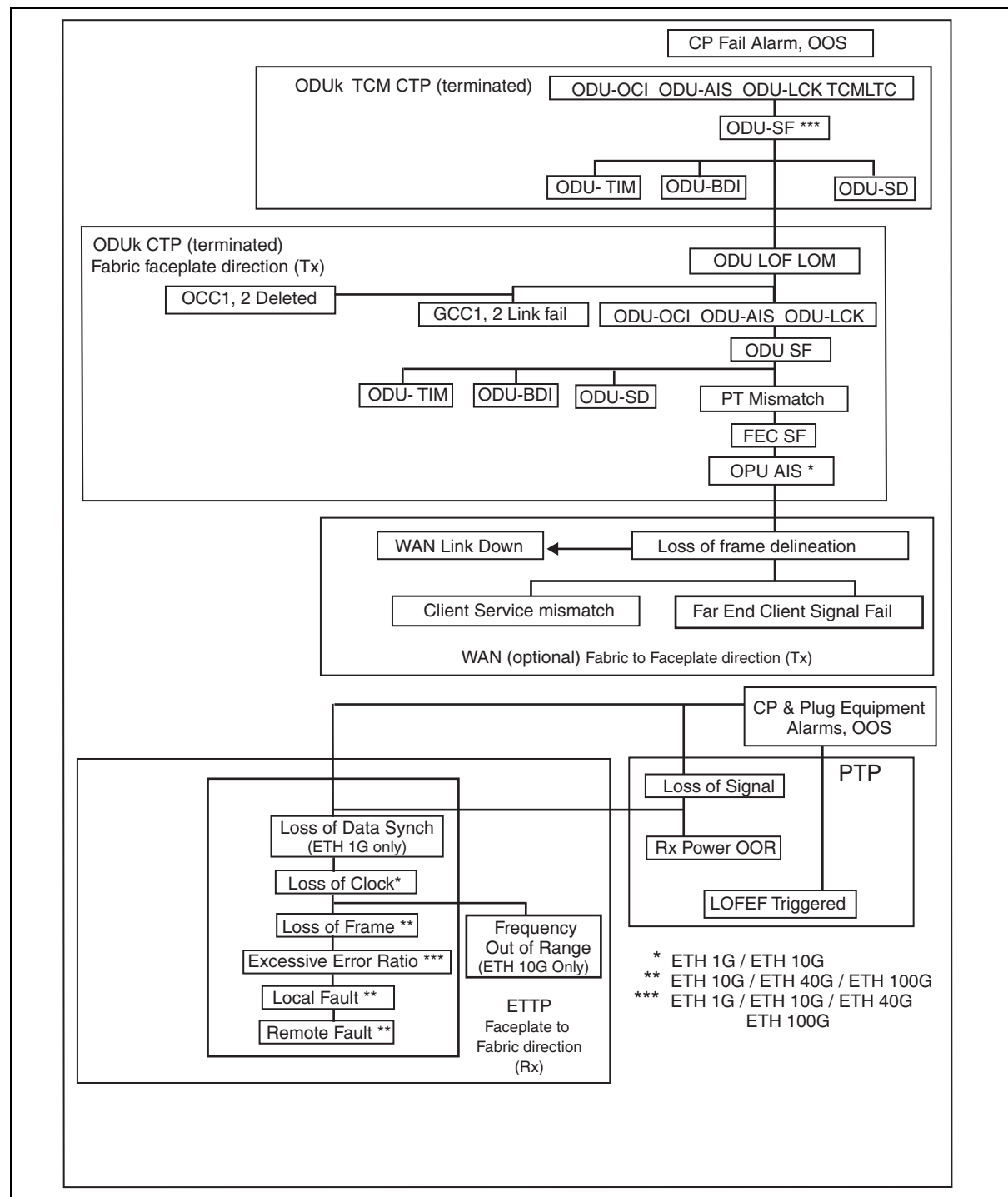
ODUi CTP OTN facility alarm hierarchy - faceplate to fabric direction (RX)

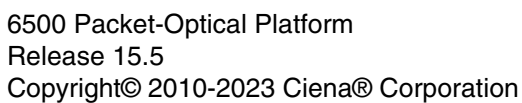


OTN facility alarm hierarchy - fabric to faceplate direction (TX)

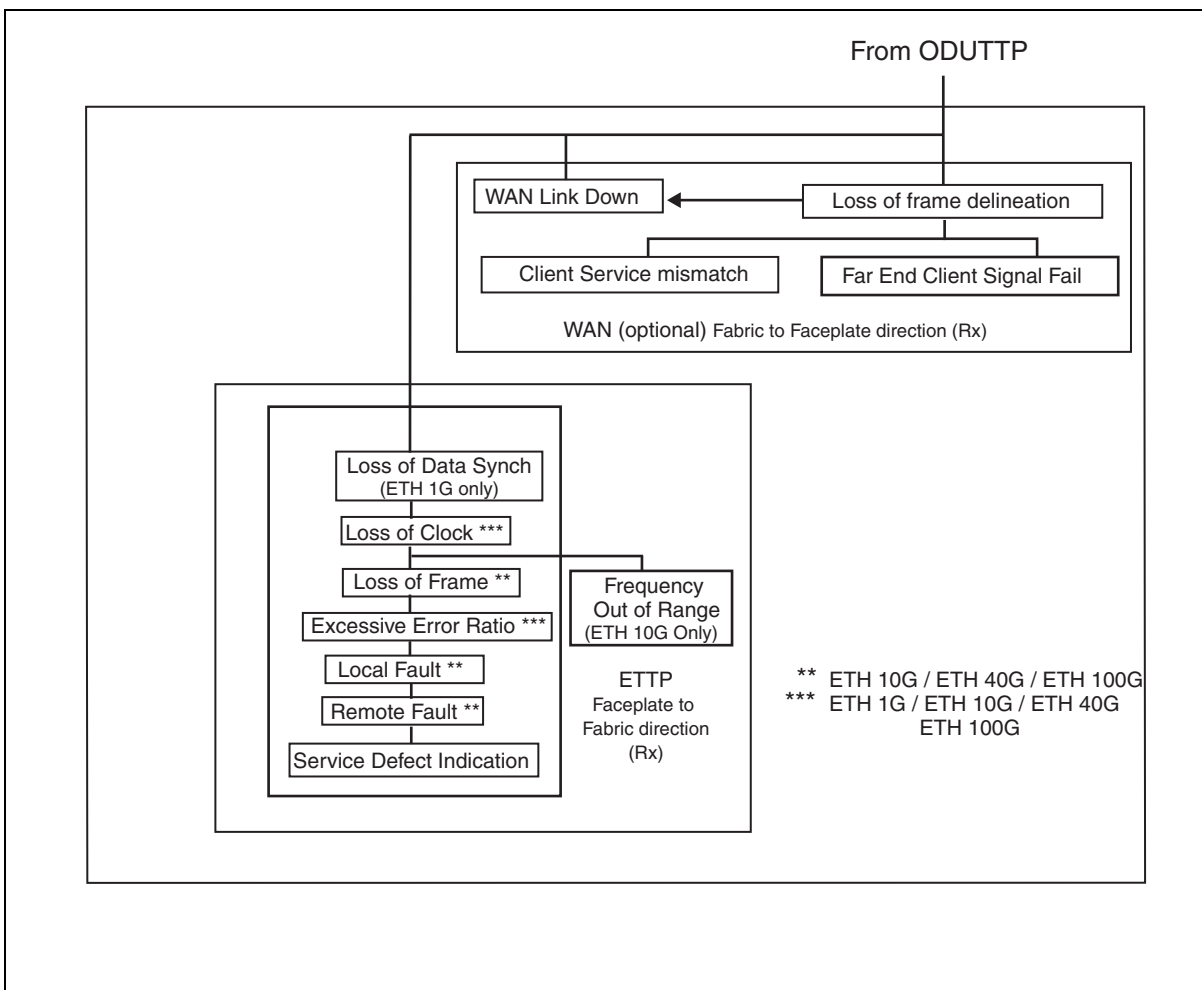


ETTP facility alarm hierarchy - monitored

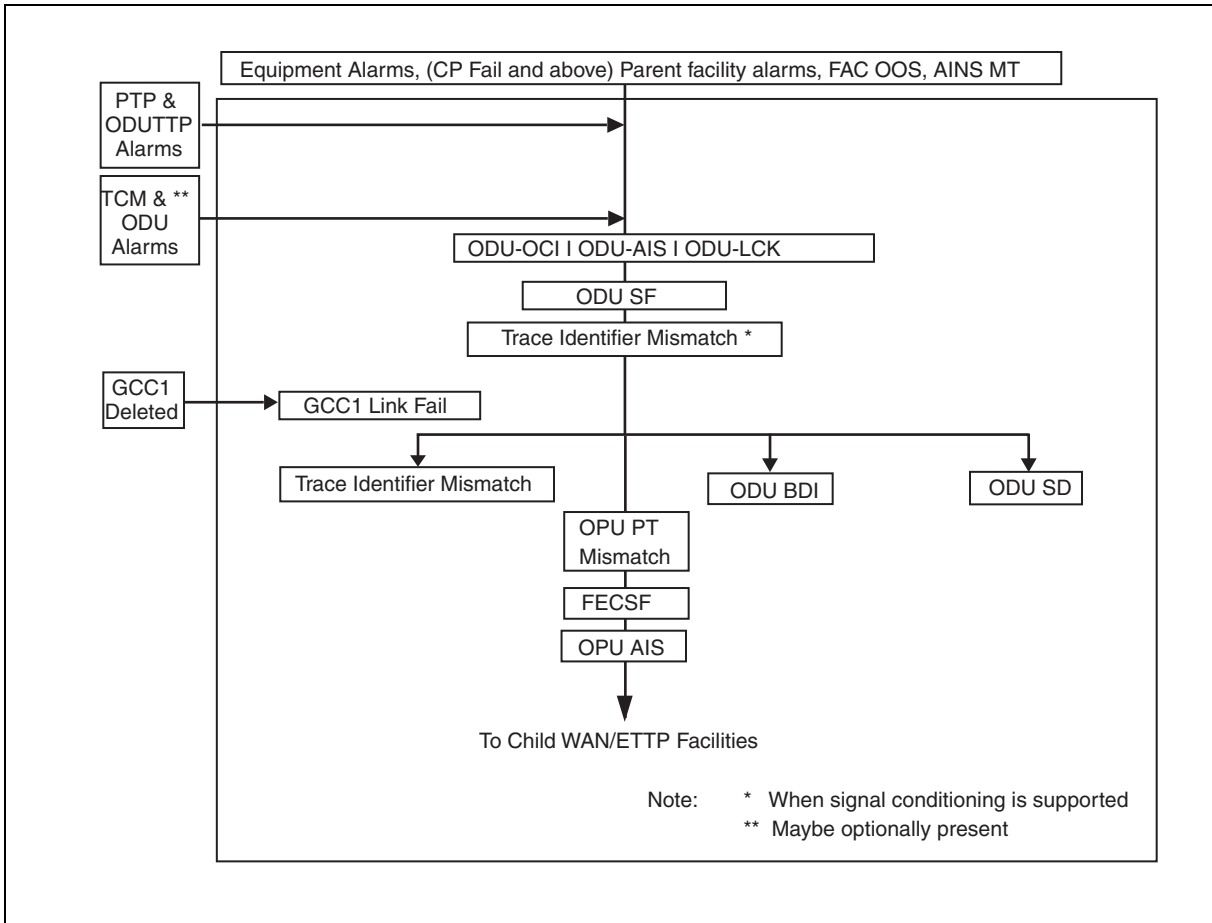




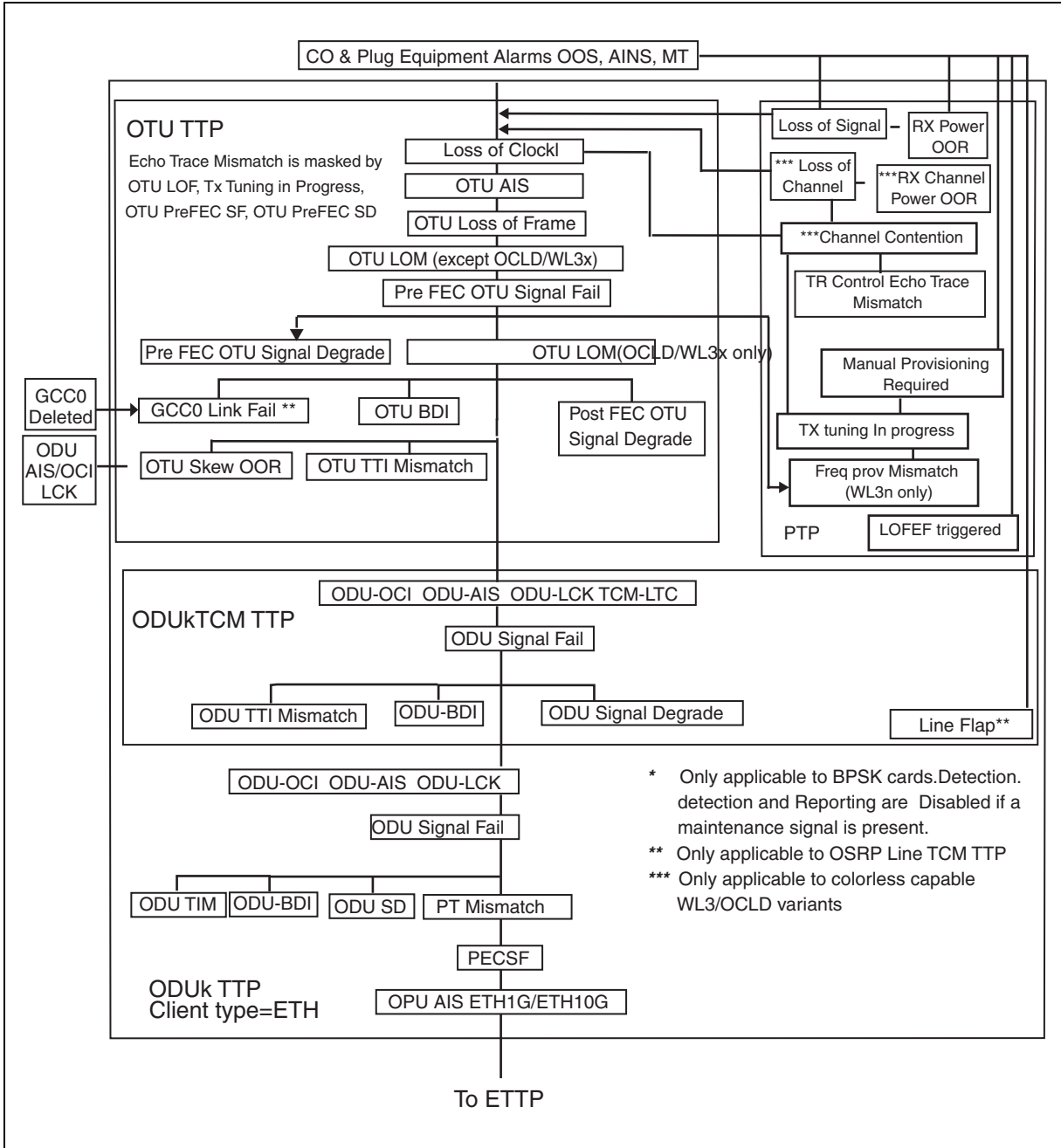
ETTP facility alarm hierarchy mapped within OTN port



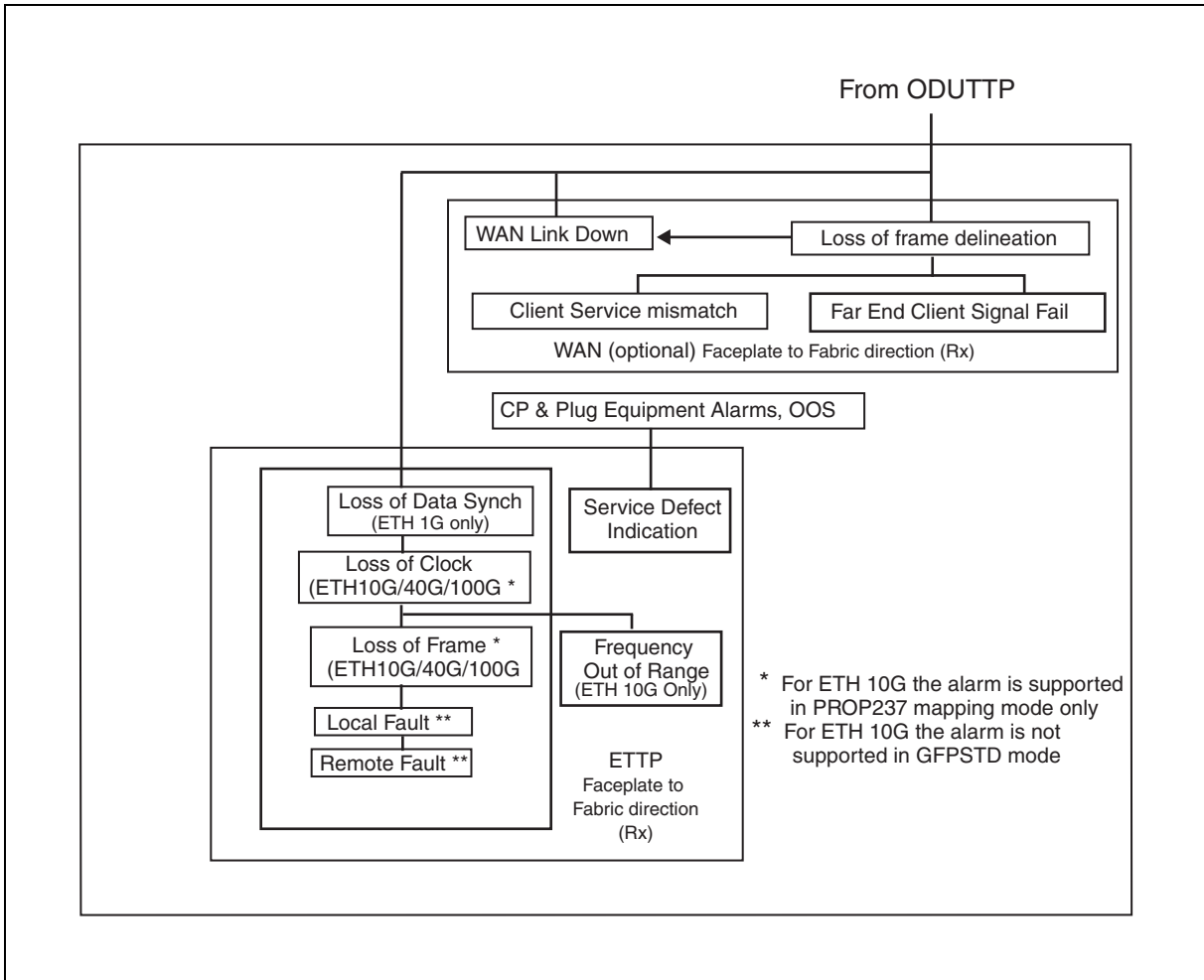
ODUTTP ETH facility alarm hierarchy



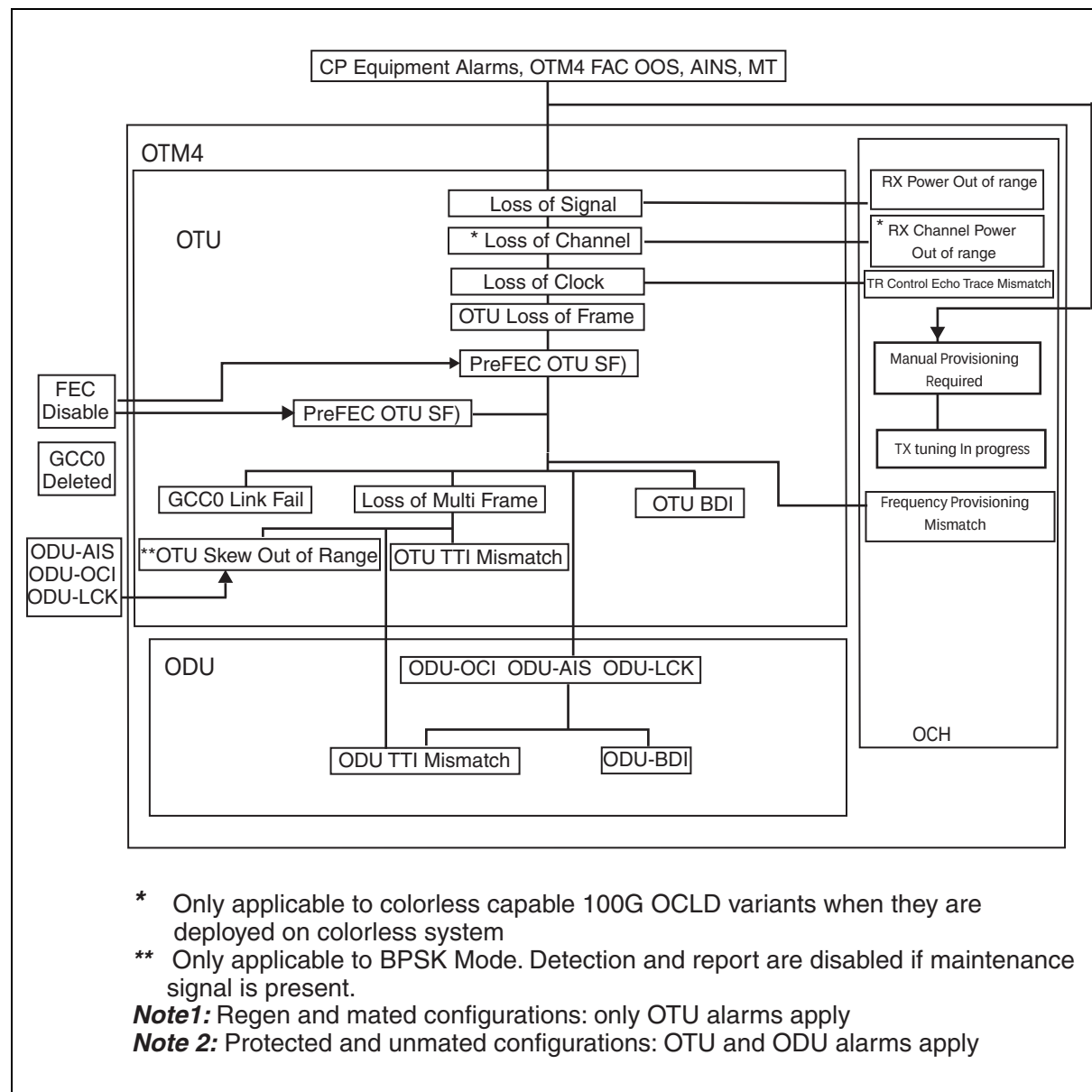
PKT facility alarm hierarchy - faceplate to fabric direction (RX)



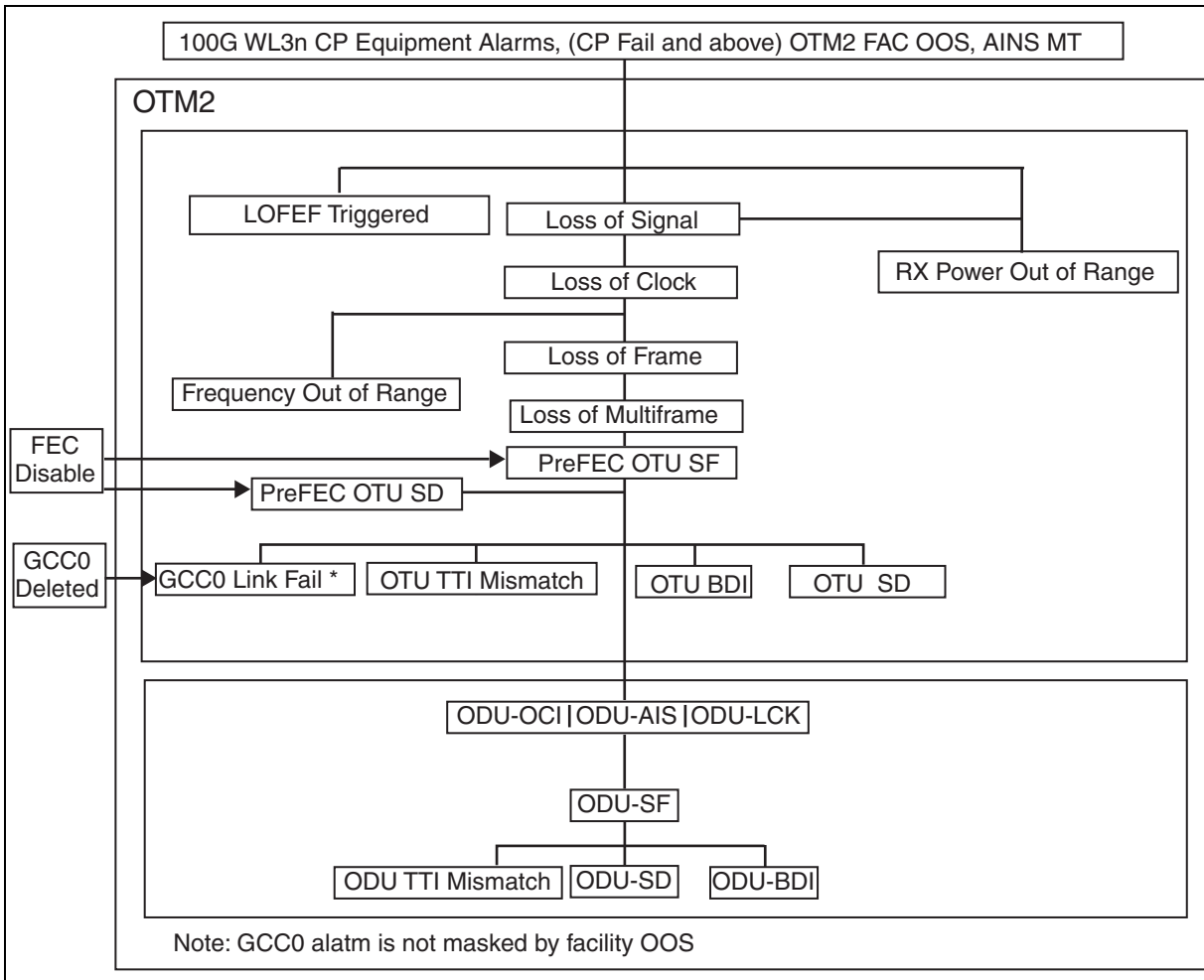
Mapped ETPP facility alarm hierarchy



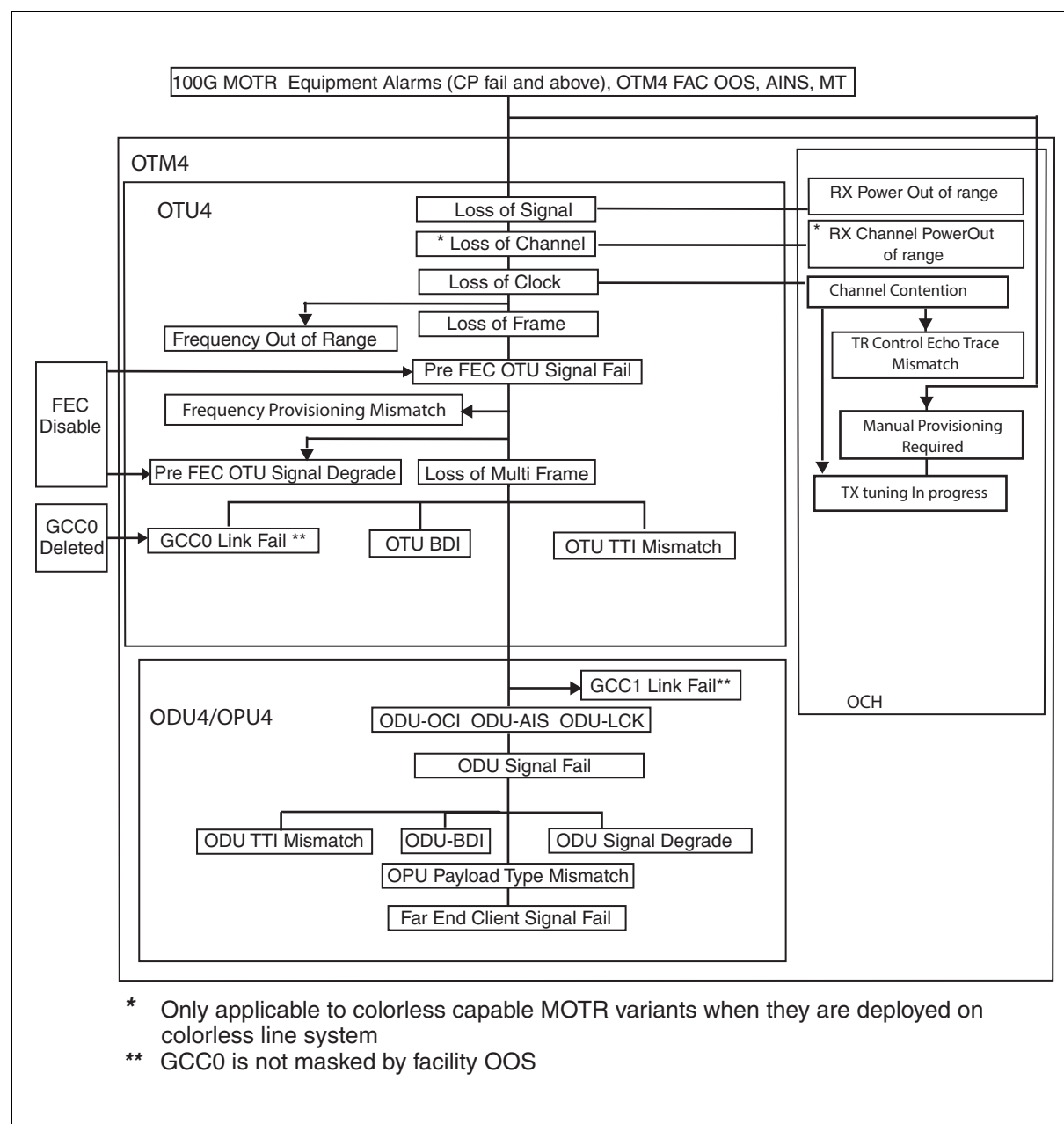
100G OCLD circuit packs line port OTM4 facility alarm hierarchy



100G WL3n MOTR circuit packs client port OTM2 facility alarm hierarchy

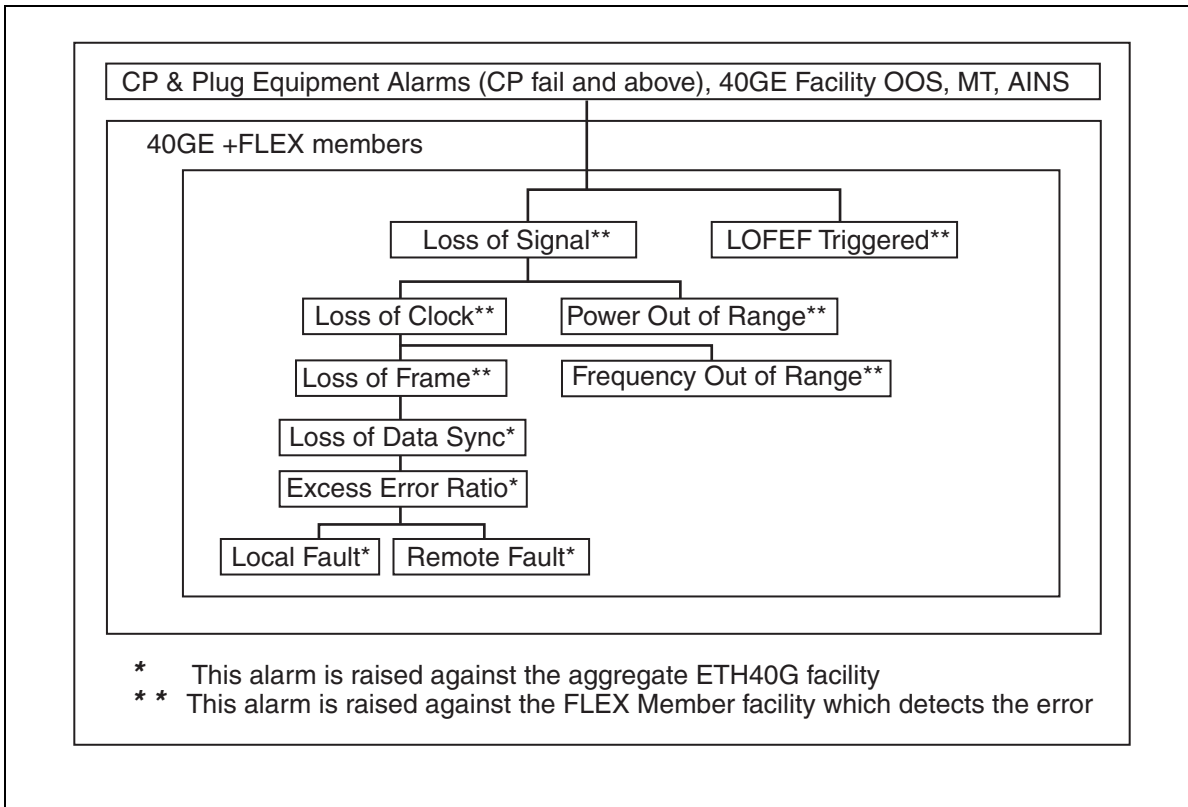


100G WL3n MOTR circuit packs line port OTM4 facility alarm hierarchy



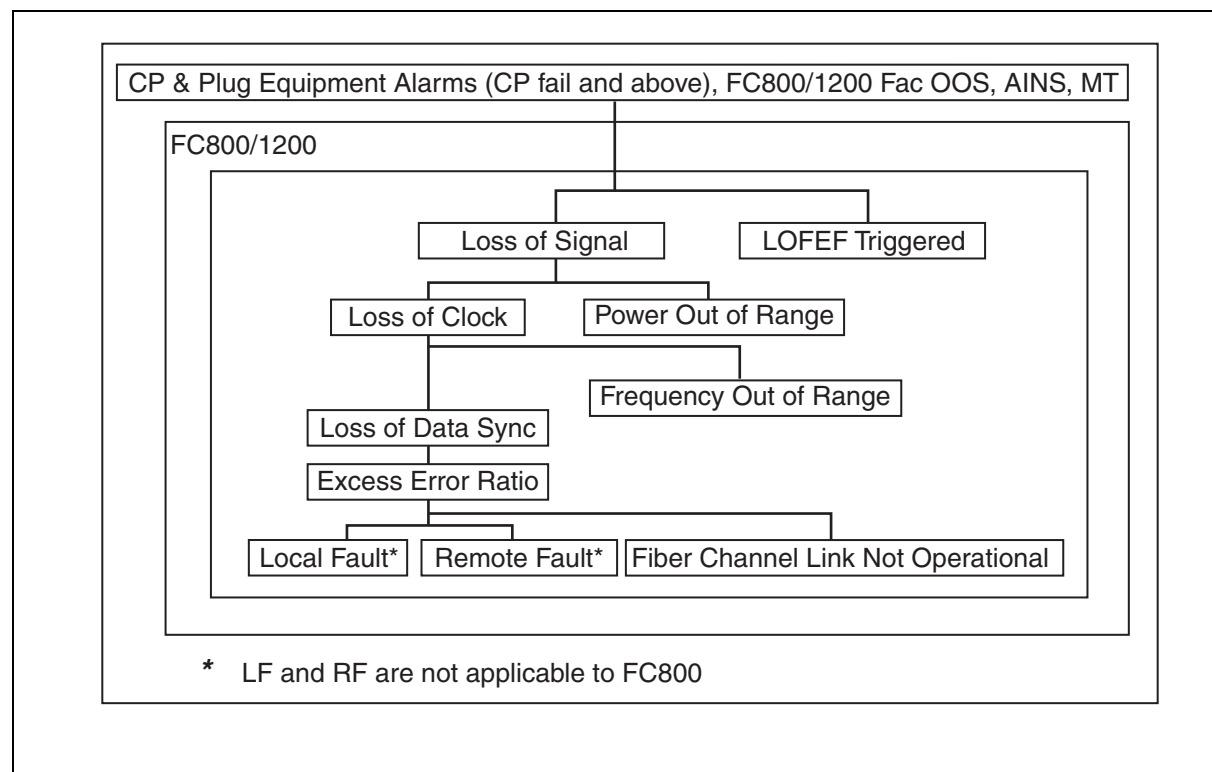
100G WL3n MOTR circuit pack ETH40G client facility alarm hierarchy

The ETH40G client facility on the client port of 100G WL3n MOTR circuit pack supports the following alarm hierarchy.



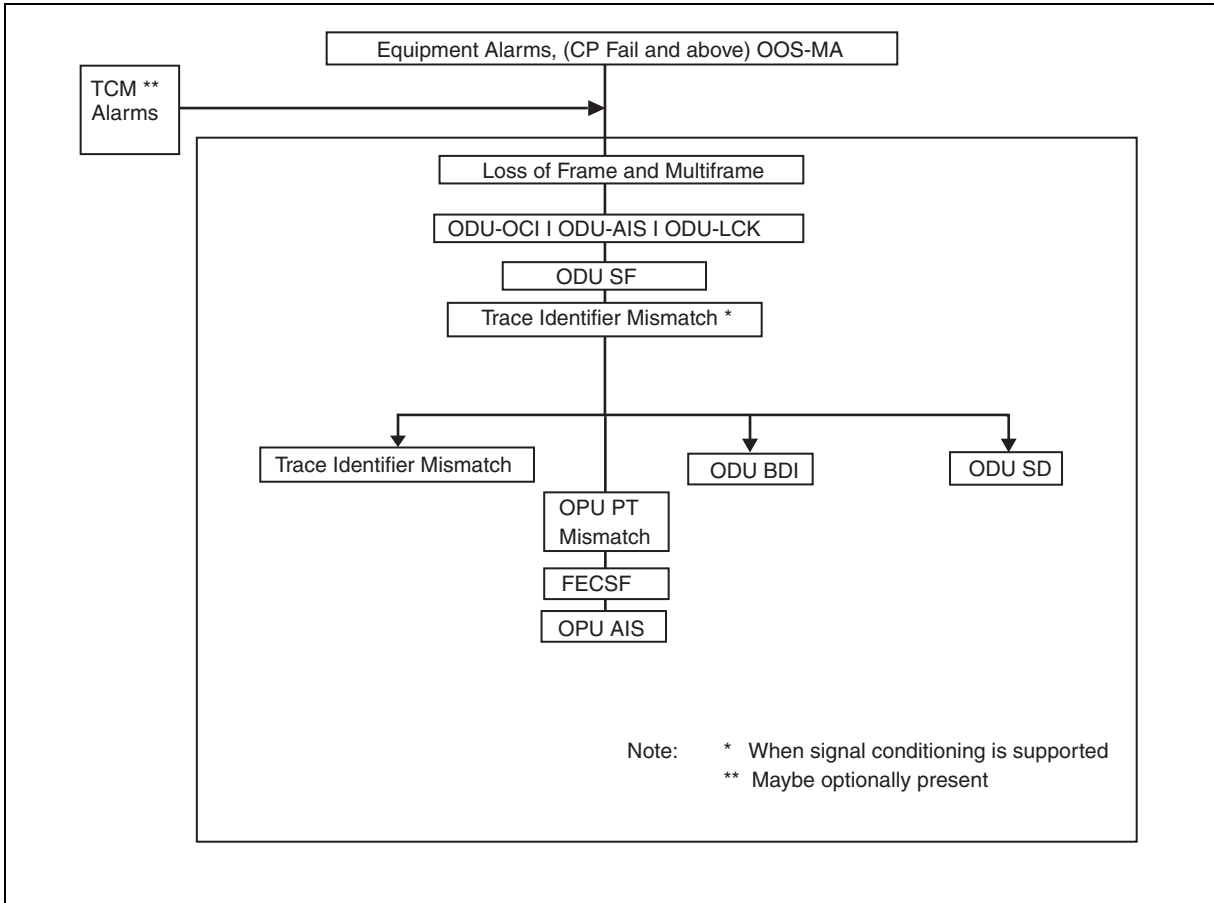
100G WL3n MOTR circuit pack Flex FC800/1200 facility alarm hierarchy

The FLEX FC800/FC1200 client facility on the client port of 100G WL3n MOTR circuit pack supports the following alarm hierarchy.



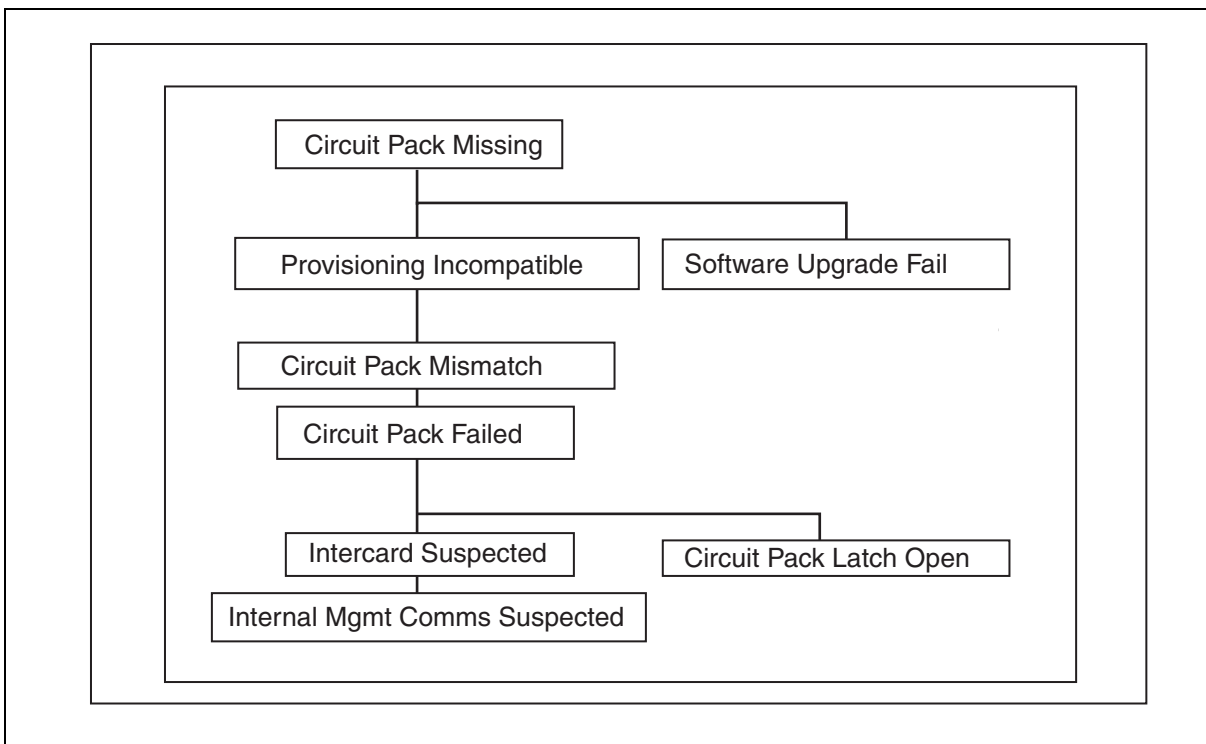
ODU Terminated CTP facility alarm hierarchy

The following indicates the ODUCTP facility alarm hierarchy.



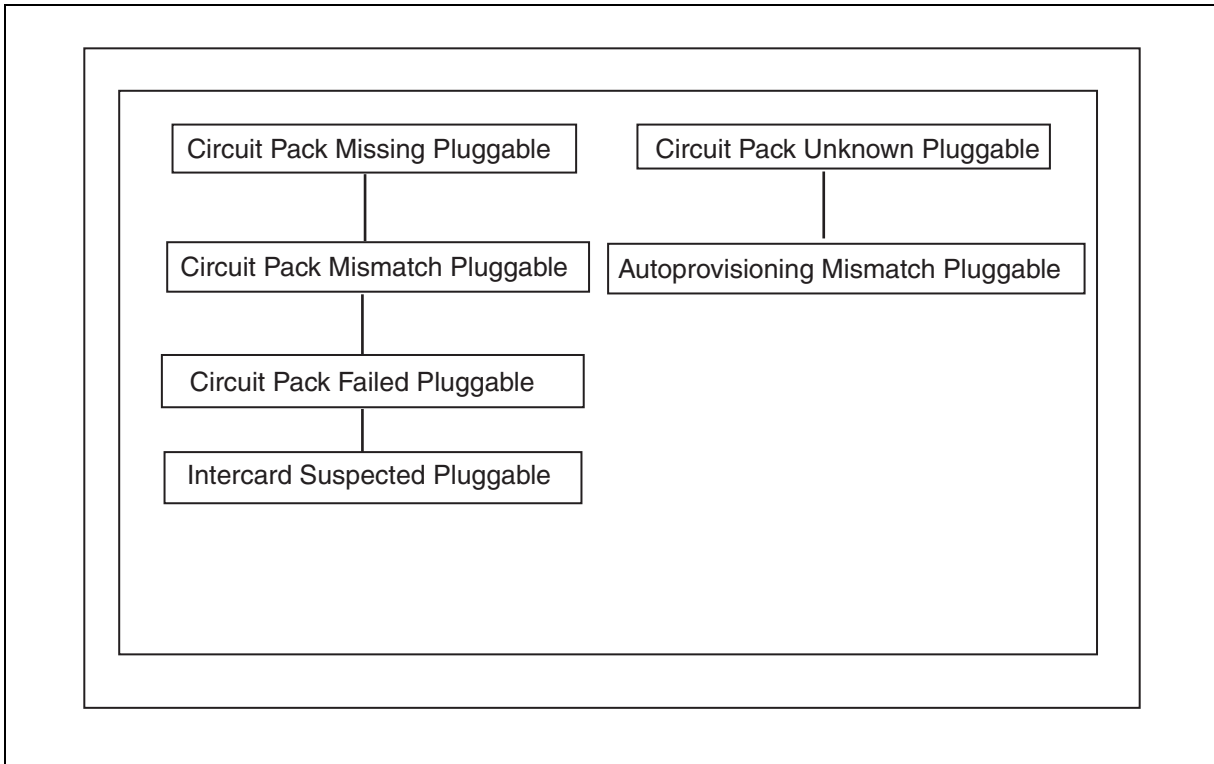
WLAi circuit pack transponder equipment alarm hierarchy

The WaveLogic Ai MOTR 56 Gbaud 4xQSFP28 (WLAi) circuit pack supports the following alarm hierarchy.



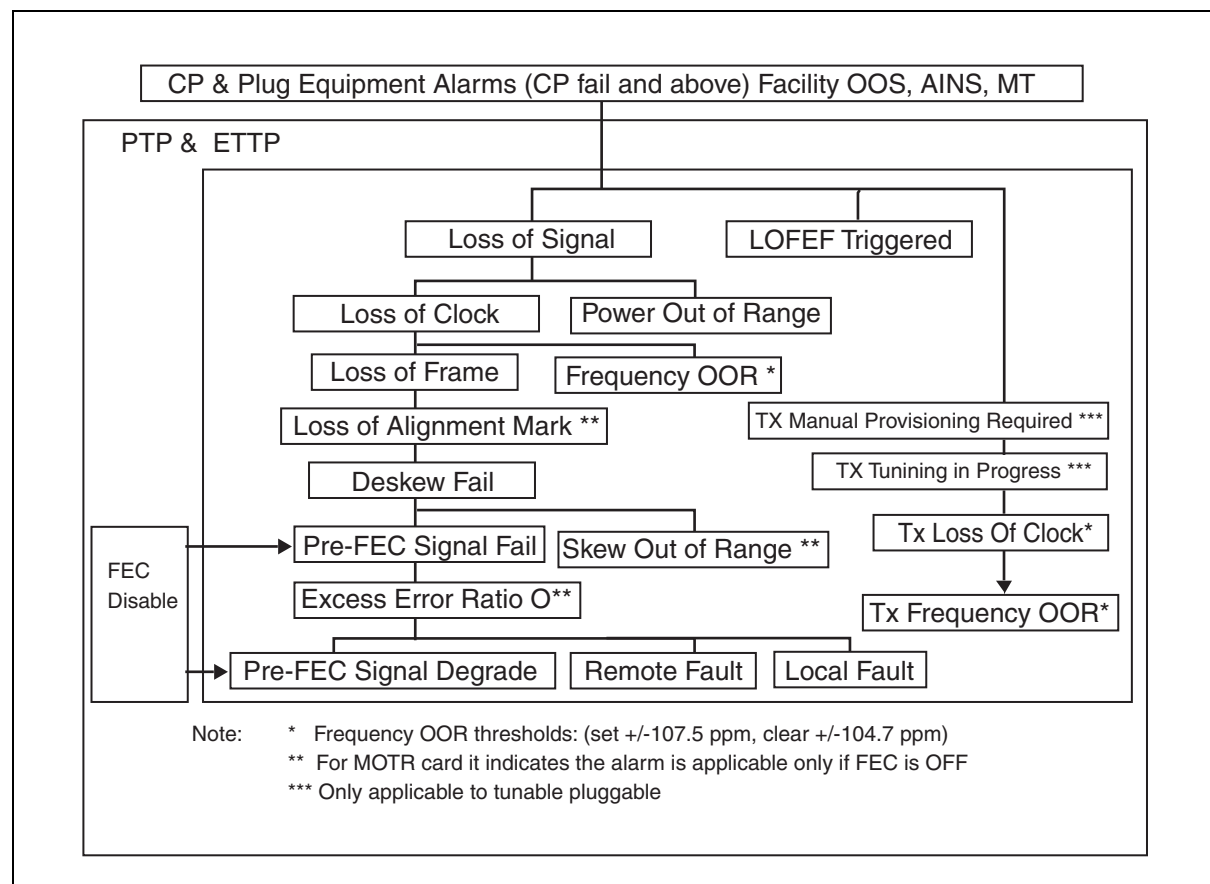
WLAi circuit pack QSFP pluggable equipment alarm hierarchy

The WLAi circuit pack supports the following QSFP pluggable equipment alarm hierarchy.



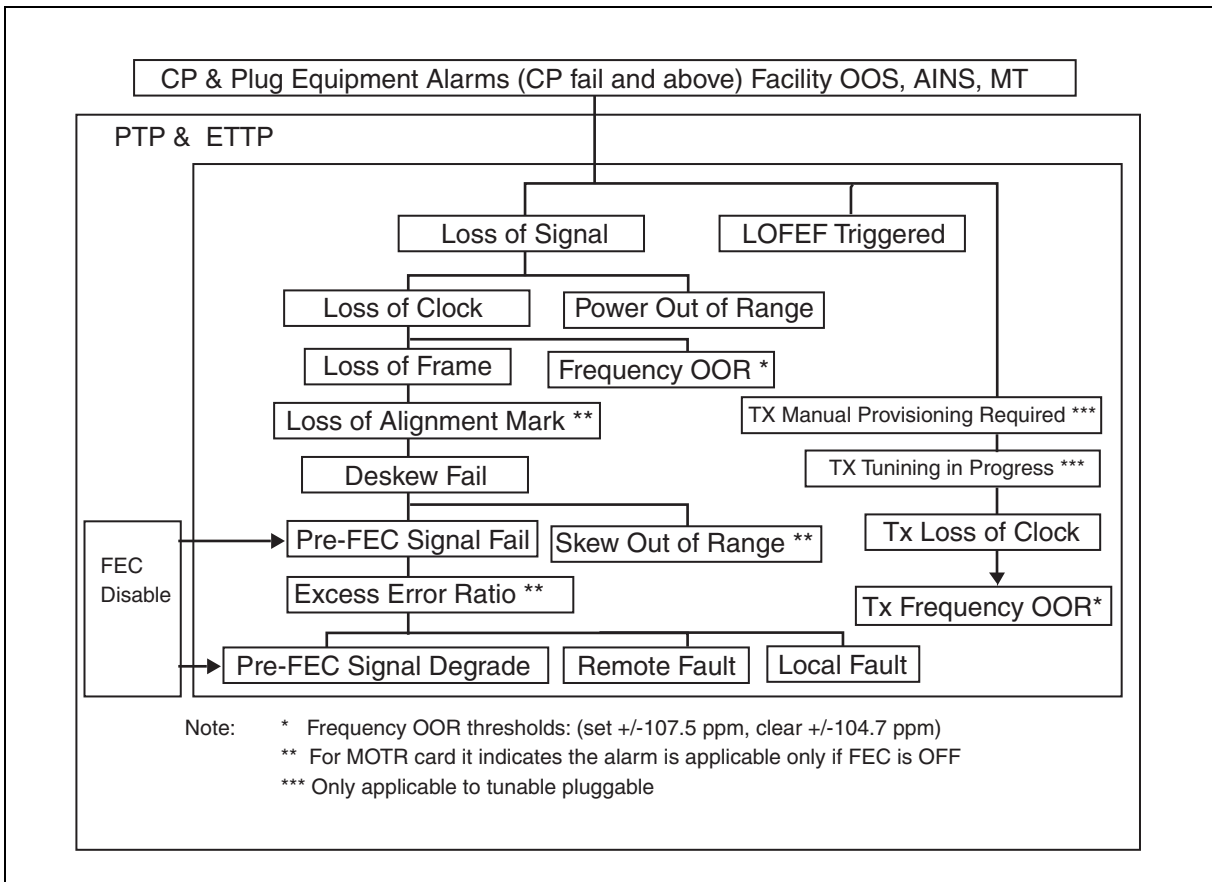
WLAi MOTR circuit pack 100GE client port alarm hierarchy

The WLAi circuit pack supports the following 100GE client port alarm hierarchy.



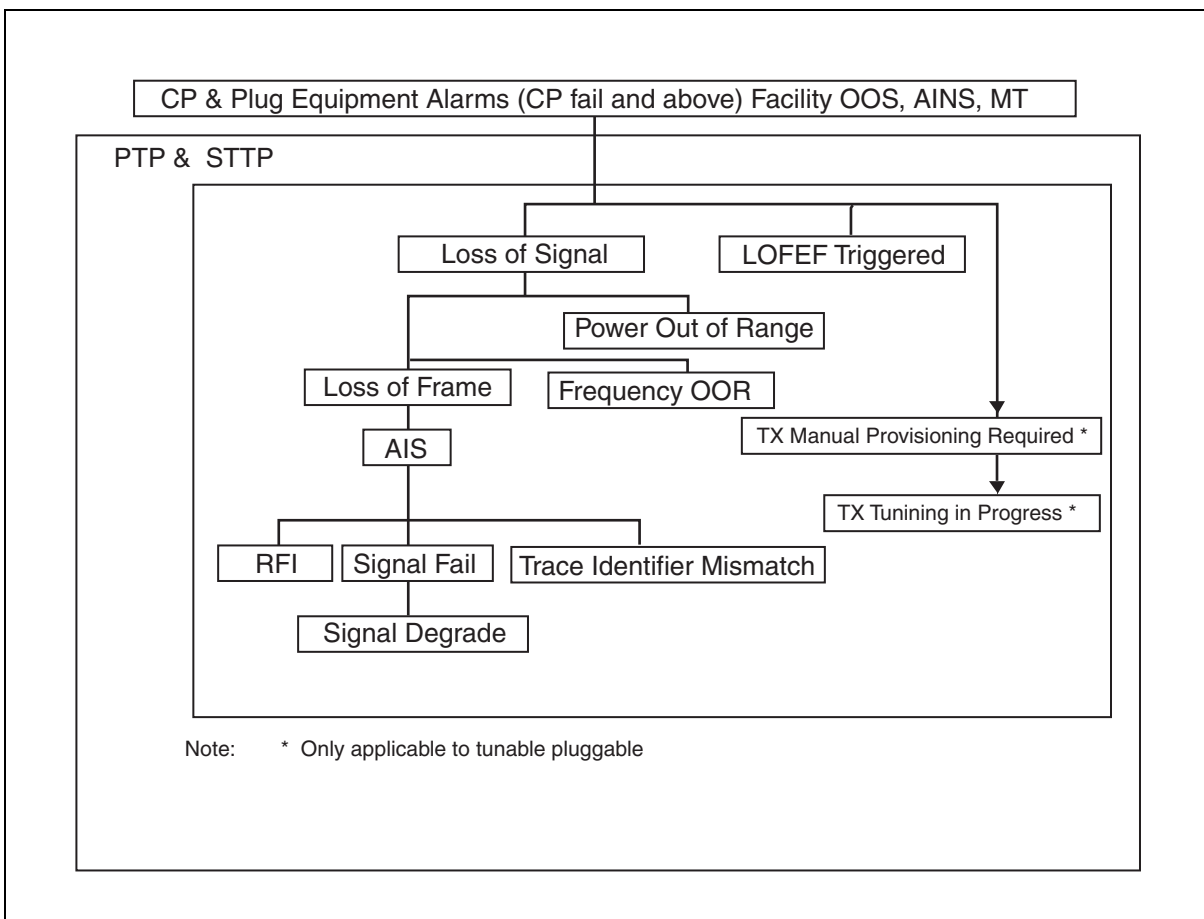
2xWLAI OTR circuit pack ETHn client port alarm hierarchy

The 2x WLAI OTR circuit pack supports the following ETHn client port alarm hierarchy



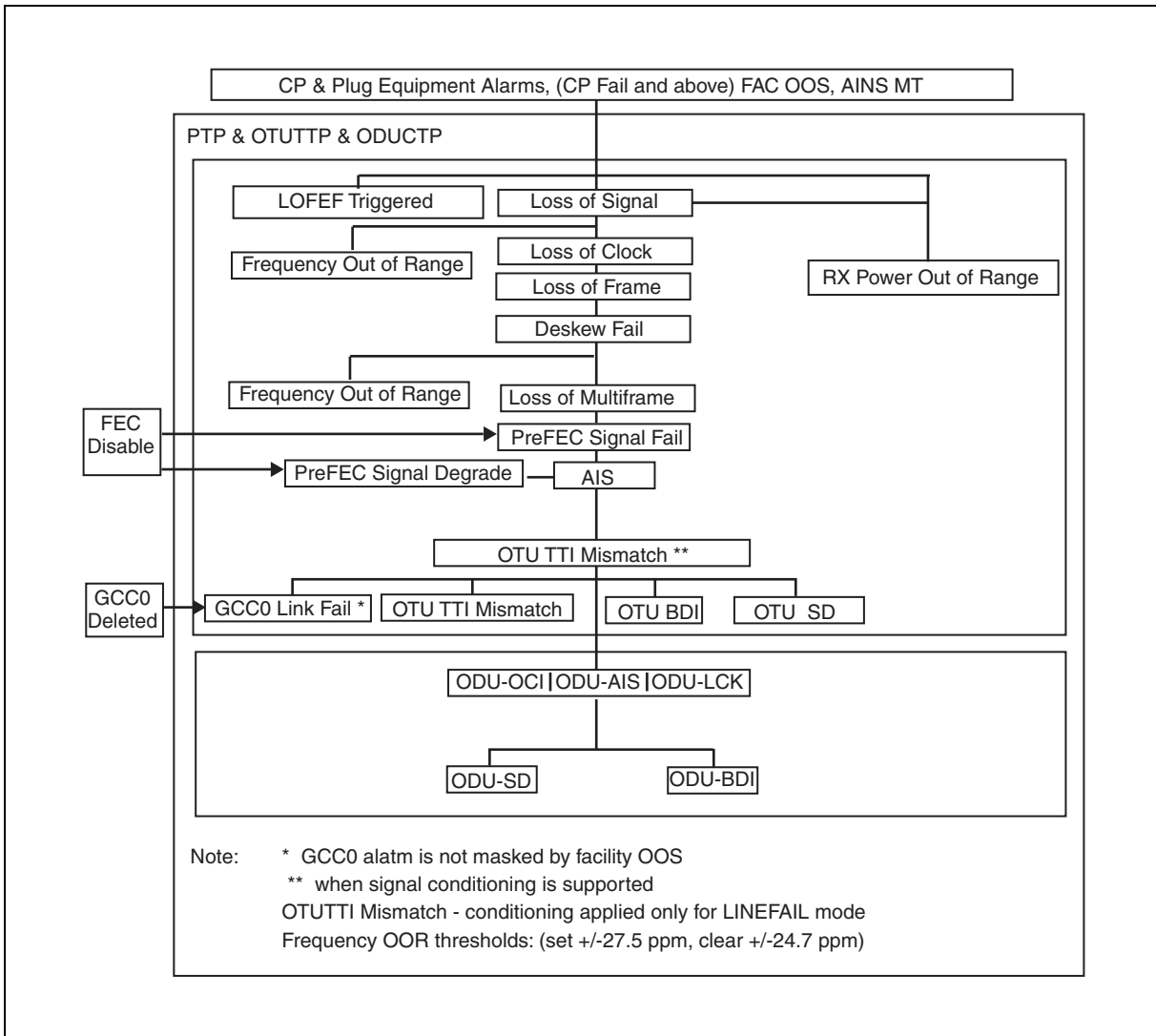
WLAi circuit pack STTP facility alarm hierarchy

The WLAi circuit pack supports the following STTP facility alarm hierarchy.



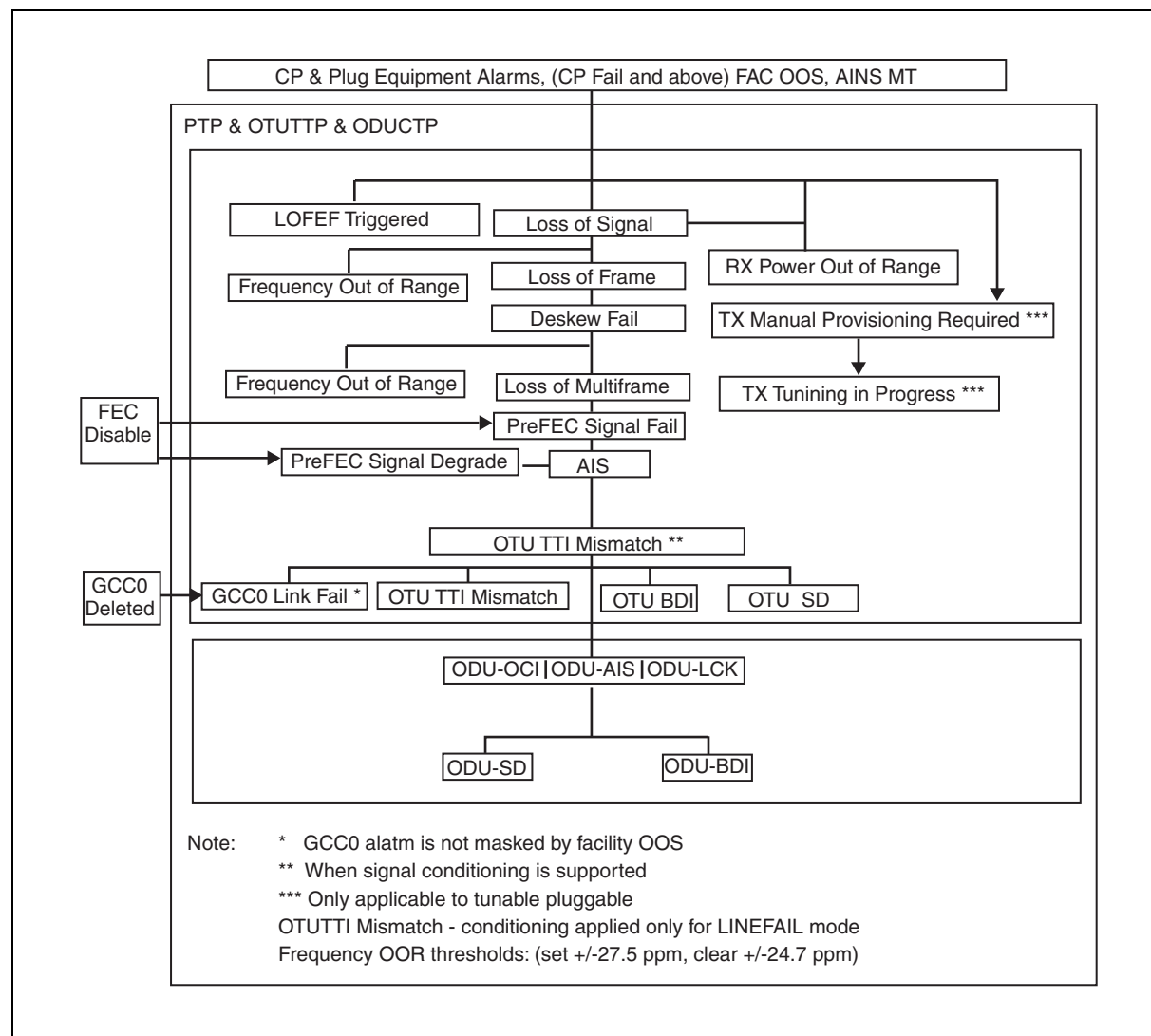
WLAi MOTR circuit pack OTU4 client port alarm hierarchy

The WLAi circuit pack supports the following OTU4 client port alarm hierarchy.



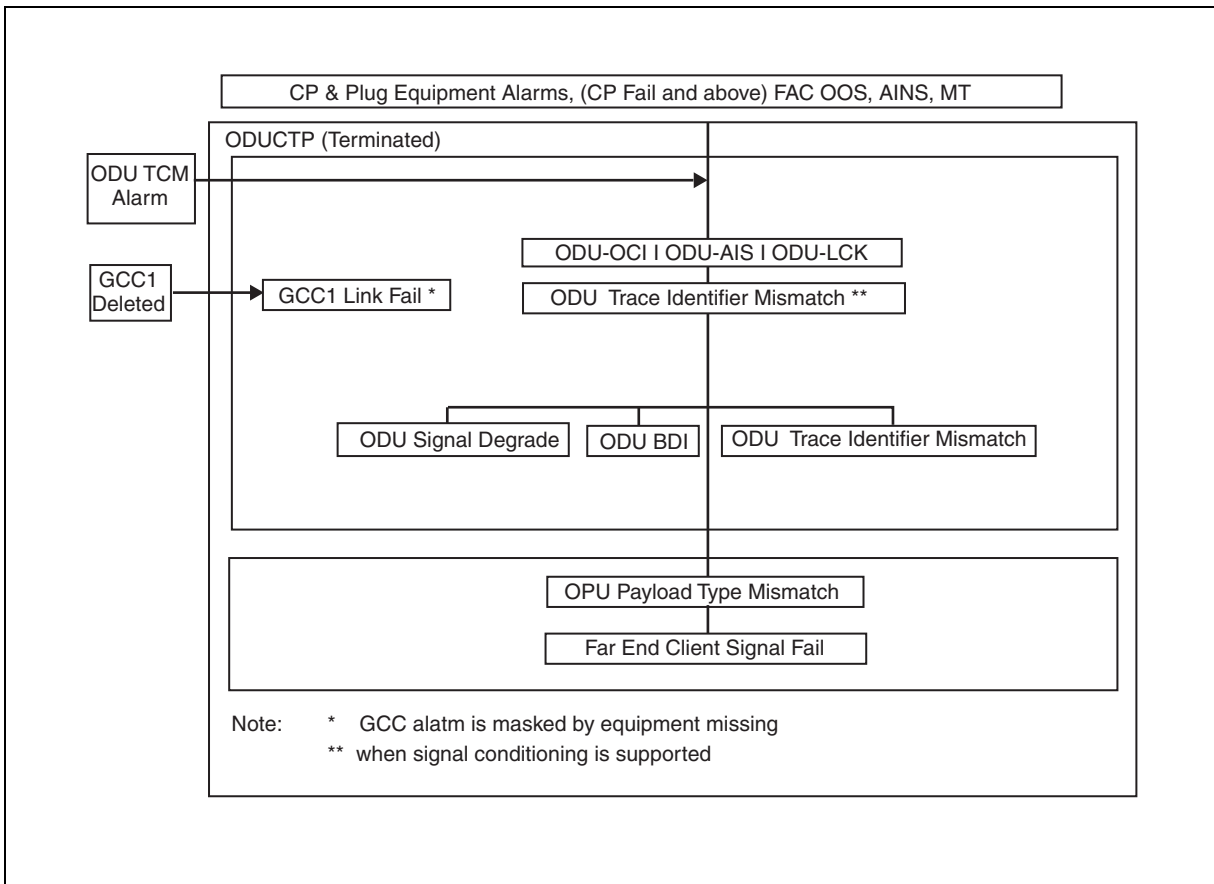
WLAi circuit pack OTUTTP and ODUCTP monitored client port alarm hierarchy

The WLAi circuit pack supports the following OTUTTP and ODUCTP monitored client port alarm hierarchy.



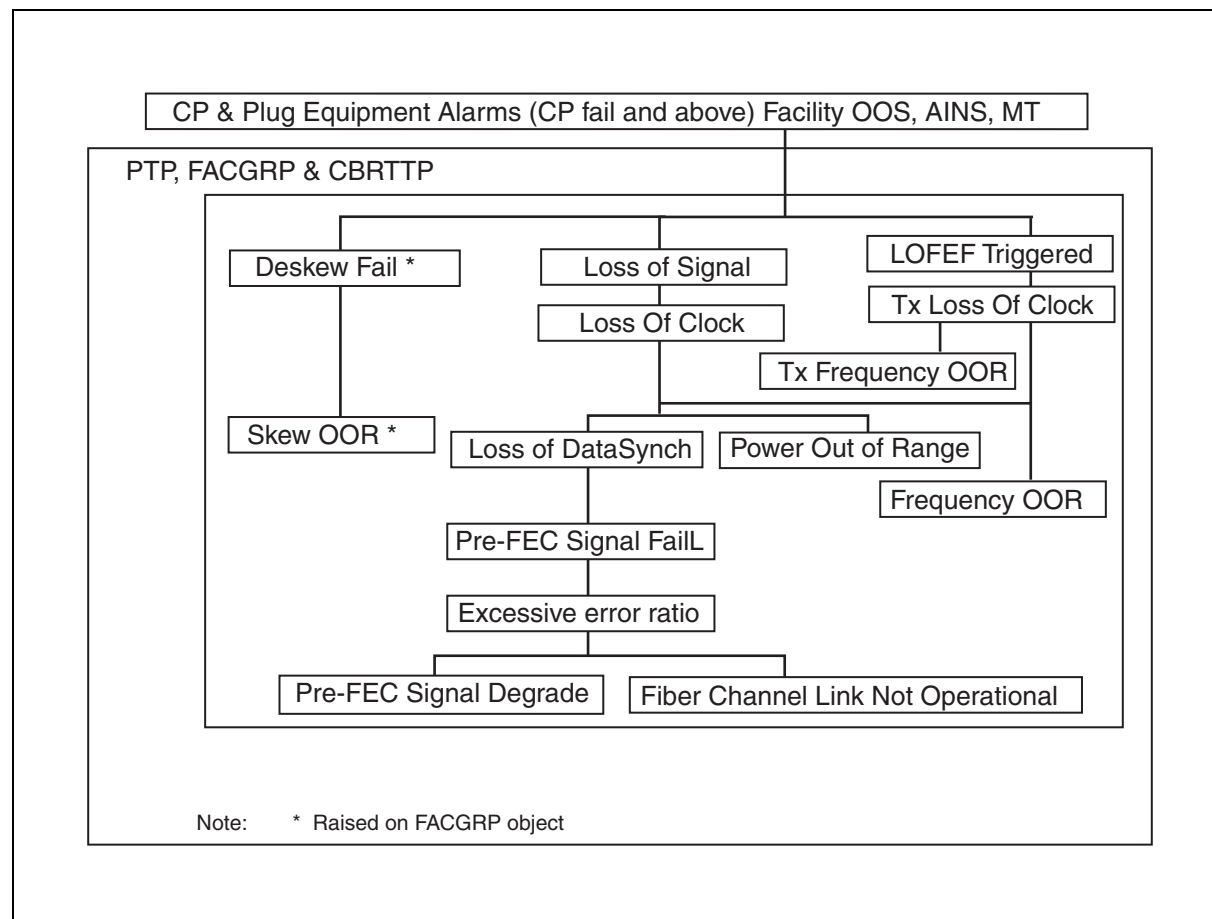
WLAi circuit pack Terminated ODUCTP facility alarm hierarchy

The WLAi circuit packs support the following ODUCTP client port alarm hierarchy.



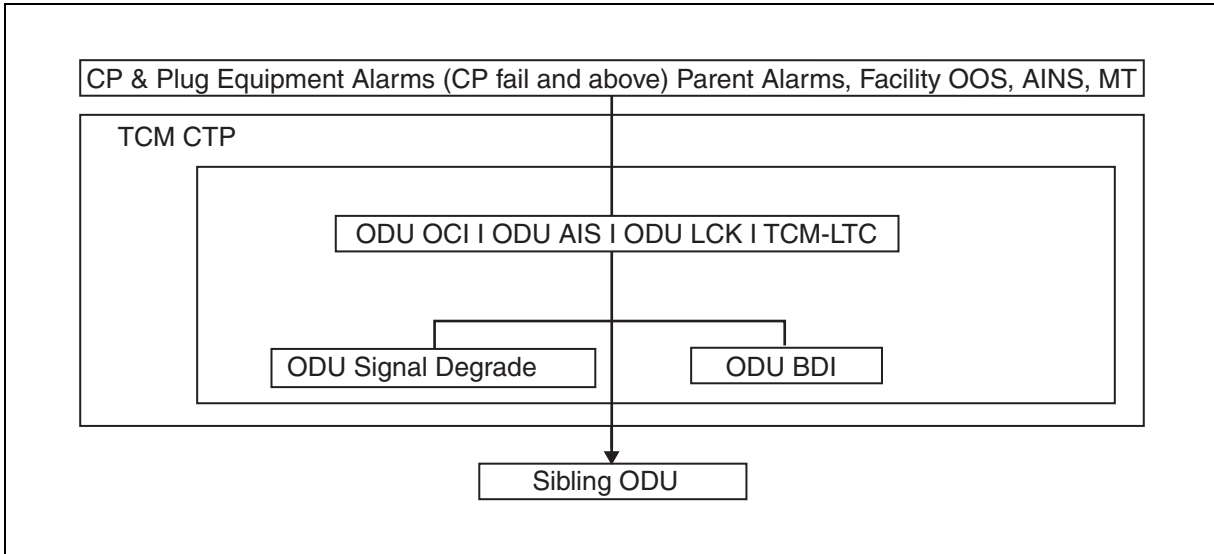
WLAi circuit pack CBRTTP facility alarm hierarchy

The WLAi circuit pack supports the following CBRTTP facility alarm hierarchy.



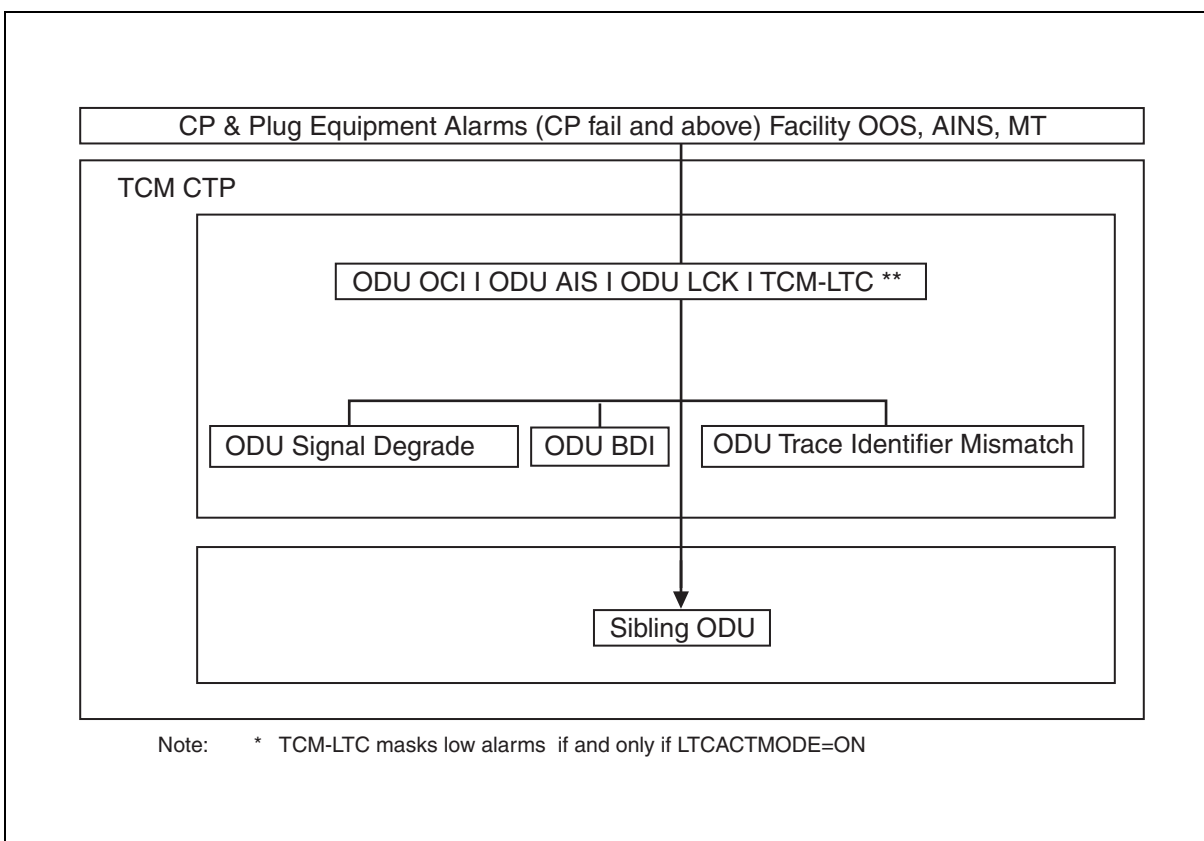
WLAi and WL5e circuit packs TCM CTP (monitor) facility alarm hierarchy

The WLAi and WL5e circuit packs support the following TCM CTP (monitor) facility alarm hierarchy.



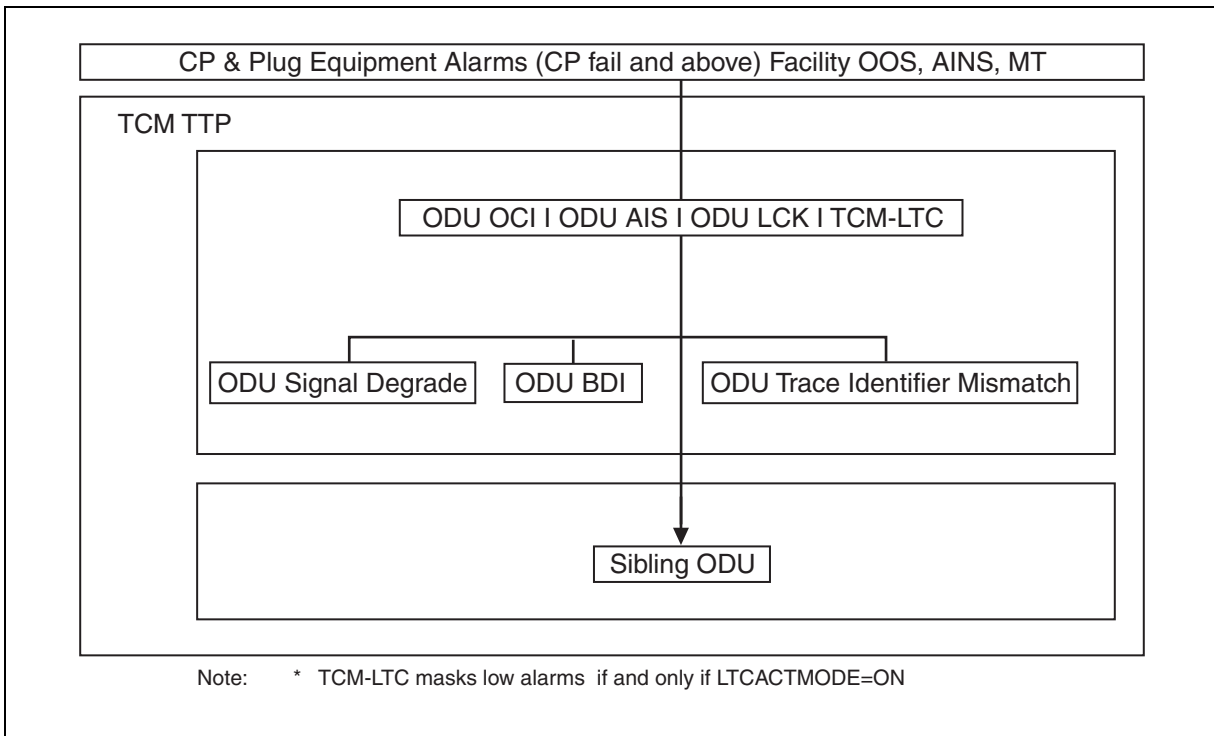
WLAi and WL5e circuit packs TCM CTP (terminated) facility alarm hierarchy

The WLAi and WL5e circuit packs support the following TCM CTP (terminated) facility alarm hierarchy.



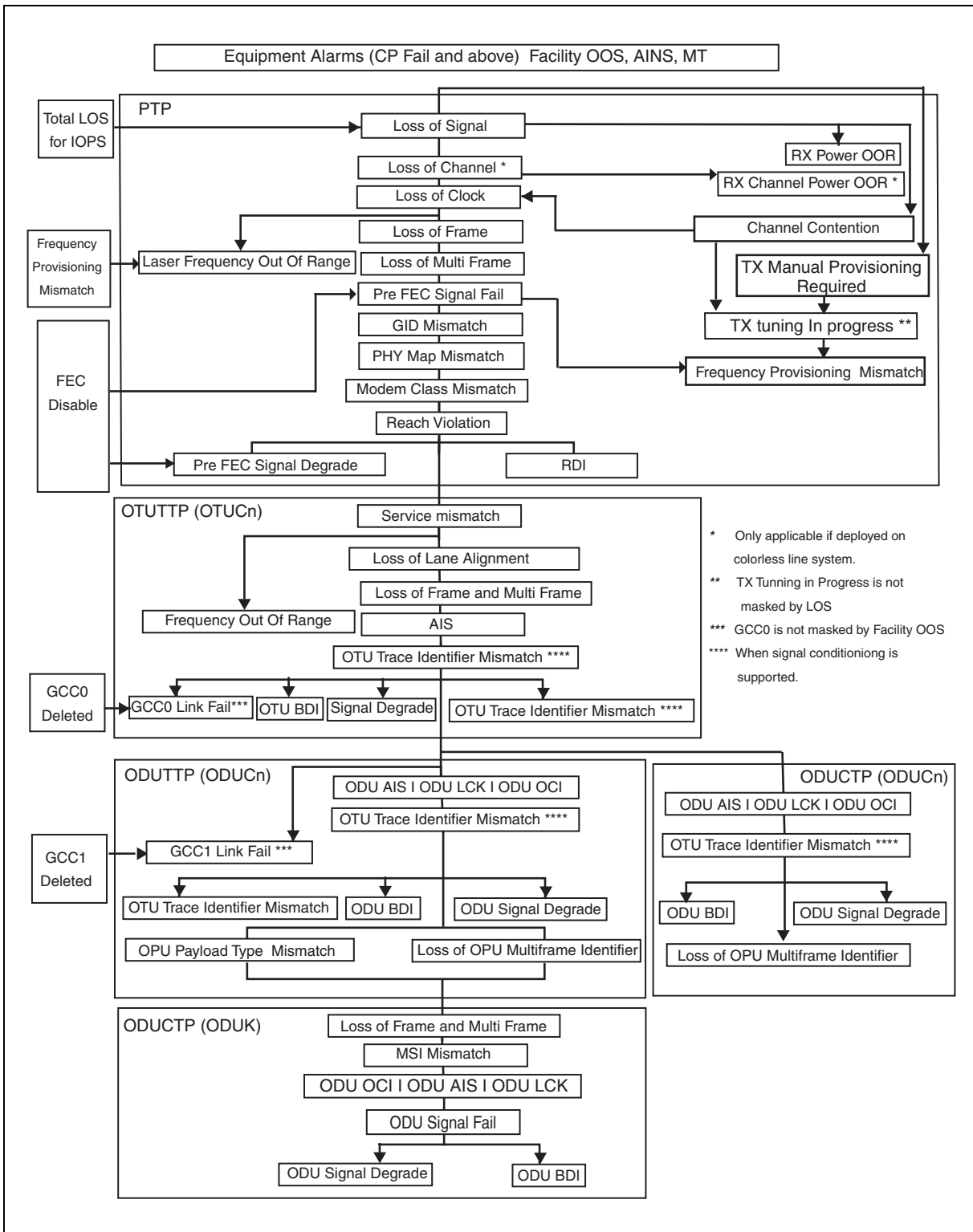
WLAi and WL5e circuit pack TCM TTP facility alarm hierarchy

The WLAi and WL5e circuit packs support the following TCMTTP facility alarm hierarchy.



WLAi circuit pack PTP and ODUCTP facility for ODUk on line port alarm hierarchy

The PTP facility and ODUCTP facility for ODUk on line port of the WLAi transponder equipment supports following alarm hierarchy.



Equipment Alarms (CP fail and above) Facility OOS, AINS, MT

Legend:

- * Only applicable if deployed on colorless line system.
- ** Tx Tuning in Progress is not masked by LOS
- *** GCC alarm is masked by equipment missing
- **** Applicable to: (1)- non-iOPS port (2)- Active PTP port for iOPS

PTP

- Loss of Signal ****
 - Loss of Channel *
 - Loss of Clock
 - Loss of Frame
 - Loss of Multiframe
 - Pre FEC Signal Fail
 - GID Mismatch
 - PHY Map Mismatch
 - Modem Class Mismatch
 - Reach Violation
- RX Power OOR
 - RX Channel Power OOR *
- Channel Contention
 - TX Manual Provisioning Required
 - TX tuning In progress **
- Frequency Provisioning Mismatch
 - Laser Frequency Out Of Range
- Rx Dispersion Post Compensation Out of Range

FEC Disable

- Pre FEC Signal Degrade
- RDI

OTUTTP (OTUCn)

- Service Mismatch
 - Loss of Frame and Multiframe
 - Loss of Lane Alignment
 - AIS
- GCC0 Deleted
 - GCC0 Link Fail***
- OTU Trace Identifier Mismatch

ODUTTP (ODUCn)

- ODU AIS | ODU LCK | ODU OCI
 - GCC1 Deleted
 - GCC1 Link Fail ****
 - OTU Trace Identifier Mismatch
 - ODU BDI
 - ODU Signal Degrade
 - OPU Payload Type Mismatch
 - Loss of OPU Multiframe Identifier

ODUCTP (ODUCn)

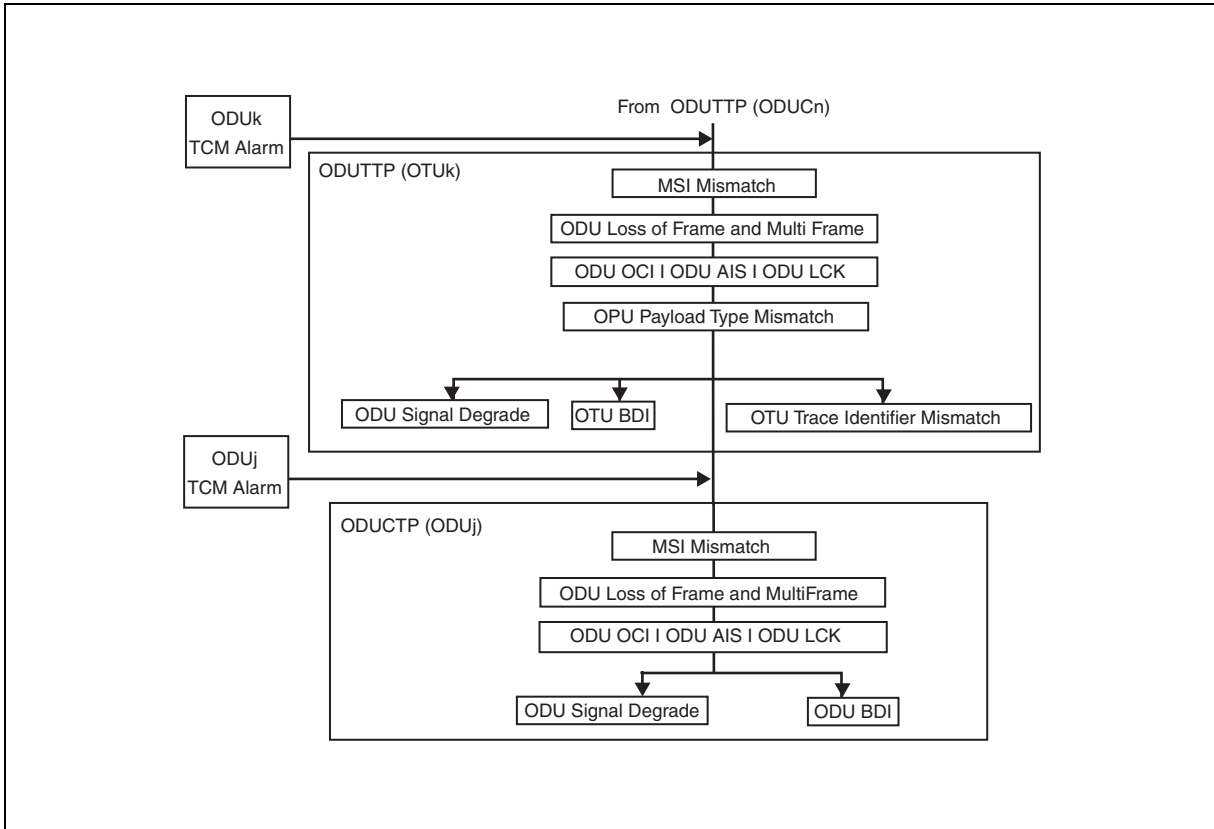
- ODU AIS | ODU LCK | ODU OCI
 - ODU BDI
 - ODU Signal Degrade

ODUCTP (ODUK)

- MSI Mismatch
 - ODU Loss of Frame and Multiframe
 - ODU OCI | ODU AIS | ODU LCK
 - ODU Signal Degrade
 - ODU BDI

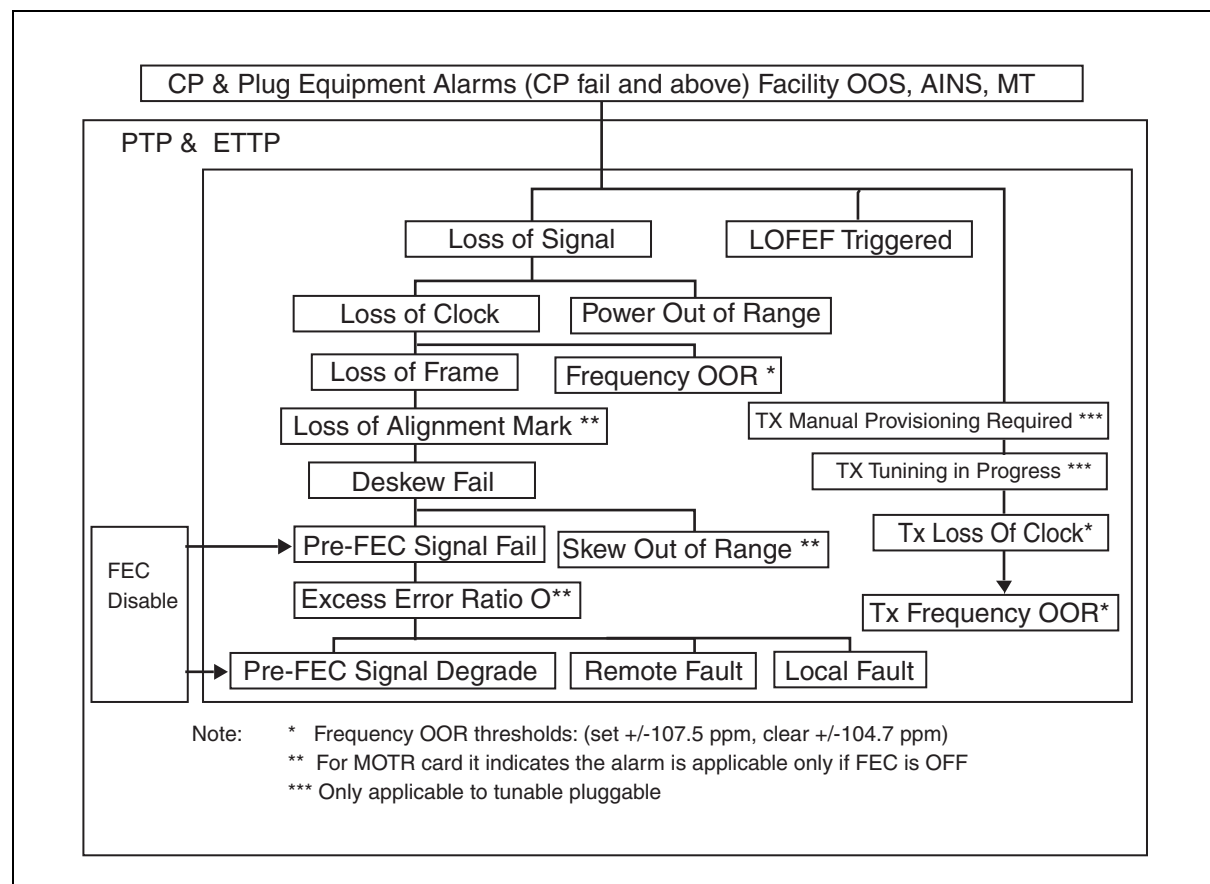
ODUCn TCM alarm

ODUK TCM alarm



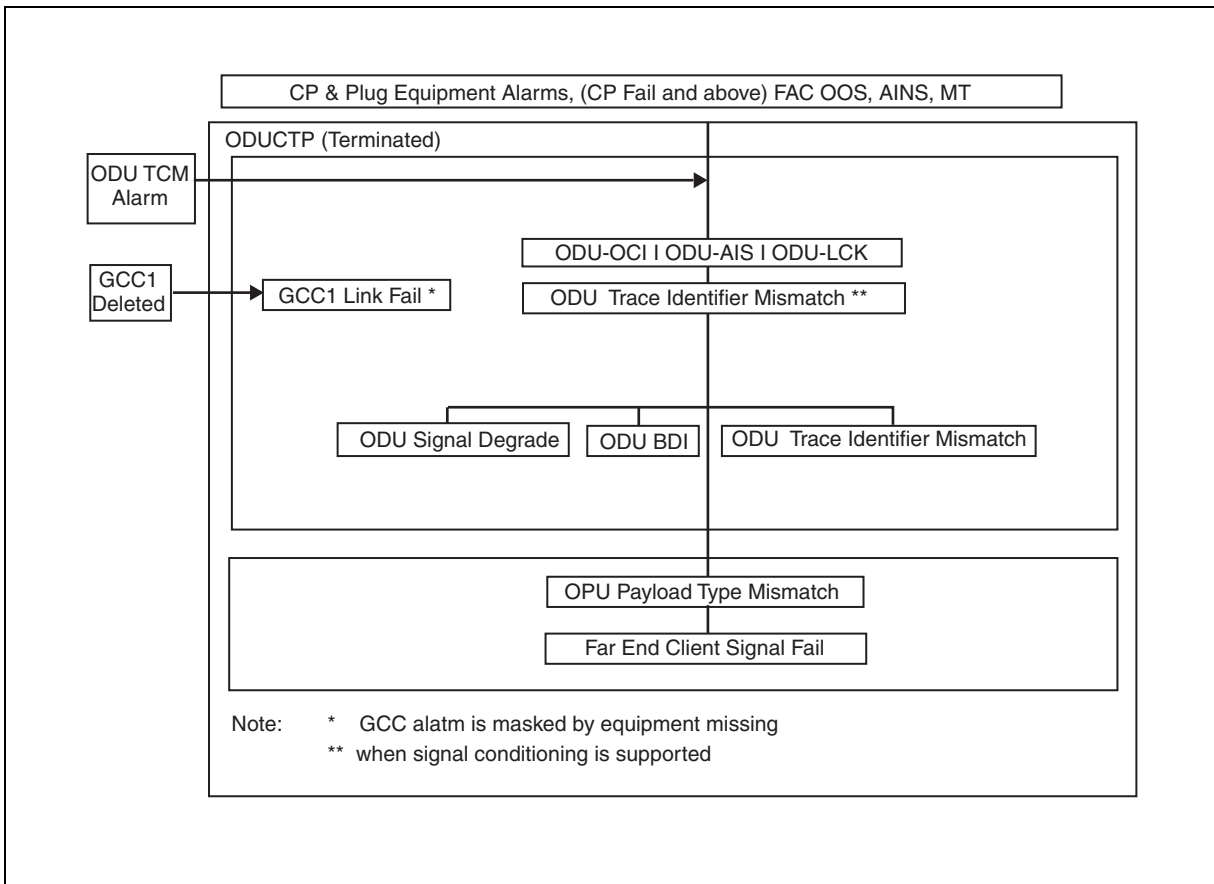
WL5e MOTR circuit pack 400GE client port alarm hierarchy

The WL5e circuit pack supports the following 400GE client port alarm hierarchy.



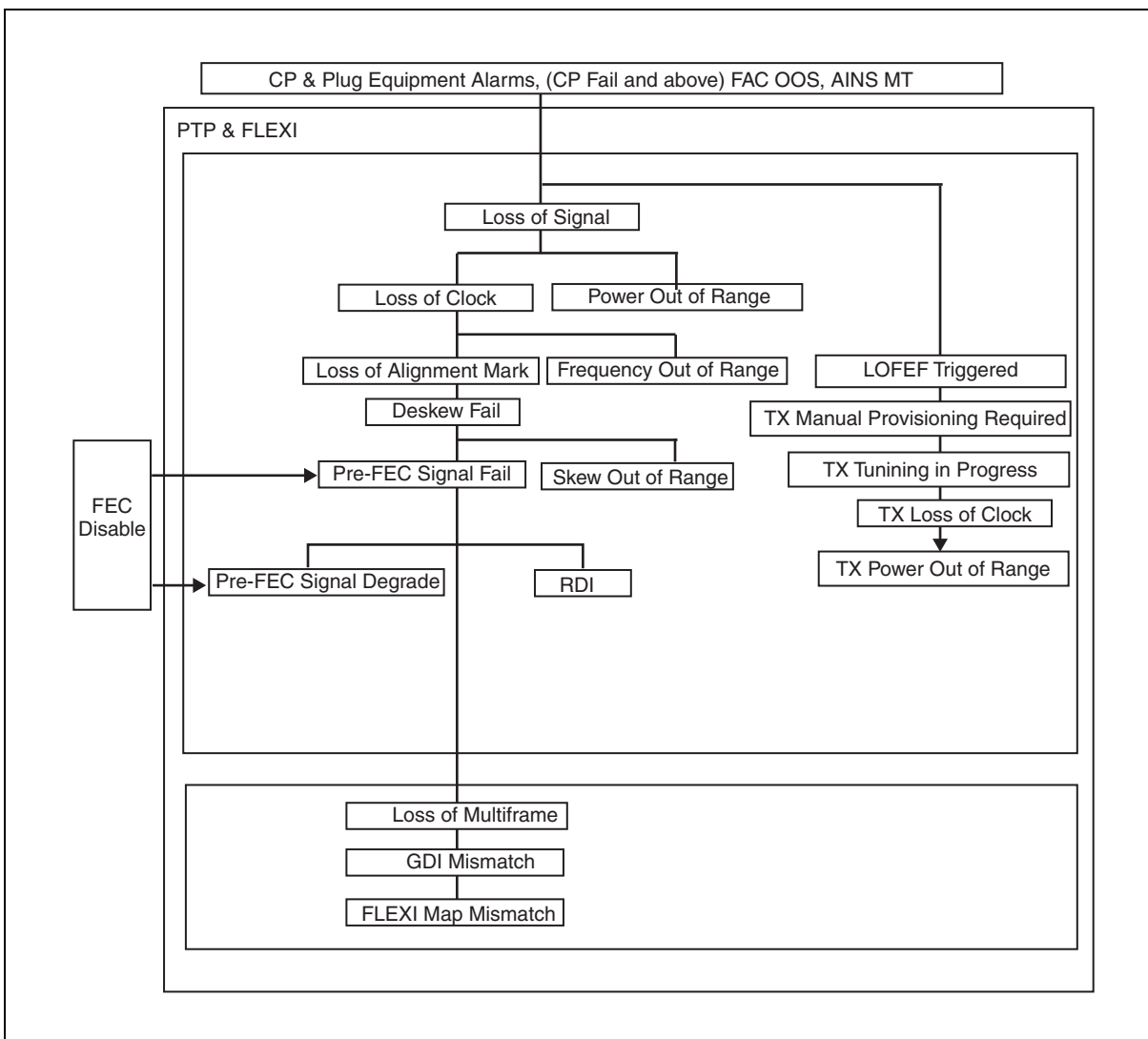
WL5e MOTR circuit pack ODUCTP facility alarm hierarchy

The WL5e MOTR circuit pack supports the following ODUCTP client port alarm hierarchy.

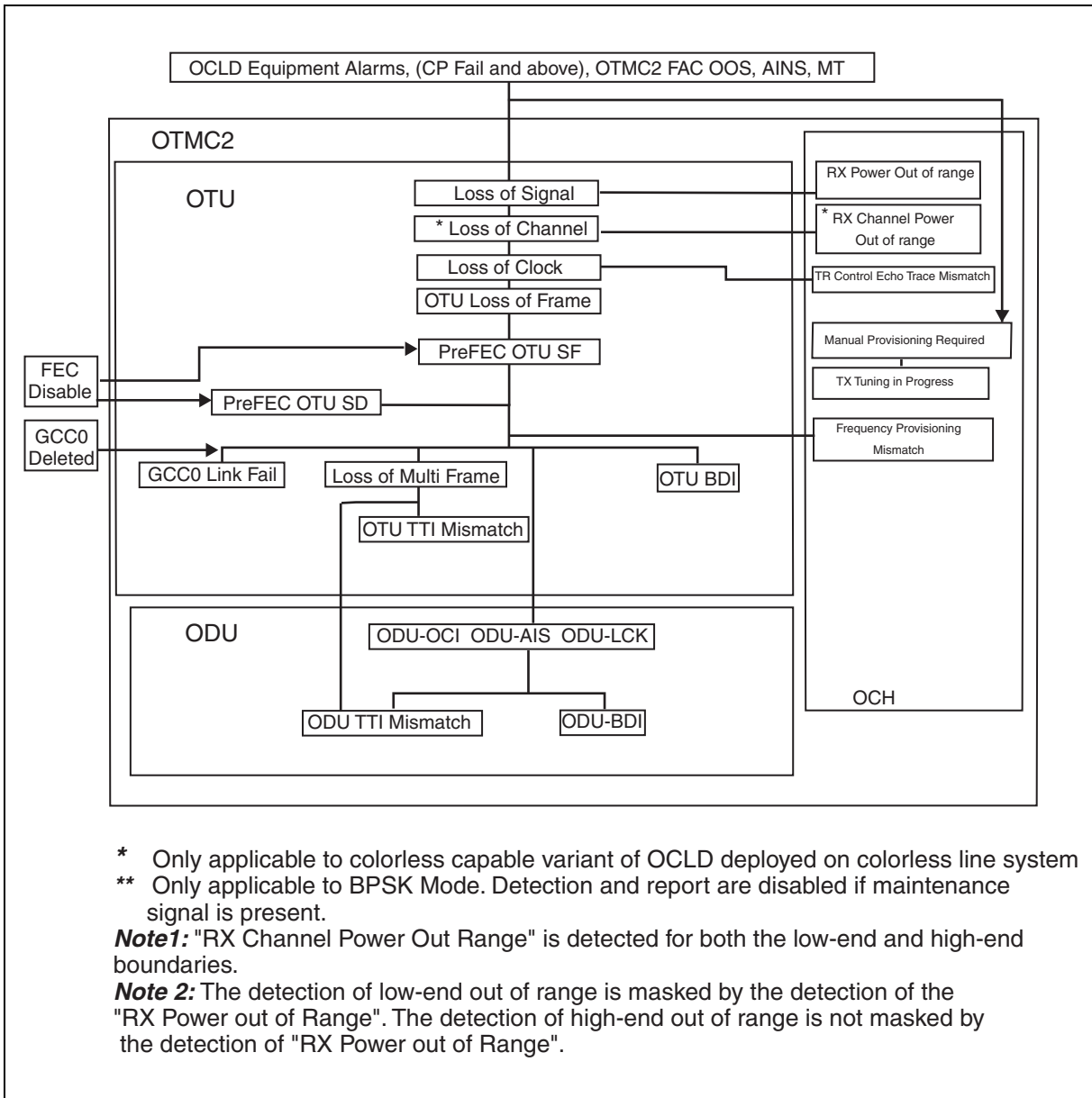


FlexO PTP facility alarm hierarchy

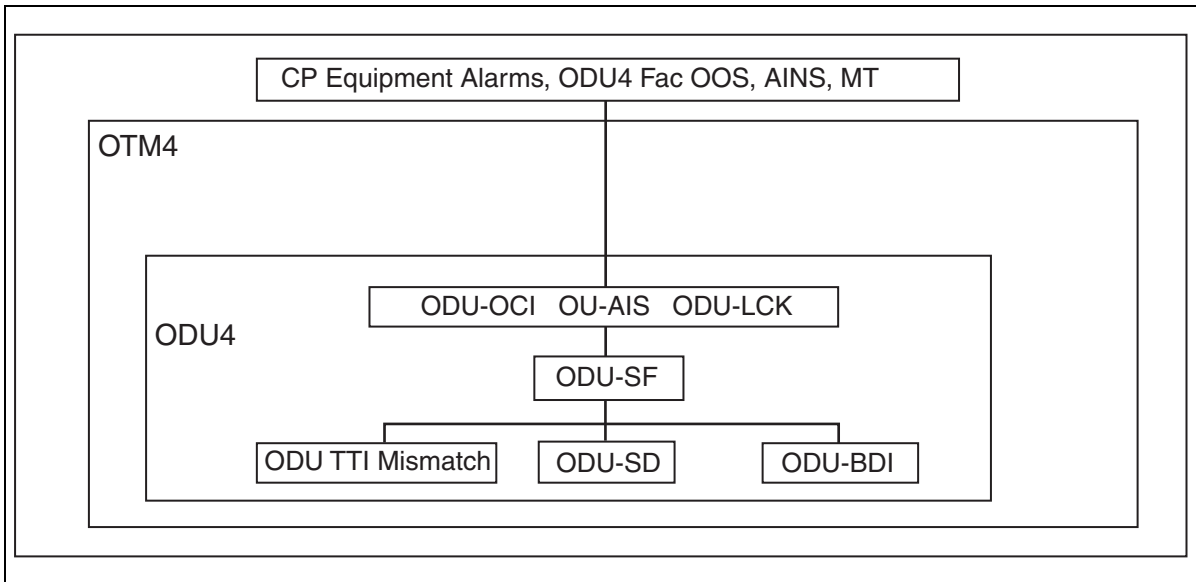
The FlexO PHY PTP facility supports the following alarm hierarchy.



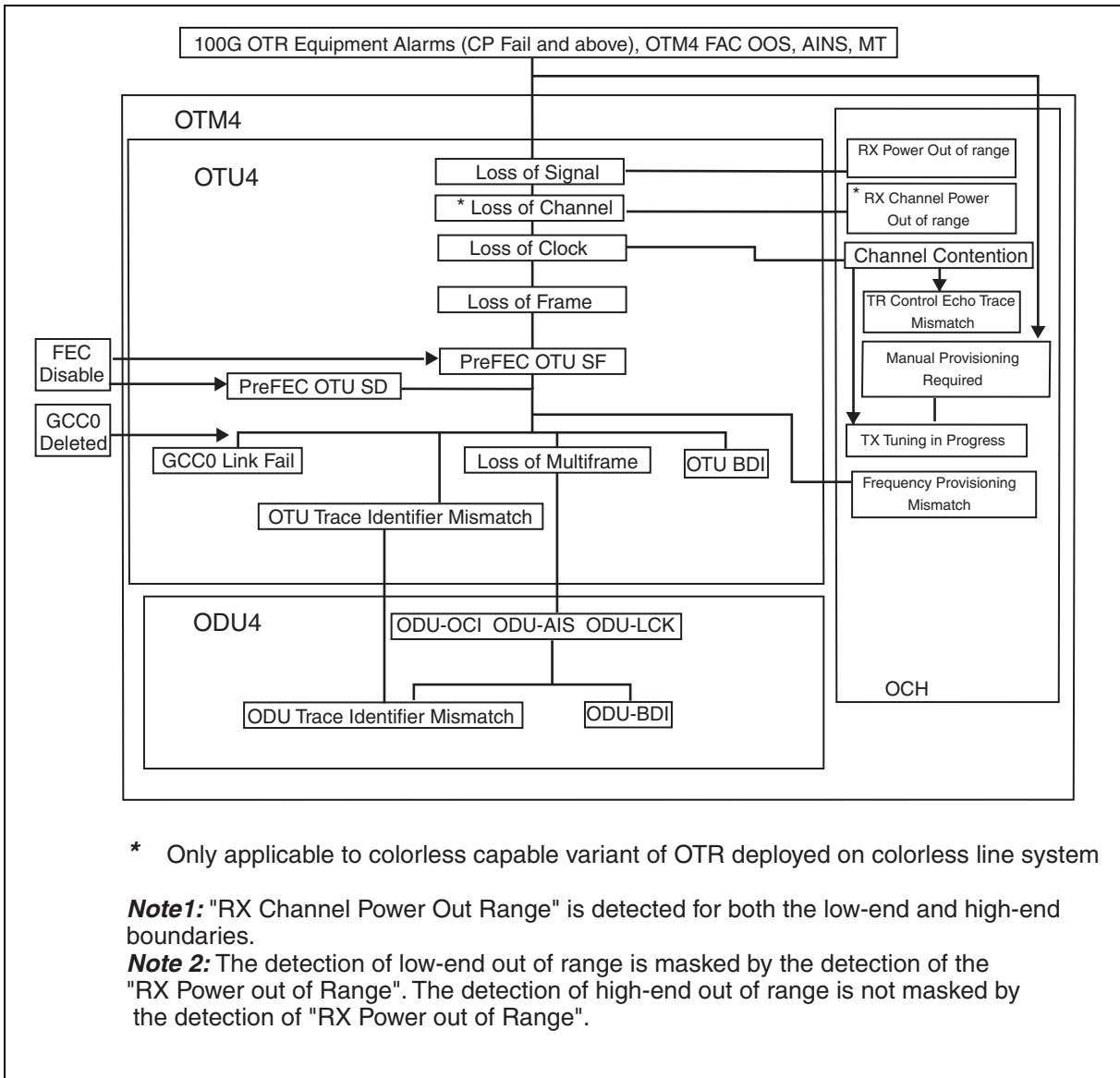
Flex3 WL3e OCLD circuit packs line port OTMC2 facility alarm hierarchy



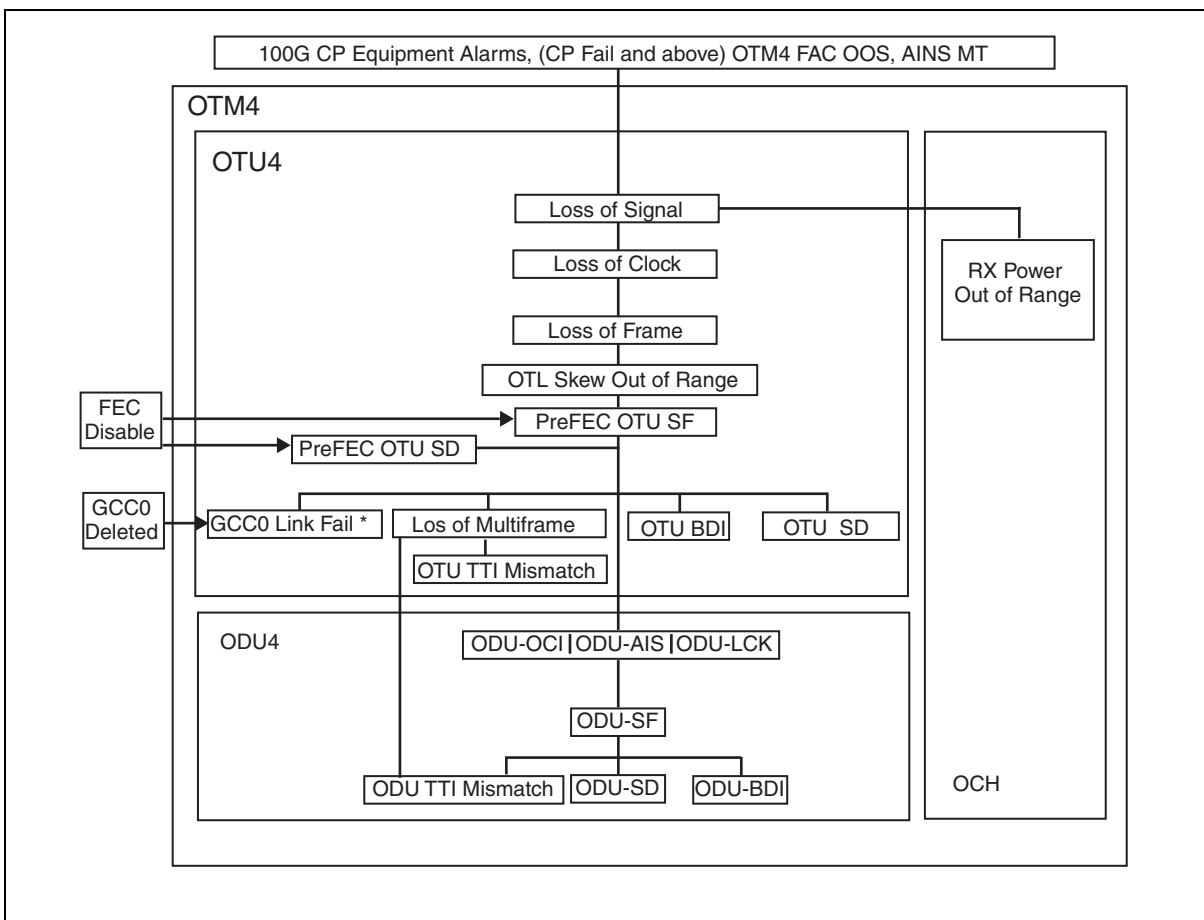
Flex3 WL3e OCLD circuit packs line port ODU4 facility in 16QAM alarm hierarchy



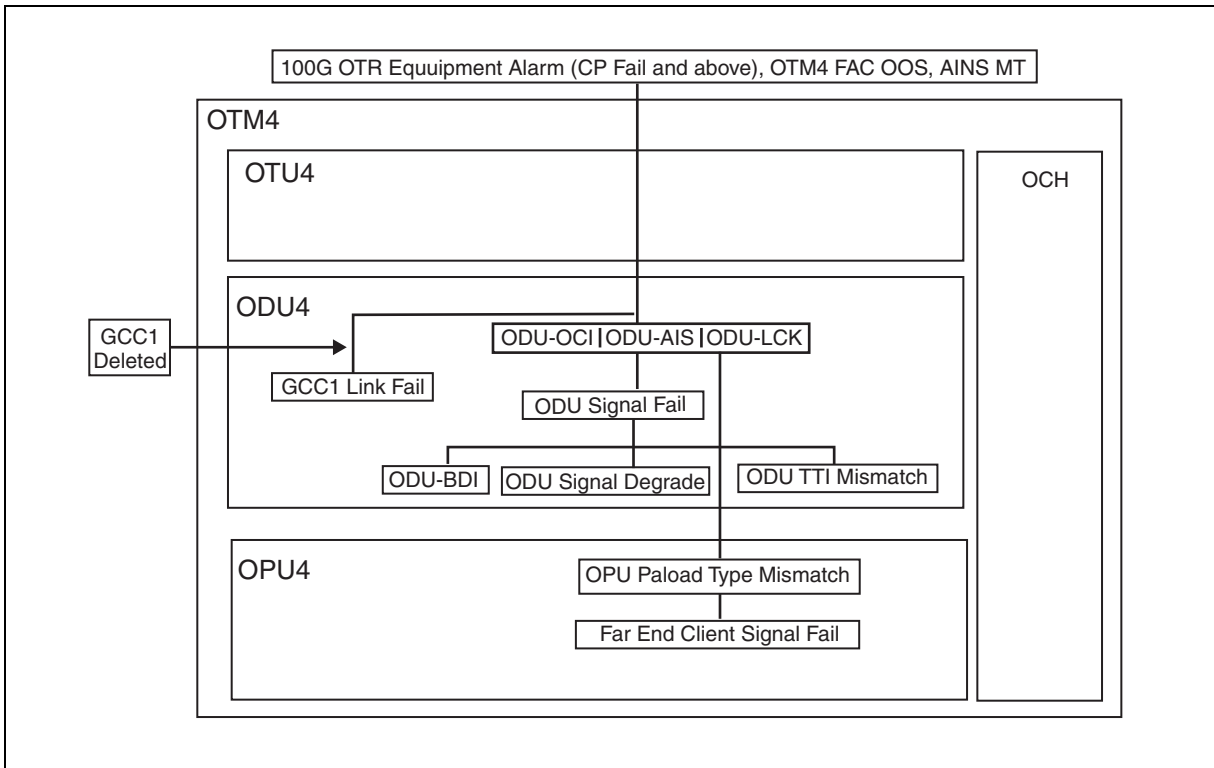
100G WL3n/WL3e OTR circuit packs line port OTM4 facility in QPSK alarm hierarchy



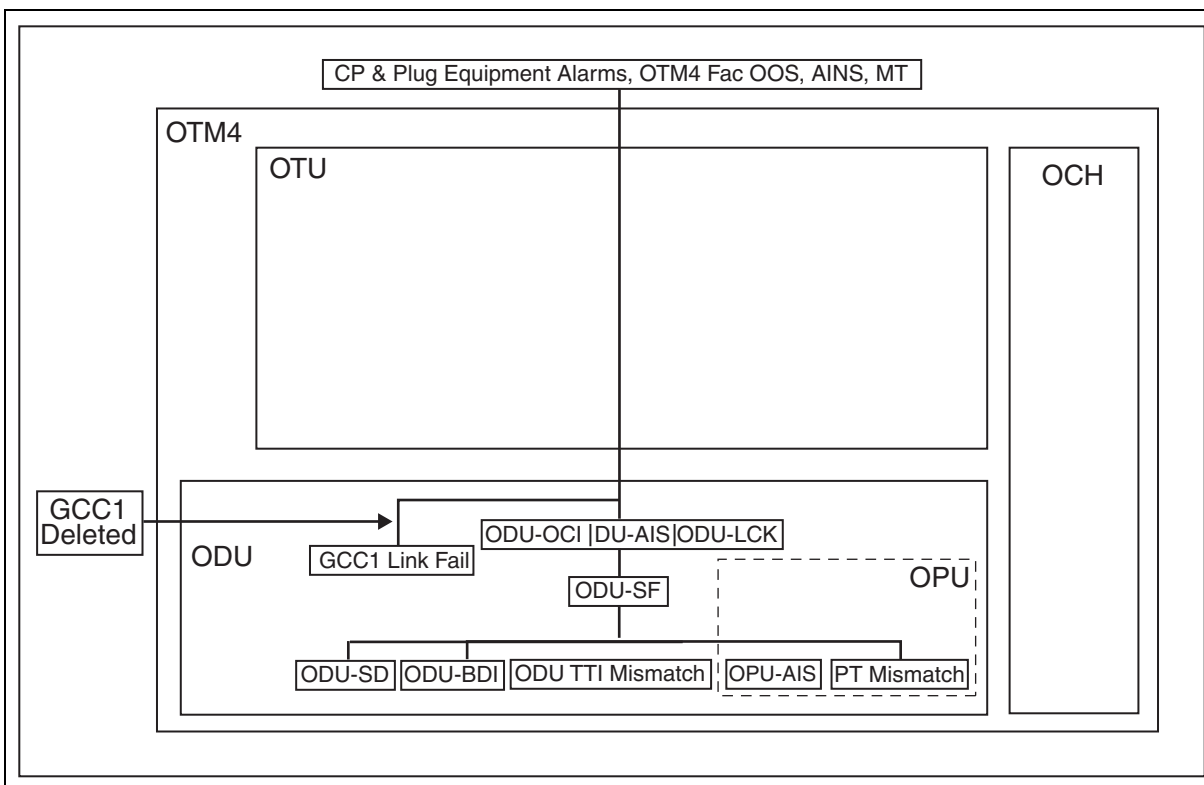
100G WL3n/WL3e OTR circuit packs client port OTM4 facility alarm hierarchy



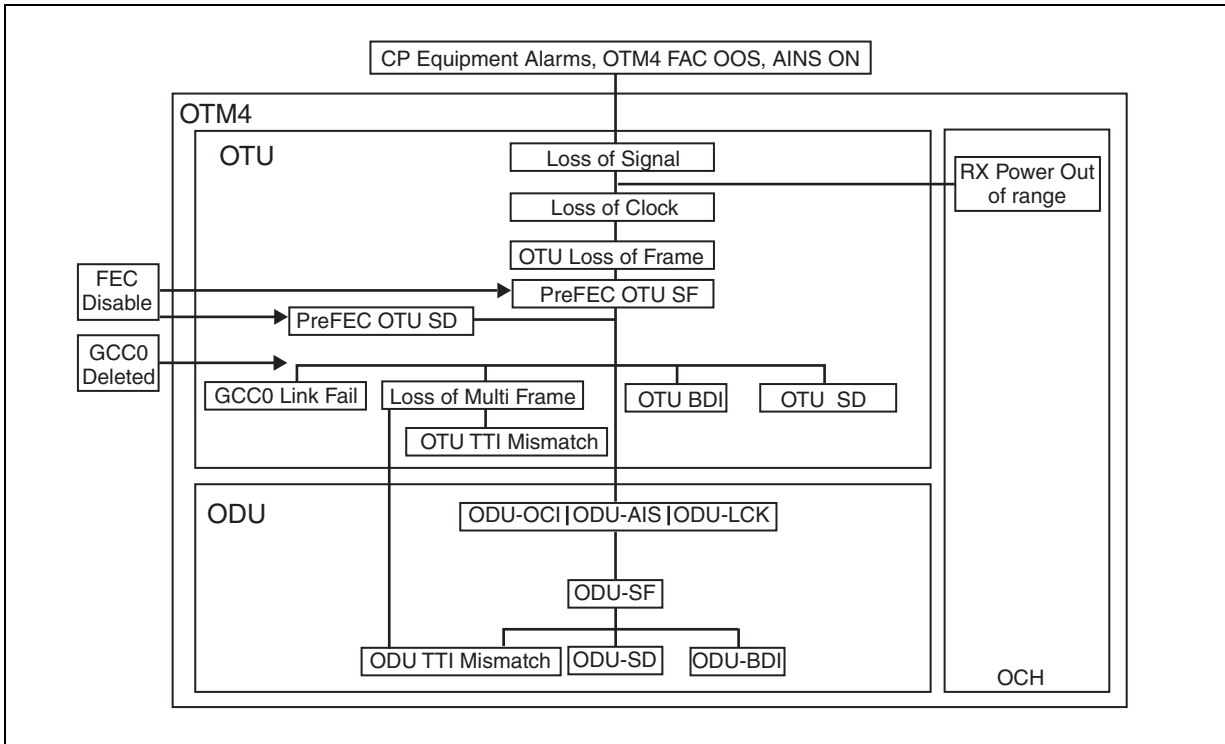
100G WL3n/WL3e OTR circuit packs OTM4 mapping facility on client port alarm hierarchy



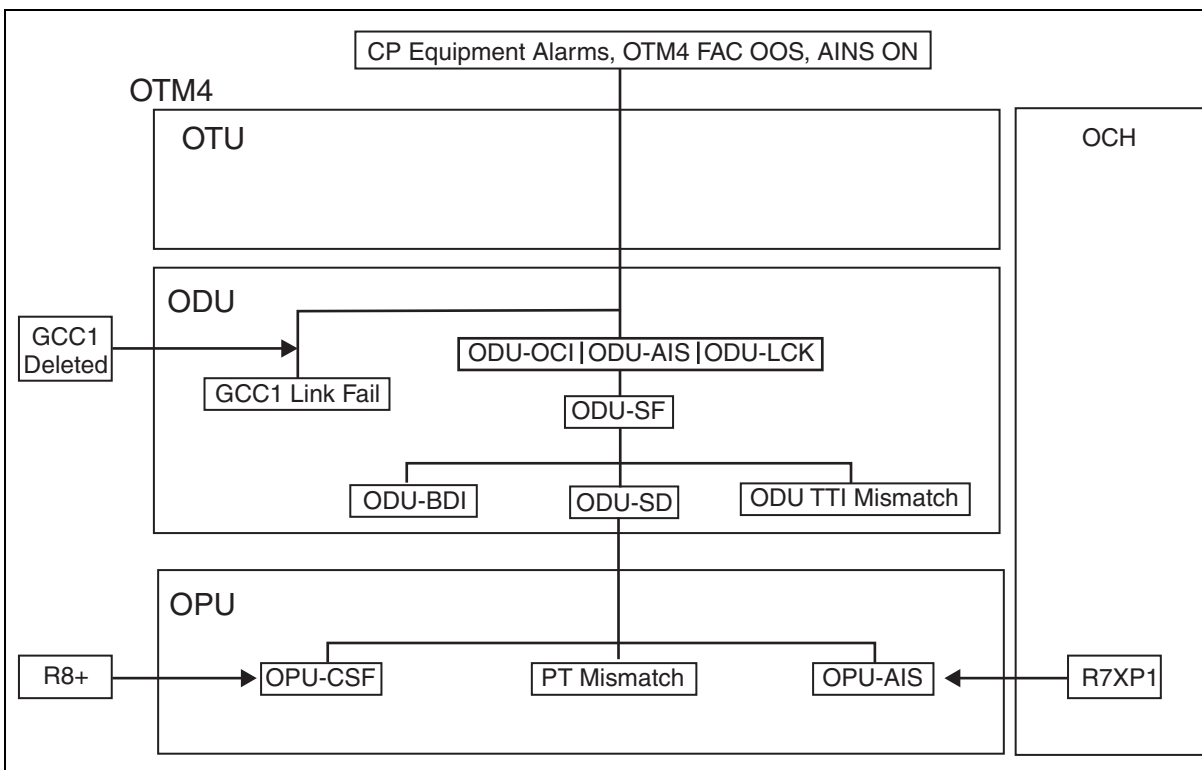
10x10GE MUX OCI circuit pack backplane port 100 OTM4 facility alarm hierarchy



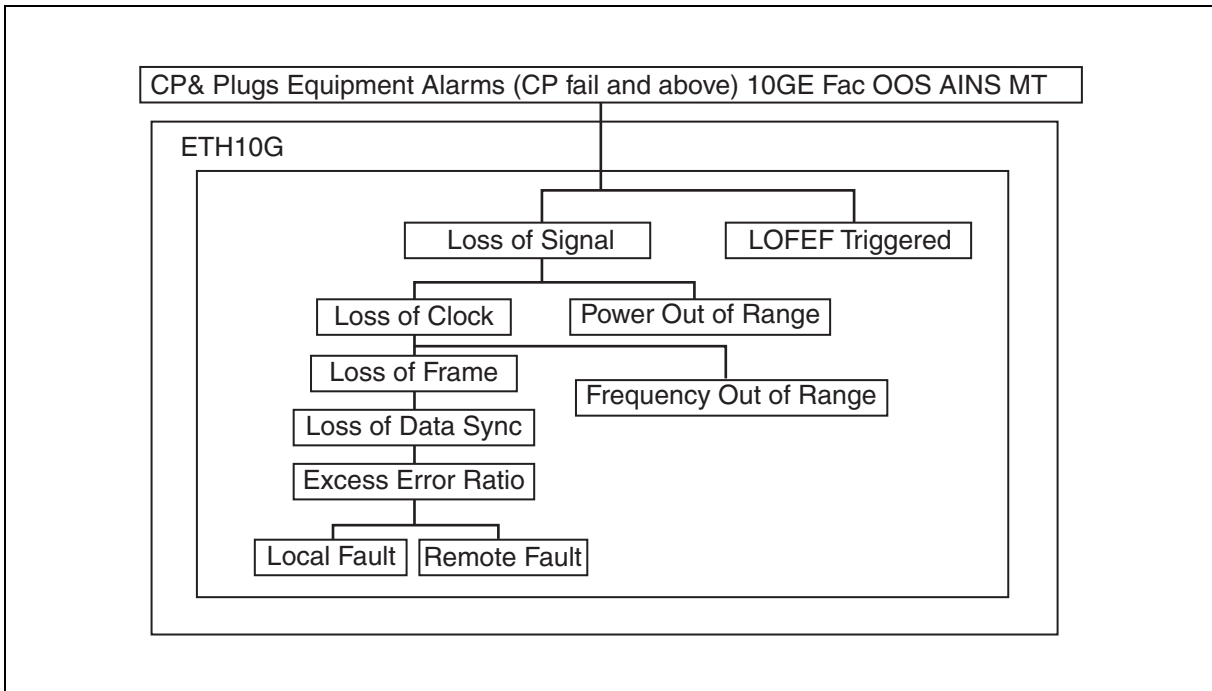
100G OCl circuit pack client port OTM4 facility alarm hierarchy



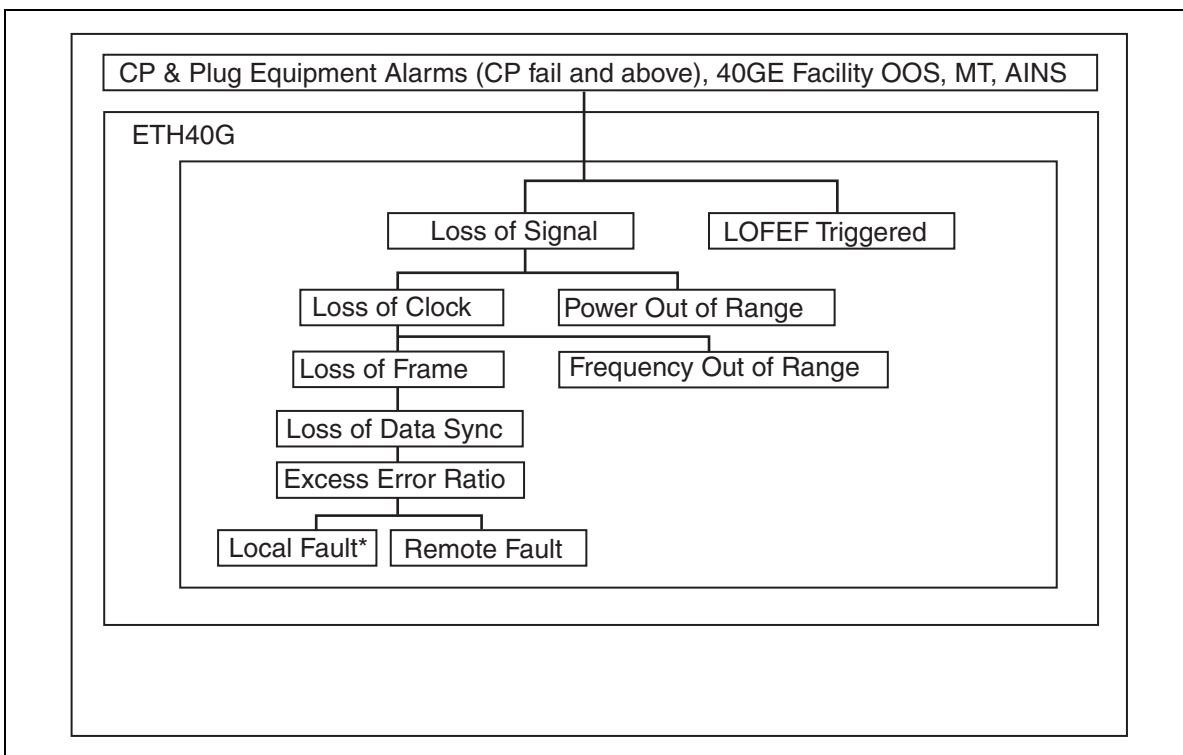
100G OCI circuit pack client port OTM4 mapping layer facility alarm hierarchy



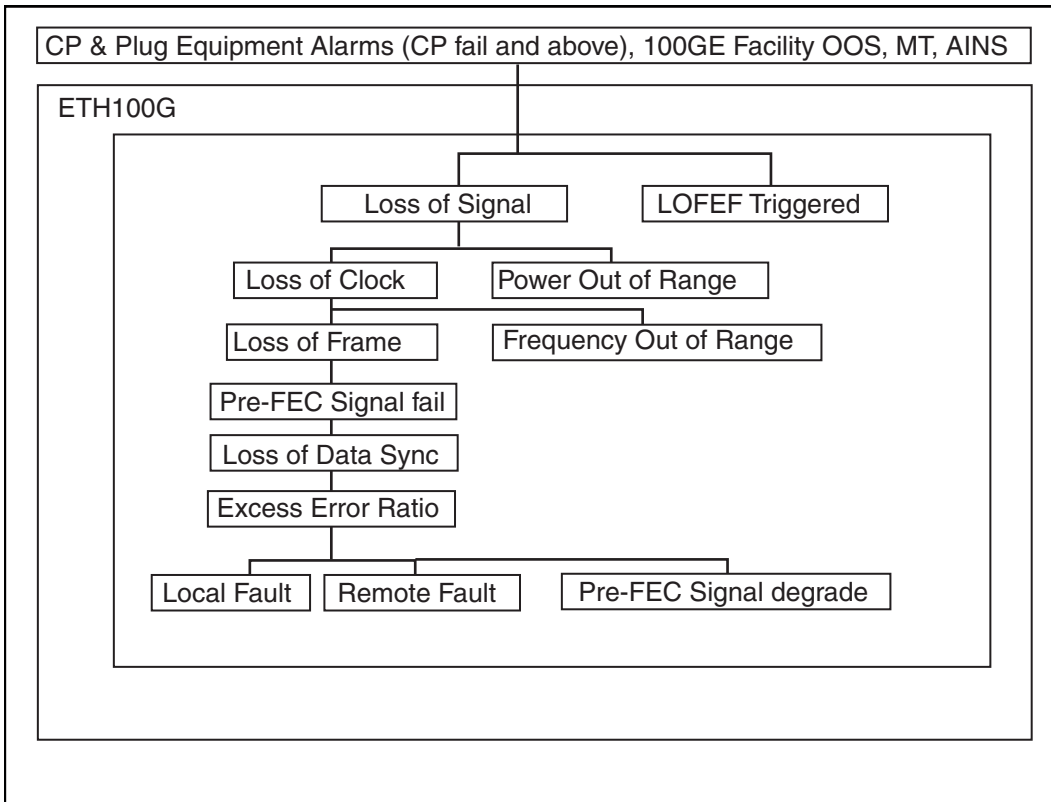
200G (2x100G/5x40G) MUX circuit pack client port ETH10G facility alarm hierarchy



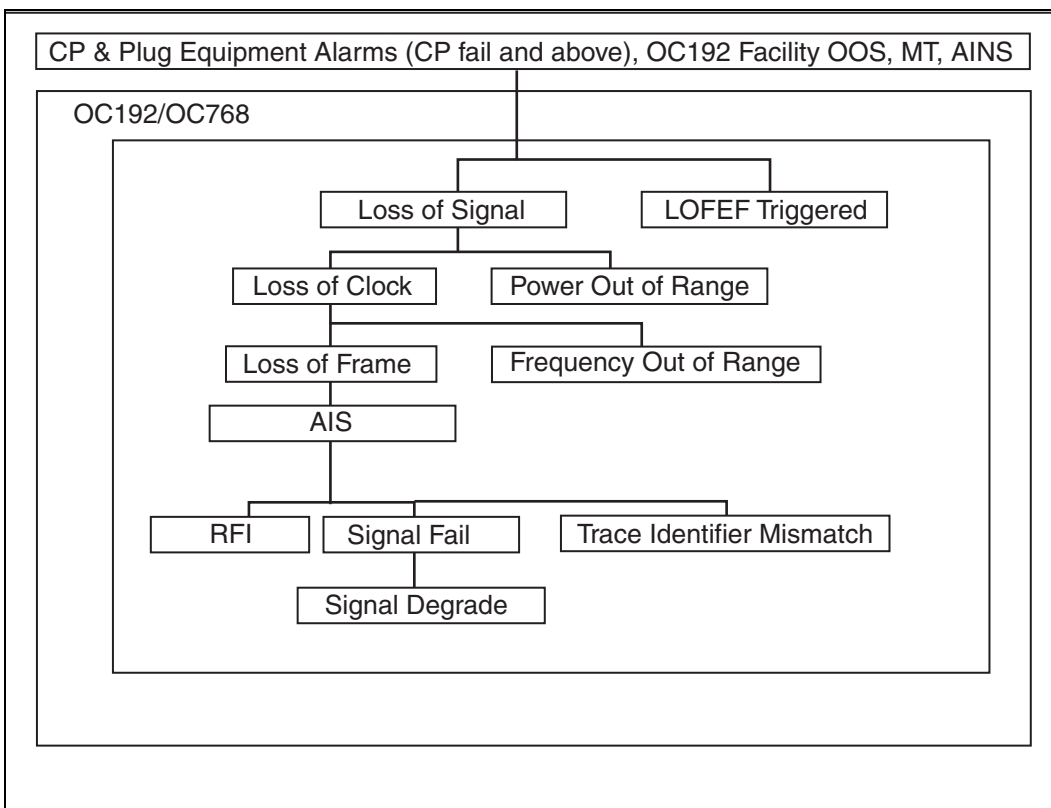
200G (2x100G/5x40G) MUX circuit pack client port ETH40G facility alarm hierarchy



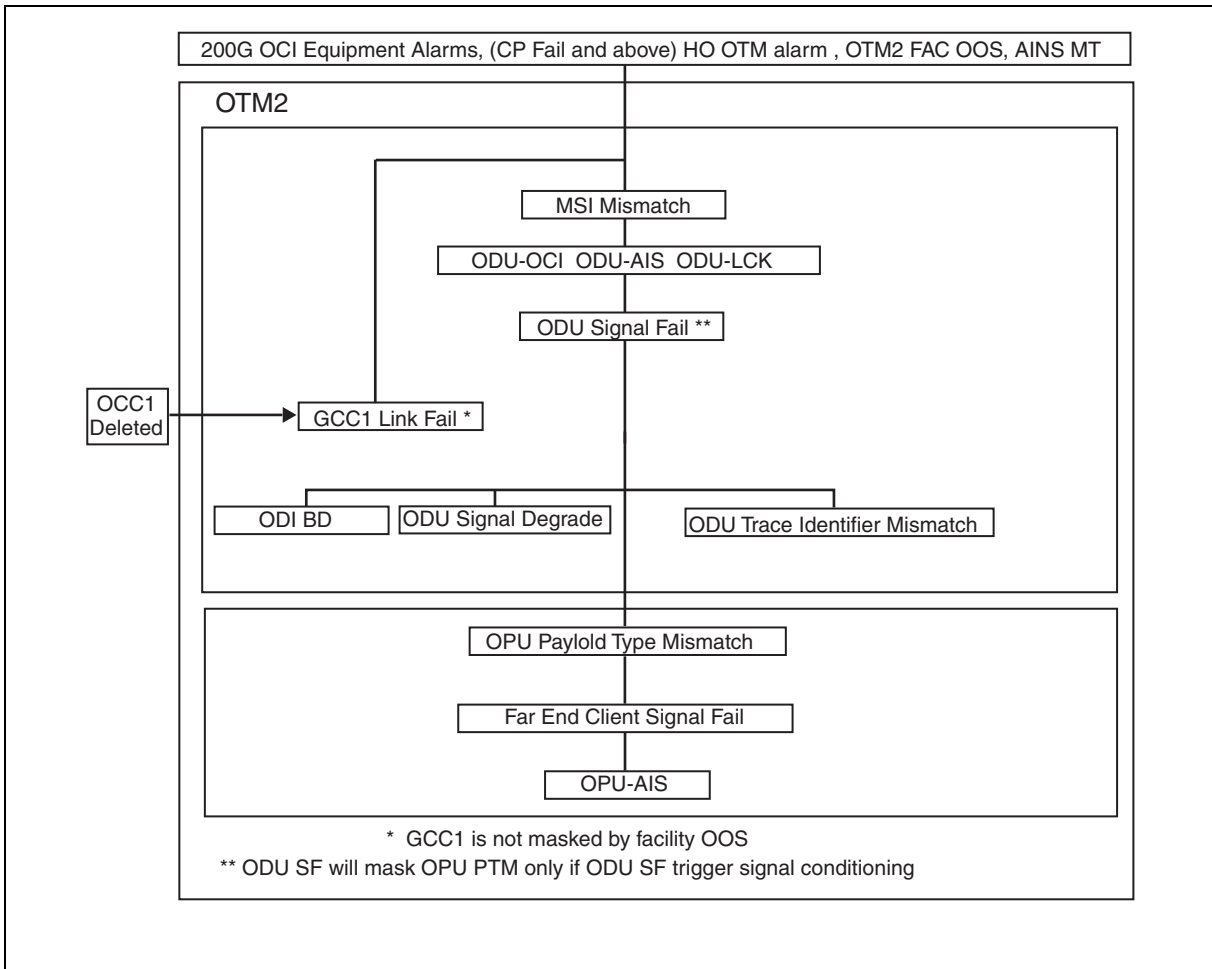
200G (2x100G/5x40G) MUX circuit pack client port ETH100G facility alarm hierarchy



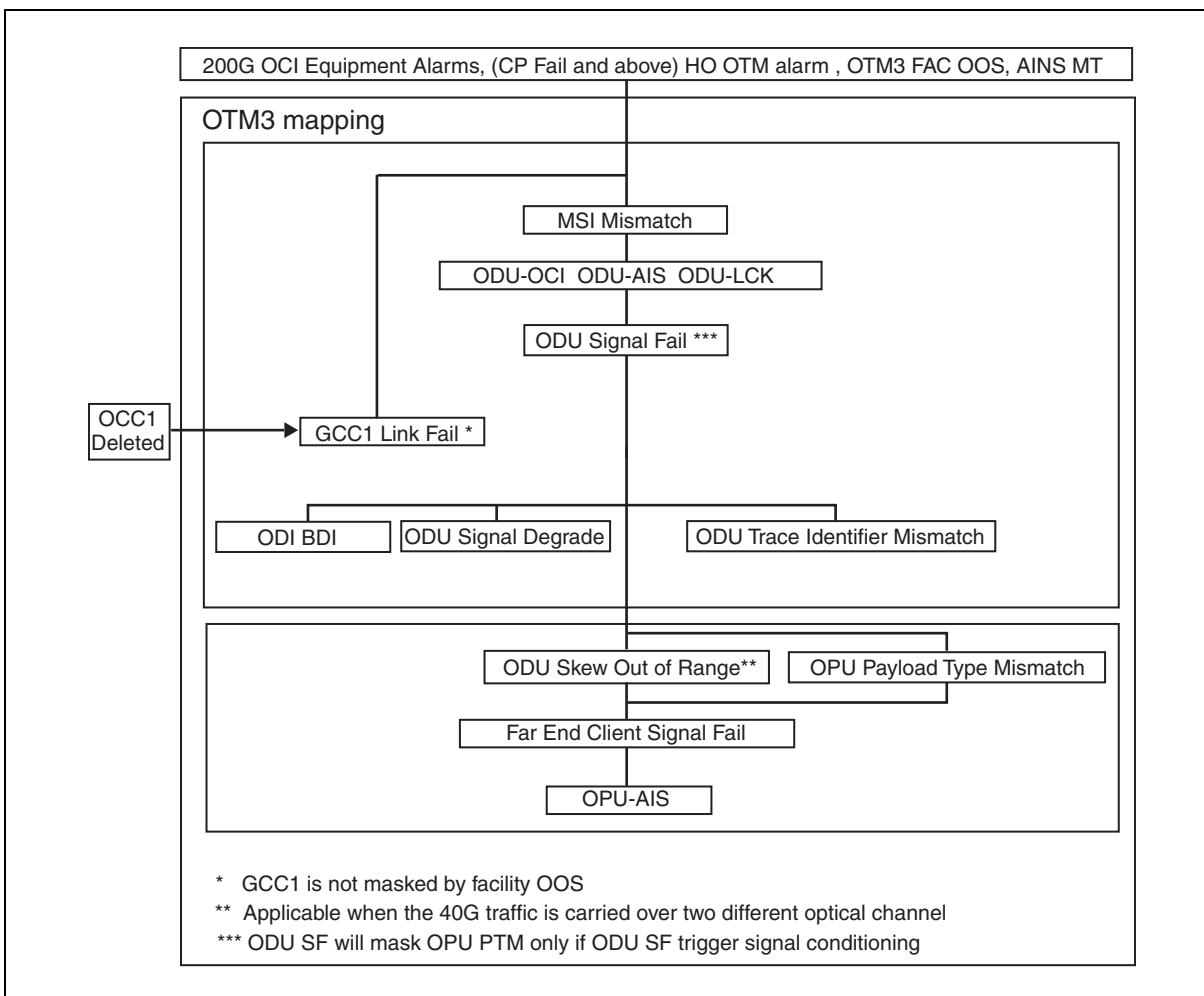
200G (2x100G/5x40G) MUX circuit pack client port OC192/OC768 facility alarm hierarchy



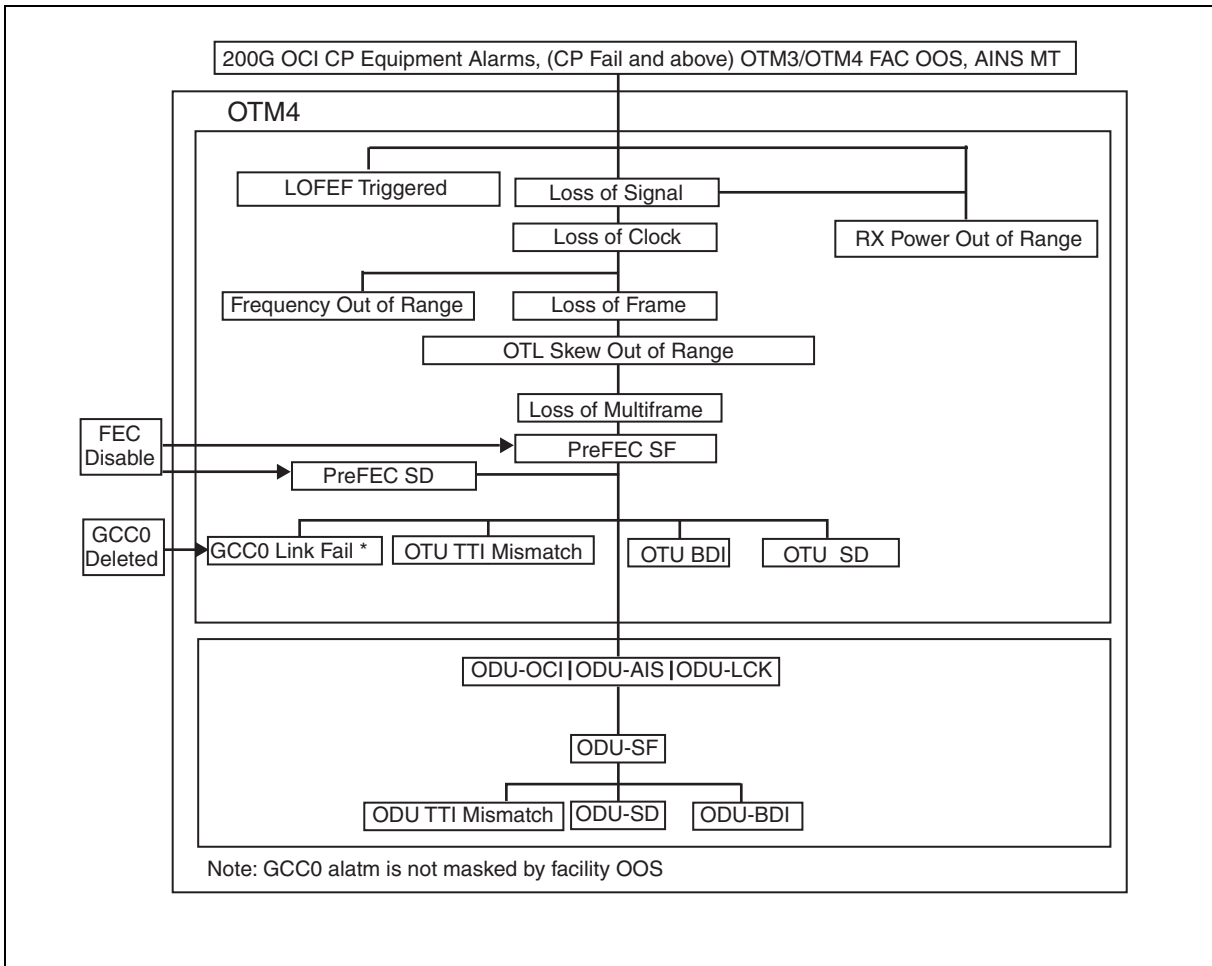
200G (2x100G/5x40G) MUX circuit pack client port OTM2 mapping layer facility alarm hierarchy



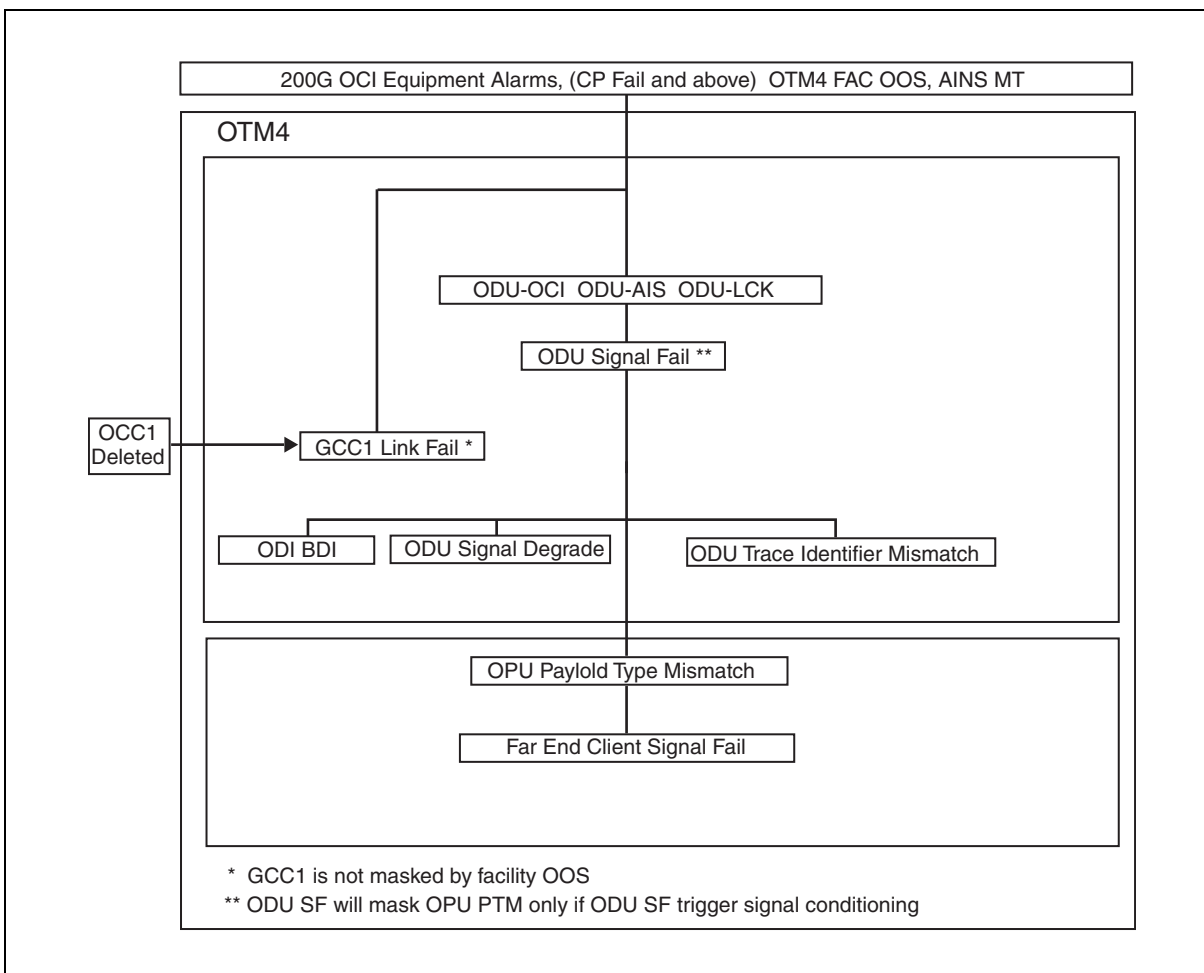
200G (2x100G/5x40G) MUX circuit pack client port OTM3 mapping layer facility alarm hierarchy



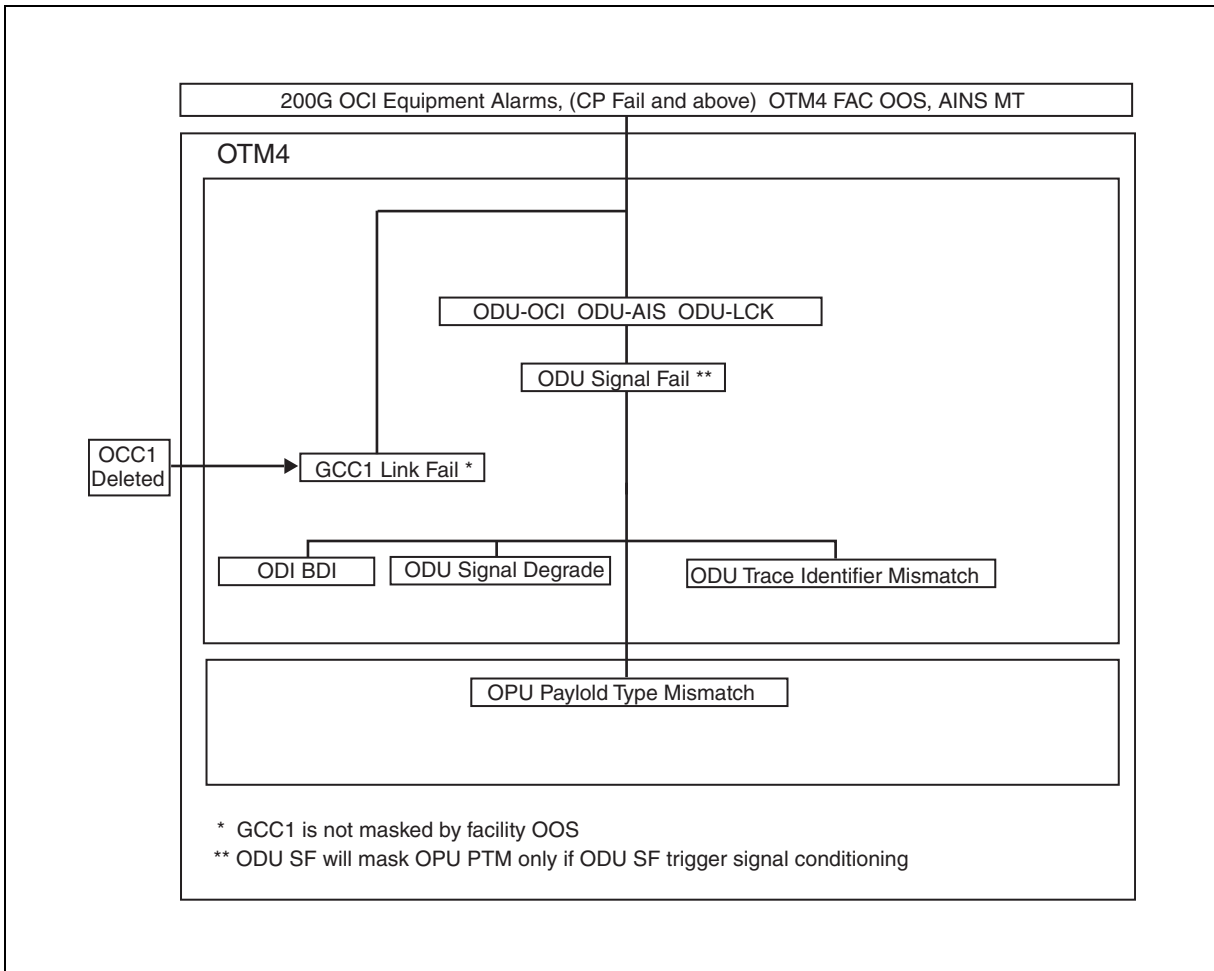
200G (2x100G/5x40G) MUX circuit pack client port OTM4 facility alarm hierarchy



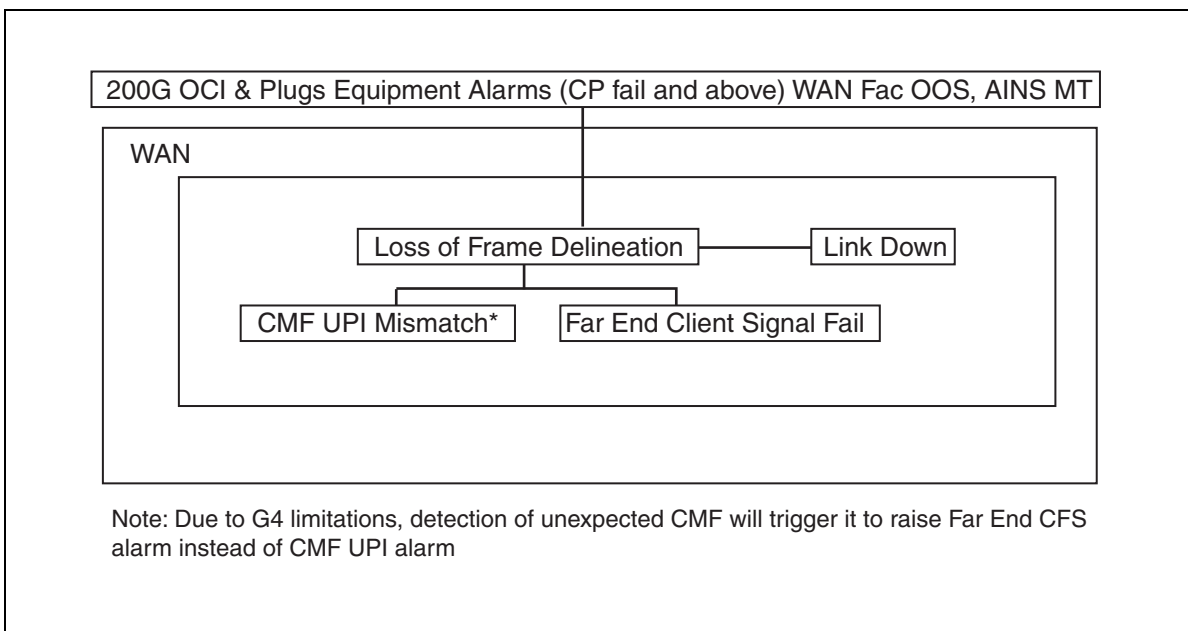
200G (2x100G/5x40G) MUX circuit pack client port OTM4 mapping layer facility alarm hierarchy



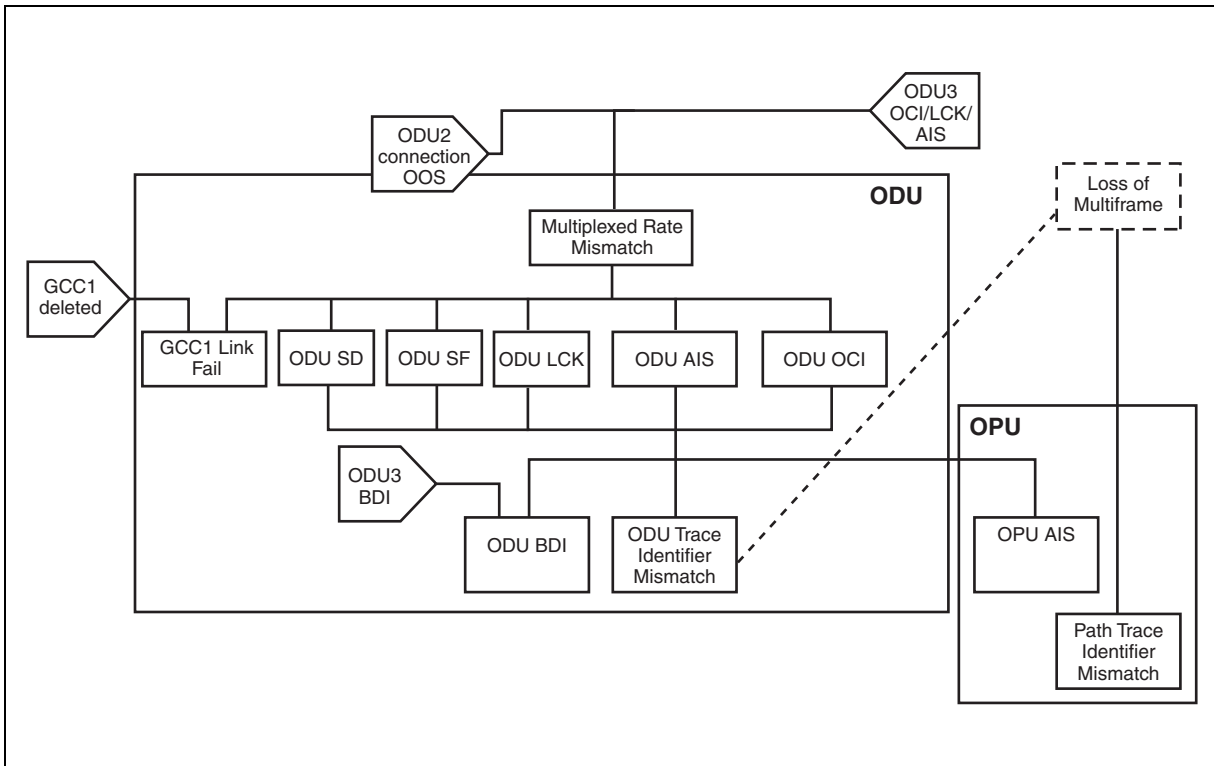
200G (2x100G/5x40G) MUX circuit pack client port OTM4 muxing layer facility alarm hierarchy



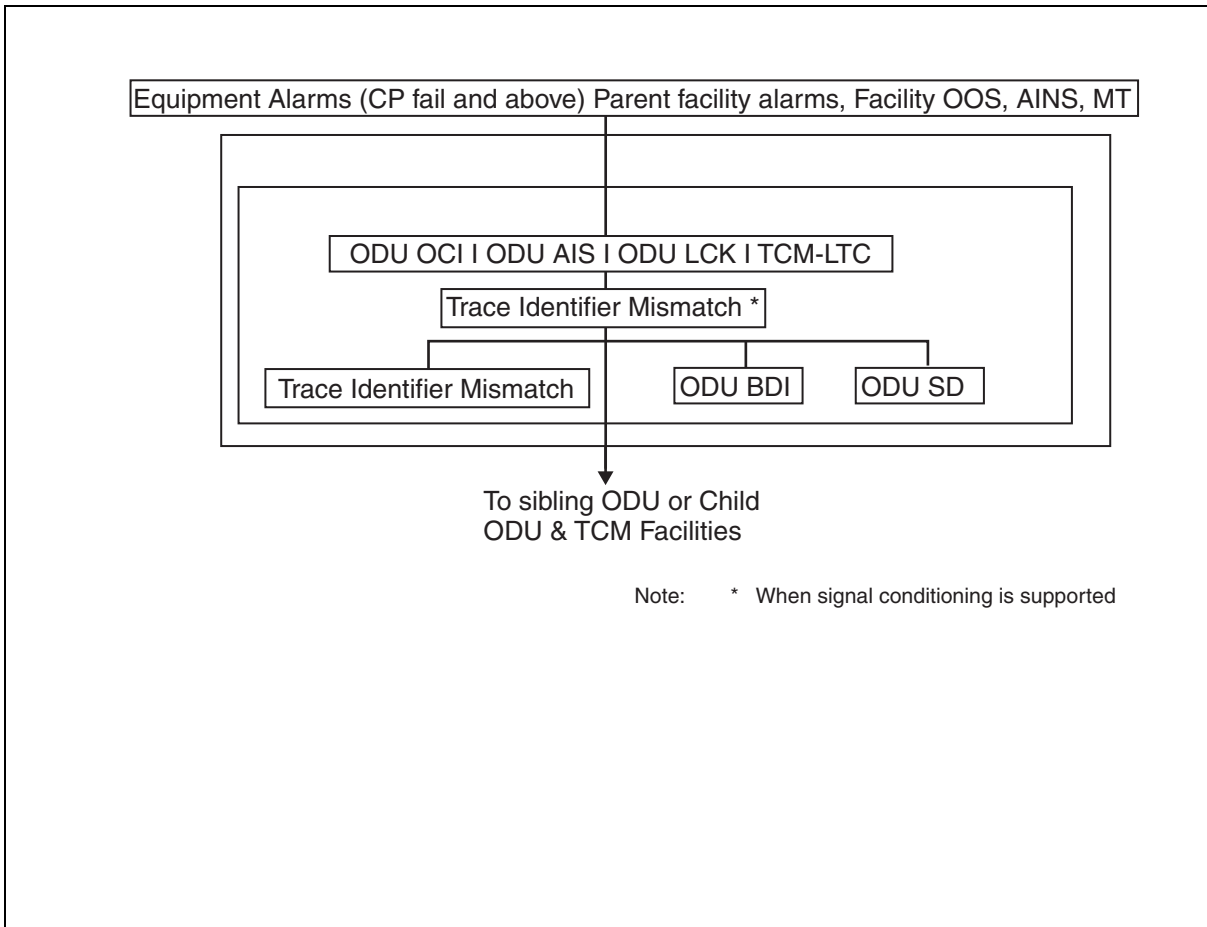
200G (2x100G/5x40G) MUX circuit pack WAN facility alarm hierarchy



OTM2 mapping facility (associated with FC1200, OC-192/STM-64, and ETH10G facility) alarm hierarchy

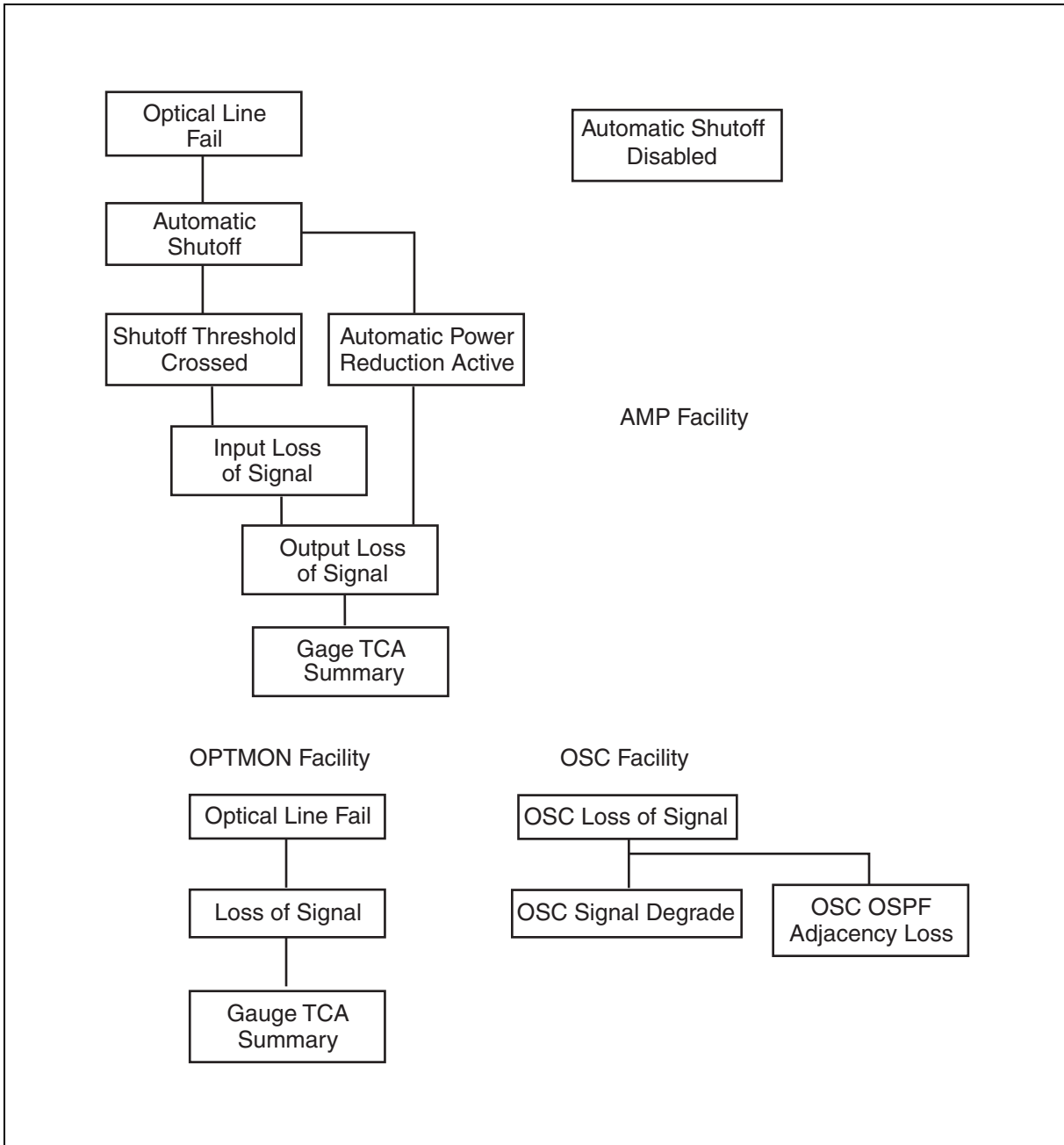


TCMCTP (terminated) facility alarm hierarchy - backplane to faceplate direction

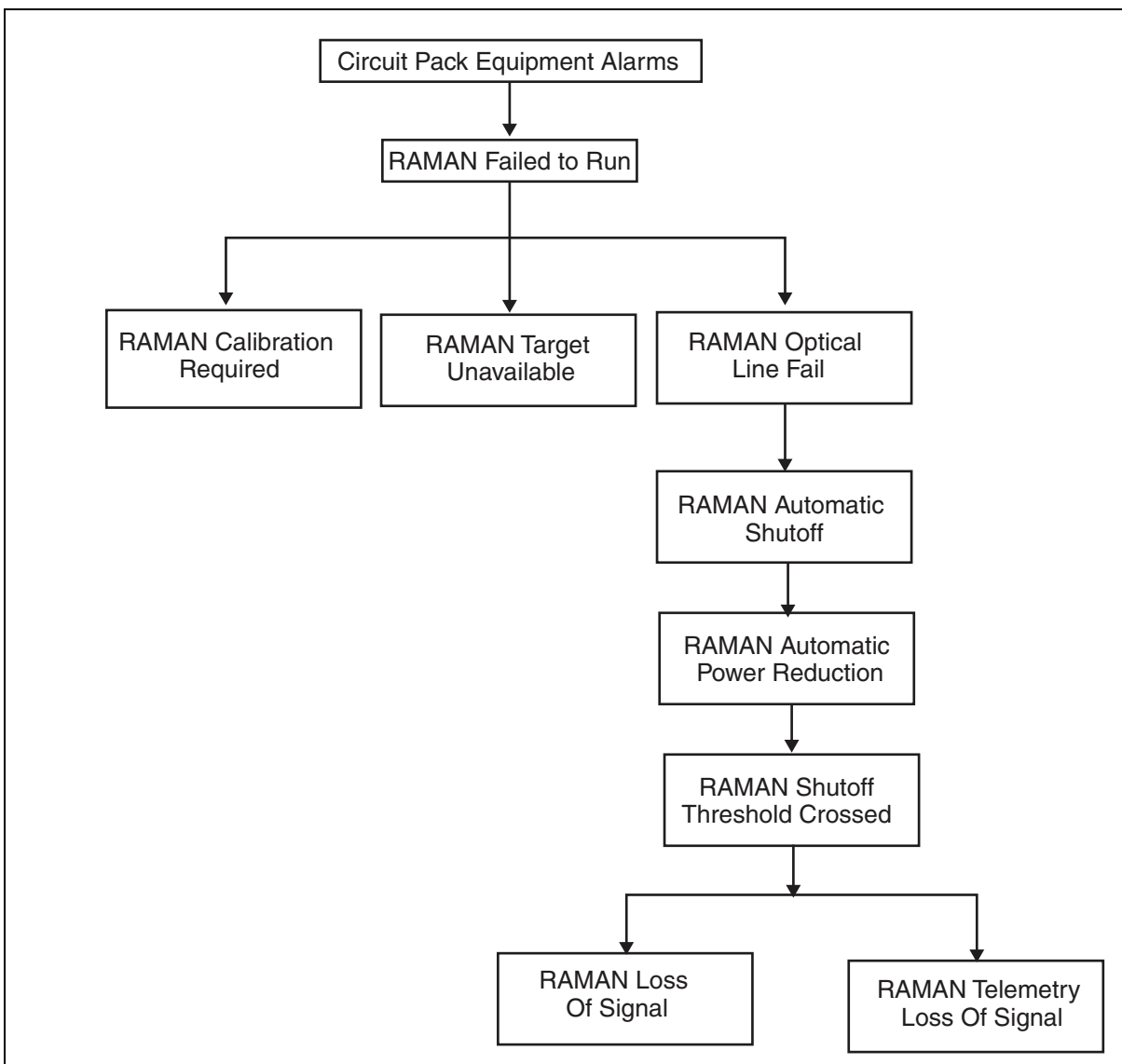


Photonic optical signal facilities alarm hierarchy

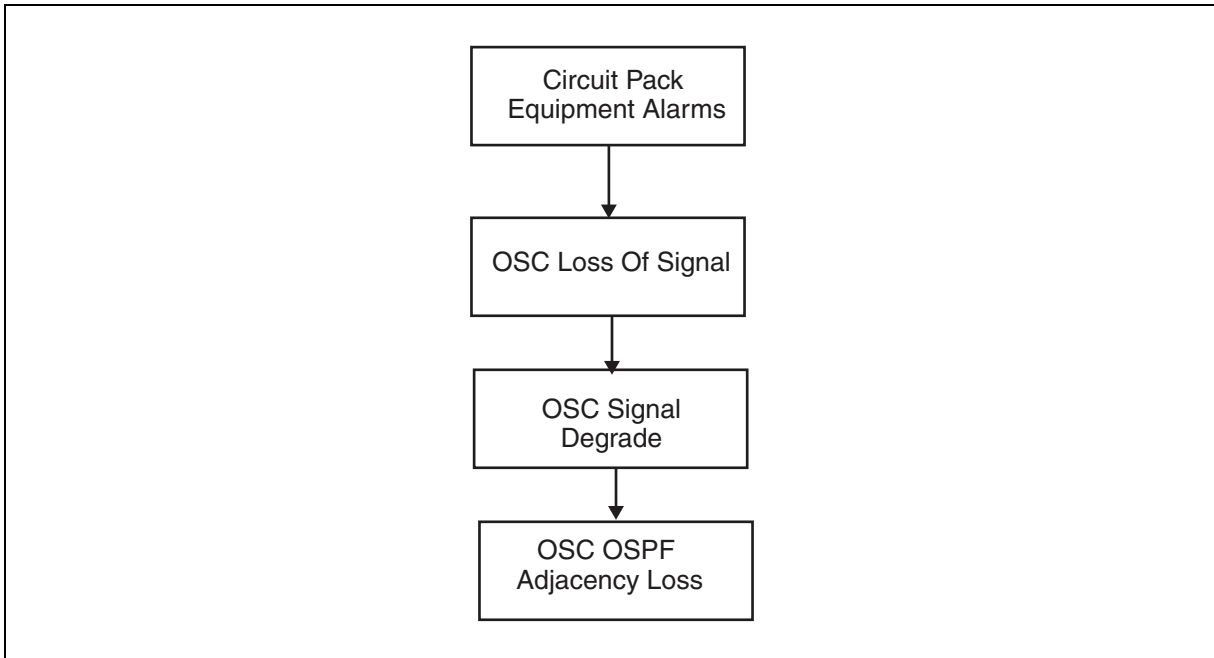
Photonic facility alarms are masked by upstream facility failures.



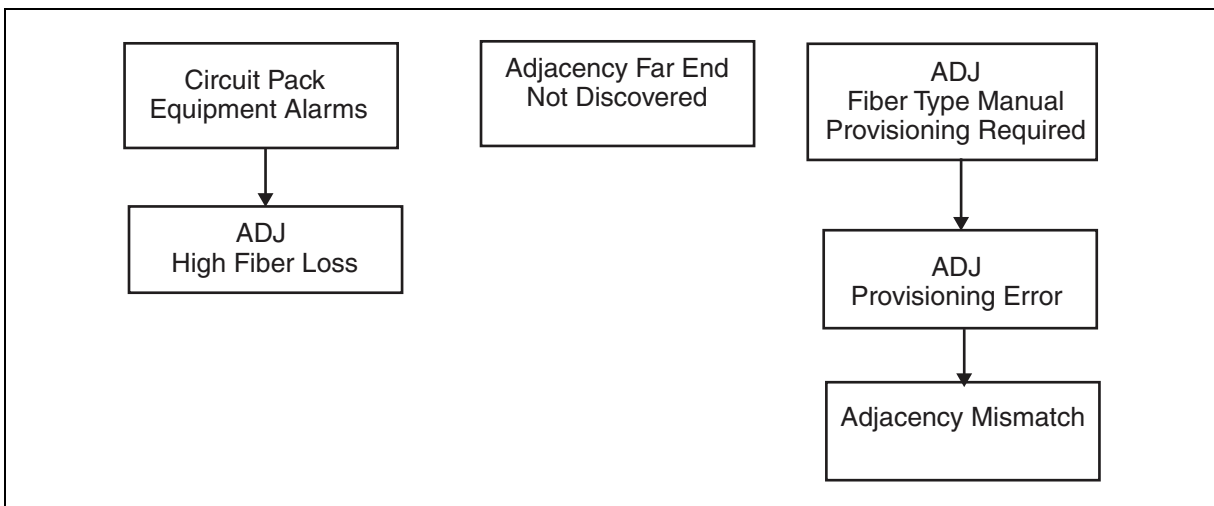
SRA circuit pack RAMAN facility alarm hierarchy



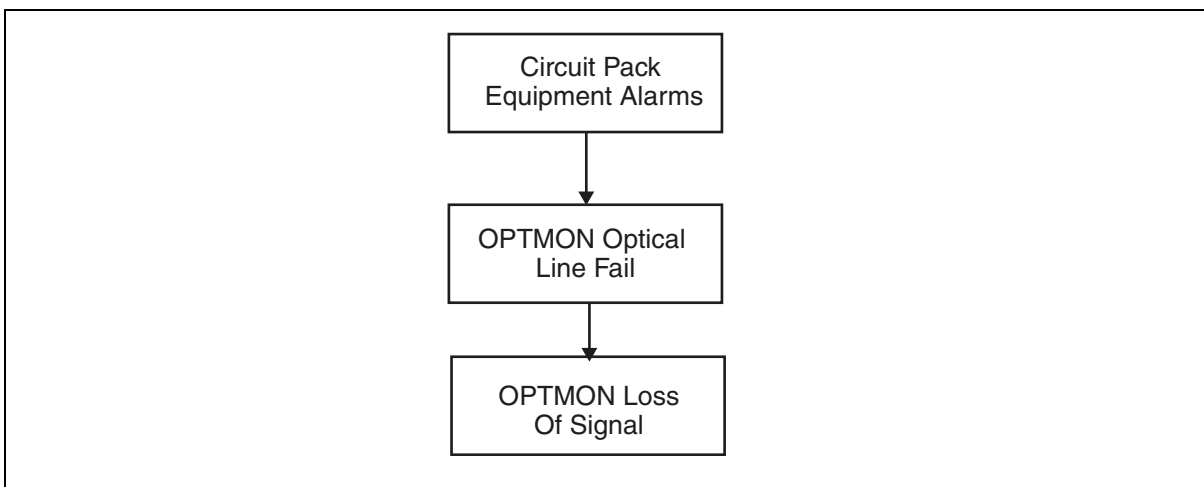
SRA circuit pack OSC facility alarm hierarchy



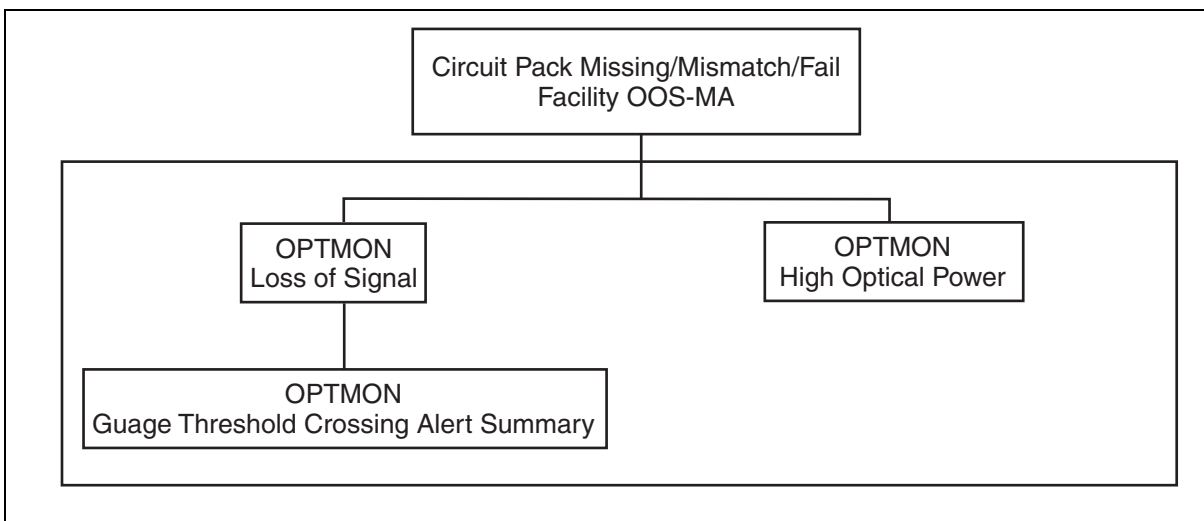
SRA circuit pack ADJ facility alarm hierarchy



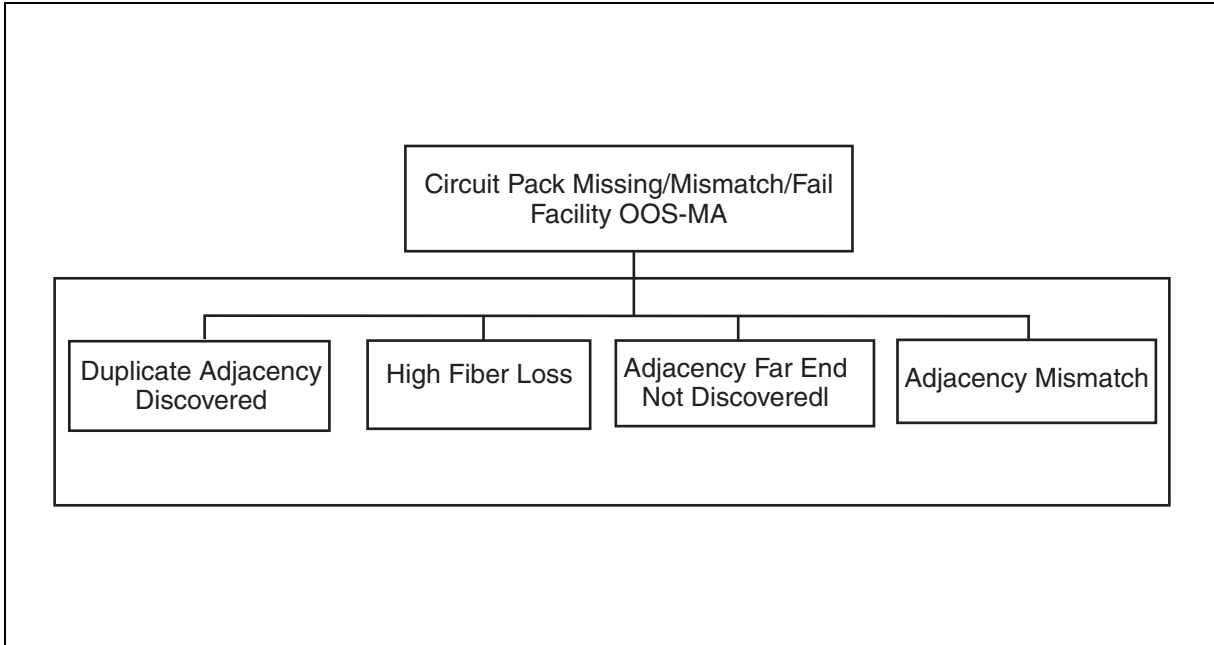
SAM and ESAM circuit packs OPTMON facility alarm hierarchy



CCMD12, MLA3 L-Band, LIM L-Band, and WSS Flex L-Band w/OPM 8x1 circuit packs OPTMON facility alarm hierarchy



CCMD12, GMD10 L-Band, MLA3 L-Band, LIM L-Band, UCS, and WSS Flex L-Band w/OPM 8x1 circuit packs adjacency facility alarm hierarchy



Alarm clearing procedures

This release of 6500 Packet-Optical Platform (6500) supports the Broadband, PKT/OTN transport and switched, and Photonic services for different Broadband, PKT/OTN I/F and Photonic circuit packs. The combination of two or all services is also supported.

For more information on the services (and the circuit packs related to each service) offered in this release, refer to *6500 Planning*, Release 15.5.

This section provides procedures for clearing single and generic alarms. Generic procedures are used for clearing more than one alarm type.

A complete list of alarms is provided in this document. Refer to the [“List of alarms”](#) to determine whether you must perform a specific or a generic alarm clearing procedure to clear the alarm.

Note: Unless otherwise specified, eMOTR in this document refers to eMOTR (NTK536AA, NTK536AB, NTK536FA, NTK536FB) and eMOTR Edge (NTK536BE) variants.

Abbreviations used in this section

AIS	Alarm Indication Signal
ALS	Automatic Laser Shutdown
AMP	Amplifier
APR	Automatic Power Reduction
APS	Automatic Protection Switching
ASE	Amplified Spontaneous Emission
BDI	Backward Defect Indication
BIP	Breaker Interface Panel
BMD	Broadband Mux/Demux

4-2 Alarm clearing procedures

CD	Colorless Directionless
CDC	Colorless Directionless Contentionless
CFM	connectivity fault management
CFP2	C form-factor pluggable
CHMON	Channel Monitoring
CLMD	C/L-Band Mux/Demux
CMD44	44 Channel Mux/Demux
CMF	Client Management Frame
CP	Circuit Pack
CPE	Customer Premise Equipment
CTM	Control timing module
DCN	Data Communications Network
DIM	Distributed I/O Module
DIP	Dual In-line Package
DOC	Domain Optical Controller
DTL	Designated Transit List
DUS	Do not Use for Synchronization
DWDM	Dense Wavelength Division Multiplexing
EBER	Excessive Bit Error
EDP	Engineering Documentation Package
ES	Errored Second
ESD	Electrostatic Discharge
ESI	External Synchronization Input
ESO	External Synchronization Output
FC	Fiber Channel
FEC	Forward Error Correction
FICON	Fiber Connection
FIM	Fiber Interconnect Module

FPGA	Field Programmable Gate Array
GCC	General Communication Channel
GE	Gigabit Ethernet
GFP-F	Generic Framing Procedure - Framed
GFP-T	Generic Framing Procedure - Transparent
HO	High Order
IS	In-Service
iVLLI	Inverse Virtual Link Loss Indicator
LAN	Local Area Network
LED	Light-Emitting Diode
LIM	Line Interface Module
LO	Low Order
LOFEF	Laser Off Far-End Fail
LOF	Loss of Frame
LOS	Loss of Signal
LSP	Label switched path
MIC	Maintenance Interface Card
MDRLA	Multi-Degree ROADM with Line Amplifiers
MLA	Midstage Line Amplifier
MPLS	Multiprotocol Label Switching
NNS	Network Name Server
NSA	Non-Service-Affecting
OBB	Optical Broadcast & Bridge
OBMD 1x8	Optical Broadband Mux/Demux
OBM	Optical Bandwidth Manager
OC	Optical Carrier
OCI	Optical Channel Interface
OCLD	Optical Channel and Laser Detector

4-4 Alarm clearing procedures

ODU	Optical Channel Data Unit
OMD4	4 Channel Optical Mux/Demux
OMX	Optical Multiplexers
OOS	Out-of-Service
OPU	Optical Channel Payload Unit
OPM	Optical Power Monitor
OSC	Optical Service Channel
OSI	Open Systems Interconnection
OSNR	Optical Signal to Noise ratio
OST	Optical System Topology
OTDR	Optical Time Domain Reflectometer
OTR	Optical Transponder
OTM	Optical Transport Module
OTN	Optical Transport Network
OTU	Optical Channel Transport Unit
PEC	Product Engineering Code
PSI	Payload Structure Identifier
RLA	ROADM with Line Amplifier
ROADM	Re-configurable Optical Add-Drop Multiplexer
RS	Regenerator Section
SA	Service-Affecting
SD	Signal Degrade
SDH	Synchronous Digital Hierarchy
SDTH	Signal Degrade Threshold
SETS	Synchronization Equipment Timing Source
SFP	small form-factor pluggable
SFP+	Enhanced small form-factor pluggable
SNCP	Subnetwork Connection Protection

SLA	Single Line Amplifier
SLAT	System Lineup and Testing
SONET	Synchronous Optical Network
SP	Shelf Processor
SPLI	Service Photonic Layer Interoperability
SSH	Secure Shell Protocol
SSM	Synchronization Status Message
SSU	Synchronization Supply Unit
STM	Synchronous Transport Module
SWT	Shelf Wavelength Topology
TACACS	terminal access controller access control system
TOD	Time of Day
TPT	Transponder protection tray
TTI	Trail Trace Identifier
UNI	Unidirectional
UPI	User Payload Identifier
UPC	User Privilege Code
VC	Virtual Container
VLLI	Virtual Link Loss Indicator
WAN	Wide Area Network
WLAi	WaveLogic Automation and Intelligence
WSS w/OPM	Wavelength Selective Switch with Optical Power Monitor
XC	Cross-Connect
XFP	10G transceiver form factor pluggable
XLA	Switchable Line Amplifier

Associated procedures

Some procedures require the user to perform procedures relating to other topics. Before performing a procedure, if necessary, ensure the information about the associated procedures is available.

All procedures assume that you have logged into the network element. Refer to the “Interface login and logout” procedures in *Administration and Security*, 323-1851-301.

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Procedure 4-1

1+1 TPT APS alarms

Protection Channel Match Fail

Alarm ID: 263, 291, 953, 1018, 1108, 1324, 1336, 1747

Probable cause

This alarm is raised when:

- the received channel ID on the protection interface circuit pack is not as expected. This is normally because of a failure in the interface circuit pack.
- the automatic protection switching (APS) communications protocol between the two optical interfaces is not working because the optical fiber is not connected to the correct slot at either end
- the protection engine does not receive APS bytes from the far-end
- the protection engine receives invalid APS bytes from the far-end
- OTM1 bi-directional protection line is connected to a LO Regen and the protection switch fails on a 8xOTN Flex MOTR, (1+8)xOTN Flex MOTR

This alarm is only raised on the OTN 1+1 line-side protection, 1+1 port TPT, and 1+1 TPT protected configurations.

For OTN protection, this alarm is applicable only if the protection scheme is 1+1 port TPT and bi-directional.

Impact

Major, non-service-affecting (M, NSA) alarm

Protection Switch Byte Fail

Alarm ID: 260, 290, 952, 1017, 1107, 1323, 1335, 1391

Probable cause

This alarm is raised when the received channel protection switching control bytes (APS bytes) on the protection interface circuit pack are not valid codes. This is normally because of a failure in the interface circuit pack or crossed fibers.

This alarm is raised on 1+1 port TPT, and 1+1 TPT protected configurations.

Procedure 4-1 (continued)
1+1 TPT APS alarms

Impact
Major, non-service-affecting (M, NSA) alarm

- Prerequisites**
- To perform this procedure, you must:
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
 - have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)

Step	Action						
1	Identify the interface circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” .						
2	Identify the protection provisioning on the circuit pack raising the alarm. Ensure that the protection scheme is 1+1 port TPT, or 1+1 TPT. Refer to “Retrieving protection parameters” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. If the protection is not 1+1 port TPT, or 1+1 TPT, contact your next level of support or your Ciena support group.						
3	Identify the protection provisioning on the far-end. Ensure that the protection scheme is 1+1 port TPT, or 1+1 TPT. If the same alarm appears at the other end, two fibers have been swapped. If a different alarm condition exists at the far-end, investigate the alarm to identify and localize the fault.						
4	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. <table><tr><td>If the original alarm was</td><td>Then go to</td></tr><tr><td>Protection Channel Match Fail</td><td>step 5</td></tr><tr><td>Protection Switch Byte Fail</td><td>step 7</td></tr></table>	If the original alarm was	Then go to	Protection Channel Match Fail	step 5	Protection Switch Byte Fail	step 7
If the original alarm was	Then go to						
Protection Channel Match Fail	step 5						
Protection Switch Byte Fail	step 7						
5	Verify that the optical fibers/cables are connected to the correct ports at each node. The working port of the protection pair at the near-end must be fibered to the working port of the protection pair at the remote end. The protection port of the protection pair at the near-end must be fibered to the protection port of the protection pair at the remote end. Use Section Trace to identify fiber connectivity. Refer to “Retrieving and editing section trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-1 (continued)

1+1 TPT APS alarms

Step	Action						
6	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 7</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Replace the interface circuit pack raising the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table. Wait 30 seconds.						
8	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 9</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Use the optical fiber connection information to determine the network element and interface circuit pack on the far-end of the optical fiber link.						
10	Replace the remote interface circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-2

Adjacency Discovery Unreliable

Alarm ID: 1072

Probable cause

This alarm is raised when a remote shelf that the SPLI feature is tracking has not communicated (UDP, TCP/SSH) with the SPLI application for more than 10 minutes. This can occur:

- if the remote shelf has changed their SiteID or TID
- if the remote shelf is isolated from the network
- if the remote shelf is constantly restarting

Note 1: For 6500, COMMsType (Transport protocol) is UDP for R11.0 and earlier. Default to TCP in R11.1 and later.

Note 2: This alarm (raised against the local shelf AID) masks all Adjacency Mismatch alarms on that shelf.

This alarm is also raised when there is a SITEGROUP mismatch on the shelf.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have a network plan or other documents that allow you to determine the SPLI connectivity
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action
1	If the alarm is raised due SITEGROUP mismatch, edit the SITEGROUP attribute to make it same on both ends. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310 for more information.
2	Determine which shelves are unreliable using the SPLI tab in the Node Information application. Find the nodes and shelves with a status of Unreliable.

Procedure 4-2 (continued)
Adjacency Discovery Unreliable

Step	Action						
	Refer to the “Displaying node information” procedure in <i>Administration and Security</i> , 323-1851-301, for more information about the Node Information application.						
3	Verify whether the TID or siteID has changed on those nodes. If changes have been made, change the TID or siteID of those nodes back to the original ones if necessary.						
4	If changes have not been made, verify the comms setting and resolve any data comms issues found. Refer to procedures in “ <i>Data Communications Planning and User Guide</i> ”, 323-1851-101.						
5	Click the Refresh button in the SPLI tab of the Node Information window to retrieve the latest statuses.						
6	If SPLI is not matching all the TID-shelves that are listed as unreliable, click on the Delete button. This removes all the unreliable entries from the table. Note that any OMD4, OMX and CMD44 Tx/Rx Adjacency with the Auto Discovered parameter set to Auto, which is associated with the remote TID-Shelf entries that are being deleted will be deprovisioned if their DOC Care parameter is set to False. As well the discovered Far End Addresses will be set to Unknown.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Contact your next level of support or your Ciena support group.						
	—end—						

Procedure 4-3

Adjacency Far End Not Discovered

Alarm ID: 538, 2172

Probable cause

This alarm is raised against an ADJ-LINE facility or the MPO ADJ facility when the adjacency cannot be automatically discovered from the far-end.

Note: This alarm can be raised against the ADJ-LINE facility but highlights a problem at the OTS LINEOUT port. In some configurations (for example, Submarine Terminal) these ports may not match. For colorless-directionless-contentionless (CDC) applications and Colorless CMD Direct Attach (CDA), the alarm can also be raised against MPO ADJ on both ends of the affected MPO connection.

Possible reasons for this far-end adjacency not being discovered include:

- the two adjacency end points have not been fibered
- the two adjacency end points have been fibered but some other problem exists with the fiber or connection
- there is a comms provisioning error, where the TID-level comms circuit is incorrectly provisioned or not provisioned at one or both ends of the alarmed span
- an upstream network element has undergone a restart operation. The alarm will clear once the restart completes
- fibers at the Line AMP NE are swapped (misconnected/crossed)
- the ADJ-LINE Expected Far End Address is incorrectly provisioned
- the appropriate OSC is not provisioned in the OTS or in the OTS Slot Sequence or is not functioning correctly
- the WSS Flex, or CCMD8x16 has undergone a warm or cold restart operation
- the WSS Flex, or CCMD8x16 has been unseated or the RJ45 cable has been pulled from FIM module
- the TIDC/IP is not provisioned
- the TID Slot Sequence provisioning (for example, Derived Expected Far End Adjacencies) does not match the actual MPO fiber connections
- COMMS are down with a remote node
- CPU has high CPU utilization

In the case of a line adjacency, the discovered provisioned expected far-end address is the Photonics circuit pack at the other end of the fiber span.

Procedure 4-3 (continued)

Adjacency Far End Not Discovered

The alarm is masked by:

- local Circuit Pack Missing or Circuit Pack Mismatch
- remote Circuit Pack Missing or Circuit Pack Mismatch (MPO ports only)
- if there is no ADJ Expected Far End Address (EFEA) (MPO ports only)

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a fiber cleaning kit (for CDC and CDA configuration, an MPO cleaning kit is needed. See the “Cleaning connectors” section in *Installation - General Information*, 323-1851-201.0)
- have the engineering documentation package (EDP) for shelf details

Step	Action				
1	<table><tr><td>If the original alarm is raised against CDC and CDA configurations</td><td>Then go to step 2</td></tr><tr><td>otherwise</td><td>step 3</td></tr></table>	If the original alarm is raised against CDC and CDA configurations	Then go to step 2	otherwise	step 3
If the original alarm is raised against CDC and CDA configurations	Then go to step 2				
otherwise	step 3				
2	For the CDC and CDA MPO, the ADJ EFEA is derived from the TID Slot Sequence provisioning. Verify that this is provisioned correctly and apply changes if required. Go to step 6 . Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.				
3	Verify and correct the adjacency provisioning information as required. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.				
4	If applicable, check the OSC facility states and confirm provisioning of the OSC in the OTS or in the Slot Sequence.				
5	<table><tr><td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has cleared	Then the procedure is complete	not cleared	go to step 6
If the original alarm has cleared	Then the procedure is complete				
not cleared	go to step 6				

Procedure 4-3 (continued)

Adjacency Far End Not Discovered

Step	Action										
6	<table> <tr> <th>If this adjacency</th><th>Then</th></tr> <tr> <td>is fibered, but a fiber break, reflective event, or disconnect is suspected</td><td>go to step 7</td></tr> <tr> <td>was provisioned for future use, and you do not want this alarm to be displayed in the active alarm list</td><td>disable this alarm so that it no longer appears in the active alarms list. Refer to Procedures for alarm provisioning and alarm profiles in this document.</td></tr> <tr> <td>was provisioned for future use and you want this alarm to be displayed in the active alarms list for informational purposes</td><td>no action is required. The procedure is complete.</td></tr> <tr> <td>requires fiberung</td><td>fiber the adjacency end points. Refer to the EDP. For logical port mappings for CCMD8x16 ports and WSS MPO ports connected to FIM, refer to “Connecting or disconnecting fiber-optic cables to or from IMs or FIMs” procedure in <i>Installation - General Information</i>, 323-1851-201.0.</td></tr> </table>	If this adjacency	Then	is fibered, but a fiber break, reflective event, or disconnect is suspected	go to step 7	was provisioned for future use, and you do not want this alarm to be displayed in the active alarm list	disable this alarm so that it no longer appears in the active alarms list. Refer to Procedures for alarm provisioning and alarm profiles in this document.	was provisioned for future use and you want this alarm to be displayed in the active alarms list for informational purposes	no action is required. The procedure is complete.	requires fiberung	fiber the adjacency end points. Refer to the EDP. For logical port mappings for CCMD8x16 ports and WSS MPO ports connected to FIM, refer to “Connecting or disconnecting fiber-optic cables to or from IMs or FIMs” procedure in <i>Installation - General Information</i> , 323-1851-201.0.
If this adjacency	Then										
is fibered, but a fiber break, reflective event, or disconnect is suspected	go to step 7										
was provisioned for future use, and you do not want this alarm to be displayed in the active alarm list	disable this alarm so that it no longer appears in the active alarms list. Refer to Procedures for alarm provisioning and alarm profiles in this document.										
was provisioned for future use and you want this alarm to be displayed in the active alarms list for informational purposes	no action is required. The procedure is complete.										
requires fiberung	fiber the adjacency end points. Refer to the EDP. For logical port mappings for CCMD8x16 ports and WSS MPO ports connected to FIM, refer to “Connecting or disconnecting fiber-optic cables to or from IMs or FIMs” procedure in <i>Installation - General Information</i> , 323-1851-201.0.										
7	Clear any alarms that indicate a fiber break or disconnect, such as: • Automatic Power Reduction Active • Automatic Shutoff • High Fiber Loss • Input Loss of Signal • Loss of Signal (OPTMON) • Optical Line Fail										
8	Verify the fibers at the Line AMP NE or MPO Adjacency facility for any reversed fiber connections and correct if necessary. Ensure the MPO connectors are firmly locked-in on both ends of the alarmed MPO cable connection.										
9	Verify that the provisioned values for ADJ-LINE Expected Far End Address Format or Expected Far End Address are correct.										
10	Edit the provisioned values as appropriate to correct the expected values. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. For CDC MPOs, edit the TID slot sequence in order to change the ADJ EFEA of the MPO. Refer to the “Editing TID slot sequences” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.										

Procedure 4-3 (continued)
Adjacency Far End Not Discovered

Step	Action
11	Check and clean any dirty fibers. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i> , 323-1851-201.0.
12	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Figure 4-1
Line adjacency far-end address examples

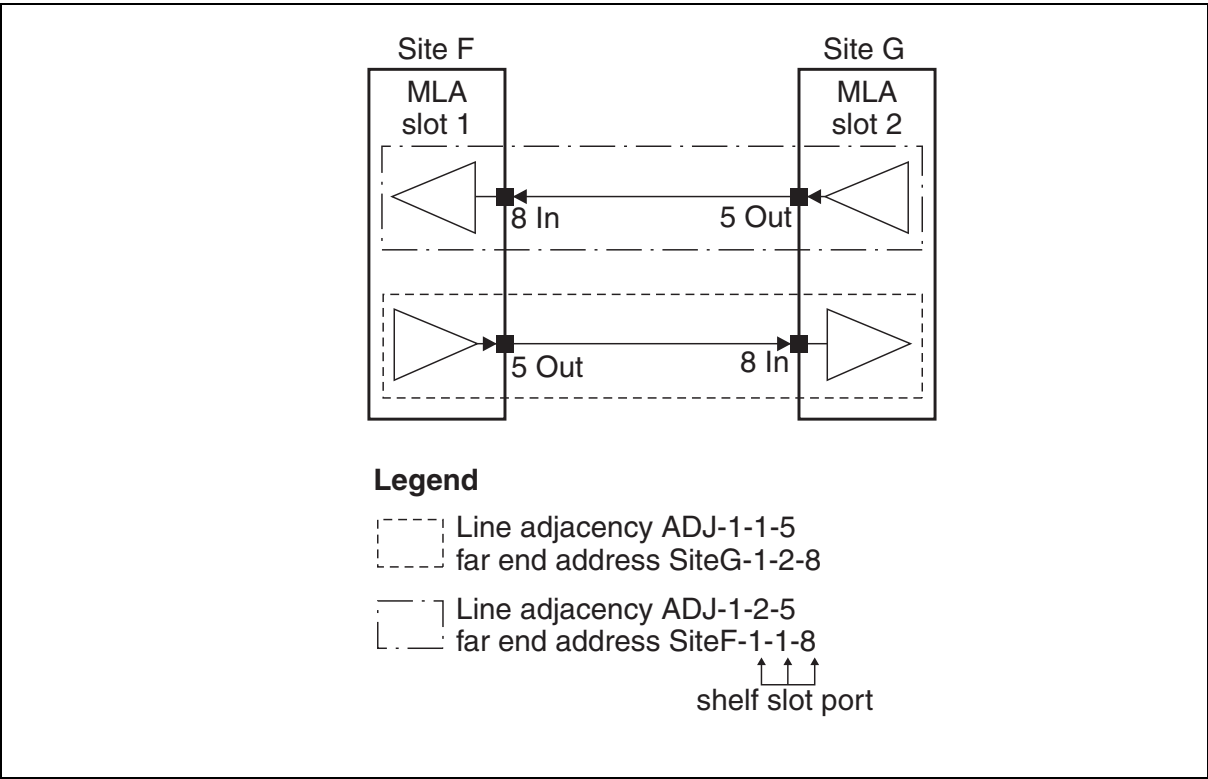
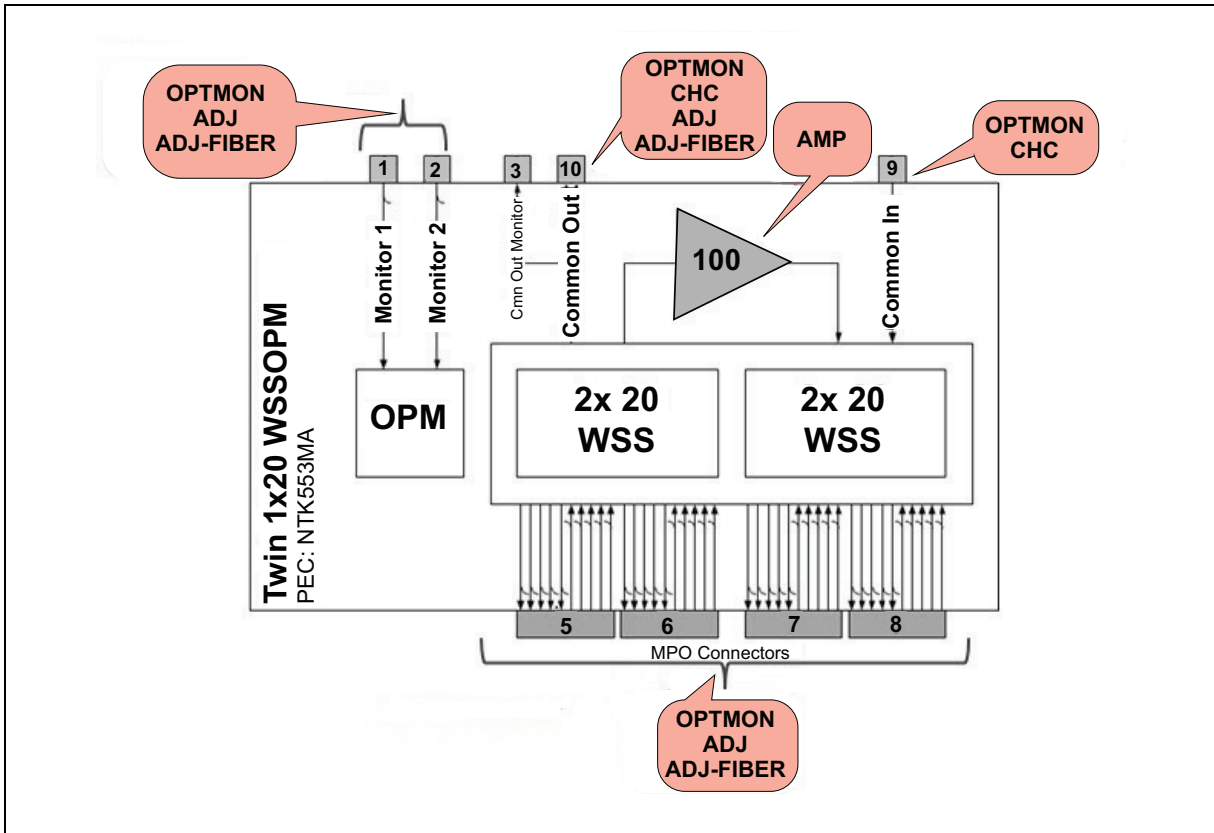


Figure 4-2
MPO adjacency example for WSS Flex C-Band w/OPM 20x1 facility model



Procedure 4-4 Adjacency Mismatch

Alarm ID: 539, 2173

Probable cause

This alarm is raised against an ADJ facility on the following ports to identify potential mis-provisioning or mis-cabling.

- Far-End address Mismatch

A mismatch on a line facing port or a port at an end of a MPO cable when Expected Far-End address and Actual Far-End Address are mismatched.

The alarm will be raised on an ADJ facility if the Adjacency Expected Far-End Address is provisioned but does not match the Actual Far-End Address of a Line facing port or a port at one end of an MPO cable. The Expected Far-End address is manually provisioned, derived by Slot Sequence application or TID Slot Sequence application.

The discovered Actual Far-End Address at one end of a MPO cable is the port of the photonic circuit pack at the other end of the MPO cable, for example, FIM, RLA, WSS Flex C-Band w/OPM 20x1, CCMD8x16, or CXM C-Band Type 1 circuit pack. See [Figure 4-3 on page 4-40](#) for an example where a cabling mistake has been made for CMD1A (port 65) and CMD1B (port 66) of FIM1, where the 2 MPO cables connecting ports 33 and 36 of CCMD8X16#1 have been swapped.

The discovered Actual Far-End Address of a line facing port is the port of the photonics circuit pack at the other end of the fiber span. Refer to [Figure 4-4 on page 4-41](#) for an example.

The alarm can be seen in the following situations:

- Mis-configuration of Expected Far-End Address
- Mis-fibering, for example, an MPO cable or line fiber
- After deleting an unused circuit pack but the MPO cables are left connecting to the deleted circuit pack
- During reconfiguration, a circuit pack is removed from TID slot sequence but an MPO cables still connected it to a FIM
- Network element adjacent to this network element is undergoing a TID consolidation reconfiguration. These alarms can be clear after the Expected Far-end Addresses are updated with the consolidated TID.

Procedure 4-4 (continued)

Adjacency Mismatch

- Channel Port of Channel Mux and Demux

- a. Transmitter/Receiver Type and Discovered Type Mismatching

This alarm is raised on ADJ facility of the Channel Port of Channel Mux and Demux modules or circuit packs if the Transmitter/Receiver Type of ADJ-TX/ADJ-RX facility on the same port is different from the discovered value when Auto Discovered is Auto and Actual Far-End address is discovered.

The alarm is raised on the ADJ-TX when the ADJ-TX Tuning Mode is provisioned to a value not supported by the transponder circuit pack.

When ADJ-TX/ADJ-RX facility has Auto Discovered set to Auto, any change to the discovered Transmitter/Receiver will be automatically populated to the Transmitter/Receiver Type if underlining conditions are valid.

- When the Transmitter/Receiver Type is provisioned, the primary state of an ADJ-TX/ADJ-RX facility is change to IS. If the equipment is OOS, the facility primary state is not allowed to be in IS and therefore Transmitter/Receiver Type will not be auto-populated with the discovered value.
- When the channel is IS and Autoroute is enabled on its OTS, photonic cross connection will be automatically created on this channel. If any conflicts preventing the photonics cross connect being created, Transmitter/Receiver Type will not be auto-populated with the discovered value.

Provisioned tuning mode on ADJ-TX is set according to associated CRS' parent MCTTP's CCMODE. If CCMODE is MODE1, provisioned tuning mode will be set to ACCELERATED. If CCMODE is MODE2 or MODE3, provisioned tuning mode will be set to FAST.

If CRS-NMCs are created on the CMD TX ADJs associated with the same OPS group, the associated MCTTPs must have same CCMODE to correctly set tuning mode on transponder and get discovered tuning mode back on CMDs. Otherwise "Adjacency Mismatch" alarm will be raised against one of CMD TX ADJ and OPS Switch ports.

When the auto-population is not possible, a mismatch is raised in the following scenarios with additional information.

Note: This alarm is masked on ADJ-TX and ADJ-RX ports when the "Adjacency Discovery Unreliable" alarm is raised.

Procedure 4-4 (continued)
Adjacency Mismatch

Note: “Fast Tuning” mode is supported on Waveserver Ai Release 1.4 and higher.

Table 4-1
Adjacency mismatch scenarios

Additional Information	Reasons
Tx/Rx DOCCARE is TRUE	The channel is DOC-managed
Discovered Tx/Rx Type 'UNKNOWN'	The Discovered Transmitter/Receiver Type is unknown due to: — Communication lost to the far-end shelf — The far-end transponder not being supported by SPLI — Temporary condition during SP restart or Upgrade.
The equipment is not in Service	The parent Channel Mux and Demux equipment is not in-service.
CRS OTS-x-x Freq=x WVL=x Blocked Freq=x CLASH Etc.	The photonic cross connect could not be auto provisioned for this channel because existing connection conflict with it — cross-connect is provisioned on the paring Tx or Rx port — If port belongs to an colored OTS with its AUTOROUTE being enabled, and with one of the following conditions: – a cross-connect for the same wavelength/Frequency is already provisioned on another port – if the same wavelength/Frequency is already provisioned on an in-service SLIC Idler facility – if the same wavelength has been manually blocked via ED-WL-RULES – If the same wavelength is used by a manual loopback on another port

b. Wavelength or Frequency Mismatching

This alarm is raised when Wavelength or Frequency is different from its corresponding discovered value when the Actual Far-End address is discovered and Sync Provision is TRUE. Additional Info in the alarm is “Discovered Wavelength Mismatch” or “Discovered Frequency Mismatch”. This mismatch alarm can be seen in the following situations

- In colored line system, the wavelength of the channel is different from the wavelength provisioned on the far-end transponder
- For colorless line system, the frequency provisioned in the photonics cross connect is different from the value on the far-end transponder

Procedure 4-4 (continued)

Adjacency Mismatch

- If the CMD is connected to an OPS, go to step “OPS” to clear the ADJ mismatch alarms on OPS switch ports first it could be one of the CRS is not provisioned.
- Note that there is temporary mismatch during
 - SP restart until SPLI rediscovers the TX Type, or
 - Reconfiguring when adding or deleting channels for the duration of the transponder laser tuning to the new provisioned wavelength/frequency

c. Insufficient Spectral Width

The alarm is raised with additional info “Insufficient Spectral Width” when Tx Minimum Spectral Width is greater than the NMC Allocated Spectral Width.

Alarm Against Switch Ports of OPS:

For OPS configurations, the same tuning mode, frequency, etc, must be provisioned on the CMD pair that is added to the OTS. If the CMDs’ adjacency parameters do not match, an ADJ MISMATCH alarm is raised.

The alarm is raised when an SPLI match is created with photonics and transponder using OPS and the value of Channel Contention Detection and Avoidance (CCDA) is changed to one of the Photonics shelf.

This alarm is raised when Discovered and Provisioned Tuning Mode, or Transmission Mode in the Additional Information section are active and do not match or when the “CCDA Setting Mismatch” is active in the Additional Information section.

Mismatch alarm is raised if any of the following parameters are different between the ADJ Tx facilities (Channel Port of Channel Mux and Demux) adjacent to a pair of OPS switch ports when their Actual Far-End addresses are both discovered. See the chart on the illustration of the related ports. The parameter name is specified in the Additional Information field of the alarm.

- ITU Center Wavelength (nm)
- Center Frequency (THz)
- Laser Centering
- Laser Centering Range

Procedure 4-4 (continued)

Adjacency Mismatch

- Equipment Profile
- OCH Line System Type
- Provisioned Transmission Mode
- Tx Tuning Mode
- Sync Provisioning
- Line Optimization Mode
- Dispersion Mode
- Pre-Compensation
- NMC Allocated Spectrum Width
- Tx Power Reduced State

Note: For CCMD8x16 with fixed-grid equipment profile, you may see temporary mismatch alarms when only one CCMD is upgraded from pre-R12.1 to R12.1 or later and the other CCMD has not upgraded yet. The upgrade to R12.1 or later changes the equipment profile of the CCMD to flexible grid. After the first upgrade is completed, the equipment profile of the upgraded CCMD is flexible grid while the equipment profile of the other CCMD is still fixed grid which causes the temporary mismatch alarm to be raised.

List of the situations:

- The specific parameters are defaulted or provisioned differently on the channel port of the Channel Mux and Demux
- Cross connections being created on one side but not the other side

If a Connection Validation loopback test is performed on two OPS protected ports, the Tx Power Reduced State parameter on both CCMD ADJ-TX facilities must be set to default (ON) value. Failure to do this will cause the ADJ-TX Mismatch alarm to be raised against the protected ports Shelf Processor Restart or 1+1 protection switch.

For the Transponder Direct Attach (TDA) configurations, this alarm is raised on the ADJ facility of WSS w/OPM Flex C-Band 9x1(NTK553LA and NTK553LB) circuit pack.

This alarm is also raised after Waveserver SNC deletion/addition operation.

The alarm can be raised momentary when the shelf processor restarts or after a 1+1 protection switch.

Procedure 4-4 (continued)
Adjacency Mismatch

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- know the correct far-end address for the alarmed adjacency

Step	Action								
1	<table> <tr> <td>If this alarm is raised against an ADJ facility on</td><td>Then go to</td></tr> <tr> <td>a line facing port or a port at one end of an MPO cable</td><td>step 2</td></tr> <tr> <td>a channel port of a Channel Mux and Demux module or circuit pack</td><td>step 6</td></tr> <tr> <td>a switch port of OPS</td><td>step 15</td></tr> </table>	If this alarm is raised against an ADJ facility on	Then go to	a line facing port or a port at one end of an MPO cable	step 2	a channel port of a Channel Mux and Demux module or circuit pack	step 6	a switch port of OPS	step 15
If this alarm is raised against an ADJ facility on	Then go to								
a line facing port or a port at one end of an MPO cable	step 2								
a channel port of a Channel Mux and Demux module or circuit pack	step 6								
a switch port of OPS	step 15								
2	Using the network planning diagram, verify if the Expected far-end address adjacency parameter is correct. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
3	<table> <tr> <td>If the actual far-end address is</td><td>Then</td></tr> <tr> <td>incorrect</td><td>go to step 4</td></tr> <tr> <td>correct</td><td>verify that the line-side fiber is correctly connected. Go to step 5.</td></tr> </table>	If the actual far-end address is	Then	incorrect	go to step 4	correct	verify that the line-side fiber is correctly connected. Go to step 5 .		
If the actual far-end address is	Then								
incorrect	go to step 4								
correct	verify that the line-side fiber is correctly connected. Go to step 5 .								
4	Edit the Expected far-end address and Expected far-end address format adjacency parameters so that they match the Actual far-end address and Actual far-end address Format listed for the ADJ-LINE facility. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
5	If the alarm does not clear, contact your next level of support or your Ciena support group. This procedure is complete.								

Procedure 4-4 (continued)
Adjacency Mismatch

Step	Action																				
6	Using the network planning diagram, verify that the actual far-end address parameter is correct. <table> <tr> <th>If the additional information is</th><th>Then go to</th></tr> <tr> <td>Tx/Rx DOCCARE is TRUE</td><td>step 7</td></tr> <tr> <td>TX/Rx ADJ or Partner ADJ EFEA (Discovered TX/RX Type Unknown)</td><td>step 8</td></tr> <tr> <td>the equipment is not in service</td><td>step 9</td></tr> <tr> <td>CRS OTS-x-x Freq=x</td><td>step 10</td></tr> <tr> <td>WVL=x Blocked</td><td></td></tr> <tr> <td>Freq=x CLASH</td><td></td></tr> <tr> <td>Discovered Wavelength Mismatch</td><td>step 12</td></tr> <tr> <td>Discovered Frequency Mismatch</td><td></td></tr> <tr> <td>Insufficient Spectral Width</td><td>step 14</td></tr> </table>	If the additional information is	Then go to	Tx/Rx DOCCARE is TRUE	step 7	TX/Rx ADJ or Partner ADJ EFEA (Discovered TX/RX Type Unknown)	step 8	the equipment is not in service	step 9	CRS OTS-x-x Freq=x	step 10	WVL=x Blocked		Freq=x CLASH		Discovered Wavelength Mismatch	step 12	Discovered Frequency Mismatch		Insufficient Spectral Width	step 14
If the additional information is	Then go to																				
Tx/Rx DOCCARE is TRUE	step 7																				
TX/Rx ADJ or Partner ADJ EFEA (Discovered TX/RX Type Unknown)	step 8																				
the equipment is not in service	step 9																				
CRS OTS-x-x Freq=x	step 10																				
WVL=x Blocked																					
Freq=x CLASH																					
Discovered Wavelength Mismatch	step 12																				
Discovered Frequency Mismatch																					
Insufficient Spectral Width	step 14																				
7	If the Transmitter Type is correct, this means the far-end transponder is not right or provisioning does not matched provisioned Transponder/Receiver type. - If transponder is not correct, replace the transponder. - If transponder is correct, please contact Ciena Support Group for further information on correcting the provisioning data on the transponder. If the Discovered Type is correct, - If the channel is currently carrying traffic with the provisioned TX Type, set the Auto Discovered parameter to "Manual". - Otherwise, remove the channel from DOC. Refer to the "Editing the DOC Settings" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.																				
8	Check if the COMM is up between this shelf and the shelf pointed by the Expected far-end address. If yes, refer to the COMMs procedures to fix the problem. Note that temporary alarm will be raise after SP restart or cleared after upgrade finishes. The alarm is raised if the Expected Far-end Address is edited from pointing to a transponder that supporting SPLI to another one that does not support SPLI. To clear the alarm, set the Auto Discovered to MANUAL.																				
9	Change the primary state of the CMD equipment to In-Service. Refer to the "Changing the primary state of a circuit pack, module, or pluggable" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.																				

Procedure 4-4 (continued)
Adjacency Mismatch

Step	Action	
10	If	Then
	a connection exist on its paired port	paired is Yes. If this is not desired, edit paired to NO
	the same wavelength/frequency is used by a manual loopback	you can select to terminate the loopback
	a cross-connect for the same wavelength/Frequency is already provisioned on another port	make sure the other cross-connect is desired
	If the same wavelength/Frequency is already provisioned on an in-service SLIC Idler facility	make sure the frequency used on the SLIC Idler is desired
11	the same wavelength has been manually blocked the wavelength provisioning Rule via ED-WL-RULES	make sure the rule is desired
	If the existing connection is correct:	
	a. If you want to keep all the expected far-end address and all the connections but do not want to see the alarm, change the Auto Discovered of ADJ facility to Manual.	
	b. If cross-connect is not meant to be auto provisioned for the whole OTS, change the Autoroute of its OTS to MANUAL.	
	c. If you decided not to use this wavelength/frequency, remove the Expected Far-End Address.	
12	If the port connects to an OPS (The actual far-end address is an OPS switch port):	
	• If the provisioned Wavelength/Frequency is 0.0, provision the cross connect on this channel (refer to the procedure on cross-connect provisioning)	
	• If the Discovered Wavelength/Frequency is 0.0, go to Step "OPS" first to clear any Adjacency mismatch alarm on OPS switch ports and this alarm can be cleared.	
	• If neither discovered nor provisioned Wavelength/Frequency are 0.0:	
	a. If the discovered wavelength/frequency is correct and the channel ports are wrong	
	— Clear the expected far-end address on the CMD channel port on the first CMD — Re-provision the expected far-end address on the correct channel port on the first CMD	

Procedure 4-4 (continued)

Adjacency Mismatch

Step	Action
	— Clear the expected far-end address on the other CMD channel port — Re-provision the expected far-end address on the correct channel port of the 2nd CMD
	b. If the discovered wavelength/frequency is incorrect — Clear the expected far-end address on both CMD channel ports — Re-provision expected far-end address on the correct CMD channel ports.
13	If the port is directly connected to the transponder (the actual far-end address is transponder port), • If the discovered Wavelength/Frequency is correct, set the Auto Discovered to manual and keep the ADJ TX Type settings. • If the discovered Wavelength/Frequency is incorrect, — clear the expected far-end address — Make sure the actual far-end address is cleared and the transponder wavelength is set to 0.0. — Re-provision the expected far-end address
14	Change the NMC Allocated Spectral Width to at least the value of Tx Minimum Spectral Width. Go to step 18 .
15	Write down the Actual Far-End Address on the ADJ facility of both Switch 1 and Switch 2 out ports.
16	Find the far-end equipment and facility using these far-end addresses.

Procedure 4-4 (continued)
Adjacency Mismatch

Step	Action																												
17	If the Additional Information of the alarm is <table> <tr> <td>ITU Center Wavelength (nm)</td><td>1- If the photonic system is colorless:</td></tr> <tr> <td>Center Frequency (THz)</td><td>• Make sure connections are provisioned on both far-end channel</td></tr> <tr> <td>NMC Allocated Spectrum Width</td><td>• make sure the specific values are the same on the two connections</td></tr> <tr> <td></td><td>2- If the photonic system is colored:</td></tr> <tr> <td></td><td>• make sure the parameters specified in the two far-end channels are the same</td></tr> <tr> <td></td><td>• Clear the expected far-end address one of the far-end port and re-provision it on the right port if necessary</td></tr> <tr> <td>Equipment Profile</td><td>Make sure the far-end CMD equipment profiles are the same</td></tr> <tr> <td>Actual Tx Power (dBm)</td><td>Make sure the far-end ADJ-TX facilities have the same value</td></tr> <tr> <td>Laser Centering</td><td></td></tr> <tr> <td>Laser Centering Range</td><td></td></tr> <tr> <td>OCH Line System Type</td><td></td></tr> <tr> <td>Provisioned Transmission Mode</td><td></td></tr> <tr> <td>Sync Provisioned</td><td></td></tr> <tr> <td>Tx Power Reduced State</td><td></td></tr> </table> Then	ITU Center Wavelength (nm)	1- If the photonic system is colorless:	Center Frequency (THz)	• Make sure connections are provisioned on both far-end channel	NMC Allocated Spectrum Width	• make sure the specific values are the same on the two connections		2- If the photonic system is colored:		• make sure the parameters specified in the two far-end channels are the same		• Clear the expected far-end address one of the far-end port and re-provision it on the right port if necessary	Equipment Profile	Make sure the far-end CMD equipment profiles are the same	Actual Tx Power (dBm)	Make sure the far-end ADJ-TX facilities have the same value	Laser Centering		Laser Centering Range		OCH Line System Type		Provisioned Transmission Mode		Sync Provisioned		Tx Power Reduced State	
ITU Center Wavelength (nm)	1- If the photonic system is colorless:																												
Center Frequency (THz)	• Make sure connections are provisioned on both far-end channel																												
NMC Allocated Spectrum Width	• make sure the specific values are the same on the two connections																												
	2- If the photonic system is colored:																												
	• make sure the parameters specified in the two far-end channels are the same																												
	• Clear the expected far-end address one of the far-end port and re-provision it on the right port if necessary																												
Equipment Profile	Make sure the far-end CMD equipment profiles are the same																												
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Laser Centering																													
Laser Centering Range																													
OCH Line System Type																													
Provisioned Transmission Mode																													
Sync Provisioned																													
Tx Power Reduced State																													
18	If the alarm did not clear, verify the other ADJ-TX parameters.																												
19	If the alarm does not clear, contact your next level of support or your Ciena support group.																												

—end—

Figure 4-3
Cabling mistake example

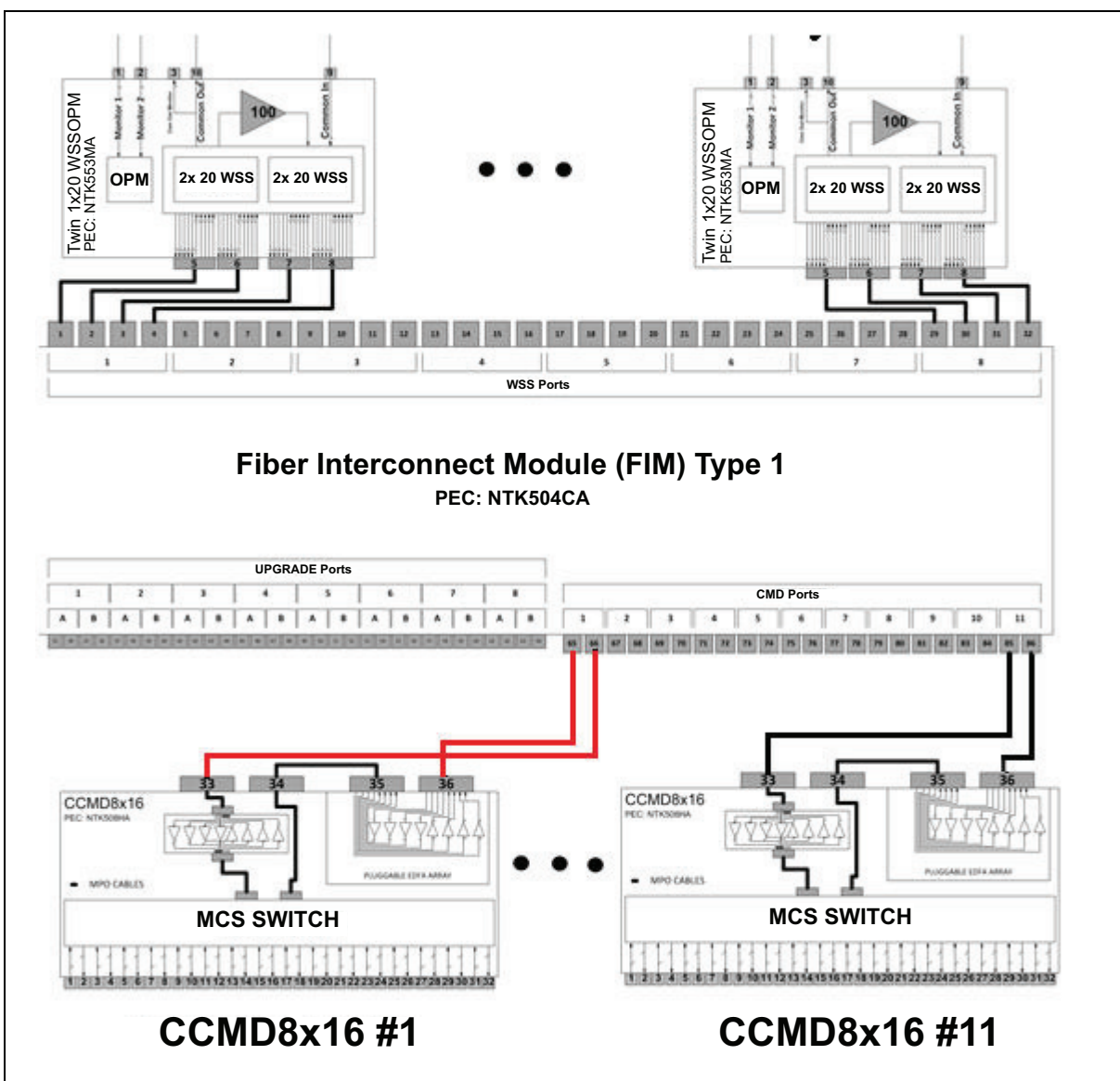
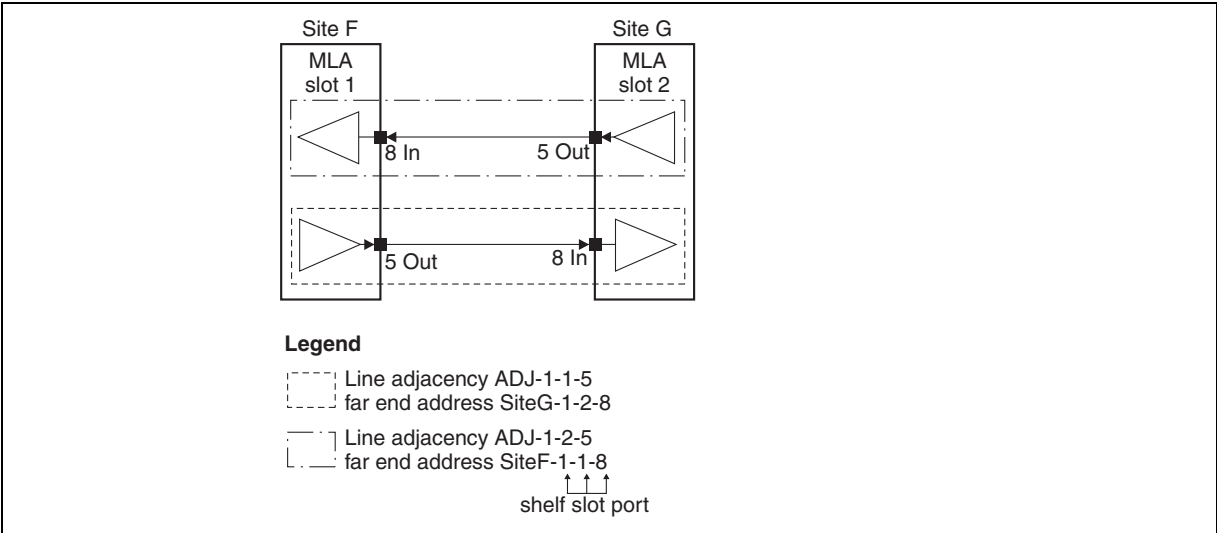


Figure 4-4
Line adjacency far-end address examples



Procedure 4-5

Adjacency Provisioning Error

Alarm ID: 707

Probable cause

This alarm is raised against an ADJ facility when the inter-shelf provisioning is missing, and there is a channel with an optical cross-connect or inter-domain passthrough (over that adjacency or over the paired adjacency) in the opposite direction.

This alarm is also raised against a DSCM adjacency facility that is provisioned in the OTS when the OTS “Enhanced Topology” attribute is Enabled and the DSCM does not have its DISP facility properly provisioned.

Note: If unpaired channels are to be used, you must provision inter-shelf adjacencies in both directions.

The conditions that raise this alarm are as follows:

- a passthrough channel is provisioned, but the far-end address does not match the discovered shelf
- an optical cross-connect is provisioned, but the far-end address does not match a shelf at the same site and in a different domain
- the adjacency type does not match the circuit pack function (for example, WSS w/OPM on a ROADM, CMD or LIM on a TOADM)
- a missing or incorrectly provisioned WSS to WSS adjacency
- a passthrough channel is provisioned at a TOADM, but a CMD for that same group is provisioned or the inter-OTS adjacencies are missing or they are incorrectly provisioned. In a TOADM configuration, the adjacency corresponding to the CMD channel OUT port for that wavelength is alarmed.
- provisioning an adjacency at only one end in an inter-OTS adjacency configuration between a DIA LIM and COADM SMD.
- if no passthrough adjacency is provisioned on a ROADM, the alarm is raised against all candidate passthrough ports. The alarm is cleared against all candidate passthrough ports when a passthrough adjacency is provisioned.

Procedure 4-5 (continued)

Adjacency Provisioning Error

- a skip channel originating from a CMD44/ROADM is provisioned as passthrough at a TOADM. This channel cannot be optically passed through the node. For more information on unavailable skip channels, refer to the table (unavailable 50 GHz channels) in the *Photonic Layer Guide*, NTRN15DA.
- there was a change in the TID of the node after some adjacencies were provisioned

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) for shelf details

Step	Action
1	If this is a ROADM site TOADM site Then go to step 2 step 6
2	Verify if the WSS-to-WSS adjacency in both directions are provisioned and correct. If not, provision the missing or correct the incorrect actual far-end address data.
3	If the facility with the alarm is provisioned properly, then verify the provisioned far-end address and go to the corresponding facility to check the provisioning. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Note: It can take up to two minutes for this alarm to clear once provisioning has been resolved.
4	If the WSS-to-WSS adjacency has not been manually provisioned or has an error is provisioned and correct Then go to step 5 step 11
5	Create or edit the WSS-to-WSS adjacency. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 11.

Procedure 4-5 (continued)

Adjacency Provisioning Error

Step	Action						
6	Verify that the CMD-to-CMD adjacencies at the end of SCMD Cascading Order are provisioned and correct. In particular, for inter-OTS adjacencies, confirm that the Upgrade OUT to Upgrade IN adjacencies for the cascading CMD are provisioned correctly. Verify that the provisioned far-end address is correct. Select Configuration->Equipment & Facility Provisioning. For intra-OTS adjacencies, confirm that the cascading SCMD order is provisioned correctly. Select the Configuration->Photonic Services->SCMD Cascading Order.						
7	<table border="0"> <tr> <td>If the CMD-to-CMD adjacency at the end of the SCO</td><td>Then go to</td></tr> <tr> <td>has an error</td><td>step 8</td></tr> <tr> <td>is provisioned and correct</td><td>step 11</td></tr> </table>	If the CMD-to-CMD adjacency at the end of the SCO	Then go to	has an error	step 8	is provisioned and correct	step 11
If the CMD-to-CMD adjacency at the end of the SCO	Then go to						
has an error	step 8						
is provisioned and correct	step 11						
8	Edit the CMD-to-CMD adjacency. See <i>Configuration - Provisioning and Operating</i> , 323-1661-310.						
9	Verify that CMDs are not provisioned for the same groups that are intended to passthrough the site. If this occurs, the system must be re-engineered to use a channel that does not drop at this site. See your EDP for more information.						
10	Verify that a skip channel is not provisioned to pass through this node. If it is de-provision the illegal channel and/or re-engineer the network. Go to step 15.						
11	<table border="0"> <tr> <td>If the alarm is raised against</td><td>Then go to</td></tr> <tr> <td>a DSCM port</td><td>step 12</td></tr> <tr> <td>the inter-OTS adjacency</td><td>step 13</td></tr> </table>	If the alarm is raised against	Then go to	a DSCM port	step 12	the inter-OTS adjacency	step 13
If the alarm is raised against	Then go to						
a DSCM port	step 12						
the inter-OTS adjacency	step 13						
12	Check the DSCM has its DISP facility properly provisioned. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
13	Check the inter-OTS adjacency is properly provisioned. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
14	Check if there is a TID changed on the local shelf. Refer to the "Displaying node information" procedure in <i>Administration and Security</i> , 323-1851-301.						
15	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-6

All License Servers Unavailable

Alarm ID: 2036

Probable cause

This alarm is raised when the 6500 shelf cannot communicate with all license server/managers.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Verify the network connection between the 6500 shelf and the license server/managers.
2	If you are using HTTPS, the correct date and time must be set on the shelf (either manually or using TOD/NTP server provisioning and synchronization). Refer to “Provisioning Time of Day servers” and “Operating a time of day synchronization” procedure in <i>Administration and Security</i> , 323-1851-301.
3	If you set the time after provisioning licensing, then perform a manual audit after setting the shelf time to clear the license server comms error. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301.
4	Verify that the license server/managers are provisioned and are not down. Refer to “Provisioning License Manager/Server information on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301.
5	It can take up to 12 hours for the alarm to clear as, in steady state, the heartbeat is run every 12 hours. Run a manual audit which will reset the heartbeat to two minutes after which the heartbeat interval will gradually increase in time to the steady state time of 12 hours. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301. The alarm will clear two minutes after the manual audit is run.
6	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-7

All Provisioned RADIUS Accounting Servers Unavailable

Alarm ID: 1518

Probable cause

This alarm is raised when no response is received from any provisioned RADIUS accounting server during a user-provisioned timeout.

This alarm is also raised when the RADIUS accounting server provisioning on the network element is incorrect.

This security alarm is raised against an SP, SPAP-2 w/ 2xOSC, or an integrated SP with an 8xOTN Flex MOTR circuit pack if applicable.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Disable the server on 6500.
2	Re-enable the server and log in or log out of the server.
3	If the alarm is raised again, disable the RADIUS accounting feature and log in or log out of the server.
4	If the alarm is raised, ensure the following RADIUS accounting server provisioning values on the network element are correct: • server IP address • server port • shared secret • timeout - if this value is too small the server may not be able to respond quickly enough Refer to the “Provisioning the primary or secondary RADIUS server” procedure in <i>Administration and Security</i> , 323-1851-301.
5	Check the status of the RADIUS accounting server. Ensure the status is ON.

Procedure 4-7 (continued)

All Provisioned RADIUS Accounting Servers Unavailable

Step	Action
6	Log in or log out of the network element. This will send a RADIUS accounting message to all provisioned RADIUS accounting servers. The alarm will clear if a response is received from the server(s) within the provisioned timeout.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-8

All Provisioned RADIUS Servers Unavailable

Alarm ID: 582

Probable cause

This alarm is raised when:

- all requests to the primary and secondary RADIUS servers of a shelf processor time out
- all requests to a RADIUS server of a shelf processor time out and only one RADIUS server has been provisioned (primary or secondary)

If the All Provisioned RADIUS servers Unavailable alarm is raised and only a single RADIUS server is provisioned (primary or secondary), provisioning the second RADIUS server will cause the All Provisioned RADIUS servers Unavailable alarm to clear and either the Primary RADIUS Server Unavailable or Secondary RADIUS Server Unavailable alarm to be raised for the original RADIUS server.

The alarm is not raised due to server time out.

If the alternate method for security is Challenge/Response, the user can log in using the challenge response generated by Site Manager. The shared secret is required to generate the correct response. If the alternate method for security is Local, the user can log in using a local user ID and password.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Ensure the primary and secondary RADIUS servers of the shelf processor are enabled and have a valid IP address. Refer to the "Provisioning the primary or secondary RADIUS server" procedure in <i>Administration and Security</i> , 323-1851-301.
2	Log into the network element again using the RADIUS authentication (centralized security administration).
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-9

All Provisioned TACACS Servers Unavailable

Alarm ID: 2045**Probable cause**

This alarm is raised when the 6500 shelf cannot communicate with TACACS servers.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
------	--------

- | | |
|---|---|
| 1 | Verify if the provisioned TACACS servers are down. The alarm clears when connectivity is re-established between the shelf and the TACACS server. |
| 2 | If the alarm does not clear, perform one of the following: <ul style="list-style-type: none">• clear the security alarms. Refer to “Clearing security alarms” procedure in this document.• provision a new IP address or Port for server 1 or server 2. Refer to the “Provisioning the TACACS+ server” procedure in <i>Administration and Security</i>, 323-1851-301• disable TACACS. Refer to <i>Administration and Security</i>, 323-1851-301 |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-10

ASE Channel Source Active

Alarm ID: 2196

Probable cause

This alarm is raised when CHC has switched from Traffic to an ASE source due to the following reasons occurred on all NMCs in an MC:

- NMCC is in Channel FDI condition.
- NMCC is in Channel Loss Of Signal condition.
- NMCC is in Channel Degrade condition.

Note 1: A channel is in FDI condition when an upstream domain is in Automatic Line Shut Off (ALSO) state.

Note 2: A channel is considered in degrade condition when the NMCC 'Derived Input Power' is less than the NMCC 'Reference Input Power Profile' by a value greater than the 'Minor Degrade Threshold'. The Minor Degrade Threshold defaults to 3 dB. Although the 'Minor Degrade Threshold' can be edited using the Site Manager Node Information application and the Systems settings, it is strongly recommended that Minor Degrade Threshold is left at its default value unless expressly informed otherwise by Ciena.

Note 3: The Channel Degrade condition has a 0.5 dB hysteresis that is considered before clearing the condition. The NMCC facility Derived Input Power and its Reference Input Power Profile must be within the threshold minus 0.5 dB to clear the condition.

Impact

Warning

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|---|
| 1 | <p>Review the Transient conditions (TC) in Site Manager "Faults -> Historical Fault Browser" to find the additional information for why ASE Channel Source Active alarm is being raised. Example TCs are:</p> <ul style="list-style-type: none">— (NMCC) ASE Source Active: Channel Loss Of Signal— (NMCC) ASE Source Active: Channel Degrade— (NMCC) ASE Source Active: Channel FDI |
|---|---|

Procedure 4-10 (continued)
ASE Channel Source Active

Step	Action										
2	Select your next step.										
3	<table> <tr> <th>If the configuration used is:</th><th>Then</th></tr> <tr> <td>Faulted local channel(s) added from SLTE Active Config(s) such as CDA (CCDM12+SMD+MLA3/XLA), COADM-DIA (CCMD12+SMD+MLA3+WSS) or GMD10-DIA (CCMD12+GMD10+MLA3+WSS) - Heal from Fiber Cut, Loss Of Signal and Channel Degrade</td><td>channels automatically switch back to Traffic. Go to step 7.</td></tr> <tr> <td>Faulted local channel(s) added from SLTE Hybrid Config such as CCMD12+GMD10 Direct Attach - Heal from Fiber Cut, Loss Of Signal and Channel Degrade.</td><td>go to step 4.</td></tr> <tr> <td>Faulted local channel(s) added from SLTE Passive Config(s) such as CMD44 Direct Attach or vCCMD64 Direct Attach - Heal from Fiber Cut, Loss Of Signal and Channel Degrade.</td><td>go to step 4.</td></tr> <tr> <td>Faulted passthrough channels for SLTE - Heal from upstream fiber faults such as Fiber Cut, Loss Of Signal, Channel Degrade and Automatic Line Shut Off (ALSO)</td><td>channels automatically switch back to Traffic. Go to step 7.</td></tr> </table>	If the configuration used is:	Then	Faulted local channel(s) added from SLTE Active Config(s) such as CDA (CCDM12+SMD+MLA3/XLA), COADM-DIA (CCMD12+SMD+MLA3+WSS) or GMD10-DIA (CCMD12+GMD10+MLA3+WSS) - Heal from Fiber Cut, Loss Of Signal and Channel Degrade	channels automatically switch back to Traffic. Go to step 7 .	Faulted local channel(s) added from SLTE Hybrid Config such as CCMD12+GMD10 Direct Attach - Heal from Fiber Cut, Loss Of Signal and Channel Degrade.	go to step 4 .	Faulted local channel(s) added from SLTE Passive Config(s) such as CMD44 Direct Attach or vCCMD64 Direct Attach - Heal from Fiber Cut, Loss Of Signal and Channel Degrade.	go to step 4 .	Faulted passthrough channels for SLTE - Heal from upstream fiber faults such as Fiber Cut, Loss Of Signal, Channel Degrade and Automatic Line Shut Off (ALSO)	channels automatically switch back to Traffic. Go to step 7 .
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Faulted passthrough channels for SLTE - Heal from upstream fiber faults such as Fiber Cut, Loss Of Signal, Channel Degrade and Automatic Line Shut Off (ALSO)	channels automatically switch back to Traffic. Go to step 7 .										
4	Switch the CHC switch selector back to Traffic using one of the following methods: a. Make a switch selector change in the WSS CHC Facility — Make note of the AID of the raised alarm using Alarm Browser (Active Alarms Application). — Make note of the CHC Target Switch Selector for the AID of the raised alarm using Site Manager Equipment & Facility Provisioning Application. — Edit the corresponding CHC Facility->Select the Switch Selector for the Traffic Source and Apply — Confirm the Switch Selector is updated — If using Multi-Select when editing CHC facilities, ensure to repeat the switch selector change until all the multi-selected CHCs are updated to the Target Switch Selector.										

Procedure 4-10 (continued)
ASE Channel Source Active

Step	Action
	b. Launch the PEQ tool — Tools->Submarine->Power Equalization — Select ASE Fault Management Tab — Select Network Element and Switch Selector — Select Show CHC — CHC AID will be highlighted in yellow if the traffic channel is pointing to ASE — Select CHC AID (No multi-select) and click on Switch to Traffic — CHC AID should no longer be highlighted
5	Confirm the Tx launch spectrum matches the baseline as follows: — In Site Manager Performance Monitoring -> PM Graph tab, select Port-5 of the SLTE DryAMP circuit pack — SDMON Tab — Baseline check-box — Compare Baseline and Current Spectrum
6	Perform a manual re-optimization of the SLTE Domain Optical Controller facility after clearing the ASE Channel Source Active alarm(s) if channel powers are less than 3 dB from expected SSC Base Target Power + Bias as follows (Automatic re-opt will occur if traffic channel delta is more than 3 dB): — Photonic Services->Photonic Channel Management — Select Photonic Controllers Tab — Select appropriate DOC and click Re-optimize
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-11

Automatic Power Reduction Active

Alarm ID: 542, 1795

Probable cause

The APR condition is caused by a reflection somewhere downstream from the amplifier. This reflection can be caused by:

- dirty optical connectors
- improper optical cable mating
- a disconnected optical fiber at the amplifier output
- an optical fiber cut
- a degraded optical fiber
- a disconnected or missing termination
- misprovisioning of an amplifier resulting in excessive power being injected into the mid-stage DSCM or fiber-plant
- backwards traveling optical signals from the WSS to the ASE loading amplifier of an SLTE
- an OSRP Line/Link Down condition triggered by fiber disconnection on intra-node fiber (except WSS-WSS fiber), AMP in shut-off state, or AMP Primary State OOS-MA

Note: OSRP provides a fault reason within the “Additional Information” field of the alarm to help with the troubleshooting.

For amplifiers, Automatic Power Reduction (APR) indicates reflection at the output. For RAMAN amplifiers, APR either indicates reflection at the input or output depending on the location of the ORL problem. When the ORL reading is not valid because the power into the backward reflective monitor tap is too low and cannot be measured accurately, the ORL PM reading(s) report “OOR”. The true ORL reading(s) cannot be determined in this case.

Note: This alarm indicates a condition on the output port of an amplifier; the alarm is raised against the Line A or Line B output port AMPMON facility of the corresponding SLA, MLA, XLA, MLA L-Band, MLA C-Band, MLA2 w/VOA, SRA, or MLA3 C-Band/MLA3 L-Band circuit packs. The alarm is raised against the Line B output port AMP facility of the corresponding RLA 5x1 and MDRLA 2x6 circuit packs. Therefore, ensure you troubleshoot the appropriate port. See the block diagrams of the amplifiers in Photonics Equipment, 323-1851-102.6.

Procedure 4-11 (continued)

Automatic Power Reduction Active

Upon alarm recovery, no OTDR trace is run (different from APR on Line A In), the RAMAN pumps are turned back on and traffic recovers. This alarm masks the “Low Optical Return Loss” alarm on the AMPMON facility.

ATTENTION

As of Release 10.2, the MuxAmp configuration disables the Automatic Power Reduction (APR) function. Certain C-band amplifier circuit packs used in a MuxAmp configuration with one or more wavelengths in the IEC 60825-1 Class 1M range (such as those from a WL3n source) may have a Class 1 Hazard level label but should have a Class 1M Hazard Level label because APR is disabled.

Circuit packs that were originally manufactured with a Hazard Level 1 warning label can be re-labeled with the Level 1M label kit (part number 415-2818-001).

The following C-band amplifier circuit packs may have a Hazard Level 1 warning label.

- NTK552AAE5, Single Line Amplifier (SLA C-Band), Revision 19 and lower
- NTK552BAE5, Midstage Line Amplifier (MLA C-Band), Revision 19 and lower
- NTK552FAE5, Midstage Line Amplifier 2 (MLA2 C-Band), Revision 17 and lower
- NTK552FB, Midstage Line Amplifier 2 (MLA2 C-Band) with variable optical attenuator (VOA), Revision 05 and lower
- NTK552GAE5, Midstage Line Amplifier 3 (MLA3 C-Band), Revision 06 and lower
- NTK553RA, ROADM with Line Amplifier (RLA) C-Band 5x1 circuit pack
- NTK553SA, MDRLA 2x6 C-Band circuit pack

For more information and the procedure to apply the Level 1M label on these circuit packs, see the section on observing product and personnel safety guidelines in *Installation - General Information*, 323-1851-201.0.

For the SLA, MLA, RLA 5x1, MDRLA 2x6, XLA, MLA L-Band, MLA C-Band, MLA3 C-Band and MLA3 L-Band circuit pack AMP facility, or the MLA2 w/VOA circuit pack VOA facility, this alarm is raised when the optical return loss (ORL) drops below a fixed threshold (17 dB). When the ORL falls below the threshold, the system reduces the amplifier output power level to minimize the danger of personal eye injury.

For the XLA, LIM, and SRA circuit packs, this alarm can be raised on the AMPMON facility. The RAMAN state can be displayed using Site Manager. Possible RAMAN states are Normal/APR/Shutoff.

Procedure 4-11 (continued)

Automatic Power Reduction Active**Impact**

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the engineering documentation package (EDP)
- have a fiber cleaning kit
- have a replacement module

Step	Action										
1	<table border="1"> <thead> <tr> <th>If</th> <th>Then</th> </tr> </thead> <tbody> <tr> <td>the alarm was raised as a result of maintenance or SLAT that has not been completed</td> <td>no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.</td> </tr> <tr> <td>the alarm was raised on ASE loading amplifier of an SLTE and traffic is down on the WSS on which alarmed ASE loading amplifier is connected</td> <td>perform a cold restart on the WSS circuit pack to clear the alarm. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor”.</td> </tr> <tr> <td>the alarm was raised on ASE loading amplifier of an SLTE and traffic is still up and running through the WSS on which alarmed ASE loading amplifier is connected”</td> <td>contact your Ciena support group.</td> </tr> <tr> <td>the alarm is unexpected</td> <td>go to step 2</td> </tr> </tbody> </table>	If	Then	the alarm was raised as a result of maintenance or SLAT that has not been completed	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.	the alarm was raised on ASE loading amplifier of an SLTE and traffic is down on the WSS on which alarmed ASE loading amplifier is connected	perform a cold restart on the WSS circuit pack to clear the alarm. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor” .	the alarm was raised on ASE loading amplifier of an SLTE and traffic is still up and running through the WSS on which alarmed ASE loading amplifier is connected”	contact your Ciena support group.	the alarm is unexpected	go to step 2
If	Then										
the alarm was raised as a result of maintenance or SLAT that has not been completed	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.										
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the alarm was raised on ASE loading amplifier of an SLTE and traffic is still up and running through the WSS on which alarmed ASE loading amplifier is connected”	contact your Ciena support group.										
the alarm is unexpected	go to step 2										
2											
Confirm that the amplifier provisioning matches what is defined in the EDP. The amplifier provisioning can be checked by using the Configuration->Equipment and Facility Provisioning screen in Site Manager. Edit the power level values as required. Refer to the “Retrieving protection parameters” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.											

Procedure 4-11 (continued)

Automatic Power Reduction Active

Step	Action	
3	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 4
4	If the alarmed amplifier is Port 8 (Line A)	Then go to step 5
	otherwise	step 9
5	Ensure that all LC/SC connectors located after the amplifier output are properly mated. Verify this on both ends of the connector-mating receptacles.	
6	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 7
7	Ensure that the termination plugs are present and are mated properly on unused ports.	
8	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 9

Procedure 4-11 (continued)

Automatic Power Reduction Active**Step Action**

9

**DANGER****Risk of laser radiation exposure**

Do not look directly into the optical beam. Invisible light can severely damage your eyes.

**CAUTION****Risk of damage to modules**

Never disconnect an optical fiber that is connected to an active or powered up optical amplifier. To disconnect or reconnect an optical fiber, make sure the optical amplifier is out of service (OOS), then disconnect or reconnect the fiber.

**CAUTION****Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

Place the alarmed AMP or RAMAN facility out of service (OOS) using the Edit button in the Configuration->Equipment and Facility Provisioning screen of Site Manager. For further instructions refer to the “Changing the primary state of a facility” procedure in *Configuration - Provisioning and Operating*, 323-1851-310. Note that for the MLA2V, you should place the amplifier with the same port number as the alarmed VOA facility OOS.

10

**CAUTION****Risk of traffic loss**

Only disconnect the output fiber of the alarmed optical amplifier. It is not necessary to disconnect any other output fibers, which could affect service. Disconnecting the Line B out fiber will impact traffic in both directions, as this triggers Automatic Laser Shut Off (ALSO).

Disconnect the output fiber of the alarmed optical amplifier, clean the output fiber and connectors at the amplifier, then reconnect the fiber.

Procedure 4-11 (continued)

Automatic Power Reduction Active

Step	Action						
11	Place the AMP or RAMAN facility back in-service (IS) using the Edit button in the Configuration->Equipment and Facility Provisioning screen of Site Manager. For further instructions refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Note that for the MLA2V, you should place the amplifier with the same port number as the alarmed VOA facility IS.						
12	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 13</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	You may have to clean a specific connector or connectors that may not be immediately connected to the alarmed amplifier output. For information on isolating connector losses, complete Procedure 2-16, “Locating a reflective event” .						
14	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-12

Automatic Shutoff

Alarm ID: 590, 942

Probable cause

This alarm is raised against an AMP facility or SRA RAMAN facility when an ALSO condition is triggered, and is raised against any amplifier that has been shut off. An ALSO condition is cleared when the OSC receives a clean signal or in case of SRA circuit pack, receives a clean telemetry signal from an upstream network element.

The RAMAN state can be displayed using Site Manager. Possible RAMAN states are Normal, APR, or Shutoff.

Note: This alarm indicates a condition on the output port of an amplifier; however, the alarm is raised against the input port of the corresponding SLA/MLA C-Band, RLA 5x1, MDRLA 2x6, MLA L-Band, MLA2 C-Band, MLA2 w/VOA, SRA, XLA, or MLA3 C-Band/MLA3 L-Band circuit pack. Therefore, ensure you troubleshoot the appropriate port.



DANGER

Risk of radiation exposure

If light is used to test the broken fiber (for example, with a light source or an OTDR), certain Automatic Laser Shut Off (ALSO) and loss of signal alarms can clear. When the shelf detects light, the alarms clear and the amplifier facility is powered up. This is an expected behavior because a shelf cannot distinguish between a light source from an optical test set and a light source from a shelf. Ensure the adjacent optical amplifiers are out of service (OOS) when performing fiber repairs.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Complete the “Optical Line Fail” alarm clearing procedure.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-13

Automatic Shutoff Compromised

Alarm ID: 1775

Probable cause

This alarm is raised when the Rx/Tx fibers of the OSC on an SRA/SAM/ESAM/RLA/MDRLA 2x6 circuit pack are crossed at one or both ends of a photonic span.

For the SRA/SAM/ESAM circuit packs, this alarm is raised when the OSC Tx fiber is removed.

The alarm is also raised when the Tx power on the SFP is below the threshold or if the transmitter is disabled.

Note: When this alarm is active, the automatic laser shutoff (ALSO) feature is disabled.



DANGER

Risk of laser radiation exposure

During this procedure the fiber plant does not have to be disrupted and the system remains a Class 1 (IEC)/Class I (FDA) product.

If the fiber downstream of the OSC becomes disconnected accidentally while the “Automatic Shutoff Disabled” alarm is active, the radiation at the exposed fiber can be at hazard level 1M (IEC 60825-2). In this situation, you must take all safety precautions appropriate to hazard level 1M (IEC 60825-2). The ORL-based APR safety mechanism remains active.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the engineering documentation package (EDP)
- have a fiber cleaning kit

Procedure 4-13 (continued)
Automatic Shutoff Compromised

Step	Action
1	Ensure that the OSC fibers are correctly connected at both ends of their spans.
2	If the alarm does not clear, ensure that the transmitter is enabled. Check for and clean any dirty fibers. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i> , 323-1851-201.0.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-14

Automatic Shutoff Disabled

Alarm ID: 1035, 1746

Probable cause

This alarm is raised on AMP or OPTMON facilities when you set the ALSO Disabled parameter to True through Site Manager for the AMP or OPTMON facility. The AMP and OPTMON facility is put in this mode for certain maintenance actions such as:

- recovery from ALSO in certain configurations
- check for ORL at SLAT Time
- during the nodal continuity testing
- initial turn up and recovery on a stretched span (for example: without OSC)

This alarm is also raised when the XLA, RLA, MDRLA 2x6 and/or SRA Gain Mode is not set as per the network planning and link budget recommendations.

This alarm is raised temporarily when DOC performs a power audit. During the power audit, DOC disables the automatic laser shutoff (ALSO) feature on the XLA so that the XLA can generate amplifier noise (ASE) for the SRA calibration. At the end of the power audit, DOC re-enables the automatic laser shutoff feature and the alarm will clear.



DANGER

Risk of laser radiation exposure

During this procedure the fiber plant does not have to be disrupted and the system remains a Class 1(IEC)/Class I (FDA) product.

If the fiber downstream of the AMP Line A output connector becomes disconnected accidentally while the “Automatic Shutoff Disabled” alarm is active, the radiation at the exposed fiber can be at hazard level 1M (IEC 60825-2). In this situation, you must take all safety precautions appropriate to hazard level 1M (IEC 60825-2). The ORL-based APR safety mechanism remains active.

Impact

Major, non-service-affecting (M, NSA) alarm

Procedure 4-14 (continued)
Automatic Shutoff Disabled

Prerequisites

To perform this procedure, you require an account with a level 4 or higher user privilege code (UPC).

Step	Action
1	Prior to running the power audit or adding channels in a dark section, ensure that the XLA switch gain mode is set properly.
2	Make sure that the maintenance work has been completed.
3	Immediately after the maintenance work has been completed, enable Auto Shutoff: • Select Facilities -> AMP/OPTMON. • Set the ALSO Disabled parameter to False in the AMP facility or OPTMON facility.
	This action clears the Automatic Shutoff Disabled alarm.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-15 Auto Protection Switch Acknowledge Time Out

Alarm IDs: 427, 428, 429, 430, 954, 1016, 1109, 1325, 1337, 1395

Probable cause

This alarm is raised for 1+1 OTN protection, 1+1 port TPT, and 1+1 TPT protection groups if an expected reverse request is not received by the tail end of a switch within 50 ms.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must have the optical fiber/cable connection information (that is, how the circuit packs on each network element connect to other network elements).

Step	Action						
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" .						
2	Ensure the fiber of the network is done properly. If not, make changes to correct the problem. If the alarm does not clear, continue with the next step.						
3	From your company records, determine the correct switch mode for the link. <table border="0"> <tr> <td>If the correct mode is</td><td>Then go to</td></tr> <tr> <td>unidirectional</td><td>step 4</td></tr> <tr> <td>bidirectional</td><td>step 5</td></tr> </table>	If the correct mode is	Then go to	unidirectional	step 4	bidirectional	step 5
If the correct mode is	Then go to						
unidirectional	step 4						
bidirectional	step 5						
4	At the local network element, change the switch mode of the optical interface pair identified in step 1 . Refer to the "Changing the protection parameters for a pair of facilities or equipment" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. Go to step 7 . Note: Changing the protection switch mode for one of the optical interface circuit packs in a pair automatically changes the protection switch mode for the other circuit pack in the pair.						
5	Use the optical fiber/cable connection information to identify the network element and optical interface modules that are on the remote end of the link.						

Procedure 4-15 (continued)

Auto Protection Switch Acknowledge Time Out

Step	Action
6	Log into the remote network element and change the mode of the optical interface modules to bidirectional.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 4-16 Autoprovisioning Mismatch

Alarm ID: 60

Probable cause

This alarm is raised when:

- a circuit pack is installed in an unprovisioned slot that does not support that circuit pack. For example, a photonic circuit pack inserted in a slot reserved by an OTS for a different EQPT type. There is no effect on shelf operations.
- more than the expected number of circuit packs on a shelf are provisioned. For example, more than four WSS w/OPM circuit packs in a 14-slot shelf type.
- a SCMD4 variant that is identical to one that is already provisioned in the OTS is inserted into a slot that is reserved for a SCMD4 in that OTS.
- a 40G, 100G or ULH circuit pack is inserted in a non-40G/100G shelf type.

When auto equipping is disabled, this alarm is not raised if you install a circuit pack in an unprovisioned slot that does not support that circuit pack.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- if necessary, obtain a replacement circuit pack or a filler card

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” .
2	Make sure you do not have more than two 2xOSC circuit packs or more than four WSS w/OPM circuit packs installed in the shelf.

Procedure 4-16 (continued)
Autoprovisioning Mismatch

Step	Action						
3	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
4	Verify whether the slot is reserved by an OTS for a different equipment type.						
5	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the alarmed slot is reserved by an OTS for a different equipment type</td><td>step 6</td></tr> <tr> <td>otherwise</td><td>step 7</td></tr> </table>	If	Then go to	the alarmed slot is reserved by an OTS for a different equipment type	step 6	otherwise	step 7
If	Then go to						
the alarmed slot is reserved by an OTS for a different equipment type	step 6						
otherwise	step 7						
6	You can edit the OTS instance using the Photonic Configuration Management application to release the slot. Then go to step 9. If you do not want to edit the OTS instance, go to step 10.						
7	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the circuit pack in the alarmed slot is a spare circuit pack you want to store in that slot</td><td>step 8</td></tr> <tr> <td>otherwise</td><td>step 10</td></tr> </table>	If	Then go to	the circuit pack in the alarmed slot is a spare circuit pack you want to store in that slot	step 8	otherwise	step 10
If	Then go to						
the circuit pack in the alarmed slot is a spare circuit pack you want to store in that slot	step 8						
otherwise	step 10						
8	Disable auto equipping for the alarmed slot. Refer to the “Enabling/disabling slot-based automatic equipping” procedure in <i>Administration and Security</i> , 323-1851-301.						
9	Reseat the circuit pack in the same slot. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 11 .						
10	Remove the circuit pack in the alarmed slot and replace it with a circuit pack that is supported in the slot, a circuit pack of the correct equipment type, or a filler card. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-17

Autoprovisioning Mismatch - Pluggable

Alarm ID: 343

Probable cause

This alarm is raised when a pluggable module installed in an unprovisioned port of a circuit pack is not supported for that circuit pack.

When auto equipping is disabled, this alarm is not raised.

Impact

Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable
Critical, service-affecting (C, SA) alarm for an active pluggable

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a supported replacement pluggable module for the corresponding circuit pack. Refer to *Pluggables Datasheets and Reference*, 323-1851-180.

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” . The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and the pluggable module port using the following format: |
|---|--|

<circuit pack>-<shelf-id>-slot#-pluggable module port#



CAUTION

Risk of damage to modules

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

Procedure 4-17 (continued)

Autoprovisioning Mismatch - Pluggable

Step	Action
2	Replace the pluggable module you identified in step 1 with a supported module. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-18

AutoRoute Configuration Mismatch

Alarm ID: 1262

Probable cause

This alarm is raised when:

- the Autoroute OTS parameter values are not provisioned consistently in the channel access OTS instances at a site and/or within a domain.
- a mismatch is detected among the channel access OTS instances at site. The detection is limited by the connectivity among the shelves at the site. For example, the availability of TR records from the other shelves.

The alarm is raised against the OTS instances where the Autoroute parameter value is Enable.

During the flexible grid conversion, if the OTS Autoroute parameter is set to Enable before the OTS “Provisioned Control Mode” parameter is changed to “Flexible Grid Capable”, software automatically changes the OTS Autoroute parameter to Disable. Then the alarm is raised at the other OTSs at the TIDc node if they have the Autoroute parameter set to Enable.

The alarm is also raised at the far-end connecting OTS if the far-end connecting OTS has the Autoroute parameter set to Enable.

For the Flexible Grid conversion, the alarm clears when the OTS “Provisioned Control Mode” parameter is changed to “Flexible Grid Capable” at the OTSs that have the alarm. If the other OTSs at the TIDc node are not converted to Flexible Grid, the requirement is that all OTSs in the network must have the OTS Autoroute parameter set to Disable.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in “Observing product and personnel safety guidelines” section in *Installation- General Information*, 323-1851-201.0.

Procedure 4-18 (continued)

AutoRoute Configuration Mismatch

Step	Action
1	Use the Site Manager “Photonic Configuration” applications to set the OTS Autoroute value to the same value in all the OTSs within the node and within each of the optical domains that this node participates. Refer to the “Editing an OTS instance in the Photonic Configuration Management application” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. The OTS Autoroute parameter is only applicable to Channel Access OTS types (for example; ROADM, DIOADM, COADM, TOADM, and GOADM. It is not applicable to other OTS types such as AMP and DGE.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-19

Backplane ID Module 1/2 Failed

Alarm IDs: 679, 680

Use this procedure to clear the following alarms:

- Backplane ID Module 1 Failed
- Backplane ID Module 2 Failed

Probable cause

This alarm is raised when the shelf identifier unit 1 or unit 2 on the backplane has failed, is missing, has invalid data, or cannot be read.

The “Backplane ID Module 2 Failed” alarm is also raised when the shelf identifier unit 2 on the backplane detects a Mismatch between SID1 and SID2.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must have an electrostatic device (ESD) ground strap.

Step	Action						
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
2	<table> <tr> <td>If</td><td>Then go to</td></tr> <tr> <td>the shelf has 2 shelf processors with protection provisioned</td><td>step 6</td></tr> <tr> <td>otherwise</td><td>step 3</td></tr> </table>	If	Then go to	the shelf has 2 shelf processors with protection provisioned	step 6	otherwise	step 3
If	Then go to						
the shelf has 2 shelf processors with protection provisioned	step 6						
otherwise	step 3						
3	Cold Restart the shelf processor. Wait five minutes for the shelf processor to restart. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor” .						
4	<table> <tr> <td>If the alarm is</td><td>Then go to</td></tr> <tr> <td>cleared</td><td>This procedure is complete.</td></tr> <tr> <td>not cleared</td><td>step 5</td></tr> </table>	If the alarm is	Then go to	cleared	This procedure is complete.	not cleared	step 5
If the alarm is	Then go to						
cleared	This procedure is complete.						
not cleared	step 5						
5	Reseat the shelf processor. Wait five minutes for the shelf processor to restart. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 15 .						

Procedure 4-19 (continued)
Backplane ID Module 1/2 Failed

Step	Action						
6	Perform a manual shelf processor protection switch. Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.						
7	<table> <tr> <th>If the alarm is</th><th>Then go to</th></tr> <tr> <td>cleared</td><td>step 8</td></tr> <tr> <td>not cleared</td><td>step 9</td></tr> </table>	If the alarm is	Then go to	cleared	step 8	not cleared	step 9
If the alarm is	Then go to						
cleared	step 8						
not cleared	step 9						
8	Perform a protection switch to switch back to the originally active SP (if needed). This procedure is complete.						
9	Cold Restart the Standby SP. Wait five minutes for the shelf processor to restart.						
10	If the alarm has not cleared, perform another manual shelf processor protection switch. Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.						
11	Cold Restart the Standby SP and wait 5 minutes.						
12	If the alarm has not cleared, re-seat or cold restart the standby shelf processor and wait five minutes. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
13	<table> <tr> <th>If the alarm is</th><th>Then</th></tr> <tr> <td>cleared</td><td>This procedure is complete.</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the alarm is	Then	cleared	This procedure is complete.	not cleared	go to step 14
If the alarm is	Then						
cleared	This procedure is complete.						
not cleared	go to step 14						
14	Reseat the active shelf processor. This will cause an automatic shelf processor protection switch. Wait five minutes for the shelf processor to restart. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor” in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-20

Bandwidth Oversubscribed

Alarm IDs: 1767

Probable cause

This alarm is raised when a Holding Priority Bandwidth on an OSRP link is reduced below zero relative to the Maximum Available Bandwidth on the link. This can occur during SNC Preemption or a Reserved Home Path Bandwidth on the link or if an OSRP line is removed/deleted from the OSRP link where the available link bandwidth would become negative.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Change the Holding Priority Bandwidth on a link to zero or greater than zero by adding more lines to the link to increase the available bandwidth or reduce the bandwidth usage by re-grooming/rerouting some SNCs on the link to a different path or different Home Path. Refer to the OSRP provisioning procedures in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-21

Battery Low

Alarm IDs: 1820

Probable cause

This alarm is raised against the 4x10G OTR w/Encryption (NTK530QE) circuit pack when the on-board battery has to be replaced.

Note: The NTK530QE variant of 4x10G OTR circuit pack has an on-board SR44 type battery that is used to protect sensitive material. Ciena recommends using the Renata 357 battery or one with similar specifications (1.55 V nominal voltage, 190 mAh capacity and an operating temperature of 0 °C to 60 °C). For security-related information on the NTK530QE, refer to *Encryption and FIPS Security Policy Overview and Procedures*, 323-1851-340.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.
- have a replacement battery

Step	Action
------	--------



CAUTION

Risk of traffic loss

Before replacing the battery, ensure the 4x10G OTR w/Encryption (NTK530QE) circuit pack is powered up in a chassis for at least one hour. Otherwise, the circuit pack will become non-functional upon removal of the battery.

- | | |
|---|---|
| 1 | Remove circuit pack from chassis. Refer to the “Replacing an optical interface circuit pack” procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. |
|---|---|

Procedure 4-21 (continued)

Battery Low

Step	Action
2	Replace the old battery with a new battery. Use a Renata 357 (Renata SR1154W) or equivalent battery for the replacement. The new battery must be put in within an hour. Refer to the “Changing the battery” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
3	 CAUTION Risk of traffic loss The 4x10G OTR (NTK530QE variant) circuit pack requires a valid Time of Day (TOD) in order to perform the certificate validation before bringing up traffic. The TOD information is received from the SP on every circuit pack restart. If the 4x10G OTR (NTK530QE variant) circuit pack is reseated while an SP is not present, then traffic will not recover automatically on this circuit pack until the SP is available to provide the system TOD. Ensure the SP is installed in the chassis before you replace the 4x10G OTR circuit pack.
	Insert the circuit pack back into the chassis.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-22

BW Lockout Configured

Alarm IDs: 1771

Probable cause

This alarm is raised when a bandwidth lockout is applied against both ends of an OSRP line intended for maintenance activities. The alarm will be cleared if either end of the line becomes bandwidth unlocked.

This alarm can be raised only for L1 OTN OSRP lines. The alarm does not apply to L0 Photonic OSRP lines and it is not raised for these lines when the condition is present.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with a level 3 or higher UPC.

Step	Action
1	Select the required network element in the navigation tree.
2	From the Configuration menu, select Control Plane: OSRP Provisioning .
3	Select the Lines tab.
4	Select the required shelf, containing link identifier, common line identifier, neighboring node, neighboring link identifier, and neighboring common line identifier from the drop-down lists.
5	Click Retrieve .
6	Select the OSRP line you want to edit from the OSRP line table.
7	Click Edit to open the Edit OSRP Line dialog box.
8	Disable (uncheck) OSRP bandwidth lockout by unchecking the Bandwidth lock out check box.
9	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-23 Cable Trace Compromised

Alarm ID: 1850

Probable cause

This alarm is raised when there is a hardware malfunction of the Smart Connect Module (SCM) in the Fiber Interconnect Module (FIM).

The alarm can also be raised against the WSS Flex C-Band w/OPM 20x1 or the CCMD8x16 circuit pack.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action		
1	Identify the circuit pack or module raising the alarm. Refer to Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm". CAUTION Risk of traffic loss Replacement of the FIM, WSS Flex C-Band w/OPM 20x1, or CCMD8x16 is service affecting. If the alarm is unexpected, contact your next level of support or your Ciena support group before replacing the circuit pack or module.		
2	<table border="0" style="width: 100%;"> <tr> <td style="width: 50%; vertical-align: top;"> If the alarm is raised against the WSS Flex C-Band w/OPM 20x1 or CCMD8x16 FIM </td> <td style="width: 50%; vertical-align: top;"> Then go to step 3 step 4 </td> </tr> </table>	If the alarm is raised against the WSS Flex C-Band w/OPM 20x1 or CCMD8x16 FIM	Then go to step 3 step 4
If the alarm is raised against the WSS Flex C-Band w/OPM 20x1 or CCMD8x16 FIM	Then go to step 3 step 4		
3	Replace the WSS Flex C-Band w/OPM 20x1 circuit pack or CCMD8x16 module. Refer to the "Replacing the WSS w/OPM or SMD circuit pack" or "Replacing a CCMD12 C-Band/L-Band, CCMD8x16 C-Band 1xCXM, C-Band or SCMD4 circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 6.		
4	Verify the Ethernet cable connectivity between the FIM and the SCM, and correct if needed.		
5	If the alarm did not clear, replace the FIM. Refer to "Replacing a Fiber Interconnect Modules (FIM)" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.		



Procedure 4-23 (continued)
Cable Trace Compromised

Step	Action
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-24

Calibration Required

Alarm ID: 1716

Probable cause

This alarm is raised when calibration of the RAMAN amplifier is required.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Perform a manual calibration on the RAMAN amplifier. The alarm clears when the calibration is successful. Refer to the “Performing a manual power audit” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
Note: This alarm also clears if the RAMAN Pump Mode is set to MANUAL or MAXGAIN. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-25

Call Records Mismatch

Alarm ID: 2193**Probable cause**

This alarm is raised when:

- call records mismatch is identified on remote node of the OSRPLINE
- loopback detected on AMP between 2 peer ROADM sites
- fiber pairs swapped between rails on a dual rail system

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	If the alarm is raised due to Loopback detected on an AMP between 2 peer ROADM sites, the alarm will be cleared once loopback clears.
2	Identify the node raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
3	If the alarm was triggered after a fiber activity was completed, verify the AMP/ROADM sites on the OSRP line to ensure the fiber pairs are not swapped between the rails on a dual rail based configuration.
4	On the alarmed node, retrieve the OSRP Lines using “Configuration -> OSRP Provisioning -> Lines” and select the OSRP Line on which the alarm is raised.
5	Click Edit to open the “Edit OSRP Line” popup menu. Change “PSNC Mismatch Action” from “Keep” to “Release”. Refer to “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330 for more information.
6	Once the alarm clears, change the “PSNC Mismatch Action” from “Release” back to “Keep”.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-26

Certificate About to Expire

Alarm ID: 1817

Probable cause

This alarm is raised against the 4x10G OTR w/Encryption (NTK530QE) and Flex3 WL3e OCLD w/ Encryption (NTK539QS/ NTK539QV variants) circuit packs to indicate a certificate in the SecureID will expire in 90 days.

For the 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD w/ Encryption circuit packs, X.509 certificates are used for authentication in datapath encryption and web access using the encryption management interface (MyCryptoTool). A Certificate Authority (CA) issues a certificate with a given expiry date.

Impact

Warning

Step	Action
1	Connect to MyCryptoTool and check the validity period of X.509 certificates on the circuit pack to determine which certificate is about to expire. Refer to the “ <i>Viewing the Data Encryption entity certificate or the Web Access entity certificate</i> ” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
2	Perform the certificate enrollment process to renew the certificate that is about to expire. a. If the Data Encryption certificate is about to expire, follow the “Performing Data Encryption Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. b. If the Web Access certificate is about to expire, follow the “Performing Web Access Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-27

Certificate Expired

Alarm ID: 1818

Probable cause

This alarm is raised against the 4x10G OTR w/Encryption (NTK530QE) and Flex3 WL3e OCLD w/ Encryption (NTK539QS/ NTK539QV variants) circuit packs to indicate a certificate in the SecureID has expired.

For the 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD w/ Encryption circuit packs, X.509 certificates are used for authentication in datapath encryption and web access using the encryption management interface (MyCryptoTool). A Certificate Authority (CA) issues a certificate with given expiry date. If the time of day on the circuit pack passes the certificate expiry date, this alarm is raised.

If the “Re-Authentication Failure Mode” is set to “Drop” when the datapath encryption certificate is expired, traffic is impacted.

If the Web Access certificate is expired, access to the MyCryptoTool is maintained.

Impact

Critical, service-affecting (C, SA) alarm

Step	Action
1	Connect to MyCryptoTool and check the validity period of X.509 certificates on the circuit pack to determine which certificate is expired. Refer to the “ <i>Viewing the Data Encryption entity certificate or the Web Access entity certificate</i> ” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
2	Perform certificate enrollment process to renew the certificate that is expired. - If the Data Encryption certificate is expired, follow the “Performing Data Encryption Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. - If the Web Access certificate is expired, follow the “Performing Web Access Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-28

Channel Contention

Alarm ID: 1860, 1870

Probable cause

This alarm is raised against the line facility of a circuit pack when the provisioned wavelength/frequency is already in use and is detected by the circuit pack.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in “Observing product and personnel safety guidelines” section in *Installation*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain a replacement circuit pack
- use an account with at least a level 3 UPC

Step	Action						
1	Identify the facility in “Channel Contention”. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	<table><tr><td>If</td><td>Then go to</td></tr><tr><td>the reported Echo Trace value is not “unknown” or the wavelength is already in use on the network</td><td>step 5</td></tr><tr><td>otherwise</td><td>step 3</td></tr></table>	If	Then go to	the reported Echo Trace value is not “unknown” or the wavelength is already in use on the network	step 5	otherwise	step 3
If	Then go to						
the reported Echo Trace value is not “unknown” or the wavelength is already in use on the network	step 5						
otherwise	step 3						

Procedure 4-28 (continued)

Channel Contention

Step	Action						
3	Check and ensure the following provisioning information is correct on the card: • Modulation Format matches the other end of the link. • ENM is matching the other end of the link. • Differential coding matches the other end of the link. • If the upstream Tx is turned on, the channel is equalized by DOC. If all the provisioning information above is correct and the alarm is still active, then disable Channel Contention Detection.						
4	<table> <tr> <th>If the original alarm is</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm is	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm is	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
5	Reprovision the facility wavelength to an available wavelength/frequency. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
6	<table> <tr> <th>If the original alarm is</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm is	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm is	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	If the alarm is not clear and the facility wavelength is not already in use on the network, disable Channel Contention Detection. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
8	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-29

Channel Controller: Failure Detected

Alarm ID: 709

Probable cause

This alarm is raised against a WSS w/OPM circuit pack or RLA 5X1/ MDRLA 2x6 when the WSS controller is unable to function properly. Conditions that can cause this alarm include:

- the WSS w/OPM circuit pack or RLA 5X1/MDRLA 2x6 has failed
- there is a loss of signal on one of the channels carried by the WSS w/OPM circuit packs or RLA 5X1/MDRLA 2x6
- the OPM is not provisioned against LIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band), MDRLA 2x6, or RLA 5X1 monitor ports (there is an adjacency provisioning error)
- the WSS w/OPM or RLA 5X1/MDRLA 2x6 circuit pack provisioning data is invalid
- the fibers to the WSS w/OPM circuit pack or RLA 5X1/MDRLA 2x6 monitor ports are crossed
- the Controller is failed or is not in RUNNING state. For example, if CCT data or adjacency data are not present, the WSS controller will raise this alarm.

Note: Use optical terminators on unused input faceplate connectors of installed WSS w/OPM circuit packs. If dust caps are used instead of optical terminators on “Switch In” ports, PMs can be reported against the ports and the ports may appear in-service.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) containing shelf details
- have a replacement circuit pack if required
- have a fiber cleaning kit
- have an electrostatic device (ESD) ground strap

Procedure 4-29 (continued)

Channel Controller: Failure Detected

Step	Action						
1	Check for and clear any active “Circuit Pack Failed” alarm if raised against the WSS w/OPM circuit pack or RLA 5x1/MDRLA 2x6.						
2	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Check for and clear any of the following adjacency alarms: - Adjacency Mismatch - Adjacency Far End Not Discovered						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Check for and clear any of the following alarms on all network elements before clearing the Channel Controller: Failure Detected alarm: - Automatic Power Reduction Active - Automatic Shutoff - Input Loss of Signal - Loss of Signal - Optical Line Fail - Output Loss of Signal - Shutoff Threshold Crossed - Channel Controller: Unexpected Loss Detected						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Verify the LIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band), MDRLA 2x6 or RLA 5x1 adjacency of the monitor port to ensure that the Expected far-end address field has the correct WSS OPM module port listed, and that the corresponding Adjacency type field has the correct adjacency type listed (OPM). Refer to the “Retrieving protection parameters” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.						

Procedure 4-29 (continued)

Channel Controller: Failure Detected

Step	Action						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Verify the derived OPM to LIM, MDRLA 2x6 or RLA 5x1 adjacency. Check the OPM adjacency of the OPM monitor port to ensure that the Expected far-end address field has the correct LIM, MDRLA 2x6 or RLA 5x1 module port listed, and that the corresponding Adjacency type field has the correct adjacency type listed (LIM). Refer to the “Retrieving protection parameters” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 11						
11	Verify that the CHC/NMCC/SSC facilities associated with the WSS w/OPM circuit pack have the correct Switch Selector provisioned. Correct any discrepancies. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	Check the Secondary State of the CHC/NMCC/SSC facilities associated with the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If only certain (not all) CHC/NMCC/SSC facilities indicate a FAF condition (facility failure), this can indicate a problem with the channel(s) added from a CMD44 module.						
14	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 15</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 15
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 15						
15	Check the shelf-to-shelf association. The Associated OTS field should list the adjacent OTS (TID-shelf-instance), and the Actual Associated OTS should be the same as the Associated OTS . Refer to the “Retrieving OTS Management, OTS Equipment, and Facility Details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-29 (continued)

Channel Controller: Failure Detected

Step	Action						
16	<table> <tr> <th>If the</th><th>Then</th></tr> <tr> <td>Associated OTS shows the incorrect adjacent OTS</td><td>correct the Associated OTS field</td></tr> <tr> <td>Actual Associated OTS is not the same as the Associated OTS</td><td>verify the inter-shelf communications</td></tr> </table>	If the	Then	Associated OTS shows the incorrect adjacent OTS	correct the Associated OTS field	Actual Associated OTS is not the same as the Associated OTS	verify the inter-shelf communications
If the	Then						
Associated OTS shows the incorrect adjacent OTS	correct the Associated OTS field						
Actual Associated OTS is not the same as the Associated OTS	verify the inter-shelf communications						
17	If necessary, repeat step 15 and step 16 on each adjacent shelf.						
18	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 19</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 19
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 19						
19	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Verify that all LIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band), MDRLA 2x6 or RLA 5x1 to OPM fibers are connected to the correct ports and that the fiber is clean. Refer to cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.						
20	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
21	Perform a warm restart on the WSS w/OPM circuit pack. Refer to Procedure 2-11, "Restarting a circuit pack or shelf processor" .						
22	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 23</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 23
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 23						

Procedure 4-29 (continued)

Channel Controller: Failure Detected

Step	Action
------	--------

23



CAUTION

Risk of traffic loss

A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic.

Perform a cold restart on the WSS w/OPM circuit pack. Refer to [Procedure 2-11, "Restarting a circuit pack or shelf processor"](#).

24

If the original alarm has	Then
cleared	the procedure is complete
not cleared	go to step 25

25

If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-30

Channel Controller: Unexpected Loss Detected

Alarm ID: 877

Probable cause

This alarm is raised when the WSS, MDRLA 2x6 or RLA 5x1 controller is unable to function properly. Conditions that can cause this alarm include:

- the WSS w/OPM, MDRLA 2x6 or RLA 5x1 circuit pack has failed
- the difference between the expected loss and the measured loss is greater than:
 - 3dB for pass-through channels
 - 3dB for adding channels with MUX structure including at least one of: CCMD12, CCMD8x16, SMD, or DIA WSS
 - 6dB for adding channels with MUX structure that does not include any of above cards (for example: CMDx directional)
- there is a loss of signal on one of the channels carried by the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1
- wrong wavelength/frequency provisioning of channels
- there is a misalignment between the ADJ-TX Facility Parameters (ADJTXTYPE, TXSIGBW3DB, TXSIGBW10DB, ADJTXCURPOW) and the actual Transponder connected to the ADJ-TX Port
- the OPM is not provisioned against LIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band), MDRLA 2x6 or RLA 5x1 monitor ports (there is an adjacency provisioning error)
- the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1 provisioned data is invalid
- the fibers to the OPM, MDRLA 2x6 or RLA 5x1 monitor ports are crossed or connected to a wrong OSC card
- the transmitter power at the CMD44 ingress port is not within +/- 3 dBm of the provisioned Max/Typical Launch Power for that Tx adjacency. The alarm is raised on managed channels that have this unexpected power level, or on inactive channels that are being manually pre-checked with this unexpected power level. This condition only applies to the CMD44 module.
- there is a fiber break between a CMD44 common out (or a BMD-2 common out, or a WSS switch output) and the corresponding OPMWSS switch input port.

Procedure 4-30 (continued)

Channel Controller: Unexpected Loss Detected

- there is a problem with the SMD switch out to CCMD12 C-Band common in fiber, the DIA LIM Line B Monitor to DIA WSS monitor in fiber, or the CCMD12 C-Band monitor to SMD OPM fiber.

Note: For a CDC configuration, the alarm is raised when there is a failure on WSS Flex C-Band w/OPM 20x1 or WSS Flex L-Band w/OPM 8x1 circuit pack. Use optical terminators on unused input faceplate connectors of installed WSS w/OPM circuit packs, MDRLA 2x6 or RLA 5x1. If dust caps are used instead of optical terminators on “Switch In” ports, PMs can be reported against the ports and the ports may appear in-service.

The alarm clears when the power level is 0.5dB higher than the config specific threshold.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 4 UPC
- have the engineering documentation package (EDP) containing shelf details
- have a replacement circuit pack if required
- have a fiber cleaning kit
- have an electrostatic device (ESD) ground strap

Step	Action
------	--------

- | | |
|---|--|
| 1 | At the WSS site, check the CHC/NMCC/SSC facility status against the affected WSS equipment. Note that in Site Manager you can click the switch selector column to sort the channels by port. |
| 2 | Check for channels with a secondary state of SGEO. Refer to the “Primary and secondary states” section in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 3 | The channels with secondary state as SEGO perform the following analysis: <ul style="list-style-type: none">— If CHCMODE = POWER on CHC Facility, compare the CTRLSSCATTEN value with LOSS value on the NMCC Facility— If CHCMODE = OPENLOOPLOSS on CHC Facility, compare the TARGLOSS value with LOSS value on the NMCC Facility— If CHCMODE = LOSS on CHC Facility, compare the ATTEN value with LOSS value on the NMCC Facility |

Procedure 4-30 (continued)

Channel Controller: Unexpected Loss Detected

Step	Action						
4	For the channels that have secondary state of SGEO, check on the corresponding ADJ-TX facility (ADJTXTYPE, TXSIGBW3DB, TXSIGBW10DB, ADJTXCURPOW) to make sure that the parameters match the actual Transponder connected to the ADJ-TX Port.						
5	For the SMD circuit pack, if all pixels on a given switch are SGEO and it is in the demux direction, check the fiber between the SMD switch out and the CCMD12 C-Band common in. Check the DIA LIM LINE B OPM fiber. Check the CCMD12 C-Band monitor to SMD OPM fiber.						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Verify the facility and equipment secondary state of the connected equipment. Refer to the "Primary and secondary states" section in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
8	If secondary states such as Supporting entity outage, FAF, Auto in-service, or SGEO exist, verify the failure. See the "Facility secondary states" table in of <i>Configuration - Provisioning and Operating</i> , 323-1851-310 for description of each failure.						
9	Check for and clear any active circuit pack Failed alarm raised against the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 11						

Procedure 4-30 (continued)

Channel Controller: Unexpected Loss Detected

Step	Action						
11	Check for and clear any of the following adjacency alarms: • Adjacency Mismatch • Adjacency Far End Not Discovered						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	Check for and clear any of the following alarms on all network elements before clearing the Channel Controller: Failure Detected alarm: • Automatic Power Reduction Active • Automatic Shutoff • Input Loss of Signal • Loss of Signal • Optical Line Fail • Output Loss of Signal • Shutoff Threshold Crossed						
14	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 15</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 15
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 15						
15	Check the wavelength provisioning of the line cards to make sure that it matches the wavelength of the CMD port the card is connected to. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. For OTM2, OTM3 or OTM4 parameters, refer to the “OTM2 facility parameters” table, “OTM3 facility parameters” table, or “OTM4 facility parameters” table in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						

Procedure 4-30 (continued)

Channel Controller: Unexpected Loss Detected

Step	Action						
16	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 17</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 17						
17	Verify the WSSLIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band, XLA), MDRLA 2x6 or RLA 5x1 adjacency of the LIM monitor port to ensure that the Expected far-end address field has the correct WSS OPM module port listed, and that the corresponding Adjacency type field has the correct adjacency type listed (OPM). Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Refer to the “ADJ/ADJ-LINE/ADJ-TX/ADJ-RX/ADJ-FIBER facility parameters” table in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
18	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 19</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 19
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 19						
19	Verify the derived OPM to WSSLIM, MDRLA 2x6 or RLA 5x1 adjacency. Check the OPM adjacency of the WSS OPM, MDRLA 2x6 or RLA 5x1 monitor port to ensure that the Expected far-end address field has the correct WSSLIM, MDRLA 2x6 or RLA 5x1 module port listed, and that the corresponding Adjacency type field has the correct adjacency type listed (LIM). Refer to the “Retrieving protection parameters” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315.						
20	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
21	Check the shelf-to-shelf association. The Associated OTS field should list the adjacent OTS (TID-shelf-instance), and the Actual Associated OTS should be the same as the Associated OTS. Refer to the “Retrieving OTS Management, OTS Equipment, and Facility Details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
22	<table> <tr> <th>If the</th><th>Then</th></tr> <tr> <td>Associated OTS shows the incorrect adjacent OTS</td><td>correct the Associated OTS field</td></tr> <tr> <td>Actual Associated OTS is not the same as the Associated OTS</td><td>verify the inter-shelf communications</td></tr> </table>	If the	Then	Associated OTS shows the incorrect adjacent OTS	correct the Associated OTS field	Actual Associated OTS is not the same as the Associated OTS	verify the inter-shelf communications
If the	Then						
Associated OTS shows the incorrect adjacent OTS	correct the Associated OTS field						
Actual Associated OTS is not the same as the Associated OTS	verify the inter-shelf communications						
23	If necessary, repeat step 21 and step 22 on each adjacent shelf.						

Procedure 4-30 (continued)

Channel Controller: Unexpected Loss Detected

Step	Action						
24	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 25</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 25
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 25						
25	Verify that all WSSLIM (SLA, MLA C-Band, MLA2 C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, LIM C-Band/LIM L-Band, XLA), MDRLA 2x6 or RLA 5x1 to WSS OPM fibers are connected to the correct ports. Refer to the EDP. Verify that the fiber is clean. Refer to “Cleaning connectors” procedure in <i>Installation - General Information</i> , 323-1851-201.0.						
26	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 27</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 27
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 27						
27	Using the Domain Optical Controller (DOC) application, check for channels that have a channel condition of “Pre-Check Fail” or “Fault Detected”. From the Site Manager window, click on Configuration - Domain Optical Controller (DOC) window and verify channel condition. If these channel conditions are the cause of this alarm, the DOC Action: Fault Detected alarm will be raised.						
28	Check the DOC logs to determine which channel has the unexpected ingress power. (From the Site Manager window, click on Configuration - Domain Optical Controller (DOC) window and then select DOC Logs button in Site Manger.) For this channel, verify (and if necessary, adjust) the power level at the source.						
29	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 30</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 30
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 30						
30	Perform a warm restart on the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1. Refer to Procedure 2-11 , “Restarting a circuit pack or shelf processor”.						
31	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 32</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 32
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 32						

Procedure 4-30 (continued)
Channel Controller: Unexpected Loss Detected

Step	Action						
32	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. Perform a cold restart on the WSS w/OPM circuit pack, MDRLA 2x6 or RLA 5x1. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor”.						
33	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 34</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 34
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 34						
34	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-31 Channel Degrade

Alarm ID: 1283, 1952

Probable cause

This alarm is raised when the WSS (or SMD), MDRLA 2x6 or RLA 5x1 facility 'Derived Input Power' is less than the value of 'Reference Input Power Profile' minus the 'Minor Degrade Threshold'.

Note: The Channel Degrade alarm has a 0.5 dB hysteresis that is considered before clearing the alarm. The NMCC facility Derived Input Power and its Reference Input Power Profile must be within the threshold minus 0.5 dB to clear the alarm.

The Minor Degrade Threshold defaults to 3 dB.

Although the minor thresholds can be user-provisioned between 0 to 30 dB using the Site Manager Node Information application and the Systems settings, it is strongly recommended that they are left at their default values unless expressly informed otherwise by Ciena.

The "Reference Input Power Profile" parameter displays a referenced or baselined value for the Derived Input Power parameter. This power profile helps determine how the channel power changes over time and whether to compensate for it. If a channel is found to be below its previously stored value by more than a threshold, a minor degrade alarm is raised and the CFS is updated for that channel, and the pixel loss is not adjusted in order to prevent an over shoot of output power when the input power returns to normal levels.

The power profile is system reset after a capacity change (a channel add or delete) as long as the facility is not in a minor degrade condition.

Procedure 4-31 (continued)
Channel Degrade

In addition, the power profiles can be user reset individually using the “Reset Power Profile” button in the Equipment and Facility Provisioning application or reset in bulk using the Reset TCA Baselines button in the DOC application. However since these actions will affect the control behavior, this should only be done after extensive troubleshooting.

**CAUTION****Risk of traffic loss**

Resetting the power profile of a low power (i.e. degraded) channel instructs the system to include it in the relative channel equalization, hence incorrectly resetting the profile can result in reducing the power of other channels, causing possible traffic impact.

ATTENTION

Clicking the Reset TCA Baselines Button from the DOC facility not only resets the 'TCA Baselines' of the applicable facilities of the whole domain but also resets the 'Reference Input Power Profile' of each and every NMCC facility, even if in a degrade state. Hence “Reset TCA Baselines” should not be used to simply attempt to clear Channel Degrade alarms.

Note the following system behavior due to a change in “Derived Input Power”:

- up to 3 dB low in derived input power results in no alarms or CFS change. Channels are allowed to be re-optimized if necessary.
- 3 to 6 dB low in derived input power results in the Channel Degrade (minor, NSA) alarm, the CFS set to “Degrade Minor”, the channel dropped from the control list, hold channel at current pixel setting.
- 6 dB low in derived input power results in the Channel Degrade (Minor, NSA) alarm, the CFS set to “Degrade Minor”, the channel dropped from control list, hold channel at current pixel setting.

Note that one Channel Degrade alarm is raised per affected channel, hence multiple alarms may be raised at a node if multiple channels are low in power.

Procedure 4-31 (continued)

Channel Degrade

The Channel Degrade alarm masks downstream Channel Degrade alarms if NLC (Network Level Correlation) is enabled, hence the alarm indicates the first ROADM to experience the power drop.

The NMCC “Channel Degrade” alarm is masked by WSS Flex or SMD circuit pack fault alarms.

Any drop in power that occurs prior to the WSS, MDRLA 2x6, RLA 5x1, or SMD input could lead to the degrade condition.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- have the optical fiber/cable connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	If NLC is not enabled, follow the channel path upstream to find the first occurrence of a “Channel Degrade” alarm.
3	If there are any other Photonic alarms active, troubleshoot those alarms in the system first before trying to troubleshoot this alarm, since they could be the cause of the power drop. For example: “High Fiber Loss” or “High Received Span Loss” alarms can cause Channel Degrades.
4	Identify and fix any potential drop in power by systematically working upstream along the channel path towards the head-end service circuit pack. If multiple Channel Degrade alarms are raised, this may identify a common source (for example, a particular WSS-WSS connection). Comparing current PM values to their reference baselines may help identify the source.

Procedure 4-31 (continued)
Channel Degrade

Step	Action				
5	<table><tr><td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has cleared	Then the procedure is complete	not cleared	go to step 6
If the original alarm has cleared	Then the procedure is complete				
not cleared	go to step 6				
6	If the source of the power drop can be explained and will not be repaired (for example, permanent fiber degradation) the input power profile of the affected WSS NMCC facility can be reset by clicking on the “Reset Power Profile” button in the Equipment and Facility Provisioning application. “Reset TCA Baseline”, under Photonic Channel Management->Photonic Controllers menu. Note 1: Reset Power profile for a NMCC will not be allowed if corresponding channel under Photonic Channel Management has Channel Source parameter as ASE. The channel source parameter will be ASE when the parent CHC facility for the NMCC has switch selector pointing to an ASE source. Note 2: Incorrect use of the profile reset function may affect other traffic - see previous section. If in doubt, contact your next level of support or your Ciena support group. ATTENTION Clicking the Reset TCA Baselines Button from the DOC facility not only resets the 'Reference Input Power Profile' of each and every NMCC facility of that domain but also resets the 'TCA Baselines' of the applicable facilities of the whole domain.				
7	If the alarm does not clear, contact your next level of support or your Ciena support group.				

—end—

Procedure 4-32

Channel Opacity Error

Alarm ID: 1433

Probable cause

This alarm is raised when a WSS w/OPM, MDRLA 2x6 or RLA 5x1 pixel of a ROADM OTS is incorrectly set to Opaque (the CHC facility Opaque parameter is set to Yes), while a DOC- managed channel is meant to be using it.

The alarm is raised against the CHC facility at the shelf where the problem is detected.

For CDC configurations, this alarm is raised for WSS Flex C-Band w/OPM 20x1 MUX, and Demux CHC facilities when the pixel is set to OPAQUE. The WSS Flex C-Band w/OPM 20x1 can be configured as part of ROADM OTS or as a CDC upgrade. For a CHC facility on a CDC upgrade WSS Flex C-Band w/OPM 20x1 circuit pack, this alarm is raised when there is a CRS provisioned on the facility but the CHC opacity is set to Opaque, regardless if the channel is DOC-managed.

The CHC “Channel Opacity Error” alarm is masked by WSS Flex or SMD circuit pack fault alarms.

Impact

Major, Service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
1	Ensure that the CHC facility against which the alarm is raised, is used by a DOC-managed channel.
2	Change the WSS w/OPM, MDRLA 2x6 or RLA 5x1 pixel Opaque setting to No. Refer to the “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-33

Circuit Pack Configuration Save Failed

Alarm ID: 1766

Probable cause

This alarm is raised when the shelf processor fails to backup L2 configuration from an eMOTR circuit pack.

Impact

Warning

Step	Action
1	No action is necessary. The NE automatically retries to save the configuration data. If the action is successful, the alarm clears.
2	If the alarm does not clear after five minutes, contact your next level of support or your Ciena support group.

—end—

Procedure 4-34

Circuit Pack Failed

Alarm IDs: 56

Probable cause

This alarm is raised in the following situations:

- the trouble detection circuits of a circuit pack detect a failure
- the shelf processor detects a major failure on another circuit pack
- a power failure on the standby SP
- a newly active SP raises the alarm against the mate SP after a SP protection switch is caused by a power outage on the previously active SP (for NTK555CAE5, NTK555EAE5, NTK555FAE5, and NTK555JA)

This alarm can also be raised against a circuit pack that is more than one slot wide and detects a failure on an intra-pack signaling interface dependent on its backplane connectors. In some cases the backplane may be at fault.

Both the alarm and LED alarm indicators can report the failure at the same time. If not, verify that a SP problem does not exist. The status LED comes on (red indicates a failure) after a circuit pack is inserted until it is completely booted. The circuit pack is not failed in this case. This LED must clear one minute after insertion.

For the eMOTR circuit pack, when one card is in a mated pair and the “Circuit Pack Failed” alarm is raised, the user can select to perform a cold restart in the Equipment Group Attribute in Site Manager.

For WSS w/OPM equipment, this alarm is raised when two or more wavelengths have a drift frequency of more than 25 GHz.

When a Circuit Pack Failed alarm is raised, some hardware may not be operational. This can cause inaccuracies in the PM counts for facilities on this circuit pack.

For the 4x10G OTR w/Encryption (NTK530QE) circuit pack, any self-test failure or signature validation failure causes the Krypto Module to lock-down and the “Circuit Pack Failed” alarm to raise. The cause of the alarm is reported and stored in persistent logs for troubleshooting purposes.

Procedure 4-34 (continued)

Circuit Pack Failed**Impact**

Critical, service-affecting (C, SA) alarm for:

- an active circuit pack in a protected configuration
- a circuit pack in an unprotected configuration with cross-connects

Minor, non-service-affecting (m, NSA) alarm for:

- an inactive or protected circuit pack in a protected configuration
- a circuit pack without cross-connects
- a working circuit pack with a protection circuit pack available or for protection circuit pack with a working circuit pack available

Broadband services

Critical, service-affecting (C, SA) alarm for:

- a working circuit pack in a 1+1 port TPT, or 1+1 TPT configuration with protection circuit pack faulty/unavailable

Minor, non-service-affecting (m, NSA) alarm for:

- an inactive circuit pack (with no cross-connects)
- an inactive or protected circuit pack in a 1+1 port TPT, or 1+1 TPT configuration

Photonic services

Critical, service-affecting (C, SA) alarm

Major, service-affecting (C, SA) alarm for XLA, SRA, SAM, and ESAM circuit packs

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a replacement circuit pack for the failed circuit pack
- have an electrostatic device (ESD) ground strap

Procedure 4-34 (continued)

Circuit Pack Failed

Step	Action
1	Perform a DGN-EQPT command. If the alarm does not clear, continue to the next step.
2	Determine the time since the “Circuit Pack Failed” alarm was raised. Design expert data is automatically saved after a “Circuit Pack Failed” condition. This will take 5 minutes for the TRIB circuit packs and 10 minutes for the SP. It is recommended that circuit packs are not replaced during this time after the “Circuit Pack Failed” alarm has raised. The design expert data will not be captured if you do not wait.
3	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. If the failure is against the shelf processor, it may not be possible to log into the network element to determine the active alarms.
4	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
5	Remove the circuit pack from the shelf. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
6	For the circuit packs that are more than one slot wide, inspect the backplane connectors to the alarmed circuit pack and the backplane pins. If the backplane pins are damaged or bent, contact your next level of support or your Ciena support group for backplane repair.
7	Replace the faulty circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
8	Retrieve all alarms and ensure the system is restored to its original state. If the original alarm has Then go to cleared step 9 not cleared step 12
9	Select Shelf Level View from the Configuration menu. (This action takes you to the Physical Shelf view of the Visualization tool) Alternatively, select Visualization from the Tools menu, and then select Physical Shelf from the drop-down menu in the Control area.
10	Ensure that the new circuit pack is displayed in the Physical Shelf view of the Visualization tool.
11	For a SLIC10 or SLIC10 Flex C-Band circuit packs this alarm is a latched alarm, which means it is not cleared on the circuit pack even if the fault is cleared. Perform a restart on the circuit pack. See Procedure 2-11, “Restarting a circuit pack or shelf processor” .

Procedure 4-34 (continued)

Circuit Pack Failed

Step	Action
12	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-35

Circuit Pack Failed - Pluggable

Alarm ID: 340

Probable cause

This alarm is raised when a provisioned SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 fails.

Impact

Critical, service-affecting (C, SA) alarm for an active pluggable
Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an optical power meter with the same optical connectors as the network element
- obtain a supported replacement SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 for the corresponding circuit pack. Refer to *Pluggables Datasheets and Reference*, 323-1851-180.

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and pluggable module port using the following format:

<circuit pack>-<shelf-id>-slot#-pluggable module port# |
| 2 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 3 | If the alarm does not clear, replace the pluggable module you identified in step 1 with a supported pluggable module. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. |
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-36

Circuit Pack Failed-Sync and Circuit Pack Failed-Traffic

Alarm ID: 88, 90

Use this procedure to clear an XC double fault scenario. If the XC double fault conditions do not exist, follow the circuit pack fail procedure. Refer to [“Circuit Pack Failed” on page 4-104](#).

Probable cause

This alarm indicates that the XC circuit packs failed in their bandwidth management or synchronization.

If a XC double fault scenario exists, follow this procedure to replace the XC circuit pack. Both circuit packs are partially functioning. If you remove either circuit pack, you will cause a shelf traffic loss.

Use this procedure to clear an XC double fault scenario. If the XC double fault conditions do not exist, follow the circuit pack fail procedure. Refer to [“Circuit Pack Failed” on page 4-104](#).

Use this procedure if on one XC circuit pack, the Circuit Pack Failed - Sync alarm is indicated by the red LED being ON; and on the other XC circuit pack, the Circuit Pack Failed - Traffic alarm is indicated by the red LED being ON.

Impact



CAUTION

Risk of traffic loss

The procedure to clear the XC double fault scenario will result in a 90 second traffic loss. This traffic loss can be up to five minutes if the replacement circuit pack requires an auto-upgrade. It is recommended that this procedure be performed during a maintenance window (when traffic is lightest), or that all traffic be routed away from the affected network element before performing this procedure.

Critical, service-affecting (C, SA) alarm if active with cross-connects
Minor, non-service-affecting (m, NSA) alarm if unprotected or with no provisioned cross-connects

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Procedure 4-36 (continued)

Circuit Pack Failed-Sync and Circuit Pack Failed-Traffic

Step	Action
1	Determine if an XC double fault scenario exists. Refer to the Probable cause section for this alarm.
2	Wait for an appropriate maintenance window when traffic is minimal. Perform step 4 through step 6 as quickly as possible to minimize traffic loss.
3	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
4	Replace the XC circuit pack raising the Circuit Pack Failed - Sync alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
5	Wait for the green active LED to come on. This normally takes 10 seconds, but can take five minutes if the circuit pack requires an auto-upgrade.
6	Replace the XC circuit pack raising the Circuit Pack Failed - Traffic alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Removing both XC circuit packs can cause the shelf processor to raise alarms against other circuit packs, such as Circuit Pack Failed alarms. If this occurs, remove the alarmed circuit pack for five seconds and then insert it.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-37

Circuit Pack Latch Open

Alarm ID: 100

Probable cause

This alarm is raised when the bottom/lower locking lever on the circuit pack is not fully closed and the circuit pack is inserted into a slot, or the latch on the circuit pack is broken.

ATTENTION

If the cross-connect circuit packs (NTK615AA/AB or NTK616AA/AB) lower latch opens and the user performs an upgrade or a switch, this may cause a traffic hit.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
------	--------

- | | |
|---|---|
| 1 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 2 | Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 3 | Ensure that the circuit pack raising the alarm is pushed all the way into its slot, until the locking levers touch their latches. |
| 4 | Lock the circuit pack into its slot by pushing the upper locking lever down and the lower lever up at the same time. |

ATTENTION

Do not force the locking levers. If the levers do not close correctly, gently re-insert the circuit pack. If the circuit pack cannot be re-inserted, remove the circuit pack and go to [step 5](#).

- | | |
|---|--|
| 5 | If the alarm does not clear, contact your next level of support or your Ciena support group. |
|---|--|

—end—

Procedure 4-38

Circuit Pack Mate Mismatch

Alarm ID: 589**Probable cause**

This alarm is raised by both cross-connect circuit packs when incompatible cross-connect circuit packs are paired in slots 7 and 8 of the 6500-7 packet-optical shelf, slots 7 and 8 of the 14-slot shelf or slots 9 and 10 of the 32-slot shelf. Slot 7 and 8 of the 14-slot shelf or slots 9 and 10 of the 32-slot shelf must contain the same type of cross-connect circuit pack, with the same PEC.

This alarm is the result of an incorrect cross-connect circuit pack replacement or installation, and is raised during cross-connect configuration or reconfiguration.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | | | | | | | |
|--|---|--|-------------------|---------------|------------------------|--------------|------------------------|
| 1 | If you are performing a cross-connect circuit pack reconfiguration procedure, complete the reconfiguration before proceeding to the next step. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545 for details. If the alarm is still present after the reconfiguration procedure is complete, go to step 2 . | | | | | | |
| 2 | Select Shelf Level View from the Configuration menu to open the network element Shelf Level View and check the details (PEC) of the cross-connect circuit packs in slots 7 and 8 of the 6500-7 packet-optical shelf, slots 7 and 8 of the 14-slot shelf or slots 9 and 10 of the 32-slot shelf.

<table><tr><td>If the PECs of both cross-connect circuit packs</td><td>Then go to</td></tr><tr><td>are different</td><td>step 3</td></tr><tr><td>are the same</td><td>step 4</td></tr></table> | If the PECs of both cross-connect circuit packs | Then go to | are different | step 3 | are the same | step 4 |
| If the PECs of both cross-connect circuit packs | Then go to | | | | | | |
| are different | step 3 | | | | | | |
| are the same | step 4 | | | | | | |
| 3 | Provision the PEC on the cross-connect circuit pack with the incorrect PEC to the correct PEC (same as the mate cross-connect circuit pack). Refer to the “Changing the provisioned PEC procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. | | | | | | |

—end—

Procedure 4-39

Circuit Pack Mismatch

Alarm ID: 36

Probable cause

This alarm is raised when a circuit pack is equipped in or inserted into a slot provisioned for:

- a circuit pack of another type
- a circuit pack of the same type that supports additional features (in this case, the circuit packs have different/mismatched PECs)

The alarm is also raised when an unsupported circuit pack is inserted into a slot that is provisioned with the same circuit pack type/identifier (AID) but the provisioned and physical PECs do not match.

During provisioning, a slot is assigned a specific facility and circuit pack type. The assignments are recorded in the provisioning database.

For CDC configurations, this alarm is raised on WSS Flex C-Band w/OPM 20x, CCMD8x16 and FIM circuit packs.

For passive modules (such as CMD44, CMD 64, CMD96, CLMD, UCS, DSCM, OMD4, OMX, OBB, OBMD 1x8, TPT, MonCS, BMD2, UBMD2, MBMD2, TPT, PPC6, passive photonic chassis), this alarm is raised when the module connected to the Access Panel External slot port does not match the provisioned module. This alarm is raised when a passive photonic module (such as a OMD4, OMD8, BS1, BS2, BS3, BS5 or OSCF) is inserted into a PPC6 sub-slot that does not match the provisioned module.

PEC editing is supported between eMOTR 4xXFP/8xSFP+ (NTK536AA and NTK536AB) circuit packs, as well as between eMOTR 4xXFP/8xSFP+/32xSFP (NTK536FA and NTK536FB) circuit packs. If you insert an NTK536AB or NTK536FB circuit pack in a slot provisioned for a NTK536AA or NTK536FA circuit pack, the “Circuit Pack Mismatch alarm will not be raised. Refer to “Changing the provisioned PEC, equipment profile, provisioning mode, or equipment mode” procedure in *Configuration - Provisioning and Operating*, 323-1851-310, for information on PEC editing.

When the physical inventory in a shelf does not match the provisioned part number, a mismatch icon appears in the Equipment and Facility Provisioning and Shelf Level View applications in Site Manager if the mismatch icon is enabled in Site Manager preference settings.

Procedure 4-39 (continued)

Circuit Pack Mismatch

Regardless of the preference setting, the “Circuit Pack Mismatch” alarm is raised when the physical and provisioned equipment are not compatible. If they are compatible but do not have matching part numbers, no alarm is raised but the mismatch icon is displayed if enabled.

By default, the mismatch icon is disabled. To enable the display of the mismatch icon, refer to *User Interface Overview and Site Manager Fundamentals*, 323-1851-195.

The alarm clears if the circuit pack in the specified slot is manually put out-of-service.

Impact

Critical, service-affecting (C, SA) alarm for an unprotected shelf processor
Minor, non-service-affecting (m, NSA) alarm for a protected shelf processor

For multi-port circuit packs with ports configured with different protection schemes, the circuit pack assumes the highest alarm severity.

Broadband services

Critical, service-affecting (C, SA) alarm for an active circuit pack
Critical, service-affecting (C, SA) alarm for a working circuit pack in 1+1 port TPT, or 1+1 TPT configuration with protection circuit pack faulty/unavailable

Minor, non-service-affecting (m, NSA) alarm for an inactive circuit pack (with no cross-connects)

Minor, non-service-affecting (m, NSA) alarm for an inactive or protected circuit pack in 1+1 port TPT, or 1+1 TPT configuration

Photonic services

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Procedure 4-39 (continued)

Circuit Pack Mismatch

Step	Action						
1	Use one of the following options to view the shelf inventory: - select Shelf Inventory from the Configuration menu - select Equipment & Facility Provisioning from the Configuration menu						
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
3	Verify the shelf inventory using one of the following methods: - If using the Shelf Inventory application, compare the slot assignments and physical PECs listed with the actual circuit packs in the shelf until you identify the mismatched circuit pack. - If using the Shelf Level View application, check if there is an "X" on an equipment graphic. The X indicates a circuit pack/module is physically present in the shelf, but the equipment provisioned for the circuit pack/module does not match what is present in the shelf. Mismatched circuit packs can be the same type, but have different PECs (for example, redundant shelf processor). In these cases, ensure PECs match. Refer to the Probable cause section for this alarm.						
4	If you are not on site, use one of the following methods to identify any mismatches between the physical PEC and provisioned PEC by comparing the: Phys. PEC and the Prov. PEC in the Circuit Pack Details tab in the Shelf Level View application Provisioned PEC in the Equipment & Facility Provisioning application and the Physical PEC in the Shelf Inventory application A mismatch can indicate a PEC provisioning error or an incorrect circuit pack is installed.						
5	<table border="0"> <tr> <td>If you have</td><td>Then go to</td></tr> <tr> <td>identified the mismatched circuit pack</td><td>step 6</td></tr> <tr> <td>not identified the mismatched circuit pack</td><td>step 8</td></tr> </table>	If you have	Then go to	identified the mismatched circuit pack	step 6	not identified the mismatched circuit pack	step 8
If you have	Then go to						
identified the mismatched circuit pack	step 6						
not identified the mismatched circuit pack	step 8						
6	Replace the mismatched circuit pack with an appropriate circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
7	<table border="0"> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						

Procedure 4-39 (continued)

Circuit Pack Mismatch

Step	Action
8	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
9	Replace the circuit pack you identified in step 8 . Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the "Module replacement procedures list" table. If the alarm clears, the circuit pack you identified in step 8 is damaged.
10	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-40

Circuit Pack Mismatch - Pluggable

Alarm ID: 342**Probable cause**

This alarm is raised when the installed pluggable module, is different from Provisioned PEC on the circuit pack.

Impact

Critical, service-affecting (C, SA) alarm for an active pluggable
Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a supported pluggable module for the corresponding circuit pack. Refer to *Pluggables Datasheets and Reference*, 323-1851-180.

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and pluggable module port using the following format: <circuit pack>-<shelf-id>-slot#-pluggable module port#
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
3	Replace the pluggable module you identified in step 1 with a supported pluggable module. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-41 Circuit Pack Missing

Alarm ID: 35, 681

Probable cause

This alarm is raised when a slot is provisioned and the following occurs:

- no circuit pack is in the designated slot
- circuit pack failure makes the circuit pack undetectable
- the access panel, MIC, or Power Input Cards A/B has been removed from the system. This alarm masks any existing alarm on that unit. For example, the Circuit Pack Missing alarm raised against Power Input Card B masks the Power Failure - B alarm.

When both the A and B power feeds to one or more zones that supply power to a circuit pack have failed, the associated “Power Failure A/B” and “Power Failure - Fuse Blown” alarms must be cleared first in order to clear this alarm. Refer to the alarm clearing procedures in this document.

Note 1: A Circuit Pack Missing alarm against an access panel masks/clears certain BITSIN and comms alarms (for example, the ILAN port, COLAN) raised against the access panel before its removal.

Note 2: For the X-Conn 1600G PKT/OTN (NTK616AA) and cross-connect 600G PKT/OTN (NTK615AA) circuit packs, it may take up to five minutes for the “Circuit Pack Missing” alarm to clear after the inventory displays the circuit pack information.

- CCMD8x16 circuit pack is missing on a shelf with CDC equipment provisioned.
- a provisioned passive module (such as CMD44, CMD 64, CMD96, CLMD, UCS, FIM, GMD10 C-Band, GMD10 L-Band, DSCM, OMD4, OMX, OBB, OBMD 1x8, TPT, MonCS, BMD-2, UBMD2, MBMD2, TPT, PPC6 Passive Photonic chassis) is disconnected from the shelf Access Panel External Slot port.
- a provisioned passive module (such as an OMD4, OMD8, BS1, BS2, BS3, BS5 or OSCF) has an open latch or is missing from a sub-slot in a PPC6 connected to an External Slot. Note that if the External Slot interface cable between the PPC6 and host shelf is disconnected or faulty, the alarm will be raised against the PPC6 external slot number, not against the passive modules sub-slot(s).

Procedure 4-41 (continued)
Circuit Pack Missing

If you change the state of the circuit pack raising this alarm to OOS, this alarm clears and the “Slot Empty” alarm is raised.

Note: In some cases, a Circuit Pack Missing alarm is simultaneously active with a Software Upgrade Failed alarm for a particular slot. This condition can be cleared either by performing a minimal delivery of the corresponding software release, or by configuring a software release server for the network element. Refer to the “Transferring a software load to a network element” procedure in *Administration and Security*, 323-1851-301.

Impact

Critical, service-affecting (C, SA) alarm for an unprotected shelf processor circuit pack

Critical, service-affecting (C, SA) alarm for an unprotected circuit pack

Critical, service-affecting (C, SA) alarm for an active circuit pack in a protected configuration.

Minor, non-service-affecting (m, NSA) alarm for a protected circuit pack

Minor, non-service-affecting (m, NSA) alarm for an inactive circuit pack or circuit pack without cross-connects.

Alarm severity depends on the following conditions:

- Alarms with Critical, service-affecting severity are raised when both circuit packs in a protection group are pulled out of their slots. In this case, a m, NSA alarm and a C, SA alarm are raised.

ATTENTION

When both XC circuit packs are removed, a Critical, service-affecting alarm is raised for each XC circuit pack removed.

- Alarms with Critical, service-affecting severity are raised when missing circuit packs are unprotected.

For multi-port circuit packs with ports configured with different protection schemes, the circuit pack assumes the highest alarm severity.

Broadband services

Critical, service-affecting (C, SA) alarm for an active circuit pack

Critical, service-affecting (C, SA) alarm for a working circuit pack in 1+1 port TPT, or 1+1 TPT configuration with protection circuit pack faulty/unavailable

Procedure 4-41 (continued)

Circuit Pack Missing

Minor, non-service-affecting (m, NSA) alarm for an inactive circuit pack (with no cross-connects)

Minor, non-service-affecting (m, NSA) alarm for an inactive or protected circuit pack in 1+1 port TPT, or 1+1 TPT configuration

Photonic services

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action												
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.												
2	Identify the circuit pack raising the alarm. Refer to the “ Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm ” procedure in this document.												
3	<table><tr><th>If</th><th>Then go to</th></tr><tr><td>both XC circuit packs are reported missing</td><td>step 5</td></tr><tr><td>a Type 6 DSCM (PECs NPAD01DB through NPAD17DB) is reported missing</td><td>step 8</td></tr><tr><td>an equipment connected to External Slots is reported missing</td><td>step 9</td></tr><tr><td>a Passive module in a PPC6 sub-slot is reported missing</td><td>step 10</td></tr><tr><td>any other circuit pack is reported missing</td><td>step 4</td></tr></table>	If	Then go to	both XC circuit packs are reported missing	step 5	a Type 6 DSCM (PECs NPAD01DB through NPAD17DB) is reported missing	step 8	an equipment connected to External Slots is reported missing	step 9	a Passive module in a PPC6 sub-slot is reported missing	step 10	any other circuit pack is reported missing	step 4
If	Then go to												
both XC circuit packs are reported missing	step 5												
a Type 6 DSCM (PECs NPAD01DB through NPAD17DB) is reported missing	step 8												
an equipment connected to External Slots is reported missing	step 9												
a Passive module in a PPC6 sub-slot is reported missing	step 10												
any other circuit pack is reported missing	step 4												
4	<table><tr><th>If</th><th>Then</th></tr><tr><td>the slot is empty</td><td>insert a circuit pack of the correct type into the slot. Go to step 7.</td></tr><tr><td>a circuit pack of the correct type is in the slot</td><td>go to step 11.</td></tr></table>	If	Then	the slot is empty	insert a circuit pack of the correct type into the slot. Go to step 7 .	a circuit pack of the correct type is in the slot	go to step 11 .						
If	Then												
the slot is empty	insert a circuit pack of the correct type into the slot. Go to step 7 .												
a circuit pack of the correct type is in the slot	go to step 11 .												
5	Insert the XC circuit pack in slot 7 of the 6500-7 packet-optical shelf, slot 7 of the 14-slot shelf, or slot 9 of the 32-slot shelf first.												

Procedure 4-41 (continued)
Circuit Pack Missing

Step	Action												
6	Insert the second XC circuit pack in slot 8 of the 6500-7 packet-optical shelf, slot 8 of the 14-slot shelf, or slot 10 of the 32-slot shelf. Slot 7 and 8 or slot 9 and 10 must contain the same type cross-connect circuit pack with the same PEC.												
7	Wait 30 seconds and retrieve all alarms to determine if the original alarm has cleared. <table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11						
If the original alarm has	Then												
cleared	the procedure is complete												
not cleared	go to step 11												
8	If a Type 6 DSCM is provisioned in slots 83 through 90, delete the associated facility and equipment and reprovision in virtual slots 91 through 99. Refer to the “Provisioning a circuit pack, module, or SFP/SFP+/XFP/CFP manually” and “Deleting a circuit pack, module, or SFP/SFP+/XFP/CFP” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 21.												
9	Verify that the cable between the shelf External Slot Inventory port and the external peripheral is connected and is not damaged. For a CDC configuration, verify that the LAN cable is connected to the FIM and is not damaged. Go to step 21.												
10	If the slot is empty, insert a passive module of the correct type in a PPC6 sub-slot. Otherwise, make sure that the module is fully inserted and the latches are tightly closed. If the card (or its replacement) in the PPC6 or the PPC6 itself cannot be inventoried after replacing the cable, the PPC6 itself must be replaced. Go to step 21.												
11	<table> <tr> <th>If the circuit pack</th><th>Then go to</th></tr> <tr> <td>is still carrying traffic and providing all services</td><td>the next step</td></tr> <tr> <td>is not carrying traffic and providing all services</td><td>step 21</td></tr> <tr> <th>If the SP</th><th>Then go to</th></tr> <tr> <td>is in a redundant configuration</td><td>step 12</td></tr> <tr> <td>is not in a redundant configuration</td><td>step 15</td></tr> </table>	If the circuit pack	Then go to	is still carrying traffic and providing all services	the next step	is not carrying traffic and providing all services	step 21	If the SP	Then go to	is in a redundant configuration	step 12	is not in a redundant configuration	step 15
If the circuit pack	Then go to												
is still carrying traffic and providing all services	the next step												
is not carrying traffic and providing all services	step 21												
If the SP	Then go to												
is in a redundant configuration	step 12												
is not in a redundant configuration	step 15												

Procedure 4-41 (continued)

Circuit Pack Missing

Step	Action						
12	Initiate a switch to the backup SP. Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. After the system recovers, check if the alarm has cleared. <table> <tr> <td>If the original alarm has</td><td>Then go to</td></tr> <tr> <td>cleared</td><td>step 13</td></tr> <tr> <td>not cleared</td><td>step 21</td></tr> </table>	If the original alarm has	Then go to	cleared	step 13	not cleared	step 21
If the original alarm has	Then go to						
cleared	step 13						
not cleared	step 21						
13	Reseat the previously active SP. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. When the SP has finished booting, initiate a switch back to the originally active SP. After the system recovers, check if the alarm has cleared. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	Initiate a switch back to the backup SP. Replace the original SP. Refer to the “Replacing the shelf processor” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Initiate a switch back to the newly inserted SP. After the system recovers, check if the alarm has cleared. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
15	Reseat the SP. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 16</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 16
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 16						
16	Reseat the circuit pack raising the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 17</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 17						

Procedure 4-41 (continued)
Circuit Pack Missing

Step	Action						
17	Replace the circuit pack against which the alarm is raised. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 18</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 18
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 18						
18	Re-install the original circuit pack into the shelf.						
19	<table> <tr> <th>If the SP</th><th>Then go to</th></tr> <tr> <td>is in a redundant configuration</td><td>step 21</td></tr> <tr> <td>is not in a redundant configuration</td><td>step 20</td></tr> </table>	If the SP	Then go to	is in a redundant configuration	step 21	is not in a redundant configuration	step 20
If the SP	Then go to						
is in a redundant configuration	step 21						
is not in a redundant configuration	step 20						
20	Replace the SP. Refer to the “Replacing the shelf processor” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
21	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-42

Circuit Pack Missing - Pluggable

Alarm ID: 339

Probable cause

This alarm is raised when a provisioned SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 is not physically installed in the circuit pack.

Impact

Critical, service-affecting (C, SA) alarm for an active pluggable transceiver
Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable transceiver

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.
- obtain a supported SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 for the corresponding circuit pack (refer to Supported SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 modules for interface circuit packs” table in *6500 Planning*, NTRN10GK)

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the “ Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm ” procedure in this document. The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 port using the following format: <circuit pack>-<shelf-id>-slot#-SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 port#
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
3	Install a supported SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 module in the port you identified in step 1 . Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-43

Circuit Pack Operational Capability Exceeded

Alarm ID: 1064

Probable cause

This alarm is raised against a 40G OCLD/Wavelength-Selective 40G OCLD, 40G UOCLD, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, or 100G OCLD circuit packs when equipment specifications are exceeded. This alarm indicates that at least one of the measurements, reach or PMD (Polarization Mode Dispersion) DGD (differential group delay), exceeded the specifications of the circuit pack. For information on operational specifications of the 40G and 100G circuit packs, refer to *6500 Planning*, NTRN10GK.

If the client is not connected yet, the severity of the alarm is NSA since it does not impact client traffic. However, it still can impact line traffic. Line signal conditioning will be applied if the condition which caused this alarm to be raised was present during optical signal acquisition (for example, after connecting fiber, or after card insertion or cold restart operation). It is possible for this alarm to be raised without impacting traffic, if traffic was already running prior to the condition being detected. Actions must be taken to clear this alarm as soon as possible.

You can confirm the presence of signal conditioning in the OTM3 PM or OTM4 PM screen where the OTU-SEFS PM count would be incrementing.

The alarm is latched even if all measurements, reach or PMD DGD do not exceed the specifications of the circuit pack and does not get re-evaluated until either a cold restart of the circuit pack or a line fault condition toggles (for example, a fiber pull and reinsertion).

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a replacement circuit pack suitable for the configuration

Procedure 4-43 (continued)

Circuit Pack Operational Capability Exceeded

Step	Action						
1	Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
3	Verify that the circuit pack is connected to the correct optical link using ODU/OTU TTI. There should be no “ODU/OTU Trace Identifier Mismatch” alarm.						
4	<table border="0"> <tr> <td>If the circuit pack</td><td>Then go to</td></tr> <tr> <td>is connected to the correct optical link</td><td>step 6</td></tr> <tr> <td>is connected to the wrong optical link</td><td>step 5</td></tr> </table>	If the circuit pack	Then go to	is connected to the correct optical link	step 6	is connected to the wrong optical link	step 5
If the circuit pack	Then go to						
is connected to the correct optical link	step 6						
is connected to the wrong optical link	step 5						
5	Disconnect the existing fiber connections and re-connect the circuit pack to the correct optical link. <table border="0"> <tr> <td>If the alarm</td><td>Then</td></tr> <tr> <td>clears</td><td>the procedure is complete</td></tr> <tr> <td>does not clear</td><td>go to step 6</td></tr> </table>	If the alarm	Then	clears	the procedure is complete	does not clear	go to step 6
If the alarm	Then						
clears	the procedure is complete						
does not clear	go to step 6						
6	Contact your next level of support to find out and order the suitable circuit pack for the configuration. Note: If the condition which triggered the “Circuit Pack Operational Capability Exceeded” alarm is not present anymore, this latched alarm could be cleared by either cold restarting the circuit pack against which the alarm is raised (Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor”) or by toggling the line in the transponder receive direction. Both actions to clear the alarm are service impacting.						
7	Replace the circuit pack with a circuit pack suitable for the configuration. Refer to <i>Fault Management - Module Replacement</i>, 323-1851-545. You can replace the 40G OCLD, Wavelength-Selective 40G OCLD, 100G WL3e OTR, or 100G OCLD circuit packs in one of the two ways as follows. - Delete the provisioning data (COMM, connection, facility and equipment). Remove the old circuit pack from the slot. Insert the new circuit pack. The equipment and facility will auto-provision. You can re-provision the connection and COMM if necessary. - Remove the old circuit pack from the slot. Insert the new circuit pack. Edit the provisioned PEC to the PEC of the new circuit pack.						

Procedure 4-43 (continued)

Circuit Pack Operational Capability Exceeded

Step	Action
	Refer to the “Reconfiguration of an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545, for the detailed procedure.
	Note: Provisioned PEC editing is not supported between any of the 40G UOCLD variants. You must replace the circuit pack with an identical circuit pack.
8	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-44

Circuit Pack Unknown

Alarm ID: 58

Probable cause

This alarm is raised in the following situations:

- when the on-board processor of a circuit pack cannot communicate with the shelf processor after you insert the circuit pack into the shelf
- when an unknown circuit pack is inserted into an unprovisioned slot
- when a circuit pack is in the wrong slot
- when both the A and B power feeds to the zone powering a circuit pack have failed “Power Failure - A or Power Failure - B” and “Power Failure - Fuse Blown” alarms have to be cleared first in order to clear this alarm. Refer to the alarm clearing procedures in this document.
- when the external equipment connected to an External Slot inventory port on the access panel is unknown
- when a module equipped in a sub-slot of connected external equipment is unknown

ATTENTION

A circuit pack in the wrong slot only occurs if the circuit pack keying is removed. Circuit packs are keyed to fit into specific slots. Do not remove the circuit pack keying for any reason.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Procedure 4-44 (continued)

Circuit Pack Unknown

Step	Action						
1	Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. <table> <tr> <td>If the circuit pack is</td><td>Then go to</td></tr> <tr> <td>a newly inserted or manually provisioned cross-connect circuit pack</td><td>step 2</td></tr> <tr> <td>not a cross-connect circuit pack</td><td>step 4</td></tr> </table>	If the circuit pack is	Then go to	a newly inserted or manually provisioned cross-connect circuit pack	step 2	not a cross-connect circuit pack	step 4
If the circuit pack is	Then go to						
a newly inserted or manually provisioned cross-connect circuit pack	step 2						
not a cross-connect circuit pack	step 4						
2	Perform protection switch of the shelf processor. Refer to “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. Do not insert any circuit packs while the shelf processor is performing a protection switch restarting.						
3	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
5	Ensure that the circuit pack reporting the alarm is supported for the software release running on the shelf. <table> <tr> <td>If the circuit pack is</td><td>Then</td></tr> <tr> <td>not supported by the software release</td><td>the circuit pack cannot be equipped in the shelf. Remove it. Go to step 6.</td></tr> <tr> <td>supported by the software release</td><td>go to step 7</td></tr> </table>	If the circuit pack is	Then	not supported by the software release	the circuit pack cannot be equipped in the shelf. Remove it. Go to step 6 .	supported by the software release	go to step 7
If the circuit pack is	Then						
not supported by the software release	the circuit pack cannot be equipped in the shelf. Remove it. Go to step 6 .						
supported by the software release	go to step 7						
6	Wait 30 seconds, and retrieve all alarms. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						

Procedure 4-44 (continued)

Circuit Pack Unknown

Step	Action								
7	Compare the circuit pack raising the alarm with the supported circuit packs for each slot on the shelf. Refer to “6500 interface circuit packs and modules” table in <i>6500 Planning</i> , NTRN10GK.								
	<table> <tr> <th>If the circuit pack raising the alarm is</th><th>Then</th></tr> <tr> <td>a passive photonics modules</td><td>go to step 8</td></tr> <tr> <td>in an unsupported slot</td><td>go to step 17</td></tr> <tr> <td>in a supported slot</td><td>the circuit pack may be damaged. Go to step 18.</td></tr> </table>	If the circuit pack raising the alarm is	Then	a passive photonics modules	go to step 8	in an unsupported slot	go to step 17	in a supported slot	the circuit pack may be damaged. Go to step 18 .
If the circuit pack raising the alarm is	Then								
a passive photonics modules	go to step 8								
in an unsupported slot	go to step 17								
in a supported slot	the circuit pack may be damaged. Go to step 18 .								
8	<table> <tr> <th>If the SP protection</th><th>Then go to</th></tr> <tr> <td>is provisioned</td><td>step 9</td></tr> <tr> <td>is not provisioned</td><td>step 14</td></tr> </table>	If the SP protection	Then go to	is provisioned	step 9	is not provisioned	step 14		
If the SP protection	Then go to								
is provisioned	step 9								
is not provisioned	step 14								
9	Perform a protection switch to the other SP.								
10	If the alarm remains cleared after 10 minutes, reseal the original active SP. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. If the alarm is raised again, go to step 17 .								
11	Perform a protection switch back to the original active SP. If the alarm remains cleared after 10 minutes, this procedure is complete.								
12	If the alarm remains active, replace this original active SP. Refer to the “Replacing the shelf processor” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. If the alarm is raised again, go to step 17 . If the alarm remains cleared after 10 minutes, this procedure is complete.								
13	If the alarm is raised again, then go to step 17 .								
14	Reseat the SP. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. If the alarm remains cleared after 10 minutes, then this procedure is complete.								
15	If the alarm is raised again, replace this SP. Refer to the “Replacing the shelf processor” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. If the alarm remains cleared after 10 minutes, this procedure is complete.								
16	If the alarm did not clear, continue to the next step.								
17	Replace the circuit pack raising the alarm with a circuit pack supported in that slot. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table. Go to step 19 .								

Procedure 4-44 (continued)
Circuit Pack Unknown

Step	Action
18	Replace the circuit pack with an identical circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table.
19	Wait 30 seconds and retrieve all alarms.
	If the original alarm
	Then
	is cleared the procedure is complete. If you replaced a circuit pack in step 18 , the circuit pack you replaced is damaged.
	is not cleared contact your next level of support or your Ciena support group
	—end—

Procedure 4-45

Circuit Pack Unknown - Pluggable

Alarm ID: 341

Probable cause

This alarm is raised when an unrecognized pluggable is installed in an unprovisioned port.

The Circuit Pack Unknown - Pluggable alarm cannot be disabled. Use the following steps to clear the alarm.

Impact

Critical, service-affecting (C, SA) alarm for an active pluggable

Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable

Prerequisites

To perform this procedure, you must

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a supported pluggable for the corresponding circuit pack (refer to Supported pluggable modules for interface circuit packs” table in *6500 Planning*, NTRN10GK)

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and pluggable port using the following format:

<circuit pack>-<shelf-id>-slot#-pluggable port# |
| 2 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 3 | Replace the pluggable you identified in step 1 with a supported pluggable. Refer to the procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. |
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-46

Client Service Mismatch

Alarm ID: 696, 869, 1245

Probable cause

This alarm is raised against a WAN facility of a 2x10G OTR, 4x10G OTR, 40G MUX OCI, 8xOTN Flex MOTR, (1+8)xOTN Flex MOTR, 2xCFP2 OTN Flex MOTR, eMOTR, 10x10GE MUX, 4x10G MUX, 16xFLEX OTN I/F, or 40G MUX OCI circuit pack

The alarm is raised when the GFP UPI Tx byte provisioned on the remote circuit pack does not match the GFP UPI expected Byte on the local circuit pack. When this alarm is active, traffic from the far-end is lost.

The alarm point is identified at the generic framing procedure (GFP) level to indicate a provisioning mismatch between the near-end and far-end facility provisioning. For example, one end is configured to preserve the preamble and the other is configured to discard the preamble. The GFP user payload identifier (UPI) byte is used for this purpose.

When using 10GE with a mapping of GFPMACOSTR, GFPMACOSTR192, or GFPMACOSTR64, frames which have a GFP-UPI mismatch will be discarded by the OTN Mapper and the Client Service Mismatch alarm will be raised. This applies to both ordered set frames (UPI_O) and data frames (UPI_D).

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must:

- have the network connection information (that is, how the optical modules on each network element connect to other network elements)
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Procedure 4-46 (continued)

Client Service Mismatch

Step	Action
1	Trace the cross-connect information to determine the corresponding facility of the far-end circuit pack.
2	Ensure that the nodal cross-connects are provisioned correctly.
3	Determine if the client facility provisioned at the corresponding far-end circuit pack matches the near-end. Correct any mismatches. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	Determine if the corresponding far-end facility parameters match the parameters at the end reporting the Client Service Mismatch alarm. Correct any mismatches. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-47

CMF UPI Mismatch

Alarm ID: 1377

Probable cause

This alarm is raised for the WAN facility under the following conditions:

- At least three consecutive GFP layer Client Management Frames (CMF) UPI received with a value equal to 83/84/85 when the ETH10G WAN facility Transparent Ordered Sets parameter is G.7041 CMF.
- At least three consecutive GFP layer CMF UPI received with a value equal to 03/04/05 when the ETH10G WAN facility Transparent Ordered Sets parameter is Legacy CMF.

When the ETH10G WAN facility Order Sets Transparency parameter is G.7041 CMF, the CMF UPI values transmitted and expected are as follows for ordered sets:

- CMF-LF(FDI) = 04
- CMF-RF(RDI) = 05
- CMF-ALL Clear (DCI) = 03

When the ETH10G WAN facility Transparent Ordered Sets parameter is Legacy CMF, the CMF UPI values transmitted and expected are as follows:

- CMF-LF(FDI) = 83
- CMF-RF(RDI) = 84
- CMF-ALL Clear (DCI) = 85

Note that when this alarm is active, Transparent Ordered Sets cannot be guaranteed and there is no signal conditioning.

Note: For the 200G (2x100G/5x40G) MUX (NTK529HA) circuit pack, detection of unexpected Client Management Frames (CMF) raises the “Far End Client Signal Fail” alarm instead of “CMF UPI Mismatch” alarm.

The 10x10G MUX (NTK529BBE5) circuit pack supports G.7041 compliant CMF and Ordered Set Transparency on the 10x10G MUX circuit pack (NTK529BBE5). This provides compliance to G.7041 Sub-clause 7.9.2.2. Mapping mode that preserves the Ordered Set information of the 10GBase-R client signal by mapping Ordered Sets into their own GFP-F frames (Client Data Frame). This mapping mode does not require the GFP CMF to communicate local, remote fault, all clear and other PCS layer ordered set information.

Procedure 4-47 (continued)

CMF UPI Mismatch

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action
1	Determine the corresponding far-end client and WAN facility parameters. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	Set the ETH10G WAN facility Transparent Ordered Sets parameter to equal values on both sides. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-48

Cold Restart Required

Alarm ID: 646

Probable cause

This alarm is raised when a new functionality is attempted to be provisioned on a circuit pack but the functionality needs to be loaded onto that circuit pack. The circuit pack must be restarted to be loaded with the new functionality. Some of reasons that can cause this alarm are as follows:

- A circuit pack loses its FPGA load from within the file system, and the FPGA load maintained on the circuit pack is older than the required load for this release.
- New Packet functionality is attempted to be enabled on PKT/OTN interfaces after a software upgrade has been done. The alarm will be cleared after the new functionality has been loaded into the circuit pack.

If this alarm is raised as part of a network element software upgrade, refer to the appropriate Software Upgrade Procedure listed in the “Software Upgrade Procedures” section of the *Planning - Ordering Information*, 323-1851-151.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-48 (continued)

Cold Restart Required

Step	Action
1	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. Perform a cold restart on the circuit pack raising the alarm. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor”.
2	If the alarm does not clear after the circuit pack restart, contact your next level of support or your Ciena support group.

—end—

Procedure 4-49

Configuration Mismatch

Alarm ID: 1415**Probable cause**

This alarm is raised if the admin weight or bundle id is different on both sides of a Optical Signaling and Routing Protocol (OSRP) link.

Impact

Minor, non-service-affecting (m, NSA) alarm

Note: Shelf function is not affected by this alarm.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Launch the OSRP Provisioning application and click on Links Tab to determine if both ends of the OSRP link have the same admin weight and bundle id assigned.
2	If different, provision the OSRP link to have the same admin weight and bundle id assigned. Refer to the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-50

Configuration Mismatch - Adv BW Limit

Alarm ID: 1416

Probable cause

This alarm is raised when there is a configuration mismatch due to advertised bandwidth (Adv BW) limit being different on adjacent OSRP links.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the Advertisement BW limit on each side of the link. Refer to the "Editing an OSRP link" procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	Ensure that both ends have the same Advertisement BW limit.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-51

Configuration Mismatch - BW Lockout

Alarm ID: 1770**Probable cause**

This alarm is raised when a bandwidth lockout is enabled on one end of an OSRP line and disabled on the other end.

If you apply a bandwidth lockout for maintenance activities to one end of the line, it is required to apply a bandwidth lockout on the other end of the line to clear the alarm.

The alarm also clears when the bandwidth lockout on both sides are matching (enabled or disabled).

It can take up to 30 seconds for this alarm to raise if one end of the OSRP line is configured and the other end is not.

Impact

Minor, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Disable the bandwidth lockout on the OSRP line or enable the bandwidth lockout at the other end of the OSRP line. Refer to the "Editing an OSRP line" procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-52

Configuration Mismatch - BW Threshold

Alarm ID: 1768

Probable cause

This alarm is raised when there is a Bandwidth Threshold mismatch configuration on both ends of an OSRP link.

The alarm clears when you provision the same Bandwidth threshold value on both sides of the link.

Impact

Minor, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Identify which Bandwidth Threshold is misconfigured (which side of the link is misconfigured). Refer to “Retrieving OSRP provisioning information” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	Provision both ends of the OSRP link with the same Bandwidth Thresholds. Refer to the “Editing an OSRP line” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-53

Configuration Mismatch - Common ID

Alarm ID: 1411**Probable cause**

This alarm is raised when the Common Line ID field of the OSRP line on either side is different.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the common id value of the OSRP line at the node raising the alarm. See the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330. |
| 2 | Verify common id value of the node at the other ends of the OSRP line and compare it to the values found in step 1 . |
| 3 | If the common id values on both ends of the OSRP line are not the same, change the values to be the same. Refer to the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330. |
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-54 Configuration Mismatch - Concatenation

Alarm ID: 1414

Probable cause

This alarm is raised when there is a concatenation mismatch between the local and the remote lines while an OSRP line is configured as standard concatenation at one node and transparent concatenation at the other node.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the port that raised the alarm. See “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” on page 2-35.
2	The 6500 nodes are configured as standard concatenation. Ensure that the remote node is a 6500. If the remote node is not a 6500 node, ensure that the node is configured as standard concatenation. Refer to non-6500 documentation for configuration information.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-55

Configuration Mismatch - Link ID

Alarm ID: 1420**Probable cause**

This alarm is raised when in a link aggregation one end of the aggregation contains OSRP lines that are part of a different OSRP link.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the OSRP lines in a link aggregation on each side of the link. See the “Adding an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330. |
| 2 | Ensure all OSRP lines in a link aggregation on each side is part of only one OSRP link. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-56

Configuration Mismatch - Node

Alarm ID: 1421

Probable cause

This alarm is raised when there is a configuration mismatch due to mismatch in aggregated OSRP line node IDs between adjacent OSRP links. The alarm raises when in a link aggregation at one end of the aggregation reports OSRP lines that are part of a different node.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the OSRP lines in a link aggregation on each side of the link. See the “Adding an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	Ensure that all OSRP lines in a link aggregation on each side are part of the same node.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-57

Configuration Mismatch - OVPN ID

Alarm ID: 1769, 1866

Probable cause

This alarm is raised when the Optical Virtual Private Network Identifier (OVPN ID) on both ends of the OSRP link do not match.

The alarm clears when you provision the same OVPN ID on both sides of the link.

Impact

Minor, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Ensure that there are no Sub-Network Connections (SNCs) on the link. Refer to “Deleting a sub-network connection” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	Provision both ends of the link with the same OVPN ID. Refer to the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-58

Configuration Mismatch - Primary State

Alarm ID: 1410

Probable cause

This alarm is raised when the administrative state of the ends of the OSRPLINE is mismatched with the remote end.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the administrative state of the OSRPLINE on each side of the link. See the “Adding an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	Ensure the administrative state of the OSRPLINE on each side is the same.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-59

Connection Validation Testing In Progress

Alarm ID: 2241**Probable cause**

This alarm is raised when the connection validation testing is in progress.

Impact

Warning, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UP.

Step	Action
1	No action is required. The alarm clears when the connection validation testing is completed.
2	If the alarm does not clear, release the test mode using the TL1 command: RLS-TESTMODE.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-60

Control Mode Conversion Failed

Alarm ID: 1956

Probable cause

This alarm is raised against an OTS facility when five attempts to convert a ROADM OTS's Actual Control Mode to Flexible Grid Capable have failed.

The alarm clears when the conversion is completed or the user cancels the conversion by setting the Provisioned Control Mode back to Fixed ITU.

Once begun, conversion is re-attempted periodically until it succeeds, even after the Control Mode Conversion Failed alarm has been raised.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action				
1	Check the DOC Logs to identify the specific problem blocking completion of the conversion. Refer to the "Displaying the DOC logs for the summary table" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.				
2	<table><tr><td>If the logs indicate that the Control Mode Conversion to Flexible Grid setup failed</td><td>Then go to step 3</td></tr><tr><td>Flexible Grid failed</td><td>step 4</td></tr></table>	If the logs indicate that the Control Mode Conversion to Flexible Grid setup failed	Then go to step 3	Flexible Grid failed	step 4
If the logs indicate that the Control Mode Conversion to Flexible Grid setup failed	Then go to step 3				
Flexible Grid failed	step 4				
3	During Provisioned Control Mode Conversion to Flexible Grid setup, the conversion may be canceled by setting the ROADM OTS's Provisioned Control Mode back to Fixed ITU. Contact Ciena for further assistance with completing the conversion to Flexible Grid control mode. This procedure is complete.				

Procedure 4-60 (continued)

Control Mode Conversion Failed

Step	Action						
4	Once conversion has started, it cannot be canceled. If the DOC Logs indicate an issue changing AMP facility provisioning, check for and clear any of the following alarms in the DOC span of control: • OSC Loss of Signal • Optical Line Fail • Incomplete Channel Topology • Adjacency Provisioning Error • OTS Provisioning Error • Equipment problems, such as “Circuit Pack Failed”, “Circuit Pack Mismatch”, “Circuit Pack Missing”, or “Circuit Pack Unknown”.						
5	<table> <tr> <th>If the alarm is</th><th>Then</th></tr> <tr> <td>cleared</td><td>this procedure is completed.</td></tr> <tr> <td>not cleared</td><td>step 6</td></tr> </table>	If the alarm is	Then	cleared	this procedure is completed.	not cleared	step 6
If the alarm is	Then						
cleared	this procedure is completed.						
not cleared	step 6						
6	If the DOC Logs indicate a failure to set OTS, NMCC or AMP facility parameters, manually set the parameter to the indicated value(s) and wait two minutes for conversion to be re-attempted.						
7	If the alarm is not cleared, check the DOC Logs again to see if the failure reason changed. If the logs indicate a failure to set different OTS, NMCC or AMP facility parameters repeat step 6 .						
8	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-61

Control Mode Conversion in Progress

Alarm ID: 1955

Probable cause

This alarm is raised when the Provisioned Control Mode of a ROADM OTS is changed from Fixed ITU to Flexible Grid Capable. The alarm clears when:

- the control mode conversion is completed
- the control mode conversion is failed, in which case the “Control Mode Conversion Failed” alarm is raised
- the user cancels the Provisioned Control Mode conversion by setting the Provisioned Control Mode back to Fixed ITU.

Once begun, conversion is re-attempted periodically until it succeeds, even after the Control Mode Conversion Failed alarm has been raised.

Impact

Warning, non-service-affecting (w, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	No action is required. The alarm clears when the conversion is completed, failed, or canceled.
2	If the alarm does not clear, verify if the DOC is In-Service and place it Out-Of-Service if needed.
3	Verify if the ROADM OTS control mode capability is Flexible Grid Capable. If the ROADM OTS control mode capability is “Fixed ITU”, contact your next level of support or your Ciena support group.

—end—

Procedure 4-62

Control Mode Provisioning Mismatch

Alarm ID: 1957

Probable cause

This alarm is raised on a ROADM OTS when the Actual Control Mode is 'Flexible Grid Capable', but the Control Mode Capability is 'Fixed ITU'. That is, the domain is not capable of supporting the control mode that the OTS is actually running. The alarm clears when the "Control Mode Capability" becomes 'Flexible Grid Capable'. The "Control Mode Capability" becomes 'Flexible Grid Capable' when the following conditions are met:

- all photonic equipment in the domain is flexible grid capable
- the OTS is provisioned with a flexible grid capable WSS
- there is only one section in the domain and the domain is complete (every OTS in the domain must be provisioned with the same OSID)
- topology software can communicate with every Line Amp node in the domain and the domain tail OTS
- the OTS at the domain tail is also a ROADM OTS
- all nodes in the domain are running a software release which supports flexible grid control
- both ROADM OTSs in the domain are in 6500 shelves equipped with either SP-2, SPAP-2 or SP-3 circuit pack
- the Reference Bandwidth of all the amplifiers in the domain are set to 12.5GHz (found in the AMP facility)

Impact

Warning

Step	Action
1	Check the conditions mentioned in the Probable cause section.
2	If the alarm cannot be cleared, contact your Ciena support group.

—end—

Procedure 4-63

Control Plane Operations Blocked

Alarm ID: 1776

Probable cause

This alarm is raised on a DOC to indicate that the control plane cannot restore or create new SNCs because DOC Auto add channels and/or DOC Auto delete channels parameters are set to disable when the corresponding OTS facility is control plane enabled (CPS is provisioned to Enable).

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action				
1	<table><tr><td>If the OTS is supposed to be control plane enabled</td><td>Then go to step 2</td></tr><tr><td>disabled</td><td>step 3</td></tr></table>	If the OTS is supposed to be control plane enabled	Then go to step 2	disabled	step 3
If the OTS is supposed to be control plane enabled	Then go to step 2				
disabled	step 3				
2	Update the DOC facility attributes DOC Auto add channels and DOC Auto delete channels parameters to enable as follows: - From the Site Manager menu, select the DOC instance Configuration->Photonic Services->Domain Optical Controller (DOC). The Domain Optical Controller (DOC) window opens. - Select the required shelf from the Shelf pull-down menu, as required. (ALL is the default.) - Click on the Start Monitoring or Refresh button. - Select the required DOC instance from the summary table. - Select the Settings tab. - Click on the Edit button. The Edit DOC dialog is displayed. - Use the pull down menus to change the DOC values Auto add channels and Auto delete channels to "Enabled". Go to step 4.				

Procedure 4-63 (continued)
Control Plane Operations Blocked

Step	Action
3	Update the CPS attribute of the OTS facility to Disable as follows: - From the Site Manager menu, select the OTS (Configuration->Photonic Services->Photonic Configuration Management). - From the OTS window, click the row with the desired OTS to highlight it and then click the Edit or Edit OTS button. The Edit OTS window opens. - From the Edit OTS window, click the CPS drop-down box and select "Disable". Refer to the "Editing an OTS instance in the Photonic Configuration Management application" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310 for more information.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-64

Control Plane System Mismatch

Alarm ID: 1773

Probable cause

This alarm is raised on a shelf when each provisioned OTS has a CPS value provisioned to Disable, but the domain to which the OTS belongs has at least one OTS within the domain with CPS provisioned to Enable.

This indicates that the CPS values for each OTS within a domain is not consistent. The event is only raised against the OTS with CPS provisioned to Disable. The alarm clears when you provision the CPS parameter on each ROADM OTS within a domain to the same value.

During the flexible grid conversion, if the OTS CPS parameter is set to Disable before the OTS “Provisioned Control Mode” parameter is changed to “Flexible Grid Capable”, software automatically changes the OTS CPS parameter to Enable. Then the alarm is raised at the far-end connecting OTS if the far-end connecting OTS has the CPS parameter set to Disable.

The alarm clears when the OTS “Provisioned Control Mode” parameter is changed to “Flexible Grid Capable” at the far-end connecting OTS.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Set the CPS parameter on each ROADM OTS within a domain to the same value.
2	Select Photonic Configuration Management from Configuration->Photonic Services .
3	In the Photonic Configuration Management window, select the required shelf from the Shelf drop-down list.

Procedure 4-64 (continued)
Control Plane System Mismatch

Step	Action
4	Select the OTS that you want to edit and click the Edit or Edit OTS button. The Edit OTS window opens — If the intention for the L0 Control Plane software is to have visibility of the optical domain managed by the OTS, each OTS in the same domain must set the CPS value Enable. — If the intention for the L0 Control Plane software is not to have visibility of the optical domain managed by the OTS, each OTS in the same domain must set the CPS value to Disable. Refer to the “Editing an OTS instance in the Photonic Configuration Management application” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310 for more information.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-65

Co-Routed SNC Degraded

Alarm ID: 1827

Probable cause

This alarm is raised when some (but not all) of the terminating Subnetwork Connection (SNC) members are in the creating or starting state. It is typically caused by a destination unreachable condition caused by insufficient bandwidth, lack of matching service classes, or lack of physical facility to the destination port.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Follow the “SNC Unavailable” alarm clearing procedure in this document. This alarm is cleared when all terminating Subnetwork. Connection (SNC) members have finished creating and starting state.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-66

Co-Routed SNC Unavailable

Alarm ID: 1826**Probable cause**

This alarm is raised when all the terminating Subnetwork Connection (SNC) members are in the creating or starting state. It is typically caused by a destination unreachable condition caused by insufficient bandwidth, lack of matching service classes, or lack of physical facility to the destination port.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Follow the “SNC Unavailable” alarm clearing procedure in this document. This alarm is cleared when any terminating Subnetwork Connection (SNC) member has finished creating and starting state.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-67

Corrupt Inventory Data

Alarm ID: 656

Probable cause

The alarm is raised against the cooling unit housing, cooling fan module, access panel, MIC, Power Input Card, external equipment connected to the remote inventory ports on the access panel and module in a sub-slot of connected external equipment which is:

- not recognized by the NE
- not fully inserted
- defective
- connected using a defective cable

Impact

Critical, service-affecting (C, SA) alarm if active on cooling unit housing or fans (when at least one other fan related alarm is present)

Major, non-service-affecting (M, NSA) alarm if active on a fan (when no other fan-related alarm is present) or when raised on the external slot inventory interface of the shelf

Minor, non-service-affecting (m, NSA) alarm if active on the access panel, MIC, or power circuit packs

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
------	--------



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

1	If the alarm is raised against	Then go to
	the cooling unit housing	step 2
	the cooling unit module (fans)	step 12
	any other unit	step 16

Procedure 4-67 (continued)
Corrupt Inventory Data

Step	Action								
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
3	Remove all fans from the cooling unit housing, one at a time (if present), starting with any fans that have alarms. Refer to the “Replacing a cooling fan module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
ATTENTION Fans with Corrupt Inventory Data or Fan Incompatible alarms must be removed from the housing before any other fans are removed.									
4	Verify that Fan Failed alarms are raised for each fan, and any existing fan alarms are cleared.								
5	Remove the cooling unit housing from the 6500 shelf. Refer to the “Replacing the cooling unit assembly” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. You cannot remove the cooling unit housing on the metro front electrical shelf, as it is an integral part of the shelf.								
6	Verify that Corrupt Inventory Data alarm for the cooling unit housing has cleared.								
7	Replace with new housing. Refer to the “Replacing the cooling unit assembly” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
8	Wait 30 seconds to verify that no new Corrupt Inventory Data alarm is raised against cooling unit housing. Fan Incompatible alarm is masked for any incompatible fans inserted after a Corrupt Inventory Data alarm has been raised against the cooling unit housing. If the cooling unit housing has corrupt inventory data upon system bootup or replacement, any incompatible fans in the housing will not have their incompatibility alarmed.								
9	Reinsert fan 1. Verify that the Fan Failed alarm has cleared. Wait at least 15 seconds before proceeding.								
10	Repeat step 9 for fan 2 and 3. Fans must be inserted one at a time, with at least 15 seconds of waiting time between fan insertions. Verify that all Fan Failed alarms clear.								
11	<table> <tr> <th>If the</th><th>Then</th></tr> <tr> <td>original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>alarm is raised against the fan module</td><td>go to step 12</td></tr> <tr> <td>Fan Incompatible alarm is raised</td><td>perform the alarm clearing procedure for “Fan Incompatible” on page 4-287</td></tr> </table>	If the	Then	original alarm has cleared	the procedure is complete	alarm is raised against the fan module	go to step 12	Fan Incompatible alarm is raised	perform the alarm clearing procedure for “ Fan Incompatible ” on page 4-287
If the	Then								
original alarm has cleared	the procedure is complete								
alarm is raised against the fan module	go to step 12								
Fan Incompatible alarm is raised	perform the alarm clearing procedure for “ Fan Incompatible ” on page 4-287								

Procedure 4-67 (continued)
Corrupt Inventory Data

Step	Action						
12	Remove alarmed fan from cooling unit housing. The Fan Failed alarm is raised for the removed fan and the Corrupt Inventory Data alarm for that fan clears.						
13	Wait at least 15 seconds. Insert a new fan into the slot. Ensure that the fan is fully inserted and the bolt is attached to the fan housing.						
14	Verify that the Fan Failed alarm has cleared and that no new alarm is raised.						
15	<table> <tr> <th>If the</th><th>Then</th></tr> <tr> <td>original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>Fan Incompatible alarm is raised</td><td>perform the alarm clearing procedure for "Fan Incompatible" on page 4-287</td></tr> </table>	If the	Then	original alarm has cleared	the procedure is complete	Fan Incompatible alarm is raised	perform the alarm clearing procedure for "Fan Incompatible" on page 4-287
If the	Then						
original alarm has cleared	the procedure is complete						
Fan Incompatible alarm is raised	perform the alarm clearing procedure for "Fan Incompatible" on page 4-287						
16	Verify that the alarmed unit is fully inserted.						
17	Verify that the cable between the shelf External Slot Inventory port and the external peripheral is connected and not damaged.						
18	If applicable, perform a restart of the alarmed unit. Refer to Procedure 2-11, "Restarting a circuit pack or shelf processor". If a restart is not possible on the unit, go to step 20.						
19	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 20</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 20
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 20						
20	Reseat the alarmed unit, or disconnect and reconnect the cable connecting the alarmed module (the cable between the shelf External Slot Inventory port and the external peripheral). Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
21	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 22</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 22
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 22						
22	Replace the alarmed unit. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. The Circuit Pack Missing alarm is raised for that unit and the Corrupt Inventory Data for that unit clears.						
23	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-68

CPU2 Warm Restart Required

Alarm ID: 1425**Probable cause**

This alarm is raised when an SP-2 Dual CPU Processor (CPU2) warm restart is required after a Photonic OSRP instance is added or deleted.

This alarm can also be caused by changes made to the OSRP for Photonic Control Plane. For example, a change of the OOB local UDP port.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

Before you perform this procedure, you must use an account with level 3 or higher UPC.

Step	Action
1	Perform a warm restart of the CPU2 of the active SP-2 Dual CPU circuit pack. See Procedure 2-11, "Restarting a circuit pack or shelf processor" .
2	If the alarm does not clear following the restart of the shelf processor, contact your next level of support or your Ciena support group.

—end—

Procedure 4-69

Cross-connection Mismatch

Alarm ID: 863

Probable cause

This alarm is raised against the shelf when:

- one or more optical cross-connections recorded in the Optical Bandwidth Manager (OBM) database are mismatched with the existing configuration in the Shelf Wavelength Topology (SWT) database
- there are SWT instances that have no corresponding photonic cross-connections
- a passthrough channel is provisioned at a TOADM, but a CMD for that same group is provisioned, or the inter-OTS adjacencies are missing or they are incorrectly provisioned
- there is a mismatch between the cross-connects and the corresponding Shelf Wavelength Topology (TOPO-SWT) instances. In particular, the following mismatch conditions are alarmed when Autoroute is disabled:
 - the TOPO-SWT instance for a provisioned local add/drop cross-connect has a routing of UNKNOWN or an owner value that is not OBM.
 - one or both TOPO-SWT instances for a provisioned spur or intra-domain passthrough cross-connect have a routing of UNKNOWN or an owner value that is not OBM.
- there is a missing provisioned local add/drop, passthrough, or spur cross-connect for an existing TOPO-SWT instance with routing and owner set.

For CDC configurations, this alarm is raised when redundant slot sequences are provisioned. Note that redundant slot sequences are not supported. The alarm clears when you deprovision the slot sequences.

For Direction Independent Access (DIA), OBM audit raises this alarm if it detects an OCH with routing OBM_DI_ADD or OBM_DI_DROP and the corresponding Tx/Rx adjacency is OOS.

A Cross-connection (CRS) Mismatch alarm with the reason “OTS not Capable of NMC CRS” will be raised against a CRS-NMC if any OTS that CRS passes through is not enhance mode capable.

A Cross-connection (CRS) Mismatch alarm with the reason “OTS not Flexible Grid Capable” will be raised against a flexible CRS-NMC if any OTS that CRS passes through is not flexible grid capable.

Procedure 4-69 (continued)
Cross-connection Mismatch

A Cross-connection (CRS) Mismatch alarm with the reason “Invalid CRS End Point” may toggle after restarting the TIDc primary shelf, as the boundary port provisioning is re-synchronized between the primary shelf and its members.

ROADM/TOADM OTS is enhance mode capable if there is only one section in the domain, all Network Elements in the domain are running a release no later than Rel 12.1, and section head and tail SPs are SP2/SPAP-2/SP-3/CTM-SP. DIA/COADM OTS is enhance mode capable if the OTS is on SP2/APAP-2/SP-3/CTM-SP shelf and the shelf release is no later than Rel 12.1.

ROADM OTS is flexible grid capable if it is enhance mode capable and its actual control mode is 12.5 GHz. TOADM OTS is flexible grid capable if it is enhance mode capable. DIA/COADM OTS is flexible grid capable if it is enhance mode capable and WSS in DIA or SMD in COADM is flexible grid capable.

The following conditions determine whether a CRS-NMC is fixed (not flexible):

- Single CRS-NMC per MCTTP.
- CRS-NMC Center Frequency is aligned with a 50 GHz wide ITU grid channel.
- CRS-NMC Width is greater than 12 GHz and not greater than 42 GHz if MCTTP width is 50 GHz, or CRS-NMC Width is greater than 12 GHz and not greater than 92 GHz if MCTTP width is 100 GHz.
- CRS-NMC must be in the center of the MCTTP.
- MCTTP width must be 50 GHz or 100 GHz if NMC center frequency is divisible by 100 GHz, or 50 GHz if NMC center frequency is divisible by 50 GHz but not 100 GHz.

Impact

Major, service-affecting (M, SA) alarm

Procedure 4-69 (continued)
Cross-connection Mismatch

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	If the alarm is raised because of “OTS not Capable of NMC CRS”, delete the CRS-NMC and provision CRS-OCH with equivalent wavelength on the same ports.
2	If the alarm is raised because of “OTS not Flexible Grid Capable”, delete the flexible CRS-NMC and provision fixed CRS-NMC, or replace any fixed grid equipment with equivalent flexible grid capable equipment and re-provision the flexible CRS-NMC. When flex grid domains interwork with 100 GHz WSS fixed grid domains, the following rules must be followed to avoid CRS Mismatch alarm: • If a Permanent SNCG is used for provisioning the channel, the SNCG Lower Frequency Filter-edge Spacing (GHz) and Upper Frequency Filter-edge Spacing (GHz) parameters are set according to the following rule: for Ciena WLAI-56 and WSAI-56 transponders, set the Lower Frequency Filter-edge Spacing (GHz) and Upper Frequency Filter-edge Spacing (GHz) parameters to 20.5; for other Ciena transponder types, set the Lower Frequency Filter-edge Spacing (GHz) and Upper Frequency Filter-edge Spacing (GHz) parameters to 31.25. • If a Permanent SNC is used for provisioning the channel, the ADJ-TX facility Lower Frequency Guard-Band (GHz) and Upper Frequency Guard-Band (GHz) parameters are set according to the following rule: for Ciena WLAI-56 and WSAI-56 transponders, set the Lower Frequency Guard-Band (GHz) and Upper Frequency Guard-Band (GHz) parameters to 41; for other Ciena transponder types, set the Lower Frequency Guard-Band (GHz) and Upper Frequency Guard-Band (GHz) parameters to 62.5. If the alarm does not clear, go to the next step.
3	Using the Photonic Connections application in Site Manager, retrieve the Photonic cross-connects with an OCH rate for the affected shelf. From the Mismatch column, note the wavelength and the direction of the connection(s) with a mismatch value other than “NONE”.
4	If possible, switch traffic away from the link with the mismatch.

Procedure 4-69 (continued)
Cross-connection Mismatch

Step	Action						
5	If the mismatch value identified in step 3 is Then go to <table> <tr> <td>TO</td><td>step 6</td></tr> <tr> <td>FROM</td><td>step 8</td></tr> <tr> <td>BOTH</td><td>step 10</td></tr> </table>	TO	step 6	FROM	step 8	BOTH	step 10
TO	step 6						
FROM	step 8						
BOTH	step 10						
6	Sort the data in the Photonic cross-connect list from step 3 by the To column. Find and record all the cross-connects that share the same value in the To column.						
7	Only one cross-connection is permitted to use a given To value. Correct the provisioning error(s) so that only a single cross-connect exists for the duplicated To value. This may require: • deleting a cross-connect. • placing an ADJ-RX facility OOS (provision the CMD44 ADJ-RX Receiver type to UNKNOWN) to delete a channel for which a derived/provisioned pass-through cross-connect was created. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Following the channel deletion, verify the channel no longer appears in channel listing of the Domain Optical Controller (DOC) application in Site Manager. Refer to the “Retrieving Domain Optical Controller summary and details view” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. • placing an ADJ-TX facility OOS (provision the CMD44 ADJ-TX Transmitter type to UNKNOWN) to delete a channel for which a derived/provisioned pass-through cross-connect was created. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Following the channel deletion, verify the channel no longer appears in channel listing of the Domain Optical Controller (DOC) application in Site Manager. Go to step 12.						

Procedure 4-69 (continued)
Cross-connection Mismatch

Step	Action				
8	Sort the data in the Photonic cross-connect list from step 3 by the From column. Find and record all the cross-connects that share the same value in the From column.				
9	The following rules apply to the cross-connects that share a given From value: - a maximum of one local drop cross-connection is allowed - a maximum of one non-broadcast pass-through cross-connection is allowed Correct the provisioning error(s) so that the cross-connections sharing the same From value comply with the rules in this step. This may require: - deleting a cross-connect. - placing an ADJ-RX facility OOS (provision the CMD44 ADJ-RX Receiver type to UNKNOWN) to delete a channel for which a derived/provisioned pass-through cross-connect was created. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Following the channel deletion, verify that the channel no longer appears in channel listing of the Domain Optical Controller (DOC) application in Site Manager. Refer to the “Retrieving Domain Optical Controller summary and details view” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - placing an ADJ-TX facility OOS (provision the CMD44 ADJ-TX Transmitter type to UNKNOWN) to delete a channel for which a derived/provisioned pass-through cross-connect was created. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Following the channel deletion, verify the channel no longer appears in the channel listing of the Domain Optical Controller (DOC) application in Site Manager. Refer to the “Retrieving Domain Optical Controller summary and details view” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 12.				
10	Follow step 6 and step 8 to resolve the mismatches (of the To and From values) identified in step 3 .				
11	If the original alarm has Then go to <table> <tr> <td>cleared</td><td>step 12</td></tr> <tr> <td>not cleared</td><td>step 15</td></tr> </table>	cleared	step 12	not cleared	step 15
cleared	step 12				
not cleared	step 15				
12	Wait 15 minutes for the next OBM audit to run.				
13	Using the Photonic Connections application in Site Manager, retrieve the Photonic cross-connects and verify that the mismatch is resolved.				

Procedure 4-69 (continued)
Cross-connection Mismatch

Step	Action						
14	Switch traffic back to the link (if you have removed it in step 4). The procedure is complete.						
15	Select Configuration->Photonic Services->Shelf Wavelength Topology and retrieve the SWT instances from all the shelves in the node.						
16	Select Configuration->Cross Connections->Photonic Connections and retrieve the cross-connection information in the node.						
17	Determine which SWT instances have missing cross-connections: - Each SWT instance of routing Drop must have a corresponding derived cross-connection. - Each SWT instance of routing Add must have a corresponding derived cross-connection. - Two SWT instances of routing Passthrough must have a corresponding cross-connection, one SWT instance for the "From OCH" portion of the cross-connection, and one SWT instance for the "To OCH" portion of the cross-connection.						
18	Select your next step for SWT instances that are missing a cross-connection. <table> <tr> <th>If this is</th><th>Then</th></tr> <tr> <td>a ROADM site</td><td>perform the same steps as when a cross-connection has a Mismatch value of From and/or To, that is, step 4 to step 14.</td></tr> <tr> <td>a TOADM</td><td>see "Adjacency Provisioning Error" on page 4-42 for the steps required to clear the alarm.</td></tr> </table>	If this is	Then	a ROADM site	perform the same steps as when a cross-connection has a Mismatch value of From and/or To, that is, step 4 to step 14 .	a TOADM	see " Adjacency Provisioning Error " on page 4-42 for the steps required to clear the alarm.
If this is	Then						
a ROADM site	perform the same steps as when a cross-connection has a Mismatch value of From and/or To, that is, step 4 to step 14 .						
a TOADM	see " Adjacency Provisioning Error " on page 4-42 for the steps required to clear the alarm.						
19	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-70

Crossed Fibers Suspected

Alarm ID: 1508

Probable cause

This alarm is raised against the ADJ facility of the associated LIM port (MON port) when the received power by the far-end is higher than the transmitting end due to a crossed fiber.

Crossed fiber is only detectable when there is a difference of at least 3 dB in the expected powers at the OPM ports. One port will have higher than expected power, the other port will have lower than expected power. The port with the higher than expected power (more than 3 dB) will raise the “Crossed Fibers Suspected” alarm. The port with lower than expected power will raise the “High Fiber loss” alarm if Fiber Loss Detection is enabled.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure

- you require an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545

Step	Action
1	Verify the adjacency of the OPM ports and the system diagrams of the shelf.
2	Trace the fiber from the OPM port to the associated monitor port on the LIM.
3	Ensure the fiber is exactly the same as the adjacencies.
4	If the fibers are crossed, uncross them.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-71

Dark Fiber Loss Measurement Disabled

Alarm ID: 1825

Probable cause

This event is raised when the Dark Fiber Loss Measurement is disabled on a shelf with CDC equipment provisioned.

If any CDC equipment was ever provisioned at least once on the shelf, this event is raised when the system parameter “Dark Fiber Loss Measurement” is set to OFF in Site Manager Node Information System tab, using the Edit command. By default, this parameter value is ON.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Open the Node Information Site Manager application and select the System tab. Click Edit then set the “Dark Fiber Loss Measurement” system parameter to On. Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i>, 323-1851-301. Note: The system parameter “Dark Fiber Loss Measurement” can be changed from a member shelf or a primary shelf (if shelf Sync is ON, the “Dark Fiber Loss Measurement” cannot be changed from a member shelf). If the system parameter is changed on a primary shelf, it will broadcast to all member shelves.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-72

Database Auto Save in Progress

Alarm ID: 1047

Probable cause

This alarm is raised when an automatic database backup is initiated.

Note: This alarm is disabled by default.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the database backup is completed.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-73

Database Integrity Fail

Alarm ID: 8**Probable cause**

This alarm is raised when the provisioning data either on the shelf processor or for a given slot is in an inconsistent or corrupted state.

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	If the alarm is raised against a specific circuit pack, Warm restart the circuit pack with this alarm. See Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document. If after 10 minutes the alarm clears, then this procedure is complete.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-74 Database Corruption

Alarm ID: 1241

Probable cause

This alarm is raised when Layer 0 Control Plane or Photonic Control Application have issues recovering its own database.

The alarm is also raised when you restore from backup and then commit fails, or if any inconsistencies are found in the SNC/SNCG during recovery.

Impact

Major, Service-affecting (M, SA) alarm

Step	Action
------	--------

- | | |
|---|---|
| 1 | Contact your next level of support or your Ciena support group. |
|---|---|

—end—

Procedure 4-75

Database Not Recovered For Slot

Alarm ID: 1390**Probable cause**

This alarm is raised if provisioning data for a provisioned circuit pack is not recovered over a system or shelf processor power-up/restart. The alarm indicates the slot for which data corruption is detected.

The alarm clears when the provisioning data is corrected or a data restore is performed as a result of clearing related higher severity alarms.

Impact

Warning

Step	Action
------	--------

- | | |
|---|--|
| 1 | No action is required. The alarm clears as a result of clearing the higher severity alarm “Database Recovery Incomplete”, that is associated with provisioning data. Refer to the “Database Recovery Incomplete” procedure in this document. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-76

Database Recovery Incomplete

Alarm ID: 205

Probable cause

This alarm is raised when the shelf processor is replaced and there is a missing, mismatched or failed circuit pack in the shelf. The alarm is the result of a data consistency audit depending on whether any equipment data was not recovered.

The data consistency audit detects problems when a shelf processor is replaced and a transport circuit pack is not able to send its data to the shelf processor. The audit runs on every restart to ensure the integrity of data such as connections, synchronization, and protection.

For each slot, the audit compares provisioned equipment known to the shelf processor to that recovered from the cards. If there is a mismatch in the comparison, this alarm is raised and then no data will be sent to the cards. Database save operations will also be blocked.

The data consistency audit requires at least one XC circuit pack, and does not run if it detects that it was run and previously found problems.

ATTENTION

Do not remove or insert any circuit packs during this procedure.

Impact

Major, non-service-affecting (M, NSA) alarm

The data transfer to the transport circuit pack is blocked until the data consistency audit is successfully completed.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-76 (continued)
Database Recovery Incomplete

Step	Action
1	Retrieve the current events and alarms. Refer to the “Viewing events” section in this document. ATTENTION If there is a Software Upgrade In Progress alarm, complete the upgrade before continuing this procedure.
2	From the list of alarms find the “Database Not Recovered For Slot” alarm. The “Database Not Recovered For Slot” alarm is raised against the slot for which data is missing.
3	Complete a database restore. Refer to the backup and restore procedures in <i>Administration and Security</i> , 323-1851-301, for more information.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-77

Database Restore in Progress

Alarm ID: 143

Probable cause

This alarm is raised when the provisioning data is being restored to a standalone shelf or to one or multiple shelves in a consolidated TID.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- be able to connect to the shelf processor
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|--|
| 1 | Wait until the currently ongoing database restore successfully completes. Refer to the “Restoring provisioning data” procedure in <i>Administration and Security</i> , 323-1851-301.

To clear the alarm and abort the database restore action, click Cancel in the Backup and Restore application. Canceling recovers the system and cleans up any backup files left in invalid states. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-78

Database Save Failed

Alarm ID: 1263

Probable cause

This alarm is raised when the shelf processor detects that a save command sent to the shelf processor fails.

This alarm can indicate that Site Manager detected a process listening on the port specified for SFTP. When Site Manager detects a process on the SFTP port, it does not launch the integrated Site Manager SFTP server, and the operation fails.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- be able to connect to the shelf processor
- use an account with at least a level 3 UPC

Step	Action						
1	If Site Manager indicates that another process is running on the specified SFTP port, stop any processes running on the specified SFTP port to allow Site Manager to launch the integrated SFTP server as follows: • on the Linux-based system running Site Manager, enter the following command: <pre>\$ kill -9 process_pid</pre>						
2	Select your next step.<table><tr><th>If the alarm</th><th>Then</th></tr><tr><td>clears</td><td>this procedure is complete</td></tr><tr><td>did not clear</td><td>go to step 3</td></tr></table>	If the alarm	Then	clears	this procedure is complete	did not clear	go to step 3
If the alarm	Then						
clears	this procedure is complete						
did not clear	go to step 3						

Procedure 4-78 (continued)
Database Save Failed

Step	Action						
3	Select your next step. <table> <tr> <th>If you want to</th><th>Then go to</th></tr> <tr> <td>clear the alarm by canceling the action</td><td>step 4</td></tr> <tr> <td>clear the alarm and retry the action</td><td>step 5</td></tr> </table>	If you want to	Then go to	clear the alarm by canceling the action	step 4	clear the alarm and retry the action	step 5
If you want to	Then go to						
clear the alarm by canceling the action	step 4						
clear the alarm and retry the action	step 5						
4	Click Cancel in the Backup and Restore application. Refer to the “Saving provisioning data” or “Restoring provisioning data” procedures in <i>Administration and Security</i>, 323-1851-301, for more information. Canceling cleans up any backup files left in invalid states. Note: The “Database Save Failed” alarm can not be cleared by Clicking “Cancel” in the Backup And Restore application when a shelf upgrade is in progress.						
5	This alarm can be cleared if you try to backup or restore the provisioning data again. The alarm clears if the backup or restore action is successful. Refer to the “Restoring provisioning data” procedure in <i>Administration and Security</i>, 323-1851-301, for more information.						
6	If the alarm does not clear, check the network element to determine whether a condition exists that can prevent a save. These conditions include: • a “Software Upgrade in Progress” alarm is active • a database save is already in progress • a “Software Mismatch” alarm is active • the software version on the shelf processor is different from the other circuit packs • a “Disk Full” alarm is active • an unsaved SAOS CLI config — Save SAOS CLI configurations from the primary PKT/OTN cross-connect/eMOTR circuit packs and then re-issue the restore operation • a corruption in the network element database is detected (indicated by a Transport Data Recovery Failed, a Database Recovery Incomplete, or a Switch Shelf ID Mismatch detected alarm) • active alarms and standing conditions are present unless you specify the backup to ignore active alarms • the NE mode is “Unknown” • Site Manager detected a process listening on the port specified for SFTP						

Procedure 4-78 (continued)

Database Save Failed

Step	Action
7	When database save is failed, TC failure messages will appear in the Site Manager. Refer to <i>Fault Management - Customer Visible Logs</i> , 323-1851-840, for more information.
8	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-79

Database Restore Failed

Alarm ID: 1264

Probable cause

This alarm is raised when the shelf processor detects that a restore command sent to the shelf processor fails.

This alarm can indicate that Site Manager detected a process listening on the port specified for SFTP. When Site Manager detects a process on the SFTP port, it does not launch the integrated Site Manager SFTP server, and the operation fails.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- be able to connect to the shelf processor
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|--|
| 1 | If Site Manager indicates that another process is running on the specified SFTP port, stop any processes running on the specified SFTP port to allow Site Manager to launch the integrated SFTP server as follows: |
|---|--|

- on the Linux-based system running Site Manager, enter the following command:

```
$ kill -9 process_pid
```

- | | |
|---|------------------------|
| 2 | Select your next step. |
|---|------------------------|

If the alarm	Then
---------------------	-------------

clears	this procedure is complete
--------	----------------------------

did not clear	go to step 3
---------------	------------------------------

Procedure 4-79 (continued)
Database Restore Failed

Step	Action						
3	Select your first step. <table> <tr> <th>If you want to</th><th>Then go to</th></tr> <tr> <td>clear the alarm by canceling the action</td><td>step 4</td></tr> <tr> <td>clear the alarm and retry the action</td><td>step 5</td></tr> </table>	If you want to	Then go to	clear the alarm by canceling the action	step 4	clear the alarm and retry the action	step 5
If you want to	Then go to						
clear the alarm by canceling the action	step 4						
clear the alarm and retry the action	step 5						
4	Click Cancel in the Backup and Restore application. Refer to the “Canceling backup and restore operation” procedure in <i>NBI Fundamentals</i>, 323-1851-165, for more information. Canceling cleans up any backup files left in invalid states.						
5	This alarm can also clear if you try to restore the provisioning data. The alarm clears if the restore action is successful. Refer to the “Restoring provisioning data” procedure in <i>Administration and Security</i>, 323-1851-301, for more information.						
6	If the alarm does not clear, check the network element to determine whether a condition exists that can prevent a restore. These conditions include: • a “Software Upgrade in Progress” alarm is active • a database save is already in progress • a “Software Mismatch” alarm is active • the software version on the shelf processor is different from the other circuit packs • a “Disk Full” alarm is active • a corruption in the network element database is detected (indicated by a Transport Data Recovery Failed, a Database Recovery Incomplete, or a Switch Shelf ID Mismatch detected alarm) • active alarms and standing conditions are present unless you specify the backup to ignore active alarms • the NE mode is “Unknown” Checklist for Database Restore specific failure (these failures appear in Site Manager when a database restore action fails). • Database Restore Failed: Invalid source — Check the database filename prefix. If the file identifier is used in the database filename then the user must use the Use filename starting with checkbox option to match with the database filename prefix. — Check the shelf number in the database filename prefix. If it does not match with the NE shelf number, then the user must use the Use filename with shelf number checkbox option to match with the database shelf number prefix.						

Procedure 4-79 (continued)

Database Restore Failed

Step	Action
	- Database Restore Failed: Backup not from this node - This failure indicates the node name saved in the database and the NE node name don't match. It is either the database was saved in the different shelf or the current NE node name had been changed after the database save. You can specify the restore to ignore the node name check by uncheck the Do not restore if data was not backed up from this NE checkbox. - Database Restore Failed: Mismatched Software Releases - Ensure the database software release is not different from the NE software release. - Database Restore Failed: XC Application Failed - This failure may indicate that the standby XC is not ready for switch of activity. If the standby XC is Out-of-Service, place the equipment In-Service and retry the Database Restore operation. For all cases, contact your next level of support or your Ciena support group. - Database Restore Failed: Incompatible S/R options - If database was saved with the communication setting option by checking the Do not restore if data was not backed up from this NE check box. Ensure this checkbox is checked on the restore. - If database was saved without checking the Do not restore if data was not backed up from this NE check box. Ensure this checkbox is not checked on the restore. - Database Restore Failed: SP Protection Not Provisioned - This failure indicates that the user tried to restore the database with no SP protection provisioned on the active SP in the mate slot (for example, slot 16 in 14-slot shelf or slot 42 in 32-slot shelf). Only the database with SP protection provisioned is allowed to be restored on SP in mate slot. - Database Restore Failed: Incompatible Shelf Assembly - This failure indicates that the user tried to restore the database from a different shelf type or backplane type (for example, an attempt to restore a 32-slot database to the 14-slot shelf). This procedure is not allowed since the database is not compatible. - Database Restore Failed: Incompatible SP - This failure indicates that the user tried to restore the SP-3 database to the NE SP-2. This procedure is not allowed since the SP-2 cannot support some SP-3 functionality. - Database Restore Failed: FTP/SFTP access denied - Ensure the IP address, directory path, userID, and password are valid for the remote host.

Procedure 4-79 (continued)

Database Restore Failed

Step Action

- Database Restore Failed: Could not connect to destination
 - Ensure there are no issues with FTP server on the remote host.
- Database Restore Failed: Could not connect to Source
 - Ensure there are no issues with FTP server on the remote host.
- Database Restore Failed: Blocked By Unsaved CLI Config
 - Save SAOS CLI configurations from the primary PKT/OTN cross-connect/eMOTR circuit packs and then re-issue the restore operation.
- Database Restore Failed: Configuration Mismatch
 - Ensure the control plane configuration on the shelf matches the control plane configuration in the backup.

An FTP server can handle up to a specified number of simultaneous FTP sessions (for example 50); when performing a Restore, the Maximum Transfer Session parameter should be set to a number that the FTP server can handle.

- 7** If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-80

Database Commit Failed

Alarm ID: 1265

Probable cause

This alarm is raised when the shelf processor detects that a Commit command sent to the shelf processor fails.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- be able to connect to the shelf processor
- use an account with at least a level 3 UPC

Step	Action						
1	Select your first step. <table><tr><td>If you want to</td><td>Then go to</td></tr><tr><td>clear the alarm by canceling the action</td><td>step 2</td></tr><tr><td>clear the alarm and retry the action</td><td>step 3</td></tr></table>	If you want to	Then go to	clear the alarm by canceling the action	step 2	clear the alarm and retry the action	step 3
If you want to	Then go to						
clear the alarm by canceling the action	step 2						
clear the alarm and retry the action	step 3						
2	Click Cancel in the Backup and Restore application. Refer to the “Restoring provisioning data” procedure in <i>Administration and Security</i> , 323-1851-301, for more information. Canceling cleans up any backup files left in invalid states.						
3	This alarm can also clear if you try to backup or restore the provisioning data again. The alarm clears if the backup or restore action is successful. Refer to the “Restoring provisioning data” procedure in <i>Administration and Security</i> , 323-1851-301, for more information.						

Procedure 4-80 (continued)

Database Commit Failed

Step	Action
4	If the alarm does not clear, check the network element to determine whether a condition exists that can prevent a commit. These conditions include: • a “Software Upgrade in Progress” alarm is active • a database save is already in progress • a “Software Mismatch” alarm is active • the software version on the shelf processor is different from the other circuit packs • a “Disk Full” alarm is active • a corruption in the network element database is detected (indicated by a Transport Data Recovery Failed, a Database Recovery Incomplete, or a Switch Shelf ID Mismatch detected alarm) • active alarms are present unless you specify the backup to ignore active alarms • the NE mode is “Unknown”
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-81 Database Save in Progress

Alarm ID: 147

Probable cause

This alarm is raised while a database save is in progress and clears when the save is completed or has failed.

Impact

Warning

Step	Action
------	--------

- | | |
|---|--|
| 1 | No action is required. The alarm clears when database save is completed or failed. |
|---|--|

—end—

Procedure 4-82

Debug Port in Use

Alarm ID: 1132**Probable cause**

This alarm is raised when a user with a UPC 4 or greater logs in to the debug port.

The “Debug Port In Use” alarm is disabled by default and can be enabled using the Site Manager Configuration->Alarms & Controls->Alarm Profiles application.

The alarm severity can also be provisioned.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | No action is required. The alarm clears when the debug port is no longer in use. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-83

Delay Measurement Enabled on Slave Node

Alarm ID: 1418

Probable cause

This alarm is raised when latency discovery is enabled on an OSRP secondary node.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Disable latency discovery on secondary node. Refer to the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-84

Delay Measurement Mismatch Capability

Alarm ID: 1417**Probable cause**

This alarm is raised when a link is up between the primary node and the secondary node and the user enables latency discovery on primary node on that link. Then the primary OSRP node will be capable of delay measurements while the secondary node is not capable of performing delay announcements.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Disable latency discovery flag on primary node. Refer to the “Editing an OSRP link” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-85

Delay Measurement Timeout

Alarm ID: 2158

Probable cause

This warning is raised when the far end OSC facility is not responding to the originating delay measurement request.

The Delay Measurement Timeout warning can be raised after the Second Invoke Upgrade stage if the OSC facility is on the 2xOSC, SRA, ESAM or SAM circuit pack or after the First Invoke Upgrade stage if the OSC facility is on the SPAP-2 w/2xOSC circuit pack.

OSC Loss of Signal, Loss of Frame, and Line RDI alarms will not trigger the OSC Delay Measurement Timeout warning.

Impact

Warning

Step	Action
1	Enable the Delay Measurement on the far end OSC. Refer to <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	If the warning did not clear, disable the Delay Measurement on the OSC that raised the warning.
3	If the warning does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-86

Deskew Fail

Alarm ID: 2032, 2103, 2151, 2211

Probable cause

This alarm is raised when the Virtual Lane (VL) Skew exceeds the device capability for the ETTP (ETH100G), PTP, FACGRP, or OTUTTP client facilities that use parallel optics, or if duplicate alignment markers are detected.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- have the fiber connection information
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- if required, obtain a replacement QSFP28
- use an account with at least a level 3 UPC

Step	Action						
1	Determine the type of client service on the subtending equipment connected to the port of the alarmed circuit pack.						
2	Determine the facility provisioned on the port of the circuit pack reporting the alarm.						
3	<table border="0"> <tr> <td>If the subtending equipment noted in step 1 is</td><td>Then go to</td></tr> <tr> <td>not provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2</td><td>step 4</td></tr> <tr> <td>provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2</td><td>step 6</td></tr> </table>	If the subtending equipment noted in step 1 is	Then go to	not provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2	step 4	provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2	step 6
If the subtending equipment noted in step 1 is	Then go to						
not provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2	step 4						
provisioned with a OTUTTP, PTP, FACGRP, or ETTP service as required by the facility noted in step 2	step 6						
4	The WLAi MOTR and WLAi MOTR w/OPS, WLAi FOTR, and WLAi FOTR w/OPS circuit packs support various client services. If possible, provision the subtending equipment for client services, otherwise contact your next level of support or your Ciena support group.						

Procedure 4-86 (continued)

Deskew Fail

Step	Action						
5	<table><tr><th>If the original alarm</th><th>Then</th></tr><tr><td>has cleared</td><td>the procedure is complete</td></tr><tr><td>has not cleared</td><td>go to step 6</td></tr></table>	If the original alarm	Then	has cleared	the procedure is complete	has not cleared	go to step 6
If the original alarm	Then						
has cleared	the procedure is complete						
has not cleared	go to step 6						
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
7	If applicable, ensure that the optical transceiver modules are the correct type. If required, replace the optical transceiver module. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-87

Disk Full alarms

Use this procedure to clear alarms associated with the shelf processor disk capacity.

Disk 75 percent Full

Alarm ID: 374

Probable cause

This alarm is raised when the disk is 75% full on the shelf processor, and clears when there is at least 30% free space.

The Disk 75 percent Full alarm is for information only and does not affect the operation of the shelf. It is recommended that you attempt to clear this alarm to prevent possible future problems if the disk becomes too full.

Impact

Minor, non-service-affecting (m, NSA) alarm

Disk 90 percent Full

Alarm ID: 375

Probable cause

This alarm is raised when the disk is 90% full on the shelf processor, and clears when there is at least 15% free space.

The Disk 90 percent Full alarm is for information only and does not affect the operation on the shelf. It is recommended that you attempt to clear this alarm to prevent possible future problems if the disk becomes too full.

Impact

Minor, non-service-affecting (m, NSA) alarm

Disk Full

Alarm ID: 146

Probable cause

This alarm is raised when the disk is full on the shelf processor.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-87 (continued)

Disk Full alarms

Step	Action
1	Delete any loads that you do not require on the disk. Refer to the “Deleting a software load” in <i>Administration and Security</i> , 323-1851-301.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-88

Disk Full - CPU2 alarms

Use this procedure to clear alarms associated with the Disk Full - CPU2.

Disk 75 percent Full - CPU2

Alarm ID: 1199

Probable cause

This alarm is raised when the disk space on CPU2 is 75% full on the shelf processor, and clears when there is at least 30% free space.

The Disk 75 percent Full - CPU2 alarm is for information only and does not affect the operation of the shelf. It is recommended that you attempt to clear this alarm to prevent possible future problems if the disk becomes too full.

Impact

Minor, non-service-affecting (m, NSA) alarm

Disk 90 percent Full - CPU2

Alarm ID: 1200

Probable cause

This alarm is raised when the disk space on CPU2 is 90% full on the shelf processor, and clears when there is at least 15% free space.

The Disk 90 percent Full alarm is for information only and does not affect the operation on the shelf. It is recommended that you attempt to clear this alarm to prevent possible future problems if the disk becomes too full.

Impact

Minor, non-service-affecting (m, NSA) alarm

Disk Full - CPU2

Alarm ID: 1201

Probable cause

This alarm is raised when the disk space on CPU2 is full on the shelf processor.

Impact

Minor, non-service-affecting (m, NSA) alarm

ATTENTION

Any time a Disk Full condition is reached, some applications or operations are blocked. For example, the system blocks upgrades, circuit pack provisioning, and initializations.

4-198 Alarm clearing procedures

Procedure 4-88 (continued)

Disk Full - CPU2 alarms

Step	Action
------	--------

- | | |
|---|---|
| 1 | Contact your next level of support or your Ciena support group. |
|---|---|

—end—

Procedure 4-89

Dispersion search Mode Scanning In Progress

Alarm ID: 2195**Probable cause**

This warning is raised on PTP facilities on the WLAi or WL5e line port to indicate that modem is performing the pre-compensation dispersion scanning for SEARCH dispersion mode.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the scanning progress is completed.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-90

DOC Action Failed: Delete

Alarm ID: 577

Probable cause

This alarm is raised when a Channel Delete command (automatic or manual) in DOC is requested and fails. Conditions that can cause this failure include:

- maintenance activities, such as module replacement, module restart, and fiber cut occurred during the Channel Delete operation
- an internal communications issue (for example, the ILAN port is not cabled)
- an optical disconnect, such as a fiber break, within the DOC span of control
- a provisioning error on the Rx or Tx adjacency
- an associated adjacency or AMP facility is deleted
- the DOC Action: Channel Delete was stopped before the deletion was completed

This alarm appears only on the affected DOC shelf.

This alarm is cleared if the **Clear DOC Alarms** button is clicked in the **DOC** application. However, this will not clear the underlying problem.

In the Enhanced Automation mode, this alarm will only clear after a successful delete channel command or after a restart, and only if there is no more partially deleted channel in the domain.

Note: It is recommended that DOC be placed OOS before inserting a standby SP. If it is not, and the DOC is set to automatically delete channels or you manually delete a channel during the standby SP insertion, the DOC Action Failed: Delete alarm can be raised after the insertion. In this case, the alarm will clear in approximately five minutes. For details on deleting channels, refer to the “Deleting optical channels” procedure in *Commissioning and Testing*, 323-1851-221.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC

Procedure 4-90 (continued)

DOC Action Failed: Delete

- have the engineering documentation package (EDP) containing provisioning details

Step	Action						
1	If maintenance activities, such as module replacement, module restart, or fiber cut occurred during the Delete channel operation, then this alarm will be active. Ensure that the maintenance activities have completed.						
2	Check for and clear active “DOC Invalid Photonic Domain” alarms using the alarm clearing procedure in this document.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Channel Controller: Unexpected Loss Detected • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Optical Line Fail • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first.						
5	<table> <tr> <th>If the original alarm has</th><th>Then go to</th></tr> <tr> <td>cleared</td><td>step 14</td></tr> <tr> <td>not cleared</td><td>step 6</td></tr> </table>	If the original alarm has	Then go to	cleared	step 14	not cleared	step 6
If the original alarm has	Then go to						
cleared	step 14						
not cleared	step 6						
6	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must be alarm free.						

Procedure 4-90 (continued)

DOC Action Failed: Delete

Step	Action						
	Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element. Refer to the “Displaying the DOC logs for the summary table” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then go to</th></tr> <tr> <td>cleared</td><td>step 14</td></tr> <tr> <td>not cleared</td><td>step 8</td></tr> </table>	If the original alarm has	Then go to	cleared	step 14	not cleared	step 8
If the original alarm has	Then go to						
cleared	step 14						
not cleared	step 8						
8	Verify, and if necessary, correct the following provisioned data (reference the EDP): - Tx and Rx adjacency parameters (Adjacency type and all power-related parameters) - Line adjacency parameter (Fiber Type) - AMP optical facility parameters (Mode, Target Gain, Target Power, and Target Peak Power) Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310, for information on how to display and edit adjacency and optical facility parameters.						
9	<table> <tr> <th>If the original alarm has</th><th>Then go to</th></tr> <tr> <td>cleared</td><td>step 14</td></tr> <tr> <td>not cleared</td><td>step 10</td></tr> </table>	If the original alarm has	Then go to	cleared	step 14	not cleared	step 10
If the original alarm has	Then go to						
cleared	step 14						
not cleared	step 10						
10	<table> <tr> <th>If the Channel Condition field displays</th><th>Then go to</th></tr> <tr> <td>a condition other than “Partially Deleted”</td><td>step 11</td></tr> <tr> <td>“Partially Deleted”</td><td>step 14</td></tr> </table>	If the Channel Condition field displays	Then go to	a condition other than “Partially Deleted”	step 11	“Partially Deleted”	step 14
If the Channel Condition field displays	Then go to						
a condition other than “Partially Deleted”	step 11						
“Partially Deleted”	step 14						
11	Click the Clear DOC Alarms button in the DOC window. This clears the alarm; however, the underlying problem is not cleared.						
12	Click the Re-optimize button, and wait until the optimization completes.						
13	<table> <tr> <th>If the optimization</th><th>Then go to</th></tr> <tr> <td>fails</td><td>Procedure 4-92, “DOC Action Failed: Optimize”</td></tr> <tr> <td>succeeds</td><td>step 14</td></tr> </table>	If the optimization	Then go to	fails	Procedure 4-92, “DOC Action Failed: Optimize”	succeeds	step 14
If the optimization	Then go to						
fails	Procedure 4-92, “DOC Action Failed: Optimize”						
succeeds	step 14						

Procedure 4-90 (continued)
DOC Action Failed: Delete

Step	Action
14	Re-attempt the Delete operation. ATTENTION Any channels that were in the delete queue are returned to their previous state when the deletion fails. These channels must be re-selected for deletion.
15	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-91

DOC Action Failed: Monitor

Alarm ID: 549

Probable cause

This alarm is raised when DOC is unable to monitor, and thus unable to determine if the Photonic Domain is optimal. Conditions that can cause this failure include:

- an internal communications issue (for example, the ILAN port is not cabled)
- an optical disconnect, such as a fiber break, within the DOC span of control
- a circuit pack or module within the DOC photonic domain was replaced or restarted

This alarm appears only on the affected DOC shelf.

This alarm is cleared if the DOC Primary State is changed to OOS. However, this will not clear the underlying problem.

Note 1: It is recommended that DOC be placed OOS before inserting a standby SP. If it is not, the DOC Action Failed: Monitor alarm can be raised after the insertion. In this case, the alarm will clear in approximately five minutes.

Note 2: When this alarm is raised, it indicates that DOC cannot determine the optimization state of the system. As a result, DOC maintains the current optimization state. If this alarm resulted from a comms issue, the optimization state displayed in DOC is most likely correct. If this alarm resulted from a system fault, such as a fiber cut, DOC may indicate the system is optimized when it is not. Once the fault is repaired, the system returns to its optimal state.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-91 (continued)
DOC Action Failed: Monitor

Step	Action						
1	If maintenance activities, such as module replacement, module restart, or fiber cut occurred while DOC automatic monitoring was running, then this alarm will be active. If this is the case, and the maintenance activities are complete, the alarm should clear autonomously during the next DOC Auto Monitor run. DOC Auto Monitor runs two minutes after the last DOC Auto Monitor operation completes.						
2	Check for and clear active alarms using procedure in this document. After the DOC Invalid Photonic Domain alarm clears, the DOC Action Failed: Monitor alarm should clear autonomously during the next DOC Auto Monitor run.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Optical Line Fail • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first. After the alarms in step 4 clear, the DOC Action Failed: Monitor alarm should clear autonomously during the next DOC Auto Monitor run.						

Procedure 4-91 (continued)

DOC Action Failed: Monitor

Step	Action						
5	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must be alarm free. Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element. Refer to the “Displaying the DOC logs for the summary table” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. After the other alarms and conditions clear, the DOC Action Failed: Monitor alarm should clear autonomously during the next DOC Auto Monitor run.						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-92

DOC Action Failed: Optimize

Alarm ID: 550

Probable cause

This alarm is raised when DOC is unable to perform a manual or automatic re-optimization. The conditions that can cause this failure include:

- maintenance activities, such as module replacement, module restart, and fiber cut occurred while DOC auto monitoring was running
- an internal communications issue (for example, the ILAN port is not cabled)
- an optical disconnect, such as a fiber break, within the DOC span of control
- a circuit pack or module within the DOC photonic domain was replaced or restarted

This alarm appears only on the affected DOC shelf.

This alarm can be raised when DOC is operating in either enhanced mode or enhanced auto monitor only mode. The alarm hold off time is about 3 to 5 minutes.

This alarm is cleared if the DOC **Primary state** is changed to OOS. However, this will not clear the underlying problem.

Note: It is recommended that DOC be placed OOS before inserting a standby SP. If it is not or you click **Re-Optimize** during the standby SP insertion, the DOC Action Failed: Optimize alarm can be raised after the insertion. In this case, the alarm will clear in approximately five minutes.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|--|
| 1 | <p>If maintenance activities, such as module replacement, module restart, or fiber cut occurred while DOC auto re-optimize was running, then this alarm will be active.</p> <p>If this is the case, and the maintenance activities are complete, the alarm should clear autonomously during the next DOC auto re-optimize run.</p> |
|---|--|

Procedure 4-92 (continued)

DOC Action Failed: Optimize

Step	Action						
2	Check for and clear active alarms using Procedure 4-96, "DOC Invalid Photonic Domain" in this document. After the DOC Invalid Photonic Domain alarm clears, the DOC Action Failed: Optimize alarm will clear autonomously during the next DOC Auto Monitor run.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Channel Controller: Unexpected Loss Detected • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Optical Line Fail • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first. After the preceding alarms clear, the DOC Action: Fault Detected alarm should clear autonomously during the next DOC auto re-optimize run.						

Procedure 4-92 (continued)
DOC Action Failed: Optimize

Step	Action						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must not have any active alarms. Use the DOC Logs window to view the DOC logs to determine whether another network element reported an error. Review the DOC logs from the other network element. After the other alarms and conditions clear, the DOC Action Failed: Optimize alarm should clear autonomously during the next DOC auto re-optimize run.						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-93

DOC Action: Fault Detected

Alarm ID: 873

Probable cause

This alarm is raised during DOC automatic fault detection, when DOC detects a fault or the action is not completed. The conditions that can cause this failure include:

- maintenance activities, such as module replacement, module restart, and fiber cut occurred while DOC monitoring was running
- a module power value crossed the operating threshold
- a module within the DOC photonic domain was replaced or restarted
- an internal communications issue (for example, the ILAN port is not cabled)
- an optical disconnect (such as a fiber break) within the DOC span of control
- malfunctioning hardware
- incorrect provisioning
- if the reference bandwidth of an AMP facility in the domain is not consistent with the actual control mode of the OTS facility

This alarm appears only on the affected DOC network element.

This alarm is cleared if the DOC **Primary state** is changed to OOS. However, this will not clear the underlying problem.

For the MLA2 w/VOA circuit pack if the VOA facility primary state is OOS, a “DOC Action: Fault Detected” alarm is raised.

This alarm can also be raised 15 to 30 minutes after a Submarine Branching unit system upgrade when alarm hold off timer expires.

Note 1: It is recommended that DOC be placed OOS before inserting the standby shelf processor (SP). If it is not, and no DOC action is being executed, the DOC Action: Fault Detected alarm can be raised after the standby SP insertion. In this case, the alarm will clear in approximately one minute.

Note 2: When this alarm is raised, the Channel Fault Status (CFS) is not necessarily updated.

Impact

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-93 (continued)

DOC Action: Fault Detected**Prerequisites**

To perform this procedure, you must:

- use an account with at least a level 4 UPC
- have the engineering documentation package (EDP) containing adjacency details

Step	Action						
1	If maintenance activities, such as module replacement, module restart, or fiber cut occurred during the fault detection operation, then this alarm will be active. Ensure that the maintenance activities have completed.						
2	If the alarm is raised after a Submarine Branching unit system upgrade, perform an SP warm restart on all the shelves in that domain (link) to clear the alarm. Refer to Procedure 2-11, "Restarting a circuit pack or shelf processor" in this document.						
3	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 4</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check for and clear any active " DOC Invalid Photonic Domain " alarm using the procedure in this document.						
5	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						

Procedure 4-93 (continued)

DOC Action: Fault Detected

Step	Action						
6	Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Channel Controller: Unexpected Loss Detected • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Optical Line Fail • GCC0 Link Failure • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first.						
7	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 8</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must be alarm free. Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element.						
9	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 10</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						

Procedure 4-93 (continued)
DOC Action: Fault Detected

Step	Action						
10	Verify, and if necessary, correct the following provisioned data (refer to the EDP): • Tx and Rx adjacency parameters (Adjacency type and all power-related parameters) • Line adjacency parameter (Fiber type) • AMP optical facility parameters (Primary state, Mode, Target Gain, Target Power, Reference Bandwidth, and Target Peak Power) • OPTMON facility parameters (Primary state) • CHC facility parameters primary state						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Run DOC fault detection again (wait at least one minute).						
13	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-94

DOC Consecutive Re-Opt Threshold Crossed

Alarm ID: 874

Probable cause

This alarm is raised when DOC detects that a re-optimization has required more iterations than the DOC Fuse Threshold. When this alarm is raised, the DOC mode is switched to “auto-monitor only”. The likely cause for a re-optimization requiring many iterations is unstable powers somewhere in the domain. This could be due to an issue with hardware or the fiber plant.

Note: The presence of this alarm does not impact the ability of the system to perform capacity changes.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 4 UPC
- have the engineering documentation package (EDP) containing provisioning details

Step	Action
1	If the alarm occurs during maintenance activities that could cause power variations, wait until these activities are complete.
2	Perform a manual re-optimization. If the re-optimization completes successfully, switch the DOC automation mode back to “enhanced mode”. This will clear the alarm.
3	Verify that there are no “Circuit Pack Fail” alarms against any Photonic equipment in the domain. If a “Circuit Pack Fail” alarm is present, replace the circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
4	Perform a manual re-optimization. If the re-optimization completes successfully switch the DOC automation mode back to “enhanced mode”. This will clear the alarm.
5	If the source of the power variations is not obvious or no variations can be found, contact your next level of support or your Ciena support group.

—end—

Procedure 4-95

DOC Domain Not Optimized

Alarm ID: 551

Probable cause

This alarm is raised when the system is not optimal, and the conditions that cause the alarm cannot be cleared by automatic re-optimization or automatic monitoring. The conditions that can cause this alarm include:

- a channel addition or deletion has failed (in the case of the alarm raised in a broadcast domain, the failed addition or deletion may have occurred within the primary domain where the traffic is broadcast)
- DOC has determined that the system is not optimal

This alarm appears only on the affected DOC shelf.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) containing adjacency details

Procedure 4-95 (continued)
DOC Domain Not Optimized

Step	Action														
1	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the "NMCC Action Failed: Add" alarm is active</td><td>"NMCC Action Failed: Add" on page 4-501</td></tr> <tr> <td>the "NMCC Action Failed: Delete" alarm is active</td><td>"NMCC Action Failed: Delete" on page 4-504</td></tr> <tr> <td>the DOC Action Failed: Monitor alarm is active</td><td>"DOC Action Failed: Monitor" on page 4-204</td></tr> <tr> <td>the DOC Action Failed: Optimize alarm is active</td><td>"DOC Action Failed: Optimize" on page 4-207</td></tr> <tr> <td>the DOC Action: Fault Detected alarm is active</td><td>"DOC Action: Fault Detected" on page 4-210</td></tr> <tr> <td>none of the above alarms are active</td><td>step 2</td></tr> </table>	If	Then go to	the "NMCC Action Failed: Add" alarm is active	"NMCC Action Failed: Add" on page 4-501	the "NMCC Action Failed: Delete" alarm is active	"NMCC Action Failed: Delete" on page 4-504	the DOC Action Failed: Monitor alarm is active	"DOC Action Failed: Monitor" on page 4-204	the DOC Action Failed: Optimize alarm is active	"DOC Action Failed: Optimize" on page 4-207	the DOC Action: Fault Detected alarm is active	"DOC Action: Fault Detected" on page 4-210	none of the above alarms are active	step 2
If	Then go to														
the "NMCC Action Failed: Add" alarm is active	"NMCC Action Failed: Add" on page 4-501														
the "NMCC Action Failed: Delete" alarm is active	"NMCC Action Failed: Delete" on page 4-504														
the DOC Action Failed: Monitor alarm is active	"DOC Action Failed: Monitor" on page 4-204														
the DOC Action Failed: Optimize alarm is active	"DOC Action Failed: Optimize" on page 4-207														
the DOC Action: Fault Detected alarm is active	"DOC Action: Fault Detected" on page 4-210														
none of the above alarms are active	step 2														
2	Wait for the re-optimization or DOC Auto Monitor run to complete, or click the Re-Optimize button to trigger an immediate re-optimization. If the re-optimization fails, follow the appropriate alarm clearing procedure(s) in this document for any alarms raised as a result of the failed optimization.														
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4								
If the original alarm has	Then														
cleared	the procedure is complete														
not cleared	go to step 4														
4	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must be alarm free. Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element.														
5	If the alarm does not clear, contact your next level of support or your Ciena support group.														

—end—

Procedure 4-96

DOC Invalid Photonic Domain

Alarm ID: 552

Probable cause

This alarm is active during system lineup and test (SLAT), and only appears on the affected DOC shelf. The alarm can be raised when a corruption of the DOC database is detected during SP restart recovery.

This alarm is also raised when the software detects that a provisioned channel exceeds this engineering limit of the neighboring domain count (21). When the provisioned channel exceeds this engineering limit, the DOC freezes to prevent DOC from acting on channels that are not properly added to the DOC Sequencer database. The alarm clears when the channel provisioning from channels that exceeded the neighboring domain is removed.

This alarm is also raised when the SSH configuration is mismatched. If the SSH configuration is mismatched, connection issues can occur which includes:

- SSH Key Exchange parameters (Host Key algorithm, Key exchange algorithm)
- SSH Cipher parameters
- SSH HMAC parameters

This alarm is raised when DOC cannot retrieve a valid Network Topology. Conditions that can cause this include:

- an internal communications issue (for example, the ILAN port is not cabled)
- an optical disconnect, such as a fiber break, within the DOC span of control
- an incorrectly provisioned shelf parameter
- an upstream circuit pack has undergone a restart operation. The alarm clears once the restart has completed
- more than two DOC shelves are provisioned within the optical system
- a channel has been optimized in the system, and an upstream Tx on the same channel has been provisioned. (To prevent this alarm from being raised, enter the second Rx adjacency for the reused wavelength before provisioning the second Tx adjacency.)

Procedure 4-96 (continued)

DOC Invalid Photonic Domain

- DOC is detecting that a DOC-controlled channel is expanded past its previous egress point. (In this case, the alarm clears when DOC detects that the expanded DOC-controlled channel is contracted back to its original egress point.)
- DOC is detecting that a DOC-controlled channel is contracted prior to its previous egress point. (In this case, the alarm clears when DOC detects that the contracted DOC-controlled channel is expanded back to its original egress point.)
- DOC is detecting that one or more DOC-controlled channels no longer has a topology present within the domain. This can occur on a database restore where there has been a capacity change since the database was collected, and where databases are restored only on a subset of nodes in the domain traversed by the modified channels.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) containing shelf details

Step	Action	
1	If the system is being SLATed	Then no action is required. The alarm clears after SLAT. The procedure is complete.
	already SLATed	go to step 2
2	Verify that both DOC site network elements are commissioned and connected to the network.	
3	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 4
4	Verify that the shelf parameters are correctly provisioned. Correct any discrepancies. Refer to the <i>Administration and Security</i> , 323-1851-301.	

Procedure 4-96 (continued)
DOC Invalid Photonic Domain

Step	Action						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	If a channel has been optimized in the system and an upstream Tx on the same channel has been provisioned, remove the Tx adjacency or add an Rx adjacency to terminate the wavelength properly. DOC clears the alarm after the topology rebuilds. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Check if any of the following alarms, which indicate an internal communications or equipment problem, are active within the DOC span of control, and clear them using the procedures in this document: • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Software Upgrade Failed • Database Integrity Fail • Database Recovery Incomplete • Database Restore in Progress • Database Restore Failed • Database Commit Failed • Duplicate IP Address • Duplicate Shelf Detected • LAN-15 Port Failure • LAN-16 Port Failure • OSPF Adjacency Loss alarms • Packet Rate Limit Exceeded • Shelf Data Missing • Software Auto-Upgrade in Progress						

Procedure 4-96 (continued)

DOC Invalid Photonic Domain

Step	Action						
9	If you have cleared any alarms in step 8, wait at least 15 minutes. <table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control: • Adjacency Far End Not Discovered • Adjacency Mismatch • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Optical Line Fail • OSC Loss of Signal Note: If the Optical Line Fail alarm is active, clear this alarm first.						
11	If you have cleared any alarms in step 10, wait at least 15 minutes. <table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Clear all other active alarms on the network elements within the DOC span of control. For optimal DOC operation, the system must be alarm free.						
13	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	Use the DOC logs to determine if database corruption was the cause. The DOC logs may show a message such as: Database corruption detected. Please restore the database or contact Ciena for support.						
15	If any database corruption is identified, restore from a backup, or contact Ciena to assist in fixing the corruption. Refer to the backup and restore procedures in <i>Administration and Security</i>, 323-1851-301, for more information.						

Procedure 4-96 (continued)
DOC Invalid Photonic Domain

Step	Action
16	If the original alarm has cleared
	Then the procedure is complete
	not cleared go to step 17
17	Use the DOC logs to determine if a DOC-controlled channel has expanded past its previous egress point. The Overall Status for the channel appears as "Network Topology unavailable". Refer to the Domain Optical Controller (DOC) procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. The DOC logs may show a message such as: <pre><date> <time> Invalid topology, controlled wavelength=<wavelength> starting in section <TID-SHELF-TX PathID> was expanded beyond its egress point.</pre> If any expanded channels are identified, contract the channel back to its original egress point by placing the corresponding ADJ-RX facility OOS (provision the CMD44 ADJ-RX Receiver type to UNKNOWN). Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. The Domain Optical Controller Destination TID-Shelf-RxPathID column (specifically the last part of the entry) of the expanded channel indicates where to edit the ADJ-RX.
18	If the original alarm has cleared
	Then the procedure is complete
	not cleared go to step 19
19	Use the DOC logs to determine if a DOC-controlled channel is contracted prior to its previous egress point. The Overall Status for the channel appears as "Network Topology unavailable". The DOC logs may show a message such as: <pre><date> <time> Invalid topology, controlled wavelength=<wavelength> starting in section <TID-SHELF-TX PathID> was contracted from section <TID-SHELF-TX PathID> to section <TID-SHELF-TX PathID>.</pre> If any contracted channels are identified, expand the channel back to its original egress point by placing the corresponding ADJ-RX facility IS. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.
20	If the original alarm has cleared
	Then the procedure is complete
	not cleared go to step 21

Procedure 4-96 (continued)

DOC Invalid Photonic Domain

Step	Action
------	--------

- | | |
|----|---|
| 21 | Use the DOC window to view the DOC logs in order to determine if the current or another network element reported an error. If another network element has reported an error, review the DOC logs and provisioning of the other network element. Refer to the “Displaying the DOC logs for the summary table” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
|----|---|

ATTENTION

If the following DOC log is reported, refer to the “Unique log Identifier: 131330” procedure in *Fault Management - Customer Visible Logs*, 323-1851-840.

<date> <time> Invalid topology, missing previously controlled wavelength=<wavelength> nm ingress=<TID-SHELF-TX PATHID>.

- | | |
|----|--|
| 22 | If the alarm does not clear, contact your next level of support or your Ciena support group. |
|----|--|

—end—

Procedure 4-97

DOC Power Audit Failed

Alarm ID: 1719

Probable cause

This alarm is raised against the DOC facility when the Domain Optical Controller (DOC) Power Audit action fails.

This alarm can be caused by the following conditions:

- Unable to collect topology
- Fiber type not provisioned on a RAMAN facility
- OTDR Trace failed on a TELEMETRY or OTDRCFG facility, and the RAMAN facility is not ready for calibration
- Optical Line Fail alarm raised against a RAMAN facility
- RAMAN or AMP facility primary state is not IS
- RAMAN or AMP facility in shutoff or APR
- Communication failure with card that has RAMAN facility

The specific cause of the alarm can be found by checking the DOC logs.

The RAMAN state can be displayed using Site Manager. Possible RAMAN states are Normal, APR, or Shutoff.

Impact

Warning

Prerequisites

To perform this procedure, you must:

- be able to connect to the network element
- use an account with at least a level 3 UPC

Step	Action
1	Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Refer to the “Displaying the DOC logs for the summary table” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

Procedure 4-97 (continued)

DOC Power Audit Failed

Step	Action						
2	Review the DOC logs from the other network element. This alarm clears when there is a successful Power Audit or when a channel has been added successfully.						
3	Verify the Office and Short traces manually for any of the following events: • Large losses/reflections that were not captured by the OTDR alarms, possibly because of the pulse width/event spacing. If events are overlapping, re-run the trace using a smaller pulse. • Check for disparate or mixed fiber types, indicated by “gainers” or exaggerated loss splices.						
4	Perform the fiber/port/patch-panel cleaning procedure(s) on the event in question.						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Ensure that all facilities in that domain are In-service. Check for and clear any of the following alarms on all the network elements that are active within the DOC span of control, and then retry the Power Audit. • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active (visible in the historical fault list) • Automatic Shutoff • Fiber Type Manual Provisioning Required • OTDR High Reflection Detected • Telemetry Loss Of Signal • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Optical Line Fail • OSC Loss of Signal • OTDR Trace Failed						

Procedure 4-97 (continued)
DOC Power Audit Failed

Step	Action
7	If the alarm does not clear Then go to and the DOC logs are indicating Calibration Firmware Error step 8 otherwise step 21
8	In the DOC logs, refer to the site that is failing the Power Audit.
9	Run a manual OTDR trace using 100ns/8000m parameters. Refer to "Performing a manual OTDR trace" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
10	Once the trace is completed, open the OTDR Graph View in Site Manager and review all events to see if there is an event that has a high loss or high reflection within the first 100m of the faceplate.
11	In the DOC logs refer to the site that is failing the Power Audit.
12	In Site Manager under equipment, select the SRA-SH-SL, then in the facility window select Raman.
ATTENTION To prevent a fiber damage, ensure that there are no OTDR alarms present and the SRA span is bookended and optically connected with no optical line and telemetry alarms present.	
13	Click on Edit and change the "Forced Shut Off" value to False. The Raman pumps will turn on.
14	Verify in the alarm window that the RAMAN does not go into APR (Automatic Power Reduction).
15	Verify the PMs for the RAMAN facility that the ORLIN-OTS (27dB) and ORLOUT-OTS (22dB) met the required thresholds.
16	Click on Edit and change the "Forced Shut Off" value to True. The Raman pumps will turn off.
17	If there are any issues (while performing step 9 to step 16), clean and scope the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0. Otherwise, go to step 18 .
18	Ensure all RAMAN and XLA primary states are in-service.
19	Repeat step 9 to step 17 to ensure the problem is resolved.
20	Re-run the power audit from the DOC window in Site Manager.
21	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-98

Domain Optical Controller Disabled

Alarm ID: 1145

Probable cause

This alarm is raised when the Domain Optical Controller (DOC) facility has been put into an out-of-service state by the user or due to an upgrade activity.

This alarm is raised against the DOC facility itself.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- be able to connect to the network element
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|--|
| 1 | Ensure that there is no software upgrade or network maintenance in progress on the network and the DOC facility can be put in-service state. |
| 2 | Manually put the DOC facility in-service from Site Manager. Refer to the "Editing the DOC Settings" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-99

Dormant Account Detected

Alarm ID: 1372

Probable cause

This alarm is raised when there is at least one user account that has become dormant. This can be verified by opening the User Profile Application in Site Manager and looking at the “Password status” column. The alarm clears when there is “NO” DORMANT account left in the system.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 4 UPC.

Step	Action								
1	Change the status of the dormant account to “Valid” account, “Disabled” account, or delete the dormant user account.								
	<table> <tr> <th>If you want to</th><th>Then go to</th></tr> <tr> <td>change the status of the dormant account to Valid</td><td>step 2</td></tr> <tr> <td>change the status of the dormant account to Disabled</td><td>step 3</td></tr> <tr> <td>Delete the dormant user account</td><td>step 4</td></tr> </table>	If you want to	Then go to	change the status of the dormant account to Valid	step 2	change the status of the dormant account to Disabled	step 3	Delete the dormant user account	step 4
If you want to	Then go to								
change the status of the dormant account to Valid	step 2								
change the status of the dormant account to Disabled	step 3								
Delete the dormant user account	step 4								
2	In Site Manager, open a User Profile Application, select DORMANT account, select “Enable” button. This will change the password status from Dormant to Valid. Go to step 5 .								
3	In Site Manager, open a User Profile Application, select DORMANT account, select “Disable” button. This will change the password status from Dormant to Disabled. Go to step 5 .								
4	Site Manager, open a User Profile Application, select DORMANT account, select “Delete” button. This will delete the Dormant user account.								
5	If the alarm does not clear, contact your next level of support or your Ciena support group.								

—end—

Procedure 4-100

Duplicate Adjacency Discovered

Alarm ID: 1071

Probable cause

This alarm is raised against an ADJ facility when two or more ports at the far-end have the same TID-SHELF-SLOT-PORT Addresses.

This alarm is also raised when you change the node name and then revert it back to the original name. The alarm clears after the SP warm restart is completed or when you delete the original name in the Site Manager.

For the Transponder Direct Attach (TDA) configurations, this alarm is raised on the ADJ facility of WSS w/OPM Flex C-Band 9x1(NTK553LA and NTK553LB) circuit pack.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 3 UPC on the shelves with a duplicate Far End Address
- have a network plan or other documents that allow you to determine the correct Far End Address

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the shelves raising this alarm. |
| 2 | Note the TID-SHELF-SLOT-PORT that the alarm is raised against. In the Active Alarms table, the TID appears in the Network Element column, and the SHELF-SLOT-PORT appears after "ADJ-" in the Unit column. |
| 3 | In the Site Manager Equipment and Facility Provisioning application, examine the ADJ facility type lists for each applicable equipment and each shelf. Search for the TID-SHELF-SLOT-PORT noted in step 2 in the Expected Far End Address column. Note all matches found for all shelves within the siteID. |
| 4 | Determine which port has the correct Far End Address and remove the duplicate entries by setting their Expected Far End Address formats to NULL or the correct value. |
| 5 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-101

Duplicate IP Address

Alarm ID: 545

Probable cause

This alarm is raised when the system detects another network element with the same Internet protocol (IP) address. The alarm occurs at the same time on all network elements that share the same IP.

This alarm will display in the Additional Information column the AID and the IP address of the interface the alarm is detected on. The additional information is only available on SP-2.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 3 UPC on the nodes with a duplicate IP
- have a network plan or other documents that allow you to determine the correct IPs

Step	Action
1	Identify the network elements raising this alarm.
2	Determine from network plans or other documents which network element has the correct IP.
3	Log into the network element with the duplicate IP and correct the IP address as follows: • From the Configuration menu, open the Comms Setting Management. • In the Comms Setting Management, click on Interfaces tab. • In the Interface Type drop down menu, select IP. • Choose the Edit button to change the IP address. If a remote log in is not possible, log in locally using the RS-232 serial modem port or the craft 10/100BT port.



CAUTION

Risk of loss of functionality

Ensure every network element has a unique IP. If you are changing the IP of a network element, ensure that the new IP is correct.

Procedure 4-101 (continued)

Duplicate IP Address

Step	Action
4	Ensure no other Duplicate IP Address alarms exist. If there are other Duplicate IP Address alarms, repeat this procedure.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-102

Duplicate IPv6 Address

Alarm ID: 1872**Probable cause**

This alarm is raised when the system detects on a specific LAN interface another network element with the same Internet protocol version 6 (IPv6) address.

Note 1: The alarm occurs only on the network element that most recently provisioned this IPv6 address. This may or may not be the network element with the incorrect address.

Note 2: This alarm displays in the Additional Information column the AID of the LAN interface the alarm is detected on.

Note 3: Once raised, the alarm is not re-raised after the following occur, even if the duplicate address is still present:

- DLT-LAN/ENT-LAN on the interface with the alarm
- Toggle (down/up) of the LAN interface with the alarm
- SP restart on the shelf with the alarm

If one of the above events occurs, the historical alarms can be consulted, or if necessary, the alarm can be re-created by deleting and re-adding the IPv6 address that cause the alarm originally.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 3 UPC on the nodes with a duplicate IPv6
- have a network plan or other documents that allow you to determine the correct IPv6 address

Step	Action
------	--------

- | | |
|---|--|
| 1 | Determine the IPv6 address of the LAN interface against which the alarm is raised. |
|---|--|

Procedure 4-102 (continued)

Duplicate IPv6 Address

Step	Action						
2	Determine from network plans or other documents which network element has the correct IPv6 address. Only one of them should have this address provisioned. <table> <tr> <td>If the NE that</td><td>Then go to</td></tr> <tr> <td>did not raise the alarm has an incorrect IPv6 address</td><td>step 3</td></tr> <tr> <td>raised the alarm has an incorrect IPv6 address</td><td>step 5</td></tr> </table>	If the NE that	Then go to	did not raise the alarm has an incorrect IPv6 address	step 3	raised the alarm has an incorrect IPv6 address	step 5
If the NE that	Then go to						
did not raise the alarm has an incorrect IPv6 address	step 3						
raised the alarm has an incorrect IPv6 address	step 5						
3	Re-provision the IPv6 address on the NE that did not raise the alarm to the correct address as follows: Log into the network element with the duplicate IPv6 address and correct the IPv6 address as follows: - From the Configuration menu, open the Comms Setting Management. - In the Comms Setting Management, click on Interfaces tab. - In the Interface Type drop down menu, select IPv6. - Choose the Edit button to change the IP address. Note: This will not clear the alarm on the NE that raised the alarm.						
4	To clear the alarm, delete this address then re-add the same address to the NE that raised the alarm. Refer to the <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 6.						
5	Re-provision the IPv6 address on the NE that raised the alarm to the correct address to clear the alarm as follows: - From the Configuration menu, open the Comms Setting Management. - In the Comms Setting Management, click on Interfaces tab. - In the Interface Type drop down menu, select IPv6. - Click the Edit button to change the IP address.						



CAUTION

Risk of loss of functionality

Ensure every network element has a unique IP. If you are changing the IPv6 address of a network element, ensure that the new IPv6 address is correct.

Procedure 4-102 (continued)

Duplicate IPv6 Address

Step	Action
6	Ensure no other Duplicate IPv6 Address alarms exist. If there are other Duplicate IPv6 Address alarms, repeat this procedure.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-103

Duplicate Primary Shelf

Alarm ID: 714

Probable cause

This alarm indicates there are duplicate primary shelves within a consolidated node, and is raised against all primary shelves within a consolidated node that are provisioned to be a primary shelf.

Note: A primary shelf will not auto-enroll or allow manual addition of new member shelves while a duplicate primary shelf exists on the same network.

Impact

Major, non-service-affecting (M, NSA)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- be able to log into both shelves in the duplicate primary condition
- have a network plan or other documents that allow you to determine the correct primary shelves

Step	Action
1	Identify the network elements raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" .
2	Determine from network plans or other documents which shelf should be the primary shelf.
3	Log into the network element that is incorrectly enabled as a primary shelf.

Procedure 4-103 (continued)
Duplicate Primary Shelf

Step	Action
4	 CAUTION Risk of loss of functionality Disabling the primary shelf will cause an administrative restart of the affected shelf. It will be unavailable for management during this time frame, and may require re-provisioning of the network element. Ensure you have removed the appropriate duplicate primary. Record all the provisioning information required to re-add the incorrectly provisioned member shelf to the consolidated node. Then delete the invalid primary shelf from the consolidated node. Refer to the “Deleting a member shelf of a consolidated node” procedure in <i>Administration and Security</i>, 323-1851-301.
5	Re-add the deleted shelf as a member shelf of the consolidated node with the Primary shelf parameter set to Disabled. Refer to the “Adding a shelf to a consolidated node” procedure in <i>Administration and Security</i>, 323-1851-301.
6	Ensure no other Duplicate Primary Shelf alarms exist. If there are other Duplicate Primary Shelf alarms, repeat this procedure for the new duplicate.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-104

Duplicate Shelf Detected

Alarm ID: 70

Probable cause

This alarm is raised when the shelf processor detects another network element with the same shelf number and TID (also referred to as node name). The alarm occurs at the same time on all network elements that share the same shelf number and TID. Each shelf processor of the network element with the same shelf number and TID detects the condition.

Note: This alarm is raised to detect invalid layer 0 network topology and is only applicable to photonic shelves (shelves with OTSes provisioned). This alarm is raised only in photonic networks and cannot be raised in the TIDS only with POTS NEs.

After a shelf processor restart, this alarm is masked for 20 minutes.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- ensure that you are the only active user logged into the network element
- be able to log into the nodes that do not have a unique shelf number and TID
- have a network plan or other documents that allow you to determine the correct shelf numbers and TIDs

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the network elements raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Determine from network plans or other documents which shelf has incorrectly provisioned information. |

Procedure 4-104 (continued)
Duplicate Shelf Detected

Step	Action						
3	Note that changing the shelf number requires decommissioning and recommissioning of the network element. If the network is a mixed 6500 and 565, 5100 and 5200 Advanced Services Platform, change the shelf number on the 5100/5200 (which requires a warm restart), instead of full recommission of the 6500.  CAUTION Risk of loss of functionality Traffic and data communications will be lost Decommissioning a shelf results in a loss of all traffic and data communications associated with the shelf that is being decommissioned.  CAUTION Risk of loss of functionality Ensure that you have identified the correct duplicate. A software configuration restart is required to properly recover, removing network visibility of the member node for the duration of the restart. <table> <tr> <th>If the</th><th>Then</th></tr> <tr> <td>shelf Name (TID) is incorrectly provisioned</td><td>correct the shelf name. Refer to the “Editing the shelf number” procedure in <i>Administration and Security</i>, 323-1851-301.</td></tr> <tr> <td>Shelf number is incorrectly provisioned</td><td>record all the provisioning information required to recommission the shelf. Decommission the shelf and re-add it with the correct shelf number. Refer to the “Deleting all shelf provisioning” procedure in <i>Administration and Security</i>, 323-1851-301.</td></tr> </table>	If the	Then	shelf Name (TID) is incorrectly provisioned	correct the shelf name. Refer to the “Editing the shelf number” procedure in <i>Administration and Security</i> , 323-1851-301.	Shelf number is incorrectly provisioned	record all the provisioning information required to recommission the shelf. Decommission the shelf and re-add it with the correct shelf number. Refer to the “Deleting all shelf provisioning” procedure in <i>Administration and Security</i> , 323-1851-301.
If the	Then						
shelf Name (TID) is incorrectly provisioned	correct the shelf name. Refer to the “Editing the shelf number” procedure in <i>Administration and Security</i> , 323-1851-301.						
Shelf number is incorrectly provisioned	record all the provisioning information required to recommission the shelf. Decommission the shelf and re-add it with the correct shelf number. Refer to the “Deleting all shelf provisioning” procedure in <i>Administration and Security</i> , 323-1851-301.						
4	Ensure no other Duplicate Shelf Detected alarms exist. If other duplicate shelf alarms exist, repeat this procedure for the new duplicate(s).						
5	If the alarm does not clear, contact your next level of support or Ciena support group.						

—end—

Procedure 4-105 Duplicate Site ID

Alarm ID: 871

Probable cause

This alarm is raised when both the site identifier and the shelf number of two or more Photonic shelves (OTS provisioned on shelf) are the same. Node Element (NE) names (TID) can either be the same or different. The alarm occurs at the same time on all network elements that share the same site identifier and shelf number. (The alarm is not raised when the site identifier is 0.) Each shelf processor detects the condition and raises the alarm against the shelf.

After a shelf processor restart, this alarm can be masked for 20 minutes.

Note: You can edit the site identifier. However, changing the shelf number requires decommissioning and recommissioning of the network element.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 4 UPC
- be able to log into the nodes that do not have a unique site identifier and shelf number
- have a network plan or other documents that allow you to determine the correct site identifiers and shelf numbers

Step	Action
1	Identify the network elements raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.

Procedure 4-105 (continued)
Duplicate Site ID

Step	Action						
2	Verify the auto discovery setting. Refer to “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If the channels are not DOC controlled, set the auto discovery to manual. Refer to “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.  CAUTION Risk of traffic loss If the Autodiscover is set incorrectly while channels are not DOC controlled, it can cause a loss of traffic.						
3	 CAUTION Risk of loss of functionality Traffic and data communications will be lost Decommissioning a shelf results in a loss of all traffic and data communications associated with the shelf that is being decommissioned.						
<table><tr><th>If the</th><th>Then go to</th></tr><tr><td>site identifier is incorrectly provisioned</td><td>step 4</td></tr><tr><td>shelf number is incorrectly provisioned</td><td>step 5</td></tr></table>		If the	Then go to	site identifier is incorrectly provisioned	step 4	shelf number is incorrectly provisioned	step 5
If the	Then go to						
site identifier is incorrectly provisioned	step 4						
shelf number is incorrectly provisioned	step 5						
4	Determine from network plans or other documents which network element has the incorrect site identifier. Edit the incorrect site identifier to be something unique from all other shelves in the network. Refer to the “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i>, 323-1851-301. Go to step 7.						
5	If the network is a mixed 6500 and 5100/5200, change the Shelf number on the 5100/5200 (which requires a warm restart), instead of a full recommission of the 6500.						
6	If the network is 6500, record all the provisioning information required to recommission the shelf. Decommission the shelf and re-provision it with the correct information. Refer to the “Deleting all shelf provisioning” procedure in <i>Administration and Security</i>, 323-1851-301, and “Commissioning a network element” procedure in <i>Commissioning and Testing</i>, 323-1851-221.						

Procedure 4-105 (continued)

Duplicate Site ID

Step	Action						
7	Verify that no other Duplicate Site ID alarms exist.						
	<table><tr><th>If</th><th>Then</th></tr><tr><td>no other Duplicate Site ID alarms exist</td><td>the procedure is complete</td></tr><tr><td>other Duplicate Site ID alarms exist</td><td>repeat step 1 to step 5 for the other duplicate(s). Go to step 8.</td></tr></table>	If	Then	no other Duplicate Site ID alarms exist	the procedure is complete	other Duplicate Site ID alarms exist	repeat step 1 to step 5 for the other duplicate(s). Go to step 8 .
If	Then						
no other Duplicate Site ID alarms exist	the procedure is complete						
other Duplicate Site ID alarms exist	repeat step 1 to step 5 for the other duplicate(s). Go to step 8 .						
8	If the alarm does not clear, contact your next level of support or Ciena support group.						

—end—

Procedure 4-106

Encryption Authentication Material Missing

Alarm ID: 1816

Probable cause

This alarm is raised against the 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit packs when the provisioning of authentication certificate on the circuit pack is not completed.

For the 4x10G OTR w/ Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit packs, X.509 certificates are used for authentication in datapath encryption and web access using the encryption management interface (MyCryptoTool). An entity certificate and corresponding signing Certificate Authority (CA) certificate have to be present on the circuit pack.

- If the enrollment of any of the material (certificate) is not completed, the alarm is raised.
- If either the Web Access Entity Certificate or the Data Encryption Certificate are not fully provisioned, the alarm is raised. If both are not fully provisioned, only one occurrence of this alarm will be raised.

If web access is using a separate X.509 Certificate than the Data Encryption Certificate, both the Data Entity Certificate Management and Web Access Certificate Management pages need to be verified to ensure the entity certificate and corresponding signing CA certificate are present.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Connect to MyCryptoTool and from the Web Access Certificate Management page, check the Web Entity Certificate Selection. Refer to the “Key Management” section of <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
2	If the Web Access is using the same X.509 Certificate as the datapath, go to step 4 .
3	Check the Enrollment status from the Web Access Certificate Management tab. If the status is “Entity Certificate is not available”, perform the certificate enrollment process. Refer to the “Performing Web Access Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.

Procedure 4-106 (continued)

Encryption Authentication Material Missing

Step	Action
4	Check the Enrollment status from the Data Entity Certificate Management tab. If the status is “Entity Certificate is not available”, perform the certificate enrollment process. Refer to the “Performing Data Encryption Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-107

Encryption Configuration Mismatch

Alarm ID: 1822, 1932, 1936, 1940

Probable cause

This alarm is raised against line facility of a 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit packs. The alarming criteria is slightly different for these circuit packs as follows:

4x10G OTR w/ Encryption (NTK530QE variant)

This alarm is raised during any of the following:

- encryption byte 1 or TCM byte selection does not match between two encryption peer ends
- the authentication failure mode does not match between two encryption peer ends
- the cryptography mode does not match between two encryption peer ends

Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants)

This alarm is raised during any of the following:

- the TCM byte selection does not match between two encryption peer ends
- the authentication failure mode does not match between two encryption peer ends

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	Select your first step: <table><tr><td>If the alarm is triggered on the</td><td>Then go to</td></tr><tr><td>4x10G OTR (NTK530QE variant)</td><td>step 2</td></tr><tr><td>Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants)</td><td>step 10</td></tr></table>	If the alarm is triggered on the	Then go to	4x10G OTR (NTK530QE variant)	step 2	Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants)	step 10
If the alarm is triggered on the	Then go to						
4x10G OTR (NTK530QE variant)	step 2						
Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants)	step 10						

Procedure 4-107 (continued)

Encryption Configuration Mismatch

Step	Action
2	Use Site Manager to verify that the Encryption TCM or Encryption Byte 1 OTM2 facility provisioning attributes on both ends of the encryption path match. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
3	If the Encryption TCM or Encryption Byte 1 OTM2 facility provisioning attributes do not match on both ends of the encryption path, use a matching byte. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
4	If the alarm did not clear, use MyCryptoTool to check the Re-Authentication Failure Mode selection (ALLOW/DROP) on both ends of the encryption path.
5	If the Re-Authentication Failure Mode settings do not match, set the OTM2 Re-Authentication Failure Mode settings on both ends of the encryption path to the same selection. Refer to the “Provisioning re-authentication failure mode, re-authentication period, and re-keying period on port(s)” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
6	If the alarm did not clear, log into the MyCryptoTool and check the cryptography mode (ECC/RSA). Refer to the “Displaying Encryption Details” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341. Note 1: The 4x10G OTR (NTK530QE variant) circuit pack running Pre-Release 11.1 software is running in RSA cryptography mode even though there was no explicit cryptography mode displayed on the MyCryptoTool. Note 2: After an upgrade to Software Release 11.1, the 4x10G OTR (NTK530QE variant) will remain in RSA mode. This cryptography mode can be provisioned to operate in either RSA mode or ECC mode. The cryptography mode selection is mutually exclusive and can be changed only when a Clear Critical Security Parameters (CSPs) action is performed on the 4x10G OTR (NTK530QE variant) circuit pack using MyCryptoTool.
7	If the cryptography modes do not match, set both ends to the same selection to clear the alarm. Refer to the “Changing the cryptography mode” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
8	Import the entity certificate that matches the new cryptographic mode. Refer to the “Importing a Data Encryption Signing CA certificate or a Web Access Signing CA certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341.
9	If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 4-107 (continued)

Encryption Configuration Mismatch

Step	Action
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Flex3 WL3e OCLD (NTK539QS/NTK539QV variants)

- | | |
|----|---|
| 10 | Use Site Manager to verify that the Encryption TCM and Encryption Byte 1 OTM4, ODU4, or OTUTTP facility provisioning attributes on both ends of the encryption path match. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340. |
| 11 | If the Encryption TCM and Encryption Byte 1 OTM4, ODU4, or OTUTTP facility provisioning attributes do not match on both ends of the encryption path, use a matching byte. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340. |
| 12 | If the alarm did not clear, use MyCryptoTool to check the Re-Authentication Failure Mode selection (ALLOW/DROP) on both ends of the encryption path. Refer to the “Displaying Encryption Details” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341. |
| 13 | If the Re-Authentication Failure Mode settings do not match, set the Re-Authentication Failure Mode settings on both ends of the encryption path to the same selection. Refer to the “Provisioning re-authentication failure mode, re-authentication period, and re-keying period on port(s)” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i> , 323-1851-341. |
| 14 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-108 Encryption Failure

Alarm ID: 1821, 1931, 1935, 1939

Probable cause

This alarm is raised against line facility of a 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit pack when the Encryption Bit Parity check fails.

The Encryption Bit Parity check fails when you choose one of the six TCM bytes on the OTN line port to be used to monitor the encryption failure and the TCM BIP8 bit parity does not match the OPU Frame data before encryption and after decryption.

This alarm can be raised momentarily in the initial setup of the datapath when the keying authentication and negotiation is taking place.

This alarm is also raised together with an Encryption Configuration Mismatch, Encryption Keying Authentication Failure, Encryption Keying Communication Failure, and Certificate Expired, in which case the cause of these alarms should be addressed in order to clear the Encryption Failure alarm.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	If any of the “Encryption Keying Communication Failure” , “Equipment Configuration Mismatch” , “Encryption Keying Authentication Failure” or “Certificate Expired” alarms are active, clear these alarms first. Refer to alarm clearing procedures in this document.
2	If the alarm is still active, there is an issue with the encrypted datapath. This can be caused by misuse of the Encryption error monitoring TCM byte by intermediate equipment in the OTN cloud. Verify the Encryption error monitoring TCM byte in intermediate equipment and correct the problem.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-109

Encryption Keying Authentication Failure

Alarm ID: 1823, 1933, 1937, 1941

Probable cause

This alarm is raised against line facility of 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit pack when the Encryption keying authentication fails between the two ends of an encryption datapath.

Encryption keying authentication can fail due to any of the following reasons:

- the peer datapath certificate has expired
- the peer datapath certificate issuer Certificate Authority (CA) is not trusted
- the peer entity certificate is on the issuer's Certificate Revocation List (CRL)

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Note: If the re-authentication failure mode is set to "ALLOW", traffic can be maintained until the next circuit pack level cold restart is performed from either end or when Encryption Keying Communication Channel Failure is detected. If the re-authentication failure mode is set to "DROP", traffic will be dropped.

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Use Site Manager to ensure the Time of Day on both ends of the encryption path are current. If the Time of Day is not current, set the correct Time of Day. Refer to the "Provisioning Time of Day servers" procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340 or "Editing the nodal general parameters" procedure in <i>Administration and Security</i> , 323-1851-301.

Procedure 4-109 (continued)

Encryption Keying Authentication Failure

Step	Action
2	Refer to <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341 to launch MyCryptoTool for each peer end to determine if any of the following conditions exist: • the Data Encryption entity certificate being received from the peer has expired. — Refer to the “Viewing the Data Encryption entity certificate or the Web Access entity certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. — Refer to the “Displaying Active Alarms and Historical Logs” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. • the issuer of the Data Encryption entity certificate being received from the peer is not on the Trusted CA list, or is not the same as the Signing CA: — Refer to the “Viewing the Data Encryption entity certificate or the Web Access entity certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. — Refer to the “Viewing the Data Encryption Signing CA certificate or the Web Access Signing CA certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. — Refer to the “Viewing a Trusted CA Certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. • a certificate being received from peer is on the Trusted CA’s CRL or on the Signing CA’s CRL. — Refer to the “Viewing a Certificate Revocation List” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341.

Procedure 4-109 (continued)

Encryption Keying Authentication Failure

Step	Action
3	If any of the above conditions exist, perform the corresponding procedure: • if the Data Encryption entity certificate being received from the peer has expired, you need to enroll a new Data Encryption Certificate on the peer. Refer to the “Performing Data Encryption Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. • if issuer (CA) of the Data Encryption entity certificate being received from the peer is not on the Trusted CA list or is not the same as the Signing CA, you need to add that CA certificate to the Trusted CA Certificate list. Refer to the “Importing a Trusted CA certificate” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341. • If the peer entity certificate is on the issuer’s (CA) Certificate Revocation List (CRL), you need to enroll a new Data Encryption Certificate on the peer. Refer to the “Performing Data Encryption Certificate Enrollment” procedure in <i>MyCryptoTool Certificate Management and Quick Start</i>, 323-1851-341.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-110

Encryption Keying Communication Failure

Alarm ID: 1824, 1934, 1938, 1942

Probable cause

This alarm is raised against line facility of 4x10G OTR w/Encryption (NTK530QE) or Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit pack when communication is lost between the two encryption peers of the encryption path due to one of the following:

- the **Encryption Byte 1** facility provisioning attribute on the Flex3 WL3e OCLD (NTK539QS/ NTK539QV variants) circuit pack is used for keying communication purpose and the user-provisioned selection does not match between both ends of the encrypted path.
- the **Encryption Byte 2** facility provisioning attribute on the 4x10G OTR w/Encryption (NTK530QE) circuit pack is used for keying communication purpose and the user-provisioned selection does not match between both ends of the encrypted path.
- the intermediate OTN equipment re-used the same byte for other purposes which prevented the chosen byte to be carried end-to-end between the two ends of the encryption path.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-110 (continued)

Encryption Keying Communication Failure

Step	Action						
1	Select your first step: <table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack</td><td>Use Site Manager to verify the encryption byte selection (Encryption Byte 1) on the OTN line channel of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i>, 323-1851-340.</td></tr> <tr> <td>If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack</td><td>Use Site Manager to verify the encryption byte selection (Encryption Byte 2) on the OTN line ports of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i>, 323-1851-340.</td></tr> </table>	If	Then	If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack	Use Site Manager to verify the encryption byte selection (Encryption Byte 1) on the OTN line channel of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.	If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack	Use Site Manager to verify the encryption byte selection (Encryption Byte 2) on the OTN line ports of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
If	Then						
If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack	Use Site Manager to verify the encryption byte selection (Encryption Byte 1) on the OTN line channel of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.						
If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack	Use Site Manager to verify the encryption byte selection (Encryption Byte 2) on the OTN line ports of both ends of the encryption path for any mis-match, and correct the byte selection if any. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.						
2	If there is no mis-match, verify the intermediate OTN equipment to ensure the selected reserved byte can be carried over end-to-end between two ends of the encrypted path.						

Procedure 4-110 (continued)

Encryption Keying Communication Failure

Step	Action						
3	Select your next step: <table><tr><th>If</th><th>Then</th></tr><tr><td>If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack</td><td>Use Site Manager to select a different Encryption Byte 1 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i>, 323-1851-340.</td></tr><tr><td>If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack</td><td>Use Site Manager to select a different Encryption Byte 2 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i>, 323-1851-340.</td></tr></table>	If	Then	If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack	Use Site Manager to select a different Encryption Byte 1 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.	If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack	Use Site Manager to select a different Encryption Byte 2 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.
If	Then						
If the alarm is on the Flex3 WL3e OCLD (NTK539QS/NTK539QV variants) circuit pack	Use Site Manager to select a different Encryption Byte 1 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.						
If the alarm is on the 4x10G OTR w/ Encryption (NTK530QE) circuit pack	Use Site Manager to select a different Encryption Byte 2 to be carried end-to-end between the two ends of the encrypted path. Refer to the “Displaying and editing the Carrier-managed encryption-related parameters on the circuit packs” procedure in <i>Encryption and FIPS Security Policy Overview and Procedures</i> , 323-1851-340.						
4	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-111

Equipment Configuration Mismatch

Alarm ID: 970

Probable cause

For a 6500 14-slot shelf type, this alarm is raised against regular flow cooling fans when circuit packs requiring high flow cooling (such as 100G WL3n MOTR, 100G OCLD, 100GE OCI, 40G+ CFP OCI, 10x10G MUX, 10x10GE MUX, 40G UOCLD, 40G OCLD, Wavelength-Selective 40G OCLD, 40G OCI, 40/43G OCI, or 40G MUX OCI circuit packs) are provisioned. This alarm masks all alarms raised against a given fan, except the Fan Failed alarm. Refer to the “Equipment provisioning validation based on shelf cooling capacity” section in “Node Information” section of *Administration and Security*, 323-1851-301, for further details.

For a 14-slot packet-optical shelf, this alarm is raised if a Type 3 high flow cooling fan module (NTK507LS or NTK507MS) is inserted in a shelf that is also equipped with Power Input Cards that are not compatible with Type 3 fans. With the exception of the center fan slot (Fan-2), Type 3 fans will not energize or light any LEDs unless there is a compatible Power Input Card in the shelf. The 2x50A Power Input Card (NTK505ES) is compatible with Type 3 fans. This alarm is also raised when mixing Type 3 and other cooling fan module types in 14-slot packet-optical shelf. This alarm will clear once the shelf is equipped with three compatible fans (either three Type 3 fans or three other matching fans). When an SP is restarted and the shelf is equipped with a mix of Type 3 and other fan types (there is a mismatch between at least two fans), the Equipment Configuration Mismatch is raised against all the three fans. For a 32-slot, 14-slot, 7-slot, 7-slot Type 2 and 2-slot Type 2, This alarm is raised against a Power Input Module that does not support the shelf powering configuration (or current rating) specified by the **Provisioned shelf current** shelf attribute (provisioned shelf power configuration and feeder amperage). The alarm is also raised in the case of two mismatched Power Input Modules. Refer to the “Equipment provisioning validation based on shelf power capacity” section in *Administration and Security*, 323-1851-301, for further details.

This alarm is raised against mismatched fan modules (fan modules with different ordering codes).

Procedure 4-111 (continued)

Equipment Configuration Mismatch

Impact

Against fans

Critical, service-affecting (C, SA) alarm if two or more fan alarms exist
Major, non-service-affecting (M, NSA) alarm if one fan alarm exists

Against Power Input Cards

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
------	--------



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

- | | | |
|---|---|--|
| 1 | If the alarm is raised against fan modules
Power Input Cards | Then go to the next step
step 13 |
| 2 | If the shelf is provisioned with circuit packs requiring high flow cooling, or such circuit packs are physically present in the shelf
not provisioned with circuit packs requiring high flow cooling, and no such circuit packs are physically present in the shelf | Then go to step 3
step 7 |
| 3 | If the circuit packs requiring high flow cooling are required
not required | Then go to step 4
step 6 |
| 4 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. | |

Procedure 4-111 (continued)

Equipment Configuration Mismatch

Step	Action						
5	Replace any regular flow the cooling fans or any Type 3 high flow cooling fans with high flow cooling fans module with a fan that has the correct PEC. Refer to the “Replacing a shelf fan module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 14 .						
6	Deprovision and remove all equipment requiring high flow cooling. Refer to the “Deleting a circuit pack, module, or SFP/XFP” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 7 .						
7	Change the fan to the regular flow fans. Refer to the “Replacing a cooling fan module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
9	Retrieve and record the Provisioned shelf current value, and the inventory information for the Power Input Modules. Refer to the “Displaying node information” procedure in <i>Administration and Security</i> , 323-1851-301.						
10	If the equipped Power Input Modules are the same type, set the Provisioned Shelf Current to a setting that is compatible with the Power Input Modules and their configuration but does not exceed the current rating of the feeders or of any equipped fuses (for the case of fused Power Input Modules). Refer to the “Determining the provisioned shelf current” procedure in <i>Administration and Security</i> , 323-1851-301.						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Remove the alarmed Power Input Module or the Power Input Module that is not compatible with the required Provisioned Shelf Current setting. Refer to the “Replacing the Power Input Card A or B” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. The “Circuit Pack Missing” alarm is raised for that Power Input Module, and the “Equipment Configuration Mismatch” alarm clears.						
13	Insert a replacement Power Input Module into the shelf. Ensure that the equipped Power Input Modules are compatible with the retrieved Provisioned Shelf Current setting in step 9 , and that all equipped Power Input Modules have the same amperage capacity (which is greater than or equal to the current associated with the Provisioned shelf current setting).						
14	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-112

Equipment Reconfiguration In Progress

Alarm ID: 1345

Probable cause

This alarm is raised to indicate the equipment reconfiguration is in progress.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Perform a warm restart on the active shelf processor to complete the reconfiguration. Wait five minutes for the shelf processor to restart. Refer to Procedure 2-11, "Restarting a circuit pack or shelf processor" .
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-113

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETTP, STTP, CBRTTP)

Use this procedure to clear alarms associated with the error alarms.

Excessive Error Ratio (ETH10G, ETH40G, ETH100G, FC800, FC1200, FLEX, ETTP, CBRTTP)

Alarm ID: 841, 1257, 1298, 1454, 2133

Probable cause

This alarm is raised against:

- an ETH10G or WAN facility of a 10x10G MUX circuit pack
- an FC1200 facility of the 2x10G OTR, 4x10G OTR, and 40G MUX OCI circuit pack
- an ETH10G or FLEX facility of the 2x10G OTR and 4x10G OTR circuit pack
- an ETH10G, ETH40G or ETH100G facility of the 200G (2x100G/5x40G) MUX circuit pack
- an ETH10G or ETH40G facility of the 100G (2xQSFP+/2xSFP+) MUX circuit pack
- an ETH40G facility of the 4x10G OTR circuit pack
- an ETTP facility of the 2xWLAi OTR, WLAi MOTR, WLAi MOTR w/OPS, WLAi FOTR, WLAi FOTR w/OPS, 16xFLEX OTN I/F, 4x10G MUX, 2xCFP2 OTN Flex MOTR, and 40G OCI circuit packs
- an ETTP facility of the eMOTR circuit pack when a High Bit Error (HIBER) condition is detected on the incoming 10GE signal

This alarm is raised against an ETH10G or ETH100G facility when at least 20 percent of the received frames are errored each second for three consecutive seconds and is raised against a WAN facility when one of the following conditions occurs:

- for GFP-F encapsulation, at least 20% of the received frames are errored each second for three consecutive seconds
- for GFP-T encapsulation, at least 20% of the received superblocks are errored each second for three consecutive seconds.

For ETH10G facilities on the 10x10G MUX circuit pack, this alarm is raised when the client 64B/66B Rx interface is detecting a BER greater than or equal to 1E-4. This occurs when 16 or more invalid sync headers are detected by the 64B/66B Rx interface within the current 125 μ s period.

Procedure 4-113 (continued)

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETTP, STTP, CBRTTP)

For the ETH40G facility on the 40G+ CFP OCI, this alarm is raised when 97 invalid 66-bit sync headers are detected within a 1.25ms window. The high BER state is exited once there are less than 97 invalid sync headers in the same window.

For the ETH100G facility on the 100G OCI, this alarm is raised when 97 invalid 66-bit sync headers are detected within a 500 μ s window. The high BER state is exited once there are less than 97 invalid sync headers in the same window.

For FC1200 facility on the 2x10G OTR, 4x10G OTR, or 40G MUX OCI circuit pack, this alarm is raised when the client 64B/66B Rx interface is detecting a BER greater than or equal to 1E-3. This occurs when 16 or more invalid sync headers are detected by the 64B/66B Rx interface within the current 125 μ s period.

The alarm clears when these conditions do not occur for 10 consecutive seconds.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Signal Degradate (ETH, ETH100, ETH10G, WAN, STTP)

Alarm ID: 866, 1465

Probable cause

This alarm is raised against:

- an ETH10G or WAN facility of a 100G OCI circuit pack
- This alarm is raised against an ETH10G facility when at least 1% of the received frames are errored each second for three consecutive seconds

This alarm is raised against WAN facility when one of the following conditions occurs:

- for GFP-F encapsulation, at least 1% of the received frames are errored each second for three consecutive seconds

Procedure 4-113 (continued)

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETPP, STTP, CBRTTP)

The alarm clears when these conditions do not occur for 10 consecutive seconds.

Impact

Minor, service-affecting (m, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Signal Fail (STTP)

Alarm ID: 1464

Probable cause

This alarm is raised when the received signal is degraded to the point where it is unusable, and the circuit pack is unable to detect the framing bytes in the received signal.

One of the following conditions causes this alarm:

- excessive attenuation
- dirty optical fibers
- dirty connectors
- improper connector seating
- transmit laser degrade

Impact

Critical, service-affecting (C, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA), if on an inactive path

Prerequisites

To perform this procedure, you must

- have an optical power meter with the same optical connectors as the network element
- if required, obtain a supported SFP optical transceiver module
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action
------	--------

- | | |
|---|--|
| 1 | Verify if there are alarms of higher order from the alarm hierarchy. Refer to “Alarm hierarchies and alarm severities” in this document. Clear any alarms of higher order on the hierarchy first using the appropriate procedures. |
|---|--|

Procedure 4-113 (continued)

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETTP, STTP, CBRTTP)

Step	Action				
2	<table> <tr> <td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has cleared	Then the procedure is complete	not cleared	go to step 3
If the original alarm has cleared	Then the procedure is complete				
not cleared	go to step 3				
3	From the Class field in the Active Alarms application, determine if the alarm is raised against an Ethernet or WAN facility.				
4	<table> <tr> <td>If this alarm is raised against an Ethernet, CBRTTP, or ETTP facility</td><td>Then go to step 5</td></tr> <tr> <td>a WAN facility</td><td>step 18</td></tr> </table>	If this alarm is raised against an Ethernet, CBRTTP, or ETTP facility	Then go to step 5	a WAN facility	step 18
If this alarm is raised against an Ethernet, CBRTTP, or ETTP facility	Then go to step 5				
a WAN facility	step 18				

Alarm raised against an Ethernet, CBRTTP, or ETTP facility

- 5 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.

**CAUTION****Risk of traffic loss**

Ensure that the correct module is identified. Removing the wrong optical fiber causes a traffic loss on an in-service facility.

**DANGER****Risk of laser radiation exposure**

Laser radiation is present on the optical fiber. Do not look into the optical fiber.

- 6 Use the optical power meter to measure the receive power on the fiber connected to the pluggables on the I/F module.
- 7 **If** the receive power on the fiber is
- | | |
|---|--|
| below the minimum receive optical power | Then go to step 8 |
| between the minimum and maximum receive optical power | step 12 |
| above the maximum receive optical power | step 13 |

Receive power is below the minimum receive optical power

- 8 Decrease the local attenuation, if equipped, to try to increase the receive power to a value above the minimum receive optical power (but below the maximum receive optical power).

Procedure 4-113 (continued)

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETTP, STTP, CBRTTP)

Step	Action				
9	<table> <tr> <td>If the adjusted receive power is still below the minimum receive optical power</td><td>Then go to step 10</td></tr> <tr> <td>within range (between the minimum and the maximum receive optical power)</td><td>step 12</td></tr> </table>	If the adjusted receive power is still below the minimum receive optical power	Then go to step 10	within range (between the minimum and the maximum receive optical power)	step 12
If the adjusted receive power is still below the minimum receive optical power	Then go to step 10				
within range (between the minimum and the maximum receive optical power)	step 12				
10	Remove the Tx optical fiber from the far-end subtending client equipment.				
11	Measure the transmit power at the far-end subtending client equipment. - If the transmit power of the far-end equipment is above the minimum launch power, the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. - If the transmit power of the far-end equipment is below the minimum launch power, there is a problem with the far-end equipment. Use your company procedure to determine and clear the problem, then go to step 14 .				

Receive power is between the minimum and the maximum receive optical power

- | | |
|----|--|
| 12 | Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers.
Go to step 14 . |
|----|--|

Receive power is above the maximum receive optical power

- | | |
|----|---|
| 13 | Add the necessary attenuation to reduce the receive power to a value between the minimum and maximum receive optical power. |
|----|---|

Determining if the alarm cleared

14	<table> <tr> <td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the SFP corresponding to the facility raising the alarm. Refer to the "Replacing a Pluggable module" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has cleared	Then the procedure is complete	not cleared	replace the SFP corresponding to the facility raising the alarm. Refer to the "Replacing a Pluggable module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has cleared	Then the procedure is complete				
not cleared	replace the SFP corresponding to the facility raising the alarm. Refer to the "Replacing a Pluggable module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.				
15	<table> <tr> <td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has cleared	Then the procedure is complete	not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has cleared	Then the procedure is complete				
not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.				
16	Clean and re-attach the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.				

Procedure 4-113 (continued)

Error alarms (ETH10G, ETH40G, ETH100G, FLEX, WAN, ETTP, STTP, CBRTTP)

Step	Action
17	If the alarm does not clear, contact your next level of support or your Ciena support group. The procedure is complete.

Alarm raised against a WAN facility

- | | |
|----|--|
| 18 | Retrieve the performance monitoring counts for the ODU paths connected to the WAN facility to determine if the path ES counts are increasing. |
| 19 | If the path ES counts are increasing, use the appropriate alarm clearing procedure to clear any ODU path alarms on the interface circuit pack. |
| 20 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-114

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Use this procedure to clear alarms associated with errors on the OC/STM line, STS/HO VC path.

Signal Degrade (OC/STM, STS/HO VC)

Alarm ID: 3, 9, 39, 151, 281, 470, 1103

Probable cause

This alarm is raised when the received STS/HO VC signal is significantly degraded.

One of the following conditions can cause this alarm:

- excessive attenuation
- dirty optical fibers
- dirty connectors
- improper connector seating
- transmit laser degrade
- threshold set too high
- incorrect or faulty cabling at source end

This alarm is raised on the OC-192/STM64 facilities when using the ETH10G to GFP to STS-192c/STM-64c to OTU2 mapping.

This alarm is raised on the OC-n/STM-n client ports with provisionable thresholds of the following Broadband circuit packs:

- 2x10G OTR (NTK530PG/NTK530PM)
- 4x10G OTR (NTK530QA/NTK530QE/NTK530QM, NTK530QP)
- 4x10G MUX OCI (NTK525CF)
- 10x10G MUX OCI (NTK529BB)
- 40G Multi-Protocol OCI (NTK529SJ)

OC-n/STM-n thresholds can be user-provisioned to a value in the range of 1E-5 to 1E-9. The default value is 1E-6.

For 2x10G OTR circuit packs, this alarm can be raised when there is a OC-192/STM-64 Out Of Frame (OOF) condition at low alarm thresholds (1E-9).

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Impact

Minor, service-affecting (m, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA), if on an inactive path

Signal Fail (OC/STM)

Alarm ID: 5, 38, 150, 283, 985

Probable cause

This alarm is raised when the client Rx interface error rate detected by the B2 BIP-8 parity byte exceeds 1E-3.

This alarm is raised when the received OC/STM signal is degraded to the point where it is unusable, and the circuit pack is unable to detect the framing bytes in the received signal.

This alarm is also raised on the OC-192/STM64 when using the ETH10G to GFP to STS-192c/STM-64c to OTU2 mapping.

One of the following conditions causes this alarm:

- excessive attenuation
- dirty optical fibers
- dirty connectors
- improper connector seating
- transmit laser degrade

This alarm is raised on the OC-n/STM-n client ports with provisionable thresholds of the following Broadband circuit packs:

- 2x10G OTR (NTK530PG/NTK530PM)
- 4x10G OTR (NTK530QA/NTK530QE/NTK530QM, NTK530QP)
- 4x10G MUX OCI (NTK525CF)
- 10x10G MUX OCI (NTK529BB)
- 40G Multi-Protocol OCI (NTK529SJ)

OC-n/STM-n thresholds can be user-provisioned to a value in the range of 1E-3 to 1E-5. The default value is 1E-3.

Impact

Critical, service-affecting (C, SA) alarm, if active or unprotected with cross-connects

Minor, non-service-affecting (m, NSA) alarm if without cross-connects

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

The network element cannot clear a Loss of Signal alarm until a framed OC-n/STM-n signal is detected. The first time an optical fiber/cable is disconnected, the Loss of Frame alarm clears and a Loss of Signal alarm is raised that will not change back to Loss of Frame when the optical fiber/cable is re-attached.

Excessive Error Rate (STS/HO VC and VT/LO VC)

Alarm ID: 127, 193, 469

Probable cause

This alarm is raised when the received STS/HO VC path is degraded to the point where it is unusable.

- This alarm is caused by one of the following conditions:
- excessive attenuation
- dirty optical fibers
- dirty connectors
- improper connector seating
- transmit laser degrade

Impact

The alarm status is service-affecting on a ring system because the OC/STM line is not protected. The system cannot determine if path protection will be successful because that occurs where the path terminates. If the protection path is available, the path-terminating network element switches to that path to protect traffic.

STS/HO VC

Major, service-affecting (M, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA) alarm, if on an inactive path

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Prerequisites

To perform this procedure, you must:

- have the optical fiber/cable connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action						
1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to “Alarm hierarchies and alarm severities” section in this document. Clear any alarms of higher order on the hierarchy first using the appropriate procedures.						
2	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 3</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
4	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
5	Log into the remote network element at the transmit end. If you cannot log in remotely from the local network element, someone must be present at the remote site.						
6	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure. The following alarms can be ignored: • RFI/RDI alarm if the local alarm is Signal Degrade or Signal Fail • Excessive Error Rate alarm if the local alarm is Excessive Error Rate						

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Step	Action						
7	Retrieve the signal degrade threshold (SDTH) and compare the SDTH with the network diagram. Refer to the “Displaying node information” procedure in <i>Administration and Security</i> , 323-1851-301.						
	<table> <tr> <th>If the provisioned SDTH</th><th>Then</th></tr> <tr> <td>matches the SDTH on the network diagram</td><td>go to step 8</td></tr> <tr> <td>does not match the SDTH on the network diagram</td><td>edit the SDTH as required. Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i>, 323-1851-301. then go to step 8.</td></tr> </table>	If the provisioned SDTH	Then	matches the SDTH on the network diagram	go to step 8	does not match the SDTH on the network diagram	edit the SDTH as required. Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301. then go to step 8 .
If the provisioned SDTH	Then						
matches the SDTH on the network diagram	go to step 8						
does not match the SDTH on the network diagram	edit the SDTH as required. Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301. then go to step 8 .						
8	Ensure that the cross-connect signal rate on the entire path matches the optical fiber/cable connection information.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to the next step</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to the next step
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to the next step						
10	 DANGER Risk of laser radiation exposure Disabling ALS causes the laser to be permanently active. Laser radiation is present on the optical fiber. Do not look into the optical fiber. If enabled, disable ALS at both ends. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
11	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
12	<table> <tr> <th>If the alarm is raised against</th><th>Then go to</th></tr> <tr> <td>optical interface</td><td>step 13</td></tr> <tr> <td>STM-1e</td><td>step 19</td></tr> </table>	If the alarm is raised against	Then go to	optical interface	step 13	STM-1e	step 19
If the alarm is raised against	Then go to						
optical interface	step 13						
STM-1e	step 19						

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Step	Action
------	--------

13



CAUTION

Risk of traffic loss

Ensure that the correct module is identified. Removing the wrong optical fiber/cable drops all traffic on the local shelf.



DANGER

Risk of laser radiation exposure

Laser radiation is present on the optical fiber. Do not look into the optical fiber.

Remove the optical fiber from the circuit pack raising the alarm and use the optical power meter to measure the receive power.

14

If the power is

Then go to

below the receiver sensitivity for this circuit pack

[step 15](#)

above the receiver sensitivity for this circuit pack

[step 18](#)

For information about circuit pack technical specifications, refer to *6500 Planning*, NTRN10GK.

Power is below the receiver sensitivity

15

Adjust the local attenuation, if equipped, to try to get the receive power above the receiver sensitivity level.

16

If the receive power after adjustment is

Then go to

still below the receiver sensitivity

[step 17](#)

above the receiver sensitivity but below the maximum receiver power

[step 18](#)

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Step	Action						
17	Remove the Tx optical fiber from the far-end circuit pack and measure the transmit power at the far-end. If the transmit power at the far-end is <table> <tr> <td>above the launch power (minimum)</td><td> Then the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 19. </td></tr> <tr> <td>below the launch power (minimum) and the circuit pack at the transmit end supports SFP/SFP+/XFP/CFP module</td><td> replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19. </td></tr> <tr> <td>below the launch power (minimum) and the circuit pack at the transmit end does not support SFPs</td><td> replace the required circuit pack at the transmit end. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19. </td></tr> </table>	above the launch power (minimum)	Then the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 19.	below the launch power (minimum) and the circuit pack at the transmit end supports SFP/SFP+/XFP/CFP module	replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19.	below the launch power (minimum) and the circuit pack at the transmit end does not support SFPs	replace the required circuit pack at the transmit end. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19.
above the launch power (minimum)	Then the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 19.						
below the launch power (minimum) and the circuit pack at the transmit end supports SFP/SFP+/XFP/CFP module	replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19.						
below the launch power (minimum) and the circuit pack at the transmit end does not support SFPs	replace the required circuit pack at the transmit end. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 19.						

Power is above the receiver sensitivity

- 18 Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers.

Determining if the alarm has cleared

19	If the original alarm has <table> <tr> <td>cleared</td><td> Then the procedure is complete </td></tr> <tr> <td>not cleared</td><td> if the circuit pack supports a pluggable module, replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. </td></tr> </table>	cleared	Then the procedure is complete	not cleared	if the circuit pack supports a pluggable module, replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.
cleared	Then the procedure is complete				
not cleared	if the circuit pack supports a pluggable module, replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.				

Procedure 4-114 (continued)

Error alarms (OC/STM, STS/HO VC, and VT/LO VC)

Step	Action						
20	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.						
21	For the optical interface, if ALS was disabled in step 10 , enable the ALS at both the near-end and far-end. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
22	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-115

ESI alarms

Use this procedure to clear alarms associated with ESI-A or ESI-B.

AIS (ESI)

Alarm ID: 104, 111

Probable cause

This alarm is raised when a XC circuit pack detects an AIS on the incoming ESI timing reference signal. The upstream equipment generates an AIS signal to tell downstream equipment that a failure occurred. This alarm indicates that the ESI source (external clock source equipment) for this shelf has a failure. This is not applicable to a 2 MHz ESI signal.

Impact

Minor, non-service-affecting (m, NSA) alarm

The ESI source is not available to the shelf. If this is the active source, a timing protection switch occurs if another source is provisioned and available. Otherwise, the shelf enters timing holdover mode.

Loss of Frame (ESI)

Alarm ID: 103, 110

Probable cause

This alarm is raised when the XC circuit pack detects a loss of frame on the incoming ESI timing reference signal.

This procedure assumes that the signal has been in service and is alarm free. Ensure that frame provisioning for the shelf and ESI source (external clock source equipment) is correct. This is not applicable to a 2 MHz ESI signal.

Impact

Minor, non-service-affecting (m, NSA) alarm

The ESI source is not available to the shelf. If this is the active source, a timing protection switch occurs when another source is provisioned and available. Otherwise, the shelf enters timing holdover mode.

Loss of Signal (ESI)

Alarm ID: 102, 108

Probable cause

This alarm is raised when the XC circuit pack cannot detect a signal on the incoming ESI timing reference.

Procedure 4-115 (continued)

ESI alarms

Impact

Minor, non-service-affecting (m, NSA) alarm

The ESI source (external clock source equipment) is not available to the shelf. If this is the active reference, a timing protection switch occurs when another source is provisioned and available. Otherwise, the shelf enters timing holdover mode.

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action						
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
2	Use an appropriate test set to determine if the ESI port has a valid E1 or 2 MHz signal. The ESI ports are on the access panel of the shelf raising the alarm. Refer to “Technical specifications” section in <i>6500 Planning</i> , NTRN10GK for information on Synchronization interface pinout for access panels.						
3	<table><tr><td>If</td><td>Then go to</td></tr><tr><td>there is an AIS, LOF or LOS condition (AIS and loss of frame do not apply to a 2 MHz signal)</td><td>step 4</td></tr><tr><td>the signal is valid</td><td>step 5</td></tr></table>	If	Then go to	there is an AIS, LOF or LOS condition (AIS and loss of frame do not apply to a 2 MHz signal)	step 4	the signal is valid	step 5
If	Then go to						
there is an AIS, LOF or LOS condition (AIS and loss of frame do not apply to a 2 MHz signal)	step 4						
the signal is valid	step 5						
4	The problem is with cabling or the external clock source equipment and the 6500 shelf is reporting a valid condition. Perform troubleshooting on the cabling or external clock source equipment according to your company procedure. The procedure is complete.						
5	Identify the timing generation references in use on the network element. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
6	Look for the alarmed timing references for Timing Generation in the Synchronization window and note the corresponding reference.						
7	Go to the site. Look at the XC circuit packs in slots 7 and 8 of the 6500-7 packet-optical shelf, 7 and 8 of the 14-slot shelf or slots 9 and 10 of the 32-slot shelf. Look at the synchronization status LED (yellow circle) that corresponds to the reference you determined.						

Procedure 4-115 (continued)

ESI alarms

Step	Action
8	Verify which XC circuit pack is the synchronization primary. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
9	If the LEDs are not off on both XC circuit packs, reseal the synchronization primary circuit pack. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Note: When you reseal the XC circuit pack, an equipment switch takes place and the other XC becomes the synchronization primary. If the problem was with the XC that was reseated, the other XC will become the primary and will be controlling the alarms and the sync LED on itself and the sync LEDs on the XC which was restarted will be off.
10	Switch the traffic back to the XC which was reseated.
11	Verify the LEDs. If the LEDs are on, replace the circuit pack that has the synchronization status LED on. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
12	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-116

Event Log Full

Alarm ID: 1523, 1524, 1525, 1526, 1632, 1633, 1634, 1635, 1890, 1900, 1910, 1920, 2141

Probable cause

This alarm is raised when the event log for the Integrated Test Set is full.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 2 UPC.

Step	Action
1	In the Test Configuration tab, click the Stop Test button to stop the test.
2	Click Clear Results to clear the test results.
3	Restart the test. Refer to the “Performing a test with the integrated test set” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-117

Excessive Input Power

Alarm ID: 1566**Probable cause**

This alarm is raised on a Fixed Gain Amplifier (FGA) when the input power exceeds maximum power by 0.5 dB at the amp input.

For the MLA L-Band circuit pack, this alarm is raised against the AMP facility if the input power exceeds -3 dBm, and clears if the power is -4 dBm or below 1 dB hysteresis.

For the XLA circuit pack, this alarm is raised when the thresholds exceeds 0.5dB above the maximum input power. The threshold in high gain is 12.5dBm and in low gain is 18.5dBm.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action
1	Measure the input power to determine the amount of exceeding power comparing to the maximum power (-5.5dBm for FGA, -3.0 dBm for L-band MLA, 18.5dBm for XLA). Refer to the PM procedures in <i>Fault Management - Performance Monitoring</i> , 323-1851-520.
2	Verify the fiber connections between modules and correct as required. Clean and then reconnect the input/output fibers, monitor ports fibers, and connectors at the corresponding equipment. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
3	Verify the provisioned parameter values of the alarmed AMP facility against the EDP. Edit power levels as required. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	Ensure that required pads are in place as per the system design.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-118

Facility Provisioning Failure

Alarm ID: 2160, 2229

Probable cause

This alarm is raised on a facility when at least one provisioned facility parameter does not align with the capability of the physical hardware that is present.

Note: This problem needs to be fixed immediately as it may prevent traffic from being recovered after a cold restart of the circuit pack.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non service-affecting (m, NSA) alarm

Prerequisites

Use an account with at least a level 3 UPC.

Step	Action						
1	Identify the facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	<table><tr><td>If the facility raising the alarm is a</td><td>Then go to</td></tr><tr><td>PTP facility</td><td>step 3</td></tr><tr><td>ETTP facility</td><td>step 4</td></tr></table>	If the facility raising the alarm is a	Then go to	PTP facility	step 3	ETTP facility	step 4
If the facility raising the alarm is a	Then go to						
PTP facility	step 3						
ETTP facility	step 4						
3	Select the PTP facility raising the alarm. Ensure that the following facility parameters contain values that are supported for the circuit pack / pluggable which is present in the slot / port by selecting options that are available in the Site Manager Equipment & Facility Provisioning application. - Provisioned Transmission Mode - Provisioned Link Optimization Mode - FEC Format Go to step 5						
4	Select the ETTP facility raising the alarm. Ensure that the “FEC Format” facility parameters contain values that are supported for the circuit pack which is present in the slot by selecting options that are available in the Site Manager Equipment & Facility Provisioning application.						
5	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-119

Facility Provisioning Mismatch

Alarm ID: 2000

Probable cause

This warning is raised on “WLAi MOTR w/ OPS” and “WLAi FOTR w/ OPS” circuit packs if any of a subset of PTP facility parameters in [Table 4-2](#) are mismatched between the pair of internal optical protection switch (iOPS) line ports (1P1/1P2). The mismatched parameter name is specified in the “Additional Information” field of this warning.

Impact

Warning

Step	Action
1	In Site Manager Configuration menu, select Equipment & Facility Provisioning and open the Provisioning Status window for 1P1 and 1P2.
2	Find the mismatched PTP parameter from the subset in Table 4-2 that is specified in the “Additional Information” field of the warning.
3	Edit the appropriate ADJ-TX facility that is pointing to PTP facility with mismatched value in question (1P1 or 1P2). <i>This will edit the appropriate 1P1 or 1P2 PTP mismatched facility attribute which clears the alarm.</i>
4	If the alarm does not clear, it can be a scenario where “Additional Information” is out of date. To resolve this, open the Provisioning Status window for 1P1 and 1P2. Compare the values in the Requested column of 1P1 and 1P2. Ensure the values of each parameter listed in Table 4-2 are the same on both 1P1 and 1P2 PTP facilities.
5	For each parameter where the values are not the same, perform the ADJ-TX editing in step 1 .

Table 4-2
PTP Facility Parameters

Requested
Transmission Mode
Link Optimization Mode
Tx Channel Spacing (GHz)
Tx Wavelength (nm)
Tx Frequency (Thz)

Table 4-2
PTP Facility Parameters (continued)

Requested
Tx Dispersion Compensation Mode
Tx Dispersion Pre-Compensation (ps/nm)
T/R Control Laser Centering Mode
Laser Centering Range (GHz)
Channel Contention Detection and Avoidance
Tx Power Reduced State
OCH Line System Type
Allocated Spectral Width (GHz)
Tuning Mode
Note 1: Requested attributes on 1P1 must be edited from the Photonics ADJ-TX that is pointing to the 1P1 PTP facility. Requested attributes on 1P2 must be edited from the Photonics ADJ-TX that is pointing to the 1P2 PTP facility.
Note 2: Refer to the “PTP facility parameters” table in <i>Configuration - Provisioning and Operating</i> , 323-1851-310 for the supported parameter values.

- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-120

Facility Provisioning Out Of Range

Alarm ID: 2224**Probable cause**

This alarm is raised on PTP facility when the provisioned Tx power value is outside the range supported by actual hardware present on the shelf.

Impact

Major, service-affecting (M, SA) alarm

Minor, non service-affecting (m, NSA) alarm

Step	Action
1	Identify the circuit pack that raised the alarm. See, Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	If the provisioned PEC is the same as the actual PEC of the hardware on the slot, then re-provision the Tx power value to make it within the Tx power specification [minimum, maximum] of the provisioned PEC. Refer to the "Technical specifications" in <i>T-Series Guide</i> , 323-1851-103.
3	If the provisioned PEC is not the same as the actual PEC of hardware present, edit the PEC to match with the actual PEC and then adjust the Tx power according to the provisioned PEC to clear the alarm. Refer to the "Changing the provisioned PEC" procedure in the <i>Configuration - Provisioning and Operating for T-Series</i> , 323-1851-311.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-121

Fan Failed

Alarm ID: 224

Probable cause

This alarm is raised when a cooling fan module is:

- not fully inserted
- provisioned but is missing
- equipped but has failed

For 14-slot shelves, this alarm is raised when the fan is missing from the shelf.

For cooling fan modules that have more than one integrated fan, the shelf will raise the alarm when the module is not inserted in the fan slot or when one or more of the integrated fans is in a failed state.

This alarm is also raised when a Type 3 high-flow cooling fan module (NTK507LS or NTK507MS) is inserted in a shelf other than the 14-slot packet-optical shelf. Because they are not compatible with the shelf, Type 3 fans will not energize or light any LEDs.

Impact

Critical, service-affecting (C, SA) alarm

Major, non-service-affecting (M, NSA) alarm

In a 14-slot shelf, the alarm will be raised as Major, non-service-affecting (M, NSA) when a single cooling fan module is missing or fails and it will escalate to Critical, service-affecting (C, SA) if more than one cooling fan module is missing or fails. For a missing or failed cooling fan module in a 2-slot, 4-slot, 7-slot, 6500-7 packet-optical, and 32-slot shelves, the alarm will be raised as Critical, service-affecting (C, SA).

If a fan module is missing for more than 60 seconds, the shelf cooling may be compromised and as a precaution, a “[High Temperature](#)” alarm will be raised as described in this document.

Procedure 4-121 (continued)
Fan Failed

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

ATTENTION

Ensure that you have the correct fan type. After removal of a cooling fan module from a 14-slot or 32-slot shelf, the cooling fan module must be replaced within one minute to prevent the circuit packs from overheating due to insufficient airflow. If the cooling fan module is removed from a 2-slot, 4-slot, 6500-7 packet-optical, or 7-slot shelf, it must be replaced within 30 seconds. Have a replacement fan module ready before removing the fan module.



CAUTION

Risk of damage to circuit packs and modules

Do not attempt to cause this alarm to be raised by removing any cooling fan modules or filler cards or by blocking the shelf air inlet or exhaust ports as it will compromise shelf cooling. A shelf with a compromised cooling system may result in circuit pack or module failures prior to the assertion of a High Temperature Warning or High Temperature alarm.

Step	Action
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
	Select your next step.
2	If the alarm is raised on a 2-slot, 4-slot, 7-slot, 6500-7 packet-optical, or 32-slot shelf
	otherwise
3	Verify that the fan module is fully inserted.
4	Make any necessary adjustments.



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

- | Step | Action |
|------|---|
| 1 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 2 | Select your next step. |
| 2 | If the alarm is raised on a 2-slot, 4-slot, 7-slot, 6500-7 packet-optical, or 32-slot shelf |
| | otherwise |
| 3 | Verify that the fan module is fully inserted. |
| 4 | Make any necessary adjustments. |

Procedure 4-121 (continued)

Fan Failed

Step	Action						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>this procedure is completed</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	this procedure is completed	not cleared	go to step 6
If the original alarm has	Then						
cleared	this procedure is completed						
not cleared	go to step 6						
6	Loosen and remove the retaining screw(s).						
7	Pull out the failed cooling fan module from the 2-slot, 6500-7 packet-optical, 7-slot, or 32-slot shelf.						
 CAUTION Risk of personal injury Use the handle on the front of the fan module to extract it. Do not hold or carry the fan modules in a manner that could cause detrimental contact to the fan blades (which will stop rotating when power is disconnected due to module extraction). You can provide extra support for the 2-slot, 4-slot shelf, 6500-7 packet-optical, or 7-slot shelf fan modules by holding the bottom of the fan module during extraction and insertion. A fan module for a 32-slot shelf should be handled by firmly grabbing the left and right sides taking care not to touch the fan blades.							
8	Insert the replacement module into the slot in the shelf until it is fully inserted and then insert and tighten the retaining screw(s).						
9	Verify that the fan green LED turns on. Go to step 24 .						

Removing the snap-on shelf cover or opening the hinged cover (if equipped)

- 10 On the hinged shelf cover, move the latches to the unlock position and rotate the shelf cover to open it.

Removing the grill/air deflector (if equipped)

- 11 On the left side of the shelf, pull on the spring-loaded pins of the grill/air deflector; at the same time pull the left end of the grill/air deflector just enough to disengage the pin from the shelf hole.

Procedure 4-121 (continued)

Fan Failed

Step	Action						
12	On the right side of the shelf, pull on the spring-loaded pins of the grill/air deflector; at the same time pull the right end of the grill/air deflector just enough to disengage the pin from the shelf hole.						
13	Pull out the grill/air deflector and store it in a safe place.						
14	Check whether the identified fan module is missing, has failed, or is not compatible with the shelf or its equipped Power Input Cards. The fan status is indicated by the LEDs on each fan module. • A green LED is lit when the module is working properly. • A red LED is lit when the fan fails but is receiving power. • No LEDs are lit if the fan control circuit is damaged or if the fan is not receiving power because it is not compatible with the shelf or its Power Input Cards. Treat a fan with no LEDs lit as a failed fan.						
15	<table> <tr> <th>If the fan module</th><th>Then go to</th></tr> <tr> <td>has failed</td><td>step 16</td></tr> <tr> <td>is missing</td><td>step 19</td></tr> </table>	If the fan module	Then go to	has failed	step 16	is missing	step 19
If the fan module	Then go to						
has failed	step 16						
is missing	step 19						

Verifying the fan module

16	Verify that the fan module is fully inserted.						
17	Make any necessary adjustments.						
18	<table> <tr> <th>If the original alarm has</th><th>Then go to</th></tr> <tr> <td>cleared</td><td>step 22</td></tr> <tr> <td>not cleared</td><td>step 19</td></tr> </table>	If the original alarm has	Then go to	cleared	step 22	not cleared	step 19
If the original alarm has	Then go to						
cleared	step 22						
not cleared	step 19						

Installing/replacing the fan module

19	Install the missing fan module(s) or replace the failed fan module(s). Refer to the "Replacing a cooling fan module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Ensure you use the correct fan type.						
20	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>go to step 21</td></tr> <tr> <td>not cleared</td><td>install the grill/air deflector (step 22 and step 23) and contact your next level of support or your Ciena support group.</td></tr> </table>	If the original alarm has	Then	cleared	go to step 21	not cleared	install the grill/air deflector (step 22 and step 23) and contact your next level of support or your Ciena support group.
If the original alarm has	Then						
cleared	go to step 21						
not cleared	install the grill/air deflector (step 22 and step 23) and contact your next level of support or your Ciena support group.						

Procedure 4-121 (continued)

Fan Failed

Step	Action
------	--------

Installing the snap-on shelf cover or closing the hinged cover (if previously equipped)

- | | |
|----|--|
| 21 | Close the hinged cover and lock it on the shelf. |
|----|--|

Installing the grill/air deflector (if previously equipped)

- | | |
|----|---|
| 22 | Pull on the spring-loaded pins of the sides grill/air deflector, insert the grill/air deflector into the shelf, and release the pins. |
| 23 | Align the pins with the holes in the sides of the shelf and push the grill/air deflector until the pins enter the holes. |
| 24 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-122

Fan Housing Missing

Alarm ID: 736

Probable cause

This alarm is raised when a cooling unit housing is physically not present in a shelf that is provisioned for fan modules.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- obtain a cooling unit housing and fan modules

Step	Action
------	--------



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

- 1 Determine if fans are required for the shelf in the current configuration. Refer to the “Cooling configurations” section in *6500 Planning*, NTRN10GK.
You cannot install a cooling unit housing for a metro front electrical shelf, as the cooling unit housing is an integral part of the shelf.
- 2 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.

Procedure 4-122 (continued)

Fan Housing Missing

Step	Action						
3	<table><tr><td>If fan modules are</td><td>Then</td></tr><tr><td>not required</td><td>deprovision the fan modules (set the fan monitor status to off). Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i>, 323-1851-301.</td></tr><tr><td>required</td><td>install a cooling unit housing and the fans in the shelf. Refer to the appropriate shelf installation procedure in the Installation technical publication specific to the respective 6500 shelf type. When the cooling unit housing is installed in the shelf, the Fan Failed alarm is raised until the fans are installed and working.</td></tr></table>	If fan modules are	Then	not required	deprovision the fan modules (set the fan monitor status to off). Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301.	required	install a cooling unit housing and the fans in the shelf. Refer to the appropriate shelf installation procedure in the Installation technical publication specific to the respective 6500 shelf type. When the cooling unit housing is installed in the shelf, the Fan Failed alarm is raised until the fans are installed and working.
If fan modules are	Then						
not required	deprovision the fan modules (set the fan monitor status to off). Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301.						
required	install a cooling unit housing and the fans in the shelf. Refer to the appropriate shelf installation procedure in the Installation technical publication specific to the respective 6500 shelf type. When the cooling unit housing is installed in the shelf, the Fan Failed alarm is raised until the fans are installed and working.						
4	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-123

Fan Incompatible

Alarm ID: 655**Probable cause**

This alarm is raised when cooling fan modules with one exhaust air direction are inserted into a cooling unit housing that has a different exhaust air direction (for example, inserting front blowing fan in rear venting cooling unit housing).

In 14-slot shelf types that have an integrated cooling unit housing (such as metro front-electrical, a converged or a packet-optical shelf) with three working fans present, where one of the fans exhausts air in a different direction (front or rear), the alarm is raised against that different fan. In a shelf with two working fans present and both fans exhaust air in a different direction, the alarm is raised against the fan in the slot that is farthest to the right.

Impact

Critical, service-affecting (C, SA) alarm if two or more fans are incompatible
Major, non-service-affecting (M, NSA) alarm if less than two fans are incompatible in the cooling unit housing

Prerequisites

To perform this procedure, you must

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- obtain a compatible fan module

Procedure 4-123 (continued)
Fan Incompatible

Step	Action
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DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

ATTENTION

After removing a cooling fan module from a 14-slot shelf, the cooling fan module must be replaced within one minute to prevent the circuit packs from overheating due to insufficient airflow. Have a replacement fan module ready before removing the fan module.

- 1 Check the shelf inventory to determine which fan is incompatible. Obtain the compatible fan for replacement. Refer to the “Displaying shelf inventory information” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.
- 2 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
- 3 Remove the alarmed fan from the shelf or the cooling unit housing. Refer to the “Replacing a cooling fan module” procedure in *Fault Management - Module Replacement*, 323-1851-545. The Fan Failed alarm is raised for the removed fan and the Fan Incompatible alarm for the fan clears.
- 4 Wait at least 15 seconds but not more than 60 seconds and then insert a new fan into the slot.
- 5 Verify that the Fan Failed alarm has cleared. Retrieve all alarms to determine if the original alarm has cleared.

—end—

Procedure 4-124

Fan Missing

Alarm ID: 223**Probable cause**

This alarm is raised on a shelf when a fan module is missing.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation- General Information*, 323-1851-201.0.

Step	Action
------	--------

**DANGER****Risk of eye injury**

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.

- 1 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
- 2 Insert the fan module into the shelf. Refer to the “Replacing a shelf fan module” procedure in *Fault Management - Module Replacement*, 323-1851-545.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-125

Far End Client Signal Fail

Alarm IDs: 695, 870, 1247, 1306, 1307, 1382, 1488, 1625, 1671, 1678, 1977, 1978

Probable cause

This alarm is raised against the OTM0/OTM1/OTM2 layer facility to indicate that there is a client signal failure such as LOS/LOF/LOL, or pluggable missing on the far end client.

This alarm is raised for an OC-192/STM-64 client mapped to OTU2 line when the FECSFMONITOR flag is enabled on the line side and when there is a P-AIS detected on the line side. The WAN Far End Client Signal Fail alarm is raised only with GFP CMF conditioning and there is a far-end client signal failure.

This alarm is also raised against an OTM3/OTM4 layer facility of a 40G OCl, 40G+ CFP OCl, or 100G OCl circuit pack when service is terminated at the near-end while the far-end CBR and Ethernet private line client signal is failed (such as LOS or LOF). The alarm is masked by line OTU-LOS/LOF/pre-FEC SF, LOMF, and ODU-AIS/LCK/OCl.

For the 100G WL3n MOTR circuit pack, this alarm is raised against the layer OTM2, OTM3 or WAN facilities when the far-end mate port for this service is experiencing a client Rx signal fault.

For the 16xFLEX OTN I/F, WLAi variants, (1+2) 100G PKT/OTN I/F, 10x10G PKT/OTN IF, 100G PKT/OTN WL3n I/F circuit pack, this alarm is raised against the WAN or ODUCTP facility when the far-end mate port for this service is experiencing a client Rx signal fault.

For the 100G (2xQSFP+/2xSFP+) MUX circuit pack, this alarm is raised against the layer OTM2, OTM3 or WAN facilities when the far-end mate port for this service is experiencing a client Rx signal fault.

For the 200G (2x100G/5x40G) MUX (NTK529HA) circuit pack, this alarm is raised against the layer OTM2, OTM3, OTM4 or WAN facilities when the far-end mate port for this service is experiencing a client Rx signal fault.

Procedure 4-125 (continued)
Far End Client Signal Fail

Note: Detection of unexpected Client Management Frames (CMF) raises the “Far End Client Signal Fail” alarm instead of “CMF UPI Mismatch” alarm.

For the 4x10G MUX circuit pack, this alarm is raised against the STTP, ETTP and WAN facilities when the far-end mate port for this service is experiencing a client Rx signal fault.

For circuit packs with WAN facilities, this alarm is raised when the far-end mate port for this service is experiencing a client Rx signal fault.

For eMOTR circuit packs, this alarm is raised against the ODUTTP, ETTP, or WAN facility when the far-end mate port has a client Rx signal fault. Note that the far-end mate port can not be a an eMOTR circuit pack, as these circuit packs do not generate a Client Signal Fail indication.

This alarm is raised at the Far end when VLLI is enabled and the LAG fails and not all members are administratively OOS.

For eMOTR circuit packs, VLLI enabled client UP MEP will show ETTP level Far End Client signal Fail when FEND UP MEP port undergoing LOS/LOL/LOF conditions. This is applicable to Flex-CCM sessions.

For S-TAG UP MEP CFM sessions, far-end UP MEP administrative change to OOS will also result in FECSF. If far-end UP MEP created on LAG port and all of its members are moved to Administratively OOS, then near-end port (or LAG lead port) will declare Far End Client Signal Fail.

Impact

Critical, service-affecting (C, SA) alarm, if the alarm is raised on an active traffic path

Minor, non-service-affecting (m, NSA) alarm, if the alarm is raised on an inactive traffic path

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Procedure 4-125 (continued)

Far End Client Signal Fail

- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- have a fiber cleaning kit
- have the engineering documentation package (EDP) containing shelf details

Step	Action
1	Verify the fibers are properly connected and not crossed, looped back, or misconnected at the corresponding far-end client circuit pack.
2	Retrieve alarms from the corresponding far-end client circuit pack. Refer to the "Retrieving active alarms for one or more network elements" procedure in this document.
3	Use the appropriate alarm clearing procedure to clear any alarms raised at the far-end client circuit pack.
4	If the alarms do not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-126

Far End Protection Line Fail

Alarm IDs: 521, 522, 523, 524, 961, 1019, 1125, 1332, 1344, 1392

Probable cause

This alarm is raised to indicate that a protection line signal failure was received in the APS bytes sent by the far-end protection engine. This indicates that the far-end protection engine cannot switch traffic to the protection line because a fault was detected downstream. This alarm is raised for a 1+1 OTN, 1+1 port TPT, or 1+1 TPT configuration protection group.

This alarm is cleared if a signal failure condition prevents the reception of valid APS bytes.

This alarm only applies to bidirectional mode protection switching.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- ensure that you have the optical fiber connection information on how the optical modules on each network element connect to other network elements

Step	Action
------	--------

- | | |
|---|--|
| 1 | Starting at the far-end network element and going upstream, check all network elements for alarms against the protection line. Refer to the “Retrieving active alarms for one or more network elements” procedure in this document. Clear all far-end alarms related to the protection line. |
| 2 | Clear all other unexpected standing alarms on the local 6500 network element by following the related trouble clearing procedures. |
| 3 | If the “Far End Protection Line Fail” alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-127

Fiber Channel Link Not operational

Alarm ID: 1258, 2132

Probable cause

This alarm is raised when fiber channel NOS primitive is being received from the subtending equipment.

This alarm is raised against the FLEX facility when the protocol is FC400 or FC800.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Step	Action						
1	Verify the transmitting subtending equipment is functioning correctly and transmitting a valid signal.						
2	Log into the remote network element at the transmit end.						
3	Retrieve all alarms from the remote network element at the transmit end.						
4	Look for an alarm message for the remote network element circuit pack connected to the original shelf. <table><tr><th>If there are</th><th>Then</th></tr><tr><td>no alarms at the transmit end</td><td>ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Then go to step 5.</td></tr><tr><td>additional alarms at the transmit end</td><td>refer to the appropriate alarm clearing procedures. Then go to step 5.</td></tr></table>	If there are	Then	no alarms at the transmit end	ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Then go to step 5 .	additional alarms at the transmit end	refer to the appropriate alarm clearing procedures. Then go to step 5 .
If there are	Then						
no alarms at the transmit end	ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Then go to step 5 .						
additional alarms at the transmit end	refer to the appropriate alarm clearing procedures. Then go to step 5 .						
5	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-128

Fiber Loss Detection Disabled

Alarm ID: 1581**Probable cause**

This alarm is raised when the “High Fiber Loss Detection Alarm” system parameter is set to Disabled on a shelf equipped with a Colorless OADM OTS. Colorless OADM OTS requires High Fiber Loss Detection.

The alarm is also raised when the system parameter is set to Disabled on a shelf equipped with a configuration that supports Dark Fiber Loss Measurement such as CDC or CDA.

The “High Fiber Loss Detection Alarm” parameter can be found in the Site Manager Node Information application under the System tab.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Enable the “High Fiber Loss Detection Alarm” system parameter. Refer to the “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-129

Fiber Type Manual Provisioning Required

Alarm ID: 907

Probable cause

This alarm is raised against a ADJ-LINE facility when the fiber type for the line adjacency is set to 'Unknownfiber type' (not provisioned), putting the line adjacency out of service.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Provision the Fiber Type value of the alarmed line adjacency to a value other than 'Unknownfiber type'. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-130

Filter Replacement Timer Expired

Alarm ID: 1580

Probable cause

This alarm is raised to indicate that you must replace the air filter in the shelf.

The default value of the replacement time interval is 24 months.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a replacement air filter to install in the empty slot

Step	Action										
1	Choose your first step.										
	<table> <tr> <th>If you want to</th><th>Then</th></tr> <tr> <td>replace the filter</td><td>step 2</td></tr> <tr> <td>set a new replacement time</td><td>step 3</td></tr> <tr> <td>reset the filter timer</td><td>step 4</td></tr> <tr> <td>disable the filter timer</td><td>step 5</td></tr> </table>	If you want to	Then	replace the filter	step 2	set a new replacement time	step 3	reset the filter timer	step 4	disable the filter timer	step 5
If you want to	Then										
replace the filter	step 2										
set a new replacement time	step 3										
reset the filter timer	step 4										
disable the filter timer	step 5										
2	Replace the air filter in the shelf. Refer to the “Replacing the shelf air filter” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 4 .										
3	Provision a new replacement time interval using Site Manager. Refer to the “Resetting the air filter replacement timer” procedure in <i>Administration and Security</i> , 323-1851-301. Go to step 6 .										
4	Reset the timer. The timer will reset to the value given by the replacement time interval. Refer to the “Resetting the air filter replacement timer” procedure in <i>Administration and Security</i> , 323-1851-301. Go to step 6 .										

Procedure 4-130 (continued)

Filter Replacement Timer Expired

Step	Action
5	Disable the timer. Refer to the “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i> , 323-1851-301. Note: Disabling and re-enabling the timer that has already expired does not cause the timer to reset. The alarm is raised after re-enabling the timer.
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-131

Flash Banks Mismatch

Alarm ID: 222**Probable cause**

This alarm is raised when an upgrade is interrupted. If an upgrade is interrupted, there can be different loads present on the shelf processor.

The alarm can also be raised during a shelf processor replacement procedure when a software load on either the flash bank of the network element is not up to date with respect to the release that is running on the network element.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	If the alarm is raised during a shelf processor replacement, it will clear once the shelf processor replacement procedure is completed.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-132

FLEXI Map Mismatch

Alarm IDs: 2206

Probable cause

This alarm is raised on the FLEXI facility of the FlexO client of WL5e MOTR circuit pack when the received FlexO Instance Identification (IID) does not match the expected IID or when the received PHY Map does not match the expected PHY Map. Each FlexO member that have the same Group Identifier (GID) must have a unique IID.

Impact

Critical, service-affecting (C, SA) alarm if not protected
Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in Installation - General Information, 323-1851-201.0
- use an account with at least a level 3 UPC

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	From the Site Manager Configuration menu, select Equipment & Facility Provisioning . Select the alarmed facility, and record the values from the Rx Instance ID , Tx Instance ID , Rx IID Mapping Bitmap , and Tx IID Mapping Bitmap columns. Refer to the “Retrieving equipment and facility details” procedure in Configuration - Provisioning and Operating, 323-1851-310.
3	If the RXIID and TXIID or the Rx IID Mapping Bitmap and Tx Then go to IID Mapping Bitmap values are not equal step 4 equal step 5
4	There is a likely a misconnected fiber. Verify that the optical fiber connections are correct on the circuit pack raising the alarm and on the upstream circuit pack. The RXIID value from step 2 indicates where the misconnection exists.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-133

Frequency Out of Range (OC192/STM64, ETH10G, ETH40G, ETH100G, ETTP, STTP, OTM3, OTUTTP, PTP)

Alarm ID: 1613, 1614, 1615, 1616, 1843, 1844, 1845, 1859, 1951, 2026, 2208

Probable cause

This alarm is raised when the frequency limits exceed the IEEE Std 802.3 Tx frequency offset specifications of +/- 20ppm for 10GBASE-W and +/-100ppm for 10GBASE-R.

More specifically, for the following 6500 transport cards: NTK525CA, NTK525CF, NTK529BB, NTK529BX, NTK530QA, NTK530QM, NTK530QP, and NTK530QE, this alarm is raised when the measured frequency of the input client clock is greater than +/-27.5ppm (OC192/STM64) or +/-107.5ppm (ETH10G and ETH40G).

For the WLAi MOTR, WLAi FOTR and WL5e MOTR circuit packs, this alarm is raised when the measured frequency of the input client clock is greater than +/-27.5ppm (OTUTTP and STTP) or +/-107.5ppm (PTP, ETTP).

The Frequency Out of Range alarm does not force conditioning.

Note: This alarm does not apply to 40G MUX OCI (NTK525CA) when ETH10G is provisioned with GFP mapping.

Impact

Major, service-affecting (M, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	If possible, clear all OC-192/STM-64, ETH10G, ETH40G, ETH100G, ETTP, STTP, OTM3, OTUTTP and PTP alarms from the network.
2	If the alarm does not clear, verify the subtending equipment to ensure that it meets the frequency specification. If the equipment does not meet the specification, change the client signal to use subtending equipment that is within specifications.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-134

Frequency Provisioning Mismatch

Alarm ID: 1814, 1885

Probable cause

This alarm is raised when the provisioned laser transmit frequency on the both ends of the link do not match.

Impact

Major, service-affecting (M, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action
------	--------

- | | |
|---|---|
| 1 | Ensure that both ends of the link are provisioned for the same channel spacing. If both ends are provisioned for gridless channel spacing, ensure that both ends are provisioned for the same frequency. If not, reprovision the transmit frequency on the PTP, OTM4 or OTUCn facility to make both ends match. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 2 | If the alarm does not clear, replace the alarmed circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-135

Gauge Threshold Crossing Alert Summary

Alarm ID: 724, 725, 726, 1034, 1972

Probable cause

This is a summary alarm for each AMP, VOA, RAMAN, OTDRCFG, and OPTMON facility, and is raised if one or more of the physical gauge power values crosses its provisioned PM threshold.

This alarm is masked by the Loss of Signal and Circuit Pack Failed alarms.

ATTENTION

PM thresholds stored in PM Profiles define the maximum deviation from the currently set baseline for a gauge power value. Resetting baselines of gauge power values is normally done by DOC, but may also be manually triggered by the user.

The typical cause for this alarm is reduced power levels on the port reporting the alarm. Conditions that can result in reduced power levels at a port include:

- a PM threshold setting that is too low for a gauge power value
- a faulty or incorrectly provisioned transmitter module
- a faulty or incorrectly provisioned receive module
- an optical signal degradation caused by a bent optical fiber or dirty optical connector
- a wrong DSCM type for a link
- improper optical cable mating
- a disconnected optical fiber at the amplifier output
- an optical fiber cut
- a disconnected or missing termination
- misprovisioning of an amplifier resulting in excessive power being injected into the mid-stage DSCM or fiber-plant
- a power value has been reported as outside of range (OOR) and the baseline has been set as out of range (OOR) and the power fluctuates to a non-OOR value, then the alarm can be raised
- new transponders are connected to DIA channels are added in a downstream domain of a branched network

Procedure 4-135 (continued)

Gauge Threshold Crossing Alert Summary

Use this alarm to resolve the causes of the threshold crossings before a service-affecting problem occurs.

Note: If this alarm is raised for amplifiers immediately after an upgrade, the alarm can be cleared by resetting the baseline.

This alarm clears automatically when all gauge readings fall within their threshold boundaries.

Note: This alarm can be cleared manually from the Site Manager PM screen by resetting the baseline.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the engineering documentation package (EDP) containing shelf details
- have a fiber cleaning kit
- obtain a replacement module or fiber patchcord, if required

Step	Action
------	--------

1	If this alarm	Then
	was raised as a result of a maintenance activity or during SLAT and it is expected	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.
	is not expected	continue with step 2
2	Verify that the PM threshold values for the alarmed facility are correctly provisioned. Adjust the value if required.	

Procedure 4-135 (continued)

Gauge Threshold Crossing Alert Summary

Step	Action	
3	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 4
4	If there are new transponders connected to or disconnected from a DIA or channels were added or deleted on downstream domains	Then manually update of TCA thresholds. See <i>Performance Monitoring</i> , 323-1851-520. If the alarm does not clear, go to step 5
	otherwise	go to step 5
5	Verify that the optical power is within range. If both the minimum and maximum values are outside of range, you can enable automatic in-service (AINS) for the facility until a valid signal is present. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If the power level is too close to the Input Loss of Signal Threshold configured in the Amp properties, the alarm is going to be toggled. In this case, edit the Threshold 1 dB below the threshold level on the Amp settings.	
6	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 7
7	Using the alarm details for the Gauge Threshold Crossing Alert Summary alarm, note the Unit and Class against which the alarm is raised.	
8	Retrieve the PMs for the Shelf, Type, and Facility based on the information noted in step 7 .	
9	From the PM application, note the facility parameter that has an Untimed value that crossed the threshold value.	

Procedure 4-135 (continued)

Gauge Threshold Crossing Alert Summary

Step	Action						
10	Check for and clear any of the following alarms on all network elements before clearing this alarm: • Automatic Power Reduction Active • Automatic Shutoff • Input Loss of Signal • Loss of Signal (OPTMON) • Optical Line Fail • OSC Loss of Signal • Output Loss of Signal • Shutoff Threshold Crossed						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Determine if there are other Gauge Threshold Crossing Alert Summary alarms raised in the network and begin by troubleshooting the most upstream Gauge Threshold Crossing Alert Summary alarm. To troubleshoot the most upstream Gauge Threshold Crossing Alert Summary alarm, verify the optical patchcord connected to the port reporting the alarm. Note: Ensure the optical patchcord is connected at both ends and that there is no problem with the optical patchcord. Clean the connectors. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.						
13	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	If the alarm is reported against the OPIN parameter of a LIM amplifier Line A In (port 8) facility (or the OPR parameter of a LIM OPTMON Line A In [port 8] facility) AND the OPR parameter of a LIM OSC A Out (port 4) facility, then verify the outside fiber plant.						
15	If the alarm is reported against the OPOUT parameter of CMD or SCMD Channel In VOA facility, make sure that the transmitting subtending equipment is functioning correctly and transmitting a valid signal.						
16	If the alarm is reported against the Common Out parameter of a SCMD4 Common In VOA facility or the OPR parameter of a SCMD4 Common In OPTMON facility, it may be a problem at the head-end site where the SCMDs are interconnected using the upgrade ports. Check the patchcords used at the upgrade ports.						

Procedure 4-135 (continued)

Gauge Threshold Crossing Alert Summary

Step	Action
17	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-136

GCC0/GCC1/GCC2 Link Failure

Alarm ID: 672, 834, 996, 1009, 1164, 1179, 1430, 1431, 1432, 1723, 1724, 1725, 1830

Probable cause

This alarm is raised when the network element communications fail on the ITU-T G.709 general communication channel (GCC0, GCC1, or GCC2) link. This alarm can also be raised if the facility (OTM1ODUCTP, ODUTTP, or OTUTTP) is physically looped back (Tx interface connected to Rx interface).

This alarm can also be raised when a connection is deleted or a fault exists on the affected path.

Note: This alarm holds true whether IPv4 or IPv6 is enabled on the shelf.

For all circuit packs, the GCC link is controlled by lower layer SDCC and not a circuit (IISIS or OSPF). Therefore, even if there are no IISIS or OSPF circuits provisioned, the GCC link is still up between near-end and far-ends and no GCC Link Failure alarm is raised.

If only one end of a GCC PPP link is provisioned while the other end is not, the link state of the PPP link at the first end will be down. Therefore, you will see this alarm raised on the SP, until the PPP link gets provisioned at the other end.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- clear all remote alarms present against the optical fiber
- use an account with at least a level 3 UPC

Procedure 4-136 (continued)
GCC0/GCC1/GCC2 Link Failure

Step	Action						
1	Identify the facility raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	Use the optical fiber connection information to identify the network element and the circuit pack that is the source of the signal reporting the alarm.						
3	Wait five minutes after the alarm was raised in case a shelf processor or interface circuit pack restart at the remote terminal caused the alarm.						
4	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	go to step 5
If	Then						
the original alarm has cleared	the procedure is complete						
not cleared	go to step 5						
5	Ensure the following comms parameters (in the Site Manager Comms Setting Management application) exist at each end of the GCC0/GCC1/GCC2 link for the circuit pack reporting the alarm: — GCC0/GCC1/GCC2 PPP interface (under Interfaces, Interface type=PPP) — GCC0/GCC1/GCC2 Lower Layer DCC/GCC interface (under Interfaces, Interface type=Lower Layer DCC/GCC) Refer to the “Retrieving communications settings” procedure in <i>Data Comms Planning & User Guide</i>, 323-1851-101.						
6	If the original alarm has cleared, the procedure is complete. Otherwise go to step 7 .						
7	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
8	Log into the remote network element using the external IP address. • If the login is successful, go to step 10. • If the login fails, go to step 9.						
9	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the remote network element is only accessible through GCC0/GCC1/GCC2</td><td>the login may not be possible, as the GCC0/GCC1/GCC2 has failed. Go to step 10</td></tr> <tr> <td>otherwise</td><td>go to step 12.</td></tr> </table>	If	Then	the remote network element is only accessible through GCC0/GCC1/GCC2	the login may not be possible, as the GCC0/GCC1/GCC2 has failed. Go to step 10	otherwise	go to step 12 .
If	Then						
the remote network element is only accessible through GCC0/GCC1/GCC2	the login may not be possible, as the GCC0/GCC1/GCC2 has failed. Go to step 10						
otherwise	go to step 12 .						
10	Replace the required circuit pack at the remote site determined in step 2. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545.						

4-310 Alarm clearing procedures

Procedure 4-136 (continued)

GCC0/GCC1/GCC2 Link Failure

Step	Action						
11	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 12</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Replace the required circuit pack at the network element originally reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
13	If the alarm is still active, the alarm can be the result of mis-fibering causing a physical loopback along the photonic layer. Check the optical fiber along the path of the channel.						
14	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-137

GID Mismatch

Alarm IDs: 2020

Probable cause

This alarm is raised when received Group ID (GID) does not match the expected Group ID.

GID is provided to identify an OTSi association to a particular group and is used to check that the OTSi belongs to the intended group and can provide alarms on mis-connections.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

Log into the network element user interface at the default view level with appropriate user access privileges.

Step	Action						
1	Identify the circuit pack or circuit packs raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	From the Site Manager Configuration menu, select Equipment & Facility Provisioning . Select the alarmed facility, and record the values from the associated Far End GID columns. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
3	<table><tr><td>If the Rx GID and Tx GID values are</td><td>Then go to</td></tr><tr><td>not equal</td><td>step 4</td></tr><tr><td>equal</td><td>step 5</td></tr></table>	If the Rx GID and Tx GID values are	Then go to	not equal	step 4	equal	step 5
If the Rx GID and Tx GID values are	Then go to						
not equal	step 4						
equal	step 5						
4	There is a likely a misconnected fiber. Verify that the optical fiber connections are correct on the circuit pack raising the alarm and on the upstream circuit pack. The associated Far End GID value from step 2 indicates where the misconnection exists.						
5	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-138

Group Loss of Signal

Alarm ID: 710, 832

Probable cause

The alarm is raised against port 1 of SCMD4, OMD4 or OMX modules (that is, the Common In port of SCMD4, OMD4 or OTS In port of OMX). For OMD4 and OMX, it is a logical alarm resulting from alarm correlation when loss of signal conditions are detected on all the line receivers connected on the optical channels of the module. This alarm is raised only when alarm correlation is turned on. For SCMD44, this alarm is raised if the LOS condition is detected on the circuit pack after the group is split.

The conditions can be caused by a faulty module or a fault that impacts all channels of the module, for example:

- a line fiber cut
- excessive power loss on the line fiber that exceeded link budget
- a disconnected, or defective fiber optic patchcord
- a dirty optical fiber connector
- a provisioning error
- a reflective event, indicated by an Automatic Power Reduction Active
- alarms at an upstream booster amplifier or pre-amplifier (DSCM)

This condition is also possible if there is a disconnected, defective patchcord or dirty optical fiber connected between the OMD4/OMx module and every line receiver.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in *Installation - General Information*, 323-1851-201.0 or *Module Replacement Procedures*, 323-1851-545
- have a network and site diagram
- have a fiber cleaning kit
- obtain a replacement circuit pack, if required

Procedure 4-138 (continued)
Group Loss of Signal

Step	Action						
1	Check for and clear any of the following alarms before clearing this alarm: • Automatic Power Reduction Active • Automatic Shutoff • Input Loss of Signal • Optical Line Fail • Output Loss of Signal • Shutoff Threshold Crossed						
2	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	<table> <tr> <td>If the alarm is raised against</td><td>Then go to</td></tr> <tr> <td>OMD4 or OMX</td><td>step 5</td></tr> <tr> <td>SCMD4</td><td>step 8</td></tr> </table>	If the alarm is raised against	Then go to	OMD4 or OMX	step 5	SCMD4	step 8
If the alarm is raised against	Then go to						
OMD4 or OMX	step 5						
SCMD4	step 8						
4	Consult the network configuration chart.						
5	Verify that the adjacency provisioning is correct for itself and all other OMD/OMX connected their upgrade/through ports.						
6	Identify all OMX/OMD4 connected through the upgrade/through ports. Check if these OMX/OMD4 are reporting G-LOS or LOS.						
7	Identify all the upstream line fiber, patchcord and equipment that are common to those groups raised LOS/G-LOS.						
8	Clean the fibers, patchcords and verify the equipment. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.						
9	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
10	Clear any upstream (either at the local to the network element reporting the alarm or other upstream remote network elements) alarms that could be causing this alarm, such as Circuit Pack Failed, Circuit Pack Missing, or Loss of Signal (OPTMON).						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-139

Hardware Subsystem Failed

Alarm ID: 1997, 1998

Probable cause

This alarm is raised when a circuit pack has a minor hardware failure and does not impact service.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Step	Action
------	--------

- | | |
|---|---|
| 1 | Run DGN-EQPT command. If the alarm does not clear, continue to the next step. |
| 2 | Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 3 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 4 | Replace the alarmed circuit pack. Refer to the related replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. |

—end—

Procedure 4-140

High Fiber Loss

Alarm ID: 1239**Probable cause**

This alarm is raised against the originating ADJ-FIBER facility or Intra-node fiber when the measured loss between this port and the far-end port is greater than either of the provisioned loss thresholds. A 0.5 dB hysteresis is also applied, which prevents the alarm from clearing until the measured loss has gone below the threshold(s) by more than 0.5 dB. The calculated fiber loss is reported against the ADJ-FIBER or Intra-node fiber facility. This alarm is supported for the following interconnections:

- LIM to WSS
- LIM to CASLIM
- CASLIM to WSS
- WSS to DROPLIM
- DROPLIM to WSS
- WSS to WSS
- MuxAmp to WSS
- LIM Monitor Out to WSS Monitor In
- IRLA port 5 connected to SAM/ESAM/SRA
- MDRLA ports 21, 23, 25, 27, 29, 31 (receive direction)
- SRA/SAM/ESAM to XLA
- XLA to SRA/SAM/ESAM
- XLA to WSS
- XLA to LIM
- XLA Monitor Out to WSS Monitor In
- Amplifier monitor port to 2-Port OPM or 2-Port OPM Flex C-Band port
- UCS to WSS Flex L-Band w/OPM 8x1

For the COADM (Colorless) OTS, the alarm is supported for the following interconnections (in addition to those described in the preceding list):

- LIM to SMD
- SMD to LIM

Procedure 4-140 (continued)

High Fiber Loss

- CCMD12 C-Band MON (port-27) to SMD OPM (ports 1-8)
- CCMD12 C-Band to SMD
- OCLD Tx to CCMD12 C-Band (alarm is raised against the CCMD12 C-Band ADJ-FIBER (receive direction) in this instance, not against the OCLD OTM port and is based on the CCMD12 C-Band)

The fiber loss is calculated between the Photonic equipment pairs (in the preceding list) and takes into account any excess loss (such as the placement of an attenuator pad) provisioned at the originating ADJ-FIBER facility, as well as any provisioned DSCM/Pad.

For a CD configuration equipped with SMD Flex C-Band 14x8, this alarm is supported for the following interconnections:

- WSS Flex C-Band w/OPM 20x1 to another WSS Flex C-Band w/OPM 20x1
- WSS Flex C-Band w/OPM 20x1 to the same WSS Flex C-Band w/OPM 20x1 passing through a MPOLC loopback connector on the FIM
- SMD Flex C-Band 14x8 to a WSS Flex C-Band w/OPM 20x1
- SMD Flex C-Band 14x8 to the same SMD Flex C-Band 14x8 passing through a MPO loopback connector on the FIM
- SMD Flex C-Band 14x8 to the same SMD Flex C-Band 14x8 passing through a LC loopback connector on the FIM
- CCMD12 common out LC port to SMD Flex C-Band 14x8 Switch input ports
- Transponder Tx to CCMD12 (alarm is raised against the CCMD12 ADJ-FIBER (receive direction) in this instance, not against the Transponder OTM port and is based on the CCMD12)

For a CDC configuration, this alarm is supported for the following interconnections:

- WSS Flex C-Band w/OPM 20x1 to another WSS Flex C-Band w/OPM 20x1
- WSS Flex C-Band w/OPM 20x1 to same WSS Flex C-Band w/OPM 20x1 passing through an MPO loopback connector

Procedure 4-140 (continued)

High Fiber Loss

- WSS Flex C-Band w/OPM 20x1 to same WSS Flex C-Band w/OPM 20x1 passing through an LC loopback connector
- WSS Flex C-Band w/OPM 20x1 to CCMD8x16
- CCMD8x16 to WSS Flex C-Band w/OPM 20x1
- WSS Flex C-Band w/OPM 20x1 to CXM C-Band Type 1
- CXM C-Band Type 1 to WSS Flex C-Band w/OPM 20x1
- CCMD8x16 to same CCMD8x16 passing through an MPO Loopback Connector
- CXM C-Band Type 1 to same CXM C-Band Type 1 passing through an MPO loopback connector
- CCMD8x16 to CXM C-Band Type 1 passing through a Jumper cable
- CXM C-Band Type 1 to CCMD8x16 passing through a Jumper cable
- CDC WSS to ROADM line-amplifier

For a CDC configuration, the fiber loss is calculated between WSS switch ports and the CCMD8x16 common ports, as well as between the Upgrade Ports on the CCMD8x16 circuit pack and the CXM C-Band Type 1 module. The fiber loss is also calculated between WSS switch ports when connected with a loopback connector.

For the control and Connection Validation applications to work properly, the system parameters for the CDC and the COADM configurations must be enabled.

For a CDA configuration equipped with WSS Flex C-Band w/OPM 20x1 and FIMs, this alarm is supported for the following interconnections:

- WSS Flex C-Band w/OPM 20x1 to another WSS Flex C-Band w/OPM 20x1 via FIMs
- WSS Flex C-Band w/OPM 20x1 to the same WSS Flex C-Band w/OPM 20x1 passing through an MPO loopback connector on the FIM-5 or FIM-6
- WSS Flex C-Band w/OPM 20x1 to the same WSS Flex C-Band w/OPM 20x1 passing through an LC loopback connector on the FIM-5
- Transponder Tx Out to CCMD12 C-Band LC Channel In port
- CCMD12 C-Band common out LC port to WSS Flex C-Band w/OPM 20x1 Switch input MPO ports
- WSS Flex C-Band w/OPM 20x1 Switch Output MPO ports to CCMD12 C-Band common in LC port

Procedure 4-140 (continued)

High Fiber Loss

For a CDA configuration equipped with WSS Flex C-Band w/OPM 9x1 and Flex SMD, this alarm is supported for the following interconnections:

- ROADM WSS Flex C-Band w/OPM 9x1 Switch Out to cascaded MLA3 Line A In
- Cascaded MLA3 Line B Out to ROADM WSS Flex C-Band w/OPM 9x1 Switch In
- Cascaded MLA3 Line A Out to Flex SMD Common In
- Flex SMD Common Out to Cascaded LIM Line B In
- CCMD12 C-Band Common Out (port 26) to Flex SMD Switch In
- CCMD12 C-Band Monitor Out (port 27) to Flex SMD OPM Monitor In (ports 1-8)
- Transponder Tx Out to CCMD12 C-Band LC Channel In port

For a Photonic configuration, this alarm is supported for the following interconnections:

- RLA to another RLA passing through FIM
- RLA to same RLA passing through FIM MPO loopback connector
- RLA to same RLA passing through FIM LC loopback connector
- RLA to “AMP4 Module” passing through FIM
- “AMP4 Module” to RLA passing through FIM
- “AMP4 Module” to same “AMP4 Module” passing through FIM MPO loopback connector
- CCMD 16x12 to “AMP4 Module” passing through an MPO cable
- “AMP4 Module” to CCMD 16x12 passing through an MPO cable
- an OSRP Line/Link Down condition triggered by fiber disconnection on intra-node fiber (except WSS-WSS fiber), AMP in shut-off state, or AMP Primary State OOS-MA

Note: OSRP provides a fault reason within the “Additional Information” field of the alarm to help with the troubleshooting.

Procedure 4-140 (continued)

High Fiber Loss

For the Optical Bridge and Broadcast (OBB) and Optical Broadband Mux/Demux (OBMD 1x8), virtual OPTMON ports are used to monitor the output power estimates and raising the alarm.

For the Transponder Direct Attach (TDA) configurations, this alarm is raised on the ADJ facility of WSS w/OPM Flex C-Band 9x1(NTK553LA and NTK553LB) circuit pack.

The “High Fiber Loss” Alarm Detection Alarm parameter in the Site Manager Node Information application and the System tab is enabled by default.

Impact

Minor, non-service-affecting (m, NSA) alarm

Major, service-affecting (M, SA) alarm

When the fiber loss exceeds the user-provisioned Fiber Loss Minor Threshold (default is 3 dB), the alarm is raised with a minor severity.

“Fiber Loss Minor Threshold” threshold possible values are 1 to 30 with 0.01 resolution.

When the fiber loss exceeds the user-provisioned Fiber Loss Major Threshold (default is 10 dB), the alarm is raised with a Major severity.

“Fiber Loss Major Threshold” threshold possible values are 1 to 30 with 0.01 resolution, but must be greater than the “Fiber Loss Minor Threshold” value.

For CDA, CDC or CD (using SMD Flex 14x8) Connection Validation, this alarm is always raised on Dark Fiber as m, NSA when the fiber loss exceeds the user-provisioned fiber loss minor threshold (default is 3 dB), even if the input power is considered Out of Range. When the fiber loss is Out of Range in this situation, the fiber loss displays “LOS”.

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a fiber cleaning kit (for CDA, CDC or CD (using SMD Flex 14x8) and COADM configurations, both MPO and LC cleaning kits are needed). See the “Cleaning connectors” section in *Installation - General Information*, 323-1851-201.0
- obtain a fiber patchcord, if required

Procedure 4-140 (continued)
High Fiber Loss

Step	Action						
1	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Check the fibers involved in the High Fiber Loss alarm (all fibers between the port where the alarm is raised and the port specified as the Far End Address for this adjacency): • verify that the fibers are connected (fully inserted and clicked-in) and not crossed, looped back, or misconnected • check and clean any dirty fibers. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i>, 323-1851-201.0. • For a CDA, CDC or CD (using SMD Flex 14x8) configuration, verify that the MPO cable for the subfiber that has the alarm and the far-end of the MPO cable are connected properly. The far-end location of the MPO cable can be determined by verifying the MPO adjacency far-end Address where the alarm is raised. Verify that the WSS and the CMD MPO cables are fibered to the FIM.						
2	<table border="0" style="width: 100%;"> <tr> <td style="width: 60%;">If the original alarm has</td><td style="width: 40%;">Then go to</td></tr> <tr> <td>raised on CDA, CDC or CD (using SMD Flex 14x8) configurations</td><td>step 3</td></tr> <tr> <td>otherwise</td><td>step 23</td></tr> </table>	If the original alarm has	Then go to	raised on CDA, CDC or CD (using SMD Flex 14x8) configurations	step 3	otherwise	step 23
If the original alarm has	Then go to						
raised on CDA, CDC or CD (using SMD Flex 14x8) configurations	step 3						
otherwise	step 23						
3	<table border="0" style="width: 100%;"> <tr> <td style="width: 60%;">If the subfiber adjacency has a far-end address that points to</td><td style="width: 40%;">Then go to</td></tr> <tr> <td>the same card</td><td>step 4</td></tr> <tr> <td>a different card</td><td>step 5</td></tr> </table>	If the subfiber adjacency has a far-end address that points to	Then go to	the same card	step 4	a different card	step 5
If the subfiber adjacency has a far-end address that points to	Then go to						
the same card	step 4						
a different card	step 5						
4	Determine which MPO/LC loopback connector is used for the subfiber on the FIM. Go to step 6 .						
5	Determine the far-end MPO adjacency from the local subfiber adjacency far-end address. Check that the MPO cable is connected properly. Determine the far-end of this MPO cable by looking at the MPO adjacency far-end address.						
6	Check that the MPO cable is connected properly to the FIM using the Site Manager Shelf Level View. Double click on the circuit pack and right click on the MPO to select “Show Fiber Topology”. Check that the Loopback connector is connected properly.						

Procedure 4-140 (continued)
High Fiber Loss

Step	Action										
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared 5 minutes after insertion of the fiber or the MPO/LC loopback connector not cleared</td><td>the procedure is complete go to step 8</td></tr> </table>	If the original alarm has	Then	cleared 5 minutes after insertion of the fiber or the MPO/LC loopback connector not cleared	the procedure is complete go to step 8						
If the original alarm has	Then										
cleared 5 minutes after insertion of the fiber or the MPO/LC loopback connector not cleared	the procedure is complete go to step 8										
8	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the alarm is raised against a WSS or RLA 5X1 sub-fiber</td><td>step 9</td></tr> <tr> <td>the alarm is raised against a CCMD or CXM C-Band Type 1 common sub-fiber</td><td>step 18</td></tr> <tr> <td>the alarm is raised against a Jumper cable connecting a CCMD8x16 to CXM C-Band Type 1</td><td>step 24</td></tr> <tr> <td>otherwise</td><td>step 24</td></tr> </table>	If	Then go to	the alarm is raised against a WSS or RLA 5X1 sub-fiber	step 9	the alarm is raised against a CCMD or CXM C-Band Type 1 common sub-fiber	step 18	the alarm is raised against a Jumper cable connecting a CCMD8x16 to CXM C-Band Type 1	step 24	otherwise	step 24
If	Then go to										
the alarm is raised against a WSS or RLA 5X1 sub-fiber	step 9										
the alarm is raised against a CCMD or CXM C-Band Type 1 common sub-fiber	step 18										
the alarm is raised against a Jumper cable connecting a CCMD8x16 to CXM C-Band Type 1	step 24										
otherwise	step 24										
9	If the WSS or RLA 5X1 subfiber loopbacks to itself, clean the loopback connector identified by “Show Fiber Topology” in step 6 . Refer to the “Cleaning connectors” section in <i>Installation - General Information</i> , 323-1851-201.0.										
10	If the alarm did not clear five minutes after insertion of the fiber, clean the MPO cables between the WSS and the FIM on both ends.										
 CAUTION Risk of traffic loss Ensure to switch any traffic that is on any of the other subfibers on the MPO, away from the ROADM.											
11	If the WSS or RLA 5X1 subfiber goes to another WSS, RLA 5X1 or CMD/CXM, clean the MPO cables between the WSS or RLA 5X1 and the FIM or the MPO cables between the CMD/CXM and the FIM. Clean the MPO cable of the newly installed circuit pack first.										
 CAUTION Risk of traffic loss Ensure to switch any traffic that is on the MPO cables. The traffic will be lost when you clean the cables.											
12	If there is a fiber adjacency to a WSS (FIM Type 1), repeat step 9 through step 11 for the WSS identified in the adjacency.										

Procedure 4-140 (continued)
High Fiber Loss

Step	Action						
13	If there is a fiber adjacency to a CMD (FIM Type 1, Type 2, Type 4), perform step 18 through step 21 for the CCMD/CXM/FIM-4 common port identified in the adjacency.						
14	<table border="0"> <tr> <td>If</td><td>Then go to</td></tr> <tr> <td>the fiber adjacency points back to itself</td><td>step 15</td></tr> <tr> <td>otherwise</td><td>step 24</td></tr> </table>	If	Then go to	the fiber adjacency points back to itself	step 15	otherwise	step 24
If	Then go to						
the fiber adjacency points back to itself	step 15						
otherwise	step 24						
15	Using the Show Fiber Topology in the Shelf Level View application in Site Manager against the alarmed MPO port, determine the FIM loopback connector (MPO or LC) to clean.						
16	Remove and clean the MPO or LC loopback as applicable.						
17	Clean the FIM port and replace the MPO or LC loopback. Go to step 28 .						
18	Clean the Common MPO port. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i> , 323-1851-201.0.						
19	Clean the MPO cable and reconnect it to the card.						
20	Clean the MPO port on the FIM.						
21	Clean the MPO cable and reconnect it to the FIM.						
22	If the alarm is raised on the CCMD8x16 subfibers and the CXM C-Band Type 1 Jumper subfibers, clean the Jumper fibers on the dark fiber or the Manual loopback path fiber as follows. a. If you performed a manual loopback in a CDC configuration and the alarm on a Jumper subfiber is raised on a dark fiber, clean the Jumper MPO cable.						



CAUTION

Risk of traffic loss

Ensure to switch traffic that is on any of the subfibers.

- b. Verify the following cables:
- MPO from the CCMD8x16 port 34 to the CXM C-Band Type 1 port 35
 - MPO from the CXM C-Band Type 1 port 36 to the FIM
 - MPO loopback connector that the subfiber passes through in the FIM
- The ordering of the cleaning must be:
- Verify the loopback connector first since this does not impact traffic
 - Verify the Jumper cable next

Procedure 4-140 (continued)
High Fiber Loss

Step	Action		
	The waiting period of five minutes does not apply here since there is no automatic test done by connection validation. Re-run the Manual Loopback test again to clear the alarm. c. When you perform a manual loopback test between a CCMD8x16 and a WSS Flex C-Band w/OPM 20x1, and the alarm is raised on one of the subfibers in the path, cancel the manual loopback test. If the alarms on the subfibers between the CMD/CXM to FIM or FIM to WSS clear, then those cables are not causing the alarm since automatic Connection Validation has validated them. Make sure you wait up to five minutes after the completion of the manual loopback before verifying this. d. If the alarms is raised on the Jumper subfibers, clean the cable. <table border="1"> <tr> <td data-bbox="542 816 688 947"> </td> <td data-bbox="703 783 1398 884"> CAUTION Risk of traffic loss Ensure to switch traffic that is on any of the subfibers. </td> </tr> </table> e. Re-run the manual connection validation test to clear the alarm and wait five minutes to ensure there are no new alarms. Go to step 28. 23 Check that the DSCM provisioning is correct. For information on provisioning the DSCM in OTS, refer to the “Retrieving OTS Management, OTS Equipment, and Facility Details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. For information on the facility provisioning for the DISP facility on DSCM, refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. To change the provisioning, refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310 or the “Editing an OTS instance” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. 24 Check Excess Loss provisioning. If there are lossy elements (patch panels and pads are typical examples of lossy elements) between the two ports, the loss of the element itself should be provisioned in the Excess Loss field. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. 25 If the reported loss is less than the Major or Minor threshold, but not by 0.5 dB, the alarm is being held on due to a 0.5 dB hysteresis. The threshold is likely too low. Adjust the threshold.		CAUTION Risk of traffic loss Ensure to switch traffic that is on any of the subfibers.
	CAUTION Risk of traffic loss Ensure to switch traffic that is on any of the subfibers.		



Procedure 4-140 (continued)

High Fiber Loss

Step	Action
26	If the reported loss is greater than 1.5 dB, this is likely a problem in the circuit packs or fiberling that must be resolved. Contact your next level of support or your Ciena support group.
27	For losses less than 1.5 dB if all other steps have been taken, clear the alarm by adjusting the threshold.
28	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-141

High Optical Power

Alarm ID: 1800, 2113

Probable cause

This alarm is raised against an OPTMON facility when the total power detected at the common IN port of the WSS Demux is higher than the threshold deemed safe for class 1M operation.

For the SMD Flex C-Band 14x8 circuit packs, this alarm is raised against an OPTMON facility when the total power detected at the common port of the CCMD12 Demux is higher than the threshold deemed safe for class 1M operation.

For the SMD8x16 circuit packs, this alarm is raised against an OPTMON facility when MPO loopback connector is missing.

For the Optical Bridge and Broadcast (OBB) circuit packs, this alarm is raised against an OPTMON facility when the total power detected at the Line or CMD IN port of the OBB is higher than the threshold deemed safe for class 1M operation.

When configured for single span trunk OPS, the alarm is raised against MLAx port 5 AMPMON if Line B output power exceeds the Amp Facility's auto-provisioned Total Power threshold.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0

Procedure 4-141 (continued)

High Optical Power

Step	Action
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ATTENTION

DO NOT disconnect fibers or remove dust caps until instructed to do so (safely), in the following steps.



DANGER

Risk of laser radiation exposure

Laser radiation is present on the optical fiber. Do not look into the optical fiber.

- | | | |
|---|---|--|
| 1 | Identify the circuit pack raising the alarm. Refer to the “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. | |
| 2 | Replace the upstream amplifier connected to the WSS, MDRLA 2x6 or RLA 5X1 port that has raised the alarm. Follow the steps in “Replacing an amplifier module” procedure in 323-1851-545. | |
| 3 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 4 |
| 4 | Replace the alarmed circuit pack. Refer to “Replacing the WSS w/OPM or SMD circuit pack”, Replacing an amplifier module or “Replacing an RLA 5x1 or MDRLA 2x6 module” in 323-1851-545. | |
| 5 | If the alarm does not clear, contact your next level of support or your Ciena support group. | |

—end—

Procedure 4-142

High Received Span Loss

Alarm ID: 1155

Probable cause

This alarm is raised against the LIM, SRA, SAM, RLA, MDRLA 2x6, or ESAM circuit packs (at the tail end of the span) when the incoming estimated Span Loss for the span is higher than the Target Span Loss plus the Span Loss Margin.

For SRA circuit packs, the Telemetry Gain Signal value is used for Span Loss calculation. Refer to the *Photonic Layer Guide*, NTRN15DA for more information.

This alarm is also raised when the XLA, SRA, MDRLA 2x6 and/or RLA Gain Mode is not set as per the network planning and link budget recommendations.

The alarm clears when the span loss drops 1.0 dB below the raise threshold. Target Span Loss and Margin are user provisionable. (Target Span Loss + Span Loss Margin) - 1 dB.

Note: Span Loss for an optical span is stored as a PM with full retrievable history. Refer to the “Retrieving performance monitoring data” in the *Fault Management - Performance Monitoring*, 323-1851-520.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a fiber cleaning kit
- obtain a fiber patchcord, if required

Procedure 4-142 (continued)
High Received Span Loss

Step	Action						
1	Prior to running the power audit or adding channels in a dark section, ensure that the XLA switch gain mode is set properly as per planning to avoid unsuccessful calibration.						
2	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Check the fibers of the alarmed facility (at the network element raising the alarm and at the upstream network element): • verify that the fibers are connected, and not crossed, looped back, or misconnected • check and clean any dirty fiber/port/patch-panel on the AID/port reporting the alarm. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i>, 323-1851-201.0.						
3	<table border="0" style="width: 100%;"> <tr> <td style="width: 60%;">If the original alarm has</td><td style="width: 40%; text-align: right;">Then</td></tr> <tr> <td>cleared</td><td style="text-align: right;">the procedure is complete</td></tr> <tr> <td>not cleared</td><td style="text-align: right;">go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	<table border="0" style="width: 100%;"> <tr> <td style="width: 60%;">If</td><td style="width: 40%; text-align: right;">Then go to</td></tr> <tr> <td>you want to clear the alarm (and disable detection) without addressing the fault</td><td style="text-align: right;">step 5</td></tr> <tr> <td>otherwise</td><td style="text-align: right;">step 6</td></tr> </table>	If	Then go to	you want to clear the alarm (and disable detection) without addressing the fault	step 5	otherwise	step 6
If	Then go to						
you want to clear the alarm (and disable detection) without addressing the fault	step 5						
otherwise	step 6						
5	Provision the Target Span Loss to 0. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
6	If the alarm is raised on Port 8 of the SRA circuit pack, verify the Office and Short traces manually for any of the following events, otherwise go to step 11: • Large losses that were not captured by the OTDR alarms, possibly because of the pulse width/event spacing. If events are overlapping, re-run the trace using a smaller pulse. • Check for disparate or mixed fiber types, indicated by “gainers” or exaggerated loss splices.						
7	Perform the fiber/port/patch-panel cleaning procedure on the event in question.						
8	If the alarm is not clear, contact your next level support to assist with alternative options to isolate the source of the high fiber loss.						

Procedure 4-142 (continued)
High Received Span Loss

Step	Action
9	If the alarm is not clear, unseat the SRA circuit pack, wait 10 seconds and reseat the SRA circuit pack. Wait for circuit pack to complete the boot cycle, then wait for the “OTDR Traces In-Progress” alarm to clear (Go/No-Go OTDR traces) for the link. Refer to “Reseating a circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545.
10	If the alarm does not clear, replace the SRA circuit pack. Refer to “Replacing the SRA circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545.
11	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-143

High Temperature

Alarm ID: 378

Probable cause

This alarm is raised against a slot when the recorded temperature is too high.

The following conditions can cause this alarm:

- The operating environment exceeds the temperature and/or altitude limits specified for the shelf. Refer to the “Operating environmental specifications for 6500 system” table in *6500 Planning*, NTRN10GK.
- There are one or more empty slots in the shelf which must be equipped with modules, circuit packs, or filler cards. Any “Circuit Pack Missing” or “Slot Empty” must be cleared first.
- The shelf cooling system is not functioning correctly. Any “Fan Failed” alarms must be cleared first.
- The shelf is operating without a cooling fan module equipped (one or more fan modules are missing or not fully inserted).
- The air exhaust port or air inlet plenum is blocked.
- The air filter has a blockage.
- The shelf exhaust air is being trapped in an enclosed space or cabinet and is re-circulating into the air inlet plenum.
- Hot exhaust air from other equipment is mixing with the shelf's intake air.
- There is a problem with a circuit pack or module in the shelf.
- For a 14-slot shelf with front exhaust fan modules, air deflector grill NTK509FA/NTK509FAE6 should not be used if the shelf is mounted in an cabinet with a front door or if the shelf is equipped with any circuit pack that has a power budget over 150W per slot (or equivalently 300W for a double-slot wide pack).

Procedure 4-143 (continued)

High Temperature

- For the extended temperature system housed in a 2-slot Type 2 chassis (NTK503LA) with the SPAP-2 w/2xOSC (NTK555NA, NTK555NB) and 1+8 OTN Flex MOTR (NTK532DE) or eMOTR (NTK536BE) circuit packs, the thresholds for the “High Temperature” alarm on the SPAP-2 w/2xOSC or 1+8 OTN Flex MOTR circuit pack (NTK532DE) is calibrated and the alarm does not raise within the extended temperature range.

Note: It may take some time for a circuit pack to cool off and the alarm to clear after the cause is corrected.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
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DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules, air filters or in proximity to the shelf air exhaust.



CAUTION

Risk of damage to circuit packs and modules

Do not attempt to cause this alarm to be raised by removing any cooling fan modules or filler cards or by blocking the shelf air inlet or exhaust ports as it will compromise shelf cooling. A shelf with a compromised cooling system may result in circuit pack or module failures prior to the assertion of a High Temperature Warning or High Temperature alarm.

- 1 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.

Procedure 4-143 (continued)

High Temperature

Step	Action				
2	Make sure that the shelf operating environment is within the specified temperature and altitude limits. Refer to the “Operating environmental specifications for 6500 system” table in <i>6500 Planning</i> , NTRN10GK.				
3	If the operating temperature for the altitude is Then <table> <tr> <td>within the specified limits</td><td>go to step 4</td></tr> <tr> <td>not within the specified limits</td><td>correct the office temperature and go to step 10</td></tr> </table>	within the specified limits	go to step 4	not within the specified limits	correct the office temperature and go to step 10
within the specified limits	go to step 4				
not within the specified limits	correct the office temperature and go to step 10				
4	Perform visual checks to ensure that the shelf air intake is not obstructed and that air is exhausting from each fan module or from the shelf's exhaust port as applicable. Verify the following: • If the shelf is configured for rear exhaust, make sure that there is clearance between the back of the shelf and any other shelf, wall, or obstruction. • Inspect the grill at the air exhaust of the cooling unit or the grill integrated into the shelf cover/door, the grill on the cooling fan module, and the grill on the air inlet plenum. If any have an accumulation of dust, vacuum the grill. • For a 14-slot shelf, make sure that the flip-down door of the fiber routing channel is closed so that it does not obstruct the air intake • For a 14-slot shelf with front exhaust fan modules that is mounted in a cabinet with a front door (which must be perforated) or that is equipped with any circuit pack that has a power budget of over 150W per slot (or equivalently 300W for a double-slot wide pack), make sure that it is not equipped with an exhaust air deflector grill (manufacturing discontinued part NTK509FA/NTK509FAE6 included with front venting cooling unit kits NTK507BA/NTK507BBE5/NTK507BCE5 and metro front electrical shelf assembly NTK503GAE5). • For a 2-slot optical Type 2, 4-slot shelf, 6500-7 packet-optical, or 7-slot shelf, ensure that the SP, AP, SPAP, or APAP-2 w/2xOSC, as applicable, are fully seated. If not, loosen the faceplate screws on the AP, SPAP, or APAP-2 w/2xOSC circuit pack, push the circuit pack in fully and tighten the screws to the prescribed torque. Refer to the applicable shelf installation procedure in the <i>Installation</i> technical publication specific to the respective 6500 shelf type.				

Procedure 4-143 (continued)
High Temperature

Step	Action						
5	Retrieve the shelf inventory or visually inspect the shelf to confirm that all slots are filled (including all circuit pack subslots and common equipment slots). Refer to the “Displaying shelf inventory information” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Note that the cross-connect slots on a 32-slot shelf do not support filler card detection and must be visually checked to confirm if they are equipped with a 2-slot x 2-row filler cards.						
6	<table> <tr> <th>If the shelf has</th><th>Then go to</th></tr> <tr> <td>all slots filled</td><td>step 8</td></tr> <tr> <td>empty slots</td><td>step 7</td></tr> </table>	If the shelf has	Then go to	all slots filled	step 8	empty slots	step 7
If the shelf has	Then go to						
all slots filled	step 8						
empty slots	step 7						
7	Re-insert any missing circuit packs or install filler cards into any empty slots. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	Remove and inspect the shelf air filter and replace it with a new one if there is a visible accumulation of dust. Refer to the “Replacing the shelf air filter” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
9	Monitor the temperatures on the circuit packs by retrieving the inventory and check whether the current temperature of all the supported circuit packs decreases. If the temperature did not decrease even after ten minutes or if the shelf processor is still reporting a current temperature of 50 °C or greater, continue to step 10 .						
10	Verify that the fan louvers are not stuck and can move freely.						
11	Replace the cooling fan module(s). Refer to the “Replacing a cooling fan module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
12	Monitor the temperatures on the cards by retrieving the inventory and check whether the current temperature of all the supported circuit packs decreases. If the temperature did not decrease even after ten minutes or if the shelf processor is still reporting a current temperature of 50 °C or greater, go to step 13 .						
13	If the alarm does not clear, it could be due to a problem with a circuit pack. Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-144

High Temperature Warning

Alarm ID: 1143

Probable cause

This alarm is raised against the slot as a warning to the user when the recorded temperature is too high. The following conditions are applicable:

- The operating environment exceeds the temperature and/or altitude limits specified for the shelf. Refer to “Operating environmental specifications for 6500 system” table in *6500 Planning*, NTRN10GK.
- There are one or more empty slots in the shelf which must be equipped with modules, circuit packs, or filler cards. Any “Circuit Pack Missing” or “Slot Empty” alarm must be cleared first.
- The shelf cooling system is not functioning correctly. Any “Fan Failed” alarms must be cleared first.
- The shelf is operating without a cooling fan module equipped (one or more fan modules are missing or not fully inserted).
- The air exhaust port or air inlet plenum is blocked.
- The air filter has a blockage.
- The shelf exhaust air is being trapped in an enclosed space or cabinet and is re-circulating into the air inlet plenum.
- Hot exhaust air from other equipment is mixing with the shelf's intake air.
- There is a problem with a circuit pack or module in the shelf.
- For a 14-slot with front exhaust fan modules, an air deflector grill should not be used if the shelf is mounted in an cabinet with a front door or if the shelf is equipped with any circuit pack that has a power budget over 150W per slot (or equivalently 300W for a double-slot wide pack).
- For the extended temperature system housed in a 2-slot Type 2 chassis (NTK503LA) with the APAP-2 w/2xOSC (NTK555NA, NTK555NB) and 1+8 OTN Flex MOTR (NTK532DE) or eMOTR (NTK536BE) circuit packs, the thresholds for the “High Temperature warning” alarm on the SPAP-2 w/2xOSC or 1+8 OTN Flex MOTR circuit pack (NTK532DE) is calibrated and the alarm does not raise within the extended temperature range.

Procedure 4-144 (continued)
High Temperature Warning

ATTENTION

If a high temperature warning condition prevails for a while, it could potentially lead to a “High Temperature” condition. Refer to [“High Temperature” on page 4-330](#) for high temperature condition information.

The high temperature warning condition is cleared on the slot automatically if the temperature on these circuit packs decreases to an acceptable level.

Temperature can be monitored through inventory retrieval, which displays the current and average temperature on the supported circuit packs. Refer to the “Displaying shelf inventory information” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.

Impact

Major, non service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
------	--------



DANGER

Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules, air filters or in proximity to the shelf air exhaust.



CAUTION

Risk of damage to circuit packs and modules

Do not attempt to cause this alarm to be raised by removing any cooling fan modules or filler cards or by blocking the shelf air inlet or exhaust ports as it will compromise shelf cooling. A shelf with a compromised cooling system may result in circuit pack or module failures prior to the assertion of a High Temperature Warning or High Temperature alarm.

Procedure 4-144 (continued)
High Temperature Warning

- 1 Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
- 2 Make sure that the shelf operating environment is within the specified temperature and altitude limits. Refer to “Environmental specification” in *6500 Planning*, NTRN10GK.
- 3

If the operating temperature for the altitude is	Then
within the specified limits	go to step 4
not within the specified limits	correct the office temperature and go to step 13
- 4 Perform visual checks to ensure that the shelf air intake is not obstructed and the air is exhausting from each fan module or from the shelf exhaust port as applicable. Verify the following:
 - If the shelf is configured for rear exhaust, make sure that there is clearance between the back of the shelf and any other shelf, wall, or obstruction.
 - For a 14-slot shelf, make sure that the flip-down door of the fiber routing channel is closed so that it does not obstruct the air intake.
 - For a 14-slot shelf with front exhaust fan modules that is mounted in a cabinet with a front door (which must be perforated) or that is equipped with any circuit pack that has a power budget of over 150W per slot (or equivalently 300W for a double-slot wide pack), make sure that it is not equipped with an exhaust air deflector grill (manufacturing discontinued part NTK509FA/NTK509FAE6 included with front venting cooling unit kits NTK507BA/NTK507BBE5/NTK507BCE5 and metro front electrical shelf assembly NTK503GAE5).
 - Inspect the grill at the air exhaust of the cooling unit or the grill integrated into the shelf cover/door, the grill on the cooling fan module, and the grill on the air inlet plenum. If any have an accumulation of dust, vacuum the grill.
 - For a 2-slot optical Type 2, 4-slot shelf, 6500-7 packet-optical, or 7-slot shelf, ensure that the SP, AP, SPAP, or APAP-2 w/2xOSC, as applicable, are fully seated. If not, loosen the faceplate screws on the AP, SPAP, or APAP-2 w/2xOSC circuit pack, push the circuit pack in fully and tighten the screws to the prescribed torque. Refer to the applicable shelf installation procedure in the *Installation* technical publication specific to the respective 6500 shelf type.

Procedure 4-144 (continued)
High Temperature Warning

Step	Action						
5	Retrieve the shelf inventory or visually inspect the shelf to confirm that all slots are filled (including all circuit pack subslots and common equipment slots). Refer to the “Displaying shelf inventory information” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Note that the cross-connect slots on a 32-slot shelf do not support filler card detection and must be visually checked to confirm if they are equipped with a 2-slot x 2-row filler card.						
6	<table> <tr> <th>If the shelf has</th><th>Then go to</th></tr> <tr> <td>all slots filled</td><td>step 8</td></tr> <tr> <td>empty slots</td><td>step 7</td></tr> </table>	If the shelf has	Then go to	all slots filled	step 8	empty slots	step 7
If the shelf has	Then go to						
all slots filled	step 8						
empty slots	step 7						
7	Re-insert any missing circuit packs or install filler cards into any empty slots. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	Remove and inspect the shelf air filter and replace it with a new one if there is a visible accumulation of dust. Refer to the “Replacing the shelf air filter” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
9	Monitor the temperatures on the circuit packs by retrieving the inventory and check whether the current temperature of all the supported circuit packs decreases. If the temperature does not decrease after ten minutes, go to step 10 . If the temperature falls to normal on all the supported circuit packs wait for 6 minutes for the alarm to clear.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared, even after all the circuit packs are at a normal temperature for more than six minutes</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared, even after all the circuit packs are at a normal temperature for more than six minutes	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared, even after all the circuit packs are at a normal temperature for more than six minutes	go to step 13						
11	If the temperature does not decrease after ten minutes or if the shelf processor is still reporting a current temperature of 50 °C or greater, go to step 12 .						
12	<table> <tr> <th>If the original alarm</th><th>Then</th></tr> <tr> <td>is not raised again six minutes after shelf processor has recovered</td><td>the procedure is complete</td></tr> <tr> <td>is raised again</td><td>go to step 13</td></tr> </table>	If the original alarm	Then	is not raised again six minutes after shelf processor has recovered	the procedure is complete	is raised again	go to step 13
If the original alarm	Then						
is not raised again six minutes after shelf processor has recovered	the procedure is complete						
is raised again	go to step 13						
13	Verify that the fan louvers are not stuck and can move freely.						
14	Replace the cooling fans module(s). Refer to the “Replacing a cooling fan module” procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						

Procedure 4-144 (continued)

High Temperature Warning

Step	Action						
15	Monitor the temperatures on the circuit packs by retrieving the inventory and see whether the current temperature decreases. If the temperature does not decrease after ten minutes or if the shelf processor is still reporting a current temperature of 50 °C or greater, go to step 17 . If the temperature falls to normal on all the supported circuit packs, wait for six minutes for the alarm to clear.						
16	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared, even after all the circuit packs are at a normal temperature for more than six minutes</td><td>go to step 17</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared, even after all the circuit packs are at a normal temperature for more than six minutes	go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared, even after all the circuit packs are at a normal temperature for more than six minutes	go to step 17						
17	If the alarm does not clear, it could be due to a problem with a circuit pack. Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-145

Home Path Not defined

Alarm ID: 1772**Probable cause**

This alarm is raised when an SNC with Reserved Home Path enabled does not have a Home Path. This can occur if the SNC Home Path is released without successfully setting up a new Home Path.

The alarm can also be raised when an SNC with Reserved Home Path is in the STARTING state. You need to ensure that the SNC can setup over dynamic routing or the DTLSET.

This alarm is raised when the Home Path OSRP line bandwidth is locked and the regroom operation of the Reserved Home Path enabled SNC (for example, make current path as home path) is failed.

The alarm will be cleared when the SNC Home Path is setup successfully or when Retain Home Path is disabled on the SNC.

Impact

Minor, non-service-affecting (m, NSA) alarm.

Prerequisites

To perform this procedure, you must:

- ensure the Primary State of the SNC to be regroomed is in-service. Refer to the “Regrooming a sub-network connection” procedure in *Configuration - Control Plane*, 323-1851-330.
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|--|
| 1 | If the alarm is raised because the OSRP line bandwidth is locked out and the SNC with Reserved Home Path enabled regroom operation failed, disable the bandwidth lockout and regroom the SNC again. Refer to the “Regrooming a sub-network connection” procedure in <i>Configuration - Control Plane</i> , 323-1851-330. |
|---|--|

Procedure 4-145 (continued)

Home Path Not defined

Step	Action
2	Otherwise, regroom the SNC for a new Home path or regroom the SNC to the current path by enabling the “Make current path as home path” check box. Refer to the “Regrooming a sub-network connection” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
3	Alternatively, disable (uncheck) the Retain Home Path checkbox when regrooming the SNC if you do not need Retain Home path functionality for this SNC. Refer to the “Regrooming a sub-network connection” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-146

Incomplete Channel Topology

Alarm ID: 1679**Probable cause**

This alarm is raised as a warning to highlight stale or leftover Shelf Wavelength Topology (SWT) provisioning (for example, orphaned Tx/Rx adjacency or unused cross-connection), which is not part of a complete NCT channel.

Such leftover provisioning can result in unintended channels being built as a result of subsequent provisioning actions. For example, an orphaned Rx adjacency causing a channel to drop unexpectedly and creating an “accidental” Drop and Continue channel.

This warning is raised against an OTS one hour after detecting one or more wavelengths in that OTS with ROUTING set to ADD, DROP, or PASSTHROUGH, and COMPLETE set to FALSE.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Retrieve the SWT provisioning using Site Manager Shelf Wavelength Topology application.
2	Find the wavelength(s) with COMPLETE flag set to FALSE and confirm these are wavelengths you were not expecting to be provisioned in the network.
3	Delete the unused SWT provisioning by putting the corresponding Tx/Rx adjacency out-of-service (for ADD/DROP routing) or by deleting the unused cross-connection (for PASSTHROUGH routing).
4	If a desired channel is unexpectedly marked incomplete, traverse the expected path of that channel upstream and downstream from that node to find the node where the channel provisioning is missing (that is, ROUTING=UNKNOWN) and add the missing provisioning at that node.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-147 Incomplete Software Lineup

Alarm ID: 92

Probable cause

This alarm is raised when a load file is missing. This can occur when a new circuit pack is inserted into the shelf, or when a shelf processor inserted into the shelf does not contain the software release that is active on the shelf, or when an active service pack release is incomplete.

This alarm can also be raised when a SP is replaced and the new SP is running a different software release than the active release on the network element.

This alarm can also be raised on a SP if the service bundle on the replacement SP does not match before the replacement.

If this alarm is raised during an upgrade activity contact your next level of support or your Ciena support group.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Verify the current load release on the shelf processor for both shelf release and possible service pack release. Refer to the "Checking the current software release" procedure in <i>Commissioning and Testing</i> , 323-1851-221.
2	Retrieve all alarms and AO on the system. Record the current state of the system.
3	Perform a software load transfer. Refer to the "Transferring a software load to a network element" procedure in <i>Administration and Security</i> , 323-1851-301. Note: A load transfer can take three hours. Contact your next level of support or your Ciena support group for assistance if necessary.
4	Perform a warm restart of the shelf processor. See Procedure 2-11 , "Restarting a circuit pack or shelf processor" procedure in this document.
5	Retrieve all alarms to ensure that the system is restored to its original state. If any alarm not recorded in step 2 is displayed, refer to the appropriate alarm clearing procedure.

Procedure 4-147 (continued)
Incomplete Software Lineup

Step	Action
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-148

Input Loss Of Signal

Alarm ID: 557

Probable cause

This alarm is raised against an AMP facility when the total input optical power to the amplifier has fallen below the provisioned input LOS threshold level.

For CDC configurations, this alarm is raised on the AMP facility of CCMD8x16, RLA 5x1, MDRLA 2x6 or CCMD12 C-Band/CCMD12 L-Band circuit packs.

The conditions that can cause the input power level to fall below the threshold level include:

- a disconnected fiber
- a dirty optical fiber connector at the receiver or adjacent transmitter
- a failure of the transmitting laser at the adjacent Tx
- a defective fiber optic patchcord
- a defective module
- an incorrect provisioned value
- a reflective event, indicated by an Automatic Power Reduction Active alarm at an upstream booster amplifier or pre-amplifier

This alarm can remain active after the fault has cleared and the original power level is restored. This occurs when the power level is lower than the user-provisioned LOS threshold plus the hysteresis value. The hysteresis value is not user provisionable and is set at 3 dB.

This alarm is masked by the “Shutoff Threshold Crossed” alarm. This alarm masks the “Output Loss Of Signal” alarm on the same amplifier facility.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a network and site diagram

Procedure 4-148 (continued)
Input Loss Of Signal

- have a fiber cleaning kit
- have the LOS threshold level for this amplifier

Step	Action						
1	Check for and clear any active Optical Line Fail alarm.						
2	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Check for and clear any active Automatic Power Reduction Active alarm at the upstream booster amplifier or pre-amplifier that is providing output power to the alarmed amplifier.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier. Then go to step 5.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier. Then go to step 5 .
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier. Then go to step 5 .						
5	Check for and clear any active Input Loss Of Signal alarms at the amplifiers farther upstream.						
6	Repeat step 1 to step 5 .						
7	<table> <tr> <th>If the alarm is raised against</th><th>Then go to</th></tr> <tr> <td>a CCMD12 C-Band, CCMD12 L-Band or CCMD8x16 circuit pack</td><td>step 8</td></tr> <tr> <td>otherwise</td><td>step 11</td></tr> </table>	If the alarm is raised against	Then go to	a CCMD12 C-Band, CCMD12 L-Band or CCMD8x16 circuit pack	step 8	otherwise	step 11
If the alarm is raised against	Then go to						
a CCMD12 C-Band, CCMD12 L-Band or CCMD8x16 circuit pack	step 8						
otherwise	step 11						
8	Ensure there are no ports switching to the Demux Degree AMP where the Connection Validation Test will be performed.						
9	Set the AMP to Amplified Spontaneous Emission (ASE) mode to generate light.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 11						
11	Verify the AMP facility parameters are correctly provisioned. If required, correct any discrepancies. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-148 (continued)
Input Loss Of Signal

Step	Action						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	Place the alarmed AMP facility out of service (OOS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
14	 DANGER Risk of laser radiation exposure Do not look directly into the optical beam. Invisible light can severely damage your eyes.  CAUTION Risk of damage to modules Never disconnect an optical fiber that is connected to an active or powered up optical amplifier. To disconnect or reconnect an optical fiber, make sure the optical amplifier is out of service (OOS), then disconnect or reconnect the fiber.  CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Clean and then reconnect the output fibers and connectors at the amplifier. Refer to the “Cleaning Connectors” section in <i>Installation - General Information</i>, 323-1851-201.0.						
15	Place the amplifier back in-service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
16	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 17</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 17						

Procedure 4-148 (continued)
Input Loss Of Signal

Step	Action						
17	Clear any upstream (either at the local to the network element reporting the alarm or other upstream remote network elements) alarms that could be causing this alarm, such as Circuit Pack Failed, Circuit Pack Missing, and Loss Of Signal. If there are no upstream alarms, then verify the optical patchcord connected to the port reporting the alarm: - ensure it is connected at both ends and that there is no problem with the optical patchcord - clean the connectors. Refer to the “Cleaning Connectors” procedure in <i>Installation - General Information</i>, 323-1851-201.0.						
18	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 19</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 19
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 19						
19	Provision the Input LOS Threshold value for the AMP facility reporting the alarm to be 5 dB less than the current value. For example, if the Input LOS Threshold is -22 dB, change the value to -27 dB. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
20	Wait at least one minute for the alarm to clear. Change the Input LOS Threshold value back to the original value.						
21	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 22</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 22
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 22						
22	Restart the circuit pack supporting the alarmed AMP facility. Refer to Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.						
23	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 24</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 24
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 24						
24	Replace the circuit pack supporting the alarmed AMP facility. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
25	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-149

Integrated Test Set Configured

Alarm ID: 1519, 1520, 1521, 1522, 1628, 1629, 1630, 1631, 1889, 1899, 1909, 1919, 2140

Probable cause

This alarm is raised when the Integrated Test Set feature has been provisioned on the ETH10G, ETH40G, ETH100G, CBRTTP, STTP, ETTP, OTUTTP, OC192/STM64, OC768/STM256, OTM2, OTM3, or OTM4 facility of 40G+ CFP OCI, 100G OCI, 100G WL3e OTR, WLAI MOTR, WLAI FOTR, 2xWLAI OTR, or 10x10G MUX circuit packs.

Impact

Warning, non-service-affecting (w, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|--|
| 1 | Select Integrated Test Set from the Configuration drop-down menu. If there is more than one possible facility on the shelf, select the appropriate facility from the facility list. Refer to the “Performing a test with the integrated test set” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310, for more information. |
| 2 | Disable the GEN & MON patterns.
<i>Gen and Mon patterns are set to No Pattern.</i> |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-150

Integrated Test Set Data Save In Progress

Alarm ID: 1531, 1532, 1533, 1534, 1664, 1665, 1666, 1667, 1891, 1911, 1921, 2142

Probable cause

This alarm is raised when there is a save in progress on a port with the Integrated Test Set provisioned.

Impact

Warning, non-service-affecting (w, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	No action is required. The alarm clears when the data save process is completed.
2	If the alarm does not clear or is unexpected, contact your next level of support or Ciena support group.

—end—

Procedure 4-151 Inter-card Suspected

Alarm ID: 73, 1867

Probable cause

This alarm is raised when a data path, control or communications path fault is detected within the product, but the fault is unable to be isolated to a specific circuit pack or to the backplane. Because the fault cannot be isolated to a specific component, the alarm will be raised against more than one circuit pack. Examples are:

- A circuit pack that has a loss of traffic or degraded signal integrity on a backplane link which interfaces to another circuit pack in the shelf. The alarm would be raised against both circuit packs.
- A communications bus (clock, parity, or interprocessor communication) failure between a shelf processor and a circuit pack. The alarm would be raised against the shelf processor and the other circuit pack.
- A fan module with one or more integrated fans running at maximum speed because the temperature is not being sensed from the access panel. The alarm would be raised against the fan module and the access panel.

ATTENTION

This alarm can be seen on adjacent eMOTR circuit packs following a cold restart of both eMOTR circuit packs. These alarms may clear automatically within 10 minutes, in which case no further action is required. Otherwise follow the clearing procedures.

Procedure 4-151 (continued)

Intercard Suspected**Impact**

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

The status of the circuit pack can be unknown because of the loss of communications. Additional failures can be service-affecting.

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
1	Identify the circuit pack(s) raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	If there are any “Circuit Pack Failed” alarms present, address those alarms first.
3	Determine the time since the “Intercard Suspected” alarm was raised. It is recommended that circuit packs are not restarted until at least 10 minutes after the “Intercard Suspected” alarm was raised so that data collection can complete for the fault, in case design investigation is required.
4	If the alarm is raised on a circuit pack which has a red LED turned on, and that circuit pack has not been replaced within the past 48 hours, then replace the circuit pack. Refer to the appropriate circuit pack replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851- 545.
5	If the alarm is raised on any of the following combinations of circuit packs, perform the actions identified below, until the alarm clears. After each step, if the alarm does not return after 10 minutes then the procedure is complete.

Procedure 4-151 (continued)

Inter-card Suspected

Step	Action
6	If the alarmed equipment is: Then perform the following: See Note 1 and Note 2 one XC and one or more circuit packs which use the XC 1- Cold Restart XC circuit pack raising the alarm 2- Replace XC circuit pack raising the alarm two XCs and only one tributary circuit pack which uses the XC 1- Cold Restart the tributary circuit pack raising the alarm 2- Replace the tributary circuit pack raising the alarm Two XCs and more than one tributary circuit pack which uses the XC 1- Cold restart the active XC 2- Replace the active XC 3- Cold restart the other XC 4- Replace the other XC 5- Cold restart alarmed tributary circuit pack with the lowest slot number 6- Replace alarmed tributary circuit pack with the lowest slot number 7- Continue with the remaining tributary circuit packs, in order of increasing slot numbers two XCs only 1- Cold restart the active XC 2- Replace the active XC 3- Cold restart the other XC 4- Replace the other XC one or more SPs and only one other circuit pack 1- Cold restart the other circuit pack raising the alarm 2- Replace other circuit pack raising the alarm one or more SPs and one or more other circuit packs 1- Cold restart the active SP 2- Replace the active SP 3- Cold restart the other SP 4- Replace the other SP

If the alarmed equipment is:	Then perform the following: See Note 1 and Note 2
a single (1+8)xOTN Flex MOTR in an even slot with traffic to another (1+8)xOTN Flex MOTR circuit pack in the adjacent, lower numbered slot	1- Cold restart the (1+8)xOTN Flex MOTR in the even slot 2- Replace the (1+8)xOTN Flex MOTR in the even slot 3- Cold restart the (1+8)xOTN Flex MOTR in the odd slot 4- Replace the (1+8)xOTN Flex MOTR in the odd slot
two Flex2 WL3/WL3e OCLD Submarine circuit packs with an equipment Profile=BPSK2x50G which have PECs that do not match	1- Replace either of the circuit packs with the other PEC, so that both PECs match
two matching transponder circuit packs which carry traffic between them, such as OCLD/OCLD pairs, OCI/OCI pairs or OCI/MUX pairs	1- Cold restart the transponder circuit pack in the higher numbered slot 2- Replace the transponder circuit pack in the higher numbered slot 3- Cold restart the transponder circuit pack in the lower numbered slot 4- Replace the transponder circuit pack in the lower numbered slot
A combination of an OCI/MUX and an OCLD circuit pack which carry traffic between them	1- Cold restart the OCI/MUX circuit pack 2- Replace both circuit packs
two or more tributary circuit packs which use the XCs	1- Cold Restart the alarmed circuit pack in the lowest slot number 2- Replace the alarmed circuit pack in the lowest slot number 3- Continue with remaining circuit packs in order of increasing slot numbers

If the alarmed equipment is:	Then perform the following: See Note 1 and Note 2
two adjacent eMOTR circuit packs which carry traffic between them	1- Cold restart the eMOTR circuit pack that was restarted the latest. 2- Cold restart the adjacent eMOTR circuit pack. See Note 3 .
a fan module and an access panel	1- Ensure that the fan and the access panel are fully inserted into their respective slots 2- Replace the access panel 3- Replace the fan module

Note 1: To perform a cold restart on the identified circuit pack, refer to the [Procedure 2-11, "Restarting a circuit pack or shelf processor"](#) procedure in this document.

Note 2: Replace a circuit pack only if it is not been replaced within the past 48 hours. To replace a circuit pack, refer to the appropriate circuit pack replacement procedure in *Fault Management - Module Replacement*, 323-1851- 545.

Note 3: To determine which eMOTR circuit pack was the latest to cold restart, verify the "Historical Fault Browser" in Site Manager to find the latest "Cold Restart Completed" event for one of the eMOTR circuit packs.

- 7 If the alarm persists after 10 minutes, or if the circuit packs had already been replaced within the past 48 hours, contact your next level of support or your Ciena support group.

—end—

Procedure 4-152

Inter-card Suspected - Pluggable

Alarm ID: 344

Probable cause

This alarm is raised when an SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 reports suspected communications bus failures. The fault may be on the circuit pack or on the pluggable module but the software is unable to determine exactly where the fault exists. The fault may also be caused if the pluggable module is not seated correctly in the host card.

Impact

Critical, service-affecting (C, SA) alarm for an active pluggable
Minor, non-service-affecting (m, NSA) alarm for an inactive pluggable

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- obtain a supported pluggable module for the corresponding circuit pack (refer to the “Supported SFP/SFP+/XFP/CFP/QSFP/QSFP+/QSFP28 modules for interface circuit packs” table in *6500 Planning*, NTRN10GK)

Step	Action
1	Identify the pluggable module raising the alarm and the circuit pack on which it exists. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	Ensure that the pluggable module is seated properly in the host card.
3	If the alarm did not clear, perform a cold restart on the circuit pack identified in step 1 . Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
4	If the alarm persists after 10 minutes, then replace the pluggable module you identified in step 1 with a supported pluggable module. Refer to the appropriate pluggable module replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.

Procedure 4-152 (continued)

Inter-card Suspected - Pluggable

Step	Action
5	If the alarm persists after 10 minutes, contact your next level of support or your Ciena support group. —end—

Procedure 4-153

Inter-card Suspected - Pluggable I/O Carrier 1/2

Alarm IDs: 1060, 1061

Probable cause

This alarm is raised against a pluggable I/O carrier when there is suspected communications bus failures with an associated circuit pack (such as 20G L2SS). This occurs when the I/O carrier cannot power up after power is enabled by the circuit pack. The pluggable I/O carrier is suspected by software based on a majority conviction algorithm.

Impact

Critical, service-affecting (C, SA) alarm for an active I/O carrier
Minor, non-service-affecting (m, NSA) alarm for an inactive I/O carrier

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- obtain a supported pluggable I/O carrier for the corresponding circuit pack (refer to *Planning*, NTRN10GK)

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. The Unit field in the Active Alarms application specifies the circuit pack, shelf ID, circuit pack slot, and pluggable port using the following format: <circuit pack>-<shelf-id>-slot#-pluggable port#
2	Ensure that I/O support is enabled.
3	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
4	Replace the I/O carrier you identified in step 1 with a supported I/O carrier. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
5	Retrieve all alarms to determine if the original alarm has cleared.

Procedure 4-153 (continued)

Inter-card Suspected - Pluggable I/O Carrier 1/2

Step	Action
6	If the alarm does not clear, replace the circuit pack you identified in step 1 . Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
7	If the alarm is still active, contact your next level of support or your Ciena support group.

—end—

Procedure 4-154

Internal Database Synch in Progress

Alarm ID: 1598

Probable cause

This alarm is raised to indicate that the SP and OTN XC (X-Conn 600G PKT/OTN [NTK615AA] and X-Conn 1600G PKT/OTN [NTK616AA] circuit packs) databases are re-synchronizing in order to maintain a consistent database. Some OTN commands might not work properly when this alarm is active.

This alarm is only applicable for OTN shelves with an OTN XC provisioned. The alarm is only raised under these scenarios:

- after an SP restart (cold and warm)
- after an SP switchover for dual SP scenarios
- after an OTN XC restart (cold and warm)
- after an XC switchover for dual XC scenarios
- after any loss of communication between the SP and OTN XC
- after a Database restore
- during an NE upgrade
- after commissioning the OSRP OTN Control Plane on the node (due to the XC SWACT that is required)

Note: The alarm is cleared automatically after the re-sync is complete. When the alarm is cleared, in Site Manager a dialog may appear which requires user acknowledgment.

Impact

Warning, non-service-affecting (w, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Procedure 4-154 (continued)

Internal Database Synch in Progress

Step	Action
1	If the alarm does not clear automatically, wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf and insert the OTN XC circuit pack. After the XC has booted up, the re-sync will occur and the alarm will clear.
2	If the alarm does not clear, restart the SP. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" in this document.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-155

Internal Mgmt Comms Suspected

Alarm ID: 438

Probable cause

This alarm is raised when the communication channel failure between two circuit packs is suspected. The alarm is raised against both circuit packs suspected of being the cause of the failure.

Impact

Minor, non-service-affecting (m, NSA) alarm, if a circuit pack is provisioned for 1+1/MSP linear or 1:N

Major, service-affecting (M, SA) alarm, if a circuit pack is unprotected or provisioned for UPSR/SNCP or 2-Fiber/4-Fiber BLSR/MS-SPRing/HERS protection

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
2	Identify the circuit packs raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” in this document.						
3	Perform a warm restart on the circuit pack raising the alarm. See Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Reseat one of the circuit packs raising the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						

Procedure 4-155 (continued)
Internal Mgmt Comms Suspected

Step	Action						
6	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Reseat the other circuit pack raising the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	<table> <tr> <td>If the circuit pack reseated in step 5 was</td><td>Then go to</td></tr> <tr> <td>not replaced in the past 48 hours</td><td>step 10</td></tr> <tr> <td>replaced in the past 48 hours</td><td>step 12</td></tr> </table>	If the circuit pack reseated in step 5 was	Then go to	not replaced in the past 48 hours	step 10	replaced in the past 48 hours	step 12
If the circuit pack reseated in step 5 was	Then go to						
not replaced in the past 48 hours	step 10						
replaced in the past 48 hours	step 12						
10	Replace the circuit pack reseated in step 5 . Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	<table> <tr> <td>If the circuit pack reseated in step 7 was</td><td>Then go to</td></tr> <tr> <td>not replaced in the past 48 hours</td><td>step 13</td></tr> <tr> <td>replaced in the past 48 hours</td><td>step 15</td></tr> </table>	If the circuit pack reseated in step 7 was	Then go to	not replaced in the past 48 hours	step 13	replaced in the past 48 hours	step 15
If the circuit pack reseated in step 7 was	Then go to						
not replaced in the past 48 hours	step 13						
replaced in the past 48 hours	step 15						
13	Replace the circuit pack reseated in step 7 . Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
14	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 15</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 15
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 15						
15	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-156

Intrusion Attempt

Alarm ID: 267

Probable cause

This alarm is raised when the shelf processor detects an intruder attempting to gain shelf access using either an IPv4 or IPv6 connection, and the maximum number of login attempts exceeds the provisioned number allowed.

The alarm is raised when an intrusion is detected and the channel/port locks. The alarm clears automatically after the lockout period expires. However, an administrator can clear the alarm manually before the lockout is cleared. If new intrusion attempts occur before this alarm is cleared, they will be logged in the Security Log and the duration of the lockout will be extended for the lockout period of the latest intrusion attempt.

ATTENTION

Clearing this alarm clears all other alarms of the SECU class (except those raised against the Primary and secondary RADIUS servers).

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must:

- be able to connect to the SP
- use an account with at least a level 4 UPC

Step	Action
1	Allow the alarm to clear itself when the lockout time expires or go to step 2 .
2	Select the alarmed network element in the navigation tree.
3	Select Clear Security Alarm from the Faults menu. Clearing the Intrusion Attempt alarm has no effect on the alarmed port. It only removes the alarm from the alarm list. The port will continue to be locked out until valid login information is delivered. The channel is locked out and no one can log in from the originating address for the duration of the lockout. The alarm clears from the network element when the channel unlocks after the provisioned elapsed time.

Procedure 4-156 (continued)

Intrusion Attempt

Step	Action
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4	Click Yes in the confirmation dialog box.
---	--

5	Follow your company policy for handling intrusion attempts.
---	---

—end—

Procedure 4-157

Invalid Site Topology

Alarm ID: 872

Probable cause

This alarm is raised against the Optical Transmission Section (OTS) entity when there are more than two OTSs with the same site identifier and optical system identifier (OSID).

Note: This alarm may be raised if there is a duplicate site provisioned within an OSID.

This alarm will also be raised whenever the number of OTSs within a site (with the same Site ID) exceeds the maximum limit. The limit is different depending to the shelf and circuit pack type in the system. For the maximum number of OTSs in each site, refer to the OTS engineering rules section in the *Data Application Guide*, NTRN15BA. The alarm will be raised against each OTS in the site.

ATTENTION

Do not use the same Tx/Rx ID in the same site, even if they are used on different shelves.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Identify the network elements raising the alarm.
2	Determine from network plans or other documents which network element has the correct site identifier and OSID.
3	Log into the network element with the duplicate site identifier and OSID.

Procedure 4-157 (continued)

Invalid Site Topology

Step	Action
------	--------

4



CAUTION

Risk of loss of functionality

Ensure every network element has a unique site identifier. If you are changing the site identifier of a network element, ensure it is unique.

Enter the correct **Site identifier** and **Optical system identifier** values are used. Refer to the “Editing the nodal shelf parameters” procedure in *Administration and Security*, 323-1851-301.

5

Wait for the network element to complete the automatic restart.

6

Ensure that no other Invalid Site Topology alarms exist. If there are other Invalid Site Topology alarms, repeat this procedure.

7

If the alarm does not clear, verify the number of OTSs in the site. Refer to the “Retrieving OTS Management, OTS Equipment, and Facility details” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.

8

Reduce the number of OTSs in the site to less than or equal to the limit.

9

If the alarm does not clear, verify that the Tx/Rx IDs are unique among all the shelves within the site.

10

If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-158 Isolated Station

Alarm ID: 815

Probable cause

This alarm is raised when the station can no longer send data traffic to any of the other stations on the RPR.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action						
1	Verify the OC-n/STM-n fiber connections related to RPR ring. Repair, clean, and reconnect the fibers as required. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.						
2	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 3</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Use the appropriate alarm clearing procedure to clear any alarms on the RPR ring including Loss Of Signal and AIS (STS Rx) alarms. If the alarms are against OC-n/STM-n circuit packs, clear these alarms first.						
4	If the WAN Link Down alarm is raised against the RPR circuit pack, clear the alarm using the appropriate alarm clearing procedure.						
5	If the alarm does not clear, ensure the WAN links on the adjacent circuit packs are not OOS.						
6	If the alarm does not clear, ensure that forced switches on the WAN links of the adjacent RPR circuit packs are not active.						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-159

LAN alarms

Use this procedure to clear alarms associated with the LAN ports.

A transient LAN port failure alarm can be raised momentarily when a LAN port is created.

ETH LAN Port Failure

Alarm ID: 536

Probable cause

This alarm is raised when the LAN port (Local Craft 10/100 BT) on the shelf processor in slot 15 or 16 of the 4-slot shelf, 6500-7 packet-optical shelf or 14 slot shelf, or slot 41 or 42 of the 32-slot shelf does not receive any Ethernet packets. This occurs if there is a Ethernet bad connection, the LAN port fails, or there is no craft terminal physically connected.

Impact

Minor, non-service-affecting (m, NSA) alarm

LAN Link Failure

Alarm ID: 1874

Probable cause

This alarm is raised when the link status of a COLAN or ILAN port is down in a network indicating that either COLAN or ILAN port lost its link/connectivity to the network and the port is not available for any further communication.

This alarm is raised on the 6500 shelves participating in a network. It tracks the link status of all the provisioned LAN ports in the network. The alarm will be automatically cleared when the link comes up for the corresponding LAN port.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Procedure 4-159 (continued)

LAN alarms

Step	Action						
1	Check the DCN information to determine if the appropriate LAN port must be enabled. <table> <tr> <td>If the LAN must be</td><td>Then go to</td></tr> <tr> <td>disabled</td><td>step 2</td></tr> <tr> <td>enabled</td><td>step 3</td></tr> </table>	If the LAN must be	Then go to	disabled	step 2	enabled	step 3
If the LAN must be	Then go to						
disabled	step 2						
enabled	step 3						
2	Disable the appropriate LAN port. Refer to the “Deleting an entry in the communications settings” procedure in <i>Data Comms Planning & User Guide</i>, 323-1851-101. Note: The ETH LAN port cannot be disabled. For the ETH LAN port Failure alarm, disable the alarm using an alarm profile. Refer to the “Editing an alarm profile” procedure in this document. Go to step 7.						
3	Check that the LAN configuration at both ends of the link are the same. Refer to the “Retrieving communications settings” procedure in <i>Data Comms Planning & User Guide</i>, 323-1851-101. If the configuration is different, edit the configuration as required. Refer to the “Editing the communications settings” procedure in <i>Data Comms Planning & User Guide</i>, 323-1851-101.						
4	At the local network element, retrieve all alarms to determine if the original alarm has cleared. <table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Ensure that the appropriate cable is correctly connected to the required LAN port on the access panel or shelf processor.						
6	If the alarm is raised against the ILAN-IN or ILAN OUT ports and the adjacent shelf for that port is a 6500 shelf, check the status of the MIC on the adjacent shelf. Ensure that the MIC is not failed or removed.						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-160

Laser Failed

Alarm ID: 1513**Probable cause**

This alarm is raised when one or more lasers in the Idler facility has failed. If one of the lasers in the Idler pair fails, both lasers in the pair will be shut off. If all IDLER facilities have Laser Failed alarms against them, the Laser Failed alarms will be replaced by a Circuit Pack Failed alarm.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

Before you perform this procedure, you must

- observe all the safety requirements described in *Installation*, 323-1851-201.0, or *Module Replacement Procedures*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	If a spare Idler facility is available on the SLIC10 or SLIC10 Flex C-Band circuit pack, deprovision the failed Idler by putting it OOS and set the wavelengths to 0.0 nm. Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
2	Provision the spare Idler to have the failed Idler's original values.						
3	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 4</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	If a spare Idler facility is not available on the SLIC10 or SLIC10 Flex C-Band circuit pack, re-evaluate the failed Idler by setting its wavelength to 0.0 nm and then back to its original wavelength. If the problem has cleared, this will trigger re-evaluation and clear the laser failed alarm.						
5	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						

Procedure 4-160 (continued)

Laser Failed

Step	Action
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf and replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
7	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-161

Laser Frequency Out Of Range

Alarm IDs: 2017, 2049

Probable cause

This alarm is raised when the difference between the nominal provisioned frequency and the actual frequency achieved via the TR controller exceeds the provisioned threshold.

For the 6500 WL3/3e/3n based circuit packs, crossing the provisioned threshold value will trigger the “Laser Frequency Out of Range” alarm. The default threshold is 1.6, but it can be between 0 and 3.0.

Impact

Major, service-affecting (M, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Step	Action
1	If possible, clear all other alarms from the network.
2	Verify the setting of the associated ADJ-Tx Laser Centering. The Laser Centering loop state has to be the same on both ends.
3	Check the frequency specification at the line port at either end of the fiber span.
4	If the alarm is not clear, verify that the equipment at the far end of the optical line meets the frequency specification. If the equipment does not meet the specification, change the far end equipment to be within specifications.
5	If the alarm did not clear, re-provision the thresholds of the line port. Refer to <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
6	If the alarm is raised against the WL3/3e/3n based circuit packs for no apparent reason, verify that the threshold value is not set to 0.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-162

License Server Time Out Of Sync

Alarm ID: 2114**Probable cause**

The License Server Time Out Of Sync alarm is raised when a 15 minute time difference is detected between the shelf client software and any license server.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Setup correct manual time or preferably NTP/TOD on 6500 shelf and both license server Linux server. Refer to “Editing the nodal general parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
2	Perform a License Audit manually on the 6500 shelf. Refer to <i>Licensing</i> , 323-1851-210 for more information. Note: To speed up the response process if there is an error related to server, licensing will let the error pass the first time and catch it in the next request.
3	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-163

Licensing Trusted Store Mismatch

Alarm ID: 2013

Probable cause

This alarm is raised when the 6500 software detects that the encrypted license database (Trusted Store) has been modified or does not contain expected shelf identifier.

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	Operate a License Audit manually or a warm restart the SP/CTM. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document. Note: This alarm will also be cleared automatically after an automatic daily License Audit (default time is 03:00).
2	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-164 License Violation

Alarm ID: 1999, 2037

Probable cause

This alarm is raised against a shelf if it is using features that require licenses but is missing at least one license.

The alarm can be raised under the following conditions:

- a license was granted for a feature while the shelf was in licensing commissioning mode and is therefore in arrears
- a license was granted for a feature as a pre-authorized feature license and is therefore in arrears
- a license was granted for a feature as a strict feature license and subsequently the license was returned to the license server/manager and not re-acquired due to a network communication error during a shelf configuration change or shelf reconfiguration and is therefore in arrears
- a license was granted for a feature as a strict or pre-authorized feature license but a loss of DCN connectivity with the license server/manager resulted in the license being reclaimed by the license server/manager and given out to another shelf before the connectivity issue was fixed and is therefore in arrears.

The 6500 shelf tracks only a count of items licensed and then raises the “License Violation” alarm when the total number of licenses acquired does not equal the total number of licensed elements.

To clear this alarm a licensing server needs to be provisioned and the shelf needs to be able to communicate properly with the licensing server. The management of licenses on a network is part of networking planning. The correct number of licenses need to be purchased and allocated for all the shelves on the network expecting to use features that require licensing.

Impact

Major, non-service-affecting (M, NSA) alarm

Critical, non-service-affecting (C, NSA) alarm, when the shelf is in arrears for 11 or more of licenses excluding license policy type with NOALARM.

Procedure 4-164 (continued)

License Violation

Step	Action
	Note: This alarm automatically clears when the licensing software retrieves the missing licenses from the license server/manager.
1	Determine the licenses that are causing the alarm by doing one of the following: Site Manager -> Configuration Menu -> Shelf Licensing -> License Information -> Arrears Or RTRV-LICENSE TL1 command and look for licenses which have counts higher than 0 for their arrears value. Those are the licenses and the counts that are required to clear the alarm.
2	Add the licenses to the license server/manager. The new licenses can be purchased and loaded onto the licensing server.
3	Once the licenses are present on the license server/manager, operate License Audit manually (in Site Manager or TL1). The alarm will also clear after the daily License Audit that occurs nightly (default 3am).
4	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-165

Line A Input OTDR High Loss Detected

Alarm ID: 2047

Probable cause

This alarm is raised as part of the Go/No-Go pass criteria for the SRA when the event loss calculated by the OTDR trace is higher than the provisioned threshold either for a single event loss or for the total loss within 20 km. This will prevent turning on the Raman pumps. Since 6500 Release 12.1, all OTDR related parameters and alarms are migrated from the port 5 Telemetry facility to the new OTDRCFG facility on port 8.

This alarm can also be raised due to a fiber type transition (FTT).

As of release 12.6, a “Result” field in the output of the OTDR event can be used to determine whether the condition was due to a single event or cumulative events. The “Result” field is empty if the OTDR trace was successful (Go condition). Refer to *TL-1 Description*, 323-1851-190, for more information.

Impact

Major, service-affecting (M, SA) alarm

Step	Action
1	Check the additional info field of the alarm for the information about the trace causing the alarm: • High Loss - Short Trace • High Loss - Office Trace • Flat Trace - Long Trace
2	It is recommended to re-run the Go/No-Go before starting any troubleshooting to make sure the issue is still present. Note: The Go/No-Go type is available for SRA using the Site Manager only if RAMAN is in Out-of-Service (OOS) state. The Go/No-Go type is available only in the Trace Type drop down menu (not in the Edit dialog box). Another option to run the Go/No-Go without having to put the Raman facility OOS is using the TL1 command as follow: <pre>OPR-OTDR:<TID>:OTDRCFG-shelf-slot-port:CTAG::GONOGOREQ=YES;</pre>
3	If the alarm is still present, based on which trace is the cause for the alarm, verify the respective trace in the OTDR screen.

Procedure 4-165 (continued)

Line A Input OTDR High Loss Detected

Step	Action
4	From the OTDR screen, select the trace raising the issue and select the View trace(s) button. The View Trace button can be selected only if a Facility is selected from the Facility Configuration and a Trace is selected from the SOR Trace.
5	In the OTDR Graph View, examine the resulting graph to determine the location of the fault(s).
6	From the graph, find the highest event loss within the first 20km by selecting the View event button. Verify the Note column to see which event is triggering the alarm.
7	If the highest loss is outside of the fiber plant, re-spliced the fiber at that location.
8	If the highest loss is due to a fiber type transition, provision an FTT following the guidelines under the “Fiber Type Transition” section in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. For more information, refer to the “Fiber Type Transition” section in the <i>Photonic Layer Guide</i>, NTRN15DA.
9	If the loss is within the fiber plant, use a fiber cleaning kit to clean all the connectors and then reconnect the fibers. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i>, 323-1851-201.0. Disconnecting and reconnecting the fiber in the fiber plant that is already connected, will trigger automatic OTDR traces to run. It is recommended to wait for the OTDR traces that were triggered automatically to be completed. If you disconnect a fiber connection and need to clean more, do all the cleaning and then reconnect the first connector that was disconnected. This will prevent multiple OTDR triggers which can take longer to complete.
10	If the span is not bookended, run the trace manually to clear the alarm.
11	If cleaning connectors fails to resolve the issue, replace faulty fiber patch cords.
12	Once the fiber is reconnected, the OTDR Traces run automatically. If the new trace passes, the alarm will clear.
13	If the alarm is still present, run the Go/No-Go on the unit.
14	If the alarm does not clear, there might be multiple failure points for Go/No-Go. If the worst loss point is corrected and the Go/No-Go fails, continue the investigation and look for the next highest loss until all the possible failure points are corrected.
15	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-166

Line A Input OTDR High Reflection Detected

Alarm ID: 2048

Probable cause

This alarm is raised as part of the Go/No-Go pass criteria for the SRA when the event reflection calculated by the OTDR trace is higher than the provisioned threshold either for a single event reflection or for the total reflection within 20 km. This will prevent turning on the Raman pumps. Since 6500 Release 12.1, all OTDR related parameters and alarms are migrated from the port 5 Telemetry facility to the new OTDRCFG facility on port 8.

As of release 12.6, a “Result” field in the output of the OTDR event can be used to determine whether the condition was due to a single event or cumulative events. The “Result” field is empty if the OTDR trace was successful (Go condition). Refer to *TL-1 Description*, 323-1851-190, for more information.

Impact

Major, service-affecting (M, SA) alarm

Step	Action
1	Check the additional info field of the alarm for the information about the trace causing the alarm: - High Reflection - Short Trace - High Reflection - Office Trace - Flat Trace - Long Trace
2	It is recommended to re-run the Go/No-Go before starting any troubleshooting to make sure the issue is still present. Note: The Go/No-Go type is available for SRA using the Site Manager only if RAMAN is in Out-of-Service (OOS) state. The Go/No-Go type is available only in the Trace Type drop down menu (not in the Edit dialog box). Another option to run the Go/No-Go without having to put the Raman facility OOS is using the TL1 command as follow: OPR-OTDR:<TID>:OTDRCFG-shelf-slot-port:CTAG::GONOGOREQ=YES;
3	If the alarm is still present, based on which trace is the cause for the alarm, the respective trace should be checked in the OTDR screen.
4	From the OTDR screen, select the trace raising the issue and select the show trace button.
5	In the OTDR Graph View, examine the resulting graph to determine the location of the fault(s).

Procedure 4-166 (continued)

Line A Input OTDR High Reflection Detected

Step	Action
6	From the graph, find the largest event Reflection within the first 20km by selecting the View event button. Verify the Note column to see which event is triggering the alarm.
7	Check proper connections at patch panels.
8	If the largest Reflection is outside of the fiber plant, re-spliced the fiber at that location.
9	If the Reflection is within the fiber plant, use a fiber cleaning kit to clean all the connectors and then reconnect the fibers. Refer to the “Cleaning connectors” section in <i>Installation - General Information</i>, 323-1851-201.0. Disconnecting and reconnecting the fiber in the fiber plant that is already connected, will trigger automatic OTDR traces to run. It is recommended to wait for the OTDR traces that were triggered automatically to be completed. If you disconnect a fiber connection and need to clean more, do all the cleaning and then reconnect the first connector that was disconnected. This will prevent multiple OTDR triggers which can take longer to complete.
10	If the span is not bookended, run the trace manually to clear the alarm.
11	If cleaning connectors fails to resolve the issue, replace faulty fiber patch cords.
12	Once the fiber is reconnected, the OTDR Traces run automatically. If the new trace passes, the alarm will clear.
13	If the alarm is still present, run the Go/No-Go on the unit.
14	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-167

Line Adjacency Manual Provisioning Required

Alarm IDs: 1945

Probable cause

This warning is raised against the ADJ-LINE facility when the equipment is provisioned as the LIM slot in an OTS and the Target Span Loss has a default value of 0 dB.

When the Target Span Loss is 0 the “High Received Span Loss” (HRSL) alarm is disabled. When the HRSL alarm is disabled, DOC’s ability to compensate for changing span losses is compromised and there is a risk of power overshoots if the span loss increases and then drops again.

Applies to all LIM equipment variants (for example, SLA, MLA, MLA2v) and to all other equipment which can be provisioned as the LIM slot (SAM, ESAM, SRA, MDRLA 2x6, and RLA).

Impact

Warning

Step	Action
------	--------

- | | |
|---|---|
| 1 | Provision the Target Span Loss to the expected value as per network link budgets. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-168

Line Flapping

Alarm ID: 1774

Probable cause

If the Line Flapping alarm is enabled, the alarm is raised against the TCMTTP facility (not any other provisioned TCMTTP on the same line) for the OSRP line of a 100G PKT/OTN XCIF, 10x10G PKT/OTN I/F, 10x10G OTN I/F, or 40G OTN XCIF circuit pack when the line is oscillating beyond the provisioned Line Flapping Alarm Threshold within the provisioned Line Flapping Alarm Raise Time. The alarm is present on the system for the affected TCMTTP for the period defined in the provisioned Line Flapping Alarm Clear Time. These Line Flapping alarm settings are applicable to all lines on the NE. For more information on the Line Flapping Alarm Clear Time, Line Flapping Alarm Raise Time, and Line Flapping Alarm Threshold parameters, refer to the “Node information” section in *Administration and Security*, 323-1851-301.

For example, if the Line Flapping Alarm Threshold is provisioned to three, the Line Flapping Alarm Raise Time is provisioned to 300 seconds, and the Line Flapping Alarm Clear Time is provisioned to 600 seconds, then three events (such as LOS) must occur within 300 seconds for the alarm to raise. The alarm will be active for 600 seconds. If the line continues to oscillate during this period and the alarm is masked by a higher priority events (see triggers below), then the alarm is raised again and the Line Flapping Alarm Clear Time timer is restarted.

The alarm can also be raised based on the following supported TCM triggers:

- LOS, LOF, or LOM condition
- TCM ODU AIS, OCI, LCK, LTC, or LOF condition
- Disabling the Line Flapping alarm or disabling/re-enabling the Line Flapping alarm clears all Line Flapping alarms on the NE
- the provisioned Line Flapping Alarm Clear Time is reached
- the provisioned Line Flapping Alarm Raise Time is reached
- the provisioned Line Flapping Alarm Threshold is reached

Impact

Major, service-affecting (M, SA) alarm
Critical, service-affecting (M, SA) alarm

Procedure 4-168 (continued)

Line Flapping**Prerequisites**

To perform this procedure, you must have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements).

Step	Action						
1	Retrieve all alarms at the transmit end of the line. Clear any higher order alarms using the appropriate procedure.						
2	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Verify that the FEC setting on the upstream circuit pack matches the FEC setting on the local circuit pack. Correct any mismatches. Refer to the "Retrieving equipment and facility details" or "Editing facility parameters" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed channel using the appropriate procedures.						
8	Verify that the Secure Overhead Mode setting on the upstream Encryption OCLD circuit pack matches the setting on the local circuit pack. Correct any mismatches. Refer to the "Retrieving equipment and facility details" or "Editing facility parameters" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
9	If the alarm does not clear, restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						

Procedure 4-168 (continued)

Line Flapping

Step	Action
10	If the alarm does not clear, disable the Line Flapping Alarm globally and re-enable the Line Flapping Alarm parameter to reset the alarm. Refer to “Editing the nodal system parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
11	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-169

Link Data Retrieval In Progress

Alarm IDs: 1884**Probable cause**

This alarm is raised when a user invokes the link data collection process for the PTP facility in the Site Manager.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the data collection process is completed.
2	If the alarm does not clear or is unexpected, contact your next level of support or Ciena support group.

—end—

Procedure 4-170

Link Data Save In Progress

Alarm IDs: 1886

Probable cause

This alarm is raised when a user invokes a link data save process.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the link data save process is completed.
2	If the alarm does not clear or is unexpected, contact your next level of support or Ciena support group.

—end—

Procedure 4-171

Link Down

Alarm IDs: 334, 865, 1975

Probable cause

This alarm is raised against

- a WAN facility of a 8xOTN Flex MOTR, (1+8)xOTN Flex MOTR, OTN Flex MOTR 2xCFP2 circuit pack
- a WAN facility of a 2x10G OTR, 4x10G OTR, (1+2) 100G PKT/OTN I/F, 10X10G PKT/OTN I/F, 10x10G MUX OCI, 100G (2xQSFP+/2xSFP+) MUX, 200G (2x100G/5x40G) MUX, 5x100G/12x40G QSFP PKT/OTN IF, 40x10G SFP+ PKT/OTN IF, or 40G MUX OCI circuit pack

This alarm is raised against a WAN facility when the administrative state of a facility is up, but the operating state of the facility is down. When the Link Down alarm is raised against the WAN facility, the link cannot carry traffic. The Link Down alarm raised against a WAN facility masks the Link Down alarm against an Ethernet or Fiber Channel facility.

When one of the following alarms is raised, the Link Down alarm is raised against the WAN facility:

- Client Service Mismatch
- Loss of Frame Delineation

ATTENTION

For electrical SFPs (NTTP61BA), if this alarm is raised and the traffic cannot be brought up, change the TXCON state to Disabled.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- use an account with at least a level 3 UPC

Step	Action
------	--------

Alarm raised against an Ethernet or Fiber Channel facility

- | | |
|---|---|
| 1 | Clear any alarms raised against the Ethernet or Fiber Channel facility of the circuit pack raising the alarm. |
| 2 | Ensure that the subtending client equipment is correctly provisioned, functioning, and transmitting a valid signal. |
| 3 | Ensure that the fiber between the subtending equipment and the Ethernet or Fiber Channel port is properly connected and is not damaged. |
| 4 | Ensure that the auto-negotiation setting provisioned on the subtending equipment matches the auto-negotiation setting provisioned on the 8xOTN Flex MOTR, (1+8)xOTN Flex MOTR circuit pack. To determine the auto-negotiation setting, refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 9 . |

Alarm raised against the WAN facility

- | 5 | Clear any alarms raised against the WAN facility of the circuit pack raising the alarm. | | | | | | |
|---------------------------|---|---------------------------|------|---------|---------------------------|-------------|------------------------------|
| 6 | <table border="1"> <thead> <tr> <th>If the original alarm has</th> <th>Then</th> </tr> </thead> <tbody> <tr> <td>cleared</td> <td>the procedure is complete</td> </tr> <tr> <td>not cleared</td> <td>go to step 7</td> </tr> </tbody> </table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 7 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 7 | | | | | | |
| 7 | Ensure that the frame checksum size of the WAN facility on the circuit pack raising the alarm and the far-end WAN facility match. Correct any mismatches. Refer to the "Retrieving equipment and facility details" or "Editing facility parameters" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 8 | If the Loss Of Signal or Link Down alarm is raised against an Ethernet or Fiber Channel facility of the far-end circuit pack, clear the alarm using the appropriate alarm clearing procedure. | | | | | | |
| 9 | If the alarm does not clear, contact your next level of support or your Ciena support group. | | | | | | |

—end—

Procedure 4-172

Link Not Protected

Alarm ID: 2265, 2266

Probable cause

When enabled, this alarm is raised against a Mesh Restorable-SNC/SNCGs during a protection routes audit when at least one of the working path links is not protected by any of the available user provisioned protection routes.

The alarm helps identify links which are single point of failure, since it raises when there is no absolutely diverse protection path available. The additional text mentions the node name and link Id in the working path that is not protected.

This alarm is disabled by default in the alarm profiles to avoid flooding the network with new alarms upon an upgrade. An exception is the ALL ENABLED profile, where the new alarm is enabled by default and when not using the ALL ENABLED profile, after an upgrade. You can enable the alarm in the primary shelf alarm profiles. You can also disable the feature at the OSRP node level using TL1, by setting the LINKPROTECTEDALARM parameter to "No".

The condition is checked every 5 minutes. Therefore, there is an expected delay between the time the state actually changes and the alarm is raised or cleared. Only the provisioned (explicit) protection Designated Transit Lists (DTLs) are checked.

While the alarm is raised, a "Link Not Protected" Transient Condition is raised/cleared for each Link that is not protected, as there can be multiple unprotected links causing this alarm.

This alarm is cleared when all links in the current working path become protected by at least one protection DTL.

Impact

Major, non-service-affecting (M, NSA)

Step	Action
------	--------

- | | |
|---|--|
| 1 | To clear the alarm perform one of the following: <ol style="list-style-type: none"> ensure that all the work path links are protected by the protection routes in the next protect routes audit. (happening every 5 minutes). send a release to SNC/SNCG or perform a mesh restore. The alarm can be raised again if the condition persists after the event. |
|---|--|

4-390 Alarm clearing procedures

Procedure 4-172 (continued)

Link Not Protected

Step	Action
	c. perform a lock on the SNC/SNCG d. perform an AUXP reboot. The alarm can be raised again if the condition persists after the event.
2	If an existing protection route is not available, provision a new protection route for the path link(s). Refer to <i>Configuration - Protection Switching</i> , 323-1851-315 for more information.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-173

Local Optical Controller Disabled

Alarm ID: 1950

Probable cause

This alarm is raised against the shelf when the Coherent Select Controller (CSCTRL) system parameter is disabled and there is at least one Coherent Select OTS provisioned on the shelf.

Impact

Warning

Step	Action
1	Enable the CSCTRL system parameter. Refer to “Enabling Coherent Select control” procedure in <i>WaveLogic Photonics Coherent Select</i> , 323-1851-980.
2	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-174

Log Collection In Progress

Alarm ID: 1048

Probable cause

This alarm is raised when a user invokes the log collection process from the TL1 interface or from the Faults > Card Logs menu in Site Manager.

This alarm is also raised when you select “Capture logs before restart” in the confirmation window of the Faults > Restart menu of the Site Manager.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the log collection process is completed.
2	If the alarm does not clear or is unexpected, contact your next level of support or Ciena support group.

—end—

Procedure 4-175

Log Save In Progress

Alarm ID: 1569**Probable cause**

This alarm is raised when a log save is in process.

For an active SP, the restart takes eight to 15 minutes. For other circuit packs, the restart takes four to 10 minutes to complete.

If after the expected time the retrieve log is not completed, Site Manager automatically issues the restart command. Refer to the [Procedure 2-11](#), “Restarting a circuit pack or shelf processor” procedure in this document for more information.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the log save process is completed.
2	If the alarm does not clear or is unexpected, contact your next level of support or Ciena support group.

—end—

Procedure 4-176 Loopback Active

Alarm IDs: 864, 898, 1696

Probable cause

This alarm is raised when a loopback is active on an OC-3/STM-1/STM-1J/STM-1e, OC-12/STM-4/STM-4J, OC-48/STM-16, OC-192/STM-64, or FC100/FC200/FC400 facility. The loopback active alarm is an important advisory message, as the unavailability of a circuit can affect maintenance performed at the same time. If more than one user logs into a network element and one user operates a loopback, the other users receive this message.

Execute loopbacks only during system testing. The facility is in out of service, maintenance (OOS-MA) mode while in loopback mode.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 3 UPC
- have the connection information (that is, how the optical/electrical modules on each network element connect to other network elements)

Step	Action
------	--------

- | | |
|---|---|
| 1 | If the alarm is raised during facility testing, no action is required. The alarm will clear once the testing is complete and the loopback is released. |
| 2 | If the alarm is unexpected, identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document. This procedure assumes that testing is complete and the circuit is ready to be released from loopback mode. |
| 3 | Release the loopback. Refer to the "Operating/releasing a loopback" procedure in Configuration - Provisioning and Operating, 323-1851-310. |

—end—

Procedure 4-177

Loopback Active - Facility

Alarm IDs: 605, 637, 833, 845, 848, 852, 948, 990, 997, 1165, 1251, 1303, 1350, 1448, 1457, 1466, 1784, 2134

Probable cause

This alarm is raised when a facility loopback is active.

Note that a mapped ETTP does not support facility or terminal loopbacks. A terminated ETTP supports facility loopback only. A monitor ETTP supports both facility and terminal loopbacks.

Perform loopbacks only during system testing.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 2 UPC
- have the fiber connection information for the line-level loopbacks (that is, how the optical modules on each network element connect to other network elements)

Procedure 4-177 (continued)

Loopback Active - Facility

Step	Action
	Note: If the alarm is raised during facility/connection testing, no action is required. The alarm will clear once the testing is complete and the loopback is released.
1	Identify the facility/connection in loopback mode. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	Release the loopback on the facility/connection identified in step 1 . For line-level loopbacks, refer to the "Operating/releasing a loopback" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. For channelized loopbacks, refer to the "Operating/releasing a channelized loopback" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

—end—

Procedure 4-178

Loopback Active - Terminal

Alarm IDs: 606, 638, 853, 935, 1159, 1166, 1214, 1215, 1216, 1217, 1242, 1305, 1449, 1458, 1467, 1740, 1785, 2135, 2159

Probable cause

This alarm is raised when a terminal line-level loopback is active.

Note that a mapped ETTP does not support facility or terminal loopbacks. A terminated ETTP supports facility loopback only. A monitor ETTP supports both facility and terminal loopbacks.

This alarm is also raised when a channelized terminal (path-level or connection) loopback is applied to a VT1.5/VC-11, VT2/VC-12, STS-1/VC-3, STS-3c/VC-4, STS-12c/VC-4-4c, STS-24c/VC-4-8c or STS-48c/VC-4-16c connection.

This alarm can be raised against an OTUTTP facility of the 10x10G PKT/OTN I/F, 100G Perform loopbacks only during system testing.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 2 UPC
- have the fiber connection information for the line-level loopbacks (that is, how the optical modules on each network element connect to other network elements).

Step	Action
1	Note: If the alarm is raised during facility/connection testing, no action is required. The alarm will clear once the testing is complete and the loopback is released. Identify the facility/connection in loopback mode. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.

Procedure 4-178 (continued)
Loopback Active - Terminal

Step	Action
2	Release the loopback on the facility/connection identified in step 1. For line-level loopbacks, refer to the “Operating/releasing a loopback” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. For channelized loopbacks, refer to the “Operating/releasing a channelized loopback” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.

—end—

Procedure 4-179

Loss of Alignment Marker

Alarm IDs: 2033, 2210

Probable cause

This alarm is raised on ETTP (ETH100G) or PTP clients that use parallel optics when the Invalid lane marker (a marker that does not match one of the encodings in Table 82–2 of IEEE Std 802.3ba) is detected on any lane.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” in this document.						
2	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
3	Log into the remote network element at the transmit end.						
4	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure.						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						

Procedure 4-179 (continued)

Loss of Alignment Marker

Step	Action						
6	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches. Refer to the "Retrieving OTM info" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed channel using the appropriate procedures.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>perform a warm restart on the circuit pack reporting the alarm. Refer to Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	perform a warm restart on the circuit pack reporting the alarm. Refer to Procedure 2-11 , "Restarting a circuit pack or shelf processor" procedure in this document.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	perform a warm restart on the circuit pack reporting the alarm. Refer to Procedure 2-11 , "Restarting a circuit pack or shelf processor" procedure in this document.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-180

Loss of Channel

Alarm IDs: 1311, 1570, 1596

Probable cause

This alarm is raised when the optical power is below the Loss of channel threshold.

For 40G/100G OCLD circuit packs, the power level must be above the LOC clear threshold value for traffic to recover. For 40G/100G OCLD circuit packs, traffic can run below the LOC threshold value. However, if traffic is lost, traffic will not recover until the power level is above the LOC clear threshold. See the “Technical specifications” tables in *WaveLogic OCLD, OCI, MUX, and Submarine Circuit Packs*, 323-1851-102.4 for the 40G/100G OCLD expected LOS alarm clear threshold values.

Impact

Critical, service-affecting (C, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	If the “Rx Channel Power Out of Range” alarm is active, clear this alarm first. Refer to “Rx Channel Power Out of Range” on page 4-664 in this document.

Procedure 4-180 (continued)

Loss of Channel

Step	Action						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.						
5	In the Site Manager Configuration menu, select the Equipment & Facility Provisioning application. Select the alarmed facility and retrieve the value from the Rx Channel Actual Power (dBm) column of the facility table. Refer to the “Retrieving optical power, wavelength, and dispersion ranges” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If the received power is out of the power monitoring range for the facility, the Rx Actual Power (dBm) column will display “OOR-HI” or “OOR-LO” instead of a numerical value.						
6	Click on the Ranges button to compare the Rx Channel Actual Power (dBm) value with the Rx Channel minimum power (dBm) and Rx Channel maximum power (dBm) values displayed.						
7	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
8	Use the optical power meter to measure the optical power level at the far-end upstream monitored point. Also, verify that the loss between the far-end transmitter and the Mux is an expected value. <table> <tr> <th>If the value of the loss is</th><th>Then</th></tr> <tr> <td>unexpected</td><td>troubleshoot the fiber or Photonic layer equipment</td></tr> <tr> <td>expected</td><td>go to step 9</td></tr> </table>	If the value of the loss is	Then	unexpected	troubleshoot the fiber or Photonic layer equipment	expected	go to step 9
If the value of the loss is	Then						
unexpected	troubleshoot the fiber or Photonic layer equipment						
expected	go to step 9						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>reseat the circuit pack reporting the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	reseat the circuit pack reporting the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	reseat the circuit pack reporting the alarm. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						

Procedure 4-180 (continued)

Loss of Channel

Step	Action
11	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-181

Loss of Clock

Alarm IDs: 840, 843, 846, 991, 998, 1167, 1255, 1296, 1442, 1452, 1460, 2016
Probable cause

This alarm is raised when the measured frequency of the input client clock exceeds the allowed frequency deviation outlined by the corresponding standard or when the provisioned line rate at the far-end circuit pack does not match the line port rate at the near-end circuit pack.

In general, this alarm is raised when the measured frequency of the input client clock exceeds the allowed frequency deviation outlined by the corresponding standard (does not improve beyond the “Frequency Out Of Range” limits) or when the provisioned line rate at the far-end circuit pack does not match the line port rate at the near-end circuit pack.

More specifically, for the following 6500 transport cards, NTK525CA, NTK525CF, NTK529BB/NTK529BX, the “Frequency OOR” alarm is raised when the measured frequency of the input client clock is greater than +/-27.5ppm (OC192/STM64) or +/-107.5ppm (ETH10G).

The Frequency OOR alarm does not force conditioning and alarm does not apply to NTK525CA when ETH10G is provisioned with GFP mapping.

Any terrestrial variant of 100G OCLD circuit packs (NTK539TAE5-NTK539TDE5 and NTK539TJE5) can interwork with any other terrestrial variant of 100G OCLD circuit packs (NTK539TAE5-NTK539TDE5 and NTK539TJE5) but the performance will be reduced to the 100G OCLD circuit pack with the lesser performance specifications. The submarine variant of 100G OCLD circuit packs (NTK539TEE5) does not interwork with any terrestrial variant of 100G OCLD circuit packs (NTK539TAE5-NTK539TDE5 and NTK539TJE5). The OTM4 facility “Loss of Clock” alarm is raised against both 100G OCLD circuit packs.

The 2x10G OTR (NTK530PG) and 4x10G OTR (NTK530QA) are not compatible for "10GE->GFP-F>SONET/SDH(AMP)-->OTU2" mapping. If this mapping type is used, these cards will not interop and the “Loss of Clock” alarm will be raised. The 4x10G OTR (NTK530QM) can be used instead of NTK530QA for this mapping.

Procedure 4-181 (continued)

Loss of Clock

Upon the differential encoding mismatch detection, “Loss of Clock” alarm is expected if SOFT differential encoding is involved at one of the two ends. This applies to all WL3/WL3e OCLDs, WL3e OTR, WL3n MOTR, and WL3n OTR circuit packs.

If there is optical signal power or sufficient amplified spontaneous emission (ASE) at the Rx, the “Loss of Signal” alarm will be deactivated and “Loss of Clock” alarm will be unmasked if present due to any of the following:

- Mismatched line CP type
- Mismatched Equipment Profile (WL3e)
- Mismatched Encoding (WL3e/WL3n)
- Mismatched Link Optimization Mode (WL3/WL3e/WL3n or WLAi)
- Mismatched Transmission Mode (WLAi)
- Misprovisioned/incorrect Path dispersion (WLAi)

Impact

Critical, service-affecting (C, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- have a fiber cleaning kit
- obtain a replacement VOA, if required
- obtain a replacement circuit pack, if required

Step	Action
------	--------

- | | |
|---|---|
| 1 | Ensure the subtending equipment is working properly and providing an error-free signal that meets the allowed frequency deviation specifications. |
|---|---|

Procedure 4-181 (continued)

Loss of Clock

Step	Action						
2	Ensure that all the optical fibers between the subtending equipment Tx port and at the alarmed circuit pack Rx port are correctly connected.						
3	Ensure that the line rate or the client type is the same for both ends of the link.						
4	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
5	Use an optical power meter to measure the received power level at the Rx port of the alarmed circuit pack. Verify if the Rx power level is within the operational threshold. Record the value. For information about circuit pack technical specifications, refer to the "Technical specifications" section in 6500 <i>Planning</i> , NTRN10GK.						
6	<table> <tr> <th>If there is</th><th>Then</th></tr> <tr> <td>a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment</td><td>re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7.</td></tr> <tr> <td>no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment</td><td>go to step 8</td></tr> </table>	If there is	Then	a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7 .	no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	go to step 8
If there is	Then						
a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7 .						
no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	go to step 8						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Use a cleaning kit to clean all the connectors between the subtending equipment Tx port and the alarmed Rx port. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.						
9	Record the operating power level after you clean each connector and compare it to the value recorded in step 5 . This will allow you to see if there is any improvement to the Rx power.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
11	Clear any upstream (either at the local to the network element reporting the alarm or other upstream remote network elements) alarms that could be causing this alarm, such as Circuit Pack Failed, Circuit Pack Missing, Channel Controller: Unexpected Loss Detected, Gauge Threshold Crossing Alert Summary, or Loss Of Signal.						

Procedure 4-181 (continued)
Loss of Clock

Step	Action						
12	Ensure there is no problem with the optical fiber plant between the alarmed circuit pack and the subtending equipment. This includes any fixed pad optical attenuators that may be faulty.						
13	Warm restart the alarmed circuit pack. Refer to the “Restarting a circuit pack or shelf processor” procedure in this document.						
14	Reseat the alarmed circuit pack. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 16</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 16
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 16						
16	Replace the alarmed circuit pack. Refer to the “Replacing an optical interface circuit pack” or “Replacing the 40G OCl, 40G+ CFP OCl, or 40/43G OCl circuit pack” procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
17	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-182

Loss Of Data Synch

Alarm IDs: 868, 1256, 1304, 1587, 2131

Probable cause

This alarm is raised against:

- an Ethernet facility of a 100G OCI, 4x10G OTR, 200G (2x100G/5x40G) MUX, or RPR circuit pack
- a FLEX facility of 8xOTN Flex MOTR, OTN Flex MOTR 2xCFP2, 4x10G OTR, or (1+8)xOTN Flex MOTR circuit pack
- an ETPP facility of a 16xFLEX OTN I/F or eMOTR circuit pack
- a CBRTTP facility of a WLAi FOTR circuit pack
- FC800/FC1200 clients of a 2x10G OTR (NTK530PME5 variant), 4x10G OTR, or FC800/FC1200 clients of 40G MUX OCI (NTK525CF) circuit pack
- ETH40G clients of a 40G+ CFP OCI (NTK529SJ) circuit pack
- a Gigabit Ethernet (GE) port on PKT I/F GE 48xSFP circuit pack (NTK642AA)

when one of the following conditions occurs:

- The circuit pack cannot establish bit synchronization or transmission word synchronization.
- The incorrect type of pluggable optical transceiver module (SX, LX, BX, or ZX) is installed.
- The client service on the subtending equipment does not match the client service provisioned on the corresponding port of the circuit pack. For example, the XFP+ of the 100G WL3n MOTR circuit pack is incorrectly connected to an OC192/STM64 optical interface when it is provisioned for an ETH10G client service.
- There is an auto-negotiation provisioning mismatch between the connected subtending equipment (a test set for example) and the corresponding port on the circuit pack.
- For ETH100G and ETH40G clients that use parallel optics, the Virtual Lane (VL) Skew exceeds the IEEE Std 802.3ba VL Skew range or if duplicate alignment markers are detected.
- For the ETPP facility of eMOTR, ETPP facility figures secondary state is Layer-2 port disabled. This phenomenon can happen due to VLLI port conditioning.

Procedure 4-182 (continued)

Loss Of Data Synch

When VLLI UP MEP port conditioned due to FEND UP MEP failure, NEND ETTP facility will have Loss Of Data Synch (for GE port and GE LAG)/Loss Of Frame (for 10G port or 10G LAG), indicating FEND port having an issue.

If UP MEP VLLI belonging port created on LAG client port then only Lead member (known to be least numbered LAG member port) shows the alarm and all member ports will declare L2 port disabled.

Similarly in case of Inverse VLLI operations, destination port instance (GE port or GE LAG) shuts off laser with Loss of Data Synch during normal operation of source port. This alarm clears when source port undergoes faults and destination port enabled by clearing LODS alarm and the same port will have “Service Defect Indication” alarm indicating Inverse VLLI conditioning is activated.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must:

- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- if required, obtain a replacement SFP/XFP
- use an account with at least a level 3 UPC

Step	Action
1	Determine the type of client service on the subtending equipment connected to the port of the circuit pack reporting the alarm.
2	Determine the facility (Ethernet, FLEX, or Fiber Channel) provisioned on the port of the circuit pack reporting the alarm.

Procedure 4-182 (continued)

Loss Of Data Synch

Step	Action						
3	<table> <tr> <td>If the subtending equipment noted in step 1 is not provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2</td><td>Then go to step 4</td></tr> <tr> <td>provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2</td><td>step 6</td></tr> </table>	If the subtending equipment noted in step 1 is not provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2	Then go to step 4	provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2	step 6		
If the subtending equipment noted in step 1 is not provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2	Then go to step 4						
provisioned with an Ethernet, FLEX, or Fiber Channel service as required by the facility noted in step 2	step 6						
4	The 40G MUX OCI (NTK525CF), 4x10G OTR, or 2x10G OTR (NTK530PME5 variants) circuit pack supports Fiber Channel client services. If possible, provision the subtending equipment for Fiber Channel client services, otherwise contact your next level of support or your Ciena support group. The 4x10G OTR, 2x10G OTR, 40G MUX, 8xOTN Flex MOTR, (1+8)xOTN Flex MOTR, OTN Flex MOTR 2xCFP2 circuit packs support FLEX client services. If possible, provision the subtending equipment for FLEX client services, otherwise contact your next level of support or your Ciena support group.						
5	<table> <tr> <td>If the original alarm</td><td>Then</td></tr> <tr> <td>has cleared</td><td>the procedure is complete</td></tr> <tr> <td>has not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm	Then	has cleared	the procedure is complete	has not cleared	go to step 6
If the original alarm	Then						
has cleared	the procedure is complete						
has not cleared	go to step 6						
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
7	If applicable, ensure that the SFP/XFP optical transceiver module is the correct type. Use an SX SFP; or SR/SW XFP for multi-mode fiber-optic cables. Use a LX, ZX, or BX SFP; or ER/EW, LR/LW or DWDM XFP for single-mode fiber-optic cables. If required, replace the pluggable optical transceiver module. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
8	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-183

Loss of Frame and Multiframe

Alarm IDs: 2025

Probable cause

The alarm is raised on an OTUTTP and ETTP client facility, and PTP and OTUTTP of the line facility of the WLAi variants when OTUCn Loss of Frame or OTUCn Loss of Multiframe is detected on any of the lanes.

Impact

Critical, service-affecting (C, SA) alarm if not protected
Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
3	Log into the remote network element at the transmit end.						
4	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure.						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						

Procedure 4-183 (continued)

Loss of Frame and Multiframe

Step	Action						
6	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches. Refer to the "Retrieving equipment and facility details" or "Editing facility parameters" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Verify that the FEC setting on the upstream circuit pack matches the FEC setting on the local circuit pack. Correct any mismatches. Refer to the "Retrieving equipment and facility details" or "Editing facility parameters" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed channel using the appropriate procedures.						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
13	Determine the optical power level into the Rx interface using one of the following methods: - Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power (dBm) value for the corresponding PTP facility. Refer to the "Retrieving OTM info" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - Use the optical power meter to measure the optical power level into the Rx interface, and verify it is within technical specifications. Refer to the "Technical section in the 6500 <i>Planning</i>, NTRN10GK specification.						

Procedure 4-183 (continued)
Loss of Frame and Multiframe

Step	Action
14	If the Rx Actual Power value or measured optical power is below the receiver sensitivity specified for the circuit pack Then go to step 15 above the receiver sensitivity specified for the circuit pack step 17 within the receiver sensitivity range specified for the circuit pack step 23
15	Remove the Tx fiber from the far-end circuit pack, and use the optical power meter to measure the transmit power at the far-end.
16	If the transmit power at the far-end is Then above the launch power (minimum) the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 27. below the launch power (minimum) Replace the required circuit pack at the transmit end. Refer to the appropriate procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 27. within the launch power range specified for the circuit pack go to step 27.
17	Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
18	If the original alarm has Then cleared the procedure is complete not cleared go to step 19
19	Review the optical plan to verify that the correct attenuator (if required) is installed between the associated DWDM source and the receiver on the alarmed circuit pack.

Procedure 4-183 (continued)

Loss of Frame and Multiframe

Step	Action						
	If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any alarms on the photonic line associated with the Loss Of Signal alarm using the appropriate procedures.						
20	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
21	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. Once the optical path is verified to be correct, perform a cold restart on the circuit pack raising the alarm. See the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						
22	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 23</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 23
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 23						
23	 CAUTION Risk of traffic loss Connecting a facility to a test set causes traffic loss on the facility.						
24	Use a test set to determine if a valid signal is on the facility.						
25	<table> <tr> <th>If the test set shows</th><th>Then</th></tr> <tr> <td>a Loss Of Frame or incorrect framing condition</td><td>the problem is in the source. Go to step 26.</td></tr> <tr> <td>otherwise</td><td>go to step 27</td></tr> </table>	If the test set shows	Then	a Loss Of Frame or incorrect framing condition	the problem is in the source. Go to step 26 .	otherwise	go to step 27
If the test set shows	Then						
a Loss Of Frame or incorrect framing condition	the problem is in the source. Go to step 26 .						
otherwise	go to step 27						

Procedure 4-183 (continued)
Loss of Frame and Multiframe

Step	Action	
26	If the source is connected to the alarmed circuit pack by a client interface	Then Perform troubleshooting on the source system according to your company procedures.
	line interface	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure. If no alarms exist on the transmit end, then perform a warm restart on the transmit circuit pack. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document. If the alarm still does not clear, then the transmit circuit pack is faulty. Replace the transmit circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
27	If the original alarm has cleared	Then the procedure is complete
	not cleared	perform a warm restart on the circuit pack reporting the alarm. See Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
28	If the original alarm has cleared	Then the procedure is complete
	not cleared	the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
29	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-184

Loss of Frame Delineation

Alarm IDs: 353, 1976

Probable cause

This alarm is raised against the WAN facility when the GFP layer cannot detect valid GFP frames.

Impact

Critical, service-affecting (C, SA) alarm if not protected
Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	<table border="0"> <tr> <td>If the alarm is raised against</td><td>Then go to</td></tr> <tr> <td>one of the following Broadband circuit packs: 2x10G OTR, 4x10G OTR, 10x10G MUX OCI, or 40G MUX OCI</td><td>step 7</td></tr> <tr> <td>otherwise</td><td>step 3</td></tr> </table>	If the alarm is raised against	Then go to	one of the following Broadband circuit packs: 2x10G OTR, 4x10G OTR, 10x10G MUX OCI, or 40G MUX OCI	step 7	otherwise	step 3
If the alarm is raised against	Then go to						
one of the following Broadband circuit packs: 2x10G OTR, 4x10G OTR, 10x10G MUX OCI, or 40G MUX OCI	step 7						
otherwise	step 3						
3	Use the appropriate alarm clearing procedure to clear any ODU path alarms on the interface circuit pack that connects to the 8xOTN Flex MOTR, (1+8)xOTN FLEX MOTR, OTN Flex MOTR 2xCFP2, eMOTR, 16xFLEX OTN I/F, or SuperMux circuit pack. These include the ODU Signal Degrade and ODU Signal Fail alarms.						
4	Trace the cross-connect information to verify there are no errors in the ODU cross-connects. Refer to the “Retrieving transponder connections” or “Retrieving OTN connections” procedure in <i>Configuration - Connections Management</i> , 323-1851-320.						
5	Use the appropriate alarm clearing procedure to clear any G.709 layer alarms raised against the line port of the circuit pack reporting the Loss of Frame Delineation alarm.						

Procedure 4-184 (continued)

Loss of Frame Delineation

Step	Action						
6	Ensure that the subtending client equipment is transmitting a valid GFP-F signal, as required. Go to step 11 .						
7	If the far-end 2x10G OTR, 4x10G OTR or 40G MUX OCI circuit pack has an OTM2 facility, clear any OTN faults (such as, LOS, LOC, LOF, LOM, ODU2-AIS, ODU2-LCK, ODU2-OCI, OPU2-AIS, or OPU2 Payload Type Mismatch) for all upstream OTM2 facilities. That is, clear all OTN alarms upstream from the far-end OTM2 port.						
8	In the Site Manager Equipment & Facility Provisioning application, verify the ETH10G facility Packet Mapping value matches at both the near-end and far-ends. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
9	<table> <tr> <th>If the Packet Mapping values</th><th>Then</th></tr> <tr> <td>are mismatched</td><td> determine if it is the near-end alarmed ETH10G facility or the far-end facility that has the incorrect Packet Mapping value. The incorrectly provisioned facility must be deleted and reprovisioned with the correct Packet Mapping value, refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Continue to step 10. </td></tr> <tr> <td>match</td><td>go to step 10</td></tr> </table>	If the Packet Mapping values	Then	are mismatched	determine if it is the near-end alarmed ETH10G facility or the far-end facility that has the incorrect Packet Mapping value. The incorrectly provisioned facility must be deleted and reprovisioned with the correct Packet Mapping value, refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Continue to step 10.	match	go to step 10
If the Packet Mapping values	Then						
are mismatched	determine if it is the near-end alarmed ETH10G facility or the far-end facility that has the incorrect Packet Mapping value. The incorrectly provisioned facility must be deleted and reprovisioned with the correct Packet Mapping value, refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Continue to step 10.						
match	go to step 10						
10	Ensure that the subtending client equipment is transmitting a valid GFP-F.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-185

Loss Of Frame, Loss Of Multiframe alarms

Use this procedure to clear the Loss of Frame, Loss of Multiframe, alarms.

ATTENTION

The G.709 protocol uses the OTU3 Multiframe Alignment Signal (MFAS) byte as alignment markers to support lane deskew when an OTU3 signal is transmitted across multiple physical lanes. This occurs on the NTK529SJ for any OTU3 facility that uses an NTTA12BA CFP as the physical equipment. Consequently, MFAS errors can occur in the bits used as lane markers, the deskew process cannot be performed, and it is impossible to reconstruct the OTU3 frame. As a result, a “Loss of Frame” (LOF) alarm is raised instead of a “Loss of Multiframe” (LOM) alarm.

Loss Of Frame (OCn/STMn)

Alarm ID: 4, 33, 247, 282, 984

Probable cause

This alarm is raised when the interface receives repeatedly errored SONET sections or SDH RS overhead bytes (A1 or A2) for three consecutive ms or more.

For 40G OCI, 40/43G OCI, 40G+ CFP OCI circuit packs, the optional 3-ms integration timer dealing with intermittent SEF defects (when monitoring for LOF) is implemented as outlined in the GR-253 SONET standard.

This alarm is caused by one of the following conditions:

- excessive attenuation
- dirty optical fibers
- dirty connectors
- improper connector seating
- transmit laser degrade

The network element cannot clear a Loss Of Signal alarm until a framed OC-n/STM-n signal is detected. The first time an optical fiber/cable is disconnected, the Loss of Frame alarm clears and a Loss Of Signal alarm is raised that will not change back to Loss of Frame when the optical fiber/cable is re-attached.

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms**Impact**

Critical, service-affecting (C, SA) alarm, if active in a protected configuration or unprotected with cross-connects

Minor, non-service-affecting (m, NSA) alarm, if inactive in a protected configuration or without cross-connects

Loss Of Frame (ETH10G, ETH40G, ETH100G, FC1200)**Alarm ID: 677, 1297****Probable cause**

This alarm is raised when the client Rx interface 64/66B PCS is unable to frame on the client signal on the:

- ETH10G of 10x10G MUX OCI, 2x10G OTR, 4x10G OTR, 4x10G MUX, or 100G (2xQSFP+/2xSFP+) MUX circuit pack
- ETH40G of 40G OCI, or 100G (2xQSFP+/2xSFP+) MUX circuit pack
- ETH100G of 100G OCI and 100G WL3e OTR circuit pack

For the 100G WL3n MOTR circuit pack, this alarm is raised when the received client traffic rate does not match the provisioned client rate (provisioning mismatch).

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Loss Of Frame (OTM1, OTM2, OTM3, OTM4, OTMC2, OTUTTP, ETTP, STTP, PTP, FLEX)**Alarm IDs: 666, 999, 1168, 1440, 1453, 1461, 1861, 2014****Probable cause**

This alarm can be raised under the following conditions:

- when the optical link exceeds the guaranteed link budget (the OSNR or the distortion penalties are in excess of the allocated margin but the Rx power is above the LOS threshold).
- when there is a fiber break on a long haul system (LH1600, CPL, 6500 Photonic services) connected to 6500 network element. The alarm is raised because the noise that is amplified through the system is so great that the power at the receiver can be higher than the Loss Of Signal threshold. Since there is no signal for the card to lock to and the power is high enough, the Loss of Frame alarm can be raised.
- when the provisioned Rx FEC setting does not match that of the incoming signal.

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

Note: Note that after removing a LOS or LOF fault condition on the 40G OCLD or Wavelength-Selective 40G OCLD Rx OTM3, 100G WL3/WL3e OCLD or 100G OCLD Rx OTM4 facility, traffic restoration can take up to nine seconds. SNC Mesh Restoration will be triggered every 15 minutes if the condition of LOC/LOF stays.

- For WLAi FOTR (NTK538Fx), this alarm can be raised on an 100GE client port under the following conditions as well:
 - Loss of Alignment Marker
 - Loss of Clock
 - Skew exceeds hardware capability

The OTM4 facility “Loss of Frame” alarm is raised against both FLEX4 WL3e OCLD circuit packs).

For an OTUTTP facility, this alarm is raised when the Rx interface repeatedly receives errored optical transport unit (OTU) overhead bytes (OA1 or OA2) for three consecutive ms or more.

For the 10x10G MUX OCI, 100G OCI, 40G OCLD and Wavelength-Selective 40G OCLD circuit packs, the optional 3-ms integration timer dealing with intermittent SEF defects (when monitoring for LOF) is implemented as outlined in the GR-253 SONET standard.

Note: The 100G WL3/WL3e OCLD (NTK539Ux), Flex2 WL3/WL3e OCLD (NTK539Bx), Flex3 WL3e OCLD (NTK539Qx), 100G WL3n MOTR (NTK538Bx), 100G WL3n OTR (NTK538Ex), and 100G WL3e OTR (NTK538Ux) circuit packs do not interwork with any variant of 100G OCLD (NTK539TxEx) circuit packs. The “Loss of Frame” alarm is raised against both end of the fiber.

For the 8xSFP OTN FLEX MOTR circuit pack, if Low Latency Mapping Procedure (LLMP) is run and is cross-connected to an OTM1 line, the alarm is raised if the far-end OTM1 line is not cross-connected to an LLMP mapped client port. This is the case with LLMP clients because the cross-connected OTM1 line runs at a higher clock rate.

For the eMOTR circuit pack, the alarm is raised when 64B/66B PCS SYNC is lost on the incoming 10GE signal.

For the ETTP facility of eMOTR circuit pack, ETTP facility figures secondary state is Layer-2 port disabled. This phenomenon can happen due to VLLI port conditioning.

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

When VLLI UP MEP port conditioned due to FEND UP MEP failure, NEND ETTP facility will have Loss Of Data Synch (for GE port and GE LAG)/Loss Of Frame (for 10G port or 10G LAG), indicating FEND port having an issue.

If UP MEP VLLI belonging port created on LAG client port, then only Lead member (known to be least numbered LAG member port) shows the alarm and all member ports will declare L2 port disabled.

Similarly, in case of Inverse VLLI operations, destination port instance (10G port or 10G LAG) shuts off laser with Loss of Frame during normal operation of source port. This alarm clears when source port undergoes faults and destination port enabled by clearing LOF alarm and the same port will have "Service Defect Indication" alarm indicating Inverse VLLI conditioning is activated.

For an ETTP facility, this alarm is raised when 64B/66B PCS is unable to frame on the client signal.

For an STTP facility, this alarm is raised when a SONET LOF is detected in the Rx direction.

Any terrestrial variant of WL3 OCLD circuit packs (NTK539UA/UB/UC/UD/UH/UJ) can interwork with any other terrestrial variant of WL3 OCLD circuit packs (NTK539UA/UB/UC/UD/UH/UJ), but the performance will be limited to the WL3 OCLD circuit pack with the lesser performance specification. The submarine variant of WL3 OCLD circuit packs (NTK539UE/NTK539UN) does not interwork with any terrestrial variant of WL3 OCLD circuit packs (NTK539UA/UB/UC/UD/UH/UJ). The OTM4 facility "Loss of Frame" alarm is raised against both circuit packs.

Any terrestrial variant of FLEX2 WL3 OCLD circuit packs (NTK539BH, and NTK539BB) can interwork with any other terrestrial variant of FLEX2 WL3 OCLD circuit packs (NTK539BH and NTK539BB), but the performance will be limited to the FLEX2 WL3 OCLD circuit pack with the lesser performance specification. The submarine variant of FLEX2 WL3/WL3e OCLD circuit packs (NTK539BE/NTK539BN) does not interwork with any terrestrial variant of FLEX2 WL3 OCLD circuit packs (NTK539BH, and NTK539BB). The OTM4 facility "Loss of Frame" alarm is raised against both circuit packs.

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

Any terrestrial variant of FLEX3 WL3e OCLD circuit packs (NTK539QJ, NTK539QK, NTK539QL, and NTK539QM) can interwork with any other terrestrial variant of FLEX3 WL3e OCLD circuit packs (NTK539QJ, NTK539QK, NTK539QL, and NTK539QM), but the performance will be limited to the FLEX3 WL3e OCLD circuit pack with the lesser performance specification. The submarine variant of FLEX3 WL3e OCLD circuit packs (NTK539QN) does not interwork with any terrestrial variant of FLEX3 WL3e OCLD circuit packs (NTK539QJ, NTK539QK, NTK539QL, and NTK539QM). The OTM4 facility “Loss of Frame” alarm is raised against both FLEX3 WL3e OCLD circuit packs.

Any terrestrial variant of 100G WL3e OTR circuit packs (NTK538UJ, NTK538UK, NTK538UL, and NTK538UM) or WLAi variants can interwork with any other terrestrial variant of 100G WL3e OTR (NTK538UJ, NTK538UK, NTK538UL, and NTK538UM) or WLAi variants circuit packs, but the performance will be limited to the 100G WL3e OTR or WLAi variants circuit pack with the lesser performance specification. The submarine variant of 100G WL3e OTR circuit packs (NTK538UN) does not interwork with any terrestrial variant of 100G WL3e OTR circuit packs (NTK538UJ, NTK538UK, NTK538UL, and NTK538UM). The OTM4 facility “Loss of Frame” alarm is raised against both 100G WL3e OTR circuit packs.

The submarine variant of FLEX4 WL3e OCLD circuit packs (NTK539FN) does not interwork with any terrestrial variant of FLEX4 WL3e OCLD circuit packs (NTK539FJ).

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm, if inactive in a protected configuration or without cross-connects

Loss Of Multiframe (OTM1, OTM2, OTM3, OTM4, OTMC2, OTUTTP, PTP, FLEXI)

Alarm IDs: 667, 1000, 1169, 1441, 2015

Probable cause

This alarm is raised when the multiframe alignment signal (MFAS) byte is out of sequence with the expected multiframe number for more than 5 consecutive OTUk frames.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms**Prerequisites**

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
3	Log into the remote network element at the transmit end.						
4	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure.						
5	<table border="0"> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table border="0"> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Verify that the FEC setting on the upstream circuit pack matches the FEC setting on the local circuit pack. Correct any mismatches. Refer to the “Retrieving equipment and facility details” or “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

Step	Action								
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 10								
10	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed channel using the appropriate procedures.								
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 12								
12	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
13	Determine the optical power level into the Rx interface using one of the following methods: - Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power (dBm) value for the facility reporting the alarm. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - Use the optical power meter to measure the optical power level into the Rx interface, and verify it is within technical specifications. Refer to the “Technical specifications” section in the 6500 <i>Planning</i>, NTRN10GK.								
14	<table> <tr> <th>If the Rx Actual Power value or measured optical power is</th><th>Then go to</th></tr> <tr> <td>below the receiver sensitivity specified for the circuit pack</td><td>step 15</td></tr> <tr> <td>above the receiver sensitivity specified for the circuit pack</td><td>step 17</td></tr> <tr> <td>within the receiver sensitivity range specified for the circuit pack</td><td>step 23</td></tr> </table>	If the Rx Actual Power value or measured optical power is	Then go to	below the receiver sensitivity specified for the circuit pack	step 15	above the receiver sensitivity specified for the circuit pack	step 17	within the receiver sensitivity range specified for the circuit pack	step 23
If the Rx Actual Power value or measured optical power is	Then go to								
below the receiver sensitivity specified for the circuit pack	step 15								
above the receiver sensitivity specified for the circuit pack	step 17								
within the receiver sensitivity range specified for the circuit pack	step 23								
15	Remove the Tx fiber from the far-end circuit pack, and use the optical power meter to measure the transmit power at the far-end.								

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

Step	Action	
16	If the transmit power at the far-end is	Then
	above the launch power (minimum)	the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 26 .
	below the launch power (minimum)	if the circuit pack at the transmit end supports SFPs/ DPOs, replace the SFP/DPO that corresponds to the facility raising the alarm. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. if the circuit pack at the transmit end does not support SFPs, replace the required circuit pack at the transmit end. Refer to the replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 26 .
	within the launch power range specified for the circuit pack	go to step 26 .
17	Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.	
18	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 19
19	Review the optical plan to verify that the correct attenuator (if required) is installed between the associated DWDM source and the receiver on the alarmed circuit pack. If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any alarms on the photonic line associated with the Loss Of Signal alarm using the appropriate procedures.	
20	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 21

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms**Step Action**

21

**CAUTION****Risk of traffic loss**

A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic.

Once the optical path is verified to be correct, perform a cold restart on the circuit pack raising the alarm. See [Procedure 2-11, "Restarting a circuit pack or shelf processor"](#) procedure in this document.

22

If the original alarm has**Then**

cleared

the procedure is complete

not cleared

go to [step 23](#)

23

**CAUTION****Risk of traffic loss**

Connecting a facility to a test set causes traffic loss on the facility.

Use a test set to determine if a valid signal is on the facility.

24

If the test set shows**Then**a Loss Of Frame or incorrect
framing condition

the problem is in the source.

go to [step 25](#).

otherwise

go to [step 26](#)

Procedure 4-185 (continued)

Loss Of Frame, Loss Of Multiframe alarms

Step	Action	
25	If the source is connected to the alarmed circuit pack by a client interface	Then Perform troubleshooting on the source system according to your company procedures.
	line interface	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure. If no alarms exist on the transmit end, then perform a warm restart on the transmit circuit pack. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document. If the alarm still does not clear, then the transmit circuit pack is faulty. Replace the transmit circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
26	If the original alarm has cleared	Then the procedure is complete
	not cleared	perform a warm restart on the circuit pack reporting the alarm. Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
27	If the original alarm has cleared	Then the procedure is complete
	not cleared	the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
28	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-186

Loss of Lane Alignment

Alarm IDs: 1856, 2174

Probable cause

For the 100G WL3n MOTR, this event is raised against the aggregated 40GE facility of a 100G WL3n MOTR when a “Loss of Signal”, “Loss of Clock”, or “Loss of Frame” alarms exists on at least one of the four of its associated FLEX member facilities.

This alarm is raised against the line OTUTTP facility of the prime 2xWLAi OTR/WLAi MOTR/WLAi FOTR circuit pack when the system detects that the lane skew between prime and member circuit packs exceeds the specified skew range from 1.5us.

Impact

Critical, service-affecting (C, SA) alarm for OTUTTP facilities

Minor, non-service-affecting (m, NSA) alarm for OTUTTP facilities

Warning for ETPP facilities

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in “Observing product and personnel safety guidelines” section in *Installation*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain a replacement circuit pack
- use an account with at least a level 3 UPC

Step	Action						
1	<table><tr><td>If the original event/alarm is raised against</td><td>Then go to</td></tr><tr><td>100G WL3n MOTR circuit pack</td><td>step 2</td></tr><tr><td>2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack</td><td>step 9</td></tr></table>	If the original event/alarm is raised against	Then go to	100G WL3n MOTR circuit pack	step 2	2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack	step 9
If the original event/alarm is raised against	Then go to						
100G WL3n MOTR circuit pack	step 2						
2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack	step 9						
2	Identify the facility that raised the event. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
3	Identify all four member FLEX facilities associated with the aggregated 40GE facility identified in step 2 .						

Procedure 4-186 (continued)

Loss of Lane Alignment

Step	Action						
4	Follow the procedure to clear the “Loss of Signal” for all four members of the FLEX facilities identified in step 3 .						
5	<table> <tr> <th>If the original event is</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original event is	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original event is	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Follow the procedure to clear the “Loss of Lock” on page 4-431 for all four members of the FLEX facilities identified in step 3 .						
7	<table> <tr> <th>If the original event is</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original event is	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original event is	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Follow the procedure to clear the “Loss Of Frame, Loss Of Multiframe alarms” on page 4-418 for all four members of the FLEX facilities identified in step 3 . Go to step 15 .						
9	Check if the length difference between the patch cord fibers for prime and member 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack is less than 40m. If not, replace them to meet this requirement.						
10	If the original alarm does not clear, check if both prime and member 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit packs go to the same CCMD. Then make them go to the same CCMD/CMD to comply with this requirement.						
11	If the original alarm does not clear, check if both prime and member 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit packs wavelength/frequency are adjacent (follow the wavelength assignment rules and Engineering Guidelines for all 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit packs documented in <i>WaveLogic OCLD, OCI, MUX, and Submarine Circuit Packs</i> , 323-1851-102.4). If not re-tune one or both wavelength/frequencies to be adjacent to make them comply with this requirement. Refer to the “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
12	If the original alarm does not clear, ensure the provisioned (TRANSMODE, LINKOPTMODE) on both prime and member circuit packs are the same. If not, make them meet with this requirement. Refer to the “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
13	If the alarm does not clear, replace the prime 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						

Procedure 4-186 (continued)

Loss of Lane Alignment

Step	Action
14	If the alarm does not clear, replace the member 2xWLAi OTR, WLAi MOTR/WLAi FOTR circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
15	If the event/alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-187

Loss of Lock

Alarm IDs: 1254

Probable cause

This alarm is raised when the FLEX Mapper is unable to lock on to a signal at the provisioned protocol rate. One of the following conditions can cause the alarm:

- the FLEX facility client protocol does not match the actual signal being received
- an optical fiber connection is degraded
- an optical fiber is bent or coiled too tightly
- the connector is dirty at the receiving 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR or at the transmitting circuit pack
- an optical patchcord is damaged
- the receiving, 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR is defective
- the transmitting subtending equipment is defective
- the optical fiber is the wrong type
- the wrong subtending equipment is connected to the client-side, or the client-side is provisioned wrong
- if there is a miniature VOA on the fiber to attenuate the signal, it may not be operating correctly

Impact

Critical, service-affecting (C, SA) alarm if not protected
Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Procedure 4-187 (continued)

Loss of Lock

Client side

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	<table> <tr> <td>If the alarm is raised against a</td><td>Then go to</td></tr> <tr> <td>client port</td><td>step 3</td></tr> <tr> <td>line port</td><td>step 18</td></tr> </table>	If the alarm is raised against a	Then go to	client port	step 3	line port	step 18
If the alarm is raised against a	Then go to						
client port	step 3						
line port	step 18						
3	Make sure the provisioned connection type and bit rate are correct.						
4	Make sure you are using the correct type of optical fiber.						
5	Make sure the optical fiber is connected correctly on the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR client-side Rx port and the subtending equipment Tx port.						
6	Make sure the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack is the correct type.						
7	Measure the received power level at the client-side Rx port of the affected circuit pack. Check to see whether the power level is within the operational threshold. Write down the value. Refer to the "Technical specifications" section in <i>Planning</i> , NTRN10GK, for minimum and maximum Rx/Tx power levels.						
	Note: You can record the power level from the Equipment—Facility window in the Site Manager.						
8	<table> <tr> <td>If the power level is</td><td>Then go to</td></tr> <tr> <td>not within the operational threshold</td><td>step 9</td></tr> <tr> <td>within the operational threshold</td><td>step 16</td></tr> </table>	If the power level is	Then go to	not within the operational threshold	step 9	within the operational threshold	step 16
If the power level is	Then go to						
not within the operational threshold	step 9						
within the operational threshold	step 16						
9	Make sure the transmitting subtending equipment is functioning correctly and transmitting a valid signal (correct bit rate and protocol).						
10	Use an optical power meter to measure the transmit power on the subtending equipment. Make sure it is working correctly and that the power of the transmitted signal is at the correct power level according to the manufacturer's specifications.						
	<table> <tr> <td>If the subtending equipment Tx laser is</td><td>Then</td></tr> <tr> <td>out of specification</td><td>repair or replace the subtending equipment. Go to step 11.</td></tr> <tr> <td>within specification</td><td>go to step 12</td></tr> </table>	If the subtending equipment Tx laser is	Then	out of specification	repair or replace the subtending equipment. Go to step 11 .	within specification	go to step 12
If the subtending equipment Tx laser is	Then						
out of specification	repair or replace the subtending equipment. Go to step 11 .						
within specification	go to step 12						

Procedure 4-187 (continued)

Loss of Lock

Step	Action	
11	If the original alarm has	Then
	cleared	this procedure is complete
	not cleared	go to step 12
12	If	Then
	there is a VOA attached to a fiber between the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR and the subtending equipment	re-adjust, check for proper functionality, and, if necessary, replace the VOA. Go to step 13 .
	there is no VOA attached to a fiber between the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR and the subtending equipment	go to step 14
13	If the original alarm has	Then
	cleared	this procedure is complete
	not cleared	go to step 14
14	Use the proper cleaning kit to clean all the connectors between the subtending equipment Tx port and the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR Rx port, and vice-versa. For information on cleaning, refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.	
	- Clean each connector separately. - Record the operating power level after you clean each connector and compare it to the value you wrote down in step 7, this will allow you to see if there is any improvement to the Rx power. - Make sure there is no problem with the optical fiber plant between the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR and the subtending equipment	
15	Clean every connector between the subtending equipment Tx port, the SC-to-LC patch panel, and the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR client-side Rx port.	
16	Reseat the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.	
	If the original alarm has	Then
	cleared	this procedure is complete
	not cleared	go to step 17

Procedure 4-187 (continued)

Loss of Lock

Step	Action						
17	Replace the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the removed circuit pack is faulty. This procedure is complete.</td></tr> <tr> <td>not cleared</td><td>the circuit pack is not the problem. Reseat the original 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack and contact your next level of support.</td></tr> </table>	If the original alarm has	Then	cleared	the removed circuit pack is faulty. This procedure is complete.	not cleared	the circuit pack is not the problem. Reseat the original 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack and contact your next level of support.
If the original alarm has	Then						
cleared	the removed circuit pack is faulty. This procedure is complete.						
not cleared	the circuit pack is not the problem. Reseat the original 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack and contact your next level of support.						

Line side

- 18 Make sure that the FEC settings of the far-end line side device matches the FEC settings on the FLEX MOTR, 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the “Retrieving and editing TR control information” procedure in *Configuration - Provisioning and Operating*, 323-1851-310. If the alarm does not clear, go to [step 19](#).
- 19 Check the corresponding upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR or, if applicable, Optical Fiber Amplifiers (OFA) for failures or alarms. Use a network topology diagram to determine the upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs and OFAs. Troubleshoot these alarms/failures before proceeding.
- 20 At the downstream shelf, use the System Manager to check if you are receiving “Loss of Lock” alarms on all (or most) 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs in the same east/west plane.
- | If the alarm appears | Then go to |
|--|-------------------------|
| on the only 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR in the east/west plane | step 21 |
| on all (or most) 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs in the same east/west plane | step 24 |
| on one of multiple 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs in the same east/west plane | step 27 |

Procedure 4-187 (continued)

Loss of Lock

Step	Action	
21	If	Then
	there are amplifiers between the upstream and downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs	there is a problem at the amplifier site, the ring requires re-equalization, or both. Contact your next level of support to troubleshoot. If troubleshooting does not clear the alarm, go to step 27 .
	there are no amplifiers between the upstream and downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs	there is a problem with the line fiber between the sites, go to step 22
22	Investigate the line fiber between the upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs and the downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs for loss or degrade.	

**CAUTION****Risk of loss of traffic across multiple wavelengths in the band**

Disturbing the line fiber affects traffic on the entire band/direction. If the traffic is protected, follow the protection switching procedures in *Configuration - Protection Switching*, 323-1851-315, to switch traffic for the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack in the east/west plane before proceeding. If the traffic is not protected, you will drop traffic.

**CAUTION****Disable Automatic Laser Shutdown**

If Automatic Laser Shutdown (ALS) is enabled, disable ALS for all shelves that will be affected by this maintenance activity. Use a network topology diagram and channel listing to determine the upstream and downstream sites that will be affected when the ring is opened. If you do not know if ALS is enabled or not, assume it is enabled. Follow the "Editing facility parameters" procedure in *Configuration - Provisioning and Operating*, 323-1851-310, to disable ALS.

Procedure 4-187 (continued)

Loss of Lock

Step	Action	
	Using a power meter, a cleaning kit, and a network topology diagram, start at the upstream site and clean every connector on the path until you reach the downstream site. Measure the power with the power meter after each cleaning to check for improvements in the line power. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0. Also, check for damaged, kinked, tightly bent fiber, or anything else that might impede the optical signal. This includes optical filters such as OMXs or C/L Splitter/Couplers. Replace any damaged fibers or filters.	
23	If the original alarm has cleared	Then this procedure is complete. Restore the traffic to both paths on a protected channel.
	not cleared	go to step 27
24	If there are amplifiers between the upstream and downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs	Then there is a problem at the amplifier site, the ring requires re-equalization, or both. Contact your next level of support.
	there are no amplifiers between the upstream and downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs	there is a problem with the line fiber between the sites, go to step 25

Procedure 4-187 (continued)

Loss of Lock

Step	Action						
25	The line fiber between the upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs and the downstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs must be investigated for loss or degrade.  CAUTION Risk of loss of traffic across multiple wavelengths in the band Disturbing the line fiber affects traffic on the entire band/direction. If the traffic is protected, follow the protection switching procedures in <i>Configuration - Protection Switching</i>, 323-1851-315, to switch traffic for the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack in the east/west plane before proceeding. If the traffic is not protected, you will drop traffic.  CAUTION Disable Automatic Laser Shutdown If Automatic Laser Shutdown (ALS) is enabled, disable ALS for all shelves that will be affected by this maintenance activity. Use a network topology diagram and channel listing to determine the upstream and downstream sites that will be affected when the ring is opened. If you do not know if ALS is enabled or not, assume it is enabled. Follow the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310, to disable ALS. Using a power meter, a cleaning kit, and a network topology diagram, start at the upstream site and clean every connector on the path until you reach the downstream site. Measure the power with the power meter after each cleaning to check for improvements in the line power. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. Also, check for damaged, kinked, tightly bent fiber, or anything else that might impede the optical signal. This includes optical filters such as OMXs or C/L Splitter/Couplers. Replace any damaged fibers or filters.						
26	<table border="0"> <tr> <td style="vertical-align: top;">If the original alarm has</td><td style="vertical-align: top;">Then</td></tr> <tr> <td>cleared</td><td>this procedure is complete. Restore the traffic to both paths on a protected channel.</td></tr> <tr> <td>not cleared</td><td>contact your next level of support</td></tr> </table>	If the original alarm has	Then	cleared	this procedure is complete. Restore the traffic to both paths on a protected channel.	not cleared	contact your next level of support
If the original alarm has	Then						
cleared	this procedure is complete. Restore the traffic to both paths on a protected channel.						
not cleared	contact your next level of support						

Procedure 4-187 (continued)

Loss of Lock

Step	Action	
27	If the path is protected	Then switch traffic off of the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the "Protection Switching" procedures in <i>Configuration - Protection Switching</i> , 323-1851-315.
	unprotected	go to step 28 .
28	 CAUTION Risk of traffic loss Performing this step may affect traffic.	
	If Automatic Laser Shutdown (ALS) is enabled, disable ALS for both the near-end and far-end shelves which contain the alarming 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack and its corresponding upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Use a network topology diagram and channel listing to determine the near-end and far-end sites. If you do not know if ALS is enabled or not, assume it is enabled. Follow the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310, to disable ALS.	
29	Using a power meter, a cleaning kit, and a network topology diagram, start at the upstream site and clean every connector on the path until you reach the downstream site. Measure the power with the power meter after each cleaning to check for improvements in the line power. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.	
30	Make sure the optical fiber is connected correctly on the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack line-side Rx port at the near-end site and on the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack line-side Tx port at the far-end site. Note: Use a network topology diagram to determine the near-end and far-end sites.	
	If the alarm	Then
	clears	you have completed the procedure. Restore the traffic to both paths on a protected channel.
	does not clear	go to step 31

Procedure 4-187 (continued)

Loss of Lock

Step	Action						
31	Check whether the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack line-side Rx power level is within the operational threshold and write down the value. Refer to the “Technical specifications” section in <i>Planning</i>, NTRN10GK, for minimum and maximum Rx/Tx power levels for the line-side specifications. Note: Record the received power level on the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. You can record the power level from the Equipment—Facility window in the Site Manager.						
32	Check the transmitted power level on the corresponding upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Check whether the transmitted power level is within the operational threshold. Refer to the “Technical specifications” section in <i>Planning</i>, NTRN10GK, for minimum and maximum Rx/Tx power levels for the line-side specifications. Note 1: You can record the power level from the Equipment—>Facility window in the Site Manager. Note 2: Use a network topology diagram to determine the upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack.						
33	<table> <tr> <td>If the transmit power level is</td><td>Then go to</td></tr> <tr> <td>not within the operational threshold</td><td>step 34</td></tr> <tr> <td>within the operational threshold</td><td>step 38</td></tr> </table>	If the transmit power level is	Then go to	not within the operational threshold	step 34	within the operational threshold	step 38
If the transmit power level is	Then go to						
not within the operational threshold	step 34						
within the operational threshold	step 38						
34	Replace the upstream 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
35	<table> <tr> <td>If the alarm</td><td>Then</td></tr> <tr> <td>clears</td><td>the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.</td></tr> <tr> <td>does not clear (and the new circuit pack is within operational specifications)</td><td>the circuit pack is not the entire problem. Go to step 36.</td></tr> </table>	If the alarm	Then	clears	the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.	does not clear (and the new circuit pack is within operational specifications)	the circuit pack is not the entire problem. Go to step 36 .
If the alarm	Then						
clears	the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.						
does not clear (and the new circuit pack is within operational specifications)	the circuit pack is not the entire problem. Go to step 36 .						
36	Reseat the near-end 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
37	<table> <tr> <td>If the alarm</td><td>Then</td></tr> <tr> <td>clears</td><td>this procedure is complete. Restore the traffic to both paths on a protected channel.</td></tr> <tr> <td>does not clear</td><td>go to step 38</td></tr> </table>	If the alarm	Then	clears	this procedure is complete. Restore the traffic to both paths on a protected channel.	does not clear	go to step 38
If the alarm	Then						
clears	this procedure is complete. Restore the traffic to both paths on a protected channel.						
does not clear	go to step 38						

Procedure 4-187 (continued)

Loss of Lock

Step	Action						
38	Use the proper cleaning kit to clean all the connectors attached to the receiving (near-end) 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack line-side Rx port and to the corresponding line-side Tx port on the transmitting (far-end) 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. Use a network topology diagram to determine the transmitting (far-end) 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Note 1: If you are using the OMX 4CH + Fiber Manager, clean both ends of the patch-cord between the OMX and the line-side 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR port. Note 2: Clean between the 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack, the SC-to-LC patch panel, and the OMX. - Clean each connector separately. - Record the operating power level after you clean each connector and compare it to the value you wrote down in step 31, this will allow you to see if there is any improvement to the Rx power.						
39	<table> <tr> <th>If the alarm</th><th>Then</th></tr> <tr> <td>clears</td><td>this procedure is complete. Restore the traffic to both paths on a protected channel.</td></tr> <tr> <td>does not clear</td><td>go to step 40</td></tr> </table>	If the alarm	Then	clears	this procedure is complete. Restore the traffic to both paths on a protected channel.	does not clear	go to step 40
If the alarm	Then						
clears	this procedure is complete. Restore the traffic to both paths on a protected channel.						
does not clear	go to step 40						
 CAUTION Risk of loss of traffic across multiple bands Do not clean any optical fibers on the optical fiber plant, or on connections to the OMX. This affects traffic over multiple bands. Clean only the connections at the specified 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit packs.							
40	Replace the near-end 8xOTN Flex MOTR, or (1+8)xOTN Flex MOTR circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.						
41	<table> <tr> <th>If the alarm</th><th>Then</th></tr> <tr> <td>clears</td><td>the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.</td></tr> <tr> <td>does not clear</td><td>the circuit pack is not the problem. Reseat the original circuit pack. Go to step 42.</td></tr> </table>	If the alarm	Then	clears	the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.	does not clear	the circuit pack is not the problem. Reseat the original circuit pack. Go to step 42 .
If the alarm	Then						
clears	the removed circuit pack is faulty. This procedure is complete. Restore the traffic to both paths on a protected channel.						
does not clear	the circuit pack is not the problem. Reseat the original circuit pack. Go to step 42 .						

Procedure 4-187 (continued)

Loss of Lock

Step	Action
42	Repeat step 36 to step 40 for the far-end site.
43	If the alarm does not clear, contact your next level of support. Note: If you set a forced switch on the path, make sure you remove the switch when the procedure is completed. If you are using ALS and disabled it during this procedure, re-enable it for each site.

—end—

Procedure 4-188

Loss of OPU Multiframe Identifier

Alarm IDs: 2035

Probable cause

This alarm is raised when loss of OPU Multiframe Identifier (OMFI) for an OPUCn frame is detected.

The value of bits 4 to 8 of the OMFI byte will be incremented each OPUCn frame to provide a 20 frame multiframe for the multiplexing of ODUk signals into the OPUCn.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
------	--------

- | | | | | | | | |
|----------------------------------|--|----------------------------------|-------------|---------|---------------------------|-------------|------------------------------|
| 1 | Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. | | | | | | |
| 2 | Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal. | | | | | | |
| 3 | Log into the remote network element at the transmit end. | | | | | | |
| 4 | Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure. | | | | | | |
| 5 | <table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 6 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 6 | | | | | | |

Procedure 4-188 (continued)
Loss of OPU Multiframe Identifier

Step	Action						
6	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches. Refer to the “Retrieving equipment and facility details” or “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Verify that the FEC setting on the upstream circuit pack matches the FEC setting on the local circuit pack. Correct any mismatches. Refer to the “Retrieving equipment and facility details” or “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed channel using the appropriate procedures.						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
13	Determine the optical power level into the Rx interface using one of the following methods: - Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power (dBm) value for the corresponding PTP facility. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - Use the optical power meter to measure the optical power level into the Rx interface, and verify it is within technical specifications. Refer to the “Technical specifications” in section in the 6500 <i>Planning</i>, NTRN10GK.						

Procedure 4-188 (continued)

Loss of OPU Multiframe Identifier

Step	Action
14	If the Rx Actual Power value or measured optical power is below the receiver sensitivity specified for the circuit pack Then go to step 15 above the receiver sensitivity specified for the circuit pack step 17 within the receiver sensitivity range specified for the circuit pack step 23
15	Remove the Tx fiber from the far-end circuit pack, and use the optical power meter to measure the transmit power at the far-end.
16	If the transmit power at the far-end is Then above the launch power (minimum) the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 27. below the launch power (minimum) Replace the required circuit pack at the transmit end. Refer to the replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Go to step 27. within the launch power range specified for the circuit pack go to step 27.
17	Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
18	If the original alarm has Then cleared the procedure is complete not cleared go to step 19
19	Review the optical plan to verify that the correct attenuator (if required) is installed between the associated DWDM source and the receiver on the alarmed circuit pack.

Procedure 4-188 (continued)
Loss of OPU Multiframe Identifier

Step	Action						
	If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any alarms on the photonic line associated with the Loss Of Signal alarm using the appropriate procedures.						
20	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 21</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 21
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 21						
21	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. Once the optical path is verified to be correct, perform a cold restart on the circuit pack raising the alarm. See Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						
22	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 23</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 23
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 23						
23	 CAUTION Risk of traffic loss Connecting a facility to a test set causes traffic loss on the facility.						
24	Use a test set to determine if a valid signal is on the facility.						
25	<table> <tr> <th>If the test set shows</th><th>Then</th></tr> <tr> <td>a Loss Of Frame or incorrect framing condition</td><td>the problem is in the source. Go to step 26.</td></tr> <tr> <td>otherwise</td><td>go to step 27</td></tr> </table>	If the test set shows	Then	a Loss Of Frame or incorrect framing condition	the problem is in the source. Go to step 26 .	otherwise	go to step 27
If the test set shows	Then						
a Loss Of Frame or incorrect framing condition	the problem is in the source. Go to step 26 .						
otherwise	go to step 27						

Procedure 4-188 (continued)
Loss of OPU Multiframe Identifier

Step	Action	
26	If the source is connected to the alarmed circuit pack by a client interface	Then Perform troubleshooting on the source system according to your company procedures.
	line interface	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure. If no alarms exist on the transmit end, then perform a warm restart on the transmit circuit pack. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document. If the alarm still does not clear, then the transmit circuit pack is faulty. Replace the transmit circuit pack. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.
27	If the original alarm has cleared	Then the procedure is complete
	not cleared	perform a warm restart on the circuit pack reporting the alarm. Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
28	If the original alarm has cleared	Then the procedure is complete
	not cleared	the receive circuit pack is faulty. Replace the circuit pack reporting the alarm. Refer to the "Replacing a n optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
29	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-189

Loss of Pointer

Alarm IDs: 465

Probable cause

This alarm is raised when one of the following conditions occurs:

- the pointer value in the SONET/SDH overhead of an STS/HO VC is out of a valid range or is not stable
- improper network synchronization
- a circuit pack has been provisioned for concatenated signals but an unequipped signal is sent

Impact

Major, service-affecting (M, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA) alarm, if on an inactive path in

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- use an account with at least a level 3 UPC
- have the optical fiber/cable connection information (that is, how the optical circuit packs on each network element connect to other network elements)

Procedure 4-189 (continued)

Loss of Pointer

Step	Action								
1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document. Clear any alarms of higher order on the hierarchy first using the appropriate procedures.								
2	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If	Then	cleared	the procedure is complete	not cleared	go to step 3		
If	Then								
cleared	the procedure is complete								
not cleared	go to step 3								
3	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.								
4	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.								
5	Log into the remote network element at the transmit end. If you cannot log in remotely from the local network element, you must travel to the remote site.								
6	Retrieve all alarms from the transmit end. If the network element at the transmit end is not a 6500 network element, use the alarm system of the far-end network element to find the problem.								
7	Look for an alarm message for the circuit pack connected to the original shelf.								
	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>there are no alarms or only a STS/VC RFI/RDI alarm at the transmit end</td><td>go to step 8</td></tr> <tr> <td>there are additional alarms of a higher order from the alarm hierarchy at the transmit end</td><td>refer to the “Alarm hierarchies and alarm severities” section in this document.</td></tr> <tr> <td>the alarm did not clear</td><td>go to step 8.</td></tr> </table>	If	Then	there are no alarms or only a STS/VC RFI/RDI alarm at the transmit end	go to step 8	there are additional alarms of a higher order from the alarm hierarchy at the transmit end	refer to the “Alarm hierarchies and alarm severities” section in this document.	the alarm did not clear	go to step 8 .
If	Then								
there are no alarms or only a STS/VC RFI/RDI alarm at the transmit end	go to step 8								
there are additional alarms of a higher order from the alarm hierarchy at the transmit end	refer to the “Alarm hierarchies and alarm severities” section in this document.								
the alarm did not clear	go to step 8 .								
8	Check that a cross-connect has been provisioned at the far-end network element, provision a cross-connect if required. Refer to the “Retrieving path connections” procedure in <i>Configuration - Connections Management</i> , 323-1851-320.								
9	At the local network element, retrieve all alarms to determine if the original alarm has cleared.								
	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If	Then	cleared	the procedure is complete	not cleared	go to step 10		
If	Then								
cleared	the procedure is complete								
not cleared	go to step 10								

Procedure 4-189 (continued)

Loss of Pointer

Step	Action						
10	Ensure that the network synchronization is correct. Refer to the “Synchronization protection” section in <i>Configuration - Protection Switching</i> , 323-1851-315.						
11	At the local network element, retrieve all alarms to determine if the original alarm has cleared. <table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 12</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Verify the connection rate for the entire path to ensure that it is correct. If the connection rate is not correct, edit the cross-connects to the correct rates. Refer to the “Editing a path connection” procedure in <i>Configuration - Connections Management</i> , 323-1851-320.						
13	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-190

Loss of Service Delineation

Alarm IDs: 1249

Probable cause

This alarm is raised when there is a lack of data frames from the network in the egress direction resulting in a client service interruption.

Impact

Major, service-affecting (M, SA) alarm if not protected
Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)

Step	Action
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	Identify if any upstream circuit pack has active alarms and then troubleshoot/clear these alarms first. If the far-end FLEX MOTR circuit pack has an OTM2 facility, clear any OTN faults (such as, LOS, LOC, LOF, LOM, ODU2-AIS, ODU2-LCK, ODU2-OCI, OPU2-AIS, or OPU2 Payload Type Mismatch) for all upstream OTM2 facilities. That is, clear all OTN alarms upstream from the far-end OTM2 port.
3	In the Site Manager Equipment & Facility Provisioning application, verify the ETH10G facility Packet Mapping value matches at both the near-end and far-ends. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

Procedure 4-190 (continued)
Loss of Service Delineation

Step	Action	
4	If the Packet Mapping values are	Then
	mismatched	determine if it is the near-end alarmed ETH10G facility or the far-end facility that has the incorrect Packet Mapping value. The incorrectly provisioned facility must be deleted and reprovisioned with the correct Packet Mapping value. Refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 5 .
	match	go to step 5
5	Ensure that the subtending client equipment is transmitting a valid GFP-F (for Gigabit Ethernet) signal, as required.	
6	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-191

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Use this procedure to clear alarms associated with the Loss Of Signal alarms.

Note that when a Loss Of Signal condition occurs on the 40G OCL circuit pack, detection and signal conditioning takes place based on Loss of Frame detection rather than on LOS. When the LOS recovers at the near-end 40G OCL, some transient alarms such as LOS, OOF, and LOF will be experienced by the subtending equipment or test set connected to the far-end 40G OCL prior to traffic recovery.

For 40G/100G OCLD circuit packs, the power level must be above the LOS clear threshold value for traffic to recover. For 40G/100G OCLD circuit packs, traffic can run below the LOS threshold value. However, if traffic is lost, traffic will not recover until the power level is above the LOS clear threshold. See the “Technical specifications” tables in *WaveLogic OCLD, OCL, MUX, and Submarine Circuit Packs*, 323-1851-102.4 for the 40G/100G OCLD expected LOS alarm clear threshold values.

ATTENTION

For electrical SFPs (NTTP61BA), if this alarm is raised and the traffic cannot be brought up, change the TXCON state to Disabled.

Loss Of Signal (OC1/OC3/12/STM1/4)

Alarm ID: 32, 248

Probable cause

This alarm is raised when the circuit pack:

- can no longer detect a signal on the optical fiber/cable
- detects a receiver overload condition (10G DWDM circuit packs only)

ATTENTION

If ALS is enabled, a Loss Of Signal will be raised at both ends of the link, as the laser is turned off under a Loss Of Signal condition. To help determine on which end the Loss Of Signal alarm must be cleared, temporarily disable ALS at both ends. Refer to the “Editing facility parameters” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Note: If a 10G signal (ETH10G or OC192/STM64) is applied to the 40G OCI (NTK529SA or NTK529SDE5) circuit pack, a “Loss Of Signal” alarm could be raised instead of the expected “Loss of Clock” alarm, due to a limitation of the 40G transponders. When an incorrect 40G signal is applied, the “Loss of Clock” alarm will be raised.

Impact

Critical, service-affecting (C, SA) alarm, if active in protected configurations or unprotected with cross-connects

Minor, non-service-affecting (m, NSA) alarm, if inactive in protected configurations, or without cross-connects

Loss Of Signal (OC48/OC192/OC768/STM16/STM64/STM256)

Alarm ID: 1, 279, 982

Loss Of Signal (ETH10G/ETH40G/ETH100G)

Alarm ID: 673, 1295

Loss Of Signal (FC100/FC200/FC400, FC800, FC1200, FLEX)

Alarm ID: 1252

Probable cause

This alarm is raised when the client Rx interface detects no light, low power or no DC (no 0/1 transition).

For ETH10G, ETH40G, and ETH100G clients using parallel optics, this alarm is raised when a Loss of Frame and low optical input power on any of the CFP optical carriers is detected.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm, if inactive

Loss Of Signal (OTM1, OTM2, OTM3, OTM4, PTP)

Alarm ID: 668, 1001, 1170, 1586

Probable cause

This alarm is raised when the Rx interface detects a Loss of Frame and the optical power is below the Loss Of Signal threshold.

The network element cannot clear a Loss Of Signal alarm until the optical power level goes above the LOS clear threshold.

Note that after removing a LOS or LOF fault condition on the 40G OCLD, Wavelength-Selective 40G OCLD Rx OTM3, or 100G OCLD Rx OTM4 facility, traffic restoration can take up to nine seconds.

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						
2	<table><tr><td>If the alarm is raised against a</td><td>Then go to</td></tr><tr><td>line port</td><td>step 3</td></tr><tr><td>client port</td><td>step 38</td></tr></table>	If the alarm is raised against a	Then go to	line port	step 3	client port	step 38
If the alarm is raised against a	Then go to						
line port	step 3						
client port	step 38						
3	If a photonic line system is used to transport the signal of the circuit pack raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms associated with the Loss Of Signal alarm using the appropriate procedures.						
4	If the signal for this circuit pack is the only signal dropping on a CMD44, check the WSS switch Out to CMD44 Common In fiber, and the CMD44 channel out to OCLD Rx fiber. Note: Use optical terminators on unused input faceplate connectors of installed WSS w/OPM circuit packs. If dust caps are used instead of optical terminators on “Switch In” ports, PMs can be reported against the ports and the ports may appear in-service.						

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action						
5	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
7	Log into the remote network element at the transmit end.						
8	Retrieve all alarms at the transmit end. Clear any higher order alarms using the appropriate procedure.						
9	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	<table> <tr> <td>If a Loss Of Signal alarm is</td><td>Then go to</td></tr> <tr> <td>raised at both the near-end and far-end</td><td>step 11</td></tr> <tr> <td>only raised at the near-end</td><td>step 15</td></tr> </table>	If a Loss Of Signal alarm is	Then go to	raised at both the near-end and far-end	step 11	only raised at the near-end	step 15
If a Loss Of Signal alarm is	Then go to						
raised at both the near-end and far-end	step 11						
only raised at the near-end	step 15						
11	Check whether ALS is enabled at the near-end and far-end. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
12	<table> <tr> <td>If ALS is</td><td>Then</td></tr> <tr> <td>enabled at both the near-end and far-end, or at only one end</td><td>go to step 13</td></tr> <tr> <td>disabled at both the near-end and far-end</td><td>perform this alarm clearing procedure at both ends. Go to step 15.</td></tr> </table>	If ALS is	Then	enabled at both the near-end and far-end, or at only one end	go to step 13	disabled at both the near-end and far-end	perform this alarm clearing procedure at both ends. Go to step 15 .
If ALS is	Then						
enabled at both the near-end and far-end, or at only one end	go to step 13						
disabled at both the near-end and far-end	perform this alarm clearing procedure at both ends. Go to step 15 .						
13	 CAUTION Risk of laser radiation exposure Disabling ALS causes the laser to be permanently active. Laser radiation is present on the optical fiber. Do not look into the optical fiber. Disable ALS at both ends. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
14	Retrieve all alarms at both ends to determine at which end the Loss Of Signal alarm is still raised.						

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action	
15	If the alarm is raised against an OTM1, OTM2, OTM3, OTM4, PTP facility another facility type	Then go to step 16 step 38
16	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.	
17	Determine the optical power level into the Rx interface using one of the following methods:	
	- Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power (dBm) value for the facility reporting the alarm. Refer to the “Retrieving optical power, wavelength, and dispersion ranges” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - Use the optical power meter to measure the optical power level into the Rx interface, and verify it is within technical specifications. Refer to the “Technical specifications” section in the 6500 <i>Planning</i>, NTRN10GK.	
 CAUTION Risk of traffic loss If MPO cables are used, ensure there is no traffic on the MPO cables before removing the cables.		
18	If the measured optical power is below the receiver sensitivity specified for the circuit pack above the receiver sensitivity specified for the circuit pack within the receiver sensitivity range specified for the circuit pack	Then go to step 19 step 21 step 25
19	Remove the Tx fiber from the far-end circuit pack, and use the optical power meter to measure the transmit power at the far-end.	

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action	
20	If the transmit power at the far-end is	Then
	above the launch power (minimum)	the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. If a photonic line system is used to transport the signal, the attenuation can be caused by the line system. Clear any alarms on the photonic line associated with the Loss Of Signal alarm using the appropriate procedures. Go to step 27 .
	below the launch power (minimum)	if the circuit pack at the transmit end supports SFPs, replace the SFP that corresponds to the facility raising the alarm. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. if the circuit pack at the transmit end does not support SFPs, replace the required circuit pack at the transmit end. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 27 .
	within the launch power range specified for the circuit pack	go to step 27 .
21	Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> 323-1851-201.0.	
22	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 23

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action						
23	Review the optical plan to verify that the correct attenuator (if required) is installed between the associated DWDM source and the receiver on the alarmed circuit pack. If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any alarms on the photonic line associated with the Loss Of Signal alarm using the appropriate procedures.						
24	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 25</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 25
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 25						
25	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes a traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. Once the optical path is verified to be correct, perform a cold restart on the circuit pack raising the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						
26	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 27</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 27
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 27						
27	For the optical interface, if ALS was disabled in step 13, re-enable ALS at the appropriate end(s). Refer to the ED-AMP command description in <i>TL 1 Description</i>, 323-1851-190. Go to step 39.						
28	Check the corresponding subtending circuit pack for failures or alarms. Troubleshoot these alarms/failures before proceeding.						
29	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 30</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 30
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 30						
30	Verify that the provisioned bit rate and protocol for the circuit pack type and pluggable modules are consistent with the subtending equipment. Correct any mismatches.						

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action								
31	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 32</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 32		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 32								
32	For multi lane interfaces, if possible, check on both the subtending circuit pack (transmit direction) and locally (receive direction) individual lane power. For Rx/Tx power values for pluggables, refer to the pluggables summary and reference tables in “Pluggables Datasheets and Reference”, 323-1851-180. Note: Validate the individual lane power using Site Manager “Equipment & Facility Provisioning” application by selecting the equipment and client facility in question and clicking the “Lane Power” button.								
33	If on the transmit interface only a subset of the lanes exhibit power outside of the expected range, then it is recommended that the transmit interface be replaced.								
34	If on the receive interface only a subset of the lanes exhibit power outside of the expected range, then the hardware issue might be on either the transmitter or the receiver.								
35	Determine the optical power level into the Rx interface using one of the following methods: - Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power (dBm) value for the facility reporting the alarm. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - Use the optical power meter to measure the optical power level into the Rx interface, and verify it is within technical specifications. Refer to the “Technical specifications” section in the 6500 <i>Planning</i>, NTRN10GK.								
36	<table> <tr> <th>If the measured optical power is</th><th>Then go to</th></tr> <tr> <td>below the Rx minimum power (dBm) value</td><td>step 37</td></tr> <tr> <td>above the Rx maximum power (dBm) value</td><td>step 38</td></tr> <tr> <td>between the Rx minimum power (dBm) and Rx maximum power (dBm) values</td><td>step 39</td></tr> </table>	If the measured optical power is	Then go to	below the Rx minimum power (dBm) value	step 37	above the Rx maximum power (dBm) value	step 38	between the Rx minimum power (dBm) and Rx maximum power (dBm) values	step 39
If the measured optical power is	Then go to								
below the Rx minimum power (dBm) value	step 37								
above the Rx maximum power (dBm) value	step 38								
between the Rx minimum power (dBm) and Rx maximum power (dBm) values	step 39								
37	Check the fiber and fiber cleanliness between the subtending equipment and the port reporting the alarm. Correct any issues found. Go to step 39.								

Procedure 4-191 (continued)

Loss Of Signal (OC/STM, ETH10G, ETH40G, ETH100G, FC, OTM1, OTM2, OTM3, OTM4, PTP, FLEX)

Step	Action						
38	The signal may require padding, or you are not using the correct pluggable type on the circuit pack reporting the alarm or on the subtending equipment. Contact your next level of support for more information. Go to step 39 .						
39	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>perform a warm restart on the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in <i>Fault Management - Alarm Clearing</i> 323-1851-543.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	perform a warm restart on the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in <i>Fault Management - Alarm Clearing</i> 323-1851-543.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	perform a warm restart on the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in <i>Fault Management - Alarm Clearing</i> 323-1851-543.						
40	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-192

Loss Of Signal (OPTMON, VOA)

Alarm IDs: 560, 553

Probable cause

This alarm is raised against an OPTMON or VOA facility when:

- the optical power transmitted and/or optical power received has fallen below the provisioned LOS threshold level of the alarmed facility
- all line cards connected to the same passive photonic Mux/Demux and subsequent cascaded Photonic Mux/Demux detected no input signal when shelf alarm correlation is turned on
- there is cross fiber on the ROADM WSS switch ports for the connected CMDs

For CDC configurations, this alarm is raised on CCMD8x16, RLA 5x1, MDRLA 2x6 or CCMD12 C-Band/CCMD12 L-Band circuit packs.

For CD configurations, this alarm is raised on SMD Flex C-Band 14x8 and CCMD12 circuit packs.

The conditions that can cause this alarm include:

- a disconnected, or defective fiber optic patchcord
- a dirty optical fiber connector
- a defective module
- defective transmitting subtending equipment
- a provisioning error in the LOS threshold of the alarmed facility
- an OSRP Line/Link Down condition triggered by fiber disconnection on intra-node fiber (except WSS-WSS fiber), AMP in shut-off state, or AMP Primary State OOS-MA

Note 1: OSRP provides a fault reason within the “Additional Information” field of the alarm to help with the troubleshooting.

Note 2: For the SMD Flex C-Band 14x8 (NTK553GC), “Loss Of Signal” alarm on the MPO level masks individual “Loss Of Signal” alarms on the input sub-fibers.

This alarm can remain active after the fault has cleared and the original power level is restored. This occurs when the power level is lower than the user-provisioned LOS threshold plus the hysteresis value. The hysteresis value is not user provisionable and is set at 3 dB.

Procedure 4-192 (continued)

Loss Of Signal (OPTMON, VOA)

In rare cases of SP restarts, the WSS w/OPM AINS state is lost and this alarm is raised against ports that have never received a valid optical signal. To clear this alarm, you can edit the Auto In-Service Time of the OPTMON facilities to re-enable the AINS state or change the primary state of the OPTMON facilities to OOS. This can be done using the Configuration->Equipment and Facility Provisioning application in Site Manager or by enabling AINS and editing the AINS Time Out parameter from the Site Manager Node Information application Edit System dialog box. Refer to the “Editing facility parameters” procedure in *Configuration - Provisioning and Operating*, 323-1851-310, or the “Editing the nodal system parameters” and “Editing the AINS default period” procedures in *Administration and Security*, 323-1851-301.

ATTENTION

In a single-channel photonic network, network level alarm correlation is unable to distinguish the following: In the Mux direction: cannot distinguish the difference between CMD44 channel input port LOS and WSS switch in LOS. In the Demux direction: cannot distinguish the difference between a CMD44 Common in LOS and a OTM facility LOS; manual fault isolation may be required to find the root cause of the fault.

ATTENTION

When the NTK592NGE5 SFP is used in spans with a span loss greater than 20.5 dB, due to SFP's lower Tx output power, a permanent LOS alarm will be active against the LIM port 4 OPTMON facility (since this SFP can support span losses up to 31.5 dB). To avoid this condition, software automatically puts the LIM port 4 OPTMON facility (OPTMON-bay- shelf-slot-4) in the OOS state (OOS-MA) when the NTK592NGE5 SFP is provisioned. This will prevent the Loss of Signal alarm from being raised. As a result, since the LIM port 4 OPTMON facility alarm never gets raised, it is not possible to infer if the problem is before the LIM or after. When the NTK592NVE5 SFP is used, the OSC Rx power can be very low under normal operating conditions (compared to the NTK592NPE6, NTK592NBE6 and NTK592NHE6 SFPs) and a permanent LOS alarm may be active against the LIM port 4 OPTMON facility even though there is no problem (since this SFP can support span losses up to 42 dB). To avoid this condition, software automatically puts the LIM port 4 OPTMON facility (OPTMON-bay-shelf-slot-4) in the OOS state (OOS-MA) when the NTK592NGE5 SFP is provisioned. This will prevent the Loss of Signal alarm from being raised. As a result, since the LIM port 4 OPTMON facility alarm never gets raised, it is not possible to infer if the problem is before the LIM. The LIM port 4 OPTMON will have a LOS threshold of “N/A”.

Procedure 4-192 (continued)

Loss Of Signal (OPTMON, VOA)

In Release 9.1 and earlier, there had to be two SPLI compliant receivers with provisioned far-end addresses reporting LOS before the OTM LOS would be masked by “Loss Of Signal” (OPTMON). As of Release 9.2, even if there is only one receiver provisioned (and with a far-end address), the OTM LOS is masked and replaced with a “Loss Of Signal” (OPTMON) on the common input port of that demux.

Additionally, any CMD with no channels will set the upgrade in port of its upstream CMD to a faulted state.

Impact

Major, service-affecting (M, SA) alarm (OPTMON and VOA)

Minor, non-service-affecting (m, NSA) alarm, if inactive (OPTMON)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a network and site diagram
- have a fiber cleaning kit
- obtain a replacement circuit pack, if required

Step	Action
1	This condition may be caused by an upstream fault. Before following this procedure, check for and clear any upstream alarms.
2	Check for and clear any of the following alarms before clearing this alarm: • Automatic Power Reduction Active • Automatic Shutoff • Input Loss Of Signal • Optical Line Fail • Output Loss Of Signal • Shutoff Threshold Crossed

Procedure 4-192 (continued)
Loss Of Signal (OPTMON, VOA)

Step	Action								
3	Verify that the OPTMON or VOA facility parameters are provisioned correctly. If necessary, correct any discrepancies. This can be done using the Configuration->Equipment and Facility Provisioning screen in Site Manager. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures and the OPTMON facility parameter table or VOA facility parameter table in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
4	<table border="0"> <tr> <td>If the alarm is raised against</td><td>Then go to</td></tr> <tr> <td>a OMD4, OMD4, OMD4, BS1, BS2, BS3, BS5, OMX, OBB, OBMD 1x8, CMD42, CMD 64, CMD96, CCMD8x16, CCMD12 C-Band, CCMD12 L-Band, or CMD44 channel input port</td><td>step 5</td></tr> <tr> <td>an RLA 5x1 or MDRLA 2x6</td><td>step 14</td></tr> <tr> <td>any other port</td><td>step 16</td></tr> </table>	If the alarm is raised against	Then go to	a OMD4, OMD4, OMD4, BS1, BS2, BS3, BS5, OMX, OBB, OBMD 1x8, CMD42, CMD 64, CMD96, CCMD8x16, CCMD12 C-Band, CCMD12 L-Band, or CMD44 channel input port	step 5	an RLA 5x1 or MDRLA 2x6	step 14	any other port	step 16
If the alarm is raised against	Then go to								
a OMD4, OMD4, OMD4, BS1, BS2, BS3, BS5, OMX, OBB, OBMD 1x8, CMD42, CMD 64, CMD96, CCMD8x16, CCMD12 C-Band, CCMD12 L-Band, or CMD44 channel input port	step 5								
an RLA 5x1 or MDRLA 2x6	step 14								
any other port	step 16								
5	If this is a single-channel photonic network, verify the optical patchcords between the CMD44 Common Out and the RLA 5x11 or MDRLA 2x6 or WSS input port, and between the WSS Common Out and the LIM Line B Input: - ensure it is connected at both ends and that there is no problem with the optical patchcord - clean the connectors. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. - ensure the transmitting equipment is functioning correctly and transmitting a valid signal Note: Use optical terminators on unused input faceplate connectors of installed WSS w/OPM circuit packs or MDRLA 2x6/RLA 5x1 module. If dust caps are used instead of optical terminators on “Switch In” ports, PMs can be reported against the ports and the ports may appear in-service.								
6	Verify the optical patchcord between the subtending equipment and the OMD4, OMD4, OMD4, BS1, BS2, BS3, BS5, OMX, OBB, OBMD 1x8, CMD42, CMD 64, CMD96, CCMD8x16, CCMD12 C-Band, CCMD12 L-Band, or CMD44 channel input port: - ensure it is connected at both ends and that there is no problem with the optical patchcord - clean the connectors. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. - ensure the transmitting subtending equipment is functioning correctly and transmitting a valid signal - check of fiber crossing when there are more than one colored CMD connected to a ROADM WSS								

Procedure 4-192 (continued)
Loss Of Signal (OPTMON, VOA)

Step	Action						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Verify the optical patchcord connected to the port reporting the alarm: - ensure it is connected at both ends and that there is no problem with the optical patchcord - clean the connectors. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	Using the Performance->Performance Monitoring->New screen of Site Manager, compare the untimed OPR-OTS power level to the LOS threshold. If the power level is greater than the threshold but by less than 3 dB, there is not enough power to clear the alarm hysteresis. Decrease the LOS threshold by 3 dB. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
11	Wait at least one minute for the Loss Of Signal alarm to clear. Change the LOS Threshold value back to the original value.						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	<table> <tr> <th>If the alarm is raised against</th><th>Then go to</th></tr> <tr> <td>an unused WSS1, MDRLA 2x6 or RLA 5x1 ingress port (the port with no adjacency provisioned)</td><td>step 14</td></tr> <tr> <td>any other port</td><td>step 16</td></tr> </table>	If the alarm is raised against	Then go to	an unused WSS1, MDRLA 2x6 or RLA 5x1 ingress port (the port with no adjacency provisioned)	step 14	any other port	step 16
If the alarm is raised against	Then go to						
an unused WSS1, MDRLA 2x6 or RLA 5x1 ingress port (the port with no adjacency provisioned)	step 14						
any other port	step 16						
14	Set the secondary state of the port to AINS using the Configuration->Equipment and Facility provisioning screen of Site Manager. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
15	Run another OTDR trace on the faulted1 MDRLA 2x6 or RLA 5x1. Refer to "Performing a manual OTDR trace" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-192 (continued)

Loss Of Signal (OPTMON, VOA)

Step	Action
16	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-193

Loss Of Synchronization Messaging Channel

Alarm IDs: 2009

Probable cause

This alarm is raised against a facility when:

- the facility is a SyncE reference
- the Sync Status Message (SSMTRANSMIT) is not disabled on the facility
- the circuit pack does not detect ESMC messages for more than 5 seconds. The missing ESMC message happens in the following scenarios:
 - when the circuit pack does not detect the signal on the fiber/cable
 - far end does not transmit ESMC packets
 - the fiber/cable cut
 - dirty fiber, cable, or connectors
 - incorrect attenuation

Impact

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all safety requirements described in *Fault Management - Module Replacement*, 323-1851-545
- have the engineering documentation package (EDP)
- have a fiber cleaning kit

Step	Action
1	Identify the module that raised the alarm. See Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	If you do not require the Sync Status Message (SSMTRANSMIT), disable the Sync Status Messaging.
3	Verify the alarm causes from the probable cause list above and clear the issue using the appropriate procedures.

Procedure 4-193 (continued)

Loss Of Synchronization Messaging Channel

Step	Action
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-194

Low Optical Return Loss at Input

Alarm IDs: 1760

Probable cause

This alarm is raised against a RAMAN facility of an SRA circuit pack when the optical return loss (ORL) at the input drops below a fixed threshold.

Low ORL indicates high reflection at the amplifier input, which can be caused by:

- dirty optical connectors
- improper optical cable mating
- a disconnected optical fiber at the SRA input
- an optical fiber cut
- a degraded optical fiber

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the engineering documentation package (EDP)
- have a fiber cleaning kit
- have a replacement module

Procedure 4-194 (continued)
Low Optical Return Loss at Input

Step	Action	
1	If this alarm	Then
	was raised as a result of a maintenance activity or during SLAT, and is expected	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.
	is unexpected	See the Attention box, then go to step 2

ATTENTION

Continuing with this procedure triggers Automatic Laser Shut Off (ALSO) and will cause traffic loss in both directions.

- 2 For the SRA circuit pack, verify that the PM threshold values for the alarmed facility are correctly provisioned. The threshold value for the ORLIN-OTS is 27dB. These parameters can be read in the PM (Performance Monitoring) of the SRA circuit pack under the Raman Type. Refer to “Retrieving equipment and facility details” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.
- 3 If the alarm has not cleared, place the RAMAN facility corresponding to the alarm and the Line A amplifier connected to the SRA circuit pack OOS.



DANGER

Risk of laser radiation exposure

Do not look directly into the optical beam. Invisible light can severely damage your eyes.



CAUTION

Risk of damage to modules

Never disconnect an optical fiber that is connected to an active or powered up optical amplifier or SRA circuit pack. To disconnect or reconnect an optical fiber, make sure the optical amplifier is out of service (OOS), then disconnect or reconnect the fiber.

Procedure 4-194 (continued)
Low Optical Return Loss at Input

**CAUTION****Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

**CAUTION****Risk of traffic loss**

Only disconnect the output fiber of the alarmed optical amplifier. It is not necessary to disconnect any other output fibers, which could affect service. Disconnecting the input fiber of the SRA will impact traffic in both directions, as it triggers ALSO.

- 4 Inspect and clean the input fiber/port/patch-panel on the AID/port reporting the ORL alarm.
- 5 Place the facilities back to IS. Refer to the “Changing the primary state of a facility” procedure in *Configuration - Provisioning and Operating*, 323-1851-310.
- 6 If the alarm has not cleared, place the facilities back to OOS and check any upstream patch panel connectors at that site.
- 7 Perform fiber/port/patch-panel cleaning procedure(s) on the far end of the AID/port reporting the alarm. (The far-end of Port 7 SRA is the XLA circuit pack).
- 8 Place the facilities back to IS.
- 9 If the alarm is not cleared, you may need to clean a specific connector or connectors that may not be immediately connected to the alarmed amplifier input. For information on isolating connector losses, complete the [“Locating a reflective event”](#) procedure in this document.
- 10 If the ORL alarm is on ORL-IN (Port 8) of the SRA circuit pack, verify the Office and Short traces manually for Large reflections that were not captured by the OTDR alarms, possibly because of the pulse width/event spacing. If events are overlapping, re-run the trace using a smaller pulse.
- 11 Perform the fiber/port/patch-panel cleaning procedure(s) on the event in question.
- 12 If the alarm is not clear, contact your next level support to assist with conducting the attenuation introduction procedure (attenuator required) to isolate the source of the reflection.

Procedure 4-194 (continued)

Low Optical Return Loss at Input

Step	Action
13	If the alarm is not clear, locate the suspect SRA circuit pack identified in previous step(s) - unseat the SRA circuit pack, count 10 secs and reseat the SRA circuit pack. Wait for circuit pack to complete boot cycle, then wait for the "OTDR Traces In-Progress" alarm clears (Go/No-Go OTDR traces) for the link. Refer to "Reseating a circuit pack" in <i>Fault Management - Module Replacement</i> , 323-1851-545.
14	If the alarm is not clear, replace the SRA circuit pack. Refer to "Replacing the SRA circuit pack" in <i>Fault Management - Module Replacement</i> , 323-1851-545.
15	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-195

Low Optical Return Loss at Output

Alarm IDs: 559, 1033, 1684, 1796

Probable cause

This alarm is raised against an AMP, RAMAN, AMPMON, or VOA facility of a LIM, RLA, MDRLA 2x6, SRA, FGA, or XLA circuit pack when the optical return loss (ORL) at the output drops below a fixed threshold.

Low ORL indicates high reflection at the amplifier output, which can be caused by:

- dirty optical connectors
- improper optical cable mating
- a disconnected optical fiber at the amplifier output
- an optical fiber cut
- a degraded optical fiber
- a disconnected or missing termination
- misprovisioning of an amplifier resulting in excessive power being injected into the mid-stage DSCM or fiber-plant

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the engineering documentation package (EDP)
- have a fiber cleaning kit
- have a replacement module

Procedure 4-195 (continued)

Low Optical Return Loss at Output

Step	Action						
1	<table> <tr> <th>If this alarm</th><th>Then</th></tr> <tr> <td>was raised as a result of a maintenance activity or during SLAT, and is expected</td><td>no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.</td></tr> <tr> <td>is unexpected</td><td>go to step 2</td></tr> </table>	If this alarm	Then	was raised as a result of a maintenance activity or during SLAT, and is expected	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.	is unexpected	go to step 2
If this alarm	Then						
was raised as a result of a maintenance activity or during SLAT, and is expected	no action is required. The alarm will clear when the maintenance activity or SLAT is completed. The procedure is complete.						
is unexpected	go to step 2						
2	For the SRA circuit pack, verify that the PM threshold values for the alarmed facility are correctly provisioned. The threshold value for the ORLOUT-OTS is 22dB. These parameters can be read in the PM (Performance Monitoring) of the SRA circuit pack under the Raman Type. Refer to "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
3	Check the alarmed amplifier power level defined in your EDP against the provisioned alarmed facility parameter values. Edit the power level values as required. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	<table> <tr> <th>If the alarmed amplifier is</th><th>Then go to</th></tr> <tr> <td>a pre-amplifier</td><td>step 6</td></tr> <tr> <td>otherwise</td><td>step 9</td></tr> </table>	If the alarmed amplifier is	Then go to	a pre-amplifier	step 6	otherwise	step 9
If the alarmed amplifier is	Then go to						
a pre-amplifier	step 6						
otherwise	step 9						
6	 DANGER Risk of laser radiation exposure Do not look directly into the optical beam. Invisible light can severely damage your eyes. Ensure that all connectors located after the amplifier output are properly mated. Verify this on both ends of the connector-mating receptacles.						
7	Ensure that the termination plugs are present and are mated properly on unused ports.						

Procedure 4-195 (continued)
Low Optical Return Loss at Output

Step	Action						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	 CAUTION Risk of damage to modules Never disconnect an optical fiber that is connected to an active or powered up optical amplifier. To disconnect or reconnect an optical fiber, make sure the optical amplifier is out of service (OOS), then disconnect or reconnect the fiber.  CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Place the alarmed facility out of service (OOS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Note: An AMPMON facility is a child facility of the corresponding AMP facility and its primary state always follows the primary state of the parent AMP. The parent AMP refers to the AMP facility on the input port of the same Line. An AMPMON facility is auto-created or deleted subsequently after the creation or deletion of the parent AMP facility.						
10	 CAUTION Risk of traffic loss Only disconnect the output fiber of the alarmed optical amplifier. It is not necessary to disconnect any other output fibers, which could affect service. Disconnecting the Line B out fiber will impact traffic in both directions, as this triggers Automatic Laser Shut Off (ALSO). Disconnect the output fiber of the alarmed optical amplifier, clean the output fiber and connectors at the amplifier, then reconnect the fiber.						
11	Place the facilities back in-service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-195 (continued)

Low Optical Return Loss at Output

Step	Action						
12	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 13</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 13						
13	You may need to clean a specific connector or connectors that may not be immediately connected to the alarmed amplifier output. For information on isolating connector losses, complete the “Locating a reflective event” procedure in this document.						
14	If the alarm is not clear, contact your next level support to assist with conducting the attenuation introduction procedure (attenuator required) to isolate the source of the reflection.						
15	If the alarm is not clear, locate the suspect SRA circuit pack identified in previous step(s) - unseat the SRA circuit pack, wait 10 seconds and reseat the SRA circuit pack. Wait for circuit pack to complete the boot cycle, then wait for the “OTDR Traces In-Progress” alarm clears (Go/No-Go OTDR traces) for the link. Refer to “Reseating a circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
16	If the alarm is not clear, replace the SRA circuit pack. Refer to “Replacing the SRA circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
17	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-196

Low Received Span Loss

Alarm ID: 1284

Probable cause

This alarm is raised against the LIM (at the tail end of the span), SRA, SAM, RLA 5x1, MDRLA 2x6, or ESAM circuit packs when the incoming estimated OSC/Telemetry Gain Span Loss for the span is lower than the Minimum Span Loss less the Span Loss Margin (Minimum Span Loss - Span Loss Margin).

The alarm clears when the Span Loss is greater than (Minimum Span Loss - Span Loss Margin) + 1 dB.

Note: Span Loss for an optical span is stored as a PM with full retrievable history. Refer to the “Retrieving performance monitoring data” in the *Fault Management - Performance Monitoring*, 323-1851-520.

For RAMAN amplifiers, the Telemetry Gain Signal value is used for Span Loss calculation. Refer to the *Photonic Layer Guide*, NTRN15DA, for more information.

It is recommended that the Low Received Span Loss detection and alarming be disabled. This alarm becomes redundant as a result of the amplifier “Minimum Gain” alarm.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- have a fiber cleaning kit
- obtain a fiber patchcord, if required

Procedure 4-196 (continued)
Low Received Span Loss

Step	Action						
1	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.						
2	Add attenuator pads as needed until the Span Loss for the link is within the Span Loss Margin by at least 1 dB (hysteresis). Note that the alarm will clear when: Measured Span Loss > Minimum Span Loss - Span Loss Margin + 1 dB (hysteresis) Note: Fixed attenuators should be added to the patch panel or within a fiber storage tray and not on the optical interface circuit packs.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>you want to clear the alarm (and disable detection) without addressing the fault</td><td>step 5</td></tr> <tr> <td>otherwise</td><td>step 6</td></tr> </table>	If	Then go to	you want to clear the alarm (and disable detection) without addressing the fault	step 5	otherwise	step 6
If	Then go to						
you want to clear the alarm (and disable detection) without addressing the fault	step 5						
otherwise	step 6						
5	Provision the Minimum Span Loss to 0. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
6	Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-197

Manual Area Address Dropped from area

Alarm ID: 804**Probable cause**

This alarm is raised when a provisioned manual area address is dropped from the computed area address set of the network. Each network element supports a maximum of three manual area addresses at one time. All network elements in the network share this information. Each network element independently calculates a set of computed area addresses from the union of all the manual area addresses provisioned on the network. The computed set is recalculated in the following situations:

- addition of a new network element to the network
- manual deletion or addition of an area address from any network element
- expiration of a local timer (maximum delay is every 10 to 15 minutes, and minimum delay is every few seconds)

This alarm can indicate a network-wide error condition in the provisioning of data communication. If an area address is dropped on a network element, isolation of the network element and a loss of OAM&P capability on the network element can occur.

Impact

Minor, non-service-affecting (m, NSA)
Enabled by default.

Prerequisites

To perform this procedure, you must have the network plan.

Step	Action
------	--------

Determine the manual area addresses provisioned in the network

- | | |
|---|--|
| 1 | To determine the manual area addresses provisioned in the network you must log into one of the network elements in the system. |
| 2 | Query all area addresses. Refer to the “Retrieving communications settings” procedure in <i>Data Communications Planning & User Guide</i> , 323-1851-101.
<i>A list of up to three addresses appears.</i> |
| 3 | Record the list of addresses in the computed set. |

Procedure 4-197 (continued)

Manual Area Address Dropped from area

Step	Action
------	--------

Determine if the manual area addresses belong in the network

Note: Determine if the area addresses belong in the network by referring to the network plan

4	If the dropped area address belongs in the network	Then contact your next level of support or Ciena support group
	does not belong in the network	go to step 5

Delete the manual area addresses that do not belong in the network

- 5 Log into the network element where the manual area address(es) that is not configured properly, is provisioned.
- 6 Delete the manual area address that does not belong in the network. Refer to the "Deleting an entry in the communications settings" procedure in *Data Communications Planning & User Guide*, 323-1851-101.
- 7 Repeat [step 5](#) and [step 6](#) to delete each manual area address that is not configured properly.

—end—

Procedure 4-198

Mapping Mismatch

Alarm ID: 1378, 1778**Probable cause**

The alarm point is identified at the G.709 level on the 40G OCI, 40G+ CFP OCI, provisioned with an OTM3 client, when the payload mapped in the G.709 signal at the upstream 40G OCI or 40G+ CFP OCI is not compatible with the expected one at the downstream 40G OCI or 40G+ CFP OCI circuit pack. The G.709 payload structure identifier (PSI) byte is used for this purpose. This condition can occur as a result of crossed fibers or as a result of incorrect provisioning. For example, on a 40G OCI (NTK529SDE5), the near-end client facility is provisioned to be OTM3 and the far-end client facility is provisioned to be OC-768.

On the 40G+ CFP OCI (NTK529SJE5), OC768 or 40GE to OTU3 handoff mappings are supported. While the 40G+ CFP OCI can support handoff mappings to OTU3, it requires the payload of the OTU3 to include the correct payload type. If an OTU3 facility at one end of a link is connected to an OC768 facility at the far-end, the OPU3 feeding the OTU3 facility must contain an OC768 Bit synchronous CBR mapped as defined in G.709, PT = 03 hex. If an OTU3 facility at one end of a link is connected to a 40GE facility at the other end, the OPU3 must contain a 40GE signal GMP mapped, PT = 07 hex.

For the 40G+ CFP OCI, this alarm is raised:

- for any Payload Type (PT) received from the line side that is not defined for this circuit pack (for example, not = 8A, 8B, 89, 20)
- if OTM3 facility mapping is ODU Transparent and the PT received is different than the PT expected
- in the event of a mismatch between the PT received on the OTU3 client port and that received from the line side
- if the NTK529SJ is used at the near-end and the NTK525CF is used at the far-end and the NTK525CF ODU2 rate is not 10.7G

This alarm is also raised on the ODUCTP facility of a 40G OTN XCIF circuit pack when the far-end PT is unsupported (not = 0x89, 0x8B or 0x8A).

Procedure 4-198 (continued)
Mapping Mismatch

ATTENTION

In a 40G client card back-to-back configuration when a 40G+ CFP OCI circuit pack (NTK529SJE5) is mated with a 40G MUX OCI circuit pack (NTK525CFE5 variant), a “Mapping Mismatch” alarm is raised on the OTU3 port of the 40G+ CFP OCI circuit pack when the 40G MUX OCI ODU2 rate is not 10.7G to help highlight that this is an unsupported configuration. See *WaveLogic OCLD, OCI, MUX, and Submarine Circuit Packs*, 323-1851-102.4, for details of the 40G client card back-to-back configuration.

Impact

Critical, service-affecting (C, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA) alarm, if on an inactive path

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the optical fiber connection information (that is, how the modules on each network element connect to other network elements)
- have the provisioning information for the client type on the 40G OCI cards

Step	Action
------	--------

- | | | | | | | | |
|--|--|---------------------------------------|-------------------|--|------------------------|----------------------|------------------------|
| 1 | <table><tr><td>If the alarm is raised against</td><td>Then go to</td></tr><tr><td>a 40G OTN XCIF, 40G OCI or 40G+ CFP OCI circuit pack</td><td>step 2</td></tr><tr><td>another circuit pack</td><td>step 3</td></tr></table> | If the alarm is raised against | Then go to | a 40G OTN XCIF, 40G OCI or 40G+ CFP OCI circuit pack | step 2 | another circuit pack | step 3 |
| If the alarm is raised against | Then go to | | | | | | |
| a 40G OTN XCIF, 40G OCI or 40G+ CFP OCI circuit pack | step 2 | | | | | | |
| another circuit pack | step 3 | | | | | | |
| 2 | Ensure that the 40G client circuit packs and associated 40G OCLD or Wavelength-Selective 40G OCLD circuit packs are connected properly. | | | | | | |

Procedure 4-198 (continued)
Mapping Mismatch

Step	Action						
3	Using connection records, determine the far-end module connected to the alarmed module. Retrieve the equipment and Facility Details for both the alarmed and remote module. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. A 40G OTN XCIF, 40G OCI, 40G+ CFP OCI/40G OCLD or Wavelength-Selective 40G OCLD pair must be connected to a 40G OTN XCIF, 40G OCI or 40G+ CFP OCI/40G OCLD or Wavelength-Selective 40G OCLD pair. From your company records, validate that the local client facility is supposed to be provisioned as an OTM3 or ODUCTP facility.						
4	If a handoff is provisioned between two 40G+ CFP OCI circuit packs, verify that the remote equipment feeding the OTU3 to one 40G+ CFP OCI circuit pack does not have an incorrect payload.						
5	Ensure that the corresponding client facility at the far-end is also provisioned as an OTM3 or ODUCTP facility.						
6	Delete and re-add the incorrect facility. Refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Ensure that no fibers are crossed. The OTU/ODU trail trace can be used to identify crossed fibers. If the OTU/ODU trail trace is provisioned, the “ ODU/OTU Trace Identifier Mismatch ” alarm is raised to indicate crossed fibers. Refer to the “Retrieving and editing trail trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 10						
10	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-199

Member Release Misaligned

Alarm ID: 1685

Probable cause

This alarm is raised to identify a member shelf that is running a different release than the primary shelf.

Impact

Major, non-service-affecting (M, NSA)

Prerequisites

Before you perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the IP address of all shelves within the consolidated TID
- have a network plan, or other documentation identifying all shelf provisioning information, including shelf function

Step	Action
------	--------

- | | |
|---|--|
| | Note: If a TIDc site is upgraded (First Invoke) shelf by shelf and the First Invoke is issued on member shelves prior to the primary shelf, this alarm is raised for all member shelves. In these cases, this alarm can be ignored. |
| 1 | Verify the current load release on the shelf processor. Refer to the “Checking the current software release” procedure in <i>Commissioning and Testing</i> , 323-1851-221. |
| 2 | Upgrade the member shelf to a software version supported by the primary shelf. Refer to the “Software Upgrade Procedures” section in <i>Administration and Security</i> , 323-1851-301. |
| 3 | If the alarm is not clear, verify If the member shelf is already running a release supported by the primary shelf and re-deliver the release to the TIDc. |
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-200

Member Shelf Mismatch

Alarm ID: 718

Probable cause

This alarm is raised on the primary shelf of a consolidated node when the member shelf's **Function provisioned** parameter seen by the primary shelf does not match the **Function actual** parameter of the member shelf. This alarm is masked by the Member Shelf Unreachable alarm.

When a member shelf is auto-enrolled into a consolidated node, the member's **Function provisioned** parameter set on the primary is equal to the **Function actual** parameter of the member shelf.

When a member shelf is manually added to a consolidated node, the **Function provisioned** parameter is provisioned in the **Node Information** application. Refer to the "Displaying member shelf information of a consolidated node" procedure in *Administration and Security*, 323-1851-301.

Impact

Major, non-service-affecting (M, NSA)

Prerequisites

Before you perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the IP address of all shelves within the consolidated TID
- have a network plan, or other documentation identifying all shelf provisioning information, including shelf function

Step	Action
1	Log into the primary shelf raising the alarm.
2	Verify that each member shelf's Function provisioned parameter detected by the primary shelf matches the corresponding Function actual parameter of the member shelf. The Secondary state parameter for the mismatched member shelf indicates "MEA". Refer to the "Displaying member shelf information of a consolidated node" procedure in <i>Administration and Security</i>, 323-1851-301.
3	After identifying the member shelf with the mismatched Function provisioned parameter, record all the provisioning information required to re-add the incorrectly provisioned member shelf to the consolidated node.

Procedure 4-200 (continued)

Member Shelf Mismatch

Step	Action
4	Delete the member shelf from the consolidated node.
5	Refer to the network plan, and re-add the member shelf to the consolidated node with the correct Function provisioned . Refer to the “Adding a shelf to a consolidated node” procedure in <i>Administration and Security</i> , 323-1851-301.
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-201

Member Shelf Unknown

Alarm ID: 719

Probable cause

This alarm is raised on the primary shelf of a consolidated node during the addition of a member shelf when a member shelf does not respond to messages from the primary shelf related to the addition. Low level communication between the shelves is still available.

Communication between a primary and member shelf can be interrupted when:

- a duplicate primary shelf exists. The Member Shelf Unknown alarm is raised on the primary shelf, which attempted to communicate with a member shelf that is already associated with another primary shelf.
- the member shelf is running a software version that is unsupported by the primary shelf
- the member shelf is provisioned with a function that is unsupported by the primary shelf

Note: If a primary or member shelf is restarted, the Member Shelf Unknown alarm is briefly raised.

Impact

Major, non-service-affecting (M, NSA)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the IP address of all shelves within the consolidated node
- have an available network plan, or other documentation identifying all shelf provisioning information, including upgrade plans

Step	Action
------	--------

- | | |
|---|---|
| 1 | Clear any Duplicate Primary Shelf alarms present. Refer to the “Duplicate Primary Shelf” on page 4-234 alarm clearing procedure in this document. |
| 2 | Log into the primary shelf. |

Procedure 4-201 (continued)

Member Shelf Unknown

Step	Action						
3	Identify which shelf/shelves is/are causing the alarm. The Secondary state parameter for the unreachable member shelf indicates “MEA”. The affected shelf/shelves may have ‘(?)’ next to the shelf number and be highlighted in cyan. Refer to the “Displaying member shelf information of a consolidated node” in <i>Administration and Security</i>, 323-1851-301. Verify the node information is synchronized and/or supported by the primary shelf. <table> <tr> <td>If the member shelf has a</td><td>Then go to</td></tr> <tr> <td>Software version unsupported by the primary shelf</td><td>step 4</td></tr> <tr> <td>Function provisioned unsupported by the primary shelf</td><td>step 5</td></tr> </table>	If the member shelf has a	Then go to	Software version unsupported by the primary shelf	step 4	Function provisioned unsupported by the primary shelf	step 5
If the member shelf has a	Then go to						
Software version unsupported by the primary shelf	step 4						
Function provisioned unsupported by the primary shelf	step 5						
4	 CAUTION Risk of communication loss Failure to synchronize the software release can result in commands not being recognized or interpreted properly by the software system, which may affect management or communications to that member shelf. Upgrade the member shelf to a software version supported by the primary shelf. Refer to the “<i>Software Upgrade Procedures</i>” section in <i>Planning - Ordering Information</i>, 323-1851-151. Go to step 7.						
5	 CAUTION Risk of loss of functionality Traffic and data communications will be lost Decommissioning a shelf results in a loss of all traffic and data communications associated with the shelf that is being decommissioned.						
6	Record all of the provisioning information required to recommission the shelf. Decommission the shelf and re-add it with the correct Function provisioned. Refer to the “Deleting all shelf provisioning” procedure in <i>Administration and Security</i>, 323-1851-301.						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-202

Member Shelf Unreachable

Alarm ID: 717**Probable cause**

This alarm is raised on a primary shelf of a consolidated node when the primary shelf cannot communicate with one or more member shelves. This can occur when the member shelf has an incorrect IP address provisioned.

Note: When a member shelf is restarted, the Member Shelf Unreachable alarm is raised for the duration of the member restart. When a primary shelf is restarted, the Member Shelf Unreachable alarm is raised briefly.

Impact

Major, non-service-affecting (M, NSA)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have a valid primary shelf provisioned for the consolidated node
- have the IP address of all shelves within the consolidated node
- have an available network plan, or other documentation identifying shelf provisioning within the consolidated node

Step	Action
1	Clear any Primary Shelf Unreachable alarm present. Refer to Procedure 4-259, "Primary Shelf Unreachable" alarm procedure.
2	Log into the primary shelf raising the alarm.
3	Retrieve the Node Information table for the consolidated node. Refer to the "Displaying member shelf information of a consolidated node" in <i>Administration and Security</i> , 323-1851-301.
4	Identify which shelf/shelves is/are causing the alarm. The Secondary state parameter for the unreachable member shelf indicates "UNEQ". The affected shelf/shelves may have '(?)' next to the shelf number and be highlighted in cyan. Refer to the "Displaying member shelf information of a consolidated node" procedure in <i>Administration and Security</i> , 323-1851-301.

Procedure 4-202 (continued)
Member Shelf Unreachable

Step	Action				
5	<table><tr><td>If within the consolidated node only one shelf displays a fault</td><td>Then the issue is with that member shelf. Go to step 6.</td></tr><tr><td>all shelves display the same fault</td><td>the primary shelf. Go to step 7.</td></tr></table>	If within the consolidated node only one shelf displays a fault	Then the issue is with that member shelf. Go to step 6 .	all shelves display the same fault	the primary shelf. Go to step 7 .
If within the consolidated node only one shelf displays a fault	Then the issue is with that member shelf. Go to step 6 .				
all shelves display the same fault	the primary shelf. Go to step 7 .				
6	Log in directly into the affected member shelf, and ensure all provisioning required for the member shelf of a consolidated node to operate is correct and without issue. Direct shelf access may be required. Refer to the “Consolidated node (TIDc)” section and the “Logging into a network element using a direct network connection to the LAN port on the shelf processor” procedure in <i>Administration and Security</i>, 323-1851-301, for more information. For example, verify and if necessary correct: • Primary state of member shelf • IP address provisioning • IP route provisioning • OAM communication • LAN/ILAN port status • OSPF provisioning • consolidated node parameter provisioning • Software release and upgrade state is applicable Refer to the “Displaying member shelf information of a consolidated node” procedure in <i>Administration and Security</i>, 323-1851-301, and the “Retrieving communications settings” procedure in <i>Data Communications Planning & User Guide</i>, 323-1851-101. If the alarm does not clear, go to step 7.				

Procedure 4-202 (continued)

Member Shelf Unreachable

Step	Action
7	Ensure all provisioning required for the primary shelf of a consolidated node to operate is correct and without issue. For example, verify and if necessary correct: • Primary state of primary shelf • IP address provisioning • IP route provisioning • OAM communication • LAN/ILAN port status • OSPF provisioning • consolidated node parameter provisioning • Software release and upgrade state is applicable Refer to the “Displaying member shelf information of a consolidated node” procedures in <i>Administration and Security</i>, 323-1851-301, and the “Retrieving communications settings” procedure in <i>Data Communications Planning & User Guide</i>, 323-1851-101.
8	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-203

Minimum Gain

Alarm ID: 1583

Probable cause

This alarm is raised when an amplifier is at its minimum extended range gain setting and the peak channel power overshoots the provisioned peak power target by 2 dB due to one of the following:

- the actual span loss is less than designed value
- wrong pad/DCSM placement at the amplifier site or upstream sites
- faulty pads which results in incorrect padding value (for example, faulty 10 dB pad only yields 5 dB attenuation)
- improper provisioning of the amplifier or upstream amplifier peak power targets (TARGPKPOW)
- the MLA2v, MDRLA 2x6 or RLA 5x1 VOA has not been optimized by DOC or has not been provisioned properly if manual provisioning is being used
- upstream DOC faults

Note: DOC checks for the Minimum Gain condition on each Auto Monitor cycle.

ATTENTION

There may be Minimum Gain alarms present at other line amplifiers on the link. Minimum Gain alarms do not mask downstream Minimum Gain alarms.

Impact

Minor, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have a valid primary shelf provisioned for the consolidated node
- have the IP address of all shelves within the consolidated node
- have an available network plan, or other documentation identifying shelf provisioning within the consolidated node

Procedure 4-203 (continued)

Minimum Gain

Step	Action
1	If there are more than one Minimum Gain alarms in the span, begin troubleshooting upstream of the first amplifier with the alarm.
2	Check if upstream DOCs have faults. If so, troubleshoot the upstream DOC faults first.
3	Check if the equipment is deployed according to network design (for example, the design requires an MLA, but an MLA2 was deployed). Make sure the correct equipment is used.
4	Check if the Peak Power Target is provisioned as designed. If not, correct it. Refer to the “Provisioning photonic parameters” and “Provisioning adjacencies” procedures in the <i>Commissioning and Testing</i> , 323-1851-221.
5	For line facing amplifiers, check if the actual span loss (measured OSC span loss includes the pads and DSCMs placed on the line) is less than the design value. If it is less, pad the span to the designed value. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
6	If there is an MLA2v amp upstream, reset the VOA on the MLA2v.
7	Check if the pads and DSCMs are deployed in the designed places. If not, correct them.
8	If the alarm does not clear, replace the pads.
9	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-204

Modem Class Mismatch

Alarm IDs: 2023

Probable cause

This alarm is raised on the PTP facility of the WLAi variants circuit pack when a line port detects mismatch of modem class between the two ends of the link.

Modem class is used to allow the user to provision the appropriate license type for Submarine or Terrestrial applications on WLAi variants line port. The Modem class license is released when the WLAi line port is deleted.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC

Step	Action
1	Verify that the Modem class at each side of the link is the same. Refer to “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	If they are different, re-provision the WLAi line port PTP to make sure the Modem Class is the same on both ends of the link. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

—end—

Procedure 4-205

MSI Mismatch (ODUTTP, ODUCTP, OTM0, OTM1, OTM2, OTM3, ODU0, ODU1, ODUFLEX, OTMFLEX)

Alarm ID: 1475, 1584, 1607, 1733, 1739, 1848

Probable cause

For OTM facilities, this alarm is raised on the client mapping layer when the client has a cross connection to an OTM port. This can be caused by:

- incorrect MSI received for the service provisioned
- incorrect BW for the Service ID
- no connection or “msi=unused” received for a service provisioned on this client port

For ODUTTP and ODUCTP facilities, this alarm is raised when the Rx interface detects no MSI is provisioned for the ODUTTP or ODUCTP. This could be caused when a far-end ODUTTP or ODUCTP is not provisioned yet or is provisioned but has a different tribPort than the local ODUTTP or ODUCTP. The tribPort for an ODUTTP or ODUCTP must be the same at both ends of the network.

This alarm is not applicable to ODUTTP/ODUCTP facilities on the OTN I/F Flex 16xSFP (NTK622AA) circuit pack.

For the (1+8)xOTN Flex MOTR (NTK532DA, NTK532DE) circuit pack, this alarm is raised when the expected Rx MSI does not match the actual Rx MSI for the corresponding facility (when the tributary slot assignment is in manual mode). In this case, conditioning is also applied in the off-ramp direction for the corresponding connection.

When the tributary slot assignment is in automatic mode, the alarm is raised against the ODU connection if the correct number of tributary port number and ODU payload type (ODTU type) cannot be found in the incoming OPU2 payload. In this case, you only have to specify the tributary port number for each ODU connection. All other relevant attributes are either derived or retrieved from the system.

The alarm is also raised when the Tributary Slot Provisioning parameter is provisioned to MANUAL and at least one Rx tribSlot matches the expected Rx tribPort, but at least one other Rx tribSlot doesn't match what is expected.

For the 100G (2xQSFP+/2xSFP+) MUX circuit pack, this alarm is raised when one end is provisioned with a 4x10G ETH facility and the other end is provisioned with a 40G ETH facility.

Procedure 4-205 (continued)

MSI Mismatch (ODUTTP, ODUCTP, OTM0, OTM1, OTM2, OTM3, ODU0, ODU1, ODUFLEX, OTMFLEX)

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	Identify the port where this alarm is being raised. Identify the OTM, ODU, ODUTTP, or ODUCTP port to which it is currently connected.						
2	Identify all the connections to the OTM, ODU, ODUTTP or ODUCTP port.						
3	<table> <tr> <td>If the original alarm is raised against an</td><td>Then go to</td></tr> <tr> <td>OTM or ODU facility</td><td>step 4</td></tr> <tr> <td>ODUTTP or ODUCTP facility</td><td>step 14</td></tr> </table>	If the original alarm is raised against an	Then go to	OTM or ODU facility	step 4	ODUTTP or ODUCTP facility	step 14
If the original alarm is raised against an	Then go to						
OTM or ODU facility	step 4						
ODUTTP or ODUCTP facility	step 14						
4	Choose one connection and identify its payload index.						
5	At both ends of this fiber: - Verify that the payload index used for the ODU0, ODU1, ODUFLEX connection to the OTM facility is the same at both the near-end and far-end. If it is not, then change one end. - If the identified far-end is an OTM to OTM facility connection, verify that the connection rate (for example, ODU0, ODU1, ODUFLEX) for this payload index is the same at this site and the near-end site. If the connection rate is different at each end then correct the connection provisioning at the OTM to OTM or ODU to ODU connection site.						
6	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	If the connection rate is the same at each end, then identify the OTM port at the site upstream from this one and go to step 5 .						
8	If the identified far-end is a OTM to FLEX or OTM1 facility connection, verify that the connection rate (for example, ODU0, ODU1, or ODUFLEX) with the payload index is the same at both ends of the fiber.						

Procedure 4-205 (continued)

MSI Mismatch (ODUTTP, ODUCTP, OTM0, OTM1, OTM2, OTM3, ODU0, ODU1, ODUFLEX, OTMFLEX)

Step	Action						
9	If the connection rate is not the same, then the protocols at each end of the fiber are mismatched and they should not be using the same payload index for their connections. Delete the connections and re-apply with the correct payload index.						
10	If the connection rate is the same and the alarm is still present, contact your next level of support or your Ciena support group.						
11	For the 100G (2xQSFP+/2xSFP+) MUX circuit pack, if one end is provisioned with a 4x10G ETH facility and the other end is provisioned with a 40G ETH facility, then delete the incorrectly provisioned facility and re-add it so that it matches the ETH40G/4x10G ETH facility mode at the other end.						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4, choose another connection and repeat step 4 to step 10</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4 , choose another connection and repeat step 4 to step 10
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4 , choose another connection and repeat step 4 to step 10						
13	If there are no more connections and the alarm is not cleared, go to step 17 .						
14	Make the tributary port provisioning match at both ends of the network.						
15	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 16</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 16
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 16						
16	Delete the ODUTTP or ODUCTP that is raising the alarm.						
17	If the alarm is not cleared, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-206

Multiplexed Rate Mismatch

Alarm ID: 1031

Probable cause

This alarm is raised against:

- an OTM2 facility of a 40G MUX OCI, 100G (2xQSFP+/2xSFP+) MUX, 200G (2x100G/5x40G) MUX, or 10x10G MUX circuit pack when a rate mismatch is detected between the local OTM2 facility and the OTM2 demultiplexed from the OTM3 or OTM4 facility. This occurs when the OTM2 rate of the 10G client provisioning at the near-end 40G MUX OCI, 100G (2xQSFP+/2xSFP+) MUX, 200G (2x100G/5x40G) MUX, or 10x10G MUX does not match the OTM2 rate of the 10G client provisioning of the same port at the far-end 40G MUX OCI, 100G (2xQSFP+/2xSFP+) MUX, 200G (2x100G/5x40G) MUX, or 10x10G MUX OCI.
- a 40G XCIF OTM2 facility when the OTU2 rate received from the backplane side does not match the expected backplane OTU2 rate.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	Identify the OTM2 facility raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	Verify if there is a difference between the Received rate from OTM2 and the Expected rate from OTM2 values in the OTM Info dialog box. Refer to the "Retrieving OTM info" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
3	<table><tr><th>If the values</th><th>Then</th></tr><tr><td>match</td><td>contact your next level of support or Ciena support group. The procedure is complete.</td></tr><tr><td>do not match</td><td>go to step 4</td></tr></table>	If the values	Then	match	contact your next level of support or Ciena support group. The procedure is complete.	do not match	go to step 4
If the values	Then						
match	contact your next level of support or Ciena support group. The procedure is complete.						
do not match	go to step 4						
4	Determine if it is the Received rate from OTM2 or the Expected rate from OTM2 value that is incorrect.						

Procedure 4-206 (continued)
Multiplexed Rate Mismatch

Step	Action	
5	If the incorrect value is the Expected rate from OTM3/OTM4	Then delete and reprovision the local OTM2 client facility to the correct rate. Refresh the OTM Info dialog box to verify the correct Expected rate from OTM2 value is displayed. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
	Received rate from the OTM3/OTM4	delete and reprovision the far-end OTM2 client facility to the correct rate. Refresh the OTM Info dialog box to verify the correct Received rate from OTM2 value is displayed. Refer to the “Retrieving OTM info” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
6	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-207

NE Mode Unknown

Alarm ID: 530

Probable cause

This alarm is raised when the network element mode has not been set and is unknown. The alarm is cleared when the user sets the network element mode during the commissioning process.

For Broadband services, this alarm can also be raised after a shelf processor replacement.

Before the shelf can be fully provisioned, the **NE mode** must be changed to SONET, SDH, as appropriate:

- for Broadband services, the mode can either be SONET or SDH
- for converged Broadband and Photonic services, the mode can either be SONET or SDH
- for Photonic services, (even though the mode is not relevant to Photonics) the mode can either be SONET or SDH

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Provision the network element mode. Refer to the “Editing the nodal general parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-208

NMCC Action Failed: Add

Alarm ID: 2169

Probable cause

This alarm is raised when a Channel Add command (automatic or manual) in NMCC is requested and fails. Conditions that can cause this failure include:

- maintenance activities, such as module replacement, module restart, and fiber cut occurred during the Channel Add command
- an internal communications issue (for example, the ILAN port is not cabled)
- a site to site communication issue (for example, OSC communications problems caused by mis-fibering of the OSC plugs)
- a fiber disconnection within the site, such as the fiber between WSS to WSS of two ROADMs OTS, or the fiber from WSS common output port to Booster AMP input port of the ROADMs OTS.
- a provisioning error on the Rx or Tx adjacency
- an associated adjacency or AMP facility is deleted
- the NMCC Action: Channel Add was stopped before the addition was completed
- provisioning mismatch between the SSC facility reference bandwidth and the OTS reference bandwidth
- Unaccounted loss in the path of the channel
- Cross Fibering between the TX and RX port of the CMD

The alarm raises separately for each channel that failed to add.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) containing provisioning details

Procedure 4-208 (continued)

NMCC Action Failed: Add

Step	Action
1	If maintenance activities, such as module replacement, module restart, or fiber cut occurred during the Add channel operation, then this alarm will be active. Ensure that the maintenance activities have completed.
2	Check for and clear any of the following alarms on all the network elements that are active within the NMCC: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Channel Controller: Unexpected Loss Detected • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Loss of Signal (LOS) • Cross fiber suspected • High Fiber Loss alarm (Major/SA) • Input Loss of Signal • Output Loss of Signal • Channel Opacity Error • DOC Power Audit Failed • Optical Line Fail • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first.
3	Find the channel that failed to add. On the NMCC facility corresponding to the channel compare the Derived Input Power with the Expected Input power for the channel on the NMCC Facility corresponding to the channel. In the event of a large delta, check for excess loss in the channel path or if the TX and RX fibers are crossed.
4	Clear all other active alarms on the network elements within the NMCC. For optimal NMCC operation, the system must be alarm free.

Procedure 4-208 (continued)

NMCC Action Failed: Add

Step	Action
	Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element.
5	Verify, and if necessary, correct the following provisioned data (reference the EDP): • Tx and Rx adjacency parameters (Adjacency type and all power-related parameters) • Line adjacency parameter (Fiber Type) • AMP optical facility parameters (Mode, Target Gain, Target Power, Target Peak Power, and Reference Bandwidth)
6	Check the subtending connections to the CMD44. Make sure that the connections, power levels and wavelengths are good.
7	After step 4 , step 5 and step 6 , wait for about 5 minutes. The DOC will attempt to automatically re-add the channel. If the previous failure has been cleared, DOC should be able to successfully add the channel. Subsequently, the "NMCC Action Failed: Add" alarm will be cleared.
8	If the NMCC alarm remains active, go to step 9 .
9	Delete the channels that were unsuccessfully added under NMCC control. If the delete operation fails, perform a Force Delete operation (service-affecting action). Note: You must successfully delete the channels that were unsuccessfully added under NMCC control before re-attempting to add them.
10	Re-attempt to add the channels.
11	If the alarm does not clear, or clears then returns, or the channels cannot be added, contact your next level of support or your Ciena support group.

—end—

Procedure 4-209

NMCC Action Failed: Delete

Alarm ID: 2170

Probable cause

This alarm is raised when a Channel Delete command (automatic or manual) in NMCC is requested and fails. Conditions that can cause this failure include:

- maintenance activities, such as module replacement, module restart, and fiber cut occurred during the Channel Delete operation
- an internal communications issue (for example, the ILAN port is not cabled)
- a site to site communication issue (for example, OSC communications problems caused by mis-fibering of the OSC plugs)
- an optical disconnect, such as a fiber break, within the NMCC
- a provisioning error on the Rx or Tx adjacency
- an associated adjacency or AMP facility is deleted
- the NMCC Action: Channel Delete was stopped before the deletion was completed

The alarm raises separately for each channel that failed to delete.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the engineering documentation package (EDP) containing provisioning details

Procedure 4-209 (continued)

NMCC Action Failed: Delete

Step	Action
1	If maintenance activities, such as module replacement, module restart, or fiber cut occurred during the Delete channel operation, then this alarm will be active. Ensure that the maintenance activities have completed.
2	Check for and clear any of the following alarms on all the network elements that are active within the NMCC: • Adjacency Far End Not Discovered • Adjacency Provisioning Error • Automatic Power Reduction Active • Automatic Shutoff • Channel Controller: Unexpected Loss Detected • Circuit Pack Failed • Circuit Pack Mismatch • Circuit Pack Missing • Circuit Pack Unknown • Gauge Threshold Crossing Alert Summary • Loss of Signal (LOS) • Cross fiber suspected • Channel Opacity Error • Optical Line Fail • OSC Loss of Signal • Shutoff Threshold Crossed Note: If the Optical Line Fail alarm is active, clear this alarm first.
3	Clear all other active alarms on the network elements within the NMCC. For optimal NMCC operation, the system must be alarm free. Use the DOC Logs window to view the DOC logs to determine if another network element reported an error. Review the DOC logs from the other network element.

Procedure 4-209 (continued)
NMCC Action Failed: Delete

- | Step | Action |
|------|---|
| 4 | Verify, and if necessary, correct the following provisioned data (reference the EDP): <ul style="list-style-type: none">• Tx and Rx adjacency parameters (Adjacency type and all power-related parameters)• Line adjacency parameter (Fiber Type)• AMP optical facility parameters (Mode, Target Gain, Target Power, and Target Peak Power) |
| 5 | After step 3 and step 4 , wait for about 5 minutes. The DOC will attempt to automatically re-delete the channel. If the previous failure has been cleared, DOC will be able to successfully delete the channel. Subsequently, the “NMCC Action Failed: delete” alarm will be cleared. |
| 6 | If the NMCC alarm remains active, perform step 7 . |
| 7 | Re-attempt the Delete operation. |

ATTENTION

Any channels that were in the delete queue are returned to their previous state when the deletion fails. These channels must be re-selected for deletion.

- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-210

No Protection Route Available

Alarm ID: 2263, 2264

Probable cause

This alarm is raised for MR-SNC/SNCGs during a protection routes audit when none of the user provisioned protected routes are available.

The audit is performed every 5 minutes. Therefore, there is an expected delay between the time the state actually changes and the alarm is raised or cleared.

The alarm is disabled by default in the alarm profiles to avoid flooding the network with new alarms upon an upgrade.

An exception is the ALL ENABLED profile, where the new alarm is enabled by default or when not using the ALL ENABLED profile, after an upgrade. You can enable the alarm in the primary shelf alarm profiles.

Impact

Major, non-service-affecting (M, NSA)

Step	Action
1	Ensure at least one SNC/SNCG protection route is available on the next protect routes audit (happening every 5 minutes).
2	Select the SNC/G from the Configuration->Sub Network Connections application in Site Manager.
3	Click on "Routes" to determine the current working path.
4	Under the Routing Info tab, verify the "Routing List" and select that Routing List from the Configuration->Routing Profiles application in Site Manager.
5	Select the Routes in that Routing List under the "Routes" tab that are not currently the working Route.
6	Note the nodes and links that are part of the other Routes in that Routing List.
7	Log into the nodes shown in the "Hops Details" and check for alarms related to those links.
8	Clear the link related alarms on those nodes to bring the protection route back into service.
9	If the alarm is not clear, send a release to SNC/SNCG. <i>The alarm can be raised again if condition persists after the event.</i>
10	If the alarm is not clear, perform a lock on the SNC/SNCG.

Procedure 4-210 (continued)

No Protection Route Available

Step	Action
11	If the alarm is not clear, perform an AUXP restart. <i>The alarm can be raised again if condition persists after the event.</i>
12	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-211

Number of Level 1 NEs Exceeded

Alarm ID: 805**Probable cause**

This alarm is raised when the number of network elements supported in the data communication channel (GCC/DCC) domain exceeds the maximum supported. The maximum number of network elements supported in the GCC/DCC domain is 150. This alarm applies only when iISIS is provisioned on the GCC/DCC channels.

Impact

Minor, non-service-affecting (m, NSA)

Step	Action
------	--------

- | | |
|---|---|
| 1 | Re-engineer the network to reduce the number of network elements in the Level 1 routing area. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-212

OCH Link Data Retrieval In Progress

Alarm IDs: 1529

Probable cause

This event is raised when there is a Link Data retrieval session in progress.

Impact

Warning

Step	Action
1	No action is required. The alarm clears once the retrieval session is complete.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-213

OCH Link Data Save In Progress

Alarm IDs: 1530**Probable cause**

This event is raised when there is a Link Data save in progress.

Impact

Warning

Step	Action
1	No action is required. The event clears once the save session is complete.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-214

ODU LCK

Alarm IDs: 962, 994, 1162, 1470, 1481, 1482, 1497, 1498, 1610, 1731

Probable cause

This alarm is raised when the Rx interface detects an ODU layer (path monitoring) LCK (locked). The upstream Tx interface is sending ODU layer LCK. This can occur when the upstream OTM1 (for OTN Flex MOTR circuit packs), facility is in the OOS state.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Verify if any upstream facilities are Out Of Service. |
| 2 | If the facilities are supposed to be In Service and are OOS, place them In-Service.

Placing the facilities In Service will clear the alarm. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-215

ODU Loss of Frame and Multiframe

Alarm IDs: 1474, 1489, 1490, 1734

Probable cause

This alarm is raised when there is a mismatch in the configuration of the equipment at the near end and far end of the network.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Verify if the ODU services are provisioned for the same rate at both ends of the network and correct the rate accordingly. For example you cannot have a service provisioned for an ODU2 at one end and an ODU2e at the other end, or an ODU Flex at one end and a different ODU rate at the other end. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-216

ODU OCI

Alarm IDs: 963, 995, 1163, 1471, 1485, 1486, 1499, 1500, 1609, 1732

Probable cause

This alarm is raised when the Rx interface detects that there is a cross-connection missing along the connection path for this service. The upstream Tx interface is sending ODU layer OCI. This can occur when the upstream, ODUCTP, ODU0, ODU1, ODUFLEX, OTM0, OTM1, facility is not connected.

Impact

Minor, non-service-affecting (m, NSA) alarm

Critical, service-affecting (C, SA) alarm

Step	Action
1	Ensure that there are cross-connects provisioned at every piece of equipment along the path.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-217

ODU Skew Out Of Range

Alarm ID: 1948

Probable cause

This alarm is raised against the OTM3 facility of a 200G (2x100G/5x40G) MUX (NTK529HA) circuit pack when the system detects that the virtual lane skew exceeds the specified skew range from G.798.

Impact

Critical, service-affecting (C, SA) alarm, unprotected
Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain the replacement fiber for connection between the subtending equipment and the QSFP28
- if required, obtain a replacement QSFP28 module
- use an account with at least a level 3 UPC

Step	Action						
1	Determine the type of the connection fiber between the subtending equipment and the QSFP28 module based on QSFP28 type.						
2	<table><tr><td>If the fiber type is</td><td>Then go to</td></tr><tr><td>SMF</td><td>step 3</td></tr><tr><td>MMF</td><td>step 8</td></tr></table>	If the fiber type is	Then go to	SMF	step 3	MMF	step 8
If the fiber type is	Then go to						
SMF	step 3						
MMF	step 8						
3	Ensure the Integrated Test Feature (ITS) is available on the 200G (2x100G/5x40G) MUX and operating in Test Set mode. Refer to the “Performing a test with the Integrated Test Set” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-217 (continued)

ODU Skew Out Of Range

Step	Action
4	Ensure the Rx/Tx port on the QSFP28 is fiber loop backed using a patch cord fiber.
5	If the alarm does not clear, replace the QSFP28. Refer to the “Replacing a Pluggable module” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.
6	If the original alarm has cleared, then restore the original configuration and the procedure is complete. Otherwise, go to step 7 .
7	If the original alarm does not clear, it indicates the problem is caused by subtending equipment. Troubleshoot the subtending equipment. Go to step 14 .
8	Ensure that all parallel fibers connected between the subtending equipment and QSFP28 have the same length. If not, then replace them to make them comply with this requirement.
9	If the original alarm does not clear, ensure the Integrated Test Feature (ITS) is available on the 200G (2x100G/5x40G) MUX OCI circuit pack and operating in Test Set mode.
10	Ensure the Rx/Tx port on the QSFP28 is fiber loop backed using a special MPO24 loopback fiber.
11	If the alarm does not clear, replace the QSFP28. Refer to the “Replacing a Pluggable module” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.
12	If the original alarm has cleared, then restore the original configuration with the new QSFP28 and the procedure is complete. Otherwise, go to step 13 .
13	If the original alarm does not clear, it indicates the problem is caused by subtending equipment. Troubleshoot the subtending equipment.
14	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-218

ODU Signal Degrade

Alarm ID: 1476, 1491, 1492, 1503, 1504, 1611, 1762, 1781, 1787, 1863

Probable cause

This alarm is raised when the Rx interface detects an ODU layer signal degrade.

For ODU port types on eMOTR in Layer 2 Extended mode and PKT/OTN circuit packs crossing the Signal Degrade threshold could be configured to trigger a switchover within an MPLS/MPLS-TP network. For details refer to *SAOS-based Packet Services Configuration*, 323-1851-630.

For 6500 Release 10.05 and higher, a Signal Degrade on a line port triggers a switch in TPT and SNCP Protection configurations for 4x10G OTR (NTK530QA, NTK530QM, NTK530QP) and 1+8xOTN Flex MOTR (NTK532DA, NTK532DE) circuit packs. A Signal Degrade on the client ports only triggers an alarm (no protection switch occurs). The ODU Signal Degrade threshold is fixed at 10E-9 or 10E-8.

Impact

Major, service-affecting (M, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Critical, service-affecting (C, SA) alarm

The alarm will become Critical, service-affecting if:

- a single VCE has been configured on the VCS
- the L2 VCE or EVPL is configured only at one end (the end that has the L2 VCE or EVPL connection) of the OTM2 line
- the L2 VCE endpoint map has not been configured on both 10GE child facilities of the OTM2 line

Procedure 4-218 (continued)
ODU Signal Degrade

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the network connection information (that is, how the interface circuit packs on each network element connect to other network elements)

Step	Action
------	--------

1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document. First, clear any alarms of higher order in the hierarchy on the 6500 system or the subtending equipment using the appropriate procedures. You can also refer to the signal conditioning section in this document to help troubleshoot secondary alarms.
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2	At the local network element, retrieve all alarms to determine if the original alarm has cleared.
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3	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
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4	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.
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Remove the optical fiber from the circuit pack raising the alarm and use the optical power meter to measure the receive power.

5	<table><thead><tr><th>If the power is</th><th>Then go to</th></tr></thead><tbody><tr><td>below the receiver sensitivity for this circuit pack</td><td>step 6</td></tr><tr><td>above the receiver sensitivity for this circuit pack</td><td>step 9</td></tr></tbody></table>	If the power is	Then go to	below the receiver sensitivity for this circuit pack	step 6	above the receiver sensitivity for this circuit pack	step 9
If the power is	Then go to						
below the receiver sensitivity for this circuit pack	step 6						
above the receiver sensitivity for this circuit pack	step 9						

For information about circuit pack technical specifications, refer to the *Planning - Ordering information*, 323-1851-151.

Power is below the receiver sensitivity

6	Adjust the local attenuation, if equipped, to try to get the receive power above the receiver sensitivity level.
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Procedure 4-218 (continued)
ODU Signal Degrade

Step	Action	
7	If the receive power after adjustment is still below the receiver sensitivity	Then go to step 8
	above the receiver sensitivity but below the maximum receiver power	step 9
8	Remove the Tx optical fiber from the far-end circuit pack and measure the transmit power at the far-end.	
	If the transmit power at the far-end is	Then
	above the launch power (minimum)	the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 10 .
	below the launch power (minimum)	replace the module that corresponds to the facility raising the alarm. Refer to the "Replacing a Pluggable module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 10 .

Power is above the receiver sensitivity

- 9 Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers.

Determining if the alarm has cleared

10	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	replace the module that corresponds to the facility raising the alarm. Refer to the "Replacing a pluggable module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.

Procedure 4-218 (continued)

ODU Signal Degrade

Step	Action								
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i>, 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.								
12	Retrieve all alarms from the remote network element at the transmit end.								
13	Look for an alarm message for the remote network element circuit pack connected to the original shelf.								
14	<table> <tr> <th>If there are</th><th>Then</th></tr> <tr> <td>no alarms at the transmit end</td><td>ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 15.</td></tr> <tr> <td>additional alarms at the transmit end</td><td>refer to the appropriate alarm clearing procedures. Go to step 15.</td></tr> </table>	If there are	Then	no alarms at the transmit end	ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 15 .	additional alarms at the transmit end	refer to the appropriate alarm clearing procedures. Go to step 15 .		
If there are	Then								
no alarms at the transmit end	ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 15 .								
additional alarms at the transmit end	refer to the appropriate alarm clearing procedures. Go to step 15 .								
15	At the local network element, retrieve all alarms to determine if the original alarm has cleared.								
16	<table> <tr> <th>If the original alarm</th><th>Then</th></tr> <tr> <td>has cleared</td><td>the procedure is complete</td></tr> <tr> <td>raised was against OTUTTP or ODUTTP facility and the alarm has not cleared</td><td>go to step 17</td></tr> <tr> <td>raised was otherwise, and the alarm has not cleared</td><td>go to step 18</td></tr> </table>	If the original alarm	Then	has cleared	the procedure is complete	raised was against OTUTTP or ODUTTP facility and the alarm has not cleared	go to step 17	raised was otherwise, and the alarm has not cleared	go to step 18
If the original alarm	Then								
has cleared	the procedure is complete								
raised was against OTUTTP or ODUTTP facility and the alarm has not cleared	go to step 17								
raised was otherwise, and the alarm has not cleared	go to step 18								
17	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches.								
18	Contact your next level of support or your Ciena support group.								

—end—

Procedure 4-219

ODU Signal Fail

Alarm ID: 1761, 1782, 1786, 1789, 1790, 1791, 1792, 1793, 1794, 1862

Probable cause

This alarm is raised when the BIP-8 error is higher than the provisioned value on the alarmed circuit pack.

Impact

Critical, service-affecting (C, SA) alarm

Major, service-affecting (M, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the network connection information (that is, how the interface circuit packs on each network element connect to other network elements)

Step	Action
1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document. First, clear any alarms of higher order in the hierarchy on the 6500 system or the subtending equipment using the appropriate procedures. You can also refer to the signal conditioning section in this document to help troubleshoot secondary alarms.
2	At the local network element, retrieve all alarms to determine if the original alarm has cleared.
3	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
4	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal. Remove the optical fiber from the circuit pack raising the alarm and use the optical power meter to measure the receive power.

Procedure 4-219 (continued)
ODU Signal Fail

Step	Action	
5	If the power is	Then go to
	below the receiver sensitivity for this circuit pack	step 6
	above the receiver sensitivity for this circuit pack	step 9
For information about circuit pack technical specifications, refer to the <i>Planning - Ordering information</i> , 323-1851-151.		

Power is below the receiver sensitivity

6	Adjust the local attenuation, if equipped, to try to get the receive power above the receiver sensitivity level.	
7	If the receive power after adjustment is	Then go to
	still below the receiver sensitivity	step 8
	above the receiver sensitivity but below the maximum receiver power	step 9
8	Remove the Tx optical fiber from the far-end circuit pack and measure the transmit power at the far-end.	
	If the transmit power at the far-end is	Then
	above the launch power (minimum)	the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Use your company procedure to determine and clear the problem. Go to step 10 .
	below the launch power (minimum)	replace the module that corresponds to the facility raising the alarm. Refer to the "Replacing a Pluggable module" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 10 .

Power is above the receiver sensitivity

9	Clean all connections at both ends of the optical fiber link following your company standards and re-attach the optical fibers.
---	---

Procedure 4-219 (continued)
ODU Signal Fail

Step	Action
Determining if the alarm has cleared	
10	If the original alarm has cleared
	Then the procedure is complete
11	If the original alarm has not cleared
	Then replace the module that corresponds to the facility raising the alarm. Refer to the “Replacing a pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
12	If the original alarm has cleared
	Then the procedure is complete
13	If the original alarm has not cleared
	Then replace the circuit pack reporting the alarm. Refer to the equipment replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Then, clean and re-attach both optical fibers. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
12	Retrieve all alarms from the remote network element at the transmit end.
13	Look for an alarm message for the remote network element circuit pack connected to the original shelf.
14	If there are no alarms at the transmit end
	Then ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 15 .
14	If there are additional alarms at the transmit end
	Then refer to the appropriate alarm clearing procedures. Go to step 15 .

Procedure 4-219 (continued)

ODU Signal Fail

Step	Action								
15	At the local network element, retrieve all alarms to determine if the original alarm has cleared.								
16	<table><tr><th>If the original alarm</th><th>Then</th></tr><tr><td>has cleared</td><td>the procedure is complete</td></tr><tr><td>raised was against OTUTTP or ODUTTP facility and the alarm has not cleared</td><td>go to step 17</td></tr><tr><td>raised was otherwise, and the alarm has not cleared</td><td>go to step 18</td></tr></table>	If the original alarm	Then	has cleared	the procedure is complete	raised was against OTUTTP or ODUTTP facility and the alarm has not cleared	go to step 17	raised was otherwise, and the alarm has not cleared	go to step 18
If the original alarm	Then								
has cleared	the procedure is complete								
raised was against OTUTTP or ODUTTP facility and the alarm has not cleared	go to step 17								
raised was otherwise, and the alarm has not cleared	go to step 18								
17	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches.								
18	Contact your next level of support or your Ciena support group.								

—end—

Procedure 4-220

ODU/OTU Trace Identifier Mismatch

Alarm IDs: 670, 836, 1003, 1010, 1172, 1180, 1447, 1472, 1483, 1501, 1502, 1743

Probable cause

This alarm is raised when the trail trace identifier (TTI) value received by the facility differs from the expected provisioned TTI value. This can be caused by incorrect fiberling or incorrect provisioning of the TTI value.

ODU/OTU TTI mismatch detection are independent of each other. Mismatches of TTI can be declared based on the comparison of the Actual received TTI and Expected received TTI values of the following components:

- Source Access Point Identifier (SAPI) only
- Destination Access Point Identifier (DAPI) only
- SAPI + DAPI
- Operator specific (default)

The TCM TTI mismatch detection is based on the comparison of the Actual received TTI and Expected received TTI values of the following components:

- Source Access Point Identifier (SAPI) only (default)
- Destination Access Point Identifier (DAPI) only
- SAPI + DAPI

For the (1+8)xOTN FLEX MOTR circuit pack, a low order ODU Trace Identifier Mismatch alarm can be raised against the low order ODU facilities.

For the eMOTR circuit pack, the alarm can be raised against the ODUTTP and OTUTTP facilities when the following is true:

TFMODE=ALONLY

TIMEN=SAPI, DAPI, SAPI+DAPI, or OPERATOR

Critical alarm will raise when there is mismatch in expected and receive trace of either of the following when:

TFMODE=ALONLY

TIMEN=SAPI, DAPI, SAPI+DAPI, or OPERATOR

Procedure 4-220 (continued)

ODU/OTU Trace Identifier Mismatch

For the 16xFLEX OTN I/F circuit pack, BDI is sent in response to TTI Mismatch if the following is true:

TFMODE=LINEFAIL or ALMONLY
TIMEN=SAPI, DAPI, SAPI+DAPI, or OPERATOR

For the 40G OTN XCIF circuit pack, BDI is sent in response to TTI Mismatch if the following is true:

TFMODE=LINEFAIL
TIMEN=SAPI, DAPI, SAPI+DAPI

For the 10x10G PKT/OTN I/F and 100G PKT/OTN XCIF circuit packs, BDI is sent in response to TTI Mismatch at the PM and TCM layers if the following is true:

TFMODE=LINEFAIL or ALMONLY
TIMEN=SAPI, DAPI, or SAPI+DAPI

Note: The LINEFAIL mode for SAPI and DAPI mismatch is only supported on 16xFLEX OTN I/F and 40G OTN XCIF circuit packs.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

The default severity for the “ODU Trace Identifier Mismatch” alarm is:

- m, NSA when ODUTIMEN is set to OPERATOR and ODUTFMODE is ALMONLY
- m, NSA when ODUTIMEN is set to SAPI and/or DAPI and ODUTFMODE is ALMONLY
- m, NSA when ODUTIMEN is set to SAPI and/or DAPI and ODUTFMODE is LINEFAIL on a protected path
- C, SA when ODUTIMEN is set to SAPI and/or DAPI and ODUTFMODE is LINEFAIL on an active/working path

The default severity for the “OTU Trace Identifier Mismatch” alarm is:

- m, NSA when OTUTIMEN is set to OPERATOR and TFMODE is ALMONLY or LINEFAIL
- m, NSA when OTUTIMEN is set to SAPI and/or DAPI and TFMODE is ALMONLY

Procedure 4-220 (continued)

ODU/OTU Trace Identifier Mismatch

- m, NSA when OTUTIMEN is set to SAPI and/or DAPI and TFMODE is LINEFAIL on a protected path
- C, SA when OTUTIMEN is set to SAPI and/or DAPI and TFMODE is LINEFAIL on an active/working path

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the optical fiber connection information (that is, how the modules on each network element connect to other network elements)
- use an account with at least a level 3 UPC

Step	Action						
1	Use the optical fiber connection information to identify the transmit and receive sites of the alarmed signal.						
2	Verify that the optical fiber connections are correct on the circuit pack that is raising the alarm and on the upstream circuit pack.						
3	If the channel is going over a Photonic line system, verify the fiber along the channel's path end-to-end.						
4	From the Site Manager Configuration menu, select the Equipment & Facility Provisioning application, and use the Trail Trace dialog box to retrieve the TTI values for the optical facilities at the transmit network element and at the alarmed receive network element. Refer to the “Retrieving and editing trail trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
5	Click Refresh in the Trail Trace dialog box.						
6	Select your next step.						
	<table> <tr> <th>If the alarm</th><th>Then</th></tr> <tr> <td>clears</td><td>this procedure is complete</td></tr> <tr> <td>does not clear</td><td>go to step 7</td></tr> </table>	If the alarm	Then	clears	this procedure is complete	does not clear	go to step 7
If the alarm	Then						
clears	this procedure is complete						
does not clear	go to step 7						
7	Compare the outgoing TTI value of the transmit signal at the transmit network element with the expected TTI value of the receive signal at the receive network element.						

Procedure 4-220 (continued)

ODU/OTU Trace Identifier Mismatch

Step	Action
------	--------

	If the values are different, change the expected TTI value of the receive signal to match the outgoing TTI value of the transmit signal. Refer to the “Retrieving and editing trail trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
--	--

- | | |
|----------|--|
| 8 | If the alarm does not clear, contact your next level of support or your Ciena support group. |
|----------|--|

—end—

Procedure 4-221

Optical Line Fail

Alarm IDs: 541, 610, 1718

Probable cause

This alarm is usually raised when there is a fiber break or an intermediate connector disconnect between neighboring sites. The alarm is raised against RAMAN, OPTMON and AMP facilities optically downstream of the fiber cut/disconnect.

For the SRA amplifiers, this alarm is raised when no OSC and no Telemetry Gain (TG) signal is received. Depending on the OSRP SNC configuration, the Optical Line Fail alarm can trigger a restoration of SNC services.

ATTENTION

For SRA and ESAM circuit packs, when there is an “optical line” condition on the “Line A In Port”, an “Automatic OTDR Trace” is triggered and a short and long trace OTDR is performed. During the OTDR Trace, an “OTDR Trace in Progress” alarm is raised. If there is an issue in the first 20 km of the Line A In (port 8), the “Line A Input OTDR High Reflection Detected” or “Line A Input OTDR High Loss Detected” alarm is raised.

The alarm is masked when you place the AMP and/or RAMAN facilities OOS.

The Optical Line Failure condition invokes “Automatic Laser ShutOff” (ALSO), which is a regulatory safety requirement that automatically shuts down the amplifier optically upstream of the cut/disconnect and in the opposite direction. The “Automatic Shutoff” alarm will also be present.

Whether one or both fiber directions have been cut will depend on whether one or two OLF alarms are present across the fiber span.



DANGER

Risk of radiation exposure

If light is used to test the broken fiber (for example, with a light source or an OTDR), certain Automatic Laser Shut Off (ALSO) and Loss Of Signal alarms can clear. When the shelf detects light, the alarms clear and the amplifier facility is powered up. This is an expected behavior because a shelf cannot distinguish between a light source from an optical test set and a light source from a shelf.

Ensure the adjacent optical amplifiers are out of service (OOS) when performing fiber repairs.

Procedure 4-221 (continued)
Optical Line Fail

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a fiber cleaning kit
- have a power meter and a power source

Step	Action
------	--------

- | | |
|---|--|
| 1 | <p>Use system design data to determine the controlling DOC instance for the link that is raising the alarm. Note that in case of bidirectional failures, there will be one DOC instance for each direction and both should be taken OOS.</p> <p>Change the DOC Primary state to out of service (OOS). Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.</p> |
|---|--|

Manually disabling the required facilities in the opposite direction

2



DANGER

Risk of laser radiation exposure

Do not look directly into the optical beam. Invisible light can severely damage your eyes.



CAUTION

Risk of damage to modules

Never disconnect an optical fiber that is connected to an active or powered-up optical amplifier. To disconnect or reconnect an optical fiber, ensure the optical amplifier is OOS, then disconnect or reconnect the fiber.

Procedure 4-221 (continued)
Optical Line Fail

Step	Action
CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.	
3	For the node with the OLF alarm (the downstream node), select the AMP and the SRA (if present) circuit pack showing the associated “Automatic Shutoff” alarm from the Configuration, Facility & Provisioning screen in the Site Manager.
4	In the Facility Type drop-down box, select “AMP” if not already selected. Select the specific AMP facility showing the “Automatic Shutoff” alarm. Click the Edit button and change the primary state to OOS. If there is an SRA circuit pack at the same site, place the RAMAN facility OOS
Determining the neighboring node	
5	Select the circuit pack that has the OLF alarm (if the OLF is raised against an OPTMON facility, then this will be a different circuit pack than selected in step 2).
6	In the Facility Type drop-down box, select “ADJ-LINE”. Record the node name (TID) reported for the “Actual Far-End Address” of this facility. This is the neighboring (upstream) node.
Manually disabling the required facilities upstream	
7	Log into the node identified in step 6 . Select the AMP showing the Automatic Shutoff alarm from the Configuration, Facility & Provisioning screen in Site Manager.
8	In the Facility Type drop-down box, select “AMP” if not already selected. Select the specific AMP facility showing the Automatic Shutoff alarm, click the edit button and change the primary state to OOS.
9	If there is an SRA circuit pack at the same site, place the RAMAN facility OOS.
Note: In the case of a dual fiber cut (for example, both directions of the same span), there will also be an OLF alarm present on this node. In this case it is unnecessary to perform steps step 2 to step 9 again. The amplifiers and RAMAN pumps (if present) across the affected span are now in a safe state to perform troubleshooting (that is, they will not re-enable during the troubleshooting process when an operator may be attempting to clean a fiber).	



CAUTION

Risk of damage to modules

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

Procedure 4-221 (continued)

Optical Line Fail

Step	Action								
Locating the fiber fault									
10	Select your next step.								
	<table> <tr> <th>If an</th><th>Then go to</th></tr> <tr> <td>OTDR is available</td><td>step 11</td></tr> <tr> <td>ESAM or SRA is available</td><td>step 12</td></tr> <tr> <td>OTDR is not available</td><td>step 14</td></tr> </table>	If an	Then go to	OTDR is available	step 11	ESAM or SRA is available	step 12	OTDR is not available	step 14
If an	Then go to								
OTDR is available	step 11								
ESAM or SRA is available	step 12								
OTDR is not available	step 14								
11	Locate the fiber break using the OTDR. This can be performed from the downstream node by connecting the OTDR to the fiber coming into port 8 of the AMP in OLF, or from upstream by connecting the OTDR to the fiber coming from port 5 of the AMP in Automatic Shutoff. Go to step 13 .								
12	Using the OTDR functionality built into the circuit pack, view the OTDR trace. Refer to the “Downloading a SOR file” and “Displaying an OTDR trace graphically” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310, to locate the fault.								
13	Select your next step.								
	<table> <tr> <th>If the OTDR indicates the break is at the</th><th>Then</th></tr> <tr> <td>mid span</td><td>end this procedure and contact the relevant operations group</td></tr> <tr> <td>near or far-end of the span</td><td>go to step 14</td></tr> </table>	If the OTDR indicates the break is at the	Then	mid span	end this procedure and contact the relevant operations group	near or far-end of the span	go to step 14		
If the OTDR indicates the break is at the	Then								
mid span	end this procedure and contact the relevant operations group								
near or far-end of the span	go to step 14								
14	Check the fiber connection manually at the downstream node. At the downstream node, systematically clean each optical connection between the fiber plant egress / patch panel towards port 8 of the AMP showing the OLF alarm. Replace any obviously damaged patchcords.								
15	Use a separate power source and power meter to measure the loss between the fiber plant ingress and port 8 of the AMP.								
16	Check the fiber connections at the upstream node. At the upstream node, systematically clean each optical connection between port 5 of the AMP showing the Automatic Shutoff alarm and the fiber plant ingress / patch panel. Replace any obviously damaged patchcords.								
17	Use a separate power source and power meter to measure the loss between port 5 of the AMP and the ingress of the fiber plant / patch panel.								
18	Reconnect, replace, or repair the fiber. If there is an AMP present, run manual OTDR traces to validate the fault. Refer to the “Performing a manual OTDR trace” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310, to locate the fault.								

Procedure 4-221 (continued)
Optical Line Fail

Step	Action						
19	For the ESAM or SRA, once the fault has been cleared, run a manual OTDR trace, and reset the baseline. Refer to the “Setting a SOR file as baseline” procedure in <i>n Configuration - Provisioning and Operating</i> , 323-1851-310.						
20	For the AMP and RAMAN (if present) facilities in step 2 and step 9 , change the primary state to IS using the same method.						
21	Select your next step.						
	<table> <tr> <th>If the affected fiber span</th><th>Then go to</th></tr> <tr> <td>is on a stretched span</td><td>“Recovering from a fiber cut on a stretched span” on page 4-533</td></tr> <tr> <td>otherwise</td><td>step 22</td></tr> </table>	If the affected fiber span	Then go to	is on a stretched span	“Recovering from a fiber cut on a stretched span” on page 4-533	otherwise	step 22
If the affected fiber span	Then go to						
is on a stretched span	“Recovering from a fiber cut on a stretched span” on page 4-533						
otherwise	step 22						
22	Change the DOC Primary state back to in service (IS). Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
23	Verify if the alarm has cleared. If the DOC Automation Mode is not set to Auto Re-optimize as Necessary, click on the Re-Optimize button in the DOC facility screen. Refer to the “Re-optimizing channels” procedure in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
	Note: Under double fault scenarios in which there is an OSC facility failure in conjunction with the Optical Line Failure, the ALSO condition may not clear upon repair of the fiber. If the alarm does not clear, follow the troubleshooting guidelines in Procedure 4-225, “OSC Loss Of Signal” on page 4-542, keeping in mind that the OSC Loss Of Signal will be masked in this situation by the Optical Line Fail alarm.						
24	If the alarm does not clear and the special case above has been ruled out, contact your next level of support or your Ciena support group.						
	This procedure is complete.						

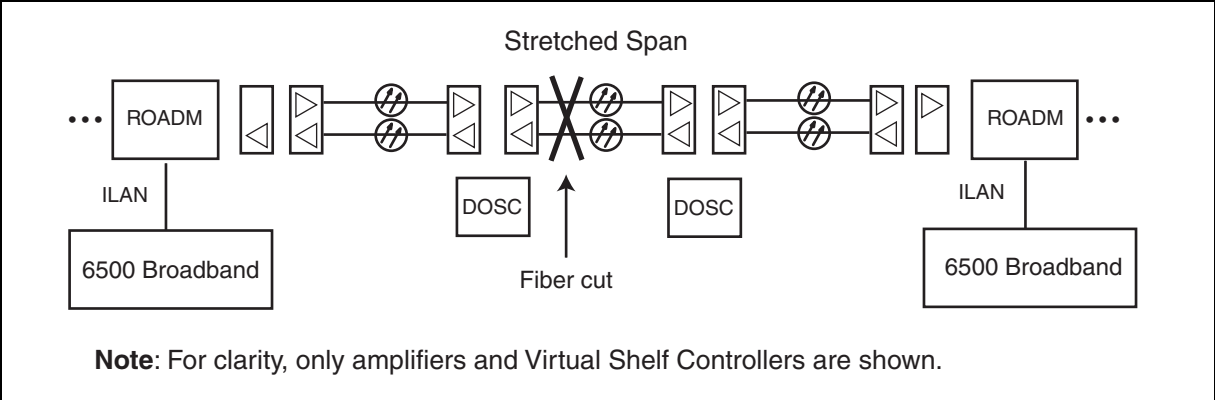
Recovering from a fiber cut on a stretched span

Note: The following procedure is used to recover line traffic after a fiber cut has been repaired on a span where the span loss exceeds the OSC budget. On such spans, the OSC cannot be used for communication across the span, so the data communication traffic that is normally carried by the OSC is usually carried over an out-of-band link, such as 6500 GCC0 or an OM5000 MOTR channel being carried as traffic. Since the link is down, this communications channel is also down. Because of this, the amplifiers cannot coordinate between themselves to bring the line up safely. See [Figure 4-5](#) for a configuration example. To bring this span back up, the amplifiers on the span must be manually restored by performing this procedure.

Procedure 4-221 (continued)
Optical Line Fail

Step	Action
	This procedure requires either remote communication to the NEs on both ends of the span that does not require the link between them or personnel on site at each site to perform the following steps.

Figure 4-5
Recovering from a fiber cut on a stretched span



Procedure 4-221 (continued)
Optical Line Fail

Step	Action
DANGER Risk of laser radiation exposure During this procedure the fiber plant does not have to be disrupted and the system remains a Class 1(IEC)/ Class I (FDA) product. If the fiber downstream of the MLA2 C-Band, MLA2 w/ VOA, SRA, SAM, ESAM, XLA, or MLA3 C-Band/MLA3 L-Band Line A output connector becomes disconnected accidentally while the Amp Auto Shutoff Disabled feature is active, the radiation at the exposed fiber can be at hazard level 1M (IEC 60825-2). In this situation, you must take all safety precautions appropriate to hazard level 1M (IEC 60825-2). The ORL based APR safety mechanism remains active.	

Procedure 4-221 (continued)

Optical Line Fail

Step	Action
27	When the post-amplifiers on each side of the stretched span have come out of the automatic shutoff condition, use the Performance Monitoring application to calculate the span loss after the repair (Head end amplifier output power - Tail end amplifier input power). Span Loss for an optical span is also stored as a PM. Make sure the span loss is within specifications; if not, take the appropriate actions.
28	Allow the system a few minutes to provide complete internal communications restoration and visibility of the domain from both sides of the stretched span. To verify data communication continuity, log into the network elements on the two ends of the stretched span and confirm all GCC and SONET DCC communications alarms have cleared.
29	After the link has been restored in both directions, enable automatic shutoff on the stretched span as follows: - Using Hyperterm or an equivalent program, open a telnet session on each network element in the stretched span and log into the NE using the following command: <pre>ACT-USER::<ADMIN Level User>:1::<ADMIN Level password>;</pre> - Set the Line A (port 8) amplifier shutoff threshold back to their initial values by entering: <pre>ED-AMP::AMP-<shelf>-<slot>-8:<CTAG>:::SHUTTHRES=-40;;</pre> For normal operation of stretched span applications with valid optical safety feature operation, the shutoff thresholds for the Line A AMP facilities must be provisioned to -40 dBm. Do not provision this threshold below -40 dBm. - Log out using the following command: <pre>CANC-USER::<ADMIN Level User>:<CTAG>;</pre>
30	Change the DOC Primary state back to in service (IS). Refer to the “Editing the DOC Settings” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
31	If the DOC Automation Mode is not set to Enhanced, click on the Re-Optimize button in the DOC facility screen. Refer to the “Re-optimizing channels” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
32	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-222

Optical Signal Failed

Alarm IDs: 2255, 2259

Probable cause

This alarm is raised against the PTP or OTM facility on transponder line ports when a fault such as LOS, LOC, LOF, Pre-FEC SF, LOM, LOLA, are detected on the optical line.

Impact

Warning, non-service-affecting (W, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Clear the line faults that trigger the alarm. Refer to appropriate alarm clearing procedures in this document for each fault.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-223

Optimization Scanning in Progress

Alarm ID: 1813, 1877

Probable cause

This event is raised when there is a Tx dispersion pre-compensation optimization scanning session in progress.

Impact

Warning, non-service-affecting (m, NSA)

Step	Action
1	No action is required. The alarm clears once the optimization scanning session is complete.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-224

OPU Payload Type Mismatch

Alarm IDs: 837, 1012, 1182, 1473, 1484, 1622, 1669

Probable cause

The alarm point is identified at the G.709 level and is raised when the payload mapped in the G.709 signal is not the same in the Tx and Rx direction. The G.709 payload structure identifier (PSI) byte is used for this purpose. This condition can occur as a result of crossed fibers or as a result of incorrect provisioning (for example, the near-end client facility is provisioned to be OC-192 and the far-end client facility is provisioned to be ETH10G).

The alarm is raised when the far-end client facility does not match. For example, if a 40G OCI/40G OCLD pair is connected to a 40G MUX OCI/40G OCLD pair, or if a 40G MUX OCI NTK525CAE5/40G OCLD pair is connected to a 40G MUX OCI NTK525CFE5/40G OCLD pair, the alarm will be raised on both client circuit packs.

For the 40G XCIF OTM2 facility this alarm is raised when the incoming Payload Type (PT) does not match the expected PT.

For the 10X10GE MUX (NTK529BA), this alarm will be raised if a new ODU4 connection is created in Release 9.0 and above and if the remote 10X10GE MUX has a ODU4 connection that was created pre-Release 9.0.

ATTENTION

For OTM4 facilities, some 100G OCI circuit packs use a Ciena proprietary mapping while others use a standard G.709 GMP mapping. The Ciena proprietary mapping and the standard GMP mapping use different OPU4 overhead bytes definitions. As a result, OCI mismatch scenarios involving a mix of the two mapper types will result in the "Payload type actual received (in Hex)" value (displayed in the OTM INFO window of Site Manager) equal to 0, and not the value transmitted by the far-end OCI. Circuit packs which use the standard GMP mapping include the 10x10G MUX (NTK529BB), the 100G OCI (NTK529AC), and the 100GE OCI (NTK529AA) in R9.0. Circuit packs which use the Ciena proprietary mapping include the 10x10GE MUX (NTK529BA) and the 100GE OCI (NTK529AA) prior to R9.0. The current mapping being used on a 100G OCI can be found in the "Packet Mapping" attribute of the ETH100G facility. Refer to the "Retrieving equipment and facility details" procedure in *Configuration - Provisioning and Operating*, 323-1851-310.

Procedure 4-224 (continued)
OPU Payload Type Mismatch

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the optical fiber connection information (that is, how the modules on each network element connect to other network elements)

Step	Action
------	--------

- | | | | | | |
|--|---|--|--|----------------------|------------------------|
| 1 | <table><tr><td>If the alarm is raised against a 40G circuit pack</td><td>Then go to step 2</td></tr><tr><td>another circuit pack</td><td>step 4</td></tr></table> | If the alarm is raised against a 40G circuit pack | Then go to step 2 | another circuit pack | step 4 |
| If the alarm is raised against a 40G circuit pack | Then go to step 2 | | | | |
| another circuit pack | step 4 | | | | |
| 2 | Ensure that the 40G client circuit packs are the same type. A 40G OCI/40G OCLD or Wavelength-Selective 40G OCLD pair must be connected to a 40G OCI/40G OCLD or Wavelength-Selective 40G OCLD pair and a 40G MUX OCI NTK525CAE5/40G OCLD or Wavelength-Selective 40G OCLD must be connected to the same 40G MUX OCI NTK525CAE5/40G OCLD or Wavelength-Selective 40G OCLD pair at the far-end (same for NTK525CF OCI). OCI to MUX interworking is not supported. | | | | |
| 3 | Check the MUX OCI PEC type, you can check the equipment inventory screen in Site Manager or the faceplate of the circuit pack (Do not check by looking at the equipment provisioning screen in Site Manager.) | | | | |
| 4 | For a 10X10GE MUX (NTK529BA), if a new ODU4 connection is created in Release 9.0 and above, delete and re-add the ODU4 connection associated with the remote 10X10GE MUX. | | | | |

Procedure 4-224 (continued)
OPU Payload Type Mismatch

Step	Action						
5	Ensure that the remote NE also has Release 9.0 or above installed when the connection is deleted and re-added. If both ends do not have the Release 9.0 or above software load, contact your next level of support or your Ciena support group. Using fiber and connection records, determine the far-end module connected to the alarmed module. Retrieve the equipment and Facility Details for both the alarmed and remote module. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
6	Ensure that the corresponding client facility at the far-end is provisioned as either the same type (ETH10G, OC192, STM64) as the client facility at the end reporting the OPU Payload Type Mismatch alarm or is provisioned as an OTU facility. Note that if the OTSC at the remote end is provisioned for an OTU facility, then the remote facility setting the OPU Payload type will not be that module, but the upstream equipment where the OPU is generated and access to it will be required to clear the alarm.						
7	If the facility types are not OTU facility and are different, determine the incorrect facility type from your company records and delete and re-add the incorrect facility. Refer to the “Deleting a facility from an equipment” and “Adding a facility to an equipment” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Ensure that no fibers are crossed. The OTU/ODU trail trace can be used to identify crossed fibers. If the OTU/ODU trail trace is provisioned, the “ ODU/OTU Trace Identifier Mismatch ” alarm is raised to indicate crossed fibers. Refer to the “Retrieving and editing trail trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
10	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-225

OSC Loss Of Signal

Alarm ID: 556

Probable cause

This alarm is raised against an OSC facility when the optical signal for the facility falls below a fixed threshold. One or more of the following conditions can raise this alarm:

- a defective or dirty fiber linking the OSC ports
- a misconnected or disconnected fiber at the OSC port of the upstream neighbor
- a disabled or out-of-service OSC link on the upstream neighbor
- a powered down upstream neighbor

A cold restart and power cycles of the SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC or shelf will cause a Loss Of Signal condition and this alarm will be raised. This alarm should clear when the cold restart or power cycle completes. In rare cases, this alarm does not clear even after the Loss Of Signal condition clears. Performing [step 7](#) and [step 24](#) of this procedure should clear this alarm.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a fiber cleaning kit
- obtain a replacement pluggable, circuit pack, or fiber patchcord, if required

Procedure 4-225 (continued)
OSC Loss Of Signal

Step	Action				
1	Attempt to log into the network element at the upstream (source) end of the alarmed OSC link. If the NE does not respond, ask on-site personnel to confirm the NE is powered up and operational. If the far-end network element is Then <table> <tr> <td>powered down</td><td>the OSC Loss Of Signal alarm is a result of this condition and will clear when this condition no longer exists.</td></tr> <tr> <td>not powered down</td><td>go to step 2</td></tr> </table>	powered down	the OSC Loss Of Signal alarm is a result of this condition and will clear when this condition no longer exists.	not powered down	go to step 2
powered down	the OSC Loss Of Signal alarm is a result of this condition and will clear when this condition no longer exists.				
not powered down	go to step 2				
2	On the shelf with the alarmed facility, check port 4 OPTMON of the LIM, SRA, SAM, or ESAM that is associated with the alarmed OSC facility. If there is significant power on port 4 but there is no power at the OSC Rx port, then the fault is local to the site with the alarmed OSC. Power levels can be checked using the Performance->Performance Monitoring->New screen in Site Manager.				
3	Using Site Manager Configuration->Equipment and Facility Provisioning screen check that the upstream OSC facility is provisioned. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If the OSC facility is out of service, restore it to service. Refer to the "Changing the primary state of a facility" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If equipment alarms are active against it, clear those alarms before returning to this procedure.				
4	If the original alarm has Then <table> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	cleared	the procedure is complete	not cleared	go to step 5
cleared	the procedure is complete				
not cleared	go to step 5				
5	Perform a warm restart on the local and upstream SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC circuit packs using the Fault->Restart menu of Site Manager. Refer to Procedure 2-11 , "Restarting a circuit pack or shelf processor" procedure in this document.				
6	If the original alarm has Then <table> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	cleared	the procedure is complete	not cleared	go to step 7
cleared	the procedure is complete				
not cleared	go to step 7				
7	Using Site Manager Configuration->Equipment and Facility Provisioning screen provision the primary state of the OSC facility at the upstream NE to OOS and then back to IS.				

Procedure 4-225 (continued)
OSC Loss Of Signal

Step	Action								
8	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	go to step 9		
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared	go to step 9								
9	 CAUTION Risk of damage to modules Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module. Have on-site personnel disconnect the Rx fiber from the OSC SFP+ SRA, SAM, ESAM, or SPAP-2 w/2xOSC reporting the alarm and using a power meter, measure the Rx power to determine if light is being received. Clean and reconnect the fiber to the Rx of the. See the minimum Rx values as follows: • -34 dBm for Short reach SFP (NTK592NPE6) • -34 dBm for Standard reach SFP (NTK592NBE6) • -37 dBm for Premium reach SFP (NTK592NHE6) • -44 dBm for Long reach SFP (NTK592NGE5) • -44 dBm for Long reach SFP (NTK592NVE5)								
10	If a DWDM SFP (NTK592NR) is used in conjunction with the OSC Filter (1516.9 nm) module (NTK504BA), check the fibering of the OSC filters at each end of the link. Verify that the power going into the filter from the Tx of the SFP is between 3 dBm and 6 dBm and the power coming out of the filter going into the Rx of the SFP is above -43 dBm and below -7 dBm. See the "Technical specifications" section in <i>Planning</i>, NTRN10GK, for more information.								
11	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>Rx power is below the minimum value</td><td>go to step 18</td></tr> <tr> <td>Rx power is above the minimum value</td><td>go to step 12</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	Rx power is below the minimum value	go to step 18	Rx power is above the minimum value	go to step 12
If	Then								
the original alarm has cleared	the procedure is complete								
Rx power is below the minimum value	go to step 18								
Rx power is above the minimum value	go to step 12								
12	Reseat the SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.								

Procedure 4-225 (continued)
OSC Loss Of Signal

Step	Action								
13	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	go to step 14		
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared	go to step 14								
14	Replace the Rx OSC pluggable raising the alarm. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
15	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 16</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	go to step 16		
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared	go to step 16								
16	Replace the SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
17	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group		
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared	contact your next level of support or your Ciena support group								
18	Have on-site personnel disconnect the Tx fiber from the upstream SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC sourcing the OSC channel at the upstream site and using a power meter, measure the Tx power to determine if light is being transmitted. Clean and reconnect the fiber to the Tx of the 2xOSC or SPAP-2 w/2xOSC circuit pack. See the minimum Tx values as follows: • -5 dBm for Short reach SFP (NTK592NPE6) • 0 dBm for Standard reach SFP (NTK592NBE6) • 1 dBm for Premium reach SFP (NTK592NHE6) • -7.5 dBm for Long reach SFP (NTK592NGE5)								
19	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>Tx power is below the minimum value</td><td>go to step 20</td></tr> <tr> <td>Tx power is above the minimum value</td><td>go to step 22</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	Tx power is below the minimum value	go to step 20	Tx power is above the minimum value	go to step 22
If	Then								
the original alarm has cleared	the procedure is complete								
Tx power is below the minimum value	go to step 20								
Tx power is above the minimum value	go to step 22								

Procedure 4-225 (continued)

OSC Loss Of Signal

Step	Action						
20	Check the fibers of the alarmed OSC facility (at the network element raising the alarm and at the upstream network element): • verify the fibers are connected, and not crossed, looped back, or misconnected • check and clean any dirty fibers. Refer to Cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0. If no line channels yet exist, verify the line fiber connections.						
21	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or your Ciena support group						
22	Reseat the SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC at the upstream site (sourcing the signal). Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
23	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 24</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 24
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 24						
24	Replace the Tx OSC pluggable at the upstream site. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
25	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 26</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 26
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 26						
26	Replace the SRA, SAM, ESAM, 2xOSC or SPAP-2 w/2xOSC circuit pack at the upstream site. Refer to the “Replacing an RLA 5x1 or MDRLA 2x6 module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
27	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or your Ciena support group						

—end—

Procedure 4-226

OSC RFI

Alarm ID: 1871**Probable cause**

This alarm is raised when the Ethernet-over-SONET (EOS) on the far end of the OSC span detects Loss of Frame (LOF) and injects Line RDI towards the local EOS.

ATTENTION

OSC RFI alarm can be displayed differently in Active Alarms and Historical Fault Browser.

For example:

Active Alarms: OSC RFI, Historical Fault Browser: OSC RDI

Active Alarms: OSC RDI, Historical Fault Browser: OSC RFI.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Check that there are no connectivity issues at the far end. |
| 2 | <p>If the “Loss Of Frame” alarm is active at the far end of the OSC, clear the alarm. Refer to the “Loss Of Frame, Loss Of Multiframe alarms” alarm clearing procedure in this document.</p> <p>The alarm clears after the Line RDI is no longer being detected (far end stops injecting Line RDI signal).</p> |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-227

OSC Signal Degrade

Alarm ID: 711

Probable cause

This alarm is raised against an OSC facility when the OSC bit error rate (BER) crosses the provisioned threshold. The alarm clears when the BER falls below the provisioned threshold by a factor of ten. For example, if the provisioned BER is 1E-5, the average BER must drop below 1E-6 for the alarm to clear. Conditions that can cause the OSC channel BER to exceed the provisioned threshold include:

- a dirty fiber connection
- a failed OSC SFP
- an span loss beyond the specifications of the OSC SFP
- an incorrect OSC SFPs installed

This alarm is masked by the OSC Loss Of Signal alarm.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a fiber cleaning kit
- obtain a replacement circuit pack or fiber patchcord, if required

Step	Action
1	Using the Site Manager PM screen, retrieve the system measured OSC power at the amplifier by retrieving the PMs for OPTMON-<shelf>-<slot>-4. Use the reading for OPR-OTS, untimed column. Refer to the “Retrieving performance monitoring data” in the <i>Fault Management - Performance Monitoring</i> , 323-1851-520.

Procedure 4-227 (continued)
OSC Signal Degrade

Step	Action						
2	Retrieve the OSC facility that is in alarm. Refer to the “Retrieving equipment and facility details” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Look in the equipment window to determine the PEC of the OSC SFP being used.						
3	Look up the minimum Rx power rating for the PEC from step 2 . Refer to the “Technical specifications” section in <i>Planning</i> , NTRN10GK.						
4	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the reported Rx power is greater than the Receiver Sensitivity</td><td>go to step 5</td></tr> <tr> <td>less than or equal to the Rx Sensitivity</td><td>go to step 9</td></tr> </table>	If	Then	the reported Rx power is greater than the Receiver Sensitivity	go to step 5	less than or equal to the Rx Sensitivity	go to step 9
If	Then						
the reported Rx power is greater than the Receiver Sensitivity	go to step 5						
less than or equal to the Rx Sensitivity	go to step 9						

**CAUTION****Risk of damage to modules**

Wear an antistatic wrist strap to protect the equipment from static damage. Connect the wrist strap to the ESD jack on the shelf or module.

5 Remove the Rx Fiber from the OSC SFP that is raising the alarm. Measure the Rx power. Clean and replace the fiber.

6

If	Then
the original alarm has cleared	the procedure is complete
the measured Rx power is greater than the Receiver Sensitivity	go to step 7
the measured Rx power is less than or equal to the Rx Sensitivity	go to step 9

7 Replace the OSC SFP. Refer to the “Replacing a Pluggable module” procedure in *Fault Management - Module Replacement*, 323-1851-545.

8

If	Then
cleared	the procedure is complete
not cleared	contact your next level of support or your Ciena support group

9 Disconnect the fiber from the LIM that is connected to the OSC SFP Rx. Using a known good fiber and power meter, measure the power from the OSC Tx port on the LIM. Clean and replace the original fiber.

Procedure 4-227 (continued)

OSC Signal Degrade

Step	Action	
10	If	Then
	the original alarm has cleared	the procedure is complete
	the measured Rx power is greater than the Receiver Sensitivity	go to step 11
	the measured Rx power is less than or equal to the Rx Sensitivity	go to step 13
11	Install a new fiber between the LIM OSC Tx and the OSC Rx port. After installation, measure the Rx power at the Rx OSC end, clean and insert the connector.	
12	If	Then
	the original alarm has cleared	the procedure is complete
	the measured Rx power is greater than the Receiver Sensitivity	go to step 7
	the measured Rx power is less than or equal to the Rx Sensitivity	contact your next level of support or Ciena support group
13	At this point, it has been determined that the OSC power out of the LIM is below specification. Retrieve the OSC facility for the source OSC SFP at the far-end of the link. Refer to the “Retrieving equipment and facility details” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Look in the equipment window to determine the PEC of the OSC SFP being used.	
14	Confirm the same kind of SFP is being used on both ends of the span.	
15	If the PECs	Then go to
	are different	step 16
	are the same	step 19
16	An incorrect PEC is installed. Consult your company records to determine which PEC is correct. The PEC must match.	
17	Replace the OSC SFP with the proper SFP. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545 and “Changing the provisioned PEC” procedure in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310.	
18	If original alarm	Then
	has cleared	the procedure is complete
	has not cleared	restart this procedure

Procedure 4-227 (continued)
OSC Signal Degrade

Step	Action								
19	At the source site for the OSC signal, remove the Rx OSC fiber from the associated LIM module and measure the Tx power into the LIM from the OSC SFP. Clean and replace the fiber.								
20	Look up the minimum Tx Output power for the SFP PEC in the “Technical specifications” section in <i>Planning</i> , NTRN10GK.								
21	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>the measured Tx power is greater than the Min Tx Output Power</td><td>go to step 28</td></tr> <tr> <td>the measured Tx power is less than or equal to the Min Tx Output Power</td><td>go to step 22</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	the measured Tx power is greater than the Min Tx Output Power	go to step 28	the measured Tx power is less than or equal to the Min Tx Output Power	go to step 22
If	Then								
the original alarm has cleared	the procedure is complete								
the measured Tx power is greater than the Min Tx Output Power	go to step 28								
the measured Tx power is less than or equal to the Min Tx Output Power	go to step 22								
22	Unplug the Tx fiber from the OSC SFP. Using a known good fiber and power meter, measure the power from the OSC SFP Tx port. Clean and replace the original fiber.								
23	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>the measured Tx power is greater than the Min Tx Output Power</td><td>go to step 26</td></tr> <tr> <td>the measured Tx power is less than or equal to the Min Tx Output Power</td><td>go to step 24</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	the measured Tx power is greater than the Min Tx Output Power	go to step 26	the measured Tx power is less than or equal to the Min Tx Output Power	go to step 24
If	Then								
the original alarm has cleared	the procedure is complete								
the measured Tx power is greater than the Min Tx Output Power	go to step 26								
the measured Tx power is less than or equal to the Min Tx Output Power	go to step 24								
24	Replace the OSC SFP. Refer to the “Replacing a Pluggable module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
25	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	contact your next level of support or your Ciena support group								
26	The fiber between the LIM and OSC is inducing extra loss. Install new fiber between the LIM OSC Rx and the OSC Tx port. After installation, measure the Rx power at the Rx OSC port of the LIM, clean and insert the connector.								
27	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>the measured Rx power is greater than the Receiver Sensitivity</td><td>go to step 7</td></tr> <tr> <td>the measured Rx power is less than or equal to the Rx Sensitivity</td><td>contact your next level of support or Ciena support group</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	the measured Rx power is greater than the Receiver Sensitivity	go to step 7	the measured Rx power is less than or equal to the Rx Sensitivity	contact your next level of support or Ciena support group
If	Then								
the original alarm has cleared	the procedure is complete								
the measured Rx power is greater than the Receiver Sensitivity	go to step 7								
the measured Rx power is less than or equal to the Rx Sensitivity	contact your next level of support or Ciena support group								

Procedure 4-227 (continued)

OSC Signal Degrade

Step	Action
28	At this point, the power into the link meets specifications, yet the power leaving the link does not. This can be caused by any of the following: • a high span loss • a dirty connector on inside of the LIM module or line side of the LIM module • an OSC filter failure on the LIM module • an incorrect OSC SFP pair for the span loss.
29	Troubleshooting of these items may be traffic affecting. Contact your next level of support or Ciena support group.

—end—

Procedure 4-228

OSPF Adjacency Loss alarms

Use this procedure to clear the following alarms associated with OSPF circuits provisioned on COLAN-A/COLAN-X, GCC, ILAN-IN/LAN-OUT, or OSC interfaces.

Conditions that can prevent the establishment of an OSPF adjacency include:

- no OSPF circuit provisioned at the far-end of the link
- the OSPF circuit parameters provisioned at both ends of the link do not match

The OSPF Adjacency Loss alarms apply to IPv4/OSPFv2 comms only.

COLAN-A OSPF Adjacency Loss

Alarm ID: 978

COLAN-X OSPF Adjacency Loss

Alarm ID: 979

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on a COLAN-A/COLAN-X interface, and the adjacency cannot be established.

SECTION/RS DCC OSPF Adjacency Loss

Alarm ID: 1036, 1037, 1038, 1039, 1131, 1356, 1357, 1358, 1359, 1360

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on a DCC interface, and the adjacency cannot be established.

LINE/MS DCC OSPF Adjacency Loss

Alarm ID: 1313, 1314, 1315, 1316, 1317

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on a Line DCC interface, and the adjacency cannot be established.

GCC0/GCC1/GCC2 OSPF Adjacency Loss

Alarm ID: 1040, 1041, 1144, 1318, 1319, 1320, 1361, 1362, 1363, 1427, 1428, 1429, 1431, 1726, 1727, 1728

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on a GCC0/GCC1/GCC2 interface, and the adjacency cannot be established.

Procedure 4-228 (continued)

OSPF Adjacency Loss alarms

ILAN-IN OSPF Adjacency Loss

Alarm ID: 976

ILAN-OUT OSPF Adjacency Loss

Alarm ID: 977

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on an ILAN-IN/ILAN-OUT interface, and the adjacency cannot be established.

OSC OSPF Adjacency Loss

Alarm ID: 975

Probable cause

This alarm is raised when there is an OSPF circuit provisioned on an OSC interface, and the adjacency cannot be established.

This alarm is also raised when the OSC fibers between the OSC1 and OSC2 at the Line AMP NE are swapped (misconnected/crossed).

Impact

Minor, non-service-affecting (m, NSA)

OSPFv2 Adjacency Loss

Alarm ID: 1873

Probable cause

This alarm is raised when there is an OSPFv2 circuit provisioned on an interface (COLAN, ILAN, OSC, GCC, DCC) but the adjacency cannot be established.

Impact

Minor, non-service-affecting (m, NSA)

OSPFv3 Adjacency Loss

Alarm ID: 1929

Probable cause

This alarm is raised when there is an OSPFv3 circuit provisioned on an interface (COLAN, ILAN, OSC, GCC, DCC) but the adjacency cannot be established.

Note that the OPSFv3 circuits apply to IPv6 networks only. The OSPFv3 and OSPFv2 alarms may both get raised together if both are applied to the same facility (for example, COLAN-X) and a fault occurs on that affected facility.

Procedure 4-228 (continued)
OSPF Adjacency Loss alarms

ATTENTION

This alarm displays in the Additional Information column the interface AID the alarm is raised against.

Impact

Minor, non-service-affecting (m, NSA)

Step	Action						
1	Verify that there is an OSPF circuit provisioned at the far-end of the link by retrieving the OSPF circuits at the far-end of the failed facility. Ensure there is an entry for the failed facility. Refer to the “Retrieving communications settings” procedure in <i>Data Comms Planning & User Guide</i>, 323-1851-101. When troubleshooting the OSPF adjacency loss alarms, make sure to compare the similar protocols on each end of the link. For example, either OSPFv2 or OSPFv3. Each protocol has its own tab in Site Manager, make sure to select the correct tab. Note: OSPFv2 and OSPFv3 are not compatible with each other. Therefore the settings in one protocol do not affect the other.						
2	<table> <tr> <th>If a far-end OSPF circuit</th><th>Then go to</th></tr> <tr> <td>exists</td><td>step 5</td></tr> <tr> <td>does not exist</td><td>step 3</td></tr> </table>	If a far-end OSPF circuit	Then go to	exists	step 5	does not exist	step 3
If a far-end OSPF circuit	Then go to						
exists	step 5						
does not exist	step 3						
3	Provision the missing OSPF circuit. Refer to the “Adding a new entry in the communications settings” in <i>Data Comms Planning & User Guide</i>, 323-1851-101.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						

Procedure 4-228 (continued)

OSPF Adjacency Loss alarms

Step	Action						
5	Retrieve the OSPF circuit parameters from both ends of the failed link and verify that all OSPF circuit parameters on both ends of the link match. If necessary, correct any discrepancies according to your company records. Refer to the “Retrieving communications settings” and “Editing the communications Settings” procedures in <i>Data Comms Planning & User Guide</i> , 323-1851-101.						
6	If the alarm has not cleared, verify the fibers at the Line AMP NE for any reversed fiber connections and correct if necessary.						
7	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>contact your next level of support or Ciena support group</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or Ciena support group						

—end—

Procedure 4-229

OSPF Max Capacity Reached

Alarm ID: 1676

Probable cause

This alarm is raised when the allocated OSPF memory exceeds 75% of the OSPF partition. The maximum partition size is 9MB. The alarm can clear autonomously if the allocated memory drops below 60% of the OSPF partition. Note that the amount of allocated memory can increase significantly during periods of OSPF route recalculations. These OSPF network elements may include 6500 and CPL shelves as well as other OSPF-capable equipment.

This alarm provides an early warning indication to prevent the OSPF network from potential failure.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	The following actions can help to reduce the amount of allocated OSPF memory: - splitting or segmenting the OSPF network - avoiding use of 6500 as Area Border Router (ABR) - reducing the amount of OSPF routing information, for example using Type-3 LSA filtering, reducing the number of external routes advertised into OSPF - eliminating routing churn Refer to “Data communications planning” in <i>Data Communications Planning and User Guide</i>, 323-1851-101.
2	If the alarm is only raised on a single 6500 shelf, it can indicate that the memory of this shelf is fragmented and the OSPF is not able to expand its memory partition. In this case, perform an SP warm restart to clear the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
3	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-230

OSRP CCI Session Down

Alarm ID: 1386

Probable cause

This alarm is raised when the OSRP communication session to the photonic subsystem has gone down. For OSRP Types of Photonic and DERIVED, this communications session is between the two CPUs of the SP-2. For OSRP of type PROV, this is within the OAMP processor of the SP-2.

This alarm is expected for Photonic Control Planes when performing any operation that causes the CPU2 to restart. The alarm should automatically clear within a minute.

Impact

Major, service-affecting (M, SA)

Major, non-service-affecting (M, NSA)

Step	Action
1	Check for “Member Shelf Unreachable” alarm. If the alarm is active, clear that alarm first. The “OSRP CCI Session Down” alarm must clear automatically.
2	If the alarm did not clear, restart CPU2 (if Photonic, else restart the SP). Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-231

OSRP CCI Session Out of Sync

Alarm ID: 1387

Probable cause

This alarm is raised when Optical Signaling and Routing Protocol (OSRP) Cross Connection Interface (CCI) session has lost synchronization with the network element.

This alarm is raised during OSRP enabling, CPU2 restart, or when synchronization is lost between the CPU2 and the OAM&P.

Impact

Major, service-affecting (M, SA)

Major, non-service-affecting (M, NSA)

Step	Action
1	This alarm is cleared by the system when CCI has synchronized the Control Plane database with the NE and the OSRP. If this alarm persists, reset the CPU2, or perform a protection switch of the SP. Refer to the "Operating a protection switch" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-232

OSRP Configuration in Progress

Alarm ID: 1745

Probable cause

This alarm is raised when an OTN OSRP instance is added/deleted. This alarm applies to OTN shelves with OTN XC provisioned.

Impact

Minor, non-service-affecting (m, NSA)

Step	Action	
1	If	Then go to
	two XCs are present	step 2
	only one XC is present	step 3
2	Perform a protection switch on the active XC. Refer to the “Operating a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. Go to step 4 .	
3	Perform a warm restart on the XC. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.	
4	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-233

OSRP Line Operationally Blocked

Alarm ID: 1764

Probable cause

This alarm is raised in the following scenarios:

- the High Received Span Loss alarm is raised (only when the “Span loss exceed action” parameter is provisioned to “Blocked” for the OSRP line)
- the Crossed Fibers Suspected alarm is raised against an OPM port
- communication failure with DGEs and AMP sites
- a Circuit Pack Failure alarm is raised for:
 - an MLA/MLA2/MLA3 C-Band/MLA3 L-Band/MLA2v/SLA/XLA or cascaded LIM (SLA) amplifier circuit-packs at ROADM OTS, at the AMP, or DGE OTS
 - a WSS included in a ROADM or a DGE OTS
- a DOC facility is OOS-MA
- DOC Auto add channels and Auto delete channels are disabled
- the DOC Invalid Photonic Domain alarm is raised
- Optical System Topology indicates TOADM or GOADM OTSs in Photonic Control Plane domain
- the OSRP line is blocked resulting from a Bandwidth lockout, call record mismatch, fiber disconnection on intra-node fiber (except WSS-WSS fiber), AMP in shut-off state, AMP Primary State OOS-MA, or Maintenance mode on the associated OSRP line. The alarm is also raised on line DOWN conditions. For information on these parameters, refer to the “L0 and L1 OSRP provisioning” section in *Configuration - Control Plane*, 323-1851-330.

Hierarchy:

- DOWN conditions have priority over BLOCKED conditions
- if DOWN conditions clears, the standing condition is cleared. If a BLOCK conditions still exists, the standing condition will be re-raised with the new state and reason. only one condition is reported to OSRP at a time.
- if it clears, then the next highest priority condition is reported to OSRP so the alarm can be re-raised with the new reason

A fault reason is provided within the alarm “Additional Information” field to help with the troubleshooting.

Procedure 4-233 (continued)

OSRP Line Operationally Blocked

- Coherent Select Control parameter set to Off in the Node Information System, on a Coherent Select node. Refer to the “Displaying node information” procedure in the *Administration and Security*, 323-1851-301.
- PSNCCSEACTION settings
- Shutoff Threshold Crossed (STC) latched on LIM Down on Line B
- additional network fault is reported on the line while the “Call Record Mismatch” alarm is present

Impact

Warning

Step	Action
1	If the line is manually blocked, un-block the line by disabling the OSRP bandwidth lockout. Refer to the “Retrieving OSRP provisioning information” and “Editing an OSRP line” procedures in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm is raised due to STC latched on LIM Line B, verify the fiber between WSS Common Out and AMP Line B In and repair/replace if needed. If the alarm does not clear, manually clear the latch (edit OSRP-LINE UNBLOCK to YES).
3	Verify and correct any of the causes listed in the “ Probable cause ”.
4	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-234

OSRP Node Operationally Blocked

Alarm ID: 1422

Probable cause

This alarm can be raised during any of the following, Photonic, or OTN Control Plane scenarios:

- Photonic Control Plane;
 - node is administratively blocked by the user
 - software upgrade
 - database backup or restore
 - warm restart of active CPU2
 - SP switchover
 - restart of an active OAM&P
- OTN Control Plane:
 - node is administratively blocked by the user
 - database restore during commit
 - warm or cold restart of active XC
 - switchover
 - warm restart of an OTN Flex MOTR or 40G OTN XCIF circuit pack

This alarm is also raised by the Photonic Control Plane when a database auto-save is in progress. See the disabled alarms list for finding occurrences of the “Database Auto Save in Progress” alarm that correlate with instances of this alarm. Note that the “Database Auto Save in Progress” alarm is disabled by default.

Impact

Minor, non-service-affecting (m, NSA)

Step	Action
1	If the node is manually blocked, un-block the node by changing the OSRP node “Admin state” to “Unlocked”. Refer to the “Retrieving OSRP provisioning information” and “Editing an OSRP node” procedures in <i>Configuration - Control Plane</i> , 323-1851-330.

Procedure 4-234 (continued)

OSRP Node Operationally Blocked

Step	Action
2	If any of the actions listed in the “Probable cause” are in progress, wait until the action is completed. Note: This alarm will not be cleared during the second invoke of the upgrade. The alarm is cleared at the end of the second invoke.
3	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-235

OSRP Port Capability Mismatch

Alarm ID: 1412**Probable cause**

This alarm is raised when there is a hierarchical level configuration mismatch.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Compare the hierarchical level of the OSRP lines in both the local and remote nodes. Refer to <i>Configuration - Control Plane</i> , 323-1851-330 for related procedures.
2	Adjust the hierarchical levels as required at the remote or local node.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-236

OTDR Trace In Progress

Alarm ID: 1712, 1964

Probable cause

This alarm is raised when the OTDR trace is in progress.

All traffic-related alarms are masked when this alarm is raised.

Impact

Warning

Step	Action
1	No action is required. The alarm clears when the trace completes automatically.
2	You can also stop the current trace in progress by pressing the “Stop Trace(s)” button in the Site Manager OTDR window to clear the alarm. Refer to “Performing a manual OTDR trace” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-237

OTL Skew Out Of Range

Alarm ID: 1681, 1682, 1683

Probable cause

This alarm is raised against the OTM3/OTUTTP client facility of a 40G+ CFP OCI circuit pack, or OTM4/OTUTTP client facility of a 100G OCI (NTK529AA/AC), or 100G WL3e OTR circuit pack when the system detects that the virtual lane skew exceeds the specified skew range from G.798.

Impact

Critical, service-affecting (C, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain the replacement fiber for connection between the subtending equipment and the CFP
- if required, obtain a replacement CFP module
- use an account with at least a level 3 UPC

Step	Action						
1	Determine the type of the connection fiber between the subtending equipment and the CFP module based on CFP type (NTTA01FB, NTTA01BJ, NTTA03BJ, and NTTA02BJ are SMF, NTTA03AA is MMF).						
2	<table border="0"> <tr> <td>If the fiber type is</td> <td>Then go to</td> </tr> <tr> <td>SMF</td> <td>step 3</td> </tr> <tr> <td>MMF</td> <td>step 8</td> </tr> </table>	If the fiber type is	Then go to	SMF	step 3	MMF	step 8
If the fiber type is	Then go to						
SMF	step 3						
MMF	step 8						
3	Ensure the Integrated Test Feature (ITS) is available on the 40G+ CFP OCI, 100G WL3e OTR, or 100G OCI and operating in Test Set mode. Refer to the “Performing a test with the Integrated Test Set” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-237 (continued)

OTL Skew Out Of Range

Step	Action
4	Ensure the Rx/Tx port on the CFP is fiber loop backed using a patch cord fiber.
5	If the alarm does not clear, replace the CFP. Refer to the “Replacing a Pluggable module” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.
6	If the original alarm has cleared, then restore the original configuration and the procedure is complete. Otherwise, go to step 7 .
7	If the original alarm does not clear, it indicates the problem is caused by subtending equipment. Troubleshoot the subtending equipment. Go to step 14 .
8	Ensure that all parallel fibers connected between the subtending equipment and CFP have the same length. If not, then replace them to make them comply with this requirement.
9	If the original alarm does not clear, ensure the Integrated Test Feature (ITS) is available on the 40G+ CFP OCI, 100G WL3e or 100G OCI circuit pack and operating in Test Set mode.
10	Ensure the Rx/Tx port on the CFP is fiber loop backed using a special MPO24 loopback fiber.
11	If the alarm does not clear, replace the CFP. Refer to the “Replacing a Pluggable module” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.
12	If the original alarm has cleared, then restore the original configuration with the new CFP and the procedure is complete. Otherwise, go to step 13 .
13	If the original alarm does not clear, it indicates the problem is caused by subtending equipment. Troubleshoot the subtending equipment.
14	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-238

OTS Provisioning Error

Alarm ID: 974

Probable cause

This alarm is raised against the OTS entity when there is an OTS provisioning error detected. This includes either a discrepancy between an OTS and its associated OTS (that is, an intra-node OTS neighbor), or between an OTS and its OSC neighbor (that is, an inter-node OTS neighbor).

This alarm can be raised in the following scenarios for intra-node OTS neighbors:

- the provisioned associated OTS (**Provisioned OTS** parameter) of the OTS entity does not match the discovered associated OTS (**Actual Associated OTS**)
- the OTS and its discovered associated OTS are not in the same Optical System Identifier (OSID)
- the OTS and its discovered associated OTS have different Configuration Types (for example, one Channel Access and one Amplifier)
- the OTS and its discovered associated OTS have Tx Path Identifiers of the same parity (that is, either both are odd or both are even)
- the OTS and its discovered associated OTS do not have the same Site Identifier
- fibers at the Line AMP NE are swapped (misconnected/crossed)

This alarm is raised when the OTS is provisioned using the Photonic Configuration Management application and the OTS “OSC Required” parameter is set to **True** and the OTS OSC slot/port is set to None. To clear the alarm, you must either provision the OTS OSC slot/port or set the OTS “OSC Required” parameter to **False**. The following actions raise the alarm for the Photonic Control Plane:

- the initial provisioning of a Dynamic Gain Equalizer (DGE) OTS (where no WSS-to-WSS adjacency exists)
- the reconfiguration of an OTS from CHA to DGE

Procedure 4-238 (continued)

OTS Provisioning Error

This alarm can be raised in the following scenarios for inter-node OTS neighbors, when the OTS and its OSC neighbor:

- are at same site
- are not in the same OSID
- have Tx Path Identifiers of the same parity (that is, either both are odd or both are even)
- OSC fibers are swapped or misconnected

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | | | | | | | |
|----------------------------------|---|----------------------------------|-------------|---------|---------------------------|-------------|------------------------------|
| | Note: The provisioning of the Provisioned OTS parameter value is optional; however, if it is provisioned, then it must match the discovered associated OTS. An associated OTS will only be present if there is a second OTS sharing the same OSID (Optical System ID) on the same NE. | | | | | | |
| 1 | Retrieve the OTS details for the alarmed OTS. Refer to the “Retrieving OTS Management, OTS Equipment, and Facility details” and “Editing an OTS instance in the Photonic Configuration Management application” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 2 | Verify the Provisioned OTS parameter value of the alarmed OTS entity matches the Actual Associated OTS . If required, correct the Provisioned OTS value to match the Actual Associated OTS . Refer to the “Editing an OTS instance in the Photonic Configuration Management application” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 3 | <table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 4</td></tr></table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 4 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 4 | | | | | | |

Procedure 4-238 (continued)
OTS Provisioning Error

Step	Action						
4	Retrieve the OTS details for the OTS associated with the far-end OSC module and port. Refer to the “Retrieving OTS and slot sequence information” procedure in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Note: The OTS on the far-end of the OSC from the local OSC must have the same OSID provisioned.						
5	If the OTS is provisioned using the Photonic Configuration Management application and the OTS “OSC Required” parameter is set to True , you must provision the OTS OSC slot/port. If the OSC is not required, set the OTS “OSC Required” parameter to False . See the Photonic Configuration Management procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Verify the Optical System Identifier parameter value of the alarmed OTS entity is the same as that of its OSC neighbor. If required, correct the incorrect Optical System Identifier value. Refer to the “Retrieving OTS and slot sequence information” and “Editing an OTS instance in the Photonic Configuration Management application” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>see the following Note, then go to step 9</td></tr> </table> Note: An associated OTS will only be present if there is a second OTS sharing the same OSID (Optical System ID) on the same NE.	If the original alarm has	Then	cleared	the procedure is complete	not cleared	see the following Note, then go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	see the following Note, then go to step 9						
9	Verify the Configuration Type parameter value of the alarmed OTS entity matches that of its discovered associated OTS. That is, ensure they are both provisioned to either Channel Access or Amplifier or DGE. If required, correct the mismatched Configuration Type value. You will need to delete and re-provision the mismatched OTS. Refer to the “Deleting an OTS instance” and “Provisioning an OTS instance” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Refer to the “Deleting an OTS instance”, “Creating an OTS instance with basic parameters”, and “Editing an OTS instance in the Photonic Configuration Management application” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						

Procedure 4-238 (continued)

OTS Provisioning Error

Step	Action				
10	If the original alarm has Then <table> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>see the following Note, then go to step 11</td></tr> </table> Note: An associated OTS will only be present if there is a second OTS sharing the same OSID (Optical System ID) on the same NE. Tx Path IDs must be all even in one direction in the system and odd in the other. For more information on Tx Path IDs, refer to the <i>Photonic Layer Guide</i>, NTRN15DA.	cleared	the procedure is complete	not cleared	see the following Note, then go to step 11
cleared	the procedure is complete				
not cleared	see the following Note, then go to step 11				
11	Verify the Tx Path Identifier parity of the alarmed OTS entity is different than that of its discovered associated OTS. That is, ensure one is even and the other one is odd. If required, correct the incorrect Tx Path value. You will have to delete and re-provision the incorrect OTS. Refer to the “Deleting an OTS instance” and “Provisioning an OTS instance” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Refer to the “Deleting an OTS instance”, “Creating an OTS instance with basic parameters”, and “Editing an OTS instance in the Photonic Configuration Management application” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.				
12	If the original alarm has Then <table> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>see the following Note, then go to step 13</td></tr> </table> Note: Tx Path IDs must be all even in one direction in the system and odd in the other. For more information on Tx Path IDs, refer to the <i>Photonic Layer Guide</i>, NTRN15DA.	cleared	the procedure is complete	not cleared	see the following Note, then go to step 13
cleared	the procedure is complete				
not cleared	see the following Note, then go to step 13				
13	Verify the Tx Path Identifier parity of the alarmed OTS entity is different than that of its OSC neighbor. That is, ensure that one is even and the other is odd. If required, correct the misprovisioned Tx Path value. You will have to delete and re-provision the incorrect OTS. Refer to the “Deleting an OTS instance” and “Provisioning an OTS instance” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Refer to the “Deleting an OTS instance”, “Creating an OTS instance with basic parameters”, and “Editing an OTS instance in the Photonic Configuration Management application” procedures in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.				

Procedure 4-238 (continued)
OTS Provisioning Error

Step	Action						
14	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>see the following Note, then go to step 15</td></tr> </table> Note: All shelves in the same site should have the same Site Identifier value.	If the original alarm has	Then	cleared	the procedure is complete	not cleared	see the following Note, then go to step 15
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	see the following Note, then go to step 15						
15	Verify the Site Identifier parameter value of the network element containing the alarmed OTS entity matches the Site Identifier parameter value of the network element containing the discovered associated OTS. The site ID is found on Site Manager by choosing Configuration, Node Information, then selecting the Node Information tab. Click on the shelf desired, then click on the “shelf” tab in the lower window. Refer to the “Displaying node information” procedure in the <i>Administration and Security</i>, 323-1851-301. If required, correct the incorrect Site Identifier value. Refer to the “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i>, 323-1851-301.						
16	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>see the following Note, then go to step 17</td></tr> </table> Note: The NE at the far-end of the OSC link must have a different Site ID.	If the original alarm has	Then	cleared	the procedure is complete	not cleared	see the following Note, then go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	see the following Note, then go to step 17						
17	Verify the Site Identifier parameter value of the network element containing the alarmed OTS entity is different than the Site Identifier parameter value of the network element containing the OSC neighbor. The site ID is found on Site Manager by choosing Configuration, Node Information, then selecting the Node Information tab. Click on the required shelf, then click on the “shelf” tab in the lower window. Refer to the “Displaying node information” procedure in the <i>Administration and Security</i>, 323-1851-301.						
18	If required, correct the incorrect Site Identifier value. Refer to the “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i>, 323-1851-301.						
19	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 20</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 20
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 20						
20	Verify the fibers at the Line AMP or DGE for any reversed fiber connections and correct if necessary.						
21	Verify the OSC fiber connections and correct if necessary.						

Procedure 4-238 (continued)

OTS Provisioning Error

Step	Action
22	Check the Actual Far End Address on the associated line-facing LIM, and compare it to the provisioned Expected Far End Address. If the Actual Far End Address on the associated line-facing LIM does not match its provisioned Expected Far End Address, verify proper OSC fiber connectivity to the LIMs.
23	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-239

OTU Signal Degrade

Alarm ID: 835, 1220, 1383, 1445

Probable cause

This alarm is raised when the Rx interface error rate detected by the Section Monitoring Error Detection Code (BIP-8) exceeds $8\text{E}-10$. The alarm clears when the error rate is $1\text{E}-10$ or better. This alarm applies to OTUTTP, OTMC2, OTM1, OTM2, OTM3, and OTM4 facilities.

The OTU Signal Degrade alarm is raised when the error threshold of $1\text{E}-9$ is reached. This is done to ensure that the alarm is raised before the bit error rate reaches $1\text{E}-9$.

For 8xOTN Flex MOTR and (1+8)x OTN Flex MOTR circuit packs, this alarm is raised when the Error detection Code (BIP-8) exceeds $1\text{E}-7$. The alarm clears when BIP-8 errors are below $1\text{E}-7$.

100G WL3e OTR and 100G OCI circuit packs support Post-FEC OTU4 Signal Degrade error calculation in Broadband mode. Post-FEC OTU4 Signal Degrade threshold is provisionable from $1\text{E}-6$ to $1\text{E}-9$ (default).

10x10G OCI circuit packs support Post-FEC OTU2 Signal Degrade on the OTM2 client port in Broadband mode. Post-FEC OTU2 Signal Degrade threshold is provisionable from $1\text{E}-6$ to $1\text{E}-9$ (default).

10x10G PKT/OTN I/F and 48xGE PKT I/F circuit packs support Post-FEC OTU4 Signal Degrade error calculation in POTS Mode.

Procedure 4-239 (continued)
OTU Signal Degrade

Impact

Major, service-affecting (M, SA) alarm, unprotected
Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other networks)
- have an optical power meter with the same optical connectors as the network element
- have attenuation pads

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Check the corresponding subtending circuit pack for failures or alarms. Troubleshoot these alarms/failures before proceeding. |
| 3 | Make sure that the circuit pack type and SFP/XFP/CFP type is consistent with the subtending equipment (bit-rate and protocol and FEC settings). Refer to the “Retrieving equipment and facility details” in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 4 | Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power value for the facility reporting the alarm. Refer to the “Retrieving equipment and facility details” in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 5 | Select the alarmed OTU facility in the Site Manager Equipment and Facility Screen, click on the Ranges button to display the specifications for that SFP/XFP/CFP. Compare the Rx Actual Power value for the facility with the Rx minimum power and Rx maximum power values displayed. |
| 6 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |

Procedure 4-239 (continued)
OTU Signal Degrade

Step	Action								
7	Make sure the signal optical power level into the Rx interface is within specification. <table> <tr> <td>If the signal level is</td><td>Then go to</td></tr> <tr> <td>greater than the Rx maximum value</td><td>step 8</td></tr> <tr> <td>less than the Rx minimum value</td><td>step 10</td></tr> <tr> <td>within specifications</td><td>step 13</td></tr> </table>	If the signal level is	Then go to	greater than the Rx maximum value	step 8	less than the Rx minimum value	step 10	within specifications	step 13
If the signal level is	Then go to								
greater than the Rx maximum value	step 8								
less than the Rx minimum value	step 10								
within specifications	step 13								
8	The signal may require padding, or an incorrect SFP/XFP/CFP type is used on the circuit pack reporting the alarm or on the subtending equipment. Confirm the proper SFP/XFP/CFPs are installed and add attenuation pads as needed until the Rx power is within the reported specification.								
9	<table> <tr> <td>If the original alarm</td><td>Then</td></tr> <tr> <td>clears</td><td>procedure is complete</td></tr> <tr> <td>does not clear</td><td>go to step 13</td></tr> </table>	If the original alarm	Then	clears	procedure is complete	does not clear	go to step 13		
If the original alarm	Then								
clears	procedure is complete								
does not clear	go to step 13								
10	The fiber may be dirty, the fiber connectors may not be fully seated, or an incorrect SFP/XFP/CFP type is used on the circuit pack reporting the alarm or on the subtending equipment. Confirm the proper SFP/XFP/CFP are installed, clean all fiber connectors and ensure they are fully seated.								
11	<table> <tr> <td>If</td><td>Then</td></tr> <tr> <td>the original alarm clears</td><td>procedure is complete</td></tr> <tr> <td>power is within specifications</td><td>go to step 13</td></tr> <tr> <td>power is still below specifications</td><td>there is an issue with the link or source fault. Contact your next level of support or Ciena support group.</td></tr> </table> Recheck the Rx power by retrieving the Rx Actual Power value for the facility reporting the alarm again. (Select the "Retrieve Facility" button to reset to current values.)	If	Then	the original alarm clears	procedure is complete	power is within specifications	go to step 13	power is still below specifications	there is an issue with the link or source fault. Contact your next level of support or Ciena support group.
If	Then								
the original alarm clears	procedure is complete								
power is within specifications	go to step 13								
power is still below specifications	there is an issue with the link or source fault. Contact your next level of support or Ciena support group.								
12	Replace the SFP/XFP/CFP that supports the facility raising the alarm. Refer to <i>Fault Management - Module Replacement</i> , 323-1851-545.								
13	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 14								

Procedure 4-239 (continued)

OTU Signal Degrade

Step	Action						
14	Replace the upstream SFP/XFP/CFP or source module. Refer to <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or your Ciena support group						

—end—

Procedure 4-240

OTU Signal Fail (OTM, OTM2, OTUTTP)

Alarm ID: 1763, 1783, 1788

Probable cause

This alarm is raised when the Post-FEC error is higher than the provisioned value. This alarm applies to 10x10G MUX, 100G OCI, 100G OCLD, and 10x10G PKT/OTNIF circuit packs.

10x10G MUX circuit packs support Post-FEC OTU2 Signal Fail on the OTM2 client port in Broadband and POTS mode. The Post-FEC OTU2 Signal Fail threshold is provisionable from 1E-6 (default) to 1E-9 and 0. Note that Post-FEC OTU2 Signal Fail is disabled if the OTU Signal Fail Threshold value is set to 0.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the network connection information (that is, how the interface circuit packs on each network element connect to other network elements)

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Check the corresponding subtending circuit pack for failures or alarms. Troubleshoot these alarms/failures before proceeding. |
| 3 | Make sure that the circuit pack type and SFP/XFP/CFP type is consistent with the subtending equipment (bit-rate and protocol and FEC settings). Refer to the “Retrieving equipment and facility details” in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |

Procedure 4-240 (continued)

OTU Signal Fail (OTM, OTM2, OTUTTP)

Step	Action								
4	Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power value for the facility reporting the alarm. Refer to the “Retrieving equipment and facility details” in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
5	Select the alarmed OTU facility in the Site Manager Equipment and Facility Screen, click on the Ranges button to display the specifications for that SFP/XFP/CFP. Compare the Rx Actual Power value for the facility with the Rx minimum power and Rx maximum power values displayed.								
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
7	Make sure the signal optical power level into the Rx interface is within specification.								
	<table> <tr> <th>If the signal level is</th><th>Then go to</th></tr> <tr> <td>greater than the Rx maximum value</td><td>step 8</td></tr> <tr> <td>less than the Rx minimum value</td><td>step 10</td></tr> <tr> <td>within specifications</td><td>step 13</td></tr> </table>	If the signal level is	Then go to	greater than the Rx maximum value	step 8	less than the Rx minimum value	step 10	within specifications	step 13
If the signal level is	Then go to								
greater than the Rx maximum value	step 8								
less than the Rx minimum value	step 10								
within specifications	step 13								
8	The signal may require padding, or an incorrect SFP/XFP/CFP type is used on the circuit pack reporting the alarm or on the subtending equipment. Confirm the proper SFP/XFP/CFPs are installed and add attenuation pads as needed until the Rx power is within the reported specification.								
9	<table> <tr> <th>If the original alarm</th><th>Then</th></tr> <tr> <td>clears</td><td>procedure is complete</td></tr> <tr> <td>does not clear</td><td>go to step 13</td></tr> </table>	If the original alarm	Then	clears	procedure is complete	does not clear	go to step 13		
If the original alarm	Then								
clears	procedure is complete								
does not clear	go to step 13								
10	The fiber may be dirty, the fiber connectors may not be fully seated, or an incorrect SFP/XFP/CFP type is used on the circuit pack reporting the alarm or on the subtending equipment. Confirm the proper SFP/XFP/CFP are installed, clean all fiber connectors and ensure they are fully seated.								
11	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm clears</td><td>procedure is complete</td></tr> <tr> <td>power is within specifications</td><td>go to step 13</td></tr> <tr> <td>power is still below specifications</td><td>there is an issue with the link or source fault. Contact your next level of support or Ciena support group.</td></tr> </table>	If	Then	the original alarm clears	procedure is complete	power is within specifications	go to step 13	power is still below specifications	there is an issue with the link or source fault. Contact your next level of support or Ciena support group.
If	Then								
the original alarm clears	procedure is complete								
power is within specifications	go to step 13								
power is still below specifications	there is an issue with the link or source fault. Contact your next level of support or Ciena support group.								
	Recheck the Rx power by retrieving the Rx Actual Power value for the facility reporting the alarm again. (Select the “Retrieve Facility” button to reset to current values.)								

Procedure 4-240 (continued)

OTU Signal Fail (OTM, OTM2, OTUTTP)

Step	Action						
12	Replace the SFP/XFP/CFP that supports the facility raising the alarm. Refer to <i>Fault Management - Module Replacement</i> , 323-1851-545.						
13	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 14</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	Replace the upstream SFP/XFP/CFP or source module. Refer to <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>contact your next level of support or your Ciena support group</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or your Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or your Ciena support group						

—end—

Procedure 4-241

Output Loss Of Signal

Alarm ID: 572, 1797

Probable cause

This alarm is raised against an AMP or AMPMON facility when the total output optical power from the amplifier has fallen below the provisioned output LOS threshold value.

The conditions that can cause the output power level to fall below the threshold level include:

- a disconnected fiber
- a dirty optical fiber connector
- a defective fiber optic patchcord
- a defective module
- incorrect provisioning data
- a reflective event (indicated by Automatic Power Reduction Active alarm at an upstream booster amplifier or pre-amplifier)

This alarm can remain active after the fault has cleared and the original power level is restored. This occurs when the power level is lower than the user-provisioned LOS threshold plus the hysteresis value. The hysteresis value is not user provisionable and is set at 3 dB.

The “Output Loss Of Signal” alarm suppresses the “VOA Output LOS” alarm.

This alarm is masked by “Input Loss Of Signal” alarm on the same amplifier facility.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a fiber cleaning kit

Procedure 4-241 (continued)

Output Loss Of Signal

- have a network and site diagram
- have the LOS threshold level value for this amplifier

Step	Action						
1	Check for and clear any active Automatic Power Reduction Active alarm at the upstream booster amplifier or pre-amplifier that is providing output power to the alarmed amplifier.						
2	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	repeat step 1 for each booster amplifier or pre-amplifier farther upstream from the alarmed amplifier						
3	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Verify the AMP facility parameters are correctly provisioned. If required, correct any discrepancies. Refer to the “Retrieving equipment and facility details” and “Editing facility parameters” procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
5	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
7	Clear any upstream (either at the local to the network element reporting the alarm or other upstream remote network elements) alarms that could be causing this alarm, such as Circuit Pack Failed, Circuit Pack Missing, and Loss Of Signal. If there are no upstream alarms, then verify the optical patchcord connected to the port reporting the alarm: • ensure it is connected at both ends and that there is no problem with the optical patchcord • clean the connectors. Refer to the cleaning connectors procedures in <i>Installation - General Information</i>, 323-1851-201.0.						

Procedure 4-241 (continued)

Output Loss Of Signal

Step	Action						
8	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Provision the Output LOS threshold value for the AMP facility reporting the alarm to be 5 dB less than the current value. For example, if the Output LOS threshold is -6 dB, change the value to -11 dB. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
10	Wait at least one minute for the alarm to clear. Change the Output LOS threshold value back to the original value.						
11	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	Restart the LIM, MDRLA 2x6, RLA 5x1 or XLA supporting the alarmed AMP facility. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” in this document.						
13	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	Replace the LIM, MDRLA 2x6, RLA 5x1 or XLA supporting the alarmed AMP facility. Refer to the “Replacing the amplifier modules” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-242

Packet Configuration Integrity Fail

Alarm ID: 1943

Probable cause

This alarm is raised when at least one PKT/OTN circuit pack configuration does not match the OTN XC configuration.

This alarm is also raised when the configuration for either a stand alone eMOTR circuit pack or the eMOTR equipment group does not match the configured hardware. The mismatch can occur due to failures such as, LM fault resulting in restart of the sub-system or incomplete configuration pushed in stand-alone eMOTR restart.

This alarm can also be raised on eMOTR circuit pack after an upgrade. This alarm condition can get raised on eMOTR when configuration in hardware does not match configuration on software during warm restart/upgrade.

A cold restart of the circuit pack that raised the alarm can clear the alarm.

Impact

Major, non-service-affecting (M, NSA) alarm

Critical, service-affecting (C, SA) alarm

Step	Action				
1	<table><tr><td>If the alarm is raised against an eMOTR</td><td>Then go to step 2</td></tr><tr><td>otherwise</td><td>step 5</td></tr></table>	If the alarm is raised against an eMOTR	Then go to step 2	otherwise	step 5
If the alarm is raised against an eMOTR	Then go to step 2				
otherwise	step 5				
2	<table><tr><td>If the alarm is raised against a specific Line Module</td><td>Then go to step 3</td></tr><tr><td>Control Module</td><td>step 4</td></tr></table>	If the alarm is raised against a specific Line Module	Then go to step 3	Control Module	step 4
If the alarm is raised against a specific Line Module	Then go to step 3				
Control Module	step 4				

Procedure 4-242 (continued)

Packet Configuration Integrity Fail

Step	Action
------	--------

- | | |
|---|--|
| 3 | <p>Perform a cold restart on that specific slot to recover the system as follows:</p> <ul style="list-style-type: none">• Login to the eMOTR CLI using a user with superuser account privileges.• Use the CLI command “module show” to display the state of all modules. If the XC has an Operstate of “Enabled”, then the “*” should always be there if there is a PCIF against that XC.• For each module that shows OperState as “Faulted*”, perform a cold restart from TL1. Go to step 10. |
|---|--|

ATTENTION

Do not perform a warm restart to clear this alarm.

Note: When the “Packet Configuration Integrity Fail” alarm is raised against eMOTR, the “config save” CLI command will be blocked.

- | | |
|---|---|
| 4 | <p>Perform a cold restart on all circuit packs in equipment group. Go to step 10.</p> |
|---|---|

ATTENTION

Do not perform a warm restart to clear this alarm.

- | | |
|---|---|
| 5 | <p>Perform a XC protection switch. Continue to step 6.</p> |
| 6 | <p>If the alarm did not clear, login to the packet CLI using a user with superuser account privileges.</p> |
| 7 | <p>Use the CLI command “module show” to display the state of all modules.</p> <p>No new configuration or existing configuration should be created or modified when this condition exists.</p> |



CAUTION

Risk of interruption to service

The following steps will cause a temporary traffic outage related to the specified Line Module (LM).

- | | |
|---|---|
| 8 | <p>For PKT/OTN circuit packs, each module with an “OperState” of “Enabled”:</p> <ul style="list-style-type: none">• Issue the CLI command “module show slot <LM>”.• Issue the “module reload slot <LM>” command if “reload to restore config integrity” is indicated for the “OperState” |
|---|---|

Procedure 4-242 (continued)

Packet Configuration Integrity Fail

Step	Action
	Once the integrity of all LMs is restored, the alarm will be cleared. If the module reload fails, "module show slot <LM>" for that LM will continue to show "reload to restore config integrity".
9	If the module reload fails multiple times, cold restart the Line Module to restore integrity. The command output will show "cold restart to restore config integrity" in this case.
10	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-243

Packet Rate Limit Exceeded

Alarm ID: 591

Probable cause

This event is raised when the aggregate number of packets received by the shelf processor (SP) system during each time period crosses a pre-determined threshold. The event is raised to warn that the CPU can be locked up due to the high receiving packet rate. Network issues such as broadcast storms are the most common causes of this event.

Conditions that can trigger this alarm include:

- network issues such as broadcast storms
- internal faults such as a babbling module
- load delivery, particularly on gateway network element (GNE) shelves that use software forwarding, including NAT and Private-IP configurations

Impact

Warning

Step	Action						
1	In general, no action is required. The network element communications software clears this alarm automatically when the receiving packets rate is lower than the threshold or when load deliveries complete.						
2	<table><tr><th>If</th><th>Then</th></tr><tr><td>this is a network issue</td><td>examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)</td></tr><tr><td>the alarm does not clear</td><td>it can be an internal fault. Contact your next level of support or your Ciena support group.</td></tr></table>	If	Then	this is a network issue	examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)	the alarm does not clear	it can be an internal fault. Contact your next level of support or your Ciena support group.
If	Then						
this is a network issue	examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)						
the alarm does not clear	it can be an internal fault. Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-244

Packet Rate Limit Exceeded - CPU2

Alarm ID: 1267

Probable cause

This event is raised when the aggregate number of packets received by the shelf processor (SP) system during each time period crosses a pre-determined threshold. The event is raised to warn that the CPU2 can be locked up due to the high receiving packet rate. Network issues such as broadcast storms are the most common causes of this event.

Conditions that can trigger this alarm include:

- network issues such as broadcast storms
- internal faults such as a babbling module
- load delivery, particularly on gateway network element (GNE) shelves that use software forwarding, including NAT and Private-IP configurations

Impact

Warning

Step	Action						
1	In general, no action is required. The network element communications software clears this alarm automatically when the receiving packets rate is lower than the threshold or when load deliveries complete.						
2	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>this is a network issue</td><td>examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)</td></tr> <tr> <td>the alarm does not clear</td><td>it can be an internal fault. Contact your next level of support or your Ciena support group.</td></tr> </table>	If	Then	this is a network issue	examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)	the alarm does not clear	it can be an internal fault. Contact your next level of support or your Ciena support group.
If	Then						
this is a network issue	examine the network configuration, and identify and remove the trouble traffic source (for example, a broadcast storm)						
the alarm does not clear	it can be an internal fault. Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-245

Payload Label Mismatch

Alarm IDs: 456

Probable cause

This alarm is raised when the incoming signal label does not match the expected signal label.

Impact

Major, service-affecting (M, SA) alarm, if on active path

Minor, non-service-affecting (m, NSA) alarm, if on inactive path in a UPSR/
SNCP configuration

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- have the network connection information (that is, how the optical modules on each network element connect to other network elements)
- use an account with at least a level 3 UPC

Step	Action						
1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document.						
2	Clear any alarms of higher order using the appropriate procedures.						
3	<table><tr><th>If the original alarm</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 4</td></tr></table>	If the original alarm	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm	Then						
cleared	the procedure is complete						
not cleared	go to step 4						

Procedure 4-245 (continued)
Payload Label Mismatch

Step	Action								
4	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.								
5	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
6	Use the network connection information to identify the transmit and receive ends of the alarmed signal.								
7	<table> <tr> <th>If the network element is</th><th>Then go to</th></tr> <tr> <td>not connected to a 6500 network element at the far-end</td><td>step 8</td></tr> <tr> <td>part of a mid-span meet and the far-end network element is from another vendor</td><td>step 8</td></tr> <tr> <td>connected to another 6500 network element</td><td>step 9</td></tr> </table>	If the network element is	Then go to	not connected to a 6500 network element at the far-end	step 8	part of a mid-span meet and the far-end network element is from another vendor	step 8	connected to another 6500 network element	step 9
If the network element is	Then go to								
not connected to a 6500 network element at the far-end	step 8								
part of a mid-span meet and the far-end network element is from another vendor	step 8								
connected to another 6500 network element	step 9								
8	Use the alarm system of the other vendor to find and correct the problem. Go to step 15 .								
9	Log into each of the network elements on the path as required, and verify that all cross-connects are correctly provisioned. Make any required changes. Refer to the "Retrieving path connections" procedure in <i>Configuration - Connections Management</i> , 323-1851-320.								
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 11								
11	Retrieve and record the signal label messages at the transmit and the receive network. Refer to the "Retrieving and editing path provisioning" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
12	Ensure that the outgoing signal label message (Transmitted) at the transmit end matches the incoming expected signal label message (Expected Rx) at the receive end.								
13	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 14								
14	Replace the circuit pack raising the alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545, and select the proper procedure for the module raising the alarm.								
15	If the alarm does not clear, contact your next level of support or your Ciena support group.								

—end—

Procedure 4-246

PHY Map Mismatch

Alarm IDs: 2021

Probable cause

This alarm is raised when at least one of the below is true.

- RX PID does not match Tx PID
- Rx PHYMAP does not match Tx PHYMAP

Each OTSi in a Group Identifier (GID) must have a unique PID.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | | | | | | | |
|--|--|--|-------------------|-----------|------------------------|-------|------------------------|
| 1 | Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. | | | | | | |
| 2 | From the Site Manager Configuration menu, select Equipment & Facility Provisioning . Select the alarmed facility, and record the values from the Rx PID , Trace Tx , and Associated Far End Rx PID columns. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 3 | <table><tr><td>If the Rx PID and Trace Tx values are</td><td>Then go to</td></tr><tr><td>not equal</td><td>step 4</td></tr><tr><td>equal</td><td>step 5</td></tr></table> | If the Rx PID and Trace Tx values are | Then go to | not equal | step 4 | equal | step 5 |
| If the Rx PID and Trace Tx values are | Then go to | | | | | | |
| not equal | step 4 | | | | | | |
| equal | step 5 | | | | | | |
| 4 | There is a likely a misconnected fiber. Verify that the optical fiber connections are correct on the circuit pack raising the alarm and on the upstream circuit pack. The Associated Far End Rx PID value from step 2 indicates where the misconnection exists. | | | | | | |
| 5 | If the alarm does not clear, contact your next level of support or your Ciena support group. | | | | | | |

—end—

Procedure 4-247

Pluggable I/O Carrier 1/2 Fail

Alarm IDs: 1054, 1055**Probable cause**

This alarm is raised when a port card (such as 20G L2SS) is provisioned with an associated pluggable I/O carrier 1/2, and the pluggable I/O carrier 1/2 fails, or when the port card detects that the power for the SFPs is not available or under voltage on the pluggable I/O carrier 1/2 after the card has enabled it.

Impact

Major, service-affecting (M,SA) alarm for a pluggable I/O carrier associated with an in-service circuit pack with in-service facilities with cross-connects, unprotected

Minor, non-service-affecting (m, NSA) alarm for a pluggable I/O carrier associated with an out-of-service circuit pack or an in-service circuit pack without cross-connects or protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- obtain a replacement pluggable I/O carrier

Step	Action
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
2	Replace the failed pluggable I/O carrier with a new one. Refer to the “Replacing the Front I/O Panel (Multi-Service)” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545, for instructions.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-248

Pluggable I/O Carrier 1/2 Missing

Alarm IDs: 1052, 1053

Probable cause

This alarm is raised when a port card (such as a 20G L2SS) is provisioned with an associated pluggable I/O carrier 1/2, and the pluggable I/O carrier 1/2 is missing, or there is a problem with the presence lines to the I/O carrier 1/2 using the pluggable I/O panel or SP.

You may also see this alarm momentarily right after a shelf processor restart. If the alarm condition does not exist, this alarm will clear after a couple of seconds.

Impact

Major, service-affecting (M, SA) alarm for a pluggable I/O carrier associated with an in-service circuit pack with in-service facilities with cross-connects, unprotected

Minor, non-service-affecting (m, NSA) alarm for a pluggable I/O carrier associated with an out-of-service circuit pack or an in-service circuit pack without cross-connects or protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- obtain a supported pluggable I/O carrier

Step	Action
------	--------

- | | |
|---|---|
| 1 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 2 | Install the required pluggable I/O carrier or replace the existing pluggable I/O carrier with the type that is required by the circuit pack. Refer to the “Replacing the Front I/O Panel (Multi-Service)” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545, for instructions. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-249

Pluggable I/O Carrier 1/2 Unknown

Alarm IDs: 1056, 1057

Probable cause

This alarm is raised when the port card (such as a 20G L2SS) is provisioned with an associated pluggable I/O carrier 1/2, and the physically present pluggable I/O carrier 1/2 has a PEC not supported by software, or when the SP has problems accessing the I/O carrier 1/2 1-wire EEPROM.

Impact

Major, service-affecting (M, SA) alarm for a pluggable I/O carrier associated with an in-service circuit pack with in-service facilities with cross-connects, unprotected

Minor, non-service-affecting (m, NSA) alarm for a pluggable I/O carrier associated with an out-of-service circuit pack or an in-service circuit pack without cross-connects or protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- obtain a supported pluggable I/O carrier

Step	Action
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
2	Replace the pluggable I/O carrier with the type that is required by the circuit pack. Refer to the “Replacing the Front I/O Panel (Multi-Service)” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545, for instructions.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-250

Power Failure

Alarm IDs:1686

Probable cause

This alarm is raised when the shelf processor detects a power input card reporting low or no voltage on its output.

This alarm is only applicable to the 7-slot Type 2 shelf equipment with AC Power Input Cards (NTK505RA).

Impact

Minor, non-service-affecting (m, NSA) alarm

Critical, Service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- have a standard multimeter
- have a spare Power Input Card to replace failed Power Input Card in the shelf

Step	Action
1	Open the Shelf Inventory Site Manager application and verify that the Power Input Card associated with the alarm is listed. Refer to the “Displaying shelf inventory information” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
3	At the shelf faceplate, identify the Power Input Card that is the source of the alarm. If power is not reaching the system through the Power Input Card (due to a tripped breaker or externally disconnected power feed), the “Power OK” LED on the faceplate of the Power Input Card is off.

Procedure 4-250 (continued)

Power Failure

Step	Action	
4	If the LED is on	Then contact your next level of support or your Ciena support group.
	off	go to step 7
5	 DANGER Risk of electrical shock and short circuit Ensure that there is no chance of coming in contact with conductors in the power cables of the Power Input Card. Electrical shock or short circuit can result. Make sure the AC power cable is connected to the power card's input connector and that the plug end is connected to a power source. For the AC power source, you can measure the power using an AC power tester.	
6	If the original alarm is within the AC input range	Then go to step 7
	not in the AC input range	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 3 .
7	Replace the Power Input Card. Refer to the “Replacing the Power Input Card” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.	
8	If the original alarm is cleared	Then the procedure is complete
	not cleared	go to step 9
9	Replace the shelf processor for the 7-slot Type 2 shelf. Refer to <i>Fault Management - Module Replacement</i> , 323-1851-545.	
10	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-251

Power Failure - A or Power Failure - B

Alarm IDs: 78, 79

Probable cause

This alarm is raised against a power feed when the shelf processor detects that low or no voltage exists on the A or B backplane power feed. If one power source has failed, the shelf traffic is not usually affected. Failure of more than one feed on a shelf with multiple power zones (such as the 32-slot shelf or 14-slot packet-optical shelf when equipped with 2x50A Power Input Card, may affect service if both the A and B feed for a zone have failed.

This alarm can also be raised against a fused Power Input Card when the fuse cartridge has been removed, or if the indicator fuse is not present on the card and the main fuse has blown. With Fused Power Input Cards, the “Power Failure - Fuse Blown” (m, NSA) alarm is raised against a power feed when its alarm indicator fuse is blown and there is sufficient voltage present on the associated power input terminals. A “Power Failure - Fuse Blown” will mask the “Power Failure - A/B” alarm.

As long as a shelf processor is active, a Power Failure - Low Voltage alarm will be raised to indicate a power brownout when all the power feeds to the shelf have low/no voltage.

Impact

Minor, non-service-affecting (m, NSA) alarm
Critical, service-affecting (C, SA) alarm

Note that the service-affecting alarm is applicable to the 32-slot shelf and 14-slot packet-optical shelf when equipped with 2x50A power input cards only. More specifically, for the 32-slot shelf and a 2x50A 14-slot configuration, a critical, service-affecting alarm is raised if matching power feeds for Power Input Card A and B are faulty or if only one Power Input Card is present. For all other scenarios, a Minor, non-service-affecting alarm is raised.

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a standard multimeter

Procedure 4-251 (continued)

Power Failure - A or Power Failure - B

- have a spare Power Input Card to replace each Power Input Card in the shelf
- have spare main and indicator fuses to replace (if alarm is raised against a fused Power Input Card)

You do not have to replace each Power Input Card with a new Power Input Card. Use a single spare Power Input Card of the correct type to replace each Power Input Card in turn.

Step	Action								
1	Open the Shelf Inventory application and verify that the Power Input Card associated with the alarm is listed. Refer to the “Displaying shelf inventory information” procedure in of <i>Configuration - Provisioning and Operating</i>, 323-1851-310. The Power Failure - A alarm is associated with Power Input Card A in slot 17-1 for 14-slot shelf, slot 17 for 4-slot shelf, 6500-7 packet-optical shelf or 7-slot shelf, slot 17 for a 2-slot shelf (NTK503LA shelf variant), and slot 43 for 32-slot shelf. The Power Failure - B alarm is associated with Power Input Card B in slot 17-3 for 14-slot shelf, slot 18 for 4-slot shelf, 6500-7 packet-optical shelf or 7-slot shelf, slot 17 for a 2-slot shelf (NTK503LA shelf variant), and slot 44 for 32-slot shelf.								
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
3	At the shelf faceplate, identify the Power Input Card that is the source of the alarm. If power is not reaching the system through the Power Input Card (due to a tripped breaker or externally disconnected power feed), the “Power OK” LED on the faceplate of the Power Input Card is off.								
4	<table> <tr> <th>If the LED is</th><th>Then go to</th></tr> <tr> <td>on</td><td>step 5</td></tr> <tr> <td>off</td><td>step 5</td></tr> </table>	If the LED is	Then go to	on	step 5	off	step 5		
If the LED is	Then go to								
on	step 5								
off	step 5								
5	<table> <tr> <th>If the associated Power Input Card is</th><th>Then go to</th></tr> <tr> <td>fused</td><td>step 6</td></tr> <tr> <td>breakerless</td><td>step 7</td></tr> <tr> <td>breakerless</td><td>step 5</td></tr> </table>	If the associated Power Input Card is	Then go to	fused	step 6	breakerless	step 7	breakerless	step 5
If the associated Power Input Card is	Then go to								
fused	step 6								
breakerless	step 7								
breakerless	step 5								

Procedure 4-251 (continued)

Power Failure - A or Power Failure - B

Step	Action										
6	Make sure the removable fuse cartridge is properly seated. If the original alarm clears, the procedure is complete; otherwise, replace the cartridge fuse and optional indicator fuse (if previously equipped and blown) with the same size fuse. If the original alarm clears, the procedure is complete; otherwise go to step 5 .										
7	Check if the breaker is tripped. If so, reset it to the ON position. If the original alarm clears, the procedure is complete; otherwise, go to step 5 .										
8	 DANGER Risk of electrical shock and short circuit Ensure that there is no chance of coming in contact with the power cables of the Power Input Card. Electrical shock or short circuit can result. For the DC power sources, using a multimeter, carefully measure the voltage of the power feed to the alarmed Power Input Card or at the power cable termination between the L- and L+/Return. Test points are available on some power modules for this test. For D-Sub terminated power inputs (such as those found used in 2-slot and 7-slot Type 2 shelves) you can measure the voltage at the shelf-end of the power cable termination. You must disconnect the cable and carefully measure the voltage. Ensure that the polarity as well as the voltage is correct. (-48 V dc or +24 V dc for NTK505TR Power Input Card). For the AC power source, you can measure the power using an AC power tester.										
9	<table> <tr> <th>If the feed voltage is</th><th>Then</th></tr> <tr> <td>within the normal input voltage operating range required by:</td><td>go to step 10</td></tr> <tr> <td> • -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card and the polarity is correct </td><td></td></tr> <tr> <td>the normal input voltage operating range required by:</td><td>use your company procedure to investigate and correct the fault in the power distribution system. Go to step 3.</td></tr> <tr> <td> • -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card or if the polarity is reversed </td><td></td></tr> </table>	If the feed voltage is	Then	within the normal input voltage operating range required by:	go to step 10	• -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card and the polarity is correct		the normal input voltage operating range required by:	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 3 .	• -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card or if the polarity is reversed	
If the feed voltage is	Then										
within the normal input voltage operating range required by:	go to step 10										
• -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card and the polarity is correct											
the normal input voltage operating range required by:	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 3 .										
• -48/-60 Vdc Power Input Card • 24 Vdc Power Input Card • AC Power Input Card or if the polarity is reversed											
10	Replace the power I/O module. Refer to the “Replacing the Power Input Card A or B” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.										

Procedure 4-251 (continued)

Power Failure - A or Power Failure - B

Step	Action	
11	If the original alarm is	Then
	cleared	the procedure is complete
	not cleared	go to step 12
12	Replace the 6500 shelf processor. Refer to the “Replacing the shelf processor” in the <i>Fault Management - Module Replacement</i> , 323-1851-545.	
13	If the original alarm is	Then
	cleared	the procedure is complete
	not cleared	contact your next level of support or Ciena support group

—end—

Procedure 4-252

Power Failure - Fuse Blown

Alarm ID: 1077

Probable cause

This alarm is raised against a power feed on a fused Power Input Module with a blown alarm indicator fuse as long there is sufficient voltage present on the associated power input terminals.

If there is low or no voltage on the power input terminals, the “Power Failure - Fuse Blown” alarm will be automatically masked and a “Power Failure - A/B” alarm is raised against the feed.

If the blown alarm indicator fuse is physically removed but the fuse cartridge containing a blown main fuse left in place, the “Power Failure - Fuse Blown” alarm will clear and a “Power Failure - A/B” alarm will be raised.

If the blown alarm indicator fuse is physically removed but the fuse cartridge containing a working fuse left in place with sufficient voltage applied to the associated power input terminal, the “Power Failure - Fuse Blown” alarm will clear and no “Power Failure - A/B” alarm will be raised.

If the fuse cartridge containing both the alarm indicator fuse and the main fuse is removed, the “Power Failure - Fuse Blown” alarm will clear and a “Power Failure -A/B” alarm will be raised.

Impact

Minor, non-service-affecting (m, NSA) alarm

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have spare main and indicator fuses
- have a multimeter

You do not have to replace each fuse with a new fuse. Use a single spare fuse of the correct type to replace each fuse in turn.

Procedure 4-252 (continued)
Power Failure - Fuse Blown

Step	Action								
1	Open the Shelf Inventory application and verify that the Power Input Module associated with the alarm is listed. Refer to the “Displaying shelf inventory information” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
2	Identify the Power Input Module that is the source of the alarm. Verify that the indicator fuse has blown by visually checking that the colored flag is visible through its viewing window.								
3	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
4	<table> <tr> <th>If the Power OK LED is</th><th>Then go to</th></tr> <tr> <td>off</td><td>step 5</td></tr> <tr> <td>on</td><td>step 6</td></tr> </table>	If the Power OK LED is	Then go to	off	step 5	on	step 6		
If the Power OK LED is	Then go to								
off	step 5								
on	step 6								
5	Remove the fuse cartridge from the Power Input Module and replace the indicator fuse and main fuse with one of the same amperage rating (which should be greater or equal to the provisioned shelf current. Refer to the “Determining the provisioned shelf current value” procedure in the <i>Administration and Security</i> , 323-1851-301). For details on replacing the fuse, refer to the “Replacing the main fuse and indicator fuse on the Fused Power Input Module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Note: When the fuse cartridge is removed, the Power Failure - Fuse Blown alarm clears and the Power Failure - A or Power Failure - B alarm is raised. This is expected.								
6	Replace the indicator fuse and reseal the fuse cartridge in the shelf, making sure it is properly seated. Refer to the “Replacing the main fuse and indicator fuse on the Fused Power Input Module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.								
7	<table> <tr> <th>If the “Power OK” LED is</th><th>Then</th></tr> <tr> <td>on</td><td>go to step 11</td></tr> <tr> <td>off and the indicator fuse does not blow</td><td>go to step 8</td></tr> <tr> <td>off and the indicator fuse blows</td><td>there is a shelf power issue or failed power I/O module. Go to step 10.</td></tr> </table>	If the “Power OK” LED is	Then	on	go to step 11	off and the indicator fuse does not blow	go to step 8	off and the indicator fuse blows	there is a shelf power issue or failed power I/O module. Go to step 10 .
If the “Power OK” LED is	Then								
on	go to step 11								
off and the indicator fuse does not blow	go to step 8								
off and the indicator fuse blows	there is a shelf power issue or failed power I/O module. Go to step 10 .								
8	Using a multimeter, measure the power feed to the alarmed Power Input Module. Test points are available on some power modules for this test. Ensure that the polarity as well as the voltage are correct (-48 V dc).								

4-604 Alarm clearing procedures

Procedure 4-252 (continued)

Power Failure - Fuse Blown

Step	Action						
9	<table><tr><th>If the feed voltage is</th><th>Then</th></tr><tr><td>equal to or greater than 44 V dc and the polarity is correct</td><td>go to step 10</td></tr><tr><td>less than 44 V dc or reverse polarity</td><td>use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4.</td></tr></table>	If the feed voltage is	Then	equal to or greater than 44 V dc and the polarity is correct	go to step 10	less than 44 V dc or reverse polarity	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4 .
If the feed voltage is	Then						
equal to or greater than 44 V dc and the polarity is correct	go to step 10						
less than 44 V dc or reverse polarity	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4 .						
10	Replace the power I/O module and all fuses. Refer to the “Replacing the Power Input Module A or B” and “Replacing the main fuse and indicator fuse on the Fused Power Input Module” in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-253

Power Failure - Low Voltage

Alarm ID: 77

Probable cause

This alarm is raised against the shelf when a power brownout occurs because the battery voltage to the shelf drops below a minimum level, approximately -40 V dc.

A brownout occurs at approximately -40 V dc. The shelf remains in brownout state until the voltage rises above -42 V dc. The shelf continues to carry traffic, but without alarm capacity as long as the battery voltage remains below -42 V dc. If the battery voltage drops below approximately -37 V dc, the shelf fails and stops carrying traffic.

The difference (hysteresis) in the detection and recovery voltages makes it possible for the battery voltage to drop below the brownout voltage on multiple occasions without causing the circuit packs to restart each time.

A power brownout can occur at different voltages for different circuit packs.

Impact

Critical, service-affecting (C, SA) alarm

The alarm status is critical because it is service-affecting if the battery voltage drops below -37 V dc.

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have a multimeter

Procedure 4-253 (continued)
Power Failure - Low Voltage

Step	Action						
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
2	Using a multimeter, measure the power feed to the alarmed Power Input Card. Test points are available on some power modules for this test. Ensure that the polarity as well as the voltage is correct (-48 V dc).						
3	 CAUTION Risk of traffic loss Do not reseat any circuit pack while this alarm is active. Unexpected problem conditions result when a circuit pack is resealed.						
<table><tr><th>If the feed supply is</th><th>Then</th></tr><tr><td>equal to or greater than 42 V dc and the polarity is correct</td><td>go to step 4</td></tr><tr><td>less than 42 V dc or reverse polarity</td><td>use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4.</td></tr></table>		If the feed supply is	Then	equal to or greater than 42 V dc and the polarity is correct	go to step 4	less than 42 V dc or reverse polarity	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4 .
If the feed supply is	Then						
equal to or greater than 42 V dc and the polarity is correct	go to step 4						
less than 42 V dc or reverse polarity	use your company procedure to investigate and correct the fault in the power distribution system. Go to step 4 .						
4	 CAUTION Risk of provisioning data loss and traffic loss Ensure that the voltage is correct at the shelf and that all circuit packs are operating correctly before continuing. Replacing the shelf processor on a partially functioning network element can result in a loss of provisioning data and a loss of traffic. Replace the shelf processor. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.						
5	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-254

Pre-FEC Signal degrade

Alarm IDs: 1174, 1183, 1184, 1444, 1954, 1984, 2019, 2139

Probable cause

This alarm is raised when the Rx performance is degrading.

Pre-FEC Signal Degrade threshold is user provisionable so the user can decide the pre-FEC BER threshold level.

Pre-FEC Signal Degrade condition is not an automatic protection switch trigger (unlike the Pre-FEC Signal Fail condition).

Pre-FEC Signal Degrade fault detection time depends on the dBQ setting. Once the fault is detected, there is a 2.5 seconds hold-off time before the Pre-FEC Signal Degrade alarm is raised. The Pre-FEC Signal Degrade fault clear time also depends on the dBQ setting. Once the fault is cleared, there is a 10 seconds hold-on time before the Pre-FEC Signal Degrade alarm is cleared. No signal conditioning is applied when the Pre-FEC Signal Degrade fault is declared. For more information on dBQ values and the corresponding Pre-FEC signal degrade detection/clearing times, refer to the “FEC corrections per second and pre-FEC BER values—PFEC (OTM2 NGM)” table in *Performance Monitoring*, 323-1851-520.

The **Equipment & Facility Provisioning** application of Site Manager allows users to provision/view the Pre-FEC signal degrade threshold level and to view the Pre-FEC BER for a given threshold setting in dBQ. For more information, refer to *Configuration - Provisioning and Operating*, 323-1851-310.

Impact

Major, service-affecting (M, SA) alarm on an active traffic path

Minor, non-service-affecting (m, NSA) alarm if no connections are provisioned or if on the non-active path in a protected configuration

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Procedure 4-254 (continued)

Pre-FEC Signal degrade

- have the optical fiber connection information (that is how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element

Step	Action						
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	Confirm that the FEC setting on the ETH100G, OTUTTP, ETTP, PTP, ETTP, CBRTP, OTM1, OTM2, OTM3, or OTM4 facility is the same at both ends of the link (RS10 (IEEE Std 802.3bj RS-10), Off, SCFEC (ITU-T G.975 I.4), UFEC (ITU-T G.975 I.7, RS8 (ITU-T G.709 RS-8)). Refer to the "Retrieving equipment and facility details" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
3	Check that the Pre-FEC Signal degrade Threshold value is correctly set. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
4	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.						
6	Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power value for the facility reporting the alarm. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	With the alarmed facility selected, click on the Ranges button in Site Manager and compare the Rx Actual Power value from the facility with the Rx minimum power and Rx maximum power values displayed.						
8	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
9	Ensure the signal optical power level into the Rx interface is within specification. If not, check the Photonic layer equipment, the fiber, and fiber cleanliness between the photonic layer equipment and the port reporting the alarm. If unable to get the Rx Actual value to a level that is within specifications, contact your next level of support or Ciena support group.						

Procedure 4-254 (continued)
Pre-FEC Signal degrade

Step	Action						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 11						
11	Verify that the current Pre-FEC BER is below the Pre-FEC BER Signal Degrade Threshold value. If the current Pre-FEC BER is below the threshold value, then lower the threshold to -1.00 dBQ. Lower the threshold and then set it back to its original value starting with the local end. Proceed with the remote end if needed.						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
13	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
14	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-255

Pre-FEC Signal Fail

Alarm IDs: 838, 1004, 1173, 1443, 1953, 1983, 2018, 2138

Probable cause

This alarm is raised when the Pre-FEC Signal Fail user provisionable threshold has been crossed.

For 8xOTN Flex MOTR, OTN Flex MOTR 2xCFP2, and (1+8)xOTN Flex MOTR circuit packs, this alarm is raised when an uncorrectable block error is detected. This means that the FEC setting on the facilities are mismatched.

Impact

Critical, service-affecting (C, SA) alarm if connections are provisioned
Minor, non-service-affecting (m, NSA) alarm if no connections are provisioned

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Confirm that the FEC setting on the ETH100G, OTUTTP, ETTP, PTP, CBRTTP, OTM1, OTM2, OTM3, or OTM4 facility is the same at both ends of the link (RS10 (IEEE Std 802.3bj RS-10), Off, SCFEC (ITU-T G.975 I.4), UFEC (ITU-T G.975 I.7), RS8 (ITU-T G.709 RS-8)). Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
| 3 | Check that the Pre-FEC Signal Fail Threshold value is correctly set. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |

Procedure 4-255 (continued)
Pre-FEC Signal Fail

Step	Action						
4	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 5</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
5	Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.						
6	Using the Site Manager Equipment & Facility Provisioning application under the Configuration menu, retrieve the Rx Actual Power value for the facility reporting the alarm. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
7	With the alarmed facility selected, click on the Ranges button in Site Manager and compare the Rx Actual Power value from the facility with the Rx minimum power and Rx maximum power values displayed.						
8	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
9	Ensure the signal optical power level into the Rx interface is within the specification reported. If not, check the Photonic layer equipment, the fiber, and fiber cleanliness between the photonic layer equipment and the port reporting the alarm. If unable to get the Rx Actual value to a level that is within specifications, contact your next level of support or Ciena support group.						
10	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the circuit pack reporting the alarm. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the upstream circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
12	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-256

Primary License Server Unavailable

Alarm ID: 2011

Probable cause

This alarm is raised when the 6500 shelf cannot communicate with the primary license server/manager.

The alarm is also raised when the NE time is not synchronized with the license server time.

Impact

Minor, non-service-affecting (m, nsa) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Verify the network connection between the 6500 shelf and the license server/manager. |
| 2 | Ensure that the NE and License server are both synchronized to the same time/date using NTP server synchronization. Refer to “Provisioning Time of Day servers” and “Operating a time of day synchronization” procedure in <i>Administration and Security</i> , 323-1851-301. |
| 3 | If you are using HTTPS, the correct date and time must be set on the shelf (either manually or using TOD/NTP server provisioning and synchronization). Refer to “Provisioning Time of Day servers” and “Operating a time of day synchronization” procedure in <i>Administration and Security</i> , 323-1851-301. |
| 4 | If you set the time after provisioning licensing, then perform a manual audit after setting the shelf time to clear the license server comms error. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301. |
| 5 | Verify that the primary license server/manager is provisioned and is not down. Refer to “Provisioning License Manager/Server information on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301. |
| 6 | It can take up to 12 hours for the alarm to clear as, in steady state, the heartbeat is run every 12 hours. Run a manual audit which will reset the heartbeat to two minutes after which the heartbeat interval will gradually increase in time to the steady state time of 12 hours. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301. |

Note: An audit cannot clear the alarm when there is no primary license server provisioned on an NE even if a license server is up, running and populated with licenses.

The alarm will clear two minutes after the manual audit is run.

Procedure 4-256 (continued)

Primary License Server Unavailable

Step	Action
7	Verify that the primary license server/manager is provisioned and is not down. Refer to “Provisioning License Manager/Server information on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301.
8	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-257

Primary RADIUS Accounting Server Unavailable

Alarm ID: 1516

Probable cause

This alarm is raised when no response is received from the primary RADIUS accounting server during user-provisioned timeout.

This alarm is also raised when the RADIUS accounting server provisioning on the network element is incorrect.

This security alarm is raised against a SP, SP-2, SPAP, SPAP-2 w/2xOSC, or an integrated shelf processor on the 8xOTN Flex MOTR circuit pack.

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Disable the server on 6500.
2	Re-enable the server and log in or log out of the server.
3	If the alarm is raised again, disable the RADIUS accounting feature and log in or log out of the server.
4	If the alarm is raised, ensure the following RADIUS accounting server provisioning values on the network element are correct: • server IP address • server port • shared secret • timeout - If this value is too small, the server may not be able to respond quickly enough. Refer to the "Provisioning the primary or secondary RADIUS server" procedure in <i>Administration and Security</i> , 323-1851-301.
5	Check the status of the RADIUS accounting server. Ensure the status is ON.

Procedure 4-257 (continued)

Primary RADIUS Accounting Server Unavailable

Step	Action
6	Log in or log out of the network element. This will send a RADIUS accounting message to all provisioned RADIUS accounting servers. The alarm will clear if a response is received from the server(s) within the provisioned timeout.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-258

Primary RADIUS Server Unavailable

Alarm ID: 583

Probable cause

This alarm is raised when all requests to the primary RADIUS server of a shelf processor times out and a secondary RADIUS server is provisioned. If only the primary RADIUS server is provisioned (no secondary RADIUS server provisioned) and all requests time out, the All Provisioned RADIUS servers Unavailable alarm is raised (refer to the [“All Provisioned RADIUS Servers Unavailable”](#) alarm clearing procedure in this document).

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

The alarm is not raised due to a server time out.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Make sure the primary RADIUS server of the shelf processor is enabled and has a valid IP address. Refer to the “Retrieving the centralized security administration details” procedure in <i>Administration and Security</i> , 323-1851-301.
2	Use your company information and confirm the following provisioned information is correct: • The primary RADIUS server of the shelf processor is enabled. • The proper primary RADIUS server IP address is provisioned. • The proper shared secret for the primary RADIUS server is provisioned. • The proper timeout value for the primary RADIUS server is provisioned. • For Private IP networks, ensure the proper RADIUS proxy server settings are provisioned.

Procedure 4-258 (continued)

Primary RADIUS Server Unavailable

Step	Action
	Refer to the following procedures under “Procedures and options for centralized security administration” in the <i>Administration and Security</i> , 323-1851-301.
	• Provisioning the primary or secondary RADIUS Server • Changing the shared secret for the RADIUS server • Editing the RADIUS proxy server settings
3	Log into the network element again using the RADIUS authentication (centralized security administration).
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-259

Primary Shelf Unreachable

Alarm ID: 716

Probable cause

This alarm is raised on a member shelf of a consolidated node when the member shelf cannot communicate with its primary shelf.

Impact

Major, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- have the IP address of all shelves within the consolidated node
- have a network plan, or other documentation identifying shelf provisioning within the consolidated node

Step	Action
------	--------

1	Using company records, locate the primary shelf associated with the member shelf raising the alarm. Log into the primary shelf with Site Manager to confirm the shelf is running correctly. Direct access using the SP LAN-15 port may be required to log in. Refer to the “Logging in to a network element using a remote network connection” or “Logging into a network element using a direct network connection to the LAN port on the shelf processor” procedures in <i>Administration and Security</i> , 323-1851-301.
---	--

2	Connect to and log into the 6500 CLI interface of the primary shelf. Refer to the “Starting a 6500 CLI session” procedure in <i>Administration and Security</i> , 323-1851-301.
---	---

At the 6500 CLI, issue a PING x.x.x.x command where x.x.x.x is the IP address of the remote shelf raising the alarm. Refer to the “Using the ping command” in *Administration and Security*, 323-1851-301.

If the ping is	Then go to
----------------	------------

successful	step 7
------------	------------------------

unsuccessful	step 3
--------------	------------------------

3	There is a communications issue between the two shelves. At a consolidated node, all shelves should be connected using ILAN cables. The shelves can be connected in a linear chain with all router types and the OSPF router type may be connected in a ring. Locate and log into all the shelves that are part of the consolidated node.
---	---

Procedure 4-259 (continued)
Primary Shelf Unreachable

Step	Action						
4	Physically check all ILAN cables and ports. The port connection LED should be on for all ports with a cable plugged in. If they are not, confirm that the cabled ILAN ports have been added on the Communications Interfaces LAN window and on the Communications Interfaces IP window. Confirm that Router circuits have been added for the cabled ports in Communications Routers (either OSPF or IISIS circuits, depending on router type). Refer to the "Retrieving communications settings" in <i>Data Comms Planning & User Guide</i> , 323-1851-101.						
5	Clear any communications alarms that are raised against the ILAN ports, such as Link Down or OSPF alarms.						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Log into the Primary shelf with Site Manager. Direct access using the SP LAN-15 port may be required to log in. Refer to the "Logging in to a network element using a remote network connection" or "Logging into a network element using a direct network connection to the LAN port on the shelf processor" in <i>Administration and Security</i> , 323-1851-301.						
8	Retrieve the Node Information table. Refer to the "Displaying node information" in the node information procedures in <i>Administration and Security</i> , 323-1851-301.						
9	Identify which shelf/shelves is/are causing the alarm. These are the shelves where the secondary state parameter for the unreachable shelf indicates "Unreachable". The affected shelf may have '(?)' next to its shelf number and be highlighted in cyan.						
10	<table> <tr> <th>If within the consolidated node</th><th>Then</th></tr> <tr> <td>only some member shelves displays a fault</td><td>the issue is likely those member shelves. Go to step 11.</td></tr> <tr> <td>all member shelves display the same fault</td><td>the issue is likely with the primary shelf. Go to step 12.</td></tr> </table>	If within the consolidated node	Then	only some member shelves displays a fault	the issue is likely those member shelves. Go to step 11 .	all member shelves display the same fault	the issue is likely with the primary shelf. Go to step 12 .
If within the consolidated node	Then						
only some member shelves displays a fault	the issue is likely those member shelves. Go to step 11 .						
all member shelves display the same fault	the issue is likely with the primary shelf. Go to step 12 .						
11	Directly log into the affected member shelf (shelves), and ensure all provisioning required for the member shelf of a consolidated node to operate is correct and without issue. Direct shelf access may be required. Refer to the "Consolidated node (TIDc)" section and the "Logging into a network element using a direct network connection to the LAN port on the shelf processor" procedure in <i>Administration and Security</i> , 323-1851-301, for more information.						

Procedure 4-259 (continued)
Primary Shelf Unreachable

Step	Action						
	For example, verify and if necessary correct: • Confirm that the shelf has a unique logical shelf number. • Confirm that the shelf TID matches with primary shelf. • Confirm that the TID consolidation option is enabled. • Confirm that the primary shelf is disabled. Go to step 13.						
12	Directly log into the primary shelf, and ensure all provisioning required for the primary shelf of a consolidated node to operate is correct and without issue. Direct shelf access may be required. For example, verify and if necessary correct: • Confirm that the shelf has a unique logical shelf number. • Confirm that the shelf TID matches with member shelves. • Confirm that the TID consolidation option is enabled. • Confirm that the primary shelf is enabled.						
13	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 14</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 14
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 14						
14	Perform a warm restart on the SP of the member shelf(ves). Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" in this document. Wait 10 minutes and log back in.						
15	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 16</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 16
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 16						
16	Perform a warm restart on the SP of the primary shelf. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" in this document. Wait 10 minutes and log back in.						
17	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>contact your next level of support or Ciena support group</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	contact your next level of support or Ciena support group
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	contact your next level of support or Ciena support group						

—end—

Procedure 4-260

Protection Exerciser Failed

Alarm IDs: 325, 437, 611, 612, 651, 955, 1025, 1110, 1326, 1338

Probable cause

This alarm is raised when the protection exerciser has failed to complete the exercise routine on the selected facilities.

This alarm is caused by one of the following conditions:

- a faulty circuit pack
- far-end network is unprotected

Note: The “Protection Exerciser Failed” alarm will not be raised in the following scenarios:

- A scheduled Protection Exerciser will not raise this alarm if an active protection switch is present in the ring (the scheduled protection exerciser will fail silently). The number of protection exerciser runs will still be decremented if the exerciser run count is not set to Indefinitely.
- A manual Protection Exerciser is denied if there is an active protection switch present in the ring. No alarm is raised.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Ensure that there is no active protection switch on the 1+1 TPT configuration. Refer to the “Retrieving protection status details” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.

Procedure 4-260 (continued)

Protection Exerciser Failed

Step	Action
2	If an auto switch is active, find and clear any of the following alarms if active: • OTM0, OTM1, OTM2, OTM3, OTM4Loss of Frame • Loss Of Signal • ODU AIS • ODU LCK • ODU OCI • OPU AIS • OTU Loss of Multiframe • OTU Pre-FEC Signal Fail • OTU Trace Identifier Mismatch • Protection Switch Active Ensure all line alarms and auto-switches are cleared before continuing.
3	Release all user-initiated protection switches. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315.
4	Initiate the exerciser on the selected equipment. Refer to the “Running/ inhibiting the exerciser” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-261

Protection Mode Mismatch

Alarm IDs: 951, 1020, 1334,1393

Probable cause

This alarm is raised when the received channel protection switching control bytes (APS bytes) on the protection interface circuit pack of a 1+1 OTN, or 1+1 TPT configuration show a different protection switch mode than is provisioned on the local network element.

When changing the protection switching mode from unidirectional to bidirectional, or from bidirectional to unidirectional, the Protection Mode Mismatch alarm is raised against the node that was changed to, or remained in bidirectional mode.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	Retrieve the switch mode of the interface pair. Refer to the "Retrieving protection parameters" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.
3	Determine the network element and interface modules that are on the remote end of the link.
4	Log into the remote terminal.
5	Retrieve the switch mode of the optical interface pair on the remote terminal. Refer to the "Retrieving protection parameters" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.

Procedure 4-261 (continued)

Protection Mode Mismatch

Step	Action
6	Compare the switch modes reported with each other and with the company records. • If both optic pairs report the same switch mode, contact your next level of support or your Ciena support group. • If the two switch modes are different, determine from the company records which is correct and change the switch mode of the other optical interface pair. Refer to the “Changing the protection parameters for a pair of facilities or equipment” procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. Note: Changing the protection switch mode for one of the optical interface circuit packs in a pair will automatically change the protection switch mode for the other circuit pack in the pair.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-262

Protection Scheme Mismatch

Alarm IDs: 252, 253, 254, 288, 819, 950, 1021, 1121, 1351, 1374, 1394

Probable cause

This alarm is raised when the near-end is provisioned as 1+1 and the protection scheme in the received APS bytes indicates a different protection scheme is provisioned on the far-end. This alarm is not indicative of the near-end provisioned as 1+1 and the far-end having no protection scheme provisioned.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|---|
| 1 | Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" in this document. |
| 2 | Retrieve the protection scheme of the interface pair. Refer to the "Retrieving protection parameters" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. |
| 3 | Determine the network element and interface modules that are on the remote end of the link. |
| 4 | Log into the remote terminal. |
| 5 | Retrieve the protection scheme of the optical interface pair on the remote terminal. Refer to the "Retrieving protection parameters" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. |
| 6 | Compare the protection scheme reported with each other and with the company records. <ul style="list-style-type: none">• If both optic pairs report the same protection scheme, contact your next level of support or your Ciena support group.• If the two protection scheme are different, determine from the company records which is correct and change the protection scheme of the other optical interface pair. Refer to the "Changing the protection parameters for a pair of facilities or equipment" procedure in <i>Configuration - Protection Switching</i>, 323-1851-315. |
| 7 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

Note: Changing the protection scheme for one of the optical interface circuit packs in a pair will automatically change the protection scheme for the other circuit pack in the pair.

—end—

Procedure 4-263

Protection Switch Active alarms

Use this procedure to clear alarms associated with active protection switches.

Forced Switch Active

Alarm ID: 62, 499, 500, 501, 502, 507, 958, 1014, 1122, 1329, 1341, 1404, 1405, 1406, 1603, 1738, 2162

Probable cause

This alarm is raised when a user successfully operates a forced switch on a linear, 1+1 port TPT, 1+1 TPT, or A-SNCP configuration. This alarm is cleared when the user releases the forced switch, or when a higher priority request overrides it.

Impact

Minor, non-service-affecting (m, NSA) alarm

Lockout Active

Alarm ID: 517, 518, 519, 520, 960, 1015, 1124, 1331, 1343, 1407, 1408, 1409, 1605, 2163

Probable cause

This alarm is raised when a user successfully operates a lockout on a 1+1 port TPT, or 1+1 TPT configuration. This alarm is cleared when the user releases the lockout request.

Impact

Minor, non-service-affecting (m, NSA) alarm

Manual Switch Active

Alarm ID: 959, 1013, 1342, 1401, 1402, 1403, 1604, 1737, 2161

Probable cause

This alarm is raised when a user successfully operates a manual switch on a 1+1 TPT configuration with revertive mode.

This alarm is cleared when the user releases the manual switch, or when a higher priority request overrides it.

Impact

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-263 (continued)

Protection Switch Active alarms**Prerequisites**

To perform this procedure, you require an account with at least a level 2 UPC.

Step	Action
------	--------

- | | |
|---|--|
| | Note: No action is required if a user is performing a test or maintenance operation. This alarm is a reminder not to leave a potentially service-affecting condition on the system. |
| 1 | Release the indicated protection switch. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-264

Protection Switch Complete

Alarm IDs: 490, 491, 492, 493, 498, 650, 957, 1023, 1328, 1340, 1396

Probable cause

This alarm is raised when a protection switch has completed.

For an G.8032 Ethernet Ring Protocol (G.8032 ERP), this alarm is raised when the Ringlet enters either protected state or pending state.

For G.8032 rings provisioned with an infinite wait-to-restore time, the clearing of this alarm is controlled by “G.8032 switch alarm mode” parameter found in the Site Manager Node Information application. Refer to the “Node Information - System parameters” table and “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301, for more information.

Impact

Warning

Minor, non-service affecting (m, NSA) alarm for Ringlet

Enabled by default

Step	Action
------	--------

- | 1 | Note: No action is required if a user is performing a test or maintenance operation. This alarm is a reminder not to leave a potentially service-affecting condition on the system. | | | | | | |
|---|--|----|------------|---|------------------------|-----------|------------------------|
| | Select your first step. | | | | | | |
| | <table><thead><tr><th>If</th><th>Then go to</th></tr></thead><tbody><tr><td>the alarm is raised against an G.8032 ERP</td><td>step 5</td></tr><tr><td>otherwise</td><td>step 2</td></tr></tbody></table> | If | Then go to | the alarm is raised against an G.8032 ERP | step 5 | otherwise | step 2 |
| If | Then go to | | | | | | |
| the alarm is raised against an G.8032 ERP | step 5 | | | | | | |
| otherwise | step 2 | | | | | | |
| 2 | Retrieve the traffic protection status for the alarmed circuit pack. Refer to the “Retrieving protection status details” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. Note that in an optical link provisioned for bidirectional switching, the switch source may be at the far-end of the link. | | | | | | |

Procedure 4-264 (continued)
Protection Switch Complete

Step	Action								
3	If an automatic protection switch is active, check for equipment failure, signal failure or degraded signal alarms related to this alarmed unit. Clear any alarms by following the appropriate trouble clearing procedure in this document.								
	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>no automatic switches are active</td><td>go to step 4</td></tr> <tr> <td>the original alarm has cleared</td><td>this procedure is complete</td></tr> <tr> <td>the original alarm has not cleared</td><td>go to step 4</td></tr> </table>	If	Then	no automatic switches are active	go to step 4	the original alarm has cleared	this procedure is complete	the original alarm has not cleared	go to step 4
If	Then								
no automatic switches are active	go to step 4								
the original alarm has cleared	this procedure is complete								
the original alarm has not cleared	go to step 4								
4	Determine if any user request resulted in a dropped protection switch. Check for protection switching alarms. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.								
	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>no user switches are active</td><td>go to step 5</td></tr> <tr> <td>the original alarm has cleared</td><td>this procedure is complete</td></tr> <tr> <td>the original alarm has not cleared</td><td>go to step 5</td></tr> </table>	If	Then	no user switches are active	go to step 5	the original alarm has cleared	this procedure is complete	the original alarm has not cleared	go to step 5
If	Then								
no user switches are active	go to step 5								
the original alarm has cleared	this procedure is complete								
the original alarm has not cleared	go to step 5								
5	Select your next step.								
	<table> <tr> <th>If the ringlet</th><th>Then go to</th></tr> <tr> <td>is newly provisioned</td><td>step 6</td></tr> <tr> <td>has gone from Idle to Protected state</td><td>step 11</td></tr> </table>	If the ringlet	Then go to	is newly provisioned	step 6	has gone from Idle to Protected state	step 11		
If the ringlet	Then go to								
is newly provisioned	step 6								
has gone from Idle to Protected state	step 11								
6	Launch the G.8032 ERP Management application under Configuration. For each node in the ringlet, verify the provisioning. Refer to in <i>Configuration - Connections Management</i> , 323-1851-320.								
7	For a major ring, verify that two ringlet ports are provisioned on each node.								
8	For a sub-ring, verify that a single Termination ringlet port is provisioned on each of the nodes with the shared segment. Verify that two Tandem ringlet ports are provisioned on all other nodes in the sub-ring.								
9	Verify that the Ring ID, RAPS VID and RAPS Type are consistent for all nodes in the ringlet.								
10	Verify that an RPL port is provisioned. Go to step 16 .								
11	Launch the G.8032 ERP Management application under Configuration. Refer to the “Retrieving rings and ring ports information in an G.8032 ERP” procedure in <i>Configuration - Connections Management</i> , 323-1851-320.								
12	Select the Ringlet Port Provisioning tab.								

Procedure 4-264 (continued)

Protection Switch Complete

Step	Action
13	Use the Ring and Group filter fields to limit the display to the ringlet for which the alarm was raised.
14	Click Retrieve .
15	Observe the Block status for the ringlet ports. A Blocked state on a ringlet port other than the RPL indicates a problem on the facility. Once the problem on the facility is resolved, the alarm should clear after the expiry of the Wait To Restore timer.
16	If the alarm remains active, ensure that no other alarm is active, then contact your next level of support or Ciena support group.

—end—

Procedure 4-265

Protection Switch Complete - Revertive

Alarm IDs: 489, 956, 1022, 1327, 1339, 1397, 1601, 1735

Probable cause

This alarm is raised when the protection mode is provisioned as revertive, and a protection switch has completed.

Impact

Warning

The Protection Switch Complete - Revertive alarm is not enabled by default and must be enabled by the user. To enable the alarm, set the Path Switch Event parameter to User & Auto in the System tab under the Node Information section. see “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301.

Step	Action								
	Note: No action is required if a user is performing a test or maintenance operation. This warning is a reminder not to leave a potentially service-affecting condition on the system.								
1	Check the traffic protection status. Refer to the “Retrieving protection status details” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.								
2	If an automatic protection switch is active, check for equipment failure, signal failure or degraded signal alarms related to this alarmed unit. Clear any alarm by following the appropriate trouble clearing procedure in this document.								
	<table border="1"> <thead> <tr> <th>If</th> <th>Then</th> </tr> </thead> <tbody> <tr> <td>no automatic switches are active</td> <td>go to step 3</td> </tr> <tr> <td>the original alarm has cleared</td> <td>this procedure is complete</td> </tr> <tr> <td>the original alarm has not cleared</td> <td>go to step 3</td> </tr> </tbody> </table>	If	Then	no automatic switches are active	go to step 3	the original alarm has cleared	this procedure is complete	the original alarm has not cleared	go to step 3
If	Then								
no automatic switches are active	go to step 3								
the original alarm has cleared	this procedure is complete								
the original alarm has not cleared	go to step 3								
3	Determine if a user switch (manual, forced or lockout) is active, release the switch. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.								
	<table border="1"> <thead> <tr> <th>If</th> <th>Then</th> </tr> </thead> <tbody> <tr> <td>no user switches are active</td> <td>go to step 4</td> </tr> <tr> <td>the original alarm has cleared</td> <td>this procedure is complete</td> </tr> <tr> <td>the original alarm has not cleared</td> <td>go to step 4</td> </tr> </tbody> </table>	If	Then	no user switches are active	go to step 4	the original alarm has cleared	this procedure is complete	the original alarm has not cleared	go to step 4
If	Then								
no user switches are active	go to step 4								
the original alarm has cleared	this procedure is complete								
the original alarm has not cleared	go to step 4								

4-632 Alarm clearing procedures

Procedure 4-265 (continued)

Protection Switch Complete - Revertive

Step	Action
4	If the alarm remains active, ensure that no other alarm is active.
5	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-266

Provisioning Database Freeze Enable

Alarm ID: 1046**Probable cause**

This alarm is raised against the shelf after the OPR-DBFRZ command is invoked by the user. All further provisioning commands are blocked. Refer to *TL-1 Description*, 323-1851-190, for more information about the OPR-DBFRZ command.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Initiate a warm restart on the shelf processor. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document. A warm restart is recommended, even though any type of restart would clear this alarm.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-267

Provisioning Incompatible

Alarm ID: 82

Probable cause

This alarm is raised when a circuit pack or pluggable is inserted in a slot with auto-equip enabled, and the circuit pack or pluggable cannot be provisioned in that slot for reasons such as the following:

- provisioning equipment that requires high-flow cooling equipped in a 14-slot shelf when the **Actual cooling capacity** (actual cooling capacity of the shelf) shelf attribute is set to “Low flow”. Refer to the “Equipment provisioning validation based on shelf cooling capacity” section in *Administration and Security*, 323-1851-301, for farther details.
- provisioning equipment with a power consumption value that will cause the **Calculated shelf power** (aggregate shelf power usage) and/or the Calculated shelf zone X power shelf attribute (where “X” is the zone number for a shelf with multiple power zones) to exceed the shelf/zone power threshold limit (calculated from the **Provisioned shelf current** shelf attribute). Refer to the “Equipment provisioning validation based on shelf power capacity” section in *Administration and Security*, 323-1851-301, for farther details.
- a PKT/OTN circuit pack is placed in a shelf before a PKT/OTN XC circuit pack is provisioned
- a PKTOTN is inserted and the SM is not provisioned.
- ISS autoprovisioning fails due to a lack of license on the License Server host.

Impact

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-267 (continued)
Provisioning Incompatible

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action	
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.	
2	Unseat the alarmed circuit pack from the shelf. The alarm should clear.	
3	Determine if the alarmed circuit pack is incompatible with the shelf type (for example, a 40G module in a non-40G shelf). Refer to the “Engineering rules” section in <i>Planning - Ordering information</i> , 323-1851-151, for engineering information on the module.	
4	If the circuit pack is not compatible	Then replace the circuit pack with one that is compatible with the shelf. The procedure is complete.
	is compatible	go to step 5
5	Retrieve the cooling types currently configured on the shelf. Refer to the “Displaying node information” in the <i>Administration and Security</i> , 323-1851-301.	
6	Compare the Provisioned and Actual cooling capacity with the required cooling capacity value for the new module. Refer to <i>Planning - Ordering information</i> , 323-1851-151, for engineering information on the module.	
7	Confirm the shelf provisioned cooling value is High flow for a circuit pack that requires High flow, or Low flow or High flow for a circuit pack that requires Low flow cooling.	
8	If the module is not compatible with the shelf cooling	Then replace the cooling module with one that is compatible with the shelf. The procedure is complete.
	compatible with the shelf cooling	go to the next step

Procedure 4-267 (continued)

Provisioning Incompatible

Step	Action						
9	Retrieve the shelf power budget and provisioned shelf current. Refer to the “Displaying node information” procedure in the <i>Administration and Security</i> , 323-1851-301.						
10	Determine the power requirement of the alarmed module. Refer to the “Technical specifications” section in <i>Planning</i> , NTRN10GK.						
11	Calculate the worst case shelf available power by multiplying the provisioned shelf current by 43 volts to get total watts available and subtracting the shelf power budget from the calculated available power to get the available power.						
12	<table><tr><td>If the power requirements of the alarmed module are</td><td>Then</td></tr><tr><td>greater than the available power</td><td>shelf power overload is causing the alarm. Refer to the alarm clearing steps for “Shelf Power Near Limit” on page 4-701.</td></tr><tr><td>less than the available power</td><td>contact your next level of support or Ciena support group</td></tr></table>	If the power requirements of the alarmed module are	Then	greater than the available power	shelf power overload is causing the alarm. Refer to the alarm clearing steps for “ Shelf Power Near Limit ” on page 4-701.	less than the available power	contact your next level of support or Ciena support group
If the power requirements of the alarmed module are	Then						
greater than the available power	shelf power overload is causing the alarm. Refer to the alarm clearing steps for “ Shelf Power Near Limit ” on page 4-701.						
less than the available power	contact your next level of support or Ciena support group						

—end—

Procedure 4-268

Provisioning Incompatible - Pluggable

Alarm ID: 1136

Probable cause

This alarm is raised when a pluggable is inserted in a supported slot with auto-equip enabled, and the pluggable cannot be provisioned in that slot because the provisioning equipment with a power consumption value will cause the **Calculated shelf power usage** shelf attribute (aggregate shelf power usage) to exceed the shelf power threshold limit (calculated from the **Provisioned shelf current** shelf attribute) or there is other provisioning currently in place which prevents this pluggable from being supported as is.

On a 40G+ CFP OCI circuit pack, if the sum of CFP pluggable “maximum power value” and ACTUALPOWER exceed the shelf power limit threshold, the automatically provisioning of CFP pluggable shall be blocked and the 'Provisioning Incompatible' alarm is raised against that port.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
2	Retrieve the shelf power budget and provisioned shelf current. Refer to the “Displaying node information” procedure in the <i>Administration and Security</i> , 323-1851-301.
3	Determine the power requirement of the alarmed pluggable. Refer to the “Technical specifications” section in <i>Planning</i> , NTRN10GK.
4	Calculate the worst case shelf available power by multiplying the provisioned shelf current by 43 volts to get total watts available and subtracting the shelf power budget from the calculated available power to get the available power.

Procedure 4-268 (continued)

Provisioning Incompatible - Pluggable

Step	Action						
5	<table><tr><td>If the power requirements of the alarmed module are</td><td>Then</td></tr><tr><td>greater than the available power</td><td>shelf power overload is causing the alarm. Refer to the alarm clearing steps for “Shelf Power Near Limit” on page 4-701</td></tr><tr><td>less than the available power</td><td>contact your next level of support or Ciena support group</td></tr></table>	If the power requirements of the alarmed module are	Then	greater than the available power	shelf power overload is causing the alarm. Refer to the alarm clearing steps for “Shelf Power Near Limit” on page 4-701	less than the available power	contact your next level of support or Ciena support group
If the power requirements of the alarmed module are	Then						
greater than the available power	shelf power overload is causing the alarm. Refer to the alarm clearing steps for “Shelf Power Near Limit” on page 4-701						
less than the available power	contact your next level of support or Ciena support group						

—end—

Procedure 4-269

RAMAN Failed To Turn On

Alarm ID: 944

Probable cause

This alarm is raised against a RAMAN facility of an SRA circuit pack when the handshaking between the local circuit pack and the far-end circuit pack fails. Reasons for the failure include:

- the far-end circuit pack has failed
- the Telemetry Gain signal is not present
- a disconnected fiber
- a pinched fiber connection
- a dirty optical fiber connector
- a defective fiber optic patchcord
- a defective module
- an incorrectly provisioned value

Note: After a shelf power cycle, the SRA circuit pack could take up to 20 minutes to recover. The alarm is active during this time.

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have a network and site diagram
- have a fiber cleaning kit

Impact

Major, service-affecting (M, SA) alarm

Procedure 4-269 (continued)

RAMAN Failed To Turn On

Step	Action
1	Ensure the provisioning values on the network element upstream of the SRA are correct.
2	Place the alarmed RAMAN amplifier facility out of service (OOS). Refer to the “Changing the primary state of a facility procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	Clean and then reconnect the input fibers and connectors at the RAMAN amplifier. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
4	Place the RAMAN amplifier back in-service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
5	If the alarm still persists, check the upstream connection for mating or pinched fibers.
6	Restart the SRA circuit pack supporting the alarmed RAMAN amplifier facility. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
7	Replace the SRA circuit pack supporting the alarmed RAMAN facility. Refer to the “Replacing the amplifier modules” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
8	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-270

RDI

Alarm IDs: 2022

Probable cause

For the WLAi variants circuit pack this alarm is raised against the PTP facilities when an incoming OTSi RDI from the far end network element is detected.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action
1	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
2	Use company traffic connection information to identify the source site of the signal.
3	If the network element is not connected to a 6500 network element at the far-end, or if the network element is part of a mid-span meet and the far-end network element is from another vendor, use the alarm system of the other vendor to find the problem.
4	Log into the source network element where the signal is generated. If you cannot log in remotely from the local network element, you must travel to the source site.
5	Retrieve all alarms from the remote network element at the transmit end. Refer to the corresponding alarms and events procedures in this document.

Procedure 4-270 (continued)

RDI

Step	Action	
6	If there are no alarms for the source facility	Then ensure that alarm reporting of related alarms is enabled. Refer to the “Retrieving alarm profiles” in this document. Also ensure that the equipment and facility of the remote circuit pack are in-service. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If there are masked alarms on this facility, clear them and go to step 7 .
	alarms for the source facility	refer to the appropriate alarm clearing procedures. The Remote Defect Indication alarm may be expected if other alarms are active.
7	At the local network element, retrieve all alarms to determine if the original alarm has cleared.	
	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 8
8	If the alarm does not clear, contact your next level of support or your Ciena support group.	

—end—

Procedure 4-271 Reach Violation

Alarm IDs: 2024, 2046

Probable cause

This alarm is raised on PTP facility if the actual length of the link exceeds the provisioned reach class. Once the violation is detected customer traffic will be blocked by conditioning from the on-ramp direction of the link, but GCC0 and GCC1 comms will remain up.

WLAi line ports allow users to provision the reach class for “Planned” distance.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Verify the reach class of the link. Refer to “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	Re-provision the WLAi line port PTP to make sure the actual length of the link does not exceed the provisioned reach class. Refer to “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

—end—

Procedure 4-272

Redundant Database Synch Failed

Alarm ID: 1028

Probable cause

This alarm is raised against the shelf to indicate that the shelf processor (SP) database synchronization between the two SPs failed.

This alarm is masked under the following conditions:

- SP equipment faults, indicated by alarms such as Autoprovisioning Mismatch, Circuit Pack Failed, Circuit Pack Mismatch, Circuit Pack Missing, and Circuit Pack Unknown
- communications problems between the two SPs, indicated by alarms such as Inter-card Suspected and Internal Mgmt Comms Suspected
- a load mismatch between the two SPs or any SP upgrade failed, indicated by alarms such as Software Upgrade Failed, Software Auto-Upgrade in Progress, Software Configuration Unknown, Software Delivery Incomplete, Software Mismatch, and Software Upgrade Failed
- any system fault, indicated by alarms such as Disk Full, Database Integrity failed, and Transport Data Recovery Failed
- SP redundancy is deleted

ATTENTION

In case of SP redundancy, this alarm can be raised after a warm or cold restart. The alarm will clear automatically.

Impact

Minor, non-service affecting (m, NSA) alarm

Step	Action
1	Verify that none of the conditions listed in the “ Probable cause ” section exist on the network element. Clear any active SP alarms according the alarm clearing procedures in this document.
2	Confirm the secondary SP equipment is in service. Refer to the “Retrieving equipment and facility details” in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Change the secondary SP equipment to in-service if required. Refer to the “Changing the primary state of a circuit pack, module, or Pluggable” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

Procedure 4-272 (continued)

Redundant Database Synch Failed

Step	Action
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-273

Redundant Database Synch Failed - CP

Alarm ID: 1185

Probable cause

This alarm is raised against the shelf in the Photonic Control Plane to indicate that the shelf processor (SP) database synchronization between the two SPs failed. This alarm can be raised against SPs with CPU-2, or can be raised against any SP type in a Coherent Select configuration.

This alarm is not masked for Photonic Control Plane.

ATTENTION

In case of SP redundancy, this alarm can be raised after a warm or cold restart. The alarm will clear automatically.

Impact

Minor, non-service affecting (m, NSA) alarm

Step	Action
1	Verify that none of the conditions listed in the “Probable cause” section exist on the network element. Clear any active SP alarms according to the alarm clearing procedures in this document.
2	Restart the inactive CPU2. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
3	If the restart fails, restart the active CPU2.
4	If the alarm does not clear, confirm the secondary SP equipment is in service. Refer to the “Retrieving equipment and facility details” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Change it to in-service if required. Refer to the “Changing the primary state of a circuit pack, module, or Pluggable” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-274

Redundant Database Synch in Progress

Alarm ID: 1026

Probable cause

This alarm is raised against the shelf to indicate that there is a database synchronization in progress between two shelf processors.

Impact

Warning

Step	Action
1	No action is required. The alarm clears once the synchronization is complete.
2	If the alarm does not clear, or is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-275

Redundant Release Synch Failed

Alarm ID: 1044

Probable cause

This alarm is raised against the shelf to indicate that the Shelf Processor release synchronization between the two SPs failed.

This alarm is masked under the following conditions:

- SP equipment faults, indicated by alarms such as Autoprovisioning Mismatch, Circuit Pack Failed, Circuit Pack Mismatch, Circuit Pack Missing, and Circuit Pack Unknown
- communications problems between the two SPs, indicated by alarms such as Inter-card Suspected and Internal Mgmt Comms Suspected
- a load mismatch between the two SPs or any SP upgrade failed, indicated by alarms such as Software Upgrade Failed, Software Auto-Upgrade in Progress, Software Configuration Unknown, Software Delivery Incomplete, Software Mismatch, and Software Upgrade Failed
- any system fault, indicated by alarms such as Disk Full, Database Integrity failed, and Transport Data Recovery Failed
- SP redundancy is deleted

Impact

Minor, non-service affecting (m, NSA) alarm

Step	Action
1	Verify that none of the conditions listed in the “Probable cause” section exist on the network element. Clear any active SP alarms according the alarm clearing procedures in this document.
2	Confirm the secondary SP equipment is in service. Refer to the “Retrieving equipment and facility details” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Change it to in-service if required. Refer to the “Changing the primary state of a circuit pack, module, or pluggable” in the <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-276

Redundant Release Synch in Progress

Alarm ID: 1042**Probable cause**

This alarm is raised against the shelf to indicate that there is a release synchronization in progress between the two shelf processors.

Impact

Warning

Step	Action
1	No action is required. The alarm clears once the synchronization is complete.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-277

Release Server Mismatch

Alarm ID: 1240

Probable cause

This alarm is raised when the provisioned release server does not match the current release.

Impact

Minor, non-service affecting (m, NSA) alarm

Step	Action
1	Retrieve the release server list to check the server. Refer to the “Retrieving a list of the software releases and release servers” procedure in <i>Administration and Security</i> , 323-1851-301.
2	Delete the invalid release server or enter the valid release that matches the current release. Refer to the “Setting a release server” procedure in <i>Administration and Security</i> , 323-1851-301.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-278

Release Server URL Fail

Alarm ID: 1078**Probable cause**

This alarm is raised against the shelf when the:

- release server FTP URL parameters entered are invalid (IP/location/User/Password)
- release server is down when the NE does a check
- proper load does not exist on the FTP server

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

Impact

Minor, non-service affecting (m, NSA) alarm

Step	Action
1	Retrieve the release server list to trigger a check of the server to see if it has become available. Refer to the “Retrieving a list of the software releases and release Servers” in the <i>Administration and Security</i> , 323-1851-301.
2	Delete the invalid release server URL (clear the URL to disable this feature) or enter the valid release server FTP URL parameters in the URL field and click the Set Server button. Ensure the server selected has the proper load on it at the location referred to by the URL. Refer to the “Setting a release server” procedure in the <i>Administration and Security</i> , 323-1851-301.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-279

Remote Inventory Not Supported

Alarm ID: 1029**Probable cause**

This alarm is raised when the access panel does not support external inventory slots (NTK505xA5) and an OMD4, OMX, CMD44, CMD64, CMD96, BMD2, UBMD2, MBDM2, GMD10, FIM, PPC6, TPT or DSCM module is provisioned in the access panel slots (83-90).

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
1	If the access panel PEC is NTK505MB Then the access panel may be faulty and requires replacement. Go to step 3
	not NTK505MB go to step 2
2	If the equipment provisioned in external slots (83-90) is correct Then the access panel may be faulty and require replacement. Go to step 3
	not correct go to step 4
3	Replace the access panel with an access panel that has connections for eight external slots (NTK505MB). Refer to the “Replacing the access panel” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 5 .
4	Delete the equipment that is incorrectly provisioned in the external slots. Refer to the “Deleting a circuit pack, module or pluggable” procedure in the <i>T-Series Guide</i> , 323-1851-103.

Procedure 4-279 (continued)

Remote Inventory Not Supported

Step	Action
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-280

Remote Node Unreachable

Alarm IDs: 1413

Probable cause

This alarm is an indication that local end is unable to communicate with the remote side and is raised on expiry of OSRPLINE timers (inactivity timer or remote information response timer).

A MiniHello message (LIP) is generated every 15 seconds on/for each OSRP line ("on" when inband comms is used and "for" when out-of-band (OOB) communications are used). Therefore, a node expects to receive LIPs every 15 seconds on/for each OSRP line. However, each node also has a MiniHello Inactivity Timer set basically to 3xLIP generation timer (3x15 seconds=45 seconds). If a node does not see an LIP on/for an OSRP line within the Inactivity Timer period, then it raises the Remote Node Unreachable alarm.

For the Photonic Control Plane, if out-of-band (OOB) is disabled, the alarm can be raised under the following conditions:

- The OSC facility associated with the OSRP line or OSRP link is out-of-service.
- The circuit pack supporting the OSC facility (for example, 2xOSC on a 6500 shelf) is out-of-service, missing, or rebooting.
- A TID consolidated node (TIDc) member shelf supporting the OSC facility is down.
- Communication between the TIDc primary and member shelf is down.

When OOB is enabled on the OSRP link, the alarm can be raised in the following conditions:

- The OSRP line OOB Common Identifier provisioned at each node terminating the OSRP line does not match. (This parameter is provisioned on the OSRP line, not the OSRP link.)
- The OOB Remote Node ID is incorrect. For example, it does not match the node ID of the neighboring node.
- The OOB Remote IP address is incorrect. For example, it does not match the CONTROL-shelf#-GROUP0 of the neighboring node.

Procedure 4-280 (continued)

Remote Node Unreachable

- The OOB Remote UDP port is incorrect. For example, it does not match the UDP port provisioned in the neighboring node (see the “Retrieving OSRP provisioning information” and “Editing an OSRP instance” procedures in *Configuration - Control Plane*, 323-1851-330).
- OOB comms messages are routed using OSPF. Refer to the “Photonic Control Plane data communications” section in *Data Communications Planning and User Guide*, 323-1851-101, for more information on how to setup and troubleshoot the various supported configurations.

This alarm is also raised when Interswitch Communication Channel Protocol (ISCC) communication fails on the Control Plane and the OSRP line goes down after the timeout).

Impact

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action				
1	Identify the port that raised the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.				
2	Identify the neighbor node name, supporting Termination Point (TP), neighbor containing link ID.				
3	<table> <tr> <td>If the alarm is raised against the Photonic Control Plane</td><td>Then go to step 10</td></tr> <tr> <td>otherwise</td><td>step 4</td></tr> </table>	If the alarm is raised against the Photonic Control Plane	Then go to step 10	otherwise	step 4
If the alarm is raised against the Photonic Control Plane	Then go to step 10				
otherwise	step 4				
4	If a remote node OSRP Connection was deleted or not created, recreate the OSRP Connection.				
5	<table> <tr> <td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has cleared	Then the procedure is complete	not cleared	go to step 6
If the original alarm has cleared	Then the procedure is complete				
not cleared	go to step 6				

Procedure 4-280 (continued)

Remote Node Unreachable

Step	Action						
6	Verify the datacomms setting of the port raising the alarm and compare it to the settings for the neighbor node. Both sides must have the same setting for the alarm to clear. If using in-band communications, check the DCC/GCC/OSC datacomms settings (whichever is applicable based on the provisioned Control Plane). For out-of-band communications (supported by the Photonic Control Plane only) check the DCN/SCN datacomms settings or OSC datacomms settings, depending on which is being used. For details on in-band and out-of-band datacomms, refer to the “SONET/SDH, OTN, and Photonic Control Plane considerations” section in <i>Data Communications Planning and User Guide</i>, 323-1851-101.						
7	If the ports are not configured correctly, change appropriate side to the correct setting.						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Verify if the Line Module is in a reboot state, wait five minutes for the Line Module to complete the boot cycle. Go to step 18 .						
10	Determine if OOB is used for an OSRP link or line by launching the OSRP Provisioning Site Manager application, selecting a link, and checking if OOB is enabled or disabled.						
11	<table> <tr> <th>If OOB is</th><th>Then go to</th></tr> <tr> <td>disabled</td><td>step 12</td></tr> <tr> <td>enabled</td><td>step 16</td></tr> </table>	If OOB is	Then go to	disabled	step 12	enabled	step 16
If OOB is	Then go to						
disabled	step 12						
enabled	step 16						
12	Determine the OSC facilities associated with the OSRP link or line by using the OSRP Provisioning and the Photonic Configuration Management Site Manager applications: - In the OSRP Provisioning application, select the Lines tab and lookup the “Local TP” column to determine the AID of the amplifier circuit pack associated with the OSRP line (for example, ADJ-2-2-5, which means the amplifier port in shelf 2, slot 2, port 5). - In the Photonic Configuration Management application, select the ROADM OTS that contains the amplifier AID determine in the previous step, and lookup up the OSC port associated to the OTS.						
13	Verify that the OSC facility is in-service using the Site Manager Equipment and Facility application. Place the facility in-service if it is out-of-service.						
14	Verify that there are no alarms related to the OSC facility. Refer to corresponding alarms clearing procedure if any OSC alarms are present.						

Procedure 4-280 (continued)

Remote Node Unreachable

Step	Action
15	If the OSC facility is on a TIDc member shelf, ensure that there are no alarms indicating communications issues between the TIDC primary shelf and that member shelf. For example, there should be no “ILAN Failure” and no “Member Shelf Unreachable” alarms. Go to step 18 .
16	Verify that the OSRP line OOB Common Identifier is correct. If it is incorrect, put the OSRP line out-of-service and edit the value to match the value provisioned at the neighboring node. Put the OSRP line back in-service.
17	Verify that the OSRP link Remote Node IP, Remote Node ID, and Remote Node UDP ports are correct. If they are incorrect, put the OSRP line associated to that OSRP link out-of-service and edit the values on the OSRP link. Put the OSRP line back in-service.
18	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-281

Remote Port OOS

Alarm IDs: 1069, 1248, 1847

Probable cause

This alarm is raised against a ETTP facility when there is a Tx conditioning failure due to the remote port being manually put OOS.

The alarm is raised on far end when NON-LAG UP MEP (maintenance end point) port administratively is declared OOS. However, if aggregated UP MEP port belonging all members moved to administratively OOS, then other end declares Far End client signal fail due to VLLI conditioning.

This alarm masks the “Far End Client Signal Fail” alarm on ETTP ports.

Impact

Major, service-affecting (M, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action
------	--------

- | | |
|---|--|
| 1 | Make sure the remote port is put back in-service. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-282

Remote Port Unreachable

Alarm IDs: 1246, 1368, 1846

Probable cause

This alarm is raised when the remote port on the far-end NE circuit pack is not transmitting heartbeat messages to the port on the near-end NE circuit pack which is alarming or the heartbeat messages are being impaired due to network interference. Alternatively, the port on the near-end NE circuit pack which is alarming, is not receiving the heartbeat messages from the remote port on the far-end NE circuit pack.

If the Virtual Link Loss Indicator (VLLI) is enabled, the alarm is raised on least numbered LAG/D-LAG member port when:

- Continuity Check Message (CCM) timeout occurs
- eMOTR receives bad MEPID or MAID from remote end
- there is a CCM interval error
- eMOTR receives a CCM MD level different than MEP MD level

This alarm masks the “Far End Client Signal Fail” and “Remote Port-Out of Service” alarms on ETTP ports.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545.

Step	Action						
1	Verify the remote port fiber connections. Repair, clean, and reconnect the fibers as required.						
2	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 3</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						

Procedure 4-282 (continued)

Remote Port Unreachable

Step	Action
3	Use the appropriate alarm clearing procedure to clear any alarms on the remote port.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-283

Resources Above Threshold

Alarm IDs: 2125**Probable cause**

This warning is raised when internal resources on the circuit pack have exceeded its 90% threshold.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|--|
| 1 | If this warning is raised during provisioning the PKTIWF facilities procedure, see the Historical Fault Browser in Site Manager to find an event that indicates which resource has hit has exceeded its 90% threshold. |
| 2 | If the jitter buffer memory is the resource which has exceeded its 90% threshold, delete the PKTIWF facilities which have a "Facility Provisioning Failure" alarm. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-284

Resources At Limit

Alarm IDs: 2128

Probable cause

This warning is raised when internal resources on the circuit pack have reached its 98% limit.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	If this warning is raised during provisioning the PKTIWF facilities procedure, see the Historical Fault Browser in Site Manager to find an event that indicates which resource has hit the 98% limit.
2	If the jitter buffer memory is the resource which is at its limit, delete the PKTIWF facilities which have a "Facility Provisioning Failure" alarm.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-285

Root Directory Has Reached Maximum File Entry Limit

Alarm ID: 1308**Probable cause**

This alarm is raised when the root directory on the SP has reached its maximum file entry limit.

Impact

Major, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Contact your next level of support or Ciena support group. |
|---|--|

—end—

Procedure 4-286

Rx Channel Power Out of Range

Alarm IDs: 1312, 1571, 1597

Probable cause

This alarm is raised when the Rx Channel power is outside the Rx channel minimum power and Rx channel maximum power optical power range.

ATTENTION

The “Rx Channel Power Out Of Range” alarm may be raised on a valid setup as the thresholds for this alarm are in the acceptable operating range of the Rx interface. It is unlikely that this will occur as OnePlanner will typically set the Rx interface power in a range which will not raise this alarm. However, in some cases, when the Wavelength-Selective 40G OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD, 100G WL3e OTR, or 100G WL3/WL3e OCLD circuit pack is deployed with an NGM circuit pack, the alarm may be raised. If necessary, an attenuator pad can be used to bring the power level within the alarm threshold level. For the alarm threshold levels, see the Ranges button within the Site Manager Equipment & Facility Provisioning application.

Note: The alarm clears when the channel power level is above the minimum by a margin of 1dBm.

Impact

Major, service-affecting (M, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- have attenuator pads

Procedure 4-286 (continued)

Rx Channel Power Out of Range

Step	Action						
1	Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	If the "Rx power Out of Range" alarm is active, clear this alarm first. Refer to "Rx Power Out of Range" on page 4-669 in this document.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.						
5	In the Site Manager Configuration menu, select the Equipment & Facility Provisioning application. Select the alarmed facility and retrieve the value from the Rx channel Actual Power (dBm) column of the facility table. Refer to the "Retrieving optical power, wavelength, and dispersion ranges" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If the received power is out of the power monitoring range for the facility, the Rx channel Actual Power (dBm) column will display "OOR-HI" or "OOR-LO" instead of a numerical value.						
6	Click on the Ranges button to compare the Rx Channel Actual Power (dBm) value with the Rx channel minimum power (dBm) and Rx channel maximum power (dBm) values displayed.						
7	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
8	Use the optical power meter to measure the optical power level at the far-end upstream monitored point. Also, verify that the loss between the far-end transmitter and the Mux is an expected value. <table> <tr> <th>If the value of the loss is</th><th>Then</th></tr> <tr> <td>unexpected</td><td>troubleshoot the fiber or Photonic layer equipment</td></tr> <tr> <td>expected</td><td>go to step 9</td></tr> </table>	If the value of the loss is	Then	unexpected	troubleshoot the fiber or Photonic layer equipment	expected	go to step 9
If the value of the loss is	Then						
unexpected	troubleshoot the fiber or Photonic layer equipment						
expected	go to step 9						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						

Procedure 4-286 (continued)

Rx Channel Power Out of Range

Step	Action						
10	<table><tr><th>If the original alarm has</th><th>Then</th></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-287

Rx Dispersion Post Compensation Out Of Range

Alarm IDs: 2194

Probable cause

This alarm is raised on PTP facility on the WLAi or WL5e line port when the receiver post-compensation value is beyond the supported range.

The alarm clears after editing the Path Dispersion Provisioned for AUTO dispersion mode, or after editing the value for the Tx Dispersion Pre-Compensation Provisioned for FIXED dispersion mode on the upstream circuit pack. See [Table 4-3](#) for the thresholds values.

Table 4-3

Rx Dispersion Post compensation thresholds for C-Band

Supported Circuit Pack	Alarm threshold (ns/nm) (Note 1)	Alarm clearing threshold (ns/nm) (Note 2)	Rx Dispersion post-compensation (ns/nm) target with cable repair margin (Note 3)
WaveLogic Ai MOTR/ FOTR - line port	18500 ns/nm	< 17500	< 12500
WaveLogic 5e MOTR - line port	38868	< 38368	< 33300
Note 1: Rx Dispersion Post Comp Out Of Range will be raised. Note 2: Rx Dispersion Post Comp Out Of Range alarm will be cleared. Note 3: "Rx Dispersion Post-Compensation (ns/nm)" target with repair margin over the life of cable.			

Impact

Major, service-affecting (M, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Step	Action
------	--------

- | | |
|---|---|
| 1 | From the Site Manager Configuration menu, select Equipment & Facility Provisioning . Select the alarmed facility. |
|---|---|

Procedure 4-287 (continued)

Rx Dispersion Post Compensation Out Of Range

Step	Action						
2	Select your next step. <table><tr><td>If Tx Dispersion Compensation Mode is set to</td><td>Then go to</td></tr><tr><td>AUTO</td><td>step 3</td></tr><tr><td>FIXED</td><td>step 4</td></tr></table>	If Tx Dispersion Compensation Mode is set to	Then go to	AUTO	step 3	FIXED	step 4
If Tx Dispersion Compensation Mode is set to	Then go to						
AUTO	step 3						
FIXED	step 4						
3	Make sure the value for the Path Dispersion Provisioned is set correctly. Go to step 7 .						
4	Ensure that the “Rx Dispersion Post-Compensation (ps/nm)” acquired value is less than the corresponding circuit pack alarm clearing threshold found in Table 4-3 on page 4-667 .						
5	To adjust the Rx Dispersion Post-Compensation value, you must edit the Far End Transponder Line facility that it is connected to, and decrease the Tx Dispersion Pre-Compensation value.						
6	Verify that the “Rx Dispersion Post-Compensation (ps/nm)” acquired value has decreased. Repeat step 4 until the acquired value is less than the corresponding circuit pack alarm clearing threshold found in Table 4-3 on page 4-667 .						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-288

Rx Power Out of Range

Alarm IDs: 671, 842, 844, 847, 989, 1005, 1175, 1253, 1302, 1434

Probable cause

This alarm is raised when the Rx power is outside the minimum and maximum receive optical power range. For parallel optic CFPs, this is the logical OR of Power Out of Range of all optical channels.

Note: The alarm clears when the power level is above the minimum by a margin of 1.5dBm for Total Power.

Impact

Major, service-affecting (M, SA) alarm, unprotected
Minor, non-service-affecting (m, NSA) alarm, protected

The alarm severity for the 10G AM1/AM2 circuit pack is always
Minor, non-service-affecting (m, NSA)

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” in this document. |
| 2 | Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding. |

Procedure 4-288 (continued)
Rx Power Out of Range

Step	Action								
3	In the Site Manager Configuration menu, select the Equipment & Facility Provisioning application. Select the alarmed facility and retrieve the value from the Rx Actual Power (dBm) column of the facility table. (For circuit packs that use pluggable optical modules with multi-lane interfaces, this parameter is Rx Actual High Power [dBm] or Rx Actual Low Power [dBm]). Refer to the “Retrieving optical power, wavelength, and dispersion ranges” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If the received power is out of the power monitoring range for the facility, the Rx Actual Power (dBm) column will display “OOR-HI” or “OOR-LO” instead of a numerical value.								
4	Click on the Ranges button to compare the Rx Actual Power (dBm) value (Rx Actual High Power [dBm] or Rx Actual Low Power [dBm] value for multi-lane interfaces) with the Rx minimum power (dBm) and Rx maximum power (dBm) values displayed. Record these values.								
5	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
6	Use the optical power meter to measure the optical power level into the Rx interface.								
<table> <tr> <th>If the optical power is</th><th>Then go to</th></tr> <tr> <td>less than the Rx minimum value in step 4</td><td>step 7</td></tr> <tr> <td>greater than the Rx maximum value in step 4</td><td>step 8</td></tr> <tr> <td>within the Rx minimum power to Rx maximum power range</td><td>step 11</td></tr> </table>		If the optical power is	Then go to	less than the Rx minimum value in step 4	step 7	greater than the Rx maximum value in step 4	step 8	within the Rx minimum power to Rx maximum power range	step 11
If the optical power is	Then go to								
less than the Rx minimum value in step 4	step 7								
greater than the Rx maximum value in step 4	step 8								
within the Rx minimum power to Rx maximum power range	step 11								

Procedure 4-288 (continued)
Rx Power Out of Range

Step	Action						
7	Check the fibers for damage and ensure connections are mated properly and verify the fiber cleanliness between the subtending equipment and the port reporting the alarm. Go to step 11 .						
8	The signal may require padding, or you are using an incorrect pluggable type on the circuit pack reporting the alarm or an incorrect subtending pluggable type/circuit pack type. Contact your next level of support or your Ciena support group if you require more information.						
9	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
11	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the pluggable module on the circuit pack reporting the alarm. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the pluggable module on the circuit pack reporting the alarm. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the pluggable module on the circuit pack reporting the alarm. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>replace the pluggable module on the subtending circuit pack or the subtending circuit pack. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the pluggable module on the subtending circuit pack or the subtending circuit pack. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the pluggable module on the subtending circuit pack or the subtending circuit pack. Refer to the replacement procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
13	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-289

Secondary alarms

Use this procedure to clear secondary alarms caused by higher order alarms (for example, Loss of Signal or Loss of Frame) at upstream/downstream locations or when a facility is put in the OOS state or a connection is deleted at upstream/downstream locations.

These alarms are only applicable to the OTN switching, MSPP and Broadband services. Unless otherwise indicated, an alarm applies to all three services.

AIS (OC/STM)

Alarm ID: (7, 34, 148, 285, 987)

Probable cause

This alarm is raised when the network element detects an OC-1, OC-3/STM-1/STM-1J/STM-1e, OC-12/STM-4/STM-4J, OC-48/STM-16, OC-192/STM-64 or OC-768/STM-256 AIS in the SONET/SDH line overhead

OC/STM AIS alarms are caused by one of the following conditions on the circuit pack that is the source of the alarmed signal:

- a facility is out-of-service
- a circuit pack has failed on the OC-1, OC-3/STM-1, OC-12/STM-4, OC-48/STM-16, OC-192/STM-64, OC-768/STM-256 circuit pack

Impact

Critical, service-affecting (C, SA) alarm, if active in protected configurations or unprotected with cross-connects

Minor, non-service-affecting (m, NSA) alarm if inactive in protected configurations, or without cross-connects

Procedure 4-289 (continued)

Secondary alarms**AIS (STS/HO VC)****Alarm ID:(189, 466)****Probable cause**

This alarm is raised when the network element detects STS-192c/VC-4-64c AIS in the SONET/SDH path overhead.

STS/HO VC AIS alarms are caused by one of the following conditions at the far-end or passthrough network elements:

- an incoming signal is missing or errored at the far-end
- a circuit pack is failed at the far-end
- a loss of pointer alarm is raised at a passthrough connection to the optical interface

This alarm can also be raised if an intrusive test access session is in progress. No action is required if this is the cause.

Impact

Minor, service-affecting (m, SA) alarm if on an active path

Minor, non-service-affecting (m, NSA) alarm if on an inactive path

The alarm status is minor because it is a secondary alarm that indicates a problem upstream of this network element.

AIS (OTUTTP, STTP)**Alarm ID:(1450, 1462)****Probable cause**

This alarm is raised when the network detects SONET Line-AIS in the SONET/SDH line overhead. On OTN capable circuit packs (such as eMOTR), the alarm is raised at the OTU layer.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Laser Off Far End Failure Triggered (OC48/STM16, OC192/STM64, OC768/STM256, ETH10G, ETH40G, ETH100G, OTM1, OTM2, OTM3, PTP, FC800, FC1200 for Broadband services)**Alarm ID: (676, 678, 839, 967, 988, 1219, 1243, 1285, 1301, 1375, 1438)****Probable cause**

This alarm is raised to indicate the client Tx laser is shut down after detecting a remote failure or the upstream circuit pack client facility is in the OOS state.

Procedure 4-289 (continued)

Secondary alarms

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Local Fault (ETH10G, ETH40G, ETH100G, FC800, FC1200, ETTP)

Alarm ID: (674, 1259, 1299, 1455)

Probable cause

This alarm is raised when the client Rx interface detects a 802.3ae local fault (LF).

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

ODU AIS (OTMFLEX, OTM0, OTM1, OTM2, OTM3, OTM4 for Broadband services, ODUTTP, ODUCTP, ODU0, ODU1, ODUFLEX, TCMCTP, TCMTTP)

Alarm ID: (662, 992, 1160, 1468, 1477, 1478, 1495, 1496, 1608, 1729)

Probable cause

This alarm is raised when the Rx interface detects an ODU layer (path monitoring) AIS. The upstream Tx interface is sending ODU layer AIS. This can occur when the upstream OTMFLEX, ODUCTP, OTM0, OTM2, OTM3, or OTM4 facility is in the OOS state or when the upstream circuit pack is reporting a circuit pack fail.

This alarm is also raised on all of the 40G and 100G Line cards when ODU monitoring is Yes. ODU monitoring is Yes by default when 1+1 Line protection is provisioned on 40G Line cards.

This alarm can also be raised when proper payload index assignment is not used for circuit packs that interwork with the 2.5G MOTR circuit packs (NTK530NAE5 and NTK530NCE5). For more information about proper payload index assignment when interworking with these circuit packs, refer to *Configuration - Connections Management*, 323-1851-320.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-289 (continued)

Secondary alarms**ODU BDI (OTMFLEX, ODUTTP, ODUCTP, ODU0, ODU1, ODUFLEX, TCMCTP, TCMTP)****Alarm ID: (664, 993, 1161, 1469, 1479, 1480, 1493, 1494, 1606, 1730)****Probable cause**

This alarm is raised when the Rx interface detects an ODU layer (path monitoring) BDI (Backward Defect Indication). The upstream Tx interface sends ODU BDI when its Rx interface detects ODU failures.

This alarm is also raised on all the 40G and 100G Line cards when ODU monitoring is Yes. ODU monitoring is Yes by default when 1+1 Line protection is provisioned on 40G Line cards.

When Alarm Correlation is enabled (see “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301), the behavior of this alarm is driven by the Conditioning Override parameter (see the “Editing the nodal system parameters” in *Administration and Security*, 323-1851-301) with the exception of OTMn/OTU client facilities with no Actual Far-End Address discovered by SPLI (see SPLI in *Administration and Security*, 323-1851-301) where this alarm is always enabled. This exception is to support interworking between the 6500 node and non-Ciena client equipment which does not support Alarm Correlation.

Impact

Minor, non-service-affecting (m, NSA) alarm

OPU AIS (OTM0, OTM1, OTM2, OTM3, or OTM4 for Broadband services, ODUCTP, ETTP, STTP, ODUTTP)**Alarm ID: (964, 1011, 1181, 1487, 1624, 1670, 1677)****Probable cause**

This alarm is raised when the Rx interface detects an OPU layer (path monitoring) AIS. The upstream Tx interface is sending OPU layer AIS. This can occur when the upstream client signal cannot be provided for mapping into the OPU payload. A PN11 pattern is mapped into the OPU as a client payload generic AIS.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-289 (continued)

Secondary alarms

OTU BDI (OTUTTP, OTM1, OTM2, OTM3, or OTM4 for Broadband services)

Alarm ID: (665, 1002, 1171, 1446)

Probable cause

This alarm is raised when the Rx interface detects an OTU layer (section monitoring) BDI (Backward Defect Indication). The upstream Tx interface sends OTU BDI when its Rx interface detects OTU failures.

When Alarm Correlation is enabled, (see the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301) the behavior of this alarm is driven by the Conditioning Override parameter (see the “Editing the nodal system parameters” in *Administration and Security*, 323-1851-301) with the exception of OTMn/OTU client facilities with no Actual Far-End Address discovered by SPLI (see SPLI in *Administration and Security*, 323-1851-301) where this alarm is always enabled. This exception is to support interworking between the 6500 node and non-Ciena client equipment which does not support Alarm Correlation.

Impact

Minor, non-service-affecting (m, NSA) alarm

RFI (OC)

Alarm ID: (6, 37, 149, 284, 986,)

Probable cause

This alarm is raised when the network element detects a OC/STM remote fault indication (RFI) in the SONET line overhead because of a fault on another network element, or an optical fiber/cable has been cut.

Impact

Minor, non-service-affecting (m, NSA) alarm

RFI (STS)

Alarm ID: (192, 468)

Probable cause

This alarm is raised when the network element detects an STS remote fault/defect indication (RFI) in the SONET/SDH line overhead because of a fault on another network element, or an optical fiber/cable has been cut.

Procedure 4-289 (continued)

Secondary alarms**Impact**

Minor, non-service-affecting (m, NSA) alarm

The alarm status is minor because it is a secondary alarm that indicates a problem upstream of this network element.

RFI (STTP)

Alarm ID: (1463)

Probable cause

This alarm is raised when the client Rx interface detects STTP RFI in the SONET line overhead.

Impact

Minor, non-service-affecting (m, NSA) alarm

Remote Fault (ETTP)

Alarm ID: (675, 966, 1260, 1300, 1456)

Probable cause

This alarm is raised when the client Rx interface detects 802.3ae remote fault.

Impact

Critical, service-affecting (C, SA) alarm if not protected

Minor, non-service-affecting (m, NSA) alarm if protected

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the network connection information (that is, how the interface circuit packs on each network element connect to other network elements)

Step	Action
------	--------

- | | | |
|---|--------------------------|------------------------|
| 1 | If the alarm is | Then go to |
| | RFI, BDI or Remote Fault | step 4 |
| | otherwise | step 2 |

Procedure 4-289 (continued)

Secondary alarms

Step	Action
2	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document. First, clear any alarms of higher order in the hierarchy on the 6500 system or the subtending equipment using the appropriate procedures. You can also refer to the signal conditioning section in this document to help troubleshoot secondary alarms.
3	At the local network element, retrieve all alarms to determine if the original alarm has cleared.
4	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.
5	If a Local fault alarm is raised on an eMOTR SFP+ port, ensure the rate of the incoming traffic is matched with what the port is provisioned for. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
6	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.
7	If the network element is not connected to a 6500 network element at the remote end, or if the network element is part of a mid-span meet and the remote network element is from another vendor, use the alarm system of the other vendor to find the problem.
8	Log into the remote network element at the transmit end. If you cannot log in remotely from the local network element, you must travel to the remote site.
9	Retrieve all alarms from the remote network element at the transmit end.
10	Look for an alarm message for the remote network element circuit pack connected to the original shelf.

Procedure 4-289 (continued)

Secondary alarms

Step	Action	
11	If there are no alarms at the transmit end	Then ensure that the equipment and facility or the client and line facilities of the remote circuit pack are in-service and connected. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 12 .
	additional alarms at the transmit end	refer to the appropriate alarm clearing procedures. The RFI alarm can be ignored if the local alarm is AIS, Loss of Pointer, or Network Trace Identifier Mismatch, or Unequipped. Go to step 12 .
The setting of the RFI condition as a consequence of a Network Trace Identifier Mismatch, or Unequipped alarm is user provisionable.		
12	At the local network element, retrieve all alarms to determine if the original alarm has cleared.	
13	If the original alarm has cleared	Then the procedure is complete
	raised was STS/HO VC AIS, and the alarm has not cleared	go to step 14
	raised was OC/STM AIS, and the alarm has not cleared	go to step 16
	raised was STS/HO VC RFI, and the alarm has not cleared	go to step 18
	raised was OC/STM, and the alarm has not cleared	go to step 21
	raised was against OTUTTP, ODUTTP, STTP, or ETTP facility and the alarm has not cleared	go to step 19
	raised was otherwise, and the alarm has not cleared	go to step 21
14	Log into each of the passthrough network elements and retrieve alarms.	
15	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “ Alarm hierarchies and alarm severities ” section in this document. Clear any alarms of higher order on the hierarchy first using the appropriate procedures. Go to step 20 .	

Procedure 4-289 (continued)

Secondary alarms

Step	Action						
16	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
17	Replace the local circuit pack reporting the original alarm. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 20 .						
18	Verify the fiber connection for the interface circuit pack that connects to the original shelf. Repair, clean, and reconnect the fiber as required.						
19	Verify that the provisioned line rate on the upstream circuit pack matches the line rate on the local circuit pack. Correct any mismatches. Refer to the "Retrieving OTM info" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
20	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 5</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 5
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 5						
21	Contact your next level of support or your Ciena support group.						

—end—

Procedure 4-290

Secondary License Server Unavailable

Alarm ID: 2012**Probable cause**

This alarm is raised when the 6500 shelf cannot communicate with the secondary license server/manager.

Impact

Minor, non-service-affecting (m, nsa) alarm

Step	Action
1	Verify the network connection between the 6500 shelf and the license server/manager.
2	If you are using HTTPS, the correct date and time must be set on the shelf (either manually or using TOD/NTP server provisioning and synchronization). Refer to “Provisioning Time of Day servers” and “Operating a time of day synchronization” procedures in <i>Administration and Security</i> , 323-1851-301.
3	If you set the time after provisioning licensing, then perform a manual audit after setting the shelf time to clear the license server comms error. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301.
4	Verify that the secondary license server/manager is provisioned and is not down. Refer to “Provisioning License Manager/Server information on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301.
5	It can take up to 12 hours for the alarm to clear as, in steady state, the heartbeat is run every 12 hours. Run a manual audit which will reset the heartbeat to two minutes after which the heartbeat interval will gradually increase in time to the steady state time of 12 hours. Refer to “Operating a license audit on the 6500” procedure in <i>Administration and Security</i> , 323-1851-301. The alarm will clear two minutes after the manual audit is run.
6	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-291

Secondary RADIUS Accounting Server Unavailable

Alarm ID: 1517

Probable cause

This alarm is raised when no response is received from the secondary RADIUS accounting server during user-provisioned timeout.

This alarm is also raised when the RADIUS accounting server provisioning on the network element is incorrect.

This security alarm is raised against an SP, SP-2, SPAP, SPAP-2 w/2xOSC, or an integrated shelf processor on a 8xOTN Flex MOTR circuit pack.

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Disable the server on 6500.
2	Re-enable the server and log in or log out of the server.
3	If the alarm is raised again, disable the RADIUS accounting feature and log in or log out of the server.
4	If the alarm is raised, ensure the following RADIUS accounting server provisioning values on the network element are correct: • server IP address • server port • shared secret • timeout - If this value is too small, the server may not be able to respond quickly enough. Refer to the "Provisioning the primary or secondary RADIUS server" procedure in <i>Administration and Security</i> , 323-1851-301.
5	Check the status of the RADIUS accounting server. Ensure the status is ON.

Procedure 4-291 (continued)

Secondary RADIUS Accounting Server Unavailable

Step	Action
6	Log in or log out of the network element. This will send a RADIUS accounting message to all provisioned RADIUS accounting servers. The alarm will clear if a response is received from the server(s) within the provisioned timeout.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-292

Secondary RADIUS Server Unavailable

Alarm ID: 584

Probable cause

This alarm is raised when all requests to the secondary RADIUS server of a SP times out and a primary RADIUS server is provisioned.

If only the secondary RADIUS server is provisioned (no primary RADIUS server provisioned) and all requests time out, the All Provisioned RADIUS servers Unavailable alarm is raised (refer to the [“All Provisioned RADIUS Servers Unavailable” on page 4-48](#) alarm clearing procedure in this document).

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

The alarm is not raised due to server time out.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Make sure the secondary RADIUS server of the SP is enabled and has a valid IP address. Refer to the “Retrieving the centralized security administration details” procedure in <i>Administration and Security</i> , 323-1851-301.
2	Log into the network element again using the RADIUS authentication (centralized security administration).
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-293

Secondary Service Failed

Alarm ID: 1261

Probable cause

This alarm is raised when:

- the External Slot Inventory Interface Devices have failed
- the wrong LAN cable type is used to connect to passive modules. The I2C interface supports only straight LAN cables.
- the LAN cable connected to passive modules is too long.
- internal faults for non-essential services are detected between the SP and Access Panel.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 4 UPC.

Step	Action
1	Select Active Alarms from the Faults menu in Site Manager to retrieve alarms. Refer to the alarms and events procedures in <i>Fault Management - Alarm Clearing</i> , 323-1851-543. Record the current state of the system. Note: Perform the following actions until the alarm clears. After each step, if the alarm does not return after 10 minutes then the procedure is complete. Otherwise, go to the next step.
2	If there is a passive module connected to the Access Panel, then a. If a crossover LAN cable is being used, replace the cable with a straight LAN cable. b. If the LAN cable is more than 3m in length, replace the cable with a cable that is 3m or shorter.
3	Perform a warm restart on the active shelf processor. Refer to the “Restarting a circuit pack or shelf processor” procedure in this document.

Procedure 4-293 (continued)

Secondary Service Failed

Step	Action
4	If there is a passive module connected to the Access Panel, reseal the LAN cable connecting it to the Access Panel, at both ends.
5	If 1+1 SP protection is provisioned, then perform a protection switch to the other shelf processor.
6	Reseat the original active SP. Refer to “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
7	Replace the original active SP if it has not been replaced within the past 48 hours. Refer to “Replacing a shelf processor” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
8	Perform a protection switch back to the original active SP, and reseal the original standby SP.
9	Replace the original standby SP if it has not been replaced within the past 48 hours. Refer to “Replacing a shelf processor” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
10	Reseat the Access Panel.
11	Replace the Access Panel if it has not been replaced within the past 48 hours. Refer to “Replacing an access panel” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
12	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-294

Secondary SETS Locking to Primary

Alarm ID: 586**Probable cause**

This alarm is raised while the synchronization equipment timing source (SETS) on the secondary cross-connect circuit pack is locking to the SETS on the primary cross-connect circuit pack.

On a (1+8)xOTN Flex MOTR circuit pack, this alarm is raised when the secondary (even slot) tries to lock to the primary circuit pack (odd slot). This is done on the first mate cross-connection provisioning time.

This alarm is also raised when the mate cross-connect circuit pack is OOS.

ATTENTION

The “Secondary SETS Locking to Primary” alarm is raised after a cross-connect circuit pack is inserted into the shelf during a cross-connect or optical interface circuit pack reconfiguration. The alarm indicates that the newly inserted cross-connect circuit pack is locking to the SETS on the existing cross-connect circuit pack. Wait until the alarm clears before proceeding with the cross-connect or optical interface circuit pack reconfiguration/replacement procedure.

ATTENTION

This alarm is raised on the (1+8)x OTN Flex MOTR circuit pack in an even slot number when the first connection is made between adjacent (1+8)x OTN Flex MOTR circuit packs or when one of the 2 circuit packs is cold restarted or replaced. The alarm can stay active for up to two minutes and will clear once the card in the even slot number has synchronized to the card in the odd slot number. While the alarm is active, inter-card traffic between the cards is conditioned. If the even slot card fails to synchronize to the odd slot card, this alarm is replaced with an Intercard Suspect alarm.

The alarm clears when the secondary SETS has locked successfully.

Procedure 4-294 (continued)

Secondary SETS Locking to Primary

Equipment protection of the cross-connect circuit packs is unavailable while this alarm is raised.



CAUTION

Risk of service interruption

If you place a cross-connect circuit pack out-of-service or open the latch on a cross-connect circuit pack while this alarm is raised, you can cause loss of traffic.

Impact

Warning

Step	Action
1	Log into the NE and verify the primary state of the mate cross-connect circuit pack. If the primary state is OOS-MA, change it to IS.
2	If the alarm does not automatically clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-1 Secure Erase In Progress

Alarm ID: 2237**Probable cause**

This alarm is raised when the Secure Erase action is initiated on a circuit pack.

Impact

Warning, non-service-affecting (W, NSA) alarm

Step	Action
1	No action is required. The alarm clears after the Secure Erase action is completed.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-2

Server Certificate About to Expire

Alarm ID: 1961

Probable cause

This alarm is raised when a server certificate on the SP is about to expire based on a provisioned number of days before the certificate expiry.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Connect to the NE using the Site Manager.
2	In the Security menu click on the Manage keys and choose SSL Keys .
3	On the SSL Keys screen, select the shelf whose certificate is about to expire.
4	Click on Regenerate button to regenerate the public and private keys.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-3

Server Certificate Expired

Alarm ID: 1960

Probable cause

This alarm is raised when a server certificate on the SP is expired.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Connect to the NE using the Site Manager.
2	In the Security menu click on the Manage keys and choose SSL Keys .
3	On the SSL Keys screen, select the shelf whose certificate is expired.
4	Click on Regenerate button to regenerate the public or private keys or click on the Upload Certificate to upload the user certificate. The alarm will clear when the operation is successful.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-4

Server Validation Certificate About to Expire

Alarm ID: 2254

Probable cause

This alarm is raised when a server validation certificate on the SP is about to expire based on a provisioned number of days before the certificate expiry.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Connect to the NE using the Site Manager.
2	In the Security menu click on the Manage keys and choose SSL Keys .
3	On the SSL Keys screen, select the shelf whose validation certificate is about to expire.
4	Click on Regenerate button to regenerate the public and private keys.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-5

Server Validation Certificate Expired

Alarm ID: 2253

Probable cause

This alarm is raised when a server validation certificate on the SP is expired.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Connect to the NE using the Site Manager.
2	In the Security menu click on the Manage keys and choose SSL Keys .
3	On the SSL Keys screen, select the shelf whose validation certificate is about to expire.
4	Click on Regenerate button to regenerate the public and private keys.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-6

Service Defect Indication

Alarm ID: 1741

Probable cause

This alarm is raised when the ETTP service on an eMOTR circuit pack has failed or is operationally disabled and conditioned because of VLLI/iVLLI (Inverse Virtual Link Loss Indicator).

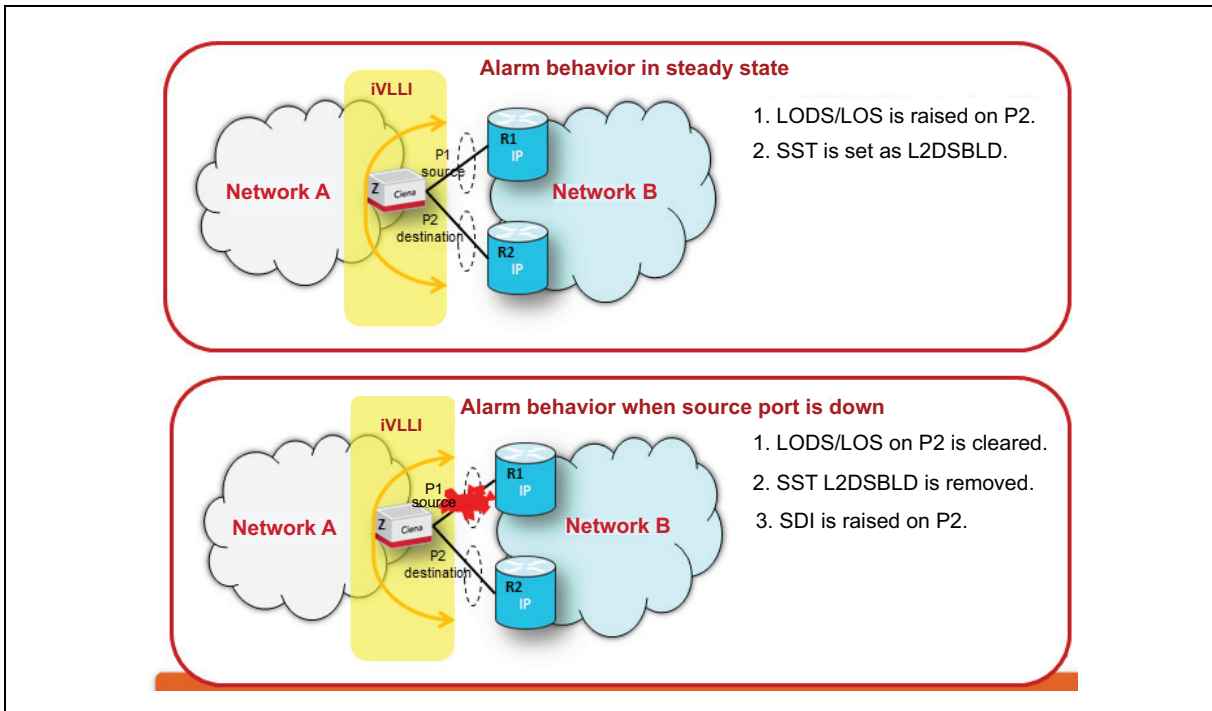
This alarm is also raised on PKT/OTN when an ETTP is conditioned because of VLLI.

This alarm is an indication from the 6500 SNMP interface that there is a L2 service-impacting alarm against the ring protection or Connectivity Fault Management (CFM) associated with the eMOTR port. The alarm is raised when a G.8032 protection switch is active on the eMOTR circuit pack.

For eMOTR circuit packs, SDI alarm is also raised by iVLLI on backup port indicating that iVLLI is active on that port (when active port goes down and backup port comes up). See [Figure 4-6](#) and [Figure 4-7 on page 4-696](#).

Procedure 4-6 (continued)
Service Defect Indication

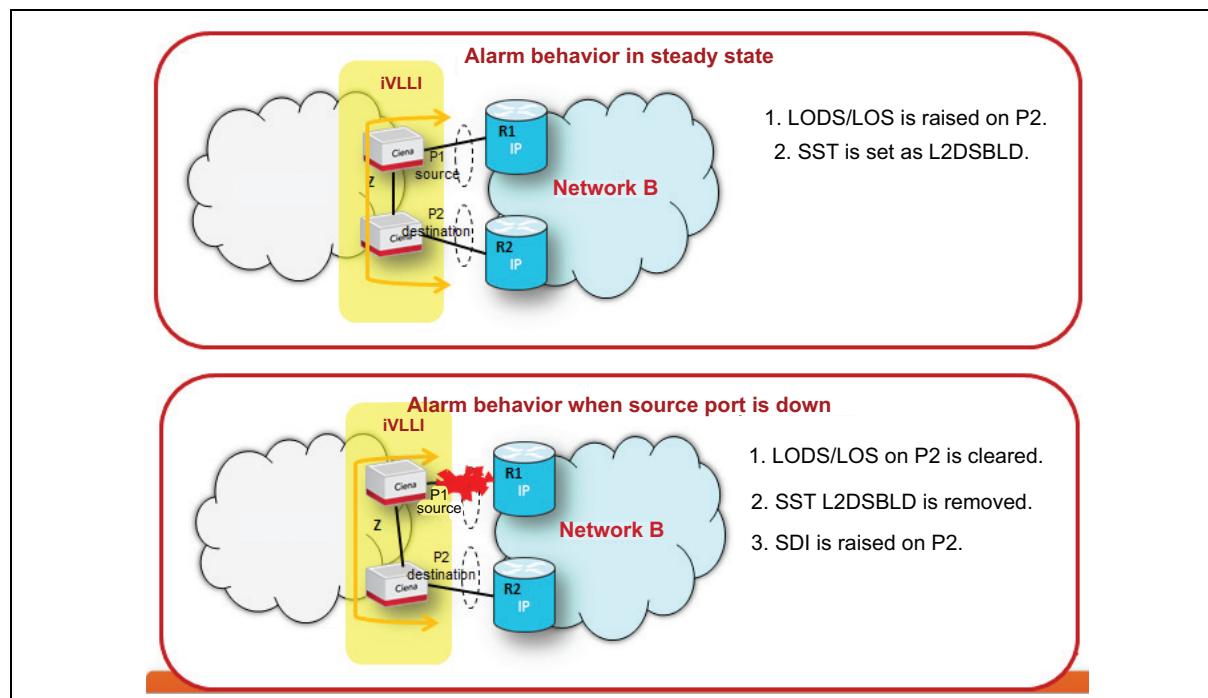
Figure 4-6
Single homed configuration



Note: The “Far End Client Signal Fail”, “Remote Port Unreachable” and “Remote Port OOS” alarms are not relevant in case of single-homed resilient handoff application.

Procedure 4-6 (continued)
Service Defect Indication

Figure 4-7
Dual homed configuration



Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Log into the SAOS CLI and examine the alarms and status of all the configured services. Refer to the <i>SAOS-based Packet Services Command Reference</i> , 323-1851-610, for information on retrieving alarm, port, Connectivity Fault Management (CFM), VLLI, and ring protection entities.

Procedure 4-6 (continued)
Service Defect Indication

Step	Action
2	Check and correct the L2 alarms and faults including: • L2 port alarms • missing or mismatched L2 ring protection provisioning at the near-end or far-end • missing or mismatched L2 CFM provisioning at the near-end or far-end • CFM faults and/or VLLI faults
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-7

Service Mismatch

Alarm ID: 2027

Probable cause

This alarm is raised against an OTUTTP facility. The alarm indicates that the OTU rate (either the n or m of the OTUCn-m) is not aligned at either end of the link.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Identify the port where this alarm is being raised and identify the OTUTTP port to which it is currently connected.
2	Verify the provisioned OTU rate (either the n or m of the OTUCn-m) at either end of the link. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the provisioned OTU rates (either the n or m of the OTUCn-m) do not match, correct any mismatches. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	If the alarm is not cleared, contact your next level of support or your Ciena support group.

—end—

Procedure 4-8

Shelf Bandwidth Near Limit

Alarm ID: 654**Probable cause**

This alarm is raised against the shelf to indicate that the provisioned total network element/shelf bandwidth is approaching the maximum allowed for the shelf.

The alarm is raised when 90% of the total capacity is reached on any cross-connect circuit pack.

Impact

Major, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Delete any cross-connects that are not required. Refer to the “Deleting path connections” procedure in <i>Configuration - Connections Management</i> , 323-1851-320. For information about how to calculate the number of cross-connects depending on the connection rate, refer to the “Connection and Connections Management” section in the <i>6500 Planning</i> , NTRN10GK.
2	If more low-order bandwidth is required, contact your network administrator to determine your course of action.

—end—

Procedure 4-9

Shelf Data Missing

Alarm ID: 570

Probable cause

This alarm is raised when the shelf does not have a shelf number provisioned.

Impact

Major, non-service-affecting (m, NSA) alarm

Prerequisites

Before you perform this procedure, you must observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0.

Step	Action
------	--------

- | | |
|---|--|
| 1 | This alarm clears when the commissioning of the shelf (specifically the shelf number) has been completed successfully. Complete the commissioning of the shelf. Refer to the “Commissioning a network element” procedures in <i>Commissioning and Testing</i> , 323-1851-221 for instructions. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-10

Shelf Power Near Limit

Alarm ID: 981

Probable cause

This alarm is raised against the shelf when the aggregate shelf power usage (**Calculated shelf power**) exceeds the shelf, feeder, fuse, or zone power limit threshold (as provisioned using the **Provisioned shelf current** shelf attribute value).

Impact

Major, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	Deprovision and remove all equipment from the shelf that is not required. Refer to the "Deleting a facility from an equipment" and "Deleting an ETH or ETH10G facility from a LAG in a 20G L2SS, L2SS, PDH gateway, or RPR circuit pack" procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
2	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						
3	Retrieve the Provisioned shelf current and Shelf current capacity (shelf power available based on equipped Power Input Cards) shelf attribute values. Refer to the "Determining the provisioned shelf current value" procedure in <i>Administration and Security</i> , 323-1851-301.						
4	<table> <tr> <th>If Provisioned shelf current is</th><th>Then</th></tr> <tr> <td>less than or equal to Shelf current capacity</td><td>go to step 8</td></tr> <tr> <td>greater than Shelf current capacity</td><td>upgrade the Power Input Cards to match or exceed the Shelf current capacity value</td></tr> </table>	If Provisioned shelf current is	Then	less than or equal to Shelf current capacity	go to step 8	greater than Shelf current capacity	upgrade the Power Input Cards to match or exceed the Shelf current capacity value
If Provisioned shelf current is	Then						
less than or equal to Shelf current capacity	go to step 8						
greater than Shelf current capacity	upgrade the Power Input Cards to match or exceed the Shelf current capacity value						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						

Procedure 4-10 (continued)

Shelf Power Near Limit

Step	Action						
6	Correct the Provisioned shelf current (if not set correctly for the actual feeders and fuses in use). To correct a zone that is over its threshold, the newly provisioned/inserted modules could be shuffled to zones that have headroom in their budget (if there are available slots and the configuration is supported). It may be necessary to shuffle multiple modules in the shelf in order to re-balance the power distribution.						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Determine the amperage of the actual power feed to the shelf. <table> <tr> <th>If the amperage of the actual power feed to the shelf is</th><th>Then</th></tr> <tr> <td>greater than equal to Provisioned shelf current</td><td>go to step 9</td></tr> <tr> <td>less than Provisioned shelf current</td><td> upgrade the power feed to a rating that can support the required power demand of the shelf, and adjust the Provisioned shelf current value to be equal to the amperage of the power feed. Go to step 10. </td></tr> </table>	If the amperage of the actual power feed to the shelf is	Then	greater than equal to Provisioned shelf current	go to step 9	less than Provisioned shelf current	upgrade the power feed to a rating that can support the required power demand of the shelf, and adjust the Provisioned shelf current value to be equal to the amperage of the power feed. Go to step 10.
If the amperage of the actual power feed to the shelf is	Then						
greater than equal to Provisioned shelf current	go to step 9						
less than Provisioned shelf current	upgrade the power feed to a rating that can support the required power demand of the shelf, and adjust the Provisioned shelf current value to be equal to the amperage of the power feed. Go to step 10.						
9	Provision the Provisioned shelf current value equal to the power feed amperage identified in step 8. Refer to the “Determining the provisioned shelf current value” procedure in <i>Administration and Security</i>, 323-1851-301. Ensure that the Provisioned shelf current value does not exceed the Shelf current capacity value.						
10	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-11

Shutoff Threshold Crossed

Alarm ID: 540

Probable cause

This alarm is raised against an AMP facility when the total input optical power to the amplifier has fallen below the provisioned Shutoff Threshold level. For CDC configurations, this alarm is raised on the AMP facility of CCMD 8x16 or CCMD12 C-Band/CCMD12 L-Band circuit packs. The conditions that can cause the input power level to fall below the threshold level include:

- a disconnected fiber
- a pinched fiber connection
- a dirty optical fiber connector
- a defective fiber optic patchcord
- a defective module
- an incorrect provisioned value
- an OSRP Line/Link Down condition triggered by fiber disconnection on intra-node fiber (except WSS-WSS fiber), AMP in shut-off state, or AMP Primary State OOS-MA

Note: OSRP provides a fault reason within the “Additional Information” field of the alarm to help with the troubleshooting.

This alarm can remain active after the fault has cleared and the original power level is restored. This occurs when the power level is lower than the user-provisioned Shutoff threshold plus the hysteresis value. The hysteresis value is not user provisionable and is set at 3 dB.

This alarm is raised on port 4 of the Fixed Gain Amplifier (FGA C-Band) (NTK552AB).

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have a network and site diagram
- have a fiber cleaning kit
- have the Shutoff threshold level for this amplifier

Procedure 4-11 (continued)
Shutoff Threshold Crossed

Impact

Major, service-affecting (M, SA) alarm

Step	Action
1	Place the alarmed AMP facility out of service (OOS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	Clean and then reconnect the input fibers and connectors at the amplifier. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.
3	Place the amplifier back in-service (IS). Refer to the “Changing the primary state of a facility” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	If the alarm still persists, check the upstream connection for mating or pinched fibers.
5	If the alarm is against port 8 of an SLA C-Band, MLA C-Band, MLA L-Band, MLA2 C-Band, MLA2 w/VOA, or MLA3 C-Band/MLA3 L-Band, check and clean the connection from the incoming line, including any patch panel connections. Refer to the “Cleaning Connectors” section in <i>Installation - General Information</i> , 323-1851-201.0.
6	If the alarm is against port 8 of an XLA circuit pack, check and clean the connection to port 7 of the SRA circuit pack and ensure the SRA circuit pack is in-service.
7	If the alarm is raised against port 6 of an MLA C-Band, MLA2 w/VOA, MLA3 C-Band/MLA3 L-Band, or XLA, check and clean the connection from the WSS common output as identified in the Table 4-4 .

Table 4-4
WSS common output ports

WSS 100 GHz w/OPM C-Band 2x1 (NTK553JAE5)	port 18
WSS 100 GHz w/OPM C-Band 2x1 (NTK553JB)	port 8
WSS 100 GHz w/OPM 5x1 (NTK553EAE5)	port 18
WSS 100 GHz w/OPM C-Band 4x1(NTK553HA)	port 12
WSS 50 GHz w/OPM 2x1 (NTK553KCE5, NTK553KAE5)	port 8
WSS 50 GHz w/OPM 9x1 (NTK553FAE5, NTK553FC)	port 22
WSS w/OPM Flex C-Band 9x1(NTK553LA)	port 22
WSS w/OPM Flex L-Band 8x1(NTK553LM)	port 20

Procedure 4-11 (continued)
Shutoff Threshold Crossed

Step	Action
8	If the alarm is against port 8 of the SLA, check and clean the connection from the relevant WSS switch port as follows: - From the Configuration, Equipment & Facility screen, select the circuit pack identified by the alarm (the provisioned PEC field should equal NTK552AA or NTK552AB: SLA C-Band). - In the Facility Type dropdown box, select "ADJ". - Locate the adjacency associated with port 5 (the Adjacency Type should equal "WSS"). - The "Expected Far End Address" gives the WSS switch port connected to the SLA in the format TID-SH-SL-PORT. Note the value after the last hyphen. - Locate the WSS port identified above. This is the connection from the SLA to the WSS switch input. Clean the fiber associated with the output of that switch (for example, if port 11 was identified above, clean port 12 of the WSS).
9	Use optical terminators on unused input faceplate connectors of installed WSS w/OPM circuit packs. If dust caps are used instead of optical terminators on "Switch In" ports, PMs can be reported against the ports and the ports may appear in-service. If the alarm is against port 4 of FGA, check and clean the connection from the incoming line.
10	Restart the LIM, MDRLA 2x6, RLA 5x1 supporting the alarmed AMP facility. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
11	Replace the LIM, MDRLA 2x6, RLA 5x1 supporting the alarmed AMP facility. Refer to the "Replacing the amplifier modules" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.
12	If the alarm does not clear, contact your next level of support or Ciena support group.

—end—

Procedure 4-12

Skew Out Of Range

Alarm ID: 1680, 1777, 2031, 2150, 2212

Probable cause

This alarm is raised when the skew between the two carriers exceeds the skew range and is detected.

For the ETTP (ETH100G), PTP, FACGRP, or OTUTTP clients that use parallel optics, this alarm is raised when the Virtual Lane (VL) Skew exceeds the IEEE Std 802.3ba VL Skew range 180ns. This is only applicable when FEC is disabled.

Impact

Major, service-affecting (M, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must;

- observe all the safety requirements described in “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain replacement patchcord fiber pairs for the Tx/Rx to Mux/Demux connection
- if required, obtain a replacement WLAi MOTR, WL5e MOTR, WLAi FOTR, Flex2 WL3 OCLD, Flex3 WL3e OCLD or a Flex4 WL3e OCLD circuit pack
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|--|
| 1 | Ensure that both prime and member WLAi MOTR, WL5e MOTR, WLAi FOTR, Flex2 WL3 OCLD, Flex3 WL3e OCLD or a Flex4 WL3e OCLD equipment are working properly with error-free signal. |
|---|--|

Procedure 4-12 (continued)
Skew Out Of Range

Step	Action						
2	Ensure that the transmitter wavelength separation between the prime and member is 200 GHz or less. If it is not, re-provision the transmitter wavelengths and rework the wavelength plan to make them comply to this rule.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Ensure that the difference in estimated fiber length for each WLAi variant, or WL3 OCLD is less than 300 meters. If it is not, ensure that the wavelengths take the same photonic path and have similar sized patchcords at each end.						
5	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.						
7	Ensure that the Tx/Rx to Mux/Demux fiber pairs on the two carriers are correctly connected with exactly equal length duplex LC to LC fibers at both sites. If they are not, replace them with exactly equal length duplex LC to LC fibers.						
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	Replace the prime circuit pack. Refer to the “Replacing an optical interface circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. If the original alarm has cleared then the procedure is complete, otherwise go to next step.						
10	Replace the member circuit pack.						
11	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-13

SLDD Adjacency Loss

Alarm ID: 1981, 1982

Probable cause

This alarm applies to the 6500 shelves that have SLDD enabled and an SLDD circuit provisioned on the alarmed interface. The alarm is raised when no SLDD adjacency can be established on the alarmed interface.

This alarm will clear on its own if it was due to missing or mis-matched provisioning at the far end, and this is resolved at the far end.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Ensure that the SLDD circuit is provisioned at the far end of the Link.
2	If the alarm does not clear, verify that the SLDD Scope ID at the two ends of the Link match.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-14

Slot Empty

Alarm ID: 379**Probable cause**

This alarm is raised when an unprovisioned or out-of-service slot is empty.

ATTENTION

You must install filler cards in all slots that do not contain a circuit pack to ensure sufficient air flow for cooling the shelf. Failure to do so can cause the shelf to exceed the maximum temperature which may cause component damage.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- obtain a filler card or a module to install in the empty slot

Step	Action
1	Identify the slot raising the alarm.
2	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
3	Insert a filler card or a circuit pack into the slot raising the alarm. Refer to the “Installing circuit packs in the 6500 shelf” procedure in <i>Installation - General Information</i> , 323-1851-201.0.
4	Place the out of service slot back to In Service.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-15

Slot Sequence Provisioning Incomplete

Alarm ID: 1832, 1833

Probable cause

This warning is raised when the Intersecting Status field for the Slot Sequence is “UNRESOLVED”.

This warning is a reminder to the user to complete the provision of its intersecting Slot Sequence.

When a Slot Sequence is provisioned, its Intersecting Status field can be set to one of the following:

- “NOT APPLICABLE”, if the Slot Sequence does not require intersecting Slot Sequence provisioning
- “UNRESOLVED”, if the Slot Sequence requires intersecting Slot Sequence provisioning
- “RESOLVED”, if its intersecting Slot Sequence is provisioned

The condition clears when the Intersecting Status field of the Slot Sequence is changed from “UNRESOLVED” to “RESOLVED” or “NOT APPLICABLE”.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	<table><tr><td>If you want to Provision the intersecting Slot Sequence to</td><td>Then go to</td></tr><tr><td>RESOLVED</td><td>step 2</td></tr><tr><td>NOT APPLICABLE</td><td>step 3</td></tr></table>	If you want to Provision the intersecting Slot Sequence to	Then go to	RESOLVED	step 2	NOT APPLICABLE	step 3
If you want to Provision the intersecting Slot Sequence to	Then go to						
RESOLVED	step 2						
NOT APPLICABLE	step 3						
2	Provision the intersecting Slot Sequence so the Intersecting Status field for Slot Sequence is changed to “RESOLVED”. Refer to the “Editing OTS slot sequences” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 4 .						
3	Ensure that the Drop and Add Sequence fields for Slot Sequence are cleared and the Intersecting Status field is changed to “NOT APPLICABLE”.						
4	If the condition does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-16

SNC Datapath Fault

Alarm ID: 1759

Probable cause

This alarm is raised when the datapath is down for an SNC because the underlying DOC channel has not achieved the end-to-end In-service/Optimized state within the provisioned “Datapath fault alarm timer period” (default is 15 minutes).

If you do not want the SNC to raise the “SNC Datapath Fault” alarm, you must set the SNC Datapath Fault timer to 0 before bringing the SNC into service.

When an SNC enters to the working state, the OSRP starts SNC monitoring by querying at each hop along the SNC for the channel state in the DOC that corresponds to the SNC. OSRP continues the query periodically until a timer expires (default is 15 minutes) or all hops are in-service. If all hops are not in-service within 15 minutes, the alarm is raised against the SNC facility.

The alarm is raised for a given SNC with properly provisioned photonic cross-connections, when receive fault is detected (for example, Loss of Frame or Loss of Clock) or when one or more underlying DOC domains in the SNC path has indicated that the channel is not optimized.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Identify the SNC that is raising the alarm. Select Sub-Network Connection from Configuration menu.
2	From the SNC list, select the SNC that raised the alarm.
3	In the Details area, click on the End to End diagnosis button to view the faults that raised the alarm.
4	If the alarm is raised due to a channel not being added properly along the path in the DOC, identify the node and view the DOC related alarms. Clear these alarms before continuing this procedure.

4-712 Alarm clearing procedures

Procedure 4-16 (continued)

SNC Datapath Fault

Step	Action
5	If the end to end diagnostic information indicates Rx Fault for the SNC on the transponder link, Loss of Signal, Loss of Frame, or Loss of Clock, then clear these alarms.
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-17

SNCG Home Path Failed - Auto Reversion Inhibited

Alarm ID: 2227

Probable cause

This alarm is raised on SNCG when auto-revert disables due to internal fault on home path.

This alarm clears when any of the following occur:

- The AUXP is restarted.
- The SNCG Administrative State is set to Locked
- The SNCG reverts to healed home-path, which was previously internal faulted. This is possible in the following cases:
 - Mesh restore to the healed home-path
 - Manual Revert to the healed home-path

Impact

Minor, non- service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Identify the node raising the alarm and note any additional information. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	Clear the network fault that triggered the mesh restoration.
3	restore the Mesh to the healed home-path. If the SNCG is provisioned to automatically revert to the home path, the alarm will clear when the SNCG reverts by meshing. Otherwise, verify that the SNCG home path is available, and manually revert the SNCG back on home path as follows: a. In Site Manager, under the Configuration drop-down menu, select Sub-Network connections window and select the Group SNC tab.

Procedure 4-17 (continued)

SNCG Home Path Failed - Auto Reversion Inhibited

Step	Action
------	--------

- | | |
|----|---|
| b. | Select the alarmed SNCG and click on the Revert button. |
|----|---|



CAUTION

Risk of traffic loss

Reverting the SNCG will cause the channel to go down for several minutes.

- | | |
|---|--|
| 4 | If the alarm does not clear, contact your next level of support or your Ciena support group. |
|---|--|

—end—

Procedure 4-18

SNCG Not On Home Path

Alarm ID: 2042

Probable cause

This alarm is raised when an SNCG is placed In-Service after a restoration event and the SNCG is not on its home path.

This alarm is a secondary alarm raised to identify services impacted by a network fault event. The alarm is raised when SNCGs recover from a fault by mesh restoring to a protection path and cleared when the SNCG is either put administratively out of service, or the SNCG reverts back to its home path.

This alarm is also raised if the SNCG is not on the home path because the user performed a Manual Switch. If the home path is available the alarm can be cleared by performing a Revert or another Manual Switch (back to home path).

After a restoration event, once the SNCG (when GROUPED=YES), transitions to IS state on a path that is not its home path, a minor “SNCG Not on Home Path” alarm will be raised.

This alarm is not enabled by default.

Impact

Warning

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Clear the network fault that triggered the mesh restoration.
2	If the SNCG is provisioned to automatically revert to the home path, the alarm will clear when the SNCG reverts itself. Otherwise, verify that the SNCG home path is available, and revert the SNCG back on home path as follows: a. In Site Manager, under the Configuration drop-down menu, select Sub-Network connections window and select the Group SNC tab.

Procedure 4-18 (continued)
SNCG Not On Home Path

Step	Action
------	--------

- | | |
|----|---|
| b. | Select the alarmed SNCG and click on the Revert button. |
|----|---|



CAUTION

Risk of traffic loss

Reverting the SNCG will cause the channel to go down for several minutes.

- | | |
|---|--|
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |
|---|--|

—end—

Procedure 4-19

SNCG Unavailable

Alarm ID: 1979

Probable cause

This alarm is raised when the Subnetwork Connection Group (SNCG) is in the creating or starting state or is frequently toggling between the working and starting states. For MR-SNCGs, it is typically caused by an underlying fault with any nodes along the path. For example, insufficient bandwidth, missing or faulty physical facility at the destination or by mis-provisioned far-end address.

Note: Due to the operating frequency range of the amplifiers, channel 1 (196.100 THz, 1528.773 nm) of the CMD48 (NTT862MA) or the CMD Type 2 (NTT862JB) is not supported.

For Permanent SNCGs, the alarm will be raised when the SNCG cannot be managed due to problems along the path (comms, CTM failed, fiber cut).

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Identify the node raising the alarm and note any additional information. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	If the SNCG raising the alarm is recently added, verify the end-to end provisioning using "Adding a sub-network connection" procedure in Configuration - Control Plane, 323-1851-330. If mis-provisioning is identified, delete and re-create the SNCG. Refer to the "Deleting a sub-network connection" and "Adding a sub-network connection" procedures in <i>Configuration - Control Plane</i> , 323-1851-330.

Procedure 4-19 (continued)

SNCG Unavailable

Step	Action						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Verify if any equipment alarms are raised against circuit packs carrying the SNCG. If there are, then clear these alarms before continuing this procedure.						
5	From the Site Manager Sub-Network Connection application, select the Group SNC tab. Select the alarmed SNCG and verify whether a current route is available (listed) in the Route Information sub-tab (in the Details area). To do this, click on the SNCG Routes button to open the Group SNC Routes dialog box. If the Current Route tab lists a route, then there is an available current route.						
6	<table> <tr> <th>If the current route is</th><th>Then go to</th></tr> <tr> <td>available</td><td>step 7</td></tr> <tr> <td>not available</td><td>step 9</td></tr> </table>	If the current route is	Then go to	available	step 7	not available	step 9
If the current route is	Then go to						
available	step 7						
not available	step 9						
7	View the provisioning SNCG screen, choose the Member SNC tab and note the member SNCs. From the SNC list in the Service Provisioning tab, note the nodes (Remote Node column) and ports (Remote EP column) of the home route.						
8	Verify if any equipment alarm exist. If there are, then, clear these alarms. Go to step 11 .						
9	From the General tab, click on the SNCG Diagnostics button to open the Group SNC Diagnostics box. If the alarm is raised with Additional information "SNC is Thrashing" for the following errors in the Event detail Column: • Port Trail Error • Preemption due to Internal Faults • Frequency Contention						
10	If the event detail column currently indicates failures or errors, stop the procedure and contact your network administrator.						
11	For the remote end point, select the Configuration drop down and select the Subnetwork Connections tab. View the Provisioning SNC screen and verify there is a valid route going to the destinations node.						
12	<table> <tr> <th>If a route is</th><th>Then</th></tr> <tr> <td>available</td><td>go to step 13</td></tr> <tr> <td>not available</td><td>establish a route</td></tr> </table>	If a route is	Then	available	go to step 13	not available	establish a route
If a route is	Then						
available	go to step 13						
not available	establish a route						

Procedure 4-19 (continued)
SNCG Unavailable

Step	Action
13	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-20

SNC Home Path Failed - Auto Reversion Inhibited

Alarm ID: 2226

Probable cause

This alarm is raised on SNC when auto-revert disables due to internal fault on home path.

This alarm clears when any of the following occur:

- The AUXP is restarted.
- The SNC Administrative State is set to Locked
- The SNC reverts to healed home-path, which was previously internal faulted. This is possible in the following cases:
 - Mesh restore to the healed home-path
 - Manual Revert to the healed home-path

Impact

Minor, non- service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Identify the node raising the alarm and note any additional information. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.
2	Clear the network fault that triggered the mesh restoration.
3	Perform a mesh restoration to the healed home-path. If the SNC is provisioned to automatically revert to the home path, the alarm will clear when the SNC reverts by meshing. Otherwise, verify that the SNC home path is available, and manually revert the SNC back on home path as follows: a. In Site Manager, under the Configuration drop-down menu, select Sub-Network connections window and select the SNC tab.

Procedure 4-20 (continued)

SNC Home Path Failed - Auto Reversion Inhibited

Step	Action
------	--------

- | | |
|--|--|
| | <p>b. Select the alarmed SNC and click on the Revert button.</p> |
|--|--|



CAUTION

Risk of traffic loss

Reverting the SNCG will cause the channel to go down for several minutes.

- | | |
|---|---|
| 4 | <p>If the alarm does not clear, contact your next level of support or your Ciena support group.</p> |
|---|---|

—end—

Procedure 4-21

SNC Not On Home Path

Alarm ID: 2041

Probable cause

This alarm is raised when a Sub-Network Connection (SNC) is off its home path.

This alarm is only applicable to L0 Photonic SNC and does not apply to L1 OTN SNC.

This alarm is a secondary alarm raised to identify services impacted by a network fault event. The alarm is raised when SNCs recover from a fault by mesh restoring to a protection path and cleared when the SNC is either put administratively out of service, or the SNC reverts back to its home path.

This alarm is also raised if the SNC is not on the home path because the user performed a Manual Switch. If the home path is available the alarm can be cleared by performing a Revert or another Manual Switch (back to home path).

After a restoration event, once the SNC (when GROUPED=NO) transitions to IS state on a path that is not its home path, a minor "SNC Not on Home Path" alarm will be raised.

This alarm is not enabled by default.

Impact

Warning

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action
1	Clear the network fault that triggered the mesh restoration.
2	If the SNC is provisioned to automatically revert to the home path, the alarm will clear when the SNC reverts itself. Otherwise, verify that the SNC home path is available, and revert the SNC back on home path as follows: In Site Manager, under the Configuration drop-down menu, select Sub-Network connections window.

Procedure 4-21 (continued)
SNC Not On Home Path

Step	Action
------	--------

- | | |
|--|--|
| | <p>b. Select the alarmed SNC and click on the Revert button.</p> |
|--|--|



CAUTION

Risk of traffic loss

Reverting the SNC will cause the channel to go down for several minutes.

- | | |
|----------|---|
| <p>3</p> | <p>If the alarm does not clear, contact your next level of support or your Ciena support group.</p> |
|----------|---|

—end—

Procedure 4-22

SNC Reservation Unavailable

Alarm ID: 1947

Probable cause

This warning is raised when there is a failure to successfully negotiate a mesh restoration route for a packet SNC provisioned with reservation.

The alarm is a warning that if the current path of a packet SNC provisioned with reservation fails, then traffic recovery will be “slow” (can take minutes) rather than “fast” (sub-second).

Impact

Warning

Step	Action
------	--------

- | | |
|---|---|
| 1 | No action is necessary. This alarm will clear once the packet SNC successfully negotiates mesh restoration route later in time. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-23

SNC Takeover Failed

Alarm ID: 1507

Probable cause

This alarm is raised when the Subnetwork Connection (SNC) has failed to take over all cross-connections from end to end.

This alarm is supported for Photonic L0 Control Plane and Photonic SNC takeover.

This alarm can also be raised for SNC takeover limiting conditions. When an explicitly routed call is configured for the purpose of path takeover, the path of the OSS service is validated. If the validation fails, the SNC is created, its status diagnostic indicates the reason of the failure and the takeover attempted SNC remains OSS owned.

Impact

Minor, non- service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Step	Action						
1	From the Site Manager Sub-Network Connection application, select the SNC tab.						
2	From the Route Information sub-tab (in the Details area), click on the “SNC Status Diagnostics” button next to the SNC state field to open the SNC Status Diagnostics dialog box.						
3	<table border="0"> <tr> <td>If the alarm is raised for</td><td>Then go to</td></tr> <tr> <td>SONET/SDH Control Plane</td><td>step 4</td></tr> <tr> <td>Photonic Control Plane</td><td>step 9</td></tr> </table>	If the alarm is raised for	Then go to	SONET/SDH Control Plane	step 4	Photonic Control Plane	step 9
If the alarm is raised for	Then go to						
SONET/SDH Control Plane	step 4						
Photonic Control Plane	step 9						
4	From within the SNC status diagnostic window, check if the Event Detail column indicates the SNC is stuck in the Starting or Down state.						

Procedure 4-23 (continued)

SNC Takeover Failed

Step	Action
5	If the SNC is in the Starting or Down state, a home route will not be available. If the SNC is in Starting state, the potential causes of the failure are a problem with the state of the Absolute Route Diversity (ARD), Max. Admin Weight (MAW), wrong routing, the regen timeslot (If the regen connection is taken over), or SNC protection class UNPROTECTED_HIGH (PRTT) provisioning, which is indicated in the Event Detail column. Follow the Procedure 4-66, "Co-Routed SNC Unavailable" alarm clearing procedure to clear this alarm.
6	If the SNC is in the Creating state, and the Event Detail column indicates an XCON validation failure, determine at which node the XCON validation has failed.
7	Determine if the failure is due to an ARD, MAW or PRTT provisioning problem. Place the SNC OOS and re-provision the SNC parameters. Re-attempt the takeover by placing the SNC back to an IS state.
8	If the failure is seen during the connect phase, delete the SNC, delete any leftover non-OSRP connections on the takeover path (if any), and then create a non-takeover SNC on the same path. Refer to the "Deleting a sub-network connection" and "Adding a sub-network connection" procedures in <i>Configuration - Control Plane</i> , 323-1851-330 or in <i>Configuration - Connections Management</i> , 323-1851-320.
9	Check the Event Detail column for indication of the problem that caused the SNC takeover to fail for the L0 Control Plane.
10	If the connection taken over is 1WAY or does not match the route specified in the Routing List, re-provision the connection to correct the problem.

Procedure 4-23 (continued)

SNC Takeover Failed

Step	Action
11	If the problem is fixed, the SNC will be taken over (there are automatic retries) and the alarm clears.
12	If the alarm did not clear, Place the SNC OOS, delete the SNC and create a new SNC with the parameters that may have been wrong on the first try. Refer to the “Deleting a sub-network connection” and “Adding a sub-network connection” procedures in <i>Configuration - Control Plane</i> , 323-1851-330 or in <i>Configuration - Connections Management</i> , 323-1851-320.
13	If you are unable to fix the problem or if the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-24

SNC Unavailable

Alarm ID: 1423

Probable cause

This alarm is raised when the terminating Subnetwork Connection (SNC) is in the creating or starting state. It is typically caused by a destination unreachable condition caused by insufficient bandwidth, lack of matching service classes, or lack of physical facility to the destination port.

This alarm is raised for P-SNC while it is in working state and the working path is down. The alarm reporting is not immediate and can take up to 15 minutes to appear.

This alarm clears when any of the following occur:

- The path defect on the OSRP line that created this condition is absent for 10 seconds.
- The SNC is deleted.
- The SNC Administrative State is set to Locked.

Note 1: For the OTN Control Plane, the SNC Name can be optionally included in all SNC related alarms. By default, SNC alarms do not include the SNC Name. Enabling of SNC Name in SNC alarms can be provisioned on a system level using the “Alarm Info” parameter. Refer to the “Editing the nodal system parameters” procedure in *Administration and Security*, 323-1851-301.

Note 2: When a remote fault condition is present for an SNC with “Release on Client Failure” (ROCF) enabled, the “SNC unavailable” alarm toggles until the remote fault condition is cleared. The toggling rate depends on the back-off timer value.

Impact

Critical, service-affecting (C, SA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 3 UPC.

Procedure 4-24 (continued)
SNC Unavailable

Step	Action						
1	Identify the node raising the alarm and note any additional information. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	If the SNC raising the alarm is recently added, delete and re-create the SNC. Refer to the "Deleting a sub-network connection" and "Adding a sub-network connection" procedures in <i>Configuration - Control Plane</i> , 323-1851-330.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Verify if any equipment alarms are raised against circuit packs carrying the SNC. If there are, then clear these alarms before continuing this procedure.						
5	From the Site Manager Sub-Network Connection application, select the SNC tab. Verify whether a current route is available in the Route Information sub-tab (in the Details area). To do this, click on the Routes button to open the SNC Routes dialog box. If the Current Route tab lists a route, then there is an available current route.						
6	<table> <tr> <th>If the current route is</th><th>Then go to</th></tr> <tr> <td>available</td><td>step 7</td></tr> <tr> <td>not available</td><td>step 9</td></tr> </table>	If the current route is	Then go to	available	step 7	not available	step 9
If the current route is	Then go to						
available	step 7						
not available	step 9						
7	View the provisioning SNC screen. From the SNC list, note the nodes (Remote Node column) and ports (Remote EP column) of the home route.						
8	Verify if any equipment alarm exist. If there are, then, clear these alarms. Go to step 11 .						
9	From the Route Information tab, click on the (...) button next to the SNC state field to open the SNC Status Diagnostics dialog box.						
10	If the event detail column currently indicates failures or errors, stop the procedure and contact your network administrator.						
11	Click Provisioning and select the SNC tab for each node. View the Provisioning SNC screen and verify there is a valid route (sufficient bandwidth that match the service class constraint) going to the destinations node.						
12	<table> <tr> <th>If a route is</th><th>Then</th></tr> <tr> <td>available</td><td>go to step 13</td></tr> <tr> <td>not available</td><td>establish a route</td></tr> </table>	If a route is	Then	available	go to step 13	not available	establish a route
If a route is	Then						
available	go to step 13						
not available	establish a route						
13	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-25

Software Auto-Upgrade in Progress

Alarm ID: 439

Probable cause

This alarm is raised if a newly inserted circuit pack is being auto-upgraded/downgraded to the active software release of the network element.

This alarm can also be raised when a shelf processor is replaced and the inserted SP is running a different software release than the active release on the network element.

For 40G OTN XCIF circuit packs, this alarm can be raised when the profile of the circuit pack is changed after a database restoration.

This alarm is also raised when the user cancels the upgrade operation.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	Wait for the auto-upgrade/downgrade of the circuit pack to complete. The auto-upgrade/downgrade of the circuit pack can take up to five minutes. For 40G OTN XCIF circuit packs, the circuit pack will go through a cold restart. It can take up to 20 minutes for the alarm to clear.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-26

Software Configuration Unknown

Alarm ID: 138

Probable cause

This alarm is raised when the SP software fails to install the software control information.

Impact

Major, non-service-affecting (m, NSA) alarm

Step	Action	
1	If	Then
	a Software Mismatch and an Incomplete Software Lineup alarm are also both active	perform the alarm clearing procedure for the “Incomplete Software Lineup” on page 4-342. If the Software Configuration Unknown alarm does not clear, go to step 2 .
	a Software Mismatch and an Incomplete Software Lineup alarm are not both active	go to step 2
2	Contact your next level of support or your Ciena support group.	
	—end—	

Procedure 4-27

Software Delivery Incomplete

Alarm ID: 453

Probable cause

This alarm is raised when a transfer of a new software release to the SP has failed.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- be able to connect to the SP
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|---|
| 1 | Click Cancel in the Release Management application. Refer to the “Upgrading a software load” procedure in <i>Administration and Security</i> , 323-1851-301.

Canceling stops the action and cleans up any files left in invalid states. |
| 2 | This alarm can also clear if you try to deliver the software release again. Try to establish and resolve the cause of the failure then retry the delivery. Refer to the “Transferring a software load to a network element” procedure in <i>Administration and Security</i> , 323-1851-301.

The alarm clears if the software release delivery is successful. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-28

Software Delivery in Progress

Alarm ID: 454**Probable cause**

This alarm is raised if a new release is being transferred to the SP.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Wait for the transfer of the software release to the SP to complete. Refer to the “Transferring a software load to a network element” procedure in <i>Administration and Security</i>, 323-1851-301. The transfer of the software release to the SP can take up to an hour. Refer to the <i>Software Upgrade Procedures</i> for information on the length of upgrades. For a list of procedures, refer to the “<i>Software Upgrade Procedures</i>” section in <i>Planning - Ordering Information</i>, 323-1851-151.
2	If you want to abort the transfer, click Cancel in the Release Management application. Canceling stops the action and cleans up any files left in invalid states.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-29 Software Mismatch

Alarm ID: 75

Probable cause

This alarm is raised if a SP is inserted in the shelf and the SP is running a different software release than the active release on the network element.

Impact

Major, service-affecting (m, SA) alarm

Major, non-service-affecting (m, NSA) alarm

Step	Action
1	Check the current release running on the network element. Refer to the “Checking the current software release” procedure in <i>Commissioning and Testing</i> , 323-1851-221.
2	If the SP is running a release that is lower than that of the shelf, the NE will have minimal functionality and display a shelf number of 0. A “Transport Data Recovery” alarm may also be present.
3	Perform an upgrade on the network element. For the upgrade “to release”, use the current release running on the network element. Refer to the <i>Software Upgrade Procedures</i> for this release listed in the “ <i>Software Upgrade Procedures</i> ” section in <i>Planning - Ordering Information</i> , 323-1851-151.
4	Ensure that the system is restored to its original state by retrieving all conditions and alarms. Clear all alarms using the appropriate alarm clearing procedure.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-30

Software Subsystem Failed

Alarm ID: 1424
Probable cause

This alarm is raised when a circuit pack has a software failure that cannot be recovered on its own or when a circuit pack (except the eMOTR circuit pack) fails to boot up after few attempts.

This alarm can be raised on an eMOTR circuit pack with a Packet Configuration Integrity Fail condition when upgraded from Release 10.2x to Release 11.1 in a standalone or mate configurations. A warm restart will clear the alarm for all cases except when the failure is because of the SDK PRAM being out of sync. See the “Packet Configuration Integrity Fail” alarm procedure.

This alarm can be raised when the secondary processor on the SP-2 is not reachable from the primary processor. This alarm applies to the SP-2 Dual CPU circuit pack.

You must refer to 6500 Packet-Optical Platform Control Plane Application Guide (NTRN71AA), for detailed information on Control Plane concepts, applications, and engineering rules supported in this release of 6500.

Note: This alarm is a diagnostics alarm to indicate the link is down. In most cases the alarm is raised during CPU2 restart and will clear once CPU2 is back in service.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	If the alarm is raised against a WLAI MOTR/FOTR, WSS w/OPM (NTK553KA/HA/JB/LA/LM/FC) or CCMD8x16 circuit pack, perform the DGN-EQPT command against the circuit pack raising the alarm. Refer to the DGN-EQPT command description in <i>TL-1 Description</i> , 323-1851-190.

Procedure 4-30 (continued)
Software Subsystem Failed

Step	Action
2	If the alarm does not clear, perform a warm restart on the circuit pack that is raising the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
3	 CAUTION Risk of traffic loss A cold restart on an unprotected circuit pack causes a traffic loss. A cold restart on an active protected circuit pack causes a protection switch that impacts traffic. If the alarm does not clear, perform a cold restart on the alarmed circuit pack. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
4	If the alarm does not clear, reseal the circuit pack. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
5	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-31

Software Upgrade Failed

Alarm ID: 124

Probable cause

This alarm is raised when a failure occurs during a manual or automatic upgrade of circuit packs.

The alarm can also be raised when there is an unsaved SAOS-based CLI configuration.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action				
1	Verify if the alarm is raised against the shelf or against individual modules.				
2	<table> <tr> <td>If the alarm is raised against the shelf</td><td>Then go to step 3</td></tr> <tr> <td>individual modules</td><td>step 4</td></tr> </table>	If the alarm is raised against the shelf	Then go to step 3	individual modules	step 4
If the alarm is raised against the shelf	Then go to step 3				
individual modules	step 4				
3	If the alarm is raised due to an unsaved SAOS-based CLI configuration, save the SAOS-based CLI configurations on the primary PKT/OTN cross-connect circuit packs and then re-issue the upgrade operation. Go to step 7 .				
4	Reseat the circuit pack raising the alarm. Refer to the “Reseating a module” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table.				
5	<table> <tr> <td>If the original alarm has cleared</td><td>Then the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 6</td></tr> </table>	If the original alarm has cleared	Then the procedure is complete	not cleared	go to step 6
If the original alarm has cleared	Then the procedure is complete				
not cleared	go to step 6				
6	Replace the circuit pack. Refer to the replacement procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545. Select the appropriate procedure from the “Module replacement procedures list” table.				
7	If the alarm does not clear, contact your next level of support or your Ciena support group.				

—end—

Procedure 4-32

Software Upgrade in Progress

Alarm ID: 74

Probable cause

This alarm is raised while a system upgrade is in progress after the initialization of the software upgrade. The alarm clears after the upgrade is complete.

This alarm is for information only. Do not perform any actions other than the upgrade activity while it is active. The alarm clears after the upgrade is complete.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | If the alarm does not clear after the upgrade is complete, contact your next level of support or your Ciena support group. |
|---|--|

—end—

Procedure 4-33

Streaming Telemetry Limit Reached

Alarm ID: 2223

Probable cause

This warning is raised when there is a continuous delay in publishing the data for raise-soak period.

This happens when threshold criteria is exceeded for hold-off period on provisioned frequency.

Threshold criteria is defined as:

$(T1/F1 + T2/F2 + \dots + Tn/Fn) < K$, where $0 < K < 1$; $T1, T2, \dots, Tn$ are the last retrieval time of publishers; $F1, F2, \dots, Fn$ are the provisioned frequency of publishers.

Threshold criteria is executed at an interval of X seconds, only if new notification is publisher. Here X is HCF($F1, F2, \dots, Fn$).

Raise-soak period is defined as maximum (3 times X, 120 seconds).

Impact.

Warning, non-service-affecting (W, NSA)

Step	Action
------	--------

- | | |
|---|---|
| 1 | Verify which system has a stable frequency (it is able to publish the required data without going under stress).
If a data is being published at more than 16 times on a particular current-frequency, then that frequency can be considered as the stable frequency for current load. |
| 2 | Change the provisioned frequency of all the publishers to the stable frequency found in step 1 .
The alarm clears after short period. |
| 3 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-34

Switch Shelf ID Mismatch Detected

Alarm ID: 269

Probable cause

This alarm is raised following a shelf processor replacement when it is not clear whether the transport cards or shelf processor have shelf mastership. This mastership indicates whether the shelf processor should retrieve the shelf provisioning data from the transport cards or deliver the provisioning data to the transport cards.

ATTENTION

Do not reseal or replace circuit packs while this alarm is active.

Impact

Major, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

1	Contact your next level of support or your Ciena support group.
---	---

—end—

Procedure 4-35

Synchronization Protection alarms

Use this procedure to clear alarms associated with synchronization protection.

No action is required if a user is performing a test or maintenance operation. This alarm is a reminder not to leave a potentially service-affecting condition on the system.

Timing Distribution Forced Switch - n Ref

Alarm ID: (389, 390, 391, 392)

Probable cause

This alarm is raised when a user issues a forced switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing distribution hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Timing Distribution Manual Switch - n Ref

Alarm ID: (2005, 2006, 2007, 2008)

Probable cause

This alarm is raised when a user issues a forced switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing distribution hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Timing Distribution Lockout - n Ref

Alarm ID: (393, 394, 395, 396)

Probable cause

This alarm is raised when a user issues a lockout switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing distribution hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Procedure 4-35 (continued)

Synchronization Protection alarms

Timing Generation Forced Switch - n Ref

Alarm ID: (413, 414, 415, 416)

Probable cause

This alarm is raised when a user issues a forced switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing generation hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Timing Generation Lockout - n Ref

Alarm ID: (417, 418, 419, 420)

Probable cause

This alarm is raised when a user issues a lockout switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing generation hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Timing Generation Manual Switch - n Ref

Alarm ID: (2001, 2002, 2003,2004)

Probable cause

This alarm is raised when a user issues a forced switch request on the **n-th** (where n can be 1, 2, 3, or 4) timing reference member of the timing distribution hierarchy.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 2 UPC.

Step	Action
1	No action is required if a user is performing a test or maintenance operation. If testing/maintenance is complete, release the indicated switch request. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.
2	If the alarm does not clear, is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-36

TACACS Server 1/2 Unavailable

Alarm ID: 2043, 2044

Probable cause

Use this procedure to clear the following alarms:

- TACACS server 1 unavailable
- TACACS server 2 unavailable

The “TACACS server 1 unavailable” alarm is raised when all requests to the server 1 times out and a server 2 is provisioned.

The “TACACS server 2 unavailable” alarm is raised when all requests to the server 2 times out and a server 1 is provisioned.

If only the Server 1 is provisioned (no server 2 provisioned) and all requests time out, the “All Provisioned TACACS Servers Unavailable” alarm is raised. Refer to the [“All Provisioned TACACS Servers Unavailable”](#) alarm clearing procedure in this document).

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must use an account with at least a level 4 UPC.

Step	Action
1	The alarm clears when connectivity is re-established between the shelf and the TACACS server.
2	If the alarm does not clear, perform one of the following: • clear the security alarms. Refer to “Clearing security alarms” procedure in this document • provision a new IP address or Port for server 1 or server 2. Refer to the “Provisioning the TACACS+ server” procedure in <i>Administration and Security</i>, 323-1851-301 • disable server 1 or server 2 • disable TACACS Refer to <i>Administration and Security</i>, 323-1851-301
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-37 Tamper Detected

Alarm ID: 1819

Probable cause

This alarm is raised when tamper condition is detected on the 4x10G OTR w/ Encryption circuit pack.

For the NTK530QE circuit pack, the FIPS security boundary is protected by a hard aluminum enclosure with anti-intrusion circuitry. Upon removal of the enclosure cover, this alarm is raised.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 2 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges

Procedure 4-37 (continued)
Tamper Detected

Step	Action
------	--------

Note: For security reasons, once a circuit pack has been tampered with, the encryption/decryption block is permanently disabled. Traffic will be down.

1



CAUTION

Risk of traffic loss

The 4x10G OTR (NTK530QE variant) circuit pack requires a valid Time of Day (TOD) in order to perform the certificate validation before bringing up traffic. The TOD information is received from the SP on every circuit pack restart.

If the 4x10G OTR (NTK530QE variant) circuit pack is reseated while an SP is not present, traffic does not recover automatically on this circuit pack until the SP is available to provide the system TOD. Ensure the SP is installed in the chassis before you reseat the 4x10G OTR circuit pack.

Replace the circuit pack. Refer to the replacement procedures in *Fault Management - Module Replacement*, 323-1851-545.

Note: Certificates must be re-enrolled on the new circuit pack using MyCryptoTool. Refer to the “Performing Data Encryption Certificate Enrollment” procedure in *Encryption and FIPS Security Policy Overview and Procedures*, 323-1851-340.

2

If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-38

Target Unachievable

Alarm ID: 1514, 1717

Probable cause

This alarm is raised when the provisioned power value for one or more lasers on the Idler facility of Submarine Line Idler 10 Channel (SLIC10 or SLIC10 Flex C-Band) circuit pack is out of range.

The alarm is also raised when the SLIC10 Flex C-Band circuit pack with a lower provisioned power is replaced with a SLIC10 Flex C-Band circuit pack (NTK554HC, rev 003). The latest hardware revision of this circuit pack does not support the same output power tuning range as the previous versions. Therefore, the new circuit pack cannot tune to the provisioned output power.

For the SRA circuit pack, this alarm is raised when the measured gain is not within 3 dB of the target gain.

Impact

Minor, non-service-affecting (m, NSA) alarm

Major, service-affecting (M, SA) alarm (for RAMAN on SRA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action	
1	If the alarm is raised against the SLIC10 or SLIC10 Flex C-Band circuit pack SRA circuit pack	Then go to step 2 step 6

Procedure 4-38 (continued)
Target Unachievable

Step	Action
2	Verify the provisioned power value of the first laser on the Idler facility (TARGPOWER1/MINPOWER1/MAXPOWER1).
3	Verify that target power1 is within the range of the minimum power1 and maximum power1. If it is out of range, change the value of target power1 such that it falls in the range. Refer to the "Provisioning photonic parameters" procedure in the <i>Photonic Equipment</i> , 323-1851-102.6.
4	Repeat step 2 and step 3 for the Idler's second laser (represented by TARGPOWER2/MINPOWER2/MAXPOWER2).
5	If the alarm is raised because the SLIC10 Flex C-Band circuit pack with a lower provisioned power is replaced with a SLIC10 Flex C-Band (NTK554HC, rev 003), modify the SLIC wavelength output power and re-align the system to clear the alarm. Refer to the <i>Submarine Networking Application Guide</i> , NTRN72AA for more information.
6	Verify the Office and Short traces manually for any of the following events: • Large losses/reflections that were not captured by the OTDR alarms, possibly because of the pulse width/event spacing. If events are overlapping, re-run the trace using a smaller pulse. • Check for disparate or mixed fiber types, indicated by "gainers" or exaggerated loss splices.
7	Perform the fiber/port/patch-panel cleaning procedure(s) on the event in question. Note: When fibers are connected and disconnected, the OTDR traces are triggered automatically. Wait until traces are completed before reconnecting or disconnecting the fiber.
8	If the alarm is not cleared, record the following values: — Raman Gain Mode — Recommended Gain — Target Gain — Calculated Gain — Discovered Fiber Type — OTDCFG manually provisioned Fiber Type (may be blank)
9	Verify the fiber type provisioning is correct. If you change the provisioned fiber type (via far end ADJ-LINE or OTDCFG facility), you MUST set the Raman calibration flag to "Not Calibrated" and re-run the Power Audit. Refer to the "Performing a manual power audit" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.

Procedure 4-38 (continued)

Target Unachievable

Step	Action						
10	<table> <tr> <th>If the alarm is</th><th>Then</th></tr> <tr> <td>cleared</td><td>this procedure is complete.</td></tr> <tr> <td>otherwise</td><td>go to step 11</td></tr> </table>	If the alarm is	Then	cleared	this procedure is complete.	otherwise	go to step 11
If the alarm is	Then						
cleared	this procedure is complete.						
otherwise	go to step 11						
11	If the “calculated gain” is significantly higher or lower than the “recommended gain”, then perform the following: Check the calculated gain and recommended gain. Determine if the calculated gain is higher or smaller than the recommended gain. Change the far-end fiber type (corresponding to the fiber closer to the Raman pumps) to compensate for the difference between the two quantities. Refer to the table below to determine which fiber would be appropriate.						

Table 4-5

Fiber efficiency

Most efficient to least efficient
DSF/LS
TWRS/TWReach
TWC/TWP
ELEAF/TeraLight
NDSF LWP
NDSF
EX2000
EX3000
ULL

- If the calculated gain is higher than the recommended gain, then a less efficient fiber type should be chosen. Therefore, choose a fiber type lower in the table with respect to the current fiber type. Note that the table entries TWRS/TWReach, EX2000, and EX3000 can be skipped. For example, if the current fiber type was NDSF LWP, you can use NDSF or ULL.
- If the calculated gain is smaller than the recommended gain, then a more efficient fiber type should be chosen. Therefore, choose a fiber higher in the table with respect to the current fiber type. Note that the table entries TWRS/TWReach, EX2000, and EX3000 can be skipped. For example, if the current fiber type was ELEAF, you can use TWC or DSF.

Procedure 4-38 (continued)

Target Unachievable

Step	Action
	— Changing fiber type will allow the calculated gain to move towards the target gain and reduce the difference between the two quantities to a value below the Target Unachievable threshold. Ultimately, this step allows to adjust to some extent the gain so that it is appropriate for the span. • Change the Calibration flag to “Not Calibrated” and re-run the calibration. If the “Target Unachievable” alarm is cleared, re-run OnePlanner and this procedure is complete.
12	If the alarm is still present after completing the steps above, increase the Target Unachievable threshold in the RAMAN facility under the SRA circuit pack that has the alarm raised against. Re-run OnePlanner.
13	If the alarm is not clear, unseat the SRA circuit pack, count 10 secs and reseat the SRA circuit pack. Wait for circuit pack to complete boot cycle, then wait for the “OTDR Traces In-Progress” alarm clears (Go/No-Go OTDR traces) for the link. Refer to “Reseating a circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545.
14	If the alarm is not clear, replace the SRA circuit pack. Refer to “Replacing the SRA circuit pack” in <i>Fault Management - Module Replacement</i> , 323-1851-545. Go to step 18 .
15	Verify the Rx PROVMCPO on the associated vCCMD64 LOC facility is provisioned as expected (Typical is -17), this translates into the SMD PORT 9 NMCC Controller Output Target.
16	If the provisioning is correct, check/clean the fiber between SMD Switch Out and the user LIM Line B In. <i>The alarm clears when there is available Target Loss to achieve the Rx PROVMCPO target.</i>
17	Head end DOC must re-optimize in order to ramp channels to NMCC Controller Target if there is a delta.
18	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-39

TCM Loss of Tandem Connection

Alarm ID: 1505, 1506, 1612

Probable cause

This alarm is raised when there is a Loss of Tandem Connection (LTC) due to provisioning the TCM facility at one end only.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Verify at which end the TCM facility is not provisioned. Refer to the “Retrieving equipment and facility details” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310323-1851-311.
2	Provision the TCM facility on the end that is not provisioned. Refer to the facility provisioning procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310323-1851-311.
3	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-40

Telemetry Loss of Signal

Alarm ID: 1715

Probable cause

This alarm is raised when the SRA RAMAN amplifier cannot receive the telemetry gain signal from upstream.

This alarm can also be raised when the upstream network element is running an OTDR trace.

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must have:

- replacement fibers
- a fiber cleaning kit

Step	Action
1	If the upstream network element is running an OTDR trace and the OTDR Trace In Progress alarm is active, wait until the OTDR trace is complete. The alarm will clear when the OTDR trace is complete.
2	If there is no OTDR trace in progress, verify that there is no "Circuit Pack Fail" alarm on the upstream SRA circuit pack and the facility is in-service.
3	Verify if there are any other alarms related to the link (for example, Shutoff Threshold Crossed, Loss of Signal, and OSC Loss of Signal) and clear the alarms.
4	If the alarm is not cleared, ensure that the far-end telemetry facility is in-service.
5	If the alarm is not cleared, verify the fiber connection for the interface circuit pack that connects to the alarmed shelf.
6	Use a fiber cleaning kit to clean all the connectors then reconnect the output fibers and connectors at the amplifier. Refer to the "Cleaning Connectors" section in <i>Installation - General Information</i> , 323-1851-201.0.
7	If the alarm does not clear, replace the fiber.
8	If the alarm does not clear, provision the circuit pack manually, run another OTDR trace, or use a different maximum trace time (the maximum provisionable value is 120 seconds).

4-752 Alarm clearing procedures

Procedure 4-40 (continued)

Telemetry Loss of Signal

Step	Action
9	If the alarm does not clear, replace the SRA amplifier.
10	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-41

Threshold AIS ESO-A/ESO-B

Alarm IDs: 265, 266

Probable cause

This alarm is raised when AIS is inserted in an ESO signal or the 2 MHz ESO signal is squelched because the signal quality of the active ESO reference (determined by the SSM) is at or below (SONET mode) or below (SDH or SDH-J mode) the provisioned threshold AIS level. The alarm can be raised against the ESO A and ESO B signals.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

Checking the provisioned threshold AIS value

- | | | | | | | | |
|---|--|---|-------------------|-----------|---------------------------|-------------|------------------------------|
| 1 | Retrieve and record the provisioned threshold AIS value for the timing distribution reference (ESO A or ESO B). Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 2 | <table border="0"> <tr> <td style="padding-right: 20px;">If the provisioned threshold AIS value is</td> <td style="text-align: right;">Then go to</td> </tr> <tr> <td>incorrect</td> <td style="text-align: right;">step 3</td> </tr> <tr> <td>correct</td> <td style="text-align: right;">step 5</td> </tr> </table> | If the provisioned threshold AIS value is | Then go to | incorrect | step 3 | correct | step 5 |
| If the provisioned threshold AIS value is | Then go to | | | | | | |
| incorrect | step 3 | | | | | | |
| correct | step 5 | | | | | | |
| 3 | If required, modify the provisioned threshold AIS value. Refer to the “Provisioning ESO parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | | | | | | |
| 4 | <table border="0"> <tr> <td style="padding-right: 20px;">If the original alarm has</td> <td style="text-align: right;">Then</td> </tr> <tr> <td>cleared</td> <td style="text-align: right;">the procedure is complete</td> </tr> <tr> <td>not cleared</td> <td style="text-align: right;">go to step 5</td> </tr> </table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 5 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 5 | | | | | | |

Checking the provisioned incoming quality level override (if applicable)

- | | |
|---|---|
| 5 | Check if there is a provisioned incoming quality level override for the active timing distribution reference. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
|---|---|

Procedure 4-41 (continued)
Threshold AIS ESO-A/ESO-B

Step	Action								
6	If the incoming quality level override for the active timing distribution reference is <table> <tr> <td>not provisioned</td><td>go to step 9</td></tr> <tr> <td>incorrectly provisioned</td><td>go to step 7</td></tr> <tr> <td>correctly provisioned and is supposed to be at or below (SONET mode) or below (SDH or SDH-J mode) the threshold AIS value you recorded in step 1</td><td>go to step 8</td></tr> <tr> <td>correctly provisioned and is supposed to be above (SONET mode) or at or above (SDH or SDH-J mode) the threshold AIS value you recorded in step 1</td><td>contact your next level of support or your Ciena support group. The procedure is complete.</td></tr> </table>	not provisioned	go to step 9	incorrectly provisioned	go to step 7	correctly provisioned and is supposed to be at or below (SONET mode) or below (SDH or SDH-J mode) the threshold AIS value you recorded in step 1	go to step 8	correctly provisioned and is supposed to be above (SONET mode) or at or above (SDH or SDH-J mode) the threshold AIS value you recorded in step 1	contact your next level of support or your Ciena support group. The procedure is complete.
not provisioned	go to step 9								
incorrectly provisioned	go to step 7								
correctly provisioned and is supposed to be at or below (SONET mode) or below (SDH or SDH-J mode) the threshold AIS value you recorded in step 1	go to step 8								
correctly provisioned and is supposed to be above (SONET mode) or at or above (SDH or SDH-J mode) the threshold AIS value you recorded in step 1	contact your next level of support or your Ciena support group. The procedure is complete.								
7	Provision the correct incoming quality level override. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Go to step 9.								
8	The quality of this timing reference signal is too low to be used as an ESO reference. Select a different ESO reference. Refer to the “Provisioning ESO references” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. The procedure is complete.								

Checking the actual quality level

9	On the network element that raised the alarm, verify the actual quality for the active timing distribution reference. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.				
10	If the actual quality is <table> <tr> <td>equal or below (SONET mode) or below (SDH or SDH-J mode) the provisioned threshold AIS value</td><td>go to step 11</td></tr> <tr> <td>equal (SONET mode) or equal or above (SDH or SDH-J mode) the provisioned threshold AIS value</td><td>contact your next level of support or your Ciena support group. The procedure is complete.</td></tr> </table>	equal or below (SONET mode) or below (SDH or SDH-J mode) the provisioned threshold AIS value	go to step 11	equal (SONET mode) or equal or above (SDH or SDH-J mode) the provisioned threshold AIS value	contact your next level of support or your Ciena support group. The procedure is complete.
equal or below (SONET mode) or below (SDH or SDH-J mode) the provisioned threshold AIS value	go to step 11				
equal (SONET mode) or equal or above (SDH or SDH-J mode) the provisioned threshold AIS value	contact your next level of support or your Ciena support group. The procedure is complete.				

Procedure 4-41 (continued)
Threshold AIS ESO-A/ESO-B

Step	Action
------	--------

Checking the source of the active timing distribution at other network elements in the network

- | | | |
|----|--|--|
| 11 | Working from the network element that raised the alarm, follow step 12 to step 17 for each network element to find the source of the problem. | |
| 12 | Verify the source of the active timing distribution reference (the alarm is raised on network element D in the example Figure 4-8 on page 4-757). | |
| 13 | Clear any synchronization alarms. | |
| 14 | If the original alarm has | Then |
| | cleared | the procedure is complete |
| | not cleared | go to step 15 |
| 15 | Check if there is a provisioned outgoing quality level override at the source of the active timing distribution reference for the original network element. Refer to the synchronization procedures in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. | |
| 16 | If the outgoing quality level override is | Then |
| | not provisioned | go to step 20 |
| | incorrectly provisioned | go to step 17 |
| | correctly provisioned and is supposed to be at or below (SONET mode) or below (SDH or SDH-J mode) the threshold AIS value you recorded in step 1 | go to step 18 |
| | correctly provisioned and is supposed to be above (SONET mode) or at or above (SDH or SDH-J mode) the threshold AIS value you recorded in step 1 | contact your next level of support or your Ciena support group. The procedure is complete. |
| 17 | Provision the correct outgoing quality level override. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
Go to step 19 . | |

Selecting a different timing reference

- | | |
|----|--|
| 18 | The quality of this timing reference signal is too low to be used as an ESO reference. Select a different ESO reference. Refer to the “Provisioning ESO references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
|----|--|

Procedure 4-41 (continued)
Threshold AIS ESO-A/ESO-B

Step	Action	
19	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 20
20	Verify the actual quality for the active timing distribution reference.	
	If the actual quality is	Then
	equal or below (SONET mode) or below (SDH or SDH-J mode) the provisioned threshold AIS value	verify the source of the active timing distribution reference, then repeat from step 11
	equal (SONET mode) or equal or above (SDH or SDH-J mode) the provisioned threshold AIS value	contact your next level of support or your Ciena support group. The procedure is complete.
—end—		

Figure 4-8 Synchronization

Procedure 4-42

TIDc Primary Authentication Server Unavailable

Alarm IDs: 2225

Probable cause

This alarm is raised when the member shelf is unable to connect to the Server resided on the Primary shelf. The root causes can be:

- Comms issue between Primary and Member shelves
- The TIDC Authentication Client on the Member shelf
- The TIDC Authentication Server on the Primary shelf

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	No action is required. this alarm informs the user that the login to a specific member shelf can not be established. The alarm will be cleared when the login to member shelf is successful.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-43

TIDc Shelf Provisioning Limit Reached

Alarm IDs: 2222

Probable cause

This alarm is raised when the total size of Target Identifier Consolidation (TIDc) is at or above threshold of the TIDc size limit which is 36.

The alarm will only be raised if the parameter 'TIDCTHRESHOLD' value is either ALM or ALM_BLOCK. This parameter value can be edited using different NBIs (gRPC, NETCONF, REST, CLI).

Impact

Warning, non-service-affecting (W, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Reduce the maximum number of the shelves in the TIDc configuration to below 36. Refer to “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-44

TIDc Shelf Provisioning Near Limit

Alarm IDs: 2221

Probable cause

This alarm is raised when the total size of Target Identifier Consolidation (TIDc) is at or above 33 (>90% threshold of the TIDc size limit which is 36).

The alarm will only be raised if the parameter 'TIDCTHRESHOLD' value is either ALM or ALM_BLOCK. This parameter value can be edited using different NBIs (gRPC, NETCONF, REST, CLI).

Impact

Warning, non-service-affecting (W, NSA)

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Reduce the maximum number of the shelves in the TIDc configuration to below 33. Refer to “Editing the nodal shelf parameters” procedure in <i>Administration and Security</i> , 323-1851-301.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-45

Timing Distribution Loss of Reference - n Ref

Alarm IDs: 385, 386, 387, 388

Probable cause

This alarm is raised when all of the following are true:

- At least one timing reference is provisioned for timing distribution.
- The **n**-th provisioned timing reference in the timing distribution hierarchy fails.

Up to four timing references can be provisioned. That is, **n** can be 1, 2, 3, or 4.

When this alarm is raised, the **n**-th timing reference is not available. A timing protection switch occurs when another timing reference is provisioned and available for timing distribution. Otherwise, an AIS is transmitted on the ESO output (DS1 and E1 signals) or the ESO signal is squelched (2 MHz signal).

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action								
1	Identify the OCn/STMn slot and port, or ESI port raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.								
2	View the provisioned timing references in the timing distribution hierarchy. Refer to the "Retrieving synchronization data for a network element" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
3	<table border="0"> <tr> <td>If</td><td>Then go to</td></tr> <tr> <td>the failed timing reference is derived from an ESI port and any alarms (such as Loss Of Signal [ESI] are raised against the ESI port)</td><td>step 4</td></tr> <tr> <td>the failed timing reference is derived from an OCn/STMn signal and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port.</td><td>step 5</td></tr> <tr> <td>no alarms are raised against the corresponding ESI port or OCn/STMn port</td><td>step 7</td></tr> </table>	If	Then go to	the failed timing reference is derived from an ESI port and any alarms (such as Loss Of Signal [ESI] are raised against the ESI port)	step 4	the failed timing reference is derived from an OCn/STMn signal and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port.	step 5	no alarms are raised against the corresponding ESI port or OCn/STMn port	step 7
If	Then go to								
the failed timing reference is derived from an ESI port and any alarms (such as Loss Of Signal [ESI] are raised against the ESI port)	step 4								
the failed timing reference is derived from an OCn/STMn signal and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port.	step 5								
no alarms are raised against the corresponding ESI port or OCn/STMn port	step 7								

Procedure 4-45 (continued)

Timing Distribution Loss of Reference - n Ref

Step	Action						
4	Clear any alarms against the ESI port associated with the failed timing distribution reference. Go to step 6 .						
5	Clear any alarms against the OCn/STMn port associated with the failed timing distribution reference.						
6	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 7</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-46

Timing Generation Entry to Freerun

Alarm IDs: 68, 971

Probable cause

This alarm is raised against CLK facility when the clock frequency is not locked and has never been locked to an external clock. This situation may occur when:

- No TMGTP facilities have been provisioned.
- No TMGTP facilities are enabled.
- No TMGTP facilities is currently or was previously in the secondary state. This situation may occur when:
 - No Announce messages are received from external clock nodes or the received Announce messages are in a different IEEE-1588 domain.
 - The external clocks are judged to be of lower quality than the local clock.
- The transmit Quality Level (QL) of the Timing Generation Clock (for example: the Synchronous Ethernet transmit QL is not PRC.

This alarm is also raised against:

- a shelf (XC circuit pack providing timing) when the timing mode is provisioned as external, line, or mixed
- a SuperMux timing group (only when SuperMux in Standalone or Dual mode) when a SuperMux circuit pack provides timing for an independent timing group
- an L2 MOTR circuit pack
- an eMOTR circuit pack

when:

- no timing generation references have been provisioned
- timing generation references have been provisioned but are unusable, and the synchronization hardware on the XC, L2 MOTR, eMOTR, or SuperMux circuit pack cannot lock any of the provisioned timing generation references

In the presence of an ODU AIS, ODU LCK, ODU OCl, or an ETH10G Remote Fault, this alarm can be raised on a line-timed Flex MOTR.

Procedure 4-46 (continued)

Timing Generation Entry to Freerun

This alarm is raised during the commissioning of the system when the timing mode is provisioned before the timing generation references are provisioned. The alarm clears once the timing generation references are provisioned during commissioning and the synchronization hardware on the XC, L2, MOTR, eMOTR, or SuperMux circuit pack locks to one of the provisioned timing generation references that is usable.

A timing generation reference is unusable if:

- the clock signal is faulty due to a facility fault (for example, Loss Of Signal, Loss of Frame, AIS)
- the synchronization status message (SSM) quality level is below the minimum (ST3/SSU-B)

A Timing Generation Entry to Freerun alarm is also raised if a forced switch is applied to a faulty timing reference source or all usable timing generation references have a lockout applied (before the XC, L2, MOTR, eMOTR, or SuperMux circuit pack can lock any of the provisioned timing generation references).

While the alarm is raised, the XC, L2, MOTR, eMOTR, or SuperMux circuit pack provides timing references of Stratum 3 (SONET)/G.813 option 1 (SDH) quality (± 4.6 ppm).

The alarm clears when:

- the synchronization hardware on the XC, L2, MOTR, eMOTR, or SuperMux circuit pack locks to one of the provisioned timing generation references which is usable
- the timing mode is changed to internal

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-46 (continued)
Timing Generation Entry to Freerun

Step	Action	
1	Select your first step.	
	If the alarm is raised against	Then go to
	CLK facility	step 2
	otherwise	step 7
2	Select your next step.	
	If	Then go to
	no TMGTP facilities have been provisioned or enabled	step 3
	no TMGTP facilities are currently or were previously in the secondary state	step 4
	the transmit QL of the Timing Generation Clock is not Primary reference clock (PRC)	step 6
3	Ensure that at least one TMGTP facility is provisioned with Packet Synchronization configured as On and Master Only configured as No. Refer to “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 17 .	
4	Check the alarm history for the Announce Receipt Timeout Expired event. This is an indication that this map can be the cause of the problem.	
5	Retrieve the sessions for the TMGTP. Refer to the “Provisioning IEEE-1588 Sync Management parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If no sessions are present, then no Announce messages are being received from the connected equipment. If a session is present, examine the contents of the Announce message to determine why it was not selected as the grandmaster. Go to step 17 .	
6	Retrieve the status of the SyncE timing references. Verify that there is a reference in the Active & Working state and that its Incoming Current Quality is PRC. Go to step 17 .	
7	Check that the correct timing mode has been provisioned. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.	
	If necessary, provision the correct timing mode. Refer to the “Provisioning the network element timing mode and references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.	
8	If the timing mode has been provisioned, check that the correct timing generation references have been provisioned. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.	

Procedure 4-46 (continued)

Timing Generation Entry to Freerun

Step	Action								
	If necessary, provision the timing generation references. Refer to the "Provisioning the network element timing mode and references" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
9	Check the synchronization protection switching to see if there are any lockouts on usable references or forced switches to unusable timing generation references. Refer to the "Retrieving synchronization protection status details" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. If there are any lockouts on usable references or forced switches to unusable timing generation references, check that the switches are still required. If the switches are not required, release them. Refer to the "Releasing a synchronization protection switch" procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.								
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared and a Timing Generation Failure to Lock alarm is present</td><td>go to step 11</td></tr> <tr> <td>not cleared and a Timing Generation Failure to Lock alarm is not present</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared and a Timing Generation Failure to Lock alarm is present	go to step 11	not cleared and a Timing Generation Failure to Lock alarm is not present	go to step 13
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared and a Timing Generation Failure to Lock alarm is present	go to step 11								
not cleared and a Timing Generation Failure to Lock alarm is not present	go to step 13								
11	Clear the Timing Generation Failure to Lock alarm. Refer to " Timing Generation Failure To Lock " on page 4-773.								
12	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 13		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	go to step 13								
13	Check the SSM status (incoming received quality and the incoming provisioned quality) of all the timing generation references. Refer to the "Retrieving synchronization data for a network element" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the incoming provisioned quality of a reference is not 'None' and is below the minimum (ST3/SSU-B)</td><td>step 14</td></tr> <tr> <td>the incoming received quality of a reference is below the minimum (ST3/SSU-B)</td><td>step 15</td></tr> <tr> <td>otherwise</td><td>step 17</td></tr> </table>	If	Then go to	the incoming provisioned quality of a reference is not 'None' and is below the minimum (ST3/SSU-B)	step 14	the incoming received quality of a reference is below the minimum (ST3/SSU-B)	step 15	otherwise	step 17
If	Then go to								
the incoming provisioned quality of a reference is not 'None' and is below the minimum (ST3/SSU-B)	step 14								
the incoming received quality of a reference is below the minimum (ST3/SSU-B)	step 15								
otherwise	step 17								

Procedure 4-46 (continued)
Timing Generation Entry to Freerun

Step	Action						
14	Check the network synchronization plan to see if the provisioned incoming quality override is correct for each timing generation reference. If necessary correct the incoming override quality level. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
15	Check the network synchronization plan to see if the incoming received quality level is expected. If not: • check and correct the outgoing quality override on the corresponding facility at the far-end network element. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. • check the far-end network element for synchronization problems that can cause the outgoing facility to transmit an SSM below the minimum (ST3/SSU-B).						
16	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 17</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 17
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 17						
17	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-47

Timing Generation Entry to Holdover

Alarm IDs: 69, 972

Probable cause

This alarm is raised against a CLK facility that was previously locked to an external clock and has lost lock. This situation can occur when:

- All TMGTP facilities have been deleted
- All TMGTP facilities have been disabled
- No TMGTP facilities is currently in the secondary state. This situation may occur when:
 - No Announce messages are received from external clock nodes or the received Announce messages are in a different IEEE-1588 domain
 - The external clocks are judged to be of lower quality than the local clock.
- The transmit Quality Level (QL) of the Timing Generation Clock (for example: the Synchronous Ethernet transmit QL is not PRC.

This alarm is also raised against:

- a shelf (XC providing timing) when the timing mode is provisioned as external, line, or mixed
- a SuperMux timing group (only when SuperMux in Standalone or Dual mode) when a SuperMux circuit pack provides timing for an independent timing group
- an L2 MOTR circuit pack
- an eMOTR circuit pack

when all the provisioned timing generation references have become unusable and can no longer be locked by the synchronization hardware on the XC, L2, MOTR, eMOTR, or SuperMux circuit pack.

A timing generation reference becomes unusable if:

- the clock signal is faulty due to a facility fault (for example, Loss Of Signal, Loss of Frame, AIS)

Procedure 4-47 (continued)

Timing Generation Entry to Holdover

- the synchronization status message (SSM) quality level is below the minimum (ST3/SSU-B)

In the presence of an ODU AIS, ODU LCK, ODU OCI, or an ETH10G Remote Fault, this alarm can be raised on a line-timed Flex MOTR.

A Timing Generation Entry to Holdover alarm is also raised if a forced switch is applied to a faulty timing reference source or all usable timing generation references have a lockout applied.

The XC, L2, MOTR, eMOTR, or SuperMux circuit pack maintains synchronization based on the last valid timing reference signal for at least 24 hours at a guaranteed holdover quality (± 0.37 ppm).

The alarm clears when one of the timing generation references becomes usable and the synchronization hardware on the XC, L2, MOTR, eMOTR, or SuperMux circuit pack locks to one of the timing references.

Impact

Warning

Prerequisites

To perform this procedure, you must

- have the network synchronization plan
- use an account with at least a level 3 UPC

Step	Action								
1	Select your first step. <table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the alarm is raised against CLK facility</td><td>step 2</td></tr> <tr> <td>otherwise</td><td>step 7</td></tr> </table>	If	Then go to	the alarm is raised against CLK facility	step 2	otherwise	step 7		
If	Then go to								
the alarm is raised against CLK facility	step 2								
otherwise	step 7								
2	Select your next step. <table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>no TMGTP facilities have been provisioned or enabled</td><td>step 3</td></tr> <tr> <td>no TMGTP facilities is currently or was previously in the secondary state</td><td>step 4</td></tr> <tr> <td>the transmit QL of the Timing Generation Clock is not Primary reference clock (PRC)</td><td>step 6</td></tr> </table>	If	Then go to	no TMGTP facilities have been provisioned or enabled	step 3	no TMGTP facilities is currently or was previously in the secondary state	step 4	the transmit QL of the Timing Generation Clock is not Primary reference clock (PRC)	step 6
If	Then go to								
no TMGTP facilities have been provisioned or enabled	step 3								
no TMGTP facilities is currently or was previously in the secondary state	step 4								
the transmit QL of the Timing Generation Clock is not Primary reference clock (PRC)	step 6								

Procedure 4-47 (continued)

Timing Generation Entry to Holdover

Step	Action
3	Ensure that at least one TMGTP facility is provisioned with Packet Synchronization configured as On and Master Only configured as No. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. Go to step 18 .
4	Check the alarm history for the Announce Receipt Timeout Expired event. This is an indication that this map can be the cause of the problem.
5	Retrieve the sessions for the TMGTP. Refer to “Provisioning IEEE-1588 Sync Management parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If no sessions are present, then no Announce messages are being received from the connected equipment. If a session is present, examine the contents of the Announce message to determine why it was not selected as the grandmaster. Go to step 18 .
6	Retrieve the status of the SyncE timing references. Verify that there is a reference in the Active & Working state and that its Incoming Current Quality is PRC. Go to step 18 .
7	Check that the correct timing mode has been provisioned. Refer to the “Provisioning the network element timing mode and references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If necessary, provision the correct timing mode. Refer to the “Provisioning the network element timing mode and references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
8	If the timing mode has been provisioned, check that the correct timing generation references have been provisioned. Refer to the “Provisioning the network element timing mode and references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If necessary, provision the timing generation references. Refer to the “Provisioning the network element timing mode and references” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
9	Check the synchronization protection switching to see if there are any lockouts on usable references or forced switches to unusable timing generation references. Refer to the “Retrieving synchronization protection status details” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315. If there are any lockouts on usable references or forced switches to unusable timing generation references, check that the switches are still required. If the switches are not required, release them. Refer to the “Releasing a protection switch” procedure in <i>Configuration - Protection Switching</i> , 323-1851-315.

Procedure 4-47 (continued)

Timing Generation Entry to Holdover

Step	Action								
10	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared and a Timing Generation Failure to Lock alarm is present</td><td>go to step 11</td></tr> <tr> <td>not cleared and a Timing Generation Failure to Lock alarm is not present</td><td>go to step 13</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared and a Timing Generation Failure to Lock alarm is present	go to step 11	not cleared and a Timing Generation Failure to Lock alarm is not present	go to step 13
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared and a Timing Generation Failure to Lock alarm is present	go to step 11								
not cleared and a Timing Generation Failure to Lock alarm is not present	go to step 13								
11	Clear the Timing Generation Failure to Lock alarm. Refer to “Timing Generation Failure To Lock” on page 4-773 .								
12	<table> <tr> <th>If</th><th>Then</th></tr> <tr> <td>the original alarm has cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 13</td></tr> </table>	If	Then	the original alarm has cleared	the procedure is complete	not cleared	go to step 13		
If	Then								
the original alarm has cleared	the procedure is complete								
not cleared	go to step 13								
13	Check the SSM status (incoming received quality and the incoming provisioned quality) of all the timing generation references. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
14	<table> <tr> <th>If</th><th>Then go to</th></tr> <tr> <td>the incoming provisioned quality of a reference is not ‘None’ and is below the minimum (ST3/SSU-B)</td><td>step 15</td></tr> <tr> <td>the incoming received quality of a reference is below the minimum (ST3/SSU-B)</td><td>step 16</td></tr> <tr> <td>otherwise</td><td>step 18</td></tr> </table>	If	Then go to	the incoming provisioned quality of a reference is not ‘None’ and is below the minimum (ST3/SSU-B)	step 15	the incoming received quality of a reference is below the minimum (ST3/SSU-B)	step 16	otherwise	step 18
If	Then go to								
the incoming provisioned quality of a reference is not ‘None’ and is below the minimum (ST3/SSU-B)	step 15								
the incoming received quality of a reference is below the minimum (ST3/SSU-B)	step 16								
otherwise	step 18								
15	Check the network synchronization plan to see if the provisioned incoming quality override is correct for each timing generation reference. If necessary correct the incoming override quality level. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
16	Check the network synchronization plan to see if the incoming received quality level is expected. If not: - check and correct the outgoing quality override on the corresponding facility at the far-end network element. Refer to the “Setting the synchronization status message override quality level” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. - check the far-end network element for synchronization problems that can cause the outgoing facility to transmit an SSM below the minimum (ST3/SSU-B)								

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Procedure 4-47 (continued)

Timing Generation Entry to Holdover

Step	Action						
17	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 18</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 18
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 18						
18	If the alarm does not clear, contact your next level of support or your Ciena support group.						
—end—							

Procedure 4-48

Timing Generation Failure To Lock

Alarm IDs: 145, 973

Probable cause

This alarm is raised against a CLK facility that loses lock to an external clock. This situation can occur when:

- All TMGTP facilities have been deleted
- All TMGTP facilities have been disabled
- No TMGTP facilities is currently in the secondary state. This situation may occur when:
 - No Announce messages are received from external clock nodes or the received Announce messages are in a different IEEE-1588 domain
 - The external clocks are judged to be of lower quality than the local clock.
- The transmit Quality Level (QL) of the Timing Generation Clock (for example: the Synchronous Ethernet transmit QL) is not PRC.

It may take a long time for a CLK facility to lock to an external clock. Therefore, the Timing Generation Failure to Lock alarm may stay active for some time after the Timing Generation Entry to Freerun or Timing Generation Entry to Holdover alarm has cleared.

This alarm is also raised against;

- a shelf (XC providing timing) when the timing mode is provisioned as external, line, or mixed
- a SuperMux timing group (only when SuperMux in Standalone or Dual mode) when a SuperMux circuit pack provides timing for an independent timing group
- an L2 MOTR circuit pack
- an eMOTR circuit pack

when:

- there is a loss of all provisioned timing generation references
- the incoming quality of all provisioned timing generation references, as determined by the SSM, is below the quality of the internal clock
- synchronization hardware on the XC, L2, MOTR, eMOTR, or SuperMux circuit pack is not locked to a timing reference

Procedure 4-48 (continued)

Timing Generation Failure To Lock

Step Action

In the presence of an ODU AIS, ODU LCK, ODU OCI, or an ETH10G Remote Fault, this alarm can be raised on a line-timed Flex MOTR.

When there is a loss of all provisioned timing references, the network element enters holdover mode. After 24 hours in holdover mode, the network element may operate in freerun mode. In holdover mode, the internal clock operates at a fixed frequency according to the last known frequency reference. In freerun mode, the network element synchronization is based on the internal Stratum 3 (SONET)/G.813 Option 1 (SDH) compliant 4.6 ppm clock.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

- 1 Select your first step.

If the alarm is raised against	Then go to
CLK facility	step 2
otherwise	step 7
- 2 Select your next step.

If	Then go to
no TMGTP facilities have been provisioned or enabled	step 3
no TMGTP facilities is currently or was previously in the secondary state	step 4
the transmit QL of the Timing Generation Clock is not Primary reference clock (PRC)	step 6
- 3 Ensure that at least one TMGTP facility is provisioned with Packet Synchronization configured as On and Master Only configured as No. Refer to the "Setting the synchronization status message override quality level" procedure in *Configuration - Provisioning and Operating*, 323-1851-310. Go to [step 13](#).
- 4 Check the alarm history for the Announce Receipt Timeout Expired event. This is an indication that this map can be the cause of the problem.

Procedure 4-48 (continued)

Timing Generation Failure To Lock

Step	Action						
5	Retrieve the sessions for the TMGTP. Refer to the “Provisioning IEEE-1588 Sync Management parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If no sessions are present, then no Announce messages are being received from the connected equipment. If a session is present, examine the contents of the Announce message to determine why it was not selected as the grandmaster. Go to step 13						
6	Retrieve the status of the SyncE timing references. Verify that there is a reference in the Active & Working state and that its Incoming Current Quality is PRC. Go to step 13 .						
7	Verify the timing generation (timing mode and references) for the network element. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
8	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 9
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 9						
9	<table> <tr> <td>If Timing Generation Loss of Reference - n Ref alarms are</td><td>Then go to</td></tr> <tr> <td>raised</td><td>step 10</td></tr> <tr> <td>not raised</td><td>step 12</td></tr> </table>	If Timing Generation Loss of Reference - n Ref alarms are	Then go to	raised	step 10	not raised	step 12
If Timing Generation Loss of Reference - n Ref alarms are	Then go to						
raised	step 10						
not raised	step 12						
10	Clear all raised Timing Generation Loss of Reference - n Ref alarms. Refer to “ Timing Generation Loss of Reference - n Ref ” on page 4-776. Wait two to five minutes.						
11	<table> <tr> <td>If the original alarm has</td><td>Then</td></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 12</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 12
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 12						
12	If the alarm does not clear, verify the quality level of the timing references to determine if the synchronization quality is lower than the network element internal clock quality. Refer to the “Retrieving synchronization data for a network element” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. If the quality level of the timing references is lower than the network element, investigate the timing reference sources.						
13	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-49

Timing Generation Loss of Reference - n Ref

Alarm IDs: 401, 402, 403, 404, 405, 406, 407, 408

Probable cause

This alarm is raised against:

- a shelf when a XC circuit pack provides timing
- a SuperMux timing group (only when SuperMux in Standalone or Dual mode) when a SuperMux circuit pack provides timing for an independent timing group
- an L2 MOTR circuit pack
- an eMOTR circuit pack

This alarm is raised when all of the following are true:

- the timing mode for the timing group is provisioned as external, line, or mixed (only line timing is supported within independent timing groups)
- at least one timing reference is provisioned for timing generation
- the **n**-th provisioned timing reference in the timing generation hierarchy fails

In the presence of an ODU AIS, ODU LCK, ODU OCI, or an ETH10G Remote Fault, this alarm can be raised on a line-timed Flex MOTR.

When a XC circuit pack provides timing, up to four timing references can be provisioned (that is, **n** can be 1, 2, 3, or 4). When a SuperMux circuit pack provides timing, up to two timing references can be provisioned (that is, **n** can be 1 or 2).

When this alarm is raised, the **n**-th timing reference is not available. A timing protection switch occurs when another timing reference is provisioned and available for timing generation. Otherwise, the shelf enters timing holdover mode.

Procedure 4-49 (continued)

Timing Generation Loss of Reference - n Ref**Impact**

Minor, non-service-affecting (m, NSA) alarm if another reference in the timing generation hierarchy is provisioned and available

Major, non-service-affecting (m, NSA) alarm if no other timing generation references are provisioned or available

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action								
1	Identify the OCn/STMn slot and port, ETTP/ODUTTP eMOTR facilities, or ESI port raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document. Note: On the XC circuit packs, the synchronization status LED (yellow circle) that corresponds to the failed reference (1, 2, 3, or 4) is lit if the corresponding reference (1, 2, 3, or 4) is lost. This condition occurs for a Loss Of Signal on an ESI port or a failed reference (Loss Of Signal, Loss of Frame, excessive bit error rate, or AIS) on optical interfaces. To view the provisioned timing references in the timing generation hierarchy, refer to the "Retrieving synchronization data for a network element" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.								
2	<table border="0"> <tr> <td>If</td><td>Then go to</td></tr> <tr> <td>the failed timing reference is derived from an ESI port and any alarms are raised against the ESI port</td><td>step 3</td></tr> <tr> <td>the failed timing reference is derived from an OCn/STMn signal, ETTP/OTUTTP eMOTR facilities, and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port</td><td>step 4</td></tr> <tr> <td>no alarms are raised against the corresponding ESI port or OCn/STMn port</td><td>step 6</td></tr> </table>	If	Then go to	the failed timing reference is derived from an ESI port and any alarms are raised against the ESI port	step 3	the failed timing reference is derived from an OCn/STMn signal, ETTP/OTUTTP eMOTR facilities, and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port	step 4	no alarms are raised against the corresponding ESI port or OCn/STMn port	step 6
If	Then go to								
the failed timing reference is derived from an ESI port and any alarms are raised against the ESI port	step 3								
the failed timing reference is derived from an OCn/STMn signal, ETTP/OTUTTP eMOTR facilities, and any alarms (such as Loss Of Signal or Loss of Frame) are raised against the corresponding OCn/STMn port	step 4								
no alarms are raised against the corresponding ESI port or OCn/STMn port	step 6								
3	Clear any alarms against the ESI port associated with the failed timing generation reference. Go to step 5 .								
4	Clear any alarms against the OCn/STMn or ETTP/OTUTTP eMOTR facilities associated with the failed timing generation reference.								

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Procedure 4-49 (continued)

Timing Generation Loss of Reference - n Ref

Step	Action						
5	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	If the alarm does not clear, contact your next level of support or your Ciena support group.						
—end—							

Procedure 4-50 TOD Server Not Provisioned

Alarm ID: 575 Probable cause

This alarm is raised when the shelf is provisioned to retrieve its date and time from a Network Time Protocol (NTP) server but the NTP server attributes have not yet been provisioned.

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

Impact Minor, non-service-affecting (m, NSA) alarm

- ### Prerequisites
- To perform this procedure you must:
- confirm that the date and time will be retrieved from an NTP server
 - know the IP addresses of and have access to the NTP servers
 - use an account with at least a level 3 UPC

Step	Action
1	Select Node Information from the Configuration menu. Select the Time of Day tab in the Node Information application. Refer to the “Displaying node information” procedure in <i>Administration and Security</i> , 323-1851-301.
2	If the date and time should be set manually Then in the Settings area, click on Edit. In the Time of Day Settings dialog box, set Status to Off to disable the Time of Day Mode (TOD Sync). The procedure is complete.
	retrieved from an NTP server complete the procedure on setting the date and time using an NTP server. Refer to the “Provisioning Time of Day servers” procedure in <i>Administration and Security</i>, 323-1851-301.
3	If the original alarm has cleared not cleared Then the procedure is complete go to step 4

Procedure 4-50 (continued)
TOD Server Not Provisioned

Step	Action						
4	Verify if the “Unable to Synchronize TOD” alarm has not been raised. If so, refer to the alarm clearing procedure for the alarm to clear the alarm.						
5	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 6</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 6
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 6						
6	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-51

TODR Reversion Inhibited

Alarm ID: 1828, 1829, 1834

Probable cause

This warning is raised when all TODR profiles, assigned to OTN Control Plane SNCP protection groups and ASNCP protection groups are disabled.

The alarm is raised for ASNCP/SNCP protection groups, when they are provisioned as Revertive.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Enable at least one of the TODR profiles assigned to the ASNCP/SNCP OTN protection group. Refer to the “Editing a TODR profile” procedure in <i>Configuration - Control Plane</i> , 323-1851-330.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-52

TOD Threshold Exceeded

Alarm ID: 335

Probable cause

The alarm is raised by the shelf processor when TOD server provided TOD varies from NE time by 10 minutes or more.

Impact

Major, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure you must:

- confirm that the date and time will be retrieved from an NTP server
- know the IP addresses of and have access to the NTP servers
- use an account with at least a level 3 UPC

Step	Action
1	Operate a time of day synchronization as follows: • Select the required network element in the navigation tree. • Select Node Information from the Configuration drop-down menu to open the Node Information window. • Select the Time Of Day tab. • Click Synchronize (in the Servers area of the window) to initiate a time of day synchronization. • If the synchronization is successful, the Detected offset field is set to 00:00:00, and the alarm clears. • Click Refresh to update the Node Information window
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-53

Topology Build Failed

Alarm ID: 1799**Probable cause**

This warning is raised to indicate that the topology build (NCT build) is failing in the background after a successful build. The build can fail due to a COMMS failure or misprovisioning.

This warning is raised against an OTS one hour after detecting the build failure.

Impact

Warning

Step	Action
1	Verify the Customer Visible Logs (CVL) for any reasons for the build failure. Refer to the DOC logs section in <i>Fault Management - Customer Visible Logs</i> , 323-1851-840, for troubleshooting information.
2	Verify the COMMS for failure.
3	Verify the provisioning and validate that the Domain is defined properly.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-54

Trace Identifier Mismatch (OCn/STMn)

Alarm IDs: 2, 137, 152, 280, 983

Probable cause

This alarm is raised when the section trace identifier value received by the facility differs from the expected provisioned section trace value. This can be the result of incorrect fiberling or incorrect provisioning of the section trace value.

For 10x10G MUX circuit packs, this alarm is masked by AIS-L.

Trace identifier mismatch detection can be enabled or disabled on a per payload instance basis. By default, detection is disabled.

The “Trace Identifier Mismatch” alarm is raised when trace identifier detection is enabled and the expected and incoming trace identifiers do not match.

Impact

Major, service-affecting (M, SA) alarm, if in Alarms on, with traffic protection mode

Major, service-affecting (M, SA) alarm, if active 1+1 port TPT, or 1+1 TPT or unprotected with cross-connects, if in Alarms on, with traffic protection mode

Minor, non-service-affecting (m, NSA) alarm, if inactive 1+1 port TPT, or 1+1 TPT, 1+1 port TPT, or 1+1 TPT, or without cross-connects, if in Alarms on, with traffic protection mode

ATTENTION

This alarm has dual severity in Alarms on, with traffic protection mode with 1+1 port TPT, or 1+1 TPT protection.

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have the optical fiber/cable connection information (that is, how the modules on each network element connect to other network elements)
- use an account with at least a level 3 UPC

Procedure 4-54 (continued)

Trace Identifier Mismatch (OCn/STMn)

Step	Action
1	Retrieve the section trace format and values on OC-n/STM-n facilities at the transmit network element, and at the alarmed receive network element on optical facilities at the transmit network element and at the alarmed receive network element. Refer to the “Retrieving and editing section trace messages” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	Compare the outgoing section trace value of the transmit signal at the transmit network element with the expected section trace value of the receive signal at the receive network element. If the values are different, change the expected section value of the receive signal to match the outgoing section trace value of the transmit signal. Refer to the “Retrieving and editing section trace messages” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. Note: If the received section trace value is null, changing the expected section trace value to null clears the alarm.
3	Click Refresh in the Section Trace dialog box.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-55

Trace Identifier Mismatch (STS/HO VC and VT/LO VC)

Alarm IDs: 458

Probable cause

This alarm is raised against a STS/HO VC path when the incoming and expected path trace values are different. This alarm is also raised if a connection mismatch or an optical fiber connection mismatch exists.

Note: The path Trace Identifier Mismatch alarm can take up to 48 seconds to be raised after the mismatch exists.

For STS/HO VC paths, use the Monitoring enabled check box in the Path Provisioning dialog box to enable or disable path trace monitoring. Refer to the “Retrieving and editing path provisioning” procedure in *Configuration - Provisioning and Operating*, 323-1851-310. By default, path trace monitoring is disabled. When path trace monitoring is enabled, this alarm is raised when the outgoing path trace message at one end and the expected path trace message at the other end are different.

Impact

Major, service-affecting (M, SA) alarm, if on active path

Minor, non-service-affecting (m, NSA) alarm, if on inactive path

Prerequisites

To perform this procedure, you must

- have the optical fiber connection information (that is, how the modules on each network element connect to other network elements)
- use an account with at least a level 3 UPC

Step	Action						
1	Verify if there are alarms of higher order from the alarm hierarchy. Refer to the “Alarm hierarchies and alarm severities” section in this document.						
2	Clear any alarms of higher order using the appropriate procedures.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.						

Procedure 4-55 (continued)

Trace Identifier Mismatch (STS/HO VC and VT/LO VC)

Step	Action
5	Use the optical fiber connection information to identify the transmit and receive ends of the STS/VC path.
6	For an STS/VC4-n path, retrieve and record the path trace messages at both ends. Refer to the “Retrieving and editing path provisioning” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
7	Ensure that path trace is provisioned correctly at each network element: • Ensure that the path trace Format parameter is provisioned to the same setting at both ends. • Ensure that the outgoing path trace string (Transmitted) at the transmit end matches the incoming expected path trace string (Expected Rx) at the receive end. • To provision path trace, ensure that the cross-connect is already provisioned. Refer to the “Retrieving path connections” procedure in <i>Configuration - Connections Management</i>, 323-1851-320.
8	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-56

Transport Data Recovery Failed

Alarm ID: 76

Probable cause

This alarm is raised when data is not recovered after a shelf processor (SP) replacement or restart. If a replacement SP is running a release lower than the current release running on the shelf, this alarm may be raised and a “Software Mismatch” alarm will also be raised. An upgrade must be performed to upgrade the software on the SP to match the software release on the shelf.

If a replacement SP is running a release higher than the current release running on the shelf, and is not within the supported upgrade release window, a “Software Mismatch” alarm will also be raised, and a downgrade must be performed to downgrade the software on the SP to match the software release on the shelf.

If this alarm is raised when the SP has the same release as the shelf, then there is an issue with the recovered data, and a database restore may be required. For example, in the case where the alarm is not caused by an SP replacement. This alarm may also be raised when a SP that has data for one shelf type is placed into a different shelf type with no other circuit pack present.

This alarm can also be raised when a “Circuit Pack Failed”, “Circuit Pack Mismatch”, “Circuit Pack Missing”, “Circuit Pack Unknown” or “Intercard Suspected” alarm or condition existed on the network element before a SP replacement.

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	Check the current release running on the network element. Refer to the “Checking the current software release” procedure in <i>Commissioning and Testing</i> , 323-1851-221.
2	If the SP is running a release that is different than that of the shelf, and a “Software Mismatch” alarm is present, go to step 3. If a Software Mismatch alarm is not present and the SP has been placed into a shelf with no other circuit pack present, go to step 4, otherwise go to step 5.

Procedure 4-56 (continued)
Transport Data Recovery Failed

Step	Action
3	Perform an upgrade on the network element. For the upgrade “to release”, use the current release running on the network element. Refer to the 6500 <i>Software Upgrade Procedures</i> for this release listed in the “ <i>Software Upgrade Procedures</i> ” section in <i>Planning - Ordering Information</i> , 323-1851-151. Go to step 5 .
4	Decommission the SP. Refer to the “Deleting all shelf provisioning information for a standalone shelf or all shelves of a consolidated node” procedure in <i>Administration and Security</i> , 323-1851-301. Go to step 6 .
5	Perform a database restore. Refer to the backup and restore procedures in <i>Administration and Security</i> , 323-1851-301.
6	Ensure that the system is restored to its original state by retrieving all conditions and alarms. Clear all alarms using the appropriate alarm clearing procedure.
7	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-57

TR Control Disabled

Alarm ID: 661

Probable cause

This alarm is raised when the user has provisioned the TR controls to the OFF state.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- use an account with at least a level 3 UPC

Step	Action
------	--------

- | | |
|---|---|
| 1 | Provision the TR control to ON. Refer to the “Retrieving and editing TR control information” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310. |
|---|---|

—end—

Procedure 4-58

TR Control Echo Trace Mismatch

Alarm IDs: 660, 1006, 1176, 1437

Probable cause

This alarm is raised when the TR control receives a TxID from the far-end receiver that does not match its own TxID. This indicates a crossed fiber. This alarm clears when TR control receives a TxID from the far-end receiver that matches its own TxID, or when TR control receives no TxID from the far-end receiver.

The alarm is also raised when there is a terminal loopback on the far-end 40G OCLD circuit pack.

This alarm applies to OTM, OTM2, OTM3, OTM4, OTMC2, or PTP facilities of the 10G WT, 10GE LAN WT, 10G OTN WT, 40G OCLD, Wavelength-Selective 40G OCLD, 40G UOCLD, 100G WL3/WL3e OCLD, Flex2 WL3/WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD, 100G WL3e OTR, and 100G OCLD circuit packs.

ATTENTION

On the 40G OCLD circuit pack, when Terminal Loopback is provisioned on the near-end, EOC comms from the far-end are not terminated on the near-end line card. Hence, the “TR Control Echo Trace Mismatch” alarm is raised against the far-end 40G OCLD circuit pack.

Impact

Critical, service-affecting (C, SA) alarm, an active circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or unprotected with cross-connects provisioned

Minor, non-service-affecting (m, NSA) alarm, an inactive circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or no cross-connects provisioned

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- use an account with at least a level 3 UPC

Procedure 4-58 (continued)

TR Control Echo Trace Mismatch

Step	Action						
1	Identify the circuit pack or circuit packs raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
2	From the Site Manager Configuration menu, select Equipment & Facility Provisioning . Select the alarmed facility, and record the values from the Echoed Trace Rx , Trace Tx , and Associated Far End Rx ID columns. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.						
3	<table><tr><td>If the Echoed Trace Rx and Trace Tx values are</td><td>Then go to</td></tr><tr><td>not equal</td><td>step 4</td></tr><tr><td>equal</td><td>step 5</td></tr></table>	If the Echoed Trace Rx and Trace Tx values are	Then go to	not equal	step 4	equal	step 5
If the Echoed Trace Rx and Trace Tx values are	Then go to						
not equal	step 4						
equal	step 5						
4	There is a likely a misconnected fiber. Verify that the optical fiber connections are correct on the circuit pack raising the alarm and on the upstream circuit pack. The Associated Far End Rx ID value from step 2 indicates where the misconnection exists.						
5	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-59

TR Control Initialization in Progress

Alarm ID: 658

Probable cause

This alarm is raised when TR control is performing the dispersion scan or OOS optimization process. This alarm indicates that the Tx/Rx pair is not in the IS or SS optimization state.

Typically, this alarm clears autonomously within five minutes and is replaced by the TR Control IS Optimization In Progress alarm. The alarm does not clear if the far-end Rx and the near-end Rx are not able to acquire a signal (BER of approximately 3.8E-3).

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management-Module Replacement*, 323-1851-545
- use an account with at least a level 3 UPC

Step	Action						
1	Wait for the alarm to clear autonomously. When it clears, the TR Control IS Optimization In Progress alarm is raised.						
2	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 3</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 3
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 3						

Procedure 4-59 (continued)

TR Control Initialization in Progress

Step	Action
3	This indicates that the far-end Rx and the near-end Rx are not able to acquire a signal (BER approximately $3.8E-3$). Check the following: • Ensure that TR Control is enabled at the upstream circuit pack. Refer to the “Retrieving and editing TR control information” procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. • Ensure that the near-end and far-end circuit packs are the same PEC. • Ensure that the photonic layer is properly optimized. Troubleshoot any photonic-related layer alarms. • Ensure the near-end and far-end circuit packs are properly fibered.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-60

TR Control IS Optimization in Progress

Alarm ID: 659**Probable cause**

This alarm is raised when TR control is performing the IS optimization process and the Pre-FEC BER has reached a certain level but the system is not optimized. That is, the system is not in the SS (steady state) optimization mode.

This alarm indicates that the Tx/Rx pair is not in the SS optimization state.

This alarm clears autonomously within five minutes, no user action is required.

Impact

Minor, non-service-affecting (m, NSA) alarm

Step	Action
------	--------

- | | |
|---|--|
| 1 | Wait for the alarm to clear autonomously. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-61

Tributary Slots Not Available

Alarm ID: 2028

Probable cause

This warning is raised against an ODUCTP facility on the WLAi line port if the allocated Tributary Slots of ODUCTP facility becomes unavailable because it exceeds the maximum capacity of the current provisioned Transmission Mode.

Impact

Warning

Step	Action
1	If this is due to downshifting of Transmission Mode, the alarm will clear after the Transmission Mode is restored to its original value and the ODUCTP that was created is deleted.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-62

Tx Frequency Out of Range

Alarm IDs: 2030, 2176, 2209

Probable cause

This alarm is raised when the network element detects an ETTP/OTUTTP/PTP signal that is transmitting to the input is out of normal frequency range. The ETTP/OTUTTP/PTP out of frequency range on the input to the system usually causes this alarm. An Rx Frequency Out of Range alarm will also be active.

Impact

Major, service-affecting (M, SA) alarm



CAUTION

Risk of traffic loss

The equipment may be unable to lock onto the ETTP/OTUTTP and traffic can be lost.

Step	Action
------	--------

- | | |
|---|--|
| 1 | If possible, clear all ETTP/OTUTTP/PTP Rx alarms from the network. |
| 2 | If the alarm does not clear, contact your next level of support or your Ciena support group. |

—end—

Procedure 4-63

Tx Loss of Clock

Alarm ID: 2029, 2165, 2207

Probable cause

This alarm is raised against the ETTP/OTUTTP facility of the WLAi MOTRPTP facility of the circuit packs when the client port Tx direction DPLL is out of clock or the clock cannot be recovered.

Impact

Critical, service-affecting (C, SA) alarm
Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the circuit packs on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element
- have a fiber cleaning kit
- obtain a replacement VOA, if required
- obtain a replacement circuit pack, if required

Step	Action
------	--------

- | | |
|---|--|
| 1 | Ensure the subtending equipment is working properly and providing an error-free signal that meets the allowed frequency deviation specifications. |
| 2 | Ensure that all the optical fibers between the subtending equipment Tx port and at the alarmed port are correctly connected. |
| 3 | Ensure that the line rate or the client type is the same for both ends of the link. |
| 4 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. |
| 5 | Use an optical power meter to measure the received power level at the Rx port of the alarmed circuit pack. Verify if the Rx power level is within the operational threshold. Record the value. |

Procedure 4-63 (continued)

Tx Loss of Clock

Step	Action						
	For information about circuit pack technical specifications, refer to the “Technical specifications” section in 6500 <i>Planning</i> , NTRN10GK.						
6	<table> <tr> <th>If there is</th><th>Then</th></tr> <tr> <td>a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment</td><td>re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7.</td></tr> <tr> <td>no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment</td><td>go to step 8</td></tr> </table>	If there is	Then	a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7 .	no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	go to step 8
If there is	Then						
a VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	re-adjust, check for proper functionality, and, if necessary, replace the VOA. Note: If the circuit pack is unprotected, adjusting the VOA may impact traffic. If necessary, route traffic to an alternate path. Go step 7 .						
no VOA attached to the fiber between the alarmed circuit pack and the subtending equipment	go to step 8						
7	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 8</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 8
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 8						
8	Use a cleaning kit to clean all the connectors between the subtending equipment Tx port and the alarmed Rx port. Refer to the cleaning connectors procedures in <i>Installation - General Information</i> , 323-1851-201.0.						
9	Record the operating power level after you clean each connector and compare it to the value recorded in step 5 . This will allow you to see if there is any improvement to the Rx power.						
10	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 11</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 11
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 11						
11	Clear any upstream (either at the local to the network element reporting the alarm or other upstream remote network elements) alarms that could be causing this alarm, such as Circuit Pack Failed, Circuit Pack Missing, Channel Controller: Unexpected Loss Detected, Gauge Threshold Crossing Alert Summary, or Loss Of Signal.						
12	Ensure there is no problem with the optical fiber plant between the alarmed circuit pack and the subtending equipment. This includes any fixed pad optical attenuators that may be faulty.						
13	Warm restart the alarmed circuit pack. Refer to the “ Restarting a circuit pack or shelf processor ” procedure in this document.						
14	Reseat the alarmed circuit pack. Refer to the “Reseating a circuit pack” procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						

Procedure 4-63 (continued)

Tx Loss of Clock

Step	Action						
15	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 16</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 16
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 16						
16	Replace the alarmed circuit pack. Refer to the “Replacing an optical interface circuit pack” procedures in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
17	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-64

Tx Loss Of Signal

Alarm IDs: 1780

Probable cause

This alarm is raised when:

- the I/O module was removed
- the I/O module is not fully inserted and locked into position
- the Tx Actual Power on the Inline Submarine Supervisory (ISS) circuit pack is below -7 dBm. The cause can be failure of the subcomponents on the Tx path in the ISS circuit pack or the connection between the subcomponents.

Impact

Critical, service-affecting (C, SA) alarm

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must

- use an account with at least a level 2 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges

Step	Action
1	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
2	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document.

Procedure 4-64 (continued)

Tx Loss Of Signal

Step	Action
3	Verify the facility Tx power level.
4	Connect an optical power meter to port 1 (Tx Monitor) and measure the Tx monitor power.
5	If the measured power showing similar power level as Tx Actual Power, then the circuit pack is faulty. Replace the faulty circuit pack. Refer to the “Replacing an optical circuit pack” procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545. Note: When the ISS C-Band circuit pack is replaced, all traffic will be lost until optical connections to the replacement circuit pack have been completed.
6	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-65

Tx Manual Provisioning Required

Alarm IDs: 669, 1007, 1177, 1435, 1673, 1675

Probable cause

This alarm is raised when a port on the circuit pack requires manual provisioning of the wavelength.

Impact

Major, service-affecting (M, SA) alarm, an active circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or unprotected with cross-connects provisioned

Minor, non-service-affecting (m, NSA) alarm, an inactive circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or no cross-connects provisioned

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- use an account with at least a level 3 UPC

Step	Action
1	Provision the wavelength on the OC192/STM64, ETH10G, PTP, OTM2 or OTM3, or OTM4 facility. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-66

Tx Power In Reduced State

Alarm IDs: 1858, 1869

Probable cause

This event is raised against the line facility of a WLAi variant, 100G WL3 OCLD, Flex2 WL3/WL3e OCLD, 100G WL3n MOTR, 100G WL3e OTR, 100G WL3n OTR, Flex2 WL3e OCLD, Flex3 WL3e OCLD, Flex4 WL3e OCLD circuit pack when Tx Power Reduction is applied on that facility.

This event can also be raised at both ends of 100G transponders when you delete a connection using the Sub-Network Connections application (edit to OOS) in Coherent Select and Flexible Grid Networks.

Note: This event is not masked when the OTM facility or the equipment is placed OOS-MA.

Impact

Warning

Prerequisites

To perform this procedure, you must:

- observe all safety requirements described in “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have the fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- if required, obtain a replacement circuit pack
- use an account with at least a level 3 UPC

Step	Action
------	--------

Note: If the event is raised when provisioning the wavelength/frequency and power for a line facility with an adjacency pointing to a Mux/Demux port that has no associated wavelength cross-connection, then no action is required. The alarm clears once the Mux/Demux port has an associated wavelength cross-connection.

- | | |
|---|---|
| 1 | Identify the facility raising the event. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Release the Tx Power Reduction on the facility identified in step 1 . |

Procedure 4-66 (continued)
Tx Power In Reduced State

Step	Action
3	If the event does not clear, contact your next level of support or your Ciena support group.
—end—	

Procedure 4-67

Tx Power Out of Range

Alarm IDs: 1779

Probable cause

This alarm is raised on the Inline Submarine Supervisory (ISS C-Band) circuit pack when the Tx Actual Power is outside the minimum and maximum output optical power range. This can be caused by failure of the subcomponents on the Tx path of the ISS C-Band circuit pack or the connections between these subcomponents.

Impact

Major, service-affecting (M, SA) alarm, unprotected

Minor, non-service-affecting (m, NSA) alarm, protected

Prerequisites

To perform this procedure, you must:

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- have an antistatic wrist strap to dissipate electrostatic charges
- have the optical fiber connection information (that is, how the optical modules on each network element connect to other network elements)
- have an optical power meter with the same optical connectors as the network element

Step	Action
------	--------

- | | |
|---|--|
| 1 | Identify the circuit pack and facility raising the alarm. Refer to the Procedure 2-12, “Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm” procedure in this document. |
| 2 | Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding. |

Procedure 4-67 (continued)
Tx Power Out of Range

Step	Action								
3	In the Site Manager Configuration menu, select the Equipment & Facility Provisioning application. Select the alarmed facility and retrieve the value from the Tx Actual Power (dBm) column of the facility table. Refer to the "Retrieving optical power, wavelength, and dispersion ranges" procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.								
4	Click on the Ranges button to compare the Tx Actual Power (dBm) value with the Tx minimum power (dBm) and Tx maximum power (dBm) values displayed.								
5	<table> <tr> <th>If the Tx Actual Power is a value</th><th>Then go to</th></tr> <tr> <td>within the Tx minimum power to Tx maximum power range</td><td>step 6</td></tr> <tr> <td>outside the minimum power to Tx maximum power range</td><td>step 9</td></tr> </table>	If the Tx Actual Power is a value	Then go to	within the Tx minimum power to Tx maximum power range	step 6	outside the minimum power to Tx maximum power range	step 9		
If the Tx Actual Power is a value	Then go to								
within the Tx minimum power to Tx maximum power range	step 6								
outside the minimum power to Tx maximum power range	step 9								
6	Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.								
7	Use the optical power meter to measure the optical power level at the last upstream monitored point, and verify that the loss from source of the signal is an expected value.								
	<table> <tr> <th>If the value of the loss is</th><th>Then</th></tr> <tr> <td>unexpected</td><td>troubleshoot the fiber or Photonic layer equipment. Go to step 8.</td></tr> <tr> <td>expected</td><td>go to step 9</td></tr> </table>	If the value of the loss is	Then	unexpected	troubleshoot the fiber or Photonic layer equipment. Go to step 8 .	expected	go to step 9		
If the value of the loss is	Then								
unexpected	troubleshoot the fiber or Photonic layer equipment. Go to step 8 .								
expected	go to step 9								
8	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>step 9</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	step 9		
If the original alarm has	Then								
cleared	the procedure is complete								
not cleared	step 9								
9	Use the optical power meter to measure the optical power level into the Tx interface.								
	<table> <tr> <th>If the optical power is</th><th>Then go to</th></tr> <tr> <td>less than the Tx minimum value in step 4</td><td>step 10</td></tr> <tr> <td>greater than the Tx maximum value in step 4</td><td>step 11</td></tr> <tr> <td>within the Tx minimum power to Tx maximum power range</td><td>step 12</td></tr> </table>	If the optical power is	Then go to	less than the Tx minimum value in step 4	step 10	greater than the Tx maximum value in step 4	step 11	within the Tx minimum power to Tx maximum power range	step 12
If the optical power is	Then go to								
less than the Tx minimum value in step 4	step 10								
greater than the Tx maximum value in step 4	step 11								
within the Tx minimum power to Tx maximum power range	step 12								

Procedure 4-67 (continued)

Tx Power Out of Range

Step	Action						
10	Check the fiber and fiber cleanliness between the subtending equipment and the port reporting the alarm. Go to step 11 .						
11	The signal may require padding or an incorrect subtending circuit pack type. Contact your next level of support or your Ciena support group if you require more information.						
12	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	restart the circuit pack reporting the alarm. Refer to the Procedure 2-11, "Restarting a circuit pack or shelf processor" procedure in this document.						
13	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	reseat the circuit pack reporting the alarm. Refer to the "Reseating a circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
14	<table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>replace the circuit pack. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i>, 323-1851-545.</td></tr></table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	replace the circuit pack. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	replace the circuit pack. Refer to the "Replacing an optical interface circuit pack" procedure in <i>Fault Management - Module Replacement</i> , 323-1851-545.						
15	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-68 TX Tuning in Progress

Alarm IDs: 663, 1008, 1178, 1436, 1515, 1672, 1674

Probable cause

This alarm is raised when Tx wavelength tuning is in progress on a port of a circuit pack.

For Submarine Line Idler 10 Channel (SLIC10 or SLIC10 Flex C-Band) circuit packs, this alarm is raised when Tx wavelength tuning is in progress for one or more lasers in the Idler facility.

Impact

Major, service-affecting (M, SA) alarm, an active circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or unprotected with cross-connects provisioned

Minor, non-service-affecting (m, NSA) alarm, an inactive circuit pack in 1+1/MSP linear, 1+1 port TPT, or 1+1 TPT configuration or no cross-connects provisioned

Step	Action
1	No action is required. This alarm clears when Tx wavelength tuning is completed successfully.
2	If the alarm does not clear, is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-69

Unable to Synchronize TOD

Alarm ID: 336

Probable cause

This alarm is raised when the following occurs:

- at least one time of day server is provisioned, and none of the provisioned time of day servers are reachable or valid
- at least one time of day server is provisioned and there is a problem with external communications from the shelf. The network element is unable to communicate with the provisioned time of day server.
- at least one time of day server is provisioned and there was a restart, upgrade, or momentary problem with external communications from the shelf, and the TOD polling has not yet taken place to re-synchronize the NE with the TOD server

ATTENTION

This alarm displays in the Additional Information column the specific TOD server the alarm is raised against. The Additional Information field is only available for SP-2.

This alarm is raised for an IPv4 and/or IPv6 provisioned server. IPv6 must be enabled if an IPv6 server is provisioned and IPv4 must be enabled if an IPv4 server is provisioned.

When a TOD Server is edited, TOD restarts its re-evaluation of which server it can reach to get the TOD, beginning from the first server in the list (server 1) to the last server. The first server that the TOD is able to reach, is set as ACTIVE. The next server the TOD can reach (after the ACTIVE one), is set to IN-ACTIVE and the servers that cannot be reached are set to UNKNOWN. When TOD can not find any reachable server (as expected) and none of the provisioned servers are reachable, it raises the “Unable to Synchronize TOD” alarm against the last server it evaluated.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Procedure 4-69 (continued)
Unable to Synchronize TOD

Step	Action						
1	Select Node Information from the Configuration menu. Select the Time of Day tab in the Node Information application. Refer to the “Displaying node information” procedure in <i>Administration and Security</i> , 323-1851-301.						
2	Ensure that the IP address for all provisioned time of day servers is correct. You can provision up to five time of day servers. Also ensure that the Settings Status is ON with the expected Polling Interval. Refer to the “Editing time of day synchronization parameters” procedure in <i>Administration and Security</i> , 323-1851-301.						
3	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 4</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 4
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 4						
4	Check if external comms are available to the shelf by trying to ping the shelf. If comms are not available, they must be restored before the network element can communicate with the TOD server. For more information refer to the “Using the ping command” procedure in <i>Administration and Security</i> , 323-1851-301.						
5	If the alarm is still active, check if the TOD server itself is unavailable by trying to ping the TOD server. If the TOD server is available and comms are working, verify if the SNTP server has been adjusted or was temporarily unavailable recently due to a restart, upgrade, or a momentary problem with external communications from the shelf.						
6	<table> <tr> <th>If the SNTP server</th><th>Then go to</th></tr> <tr> <td>has been adjusted or momentary unavailable recently</td><td>step 7</td></tr> <tr> <td>has not been adjusted or momentary unavailable recently</td><td>step 8</td></tr> </table>	If the SNTP server	Then go to	has been adjusted or momentary unavailable recently	step 7	has not been adjusted or momentary unavailable recently	step 8
If the SNTP server	Then go to						
has been adjusted or momentary unavailable recently	step 7						
has not been adjusted or momentary unavailable recently	step 8						
7	Either wait for the polling interval to occur, or operate a time of day synchronization. Refer to the “Operating a time of day synchronization” procedure in <i>Administration and Security</i> , 323-1851-301.						
8	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-70

Unassigned Channel Detected

Alarm ID: 1864

Probable cause

This alarm is raised against the OBMD 1x8 Mux port or CCMD12 Tx port when a signal is detected and no cross-connect is provisioned.

This alarm is raised against ADJ-TX facilities of an OBMD 1x8 module in a WaveLogic Photonics Coherent Select network or CCMD12 when unexpectedly high input power (that is, greater than OPTMON LOS Threshold + 3 db) is measured on an unused Tx port. An unused Tx port in this context is defined as one with no cross-connection provisioned on it.

For the OBMD 1x8D, due to the broadcast nature of the Coherent Select hardware, such unexpected input power is not blocked and propagates throughout the Coherent Select network, potentially trampling an existing channel using the same wavelength.

To clear this alarm, fibering must be verified between the alarmed port on the OBMD 1x8 module or CCMD12, and the transponder, or the input power must be dropped below the OPTMON LOS Threshold.

This alarm is also raised against ADJ facility on the MDRLA 2x6 channel in ports (21, 23, 25, 27, 29, 31) when there is no CRS provisioned and the measured OPTMON power exceeds the provisioned LOS Threshold by more than 3 dB (hysteresis).

Impact

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
------	--------

- | | |
|---|---|
| 1 | Verify the fibering between the alarmed port on the OBMD 1x8 or CCMD12 module and the transponder. Confirm the expected transponder is connected to the expected/provisioned OBMD 1x8 or CCMD12 port. |
|---|---|

Procedure 4-70 (continued)
Unassigned Channel Detected

Step	Action
2	If the ADJ-TX is connected to an SPLI-enabled WL3 transponder which supports Tx Power Reduction, verify that the transponder connected to the alarmed port on the OBMD 1x8 or CCMD12 module is not assigned to carry traffic yet. If this is the case, the Tx Power Reduction must be enabled on that transponder by setting its OTM4 or PTP facility OOS, then set the Tx Power Reduced State to On in Site Manager Equipment & Facility Provisioning, Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
3	If the transponder is not SPLI-enabled or does not support Tx Power Reduction (such as foreign wavelength), verify that the transponder connected to the alarmed port on the OBMD 1x8 or CCMD12 module is not assigned to carry traffic yet. If it is assigned to carry traffic, the laser must be manually turned off (for example, by placing the transponder facility OOS).
4	If the alarm does not clear or is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-71

Unequipped

Alarm IDs: 467

Probable cause

This alarm is raised when:

- there is an improper connection or a facility at the far-end network element is out-of-service. For example, no cross-connect is provisioned at the far-end.

ATTENTION

While provisioning, the likelihood of Unequipped alarms being raised is high. If the alarm is raised during provisioning, allow two to three minutes for the condition to clear prior to troubleshooting the alarm.

This procedure assumes that provisioning is not taking place, the system has been running without STS/HO or VC unequipped errors, and that all fail LEDs are cleared.

Impact

Major, service-affecting (M, SA) alarm, if on an active path

Minor, non-service-affecting (m, NSA) alarm, if on an inactive path

Prerequisites

To perform this procedure, you must

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0 or *Fault Management - Module Replacement*, 323-1851-545
- use an account with at least a level 3 UPC
- have the optical fiber/cable connection information (that is, how the circuit packs on each network element connect to other network elements)

Step	Action
------	--------

- | | | | | | | | |
|----------------------------------|---|----------------------------------|-------------|---------|---------------------------|-------------|------------------------------|
| 1 | Verify whether there are alarms of a higher order from the alarm hierarchy. Refer to the “ Alarm hierarchies and alarm severities ” section in this document. Clear any alarms of a higher order on the hierarchy first using the appropriate procedures. | | | | | | |
| 2 | <table><tr><td>If the original alarm has</td><td>Then</td></tr><tr><td>cleared</td><td>the procedure is complete</td></tr><tr><td>not cleared</td><td>go to step 3</td></tr></table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 3 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 3 | | | | | | |

Procedure 4-71 (continued)
Unequipped

Step	Action						
3	Identify the circuit pack raising the alarm. Refer to the Procedure 2-12, "Identifying the circuit pack, pluggable module/port, or facility that has raised an alarm" procedure in this document.						
4	Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.						
5	If the local network element is part of a mid-span meet and the remote network element is another equipment type or from another vendor, use the alarm system of the other equipment or vendor to find the problem.						
6	<table> <tr> <th>If the original alarm has</th><th>Then</th></tr> <tr> <td>cleared</td><td>the procedure is complete</td></tr> <tr> <td>not cleared</td><td>go to step 7</td></tr> </table>	If the original alarm has	Then	cleared	the procedure is complete	not cleared	go to step 7
If the original alarm has	Then						
cleared	the procedure is complete						
not cleared	go to step 7						
7	Log into each of the network elements on the required path and retrieve the cross-connects. Refer to the "Retrieving path connections" procedure , 323-1851-320.						
8	Verify that the far-end facilities of the connection are in-service. Refer to the "Retrieving equipment and facility details" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310. If the far-end facilities of the connection are out-of-service, change the facility primary state to in-service. Refer to the "Editing facility parameters" procedure in <i>Configuration - Provisioning and Operating</i>, 323-1851-310.						
9	Verify cross-connects for the entire path to ensure an end-to-end connection exists. If an end-to-end connection does not exist, provision the necessary cross-connects to make a complete end-to-end connection. Refer to the "Provisioning a bidirectional connection in a 1+1/MSP linear configuration" procedure in <i>Configuration - Connections Management</i>, 323-1851-320.						
10	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-72

Unpaired SSH Key

Alarm ID: 936

Probable cause

This alarm is raised against the shelf when the SSH keys are invalid. That is, the public and private keys are unpaired.

Impact

Major, non-service-affecting (M, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action						
1	<table><tr><td>If you want to</td><td>Then go to</td></tr><tr><td>create a new valid key pair</td><td>step 2</td></tr><tr><td>use a specific public/private key pair</td><td>step 3</td></tr></table>	If you want to	Then go to	create a new valid key pair	step 2	use a specific public/private key pair	step 3
If you want to	Then go to						
create a new valid key pair	step 2						
use a specific public/private key pair	step 3						
2	Regenerate a new public/private key pair. Refer to the “Regenerating SSH/SFTP keys” procedure in <i>Administration and Security</i> , 323-1851-301. If the alarm does not clear, go to step 4 .						
3	Enter the desired public/private key pair.						
4	If the alarm does not clear, contact your next level of support or your Ciena support group.						

—end—

Procedure 4-73

Unsupported Channel Provisioned

Alarm ID: 1944

Probable cause

This alarm is raised when a channel is provisioned and the channel is not supported by one or more equipment present in the network path the channel traverses.

After the add/drop connections are provisioned in the network, NCT tries to build the channel. If the channel is not supported by some equipment in the path, it marks the channel as unsupported and raises the alarm.

Impact

Warning

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Find out which channel is not supported in network. De-provision the unsupported channel. Note: The channel can be provisioned successfully only if the equipment that is not capable of supporting the channel is replaced by the one that can support it.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-74

Validation Certificate About to Expire

Alarm ID: 2040

Probable cause

This alarm is raised when a validation certificate in the shelf processor is about to expire based on a provisioned number of days before the certificate expiry.

The CA (Certificate Authority) issues a validation certificate with a given expiry date. If the time of day on the shelf processor passes the provisioned number of days before the certificate expiry date, this alarm is raised. The default number of days is 60 days and is provisionable from 0 to 180 days.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Connect to Network Element using Site Manager.
2	Select Manage keys from Security menu, and click on Certificates to validate Servers .
3	To clear the alarm perform one of the following: a. Upload a new certificate. b. Delete the certificate.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-75

Validation Certificate Expired

Alarm ID: 2039

Probable cause

This alarm is raised when a validation certificate in the shelf processor is expired.

The CA (Certificate Authority) issues a validation certificate with a given expiry date. If the time of day on the shelf processor passes the certificate expiry date, this alarm is raised.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Connect to Network Element using Site Manager.
2	Select Manage keys from Security menu, and click on Certificates to validate Servers .
3	To clear the alarm perform one of the following: a. Upload a new certificate. b. Delete the certificate.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-76

VOA Output LOS

Alarm ID: 1626

Probable cause

This alarm is raised when the total output optical power after the VOA has fallen below the provisioned output threshold value.

The alarm can be raised between the erbium-doped fiber amplifier (EDFA), and the VOA due to a hardware failure that could cause the power at the output of the EDFA to be very low.

When the alarm correlation is on, this alarm can be masked by the “Shutoff Threshold Crossed” alarm or the “Input Loss Of Signal” alarm.

Impact

Major-service-affecting (M, SA) alarm

Prerequisites

Before you perform this procedure, you must

- ensure you have all the documentation referenced in this procedure
- use an account with level 3 or higher user privilege code (UPC)

Step	Action
1	Ensure the alarm correlation is off.
2	If the “Shutoff Threshold Crossed” or the “Input Loss Of Signal” alarms are active, clear the alarms by improving input power into the amplifier.
3	Increase the power that goes through the VOA by reducing the VOA attenuation or increasing the amplifier gain. Refer to the “Editing facility parameters” procedure in <i>Configuration - Provisioning and Operating</i> , 323-1851-310.
4	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-77

Warm Restart Required

Alarm ID: 1426

Probable cause

This alarm is raised during MRO reconfiguration, if a card under reconfiguration has transparent DCC provisioned on one of its ports (“mated” to a port on another card on the shelf), and the reconfiguration involves a port number change (for example, line rate reconfiguration of an OC48 facility to a OC192 facility on a 20G MRO, or circuit pack reconfiguration of a 2xOC48 to a 10G 16xOCn MRO). In this case, any “DCC mate” cards of the card being reconfigured need to be warm restarted to appropriately handle the change to “their” own Slot-Based Files (SBFs).

If the DCC mate card is not restarted, then any transparent DCC link between the cards will not recover after the reconfiguration, staying permanently in a DISCONNECTED state.

This alarm can be raised against any card type which supports DCC link provisioning.

This alarm is also raised when OSRP is provisioned with type L0 Provisioning [PROV] and the shelf IP address has changed.

Impact

Minor, non-service-affecting (m, NSA) alarm

Major, service-affecting (M, SA) alarm

Prerequisites

To perform this procedure, you require an account with at least a level 3 UPC.

Step	Action
1	Perform a warm restart on the circuit pack that is raising the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
2	If the alarm does not clear, contact your next level of support or your Ciena support group.

—end—

Procedure 4-78

Wavelength Measurement Error

Alarm ID: 879

Probable cause

This alarm is raised against an OPTMON facility when the channel has been detected to have drifted by more than 25 GHz from its start of monitoring reference point.

Impact

Major, non-service-affecting (M, NSA) alarm

Step	Action
1	No action is required. This alarm clears when Tx wavelength tuning is completed successfully.
2	If the alarm does not clear, check if a “High Fiber Loss” Alarm is raised against the WSS OPTMON Port on the ADJ facility between the ROADM WSS and DIA WSS. If the alarm is present, follow the Procedure 4-140, “High Fiber Loss” in this document to clear the alarm. After the “High Fiber Loss” alarm is cleared, wait for 20 minutes for the “Wavelength Measurement Error” alarm to clear. If the alarm does not clear, continue to the next step.
3	If the alarm does not clear, perform a warm restart on the circuit pack that is raising the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
4	If the alarm does not clear after the circuit pack warm restart, warm restart the SP. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
5	If the alarm does not clear, is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-79

Wavelength Measurement Warning

Alarm ID: 878**Probable cause**

This alarm is raised against an OPTMON facility when the channel has been detected to have drifted by more than 20 GHz from its start of monitoring reference point.

This alarm can also be raised during an upgrade. If the alarm is raised during an upgrade, there is no need to back out the upgrade. Contact your next level of support or your Ciena support group.

Impact.

Minor, non-service-affecting (m, NSA) alarm

Step	Action
1	No action is required. This alarm clears when Tx wavelength tuning is completed successfully.
2	If the alarm does not clear, check if a “High Fiber Loss” Alarm is raised against the WSS OPTMON Port on the ADJ facility between the ROADM WSS and DIA WSS. If the alarm is present, follow the Procedure 4-140, “High Fiber Loss” in this document to clear the alarm. After the “High Fiber Loss” alarm is cleared, wait for 20 minutes for the “Wavelength Measurement Error” alarm to clear. If the alarm does not clear, continue to the next step.
3	If the alarm does not clear, perform a warm restart on the circuit pack that is raising the alarm. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
4	If the alarm does not clear after the circuit pack warm restart, warm restart the SP. Refer to the Procedure 2-11, “Restarting a circuit pack or shelf processor” procedure in this document.
5	If the alarm does not clear, is unexpected, contact your next level of support or your Ciena support group.

—end—

Procedure 4-80

WAYSIDE 1/2 Port Failure

Alarm IDs: 573, 574

Probable cause

This alarm is raised against a wayside channel (WSC) port facility of a SPAP-2 w/2xOSC, 2xOSC, ESAM, SAM, or SRA circuit pack when the wayside port cannot connect to another wayside port or when the port is provisioned and enabled, but the link is down.

Wayside traffic enters the system through one wayside port and exits the system through another wayside port.



CAUTION

Risk of traffic loss

A warm or cold restart of a SPAP-2 w/2xOSC, 2xOSC, ESAM, SAM, or SRA circuit pack causes a WSC traffic hit.

Impact

Minor, non-service-affecting (m, NSA) alarm

Prerequisites

To perform this procedure, you must:

- use an account with at least a level 3 UPC
- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section in *Installation - General Information*, 323-1851-201.0
- have an antistatic wrist strap to dissipate electrostatic charges
- have DCN information for the network

Step	Action
------	--------

1	Check the DCN information to determine if the wayside port should be enabled.
---	---

If the WSC facility should be	Then go to
--------------------------------------	-------------------

disabled	step 2
----------	------------------------

enabled	step 3
---------	------------------------

Procedure 4-80 (continued)

WAYSIDE 1/2 Port Failure

- | Step | Action | | | | | | |
|----------------------------------|--|----------------------------------|-------------|---------|---------------------------|-------------|------------------------------|
| 2 | Delete the alarmed WSC facility from the Comms Setting Management Site Manager application (Interfaces tab, Interface type=LAN). Refer to the “Deleting an entry in the communications settings” procedure in <i>Data Comms Planning & User Guide</i> , 323-1851-101. | | | | | | |
| | <div style="border: 1px solid black; padding: 5px; text-align: center;"> <p>ATTENTION</p> <p>By deleting the WSC facility, you lose any established connections that use this port.</p> </div> | | | | | | |
| | The procedure is complete. | | | | | | |
| 3 | Verify that the wayside configuration at both ends of the Ethernet link match. Correct any WSC LAN parameter mismatches from the Comms Setting Management Site Manager application (Interfaces tab, Interface type=LAN). Refer to the “Editing the communications settings” procedure in <i>Data Comms Planning & User Guide</i> , 323-1851-101.

For further details on wayside interface, refer to the “Wayside channel (WSC) interface” section in <i>Data Comms Planning & User Guide</i> , 323-1851-101. | | | | | | |
| 4 | <table border="0"> <tr> <td style="vertical-align: top;">If the original alarm has</td> <td style="vertical-align: top;">Then</td> </tr> <tr> <td style="border-top: 1px solid black; padding-top: 5px;">cleared</td> <td style="border-top: 1px solid black; padding-top: 5px;">the procedure is complete</td> </tr> <tr> <td style="padding-top: 5px;">not cleared</td> <td style="padding-top: 5px;">go to step 5</td> </tr> </table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 5 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 5 | | | | | | |
| 5 | Ensure both ends of the wayside cable are connected and the ports are enabled on both devices. | | | | | | |
| 6 | <table border="0"> <tr> <td style="vertical-align: top;">If the original alarm has</td> <td style="vertical-align: top;">Then</td> </tr> <tr> <td style="border-top: 1px solid black; padding-top: 5px;">cleared</td> <td style="border-top: 1px solid black; padding-top: 5px;">the procedure is complete</td> </tr> <tr> <td style="padding-top: 5px;">not cleared</td> <td style="padding-top: 5px;">go to step 7</td> </tr> </table> | If the original alarm has | Then | cleared | the procedure is complete | not cleared | go to step 7 |
| If the original alarm has | Then | | | | | | |
| cleared | the procedure is complete | | | | | | |
| not cleared | go to step 7 | | | | | | |
| 7 | Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf. | | | | | | |
| 8 | The cable connected to the wayside port may be defective. Replace the cable. | | | | | | |
| 9 | If the alarm does not clear, contact your next level of support or your Ciena support group. | | | | | | |

—end—



6500 Packet-Optical Platform

Fault Management - Alarm Clearing

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